

GNU Gnulib

updated 2026-01-01 20:18:25

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This manual is for GNU Gnulib (updated 2026-01-01 20:18:25), which is a library of common routines intended to be shared at the source level.

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Table of Contents

1	Brief Overview	1
1.1	Gnulib Basics	1
1.2	Git Checkout	1
1.3	Keeping Up-to-date	2
1.4	Contributing to Gnulib	2
1.4.1	Gnulib licensing	2
1.4.2	Indent with spaces not TABs	2
1.4.3	How to add a new module	3
1.5	Portability guidelines	4
1.5.1	C language versions	4
1.5.2	C99 features assumed by Gnulib	5
1.5.3	C99 features avoided by Gnulib	5
1.5.4	Other portability assumptions made by Gnulib	6
1.6	High Quality	7
1.6.1	Stable Branches	7
1.6.2	Writing reliable code	9
1.7	Join the GNU Project	9
2	Philosophy	11
2.1	Benefits of using Gnulib	11
2.2	Library vs. Reusable Code	11
2.2.1	Comparison with other source code libraries	11
2.3	Portability and Application Code	12
2.4	Target Platforms	13
2.4.1	Supported Platforms	13
2.4.2	Formerly Supported Platforms	14
2.4.3	Unsupported Platforms	14
2.5	Modules	15
2.6	Various Kinds of Modules	16
2.6.1	Support for ISO C or POSIX functions	16
2.6.2	Enhancements of ISO C or POSIX functions	17
2.6.3	Portable general use facilities	17
2.6.4	Reusable application code	17
2.6.5	Object oriented classes	17
2.6.6	Interfaces to external libraries	17
2.6.7	Build / maintenance infrastructure	17
2.7	Collaborative Development	17
2.8	Copyright	18
2.9	Steady Development	19
2.10	Openness	19
3	Invoking gnulib-tool	20
3.1	Finding modules	20

3.2	Initial import	20
3.3	Modified imports	24
3.4	Simple update	27
3.5	Changing your sources for use with Gnulib	27
3.5.1	-I options	27
3.5.2	Include <config.h>	27
3.5.3	Style of #include statements	27
3.6	Changing your link commands for use with Gnulib	29
3.7	Finding recommended ISO C and POSIX function substitutes ..	30
3.8	Modifying the build rules of a Gnulib import directory	30
3.9	Building directly from the top-level directory	31
3.10	Using Gnulib for both a library and a program	32
3.11	Caveat: <code>gettextize</code> and <code>autopoint</code> users	34
3.12	Handling Gnulib's own message translations	35
3.13	Integration with Version Control Systems	36
3.14	Bundling the unit tests of the Gnulib modules	37
3.15	Avoiding unnecessary checks and compilations	38
4	Writing modules	40
4.1	Source code files	40
4.2	Header files	40
4.3	Implementation files	41
4.4	Specification	41
4.5	Module description	42
4.6	Autoconf macros	45
4.7	Making proper use of <code>AC_LIBOBJ</code>	45
4.8	Unit test modules	46
4.9	Incompatible changes	47
5	Extending Gnulib	48
6	Miscellaneous Notes	49
6.1	Out of memory handling	49
6.2	Obsolete modules	49
6.3	Extra tests modules	49
6.4	Modules that modify the way other modules work	50
6.5	A C++ namespace for <code>gnulib</code>	51
6.6	License Texinfo sources	52
6.7	Building <code>gnulib</code>	53
6.8	Gnulib Git Bundle	54
7	Building the ISO C and POSIX Substitutes ..	56

8 ISO C Keyword Substitutes 57

8.1 alignof and alignas	57
8.2 bool	57
8.3 nullptr	57
8.4 static_assert	58

9 ISO C and POSIX Header File Substitutes .. 59

9.1 aio.h	59
9.2 arpa/inet.h	59
9.3 assert.h	59
9.4 complex.h	60
9.5 cpio.h	60
9.6 ctype.h	60
9.7 devctl.h	61
9.8 dirent.h	61
9.9 dlfcn.h	62
9.10 endian.h	62
9.11 errno.h	62
9.12 fcntl.h	63
9.13 fenv.h	65
9.14 float.h	65
9.15 fmtmsg.h	66
9.16 fnmatch.h	66
9.17 ftw.h	67
9.18 glob.h	67
9.19 grp.h	67
9.20 iconv.h	67
9.21 inttypes.h	68
9.22 iso646.h	68
9.23 langinfo.h	68
9.24 libgen.h	69
9.25 libintl.h	69
9.26 limits.h	69
9.27 locale.h	70
9.28 math.h	71
9.29 monetary.h	71
9.30 mqueue.h	72
9.31 ndbm.h	72
9.32 net/if.h	72
9.33 netdb.h	72
9.34 netinet/in.h	73
9.35 netinet/tcp.h	73
9.36 nl_types.h	73
9.37 poll.h	73
9.38 pthread.h	74
9.39 pwd.h	74

9.40	regex.h	74
9.41	sched.h	75
9.42	search.h	75
9.43	semaphore.h	75
9.44	setjmp.h	75
9.45	signal.h	76
9.46	spawn.h	76
9.47	stdalign.h	76
9.48	stdarg.h	78
9.49	stdatomic.h	78
9.50	stdbit.h	78
9.51	stdbool.h	79
9.52	stdckdint.h	79
9.53	stddef.h	80
9.54	stdint.h	80
9.55	stdio.h	81
9.56	stdlib.h	82
9.57	stdnoreturn.h	82
9.58	string.h	83
9.59	strings.h	83
9.60	stropts.h	84
9.61	sys/ipc.h	84
9.62	sys/mman.h	84
9.63	sys/msg.h	84
9.64	sys/resource.h	85
9.65	sys/select.h	85
9.66	sys/sem.h	85
9.67	sys/shm.h	86
9.68	sys/socket.h	86
9.69	sys/stat.h	87
9.70	sys/statvfs.h	88
9.71	sys/time.h	89
9.72	sys/timeb.h	89
9.73	sys/times.h	89
9.74	sys/types.h	89
9.75	sys/uio.h	90
9.76	sys/un.h	90
9.77	sys/utsname.h	91
9.78	sys/wait.h	91
9.79	syslog.h	91
9.80	tar.h	92
9.81	termios.h	92
9.82	tgmath.h	92
9.83	threads.h	92
9.84	time.h	93
9.85	trace.h	94
9.86	uchar.h	94

9.87	<code>ucontext.h</code>	95
9.88	<code>ulimit.h</code>	95
9.89	<code>unistd.h</code>	95
9.90	<code>utime.h</code>	96
9.91	<code>utmpx.h</code>	96
9.92	<code>wchar.h</code>	97
9.93	<code>wctype.h</code>	97
9.94	<code>wordexp.h</code>	97

10 ISO C and POSIX Function Substitutes 98

10.1	Functions in <code><aio.h></code>	98
10.1.1	<code>aio_cancel</code>	98
10.1.2	<code>aio_error</code>	98
10.1.3	<code>aio_fsync</code>	98
10.1.4	<code>aio_read</code>	99
10.1.5	<code>aio_return</code>	99
10.1.6	<code>aio_suspend</code>	99
10.1.7	<code>aio_write</code>	100
10.1.8	<code>lio_listio</code>	100
10.2	Functions in <code><arpa/inet.h></code>	100
10.2.1	<code>htonl</code>	100
10.2.2	<code>htons</code>	100
10.2.3	<code>inet_addr</code>	101
10.2.4	<code>inet_ntoa</code>	101
10.2.5	<code>inet_ntop</code>	101
10.2.6	<code>inet_pton</code>	102
10.2.7	<code>ntohl</code>	102
10.2.8	<code>ntohs</code>	102
10.3	Functions in <code><assert.h></code>	103
10.3.1	<code>assert</code>	103
10.4	Functions in <code><complex.h></code>	103
10.4.1	<code>CMPLX</code>	103
10.4.2	<code>CMPLXF</code>	103
10.4.3	<code>CMPLXL</code>	103
10.4.4	<code>cabs</code>	104
10.4.5	<code>cabsf</code>	104
10.4.6	<code>cabsl</code>	104
10.4.7	<code>cacos</code>	104
10.4.8	<code>cacosf</code>	105
10.4.9	<code>cacosh</code>	105
10.4.10	<code>cacoshf</code>	105
10.4.11	<code>cacoshl</code>	105
10.4.12	<code>cacosl</code>	106
10.4.13	<code>carg</code>	106
10.4.14	<code>cargf</code>	106
10.4.15	<code>cargl</code>	106

10.4.16	<code>casin</code>	107
10.4.17	<code>casinf</code>	107
10.4.18	<code>casinh</code>	107
10.4.19	<code>casinhf</code>	107
10.4.20	<code>casinhl</code>	108
10.4.21	<code>casinl</code>	108
10.4.22	<code>catan</code>	108
10.4.23	<code>catanf</code>	108
10.4.24	<code>catanh</code>	109
10.4.25	<code>catanhf</code>	109
10.4.26	<code>catanhl</code>	109
10.4.27	<code>catanl</code>	109
10.4.28	<code>ccos</code>	110
10.4.29	<code>ccosf</code>	110
10.4.30	<code>ccosh</code>	110
10.4.31	<code>ccoshf</code>	110
10.4.32	<code>ccoshl</code>	110
10.4.33	<code>ccosl</code>	111
10.4.34	<code>cexp</code>	111
10.4.35	<code>cexpf</code>	111
10.4.36	<code>cexpl</code>	111
10.4.37	<code>cimag</code>	112
10.4.38	<code>cimagf</code>	112
10.4.39	<code>cimagl</code>	112
10.4.40	<code>clog</code>	112
10.4.41	<code>clogf</code>	113
10.4.42	<code>clogl</code>	113
10.4.43	<code>conj</code>	113
10.4.44	<code>conjf</code>	113
10.4.45	<code>conjl</code>	114
10.4.46	<code>cpow</code>	114
10.4.47	<code>cpowf</code>	114
10.4.48	<code>cpowl</code>	114
10.4.49	<code>cproj</code>	115
10.4.50	<code>cprojf</code>	115
10.4.51	<code>cprojl</code>	115
10.4.52	<code>creal</code>	115
10.4.53	<code>crealf</code>	116
10.4.54	<code>creall</code>	116
10.4.55	<code>csin</code>	116
10.4.56	<code>csinf</code>	116
10.4.57	<code>csinh</code>	117
10.4.58	<code>csinhf</code>	117
10.4.59	<code>csinhl</code>	117
10.4.60	<code>csinl</code>	117
10.4.61	<code>csqrt</code>	118
10.4.62	<code>csqrtf</code>	118

10.4.63	<code>csqrtl</code>	118
10.4.64	<code>ctan</code>	118
10.4.65	<code>ctanf</code>	119
10.4.66	<code>ctanh</code>	119
10.4.67	<code>ctanhf</code>	119
10.4.68	<code>ctanhl</code>	119
10.4.69	<code>ctanl</code>	120
10.5	Functions in <code><ctype.h></code>	120
10.5.1	<code>isalnum</code>	120
10.5.2	<code>isalnum_l</code>	121
10.5.3	<code>isalpha</code>	121
10.5.4	<code>isalpha_l</code>	122
10.5.5	<code>isblank</code>	122
10.5.6	<code>isblank_l</code>	123
10.5.7	<code>iscntrl</code>	123
10.5.8	<code>iscntrl_l</code>	124
10.5.9	<code>isdigit</code>	124
10.5.10	<code>isdigit_l</code>	125
10.5.11	<code>isgraph</code>	125
10.5.12	<code>isgraph_l</code>	126
10.5.13	<code>islower</code>	126
10.5.14	<code>islower_l</code>	127
10.5.15	<code>isprint</code>	127
10.5.16	<code>isprint_l</code>	128
10.5.17	<code>ispunct</code>	128
10.5.18	<code>ispunct_l</code>	129
10.5.19	<code>isspace</code>	129
10.5.20	<code>isspace_l</code>	130
10.5.21	<code>isupper</code>	130
10.5.22	<code>isupper_l</code>	131
10.5.23	<code>isxdigit</code>	131
10.5.24	<code>isxdigit_l</code>	132
10.5.25	<code>tolower</code>	132
10.5.26	<code>tolower_l</code>	132
10.5.27	<code>toupper</code>	133
10.5.28	<code>toupper_l</code>	133
10.6	Functions in <code><devctl.h></code>	134
10.6.1	<code>posix_devctl</code>	134
10.7	Functions in <code><dirent.h></code>	134
10.7.1	<code>alphasort</code>	134
10.7.2	<code>closedir</code>	134
10.7.3	<code>dirfd</code>	134
10.7.4	<code>fdopendir</code>	135
10.7.5	<code>opendir</code>	135
10.7.6	<code>posix_getdents</code>	135
10.7.7	<code>readdir</code>	136
10.7.8	<code>readdir_r</code>	136

10.7.9	<code>rewinddir</code>	137
10.7.10	<code>scandir</code>	137
10.7.11	<code>seekdir</code>	137
10.7.12	<code>telldir</code>	138
10.8	Functions in <code><dlfcn.h></code>	138
10.8.1	<code>dladdr</code>	138
10.8.2	<code>dlclose</code>	138
10.8.3	<code>dLError</code>	138
10.8.4	<code>dlopen</code>	139
10.8.5	<code>dlsym</code>	139
10.9	Functions in <code><endian.h></code>	139
10.9.1	<code>be16toh</code>	139
10.9.2	<code>be32toh</code>	140
10.9.3	<code>be64toh</code>	140
10.9.4	<code>htobe16</code>	140
10.9.5	<code>htobe32</code>	140
10.9.6	<code>htobe64</code>	141
10.9.7	<code>htole16</code>	141
10.9.8	<code>htole32</code>	141
10.9.9	<code>htole64</code>	142
10.9.10	<code>le16toh</code>	142
10.9.11	<code>le32toh</code>	142
10.9.12	<code>le64toh</code>	142
10.10	Functions in <code><errno.h></code>	143
10.10.1	<code>errno</code>	143
10.11	Functions in <code><fcntl.h></code>	143
10.11.1	<code>creat</code>	143
10.11.2	<code>fcntl</code>	143
10.11.3	<code>open</code>	144
10.11.4	<code>openat</code>	145
10.11.5	<code>posix_fadvise</code>	146
10.11.6	<code>posix_fallocate</code>	146
10.12	Functions in <code><fenv.h></code>	147
10.12.1	<code>feclearexcept</code>	147
10.12.2	<code>fegetenv</code>	147
10.12.3	<code>fegetexceptflag</code>	147
10.12.4	<code>fegetmode</code>	148
10.12.5	<code>fegetround</code>	148
10.12.6	<code>feholdexcept</code>	148
10.12.7	<code>feraiseexcept</code>	148
10.12.8	<code>fesetenv</code>	149
10.12.9	<code>fesetexcept</code>	149
10.12.10	<code>fesetexceptflag</code>	149
10.12.11	<code>fesetmode</code>	150
10.12.12	<code>fesetround</code>	150
10.12.13	<code>fetestexcept</code>	150
10.12.14	<code>fetestexceptflag</code>	151

10.12.15	feupdateenv.....	151
10.13	Functions in <fmtmsg.h>.....	151
10.13.1	fmtmsg.....	151
10.14	Functions in <fnmatch.h>.....	152
10.14.1	fnmatch.....	152
10.15	Functions in <ftw.h>.....	153
10.15.1	nftw.....	153
10.16	Functions in <glob.h>.....	153
10.16.1	glob.....	153
10.16.2	globfree.....	154
10.17	Functions in <grp.h>.....	154
10.17.1	endgrent.....	154
10.17.2	getgrent.....	154
10.17.3	getgrgid.....	154
10.17.4	getgrgid_r.....	155
10.17.5	getgrnam.....	155
10.17.6	getgrnam_r.....	155
10.17.7	setgrent.....	155
10.18	Functions in <iconv.h>.....	156
10.18.1	iconv.....	156
10.18.2	iconv_close.....	156
10.18.3	iconv_open.....	156
10.19	Functions in <inttypes.h>.....	157
10.19.1	imaxabs.....	157
10.19.2	imaxdiv.....	157
10.19.3	strtoimax.....	158
10.19.4	strtoumax.....	158
10.19.5	wcstoimax.....	158
10.19.6	wcstoumax.....	159
10.20	Functions in <langinfo.h>.....	159
10.20.1	nl_langinfo.....	159
10.20.2	nl_langinfo_l.....	160
10.21	Functions in <libgen.h>.....	160
10.21.1	basename.....	160
10.21.2	dirname.....	160
10.22	Functions in <libintl.h>.....	161
10.22.1	bind_textdomain_codeset.....	161
10.22.2	bindtextdomain.....	161
10.22.3	dcgettext.....	161
10.22.4	dcgettext_l.....	162
10.22.5	dcngettext.....	162
10.22.6	dcngettext_l.....	162
10.22.7	dgettext.....	163
10.22.8	dgettext_l.....	163
10.22.9	dngettext.....	163
10.22.10	dngettext_l.....	164
10.22.11	gettext.....	164

10.22.12	gettext_l.....	164
10.22.13	ngettext.....	164
10.22.14	ngettext_l.....	165
10.22.15	textdomain.....	165
10.23	Functions in <locale.h>.....	165
10.23.1	duplocale.....	165
10.23.2	freelocale.....	166
10.23.3	getlocalename_l.....	166
10.23.4	localeconv.....	167
10.23.5	newlocale.....	167
10.23.6	setlocale.....	167
10.23.7	uselocale.....	168
10.24	Functions in <math.h>.....	169
10.24.1	acos.....	169
10.24.2	acosf.....	169
10.24.3	acosh.....	169
10.24.4	acoshf.....	170
10.24.5	acoshl.....	170
10.24.6	acosl.....	170
10.24.7	acospi.....	170
10.24.8	acospif.....	171
10.24.9	acospil.....	171
10.24.10	asin.....	171
10.24.11	asinf.....	171
10.24.12	asinh.....	172
10.24.13	asinhf.....	172
10.24.14	asinh1.....	172
10.24.15	asinl.....	172
10.24.16	asinpi.....	173
10.24.17	asinpif.....	173
10.24.18	asinpil.....	173
10.24.19	atan.....	173
10.24.20	atan2.....	173
10.24.21	atan2f.....	174
10.24.22	atan2l.....	174
10.24.23	atan2pi.....	174
10.24.24	atan2pif.....	174
10.24.25	atan2pil.....	175
10.24.26	atanf.....	175
10.24.27	atanh.....	175
10.24.28	atanhf.....	175
10.24.29	atanhl.....	176
10.24.30	atanl.....	176
10.24.31	atanpi.....	176
10.24.32	atanpif.....	176
10.24.33	atanpil.....	177
10.24.34	canonicalize.....	177

10.24.35	canonicalizef	177
10.24.36	canonicalizel	177
10.24.37	cbrt	178
10.24.38	cbrtf	178
10.24.39	cbrtl	178
10.24.40	ceil	178
10.24.41	ceilf	179
10.24.42	ceill	179
10.24.43	compoundn	179
10.24.44	compoundnf	180
10.24.45	compoundnl	180
10.24.46	copysign	180
10.24.47	copysignf	180
10.24.48	copysignl	180
10.24.49	cos	181
10.24.50	cosf	181
10.24.51	cosh	181
10.24.52	coshf	181
10.24.53	coshl	182
10.24.54	cosl	182
10.24.55	cospi	182
10.24.56	cospif	182
10.24.57	cospil	183
10.24.58	daddl	183
10.24.59	ddivl	183
10.24.60	dfmal	183
10.24.61	dmull	184
10.24.62	dsqrtl	184
10.24.63	dsubl	184
10.24.64	erf	185
10.24.65	erfc	185
10.24.66	erfcf	185
10.24.67	erfcl	185
10.24.68	erff	185
10.24.69	erfl	186
10.24.70	exp	186
10.24.71	exp10	186
10.24.72	exp10f	187
10.24.73	exp10l	187
10.24.74	exp10m1	187
10.24.75	exp10m1f	188
10.24.76	exp10m1l	188
10.24.77	exp2	188
10.24.78	exp2f	188
10.24.79	exp2l	189
10.24.80	exp2m1	189
10.24.81	exp2m1f	189

10.24.82	<code>exp2m1l</code>	189
10.24.83	<code>expf</code>	190
10.24.84	<code>expl</code>	190
10.24.85	<code>expm1</code>	190
10.24.86	<code>expm1f</code>	191
10.24.87	<code>expm1l</code>	191
10.24.88	<code>fabs</code>	191
10.24.89	<code>fabsf</code>	191
10.24.90	<code>fabsl</code>	192
10.24.91	<code>fadd</code>	192
10.24.92	<code>faddl</code>	192
10.24.93	<code>fdim</code>	192
10.24.94	<code>fdimf</code>	193
10.24.95	<code>fdiml</code>	193
10.24.96	<code>fdiv</code>	193
10.24.97	<code>fdivl</code>	193
10.24.98	<code>ffma</code>	194
10.24.99	<code>ffmal</code>	194
10.24.100	<code>floor</code>	194
10.24.101	<code>floorf</code>	194
10.24.102	<code>floorl</code>	195
10.24.103	<code>fma</code>	195
10.24.104	<code>maf</code>	195
10.24.105	<code>fmal</code>	196
10.24.106	<code>fmax</code>	196
10.24.107	<code>fmaxf</code>	196
10.24.108	<code>fmaximum</code>	196
10.24.109	<code>fmaximum_mag</code>	197
10.24.110	<code>fmaximum_magf</code>	197
10.24.111	<code>fmaximum_magl</code>	197
10.24.112	<code>fmaximum_mag_num</code>	197
10.24.113	<code>fmaximum_mag_numf</code>	198
10.24.114	<code>fmaximum_mag_numl</code>	198
10.24.115	<code>fmaximum_num</code>	198
10.24.116	<code>fmaximum_numf</code>	199
10.24.117	<code>fmaximum_numl</code>	199
10.24.118	<code>fmaximumf</code>	199
10.24.119	<code>fmaximuml</code>	199
10.24.120	<code>fmaxl</code>	200
10.24.121	<code>fmaxmag</code>	200
10.24.122	<code>fmaxmagf</code>	200
10.24.123	<code>fmaxmagl</code>	200
10.24.124	<code>fmin</code>	201
10.24.125	<code>fminf</code>	201
10.24.126	<code>fminimum</code>	201
10.24.127	<code>fminimum_mag</code>	201
10.24.128	<code>fminimum_magf</code>	202

10.24.129	fminimum_magl	202
10.24.130	fminimum_mag_num	202
10.24.131	fminimum_mag_numf	203
10.24.132	fminimum_mag_numl	203
10.24.133	fminimum_num	203
10.24.134	fminimum_numf	203
10.24.135	fminimum_numl	204
10.24.136	fminimumf	204
10.24.137	fminimuml	204
10.24.138	fminl	205
10.24.139	fminmag	205
10.24.140	fminmagf	205
10.24.141	fminmagl	205
10.24.142	fmod	206
10.24.143	fmodf	206
10.24.144	fmodl	206
10.24.145	fmul	207
10.24.146	fmull	207
10.24.147	fpclassify	207
10.24.148	frexp	207
10.24.149	frexpf	208
10.24.150	frexpl	208
10.24.151	fromfp	208
10.24.152	fromfpf	209
10.24.153	fromfpl	209
10.24.154	fromfpx	209
10.24.155	fromfpxf	209
10.24.156	fromfpxl	210
10.24.157	fsqrt	210
10.24.158	fsqrtl	210
10.24.159	fsub	210
10.24.160	fsubl	211
10.24.161	getpayload	211
10.24.162	getpayloadf	211
10.24.163	getpayloadl	212
10.24.164	hypot	212
10.24.165	hypotf	212
10.24.166	hypotl	213
10.24.167	ilogb	213
10.24.168	ilogbf	213
10.24.169	ilogbl	214
10.24.170	isfinite	214
10.24.171	isgreater	214
10.24.172	isgreaterequal	215
10.24.173	isinf	215
10.24.174	isless	215
10.24.175	islessequal	215

10.24.176	islessgreater	216
10.24.177	isnan	216
10.24.178	isnormal	216
10.24.179	isunordered	216
10.24.180	j0	217
10.24.181	j1	217
10.24.182	jn	217
10.24.183	ldexp	217
10.24.184	ldexpf	217
10.24.185	ldexpl	218
10.24.186	lgamma	218
10.24.187	lgammaf	218
10.24.188	lgammal	218
10.24.189	llogb	219
10.24.190	llogbf	219
10.24.191	llogbl	219
10.24.192	llrint	219
10.24.193	llrintf	220
10.24.194	llrintl	220
10.24.195	llround	220
10.24.196	llroundf	220
10.24.197	llroundl	220
10.24.198	log	221
10.24.199	log10	221
10.24.200	log10f	221
10.24.201	log10l	222
10.24.202	log10p1	222
10.24.203	log10p1f	222
10.24.204	log10p1l	222
10.24.205	log1p	223
10.24.206	log1pf	223
10.24.207	log1pl	223
10.24.208	log2	224
10.24.209	log2f	224
10.24.210	log2l	224
10.24.211	log2p1	225
10.24.212	log2p1f	225
10.24.213	log2p1l	225
10.24.214	logb	225
10.24.215	logbf	226
10.24.216	logbl	226
10.24.217	logf	226
10.24.218	logl	227
10.24.219	logp1	227
10.24.220	logp1f	227
10.24.221	logp1l	227
10.24.222	lrint	228

10.24.223	lrintf	228
10.24.224	lrintl	228
10.24.225	lround	228
10.24.226	lroundf	229
10.24.227	lroundl	229
10.24.228	math_errhandling	229
10.24.229	modf	229
10.24.230	modff	230
10.24.231	modfl	230
10.24.232	nan	230
10.24.233	nanf	231
10.24.234	nanl	231
10.24.235	nearbyint	231
10.24.236	nearbyintf	231
10.24.237	nearbyintl	231
10.24.238	nextafter	232
10.24.239	nextafterf	232
10.24.240	nextafterl	232
10.24.241	nextdown	232
10.24.242	nextdownf	233
10.24.243	nextdownl	233
10.24.244	nexttoward	233
10.24.245	nexttowardf	233
10.24.246	nexttowardl	234
10.24.247	nextup	234
10.24.248	nextupf	234
10.24.249	nextupl	234
10.24.250	pow	235
10.24.251	powf	235
10.24.252	powl	235
10.24.253	pown	235
10.24.254	pownf	236
10.24.255	pownl	236
10.24.256	powr	236
10.24.257	powrf	236
10.24.258	powrl	237
10.24.259	remainder	237
10.24.260	remainderf	237
10.24.261	remainderl	237
10.24.262	remquo	238
10.24.263	remquof	238
10.24.264	remquol	238
10.24.265	rint	238
10.24.266	rintf	239
10.24.267	rintl	239
10.24.268	rootn	239
10.24.269	rootnf	239

10.24.270	rootnl	240
10.24.271	round	240
10.24.272	roundeven	240
10.24.273	roundevenf	240
10.24.274	roundevenl	241
10.24.275	roundf	241
10.24.276	roundl	241
10.24.277	rsqrt	242
10.24.278	rsqrtf	242
10.24.279	rsqrtl	242
10.24.280	scalbln	242
10.24.281	scalblnf	243
10.24.282	scalblnl	243
10.24.283	scalbn	243
10.24.284	scalbnf	243
10.24.285	scalbnl	243
10.24.286	setpayload	244
10.24.287	setpayloadf	244
10.24.288	setpayloadl	244
10.24.289	setpayloadsig	244
10.24.290	setpayloadsigf	245
10.24.291	setpayloadsigl	245
10.24.292	signbit	245
10.24.293	signgam	246
10.24.294	sin	246
10.24.295	sinf	246
10.24.296	sinh	246
10.24.297	sinhf	246
10.24.298	sinhl	247
10.24.299	sinl	247
10.24.300	sinpi	247
10.24.301	sinpif	248
10.24.302	sinpil	248
10.24.303	sqrt	248
10.24.304	sqrtf	248
10.24.305	sqrtl	249
10.24.306	tan	249
10.24.307	tanf	249
10.24.308	tanh	249
10.24.309	tanhf	250
10.24.310	tanhl	250
10.24.311	tanl	250
10.24.312	tanpi	250
10.24.313	tanpif	251
10.24.314	tanpil	251
10.24.315	tgamma	251
10.24.316	tgammaf	251

10.24.317	tgammal.....	252
10.24.318	totalorder.....	252
10.24.319	totalorderf.....	252
10.24.320	totalorderl.....	253
10.24.321	totalordermag.....	253
10.24.322	totalordermagf.....	253
10.24.323	totalordermagl.....	254
10.24.324	trunc.....	254
10.24.325	truncf.....	254
10.24.326	truncl.....	255
10.24.327	ufromfp.....	255
10.24.328	ufromfpf.....	255
10.24.329	ufromfpl.....	255
10.24.330	ufromfpx.....	256
10.24.331	ufromfpxf.....	256
10.24.332	ufromfpxl.....	256
10.24.333	y0.....	256
10.24.334	y1.....	257
10.24.335	yn.....	257
10.25	Functions in <monetary.h>.....	257
10.25.1	strfmon.....	257
10.25.2	strfmon_l.....	257
10.26	Functions in <mqueue.h>.....	258
10.26.1	mq_close.....	258
10.26.2	mq_getattr.....	258
10.26.3	mq_notify.....	258
10.26.4	mq_open.....	258
10.26.5	mq_receive.....	259
10.26.6	mq_send.....	259
10.26.7	mq_setattr.....	259
10.26.8	mq_timedreceive.....	259
10.26.9	mq_timedsend.....	260
10.26.10	mq_unlink.....	260
10.27	Functions in <ndbm.h>.....	260
10.27.1	dbm_clearerr.....	260
10.27.2	dbm_close.....	260
10.27.3	dbm_delete.....	261
10.27.4	dbm_error.....	261
10.27.5	dbm_fetch.....	261
10.27.6	dbm_firstkey.....	261
10.27.7	dbm_nextkey.....	262
10.27.8	dbm_open.....	262
10.27.9	dbm_store.....	262
10.28	Functions in <net/if.h>.....	262
10.28.1	if_freenameindex.....	262
10.28.2	if_indextoname.....	263
10.28.3	if_nameindex.....	263

10.28.4	<code>if_nametoindex</code>	263
10.29	Functions in <code><netdb.h></code>	263
10.29.1	<code>endhostent</code>	263
10.29.2	<code>endnetent</code>	264
10.29.3	<code>endprotoent</code>	264
10.29.4	<code>endservent</code>	264
10.29.5	<code>freeaddrinfo</code>	264
10.29.6	<code>gai_strerror</code>	265
10.29.7	<code>getaddrinfo</code>	265
10.29.8	<code>gethostent</code>	265
10.29.9	<code>getnameinfo</code>	265
10.29.10	<code>getnetbyaddr</code>	266
10.29.11	<code>getnetbyname</code>	266
10.29.12	<code>getnetent</code>	266
10.29.13	<code>getprotobyname</code>	266
10.29.14	<code>getprotobynumber</code>	267
10.29.15	<code>getprotoent</code>	267
10.29.16	<code>getservbyname</code>	267
10.29.17	<code>getservbyport</code>	267
10.29.18	<code>getservent</code>	268
10.29.19	<code>sethostent</code>	268
10.29.20	<code>setnetent</code>	268
10.29.21	<code>setprotoent</code>	268
10.29.22	<code>setservent</code>	268
10.30	Functions in <code><netinet/in.h></code>	269
10.30.1	<code>in6addr_any</code>	269
10.30.2	<code>in6addr_loopback</code>	269
10.31	Functions in <code><nl_types.h></code>	269
10.31.1	<code>catclose</code>	269
10.31.2	<code>catgets</code>	270
10.31.3	<code>catopen</code>	270
10.32	Functions in <code><poll.h></code>	270
10.32.1	<code>poll</code>	270
10.32.2	<code>ppoll</code>	270
10.33	Functions in <code><pthread.h></code>	271
10.33.1	<code>pthread_atfork</code>	271
10.33.2	<code>pthread_attr_destroy</code>	271
10.33.3	<code>pthread_attr_getdetachstate</code>	271
10.33.4	<code>pthread_attr_getguardsize</code>	272
10.33.5	<code>pthread_attr_getinheritsched</code>	272
10.33.6	<code>pthread_attr_getschedparam</code>	272
10.33.7	<code>pthread_attr_getschedpolicy</code>	272
10.33.8	<code>pthread_attr_getscope</code>	273
10.33.9	<code>pthread_attr_getstack</code>	273
10.33.10	<code>pthread_attr_getstacksize</code>	273
10.33.11	<code>pthread_attr_init</code>	273
10.33.12	<code>pthread_attr_setdetachstate</code>	274

10.33.13	pthread_attr_setguardsize.....	274
10.33.14	pthread_attr_setinheritsched.....	274
10.33.15	pthread_attr_setschedparam.....	274
10.33.16	pthread_attr_setschedpolicy.....	275
10.33.17	pthread_attr_setscope.....	275
10.33.18	pthread_attr_setstack.....	275
10.33.19	pthread_attr_setstacksize.....	275
10.33.20	pthread_barrier_destroy.....	276
10.33.21	pthread_barrier_init.....	276
10.33.22	pthread_barrier_wait.....	276
10.33.23	pthread_barrierattr_destroy.....	276
10.33.24	pthread_barrierattr_getpshared.....	277
10.33.25	pthread_barrierattr_init.....	277
10.33.26	pthread_barrierattr_setpshared.....	277
10.33.27	pthread_cancel.....	277
10.33.28	pthread_cleanup_pop.....	278
10.33.29	pthread_cleanup_push.....	278
10.33.30	pthread_cond_broadcast.....	278
10.33.31	pthread_cond_clockwait.....	278
10.33.32	pthread_cond_destroy.....	279
10.33.33	pthread_cond_init.....	279
10.33.34	pthread_cond_signal.....	279
10.33.35	pthread_cond_timedwait.....	279
10.33.36	pthread_cond_wait.....	280
10.33.37	pthread_condattr_destroy.....	280
10.33.38	pthread_condattr_getclock.....	280
10.33.39	pthread_condattr_getpshared.....	280
10.33.40	pthread_condattr_init.....	281
10.33.41	pthread_condattr_setclock.....	281
10.33.42	pthread_condattr_setpshared.....	281
10.33.43	pthread_create.....	281
10.33.44	pthread_detach.....	282
10.33.45	pthread_equal.....	282
10.33.46	pthread_exit.....	282
10.33.47	pthread_getcpuclockid.....	282
10.33.48	pthread_getschedparam.....	283
10.33.49	pthread_getspecific.....	283
10.33.50	pthread_join.....	283
10.33.51	pthread_key_create.....	283
10.33.52	pthread_key_delete.....	284
10.33.53	pthread_mutex_clocklock.....	284
10.33.54	pthread_mutex_consistent.....	284
10.33.55	pthread_mutex_destroy.....	284
10.33.56	pthread_mutex_getprioceiling.....	285
10.33.57	pthread_mutex_init.....	285
10.33.58	pthread_mutex_lock.....	285
10.33.59	pthread_mutex_setprioceiling.....	285

10.33.60	pthread_mutex_timedlock	286
10.33.61	pthread_mutex_trylock	286
10.33.62	pthread_mutex_unlock	286
10.33.63	pthread_mutexattr_destroy	287
10.33.64	pthread_mutexattr_getprioceiling	287
10.33.65	pthread_mutexattr_getprotocol	287
10.33.66	pthread_mutexattr_getpshared	287
10.33.67	pthread_mutexattr_getrobust	288
10.33.68	pthread_mutexattr_gettype	288
10.33.69	pthread_mutexattr_init	288
10.33.70	pthread_mutexattr_setprioceiling	288
10.33.71	pthread_mutexattr_setprotocol	289
10.33.72	pthread_mutexattr_setpshared	289
10.33.73	pthread_mutexattr_setrobust	289
10.33.74	pthread_mutexattr_settype	289
10.33.75	pthread_once	290
10.33.76	pthread_rwlock_clockrdlock	290
10.33.77	pthread_rwlock_clockwrlock	290
10.33.78	pthread_rwlock_destroy	291
10.33.79	pthread_rwlock_init	291
10.33.80	pthread_rwlock_rdlock	291
10.33.81	pthread_rwlock_timedrdlock	291
10.33.82	pthread_rwlock_timedwrlock	292
10.33.83	pthread_rwlock_tryrdlock	292
10.33.84	pthread_rwlock_trywrlock	292
10.33.85	pthread_rwlock_unlock	293
10.33.86	pthread_rwlock_wrlock	293
10.33.87	pthread_rwlockattr_destroy	293
10.33.88	pthread_rwlockattr_getpshared	294
10.33.89	pthread_rwlockattr_init	294
10.33.90	pthread_rwlockattr_setpshared	294
10.33.91	pthread_self	294
10.33.92	pthread_setcancelstate	295
10.33.93	pthread_setcanceltype	295
10.33.94	pthread_setschedparam	295
10.33.95	pthread_setschedprio	295
10.33.96	pthread_setspecific	296
10.33.97	pthread_spin_destroy	296
10.33.98	pthread_spin_init	296
10.33.99	pthread_spin_lock	296
10.33.100	pthread_spin_trylock	297
10.33.101	pthread_spin_unlock	297
10.33.102	pthread_testcancel	297
10.34	Functions in <pwd.h>	297
10.34.1	endpwent	298
10.34.2	getpwent	298
10.34.3	getpwnam	298

10.34.4	getpwnam_r.....	298
10.34.5	getpwuid.....	298
10.34.6	getpwuid_r.....	299
10.34.7	setpwent.....	299
10.35	Functions in <regex.h>.....	299
10.35.1	regcomp.....	299
10.35.2	regerror.....	300
10.35.3	regexec.....	300
10.35.4	regfree.....	300
10.36	Functions in <sched.h>.....	300
10.36.1	sched_get_priority_max.....	300
10.36.2	sched_get_priority_min.....	301
10.36.3	sched_getparam.....	301
10.36.4	sched_getscheduler.....	301
10.36.5	sched_rr_get_interval.....	301
10.36.6	sched_setparam.....	302
10.36.7	sched_setscheduler.....	302
10.36.8	sched_yield.....	302
10.37	Functions in <search.h>.....	302
10.37.1	hcreate.....	302
10.37.2	hdestroy.....	303
10.37.3	hsearch.....	303
10.37.4	insque.....	303
10.37.5	lfind.....	303
10.37.6	lsearch.....	304
10.37.7	remque.....	304
10.37.8	tdelete.....	304
10.37.9	tfind.....	304
10.37.10	tsearch.....	305
10.37.11	twalk.....	305
10.38	Functions in <semaphore.h>.....	305
10.38.1	sem_clockwait.....	305
10.38.2	sem_close.....	305
10.38.3	sem_destroy.....	306
10.38.4	sem_getvalue.....	306
10.38.5	sem_init.....	306
10.38.6	sem_open.....	306
10.38.7	sem_post.....	307
10.38.8	sem_timedwait.....	307
10.38.9	sem_trywait.....	307
10.38.10	sem_unlink.....	307
10.38.11	sem_wait.....	307
10.39	Functions in <setjmp.h>.....	308
10.39.1	longjmp.....	308
10.39.2	setjmp.....	308
10.39.3	siglongjmp.....	308
10.39.4	sigsetjmp.....	309

10.40	Functions in <signal.h>.....	309
10.40.1	kill.....	309
10.40.2	killpg.....	309
10.40.3	psiginfo.....	309
10.40.4	psignal.....	310
10.40.5	pthread_kill.....	310
10.40.6	pthread_sigmask.....	310
10.40.7	raise.....	311
10.40.8	sig2str.....	311
10.40.9	sigaction.....	311
10.40.10	sigaddset.....	312
10.40.11	sigaltstack.....	312
10.40.12	sigdelset.....	312
10.40.13	sigemptyset.....	312
10.40.14	sigfillset.....	313
10.40.15	sigismember.....	313
10.40.16	signal.....	313
10.40.17	sigpending.....	314
10.40.18	sigprocmask.....	314
10.40.19	sigqueue.....	314
10.40.20	sigsuspend.....	314
10.40.21	sigtimedwait.....	315
10.40.22	sigwait.....	315
10.40.23	sigwaitinfo.....	315
10.40.24	str2sig.....	315
10.41	Functions in <spawn.h>.....	316
10.41.1	posix_spawn.....	316
10.41.2	posix_spawn_file_actions_addchdir.....	316
10.41.3	posix_spawn_file_actions_addclose.....	316
10.41.4	posix_spawn_file_actions_adddup2.....	317
10.41.5	posix_spawn_file_actions_addfchdir.....	317
10.41.6	posix_spawn_file_actions_addopen.....	317
10.41.7	posix_spawn_file_actions_destroy.....	318
10.41.8	posix_spawn_file_actions_init.....	318
10.41.9	posix_spawnattr_destroy.....	318
10.41.10	posix_spawnattr_getflags.....	318
10.41.11	posix_spawnattr_getpgroup.....	319
10.41.12	posix_spawnattr_getschedparam.....	319
10.41.13	posix_spawnattr_getschedpolicy.....	319
10.41.14	posix_spawnattr_getsigdefault.....	320
10.41.15	posix_spawnattr_getsigmask.....	320
10.41.16	posix_spawnattr_init.....	320
10.41.17	posix_spawnattr_setflags.....	320
10.41.18	posix_spawnattr_setpgroup.....	321
10.41.19	posix_spawnattr_setschedparam.....	321
10.41.20	posix_spawnattr_setschedpolicy.....	321
10.41.21	posix_spawnattr_setsigdefault.....	321

10.41.22	posix_spawnattr_setsigmask.....	322
10.41.23	posix_spawnp.....	322
10.42	Functions in <stdarg.h>.....	322
10.42.1	va_arg.....	323
10.42.2	va_copy.....	323
10.42.3	va_end.....	323
10.42.4	va_start.....	323
10.43	Functions in <stdatomic.h>.....	324
10.43.1	atomic_compare_exchange_strong.....	324
10.43.2	atomic_compare_exchange_strong_explicit.....	324
10.43.3	atomic_compare_exchange_weak.....	324
10.43.4	atomic_compare_exchange_weak_explicit.....	325
10.43.5	atomic_exchange.....	325
10.43.6	atomic_exchange_explicit.....	325
10.43.7	atomic_fetch_add.....	326
10.43.8	atomic_fetch_add_explicit.....	326
10.43.9	atomic_fetch_and.....	326
10.43.10	atomic_fetch_and_explicit.....	327
10.43.11	atomic_fetch_or.....	327
10.43.12	atomic_fetch_or_explicit.....	327
10.43.13	atomic_fetch_sub.....	328
10.43.14	atomic_fetch_sub_explicit.....	328
10.43.15	atomic_fetch_xor.....	328
10.43.16	atomic_fetch_xor_explicit.....	329
10.43.17	atomic_flag_clear.....	329
10.43.18	atomic_flag_clear_explicit.....	329
10.43.19	atomic_flag_test_and_set.....	330
10.43.20	atomic_flag_test_and_set_explicit.....	330
10.43.21	atomic_init.....	330
10.43.22	atomic_is_lock_free.....	331
10.43.23	atomic_load.....	331
10.43.24	atomic_load_explicit.....	331
10.43.25	atomic_signal_fence.....	332
10.43.26	atomic_store.....	332
10.43.27	atomic_store_explicit.....	332
10.43.28	atomic_thread_fence.....	333
10.43.29	kill_dependency.....	333
10.44	Functions in <stdbit.h>.....	333
10.44.1	stdc_bit_ceil.....	333
10.44.2	stdc_bit_floor.....	334
10.44.3	stdc_bit_width.....	334
10.44.4	stdc_count_ones.....	334
10.44.5	stdc_count_zeros.....	334
10.44.6	stdc_first_leading_one.....	335
10.44.7	stdc_first_leading_zero.....	335
10.44.8	stdc_first_trailing_one.....	335
10.44.9	stdc_first_trailing_zero.....	335

10.44.10	stdc_has_single_bit	336
10.44.11	stdc_leading_ones	336
10.44.12	stdc_leading_zeros	336
10.44.13	stdc_trailing_ones	336
10.44.14	stdc_trailing_zeros	337
10.45	Functions in <stdio.h>	337
10.45.1	asprintf	337
10.45.2	clearerr	338
10.45.3	ctermid	339
10.45.4	dprintf	339
10.45.5	fclose	340
10.45.6	fdopen	341
10.45.7	feof	341
10.45.8	ferror	341
10.45.9	fflush	341
10.45.10	fgetc	342
10.45.11	fgetpos	342
10.45.12	fgets	343
10.45.13	fileno	343
10.45.14	flockfile	343
10.45.15	fmemopen	343
10.45.16	fopen	344
10.45.17	fprintf	344
10.45.18	fputc	346
10.45.19	fputs	347
10.45.20	fread	347
10.45.21	freopen	348
10.45.22	fscanf	348
10.45.23	fseek	349
10.45.24	fseeko	349
10.45.25	fsetpos	350
10.45.26	ftell	350
10.45.27	ftello	350
10.45.28	fttrylockfile	351
10.45.29	funlockfile	351
10.45.30	fwrite	351
10.45.31	getc	352
10.45.32	getc_unlocked	352
10.45.33	getchar	353
10.45.34	getchar_unlocked	353
10.45.35	getdelim	353
10.45.36	getline	354
10.45.37	open_memstream	354
10.45.38	pclose	354
10.45.39	perror	355
10.45.40	popen	355
10.45.41	printf	355

10.45.42	putc.....	358
10.45.43	putc_unlocked.....	358
10.45.44	putchar.....	358
10.45.45	putchar_unlocked.....	359
10.45.46	puts.....	359
10.45.47	remove.....	360
10.45.48	rename.....	360
10.45.49	renameat.....	361
10.45.50	rewind.....	362
10.45.51	scanf.....	362
10.45.52	setbuf.....	363
10.45.53	setvbuf.....	363
10.45.54	snprintf.....	363
10.45.55	sprintf.....	365
10.45.56	sscanf.....	367
10.45.57	stderr.....	367
10.45.58	stdin.....	368
10.45.59	stdout.....	368
10.45.60	tmpfile.....	368
10.45.61	tmpnam.....	369
10.45.62	ungetc.....	369
10.45.63	vasprintf.....	369
10.45.64	vdprintf.....	371
10.45.65	vfprintf.....	372
10.45.66	vfscanf.....	374
10.45.67	vprintf.....	375
10.45.68	vscanf.....	377
10.45.69	vsnprintf.....	377
10.45.70	vsprintf.....	379
10.45.71	vsscanf.....	381
10.46	Functions in <stdlib.h>.....	381
10.46.1	_Exit.....	381
10.46.2	a64l.....	381
10.46.3	abort.....	382
10.46.4	abs.....	382
10.46.5	aligned_alloc.....	382
10.46.6	at_quick_exit.....	383
10.46.7	atexit.....	383
10.46.8	atof.....	384
10.46.9	atoi.....	384
10.46.10	atol.....	384
10.46.11	atoll.....	385
10.46.12	bsearch.....	385
10.46.13	calloc.....	385
10.46.14	div.....	385
10.46.15	drand48.....	386
10.46.16	erand48.....	386

10.46.17	exit.....	386
10.46.18	free.....	386
10.46.19	free_aligned_sized.....	386
10.46.20	free_sized.....	387
10.46.21	getenv.....	387
10.46.22	getsubopt.....	387
10.46.23	grantpt.....	387
10.46.24	initstate.....	388
10.46.25	jrand48.....	388
10.46.26	l64a.....	388
10.46.27	labs.....	388
10.46.28	lcong48.....	389
10.46.29	ldiv.....	389
10.46.30	llabs.....	389
10.46.31	lldiv.....	389
10.46.32	lrnd48.....	389
10.46.33	malloc.....	390
10.46.34	mblen.....	390
10.46.35	mbstowcs.....	390
10.46.36	mbtowc.....	390
10.46.37	memalignment.....	391
10.46.38	mkdtemp.....	391
10.46.39	mkostemp.....	391
10.46.40	mkstemp.....	392
10.46.41	mrnd48.....	392
10.46.42	nrnd48.....	392
10.46.43	posix_memalign.....	393
10.46.44	posix_openpt.....	393
10.46.45	ptsname.....	393
10.46.46	ptsname_r.....	394
10.46.47	putenv.....	394
10.46.48	qsort.....	395
10.46.49	qsort_r.....	395
10.46.50	quick_exit.....	395
10.46.51	rand.....	396
10.46.52	rand_r.....	396
10.46.53	random.....	396
10.46.54	realloc.....	397
10.46.55	reallocarray.....	398
10.46.56	realpath.....	399
10.46.57	secure_getenv.....	399
10.46.58	seed48.....	399
10.46.59	setenv.....	400
10.46.60	setkey.....	400
10.46.61	setstate.....	400
10.46.62	srand.....	401
10.46.63	srand48.....	401

10.46.64	<code>srandom</code>	401
10.46.65	<code>strfromd</code>	401
10.46.66	<code>strfromf</code>	402
10.46.67	<code>strfroml</code>	402
10.46.68	<code>strtod</code>	402
10.46.69	<code>strtod</code>	403
10.46.70	<code>strtod</code>	404
10.46.71	<code>strtold</code>	404
10.46.72	<code>strtoll</code>	405
10.46.73	<code>strtoul</code>	406
10.46.74	<code>strtoull</code>	406
10.46.75	<code>system</code>	406
10.46.76	<code>unlockpt</code>	407
10.46.77	<code>unsetenv</code>	407
10.46.78	<code>wcstombs</code>	407
10.46.79	<code>wctomb</code>	408
10.47	Functions in <code><string.h></code>	408
10.47.1	<code>memccpy</code>	408
10.47.2	<code>memchr</code>	408
10.47.3	<code>memcmp</code>	408
10.47.4	<code>memcpy</code>	409
10.47.5	<code>memmem</code>	409
10.47.6	<code>memmove</code>	409
10.47.7	<code>memset</code>	410
10.47.8	<code>memset_explicit</code>	410
10.47.9	<code>stpcpy</code>	410
10.47.10	<code>stpncpy</code>	411
10.47.11	<code>strcat</code>	411
10.47.12	<code>strchr</code>	411
10.47.13	<code>strcmp</code>	411
10.47.14	<code>strcoll</code>	412
10.47.15	<code>strcoll_l</code>	412
10.47.16	<code>strcpy</code>	412
10.47.17	<code>strcspn</code>	412
10.47.18	<code>strdup</code>	413
10.47.19	<code>strerror</code>	413
10.47.20	<code>strerror_l</code>	413
10.47.21	<code>strerror_r</code>	414
10.47.22	<code>strlcat</code>	415
10.47.23	<code>strncpy</code>	415
10.47.24	<code>strlen</code>	415
10.47.25	<code>strncat</code>	415
10.47.26	<code>strncmp</code>	416
10.47.27	<code>strncpy</code>	416
10.47.28	<code>strndup</code>	416
10.47.29	<code>strnlen</code>	417
10.47.30	<code>strpbrk</code>	417

10.47.31	<code>strrchr</code>	417
10.47.32	<code>strsignal</code>	417
10.47.33	<code>strspn</code>	418
10.47.34	<code>strstr</code>	418
10.47.35	<code>strtok</code>	418
10.47.36	<code>strtok_r</code>	419
10.47.37	<code>strxfrm</code>	419
10.47.38	<code>strxfrm_l</code>	419
10.48	Functions in <code><strings.h></code>	419
10.48.1	<code>ffs</code>	419
10.48.2	<code>ffsl</code>	420
10.48.3	<code>ffsll</code>	420
10.48.4	<code>strcasecmp</code>	420
10.48.5	<code>strcasecmp_l</code>	421
10.48.6	<code>strncasecmp</code>	421
10.48.7	<code>strncasecmp_l</code>	421
10.49	Functions in <code><sys/ipc.h></code>	422
10.49.1	<code>ftok</code>	422
10.50	Functions in <code><sys/mman.h></code>	422
10.50.1	<code>mlock</code>	422
10.50.2	<code>mlockall</code>	422
10.50.3	<code>mmap</code>	422
10.50.4	<code>mprotect</code>	423
10.50.5	<code>msync</code>	423
10.50.6	<code>munlock</code>	423
10.50.7	<code>munlockall</code>	424
10.50.8	<code>munmap</code>	424
10.50.9	<code>posix_madvise</code>	424
10.50.10	<code>posix_mem_offset</code>	424
10.50.11	<code>posix_typed_mem_get_info</code>	425
10.50.12	<code>posix_typed_mem_open</code>	425
10.50.13	<code>shm_open</code>	425
10.50.14	<code>shm_unlink</code>	425
10.51	Functions in <code><sys/msg.h></code>	426
10.51.1	<code>msgctl</code>	426
10.51.2	<code>msgget</code>	426
10.51.3	<code>msgrcv</code>	426
10.51.4	<code>msgsnd</code>	426
10.52	Functions in <code><sys/resource.h></code>	427
10.52.1	<code>getpriority</code>	427
10.52.2	<code>getrlimit</code>	427
10.52.3	<code>getrusage</code>	427
10.52.4	<code>setpriority</code>	427
10.52.5	<code>setrlimit</code>	428
10.53	Functions in <code><sys/select.h></code>	428
10.53.1	<code>FD_CLR</code>	428
10.53.2	<code>FD_ISSET</code>	428

10.53.3	FD_SET.....	428
10.53.4	FD_ZERO.....	429
10.53.5	pselect.....	429
10.53.6	select.....	429
10.54	Functions in <sys/sem.h>.....	430
10.54.1	semctl.....	430
10.54.2	semget.....	430
10.54.3	semop.....	430
10.55	Functions in <sys/shm.h>.....	430
10.55.1	shmat.....	430
10.55.2	shmctl.....	431
10.55.3	shmdt.....	431
10.55.4	shmget.....	431
10.56	Functions in <sys/socket.h>.....	431
10.56.1	accept.....	431
10.56.2	accept4.....	432
10.56.3	bind.....	432
10.56.4	connect.....	432
10.56.5	getpeername.....	433
10.56.6	getsockname.....	433
10.56.7	getsockopt.....	434
10.56.8	listen.....	434
10.56.9	recv.....	434
10.56.10	recvfrom.....	435
10.56.11	recvmsg.....	435
10.56.12	send.....	435
10.56.13	sendmsg.....	436
10.56.14	sendto.....	436
10.56.15	setsockopt.....	436
10.56.16	shutdown.....	436
10.56.17	socketatmark.....	437
10.56.18	socket.....	437
10.56.19	socketpair.....	437
10.57	Functions in <sys/stat.h>.....	437
10.57.1	chmod.....	437
10.57.2	fchmod.....	438
10.57.3	fchmodat.....	438
10.57.4	fstat.....	439
10.57.5	fstatat.....	439
10.57.6	futimens.....	440
10.57.7	lstat.....	441
10.57.8	mkdir.....	441
10.57.9	mkdirat.....	442
10.57.10	mkfifo.....	442
10.57.11	mkfifoat.....	442
10.57.12	mknod.....	443
10.57.13	mknodat.....	443

10.57.14	stat.....	444
10.57.15	umask.....	445
10.57.16	utimensat.....	445
10.58	Functions in <sys/statvfs.h>.....	446
10.58.1	fstatvfs.....	446
10.58.2	statvfs.....	446
10.59	Functions in <sys/time.h>.....	446
10.59.1	utimes.....	446
10.60	Functions in <sys/times.h>.....	447
10.60.1	times.....	447
10.61	Functions in <sys/uio.h>.....	447
10.61.1	readv.....	447
10.61.2	writev.....	448
10.62	Functions in <sys/utsname.h>.....	448
10.62.1	uname.....	448
10.63	Functions in <sys/wait.h>.....	448
10.63.1	wait.....	448
10.63.2	waitid.....	448
10.63.3	waitpid.....	449
10.64	Functions in <syslog.h>.....	449
10.64.1	closelog.....	449
10.64.2	openlog.....	449
10.64.3	setlogmask.....	449
10.64.4	syslog.....	450
10.65	Functions in <termios.h>.....	450
10.65.1	cfgetispeed.....	450
10.65.2	cfgetospeed.....	450
10.65.3	cfsetispeed.....	450
10.65.4	cfsetospeed.....	450
10.65.5	tcdrain.....	451
10.65.6	tcflow.....	451
10.65.7	tcflush.....	451
10.65.8	tcgetattr.....	451
10.65.9	tcgetsid.....	452
10.65.10	tcgetwinsize.....	452
10.65.11	tcsendbreak.....	452
10.65.12	tcsetattr.....	452
10.65.13	tcsetwinsize.....	453
10.66	Functions in <threads.h>.....	453
10.66.1	call_once.....	453
10.66.2	cnd_broadcast.....	453
10.66.3	cnd_destroy.....	454
10.66.4	cnd_init.....	454
10.66.5	cnd_signal.....	454
10.66.6	cnd_timedwait.....	455
10.66.7	cnd_wait.....	455
10.66.8	mtx_destroy.....	456

10.66.9	mtx_init	456
10.66.10	mtx_lock	456
10.66.11	mtx_timedlock	457
10.66.12	mtx_trylock	457
10.66.13	mtx_unlock	457
10.66.14	thrd_create	458
10.66.15	thrd_current	458
10.66.16	thrd_detach	458
10.66.17	thrd_equal	459
10.66.18	thrd_exit	459
10.66.19	thrd_join	459
10.66.20	thrd_sleep	460
10.66.21	thrd_yield	460
10.66.22	tss_create	461
10.66.23	tss_delete	461
10.66.24	tss_get	461
10.66.25	tss_set	462
10.67	Functions in <time.h>	462
10.67.1	asctime	462
10.67.2	asctime_r	462
10.67.3	clock	463
10.67.4	clock_getcpuclockid	463
10.67.5	clock_getres	463
10.67.6	clock_gettime	464
10.67.7	clock_nanosleep	464
10.67.8	clock_settime	464
10.67.9	ctime	465
10.67.10	ctime_r	466
10.67.11	daylight	466
10.67.12	difftime	466
10.67.13	getdate	467
10.67.14	getdate_err	467
10.67.15	gmtime	467
10.67.16	gmtime_r	467
10.67.17	localtime	468
10.67.18	localtime_r	468
10.67.19	mktime	469
10.67.20	nanosleep	469
10.67.21	strftime	469
10.67.22	strftime_l	470
10.67.23	strptime	470
10.67.24	time	470
10.67.25	timegm	471
10.67.26	timer_create	471
10.67.27	timer_delete	471
10.67.28	timer_getoverrun	472
10.67.29	timer_gettime	472

10.67.30	timer_settime	472
10.67.31	timespec_get	472
10.67.32	timespec_getres	473
10.67.33	timezone	473
10.67.34	tzname	473
10.67.35	tzset	474
10.68	Functions in <uchar.h>	474
10.68.1	c8rtomb	474
10.68.2	c16rtomb	475
10.68.3	c32rtomb	475
10.68.4	mbrtoc8	475
10.68.5	mbrtoc16	475
10.68.6	mbrtoc32	476
10.69	Functions in <unistd.h>	477
10.69.1	_exit	477
10.69.2	_Fork	477
10.69.3	access	477
10.69.4	alarm	478
10.69.5	chdir	478
10.69.6	chown	479
10.69.7	close	479
10.69.8	confstr	479
10.69.9	crypt	480
10.69.10	dup	480
10.69.11	dup2	480
10.69.12	dup3	481
10.69.13	encrypt	481
10.69.14	environ	482
10.69.15	execl	482
10.69.16	execle	483
10.69.17	execvp	483
10.69.18	execv	483
10.69.19	execve	484
10.69.20	execvp	484
10.69.21	faccessat	485
10.69.22	fchdir	485
10.69.23	fchown	485
10.69.24	fchownat	486
10.69.25	fdatasync	486
10.69.26	fexecve	487
10.69.27	fork	487
10.69.28	fpathconf	487
10.69.29	fsync	487
10.69.30	ftruncate	488
10.69.31	getcwd	488
10.69.32	getegid	489
10.69.33	getentropy	489

10.69.34	geteuid.....	489
10.69.35	getgid.....	490
10.69.36	getgroups.....	490
10.69.37	gethostid.....	490
10.69.38	gethostname.....	491
10.69.39	getlogin.....	491
10.69.40	getlogin_r.....	491
10.69.41	getopt.....	492
10.69.42	getpgid.....	493
10.69.43	getpgrp.....	493
10.69.44	getpid.....	493
10.69.45	getppid.....	494
10.69.46	getresgid.....	494
10.69.47	getresuid.....	494
10.69.48	getsid.....	494
10.69.49	getuid.....	495
10.69.50	isatty.....	495
10.69.51	lchown.....	495
10.69.52	link.....	496
10.69.53	linkat.....	496
10.69.54	lockf.....	496
10.69.55	lseek.....	497
10.69.56	nice.....	497
10.69.57	optarg.....	497
10.69.58	opterr.....	498
10.69.59	optind.....	498
10.69.60	optopt.....	498
10.69.61	pathconf.....	498
10.69.62	pause.....	498
10.69.63	pipe.....	499
10.69.64	pipe2.....	499
10.69.65	posix_close.....	499
10.69.66	pread.....	499
10.69.67	pwrite.....	500
10.69.68	read.....	500
10.69.69	readlink.....	501
10.69.70	readlinkat.....	501
10.69.71	rmdir.....	502
10.69.72	setegid.....	503
10.69.73	seteuid.....	503
10.69.74	setgid.....	503
10.69.75	setpgid.....	503
10.69.76	setregid.....	503
10.69.77	setresgid.....	504
10.69.78	setresuid.....	504
10.69.79	setreuid.....	504
10.69.80	setsid.....	504

10.69.81	setuid	505
10.69.82	sleep	505
10.69.83	swab	505
10.69.84	symlink	505
10.69.85	symlinkat	506
10.69.86	sync	506
10.69.87	sysconf	506
10.69.88	tcgetpgrp	507
10.69.89	tcsetpgrp	507
10.69.90	truncate	507
10.69.91	ttyname	507
10.69.92	ttyname_r	508
10.69.93	unlink	508
10.69.94	unlinkat	509
10.69.95	write	509
10.70	Functions in <utmpx.h>	510
10.70.1	endutxent	510
10.70.2	getutxent	510
10.70.3	getutxid	510
10.70.4	getutxline	511
10.70.5	pututxline	511
10.70.6	setutxent	511
10.71	Functions in <wchar.h>	511
10.71.1	btowc	511
10.71.2	fgetwc	512
10.71.3	fgetws	512
10.71.4	fputwc	512
10.71.5	fputws	513
10.71.6	fwide	513
10.71.7	fwprintf	513
10.71.8	fwscanf	514
10.71.9	getwc	515
10.71.10	getwchar	515
10.71.11	mbrlen	515
10.71.12	mbrtowc	516
10.71.13	mbsinit	516
10.71.14	mbsnrtowcs	517
10.71.15	mbsrtowcs	517
10.71.16	open_wmemstream	518
10.71.17	putwc	518
10.71.18	putwchar	518
10.71.19	swprintf	518
10.71.20	swscanf	520
10.71.21	ungetwc	520
10.71.22	vfwprintf	520
10.71.23	vfwscanf	521
10.71.24	vswprintf	522

10.71.25	vswscanf	523
10.71.26	vwprintf	523
10.71.27	vswscanf	524
10.71.28	wcpcpy	525
10.71.29	wcpncpy	525
10.71.30	wcrtomb	525
10.71.31	wscasecmp	526
10.71.32	wscasecmp_l	526
10.71.33	wscat	526
10.71.34	wcschr	526
10.71.35	wscmp	527
10.71.36	wscoll	527
10.71.37	wscoll_l	527
10.71.38	wscpy	527
10.71.39	wscspn	528
10.71.40	wcsdup	528
10.71.41	wcsftime	528
10.71.42	wcslcat	529
10.71.43	wcslcpy	529
10.71.44	wcslen	529
10.71.45	wcsncasecmp	529
10.71.46	wcsncasecmp_l	530
10.71.47	wcsncat	530
10.71.48	wcsncmp	530
10.71.49	wcsncpy	530
10.71.50	wcsnlen	531
10.71.51	wcsnrtombs	531
10.71.52	wcspbrk	532
10.71.53	wcsrchr	532
10.71.54	wcsrtombs	532
10.71.55	wcsspn	532
10.71.56	wcsstr	533
10.71.57	wcstod	533
10.71.58	wcstof	533
10.71.59	wcstok	533
10.71.60	wcstol	534
10.71.61	wcstold	534
10.71.62	wcstoll	534
10.71.63	wcstoul	535
10.71.64	wcstoull	535
10.71.65	wcswidth	535
10.71.66	wcsxfrm	535
10.71.67	wcsxfrm_l	536
10.71.68	wctob	536
10.71.69	wcwidth	536
10.71.70	wmemchr	537
10.71.71	wmemcmp	537

10.71.72	wmemcpy	537
10.71.73	wmemmove	537
10.71.74	wmemset	538
10.71.75	wprintf	538
10.71.76	wscanf	539
10.72	Functions in <wctype.h>	539
10.72.1	iswalnum	539
10.72.2	iswalnum_l	540
10.72.3	iswalpha	540
10.72.4	iswalpha_l	540
10.72.5	iswblank	541
10.72.6	iswblank_l	541
10.72.7	iswcntrl	541
10.72.8	iswcntrl_l	542
10.72.9	iswctype	542
10.72.10	iswctype_l	542
10.72.11	iswdigit	543
10.72.12	iswdigit_l	543
10.72.13	iswgraph	543
10.72.14	iswgraph_l	544
10.72.15	iswlower	544
10.72.16	iswlower_l	544
10.72.17	iswprint	544
10.72.18	iswprint_l	545
10.72.19	iswpunct	545
10.72.20	iswpunct_l	546
10.72.21	iswspace	546
10.72.22	iswspace_l	546
10.72.23	iswupper	546
10.72.24	iswupper_l	547
10.72.25	iswxdigit	547
10.72.26	iswxdigit_l	547
10.72.27	towctrans	548
10.72.28	towctrans_l	548
10.72.29	towlower	548
10.72.30	towlower_l	549
10.72.31	towupper	549
10.72.32	towupper_l	550
10.72.33	wctrans	550
10.72.34	wctrans_l	550
10.72.35	wctype	551
10.72.36	wctype_l	551
10.73	Functions in <wordexp.h>	551
10.73.1	wordexp	551
10.73.2	wordfree	552

11 Past POSIX Function Substitutes 553

11.1	_longjmp.....	553
11.2	_setjmp.....	553
11.3	_tolower.....	554
11.4	_toupper.....	554
11.5	bcmp.....	554
11.6	bcopy.....	554
11.7	bsd_signal.....	555
11.8	bzero.....	555
11.9	ecvt.....	555
11.10	fattach.....	555
11.11	fcvt.....	556
11.12	fdetach.....	556
11.13	ftime.....	556
11.14	ftw.....	556
11.15	gcvrt.....	557
11.16	getcontext.....	557
11.17	gethostbyaddr.....	557
11.18	gethostbyname.....	558
11.19	getitimer.....	558
11.20	getmsg.....	558
11.21	getpmsg.....	558
11.22	gets.....	559
11.23	gettimeofday.....	559
11.24	getwd.....	559
11.25	h_errno.....	560
11.26	index.....	560
11.27	ioctl.....	560
11.28	isascii.....	561
11.29	isastream.....	561
11.30	makecontext.....	562
11.31	mktemp.....	562
11.32	posix_trace_attr_destroy.....	562
11.33	posix_trace_attr_getclockres.....	562
11.34	posix_trace_attr_getcreatetime.....	563
11.35	posix_trace_attr_getgenversion.....	563
11.36	posix_trace_attr_getinherited.....	563
11.37	posix_trace_attr_getlogfullpolicy.....	563
11.38	posix_trace_attr_getlogsize.....	564
11.39	posix_trace_attr_getmaxdatasize.....	564
11.40	posix_trace_attr_getmaxsystemeventsz.....	564
11.41	posix_trace_attr_getmaxusereventsiz.....	565
11.42	posix_trace_attr_getname.....	565
11.43	posix_trace_attr_getstreamfullpolicy.....	565
11.44	posix_trace_attr_getstreamsize.....	565
11.45	posix_trace_attr_init.....	566

11.46	<code>posix_trace_attr_setinherited</code>	566
11.47	<code>posix_trace_attr_setlogfullpolicy</code>	566
11.48	<code>posix_trace_attr_setlogsize</code>	567
11.49	<code>posix_trace_attr_setmaxdatasize</code>	567
11.50	<code>posix_trace_attr_setname</code>	567
11.51	<code>posix_trace_attr_setstreamfullpolicy</code>	567
11.52	<code>posix_trace_attr_setstreamsize</code>	568
11.53	<code>posix_trace_clear</code>	568
11.54	<code>posix_trace_close</code>	568
11.55	<code>posix_trace_create</code>	569
11.56	<code>posix_trace_create_withlog</code>	569
11.57	<code>posix_trace_event</code>	569
11.58	<code>posix_trace_eventid_equal</code>	569
11.59	<code>posix_trace_eventid_get_name</code>	570
11.60	<code>posix_trace_eventid_open</code>	570
11.61	<code>posix_trace_eventset_add</code>	570
11.62	<code>posix_trace_eventset_del</code>	571
11.63	<code>posix_trace_eventset_empty</code>	571
11.64	<code>posix_trace_eventset_fill</code>	571
11.65	<code>posix_trace_eventset_ismember</code>	571
11.66	<code>posix_trace_eventtypelist_getnext_id</code>	572
11.67	<code>posix_trace_eventtypelist_rewind</code>	572
11.68	<code>posix_trace_flush</code>	572
11.69	<code>posix_trace_get_attr</code>	573
11.70	<code>posix_trace_get_filter</code>	573
11.71	<code>posix_trace_get_status</code>	573
11.72	<code>posix_trace_getnext_event</code>	573
11.73	<code>posix_trace_open</code>	574
11.74	<code>posix_trace_rewind</code>	574
11.75	<code>posix_trace_set_filter</code>	574
11.76	<code>posix_trace_shutdown</code>	575
11.77	<code>posix_trace_start</code>	575
11.78	<code>posix_trace_stop</code>	575
11.79	<code>posix_trace_timedgetnext_event</code>	575
11.80	<code>posix_trace_trid_eventid_open</code>	576
11.81	<code>posix_trace_trygetnext_event</code>	576
11.82	<code>pthread_attr_getstackaddr</code>	576
11.83	<code>pthread_attr_setstackaddr</code>	577
11.84	<code>pthread_getconcurrency</code>	577
11.85	<code>pthread_setconcurrency</code>	577
11.86	<code>putmsg</code>	577
11.87	<code>putpmsg</code>	578
11.88	<code>rindex</code>	578
11.89	<code>scalb</code>	578
11.90	<code>setcontext</code>	578
11.91	<code>setitimer</code>	579
11.92	<code>setpgrp</code>	579

11.93	<code>sighold</code>	579
11.94	<code>sigignore</code>	579
11.95	<code>siginterrupt</code>	580
11.96	<code>sigpause</code>	580
11.97	<code>sigrelse</code>	580
11.98	<code>sigset</code>	581
11.99	<code>swapcontext</code>	581
11.100	<code>tempnam</code>	581
11.101	<code>toascii</code>	581
11.102	<code>ualarm</code>	582
11.103	<code>ulimit</code>	582
11.104	<code>usleep</code>	582
11.105	<code>utime</code>	582
11.106	<code>vfork</code>	583
11.107	<code>wcswcs</code>	583
 12 Glibc Header File Substitutes.....		584
12.1	<code>a.out.h</code>	584
12.2	<code>aliases.h</code>	584
12.3	<code>alloca.h</code>	584
12.4	<code>ar.h</code>	585
12.5	<code>argp.h</code>	585
12.6	<code>argz.h</code>	585
12.7	<code>byteswap.h</code>	586
12.8	<code>crypt.h</code>	586
12.9	<code>envz.h</code>	586
12.10	<code>err.h</code>	587
12.11	<code>error.h</code>	587
12.12	<code>execinfo.h</code>	587
12.13	<code>fpu_control.h</code>	588
12.14	<code>fstab.h</code>	588
12.15	<code>fts.h</code>	588
12.16	<code>getopt.h</code>	589
12.17	<code>gshadow.h</code>	589
12.18	<code>ieee754.h</code>	589
12.19	<code>ifaddrs.h</code>	590
12.20	<code>link.h</code>	590
12.21	<code>malloc.h</code>	590
12.22	<code>mcheck.h</code>	591
12.23	<code>mntent.h</code>	591
12.24	<code>obstack.h</code>	591
12.25	<code>paths.h</code>	592
12.26	<code>printf.h</code>	592
12.27	<code>pty.h</code>	592
12.28	<code>resolv.h</code>	593
12.29	<code>shadow.h</code>	593

12.30	<code>sys/file.h</code>	593
12.31	<code>sys/ioctl.h</code>	594
12.32	<code>sys/random.h</code>	594
12.33	<code>sysexits.h</code>	594
12.34	<code>ttyent.h</code>	595
12.35	<code>utmp.h</code>	595

13 Glibc Function Substitutes..... 597

13.1	Glibc Extensions to <code><aio.h></code>	597
13.1.1	<code>aio_init</code>	597
13.2	Glibc <code><aliases.h></code>	597
13.2.1	<code>endaliasent</code>	597
13.2.2	<code>getaliasbyname</code>	598
13.2.3	<code>getaliasbyname_r</code>	598
13.2.4	<code>getaliasent</code>	598
13.2.5	<code>getaliasent_r</code>	598
13.2.6	<code>setaliasent</code>	599
13.3	Glibc <code><argp.h></code>	599
13.3.1	<code>argp_err_exit_status</code>	599
13.3.2	<code>argp_error</code>	599
13.3.3	<code>argp_failure</code>	599
13.3.4	<code>argp_help</code>	600
13.3.5	<code>argp_parse</code>	600
13.3.6	<code>argp_program_bug_address</code>	600
13.3.7	<code>argp_program_version</code>	600
13.3.8	<code>argp_program_version_hook</code>	601
13.3.9	<code>argp_state_help</code>	601
13.3.10	<code>argp_usage</code>	601
13.4	Glibc <code><argz.h></code>	601
13.4.1	<code>argz_add</code>	601
13.4.2	<code>argz_add_sep</code>	602
13.4.3	<code>argz_append</code>	602
13.4.4	<code>argz_count</code>	603
13.4.5	<code>argz_create</code>	603
13.4.6	<code>argz_create_sep</code>	603
13.4.7	<code>argz_delete</code>	604
13.4.8	<code>argz_extract</code>	604
13.4.9	<code>argz_insert</code>	604
13.4.10	<code>argz_next</code>	605
13.4.11	<code>argz_replace</code>	605
13.4.12	<code>argz_stringify</code>	606
13.5	Glibc Extensions to <code><arpa/inet.h></code>	606
13.5.1	<code>inet_aton</code>	606
13.5.2	<code>inet_lnaof</code>	606
13.5.3	<code>inet_makeaddr</code>	607
13.5.4	<code>inet_net_ntop</code>	607

13.5.5	<code>inet_net_pton</code>	607
13.5.6	<code>inet_netof</code>	607
13.5.7	<code>inet_network</code>	608
13.5.8	<code>inet_nsap_addr</code>	608
13.5.9	<code>inet_nsap_ntoa</code>	608
13.6	Glibc Extensions to <code><arpa/nameser.h></code>	608
13.6.1	<code>ns_name_compress</code>	608
13.6.2	<code>ns_name_ntop</code>	609
13.6.3	<code>ns_name_pack</code>	609
13.6.4	<code>ns_name_pton</code>	609
13.6.5	<code>ns_name_skip</code>	609
13.6.6	<code>ns_name_uncompress</code>	610
13.6.7	<code>ns_name_unpack</code>	610
13.7	Glibc <code><byteswap.h></code>	610
13.7.1	<code>bswap_16</code>	610
13.7.2	<code>bswap_32</code>	610
13.7.3	<code>bswap_64</code>	611
13.8	Glibc Extensions to <code><complex.h></code>	611
13.8.1	<code>clog10</code>	611
13.8.2	<code>clog10f</code>	611
13.8.3	<code>clog10l</code>	612
13.9	Glibc Extensions to <code><ctype.h></code>	612
13.9.1	<code>isctype</code>	612
13.10	Glibc Extensions to <code><dirent.h></code>	612
13.10.1	<code>getdirentries</code>	612
13.10.2	<code>scandirat</code>	613
13.10.3	<code>versionsort</code>	613
13.11	Glibc Extensions to <code><dlfcn.h></code>	613
13.11.1	<code>dladdr1</code>	613
13.11.2	<code>dlinfo</code>	613
13.11.3	<code>dlopen</code>	614
13.11.4	<code>dlvsym</code>	614
13.11.5	<code>_dl_find_object</code>	614
13.12	Glibc <code><envz.h></code>	614
13.12.1	<code>envz_add</code>	614
13.12.2	<code>envz_entry</code>	615
13.12.3	<code>envz_get</code>	615
13.12.4	<code>envz_merge</code>	615
13.12.5	<code>envz_remove</code>	616
13.12.6	<code>envz_strip</code>	616
13.13	Glibc <code><err.h></code>	617
13.13.1	<code>err</code>	617
13.13.2	<code>errx</code>	617
13.13.3	<code>verr</code>	617
13.13.4	<code>verrx</code>	618
13.13.5	<code>vwarn</code>	618
13.13.6	<code>vwarnx</code>	618

13.13.7	warn.....	618
13.13.8	warnx.....	619
13.14	Glibc Extensions to <errno.h>.....	619
13.14.1	program_invocation_name.....	619
13.14.2	program_invocation_short_name.....	619
13.15	Glibc <error.h>.....	620
13.15.1	error.....	620
13.15.2	error_at_line.....	620
13.15.3	error_message_count.....	620
13.15.4	error_one_per_line.....	621
13.15.5	error_print_progname.....	621
13.16	Glibc <execinfo.h>.....	621
13.16.1	backtrace.....	621
13.16.2	backtrace_symbols.....	622
13.16.3	backtrace_symbols_fd.....	622
13.17	Glibc Extensions to <fcntl.h>.....	623
13.17.1	fallocate.....	623
13.17.2	name_to_handle_at.....	623
13.17.3	readahead.....	623
13.17.4	open_by_handle_at.....	623
13.17.5	sync_file_range.....	624
13.18	Glibc Extensions to <fenv.h>.....	624
13.18.1	fedisableexcept.....	624
13.18.2	feenableexcept.....	624
13.18.3	fegetexcept.....	625
13.19	Glibc Extensions to <fmtmsg.h>.....	625
13.19.1	addseverity.....	625
13.20	Glibc <fstab.h>.....	626
13.20.1	endfsent.....	626
13.20.2	getfsent.....	626
13.20.3	getfsfile.....	626
13.20.4	getfsspec.....	626
13.20.5	setfsent.....	627
13.21	Glibc <fts.h>.....	627
13.21.1	fts_children.....	627
13.21.2	fts_close.....	627
13.21.3	fts_open.....	627
13.21.4	fts_read.....	628
13.21.5	fts_set.....	628
13.22	Glibc <getopt.h>.....	628
13.22.1	getopt_long.....	628
13.22.2	getopt_long_only.....	630
13.23	Glibc Extensions to <glob.h>.....	631
13.23.1	glob_pattern_p.....	631
13.24	Glibc Extensions to <gnu/libc-version.h>.....	631
13.24.1	gnu_get_libc_release.....	632
13.24.2	gnu_get_libc_version.....	632

13.25	Glibc Extensions to <code><grp.h></code>	632
13.25.1	<code>fgetgrent</code>	632
13.25.2	<code>fgetgrent_r</code>	633
13.25.3	<code>getgrent_r</code>	633
13.25.4	<code>getgrouplist</code>	633
13.25.5	<code>initgroups</code>	634
13.25.6	<code>putgrent</code>	634
13.25.7	<code>setgroups</code>	634
13.26	Glibc <code><gshadow.h></code>	635
13.26.1	<code>endsgent</code>	635
13.26.2	<code>fgetsgent</code>	635
13.26.3	<code>fgetsgent_r</code>	635
13.26.4	<code>getsgent</code>	635
13.26.5	<code>getsgent_r</code>	635
13.26.6	<code>getsgnam</code>	636
13.26.7	<code>getsgnam_r</code>	636
13.26.8	<code>putsgent</code>	636
13.26.9	<code>setsgent</code>	636
13.26.10	<code>sgetsgent</code>	636
13.26.11	<code>sgetsgent_r</code>	637
13.27	Glibc <code><ifaddrs.h></code>	637
13.27.1	<code>getifaddrs</code>	637
13.27.2	<code>freeifaddrs</code>	637
13.28	Glibc Extensions to <code><inttypes.h></code>	637
13.28.1	<code>uimaxabs</code>	637
13.29	Glibc <code><link.h></code>	638
13.29.1	<code>dl_iterate_phdr</code>	638
13.30	Glibc <code><malloc.h></code>	638
13.30.1	<code>mallinfo</code>	638
13.30.2	<code>mallinfo2</code>	638
13.30.3	<code>malloc_info</code>	639
13.30.4	<code>malloc_stats</code>	639
13.30.5	<code>malloc_trim</code>	639
13.30.6	<code>malloc_usable_size</code>	639
13.30.7	<code>mallopt</code>	640
13.30.8	<code>memalign</code>	640
13.30.9	<code>pvalloc</code>	640
13.31	Glibc Extensions to <code><math.h></code>	641
13.31.1	<code>drem</code>	641
13.31.2	<code>dremf</code>	641
13.31.3	<code>dreml</code>	641
13.31.4	<code>finite</code>	642
13.31.5	<code>finitef</code>	642
13.31.6	<code>finitel</code>	642
13.31.7	<code>gamma</code>	643
13.31.8	<code>gammaf</code>	643
13.31.9	<code>gammal</code>	643

13.31.10	isinf	644
13.31.11	isinfl	644
13.31.12	isnanf	644
13.31.13	isnanl	645
13.31.14	j0f	645
13.31.15	j0l	645
13.31.16	j1f	646
13.31.17	j1l	646
13.31.18	jnf	646
13.31.19	jnl	647
13.31.20	lgamma_r	647
13.31.21	lgammaf_r	648
13.31.22	lgammal_r	648
13.31.23	matherr	648
13.31.24	pow10	649
13.31.25	pow10f	649
13.31.26	pow10l	649
13.31.27	scalbf	650
13.31.28	scalbl	650
13.31.29	significand	650
13.31.30	significandf	651
13.31.31	significandl	651
13.31.32	sincos	651
13.31.33	sincosf	652
13.31.34	sincosl	652
13.31.35	y0f	652
13.31.36	y0l	653
13.31.37	y1f	653
13.31.38	y1l	653
13.31.39	ynf	654
13.31.40	ynl	654
13.32	Glibc <mcheck.h>	654
13.32.1	mcheck	654
13.32.2	mcheck_check_all	655
13.32.3	mcheck_pedantic	655
13.32.4	mprobe	655
13.32.5	mtrace	656
13.32.6	muntrace	656
13.33	Glibc <mntent.h>	656
13.33.1	addmntent	656
13.33.2	endmntent	656
13.33.3	getmntent	657
13.33.4	getmntent_r	657
13.33.5	hasmntopt	657
13.33.6	setmntent	658
13.34	Glibc Extensions to <netdb.h>	658
13.34.1	endnetgrent	658

13.34.2	gethostbyaddr_r	658
13.34.3	gethostbyname2	659
13.34.4	gethostbyname2_r	659
13.34.5	gethostbyname_r	659
13.34.6	gethostent_r	660
13.34.7	getnetbyaddr_r	660
13.34.8	getnetbyname_r	660
13.34.9	getnetent_r	660
13.34.10	getnetgrent	660
13.34.11	getnetgrent_r	661
13.34.12	getprotobyname_r	661
13.34.13	getprotobynumber_r	661
13.34.14	getprotoent_r	662
13.34.15	getservbyname_r	662
13.34.16	getservbyport_r	662
13.34.17	getservent_r	662
13.34.18	herror	663
13.34.19	hstrerror	663
13.34.20	innetgr	663
13.34.21	rcmd	663
13.34.22	rcmd_af	664
13.34.23	rexec	664
13.34.24	rexec_af	664
13.34.25	rresvport	664
13.34.26	rresvport_af	665
13.34.27	ruserok	665
13.34.28	ruserok_af	665
13.34.29	setnetgrent	665
13.35	Glibc <netinet/ether.h>	666
13.35.1	ether_aton	666
13.35.2	ether_aton_r	666
13.35.3	ether_hostton	666
13.35.4	ether_line	666
13.35.5	ether_ntoa	667
13.35.6	ether_ntoa_r	667
13.35.7	ether_ntohost	667
13.36	Glibc Extensions to <netinet/in.h>	667
13.36.1	bindresvport	667
13.36.2	getipv4sourcefilter	668
13.36.3	getsourcefilter	668
13.36.4	inet6_option_alloc	668
13.36.5	inet6_option_append	668
13.36.6	inet6_option_find	668
13.36.7	inet6_option_init	669
13.36.8	inet6_option_next	669
13.36.9	inet6_option_space	669
13.36.10	inet6_opt_append	669

13.36.11	inet6_opt_find.....	669
13.36.12	inet6_opt_finish.....	670
13.36.13	inet6_opt_get_val	670
13.36.14	inet6_opt_init.....	670
13.36.15	inet6_opt_next.....	670
13.36.16	inet6_opt_set_val	670
13.36.17	inet6_rth_add.....	671
13.36.18	inet6_rth_getaddr	671
13.36.19	inet6_rth_init.....	671
13.36.20	inet6_rth_reverse	671
13.36.21	inet6_rth_segments	671
13.36.22	inet6_rth_space.....	672
13.36.23	setipv4sourcefilter.....	672
13.36.24	setsourcefilter.....	672
13.37	Glibc <obstack.h>	672
13.37.1	obstack_alloc_failed_handler.....	672
13.37.2	obstack_exit_failure.....	672
13.37.3	obstack_free	673
13.37.4	obstack_printf.....	673
13.37.5	obstack_vprintf.....	675
13.38	Glibc <printf.h>	676
13.38.1	parse_printf_format.....	676
13.38.2	printf_size.....	677
13.38.3	printf_size_info.....	677
13.38.4	register_printf_function	677
13.38.5	register_printf_modifier	677
13.38.6	register_printf_specifier	678
13.38.7	register_printf_type.....	678
13.39	Glibc Extensions to <pthread.h>	678
13.39.1	pthread_attr_getaffinity_np.....	678
13.39.2	pthread_attr_setaffinity_np.....	678
13.39.3	pthread_attr_getsigmask_np.....	679
13.39.4	pthread_attr_setsigmask_np.....	679
13.39.5	pthread_clockjoin_np.....	679
13.39.6	pthread_getaffinity_np.....	679
13.39.7	pthread_getattr_default_np.....	680
13.39.8	pthread_getattr_np.....	680
13.39.9	pthread_getname_np.....	680
13.39.10	pthread_gettid_np.....	681
13.39.11	pthread_kill_other_threads_np.....	681
13.39.12	pthread_rwlockattr_getkind_np.....	681
13.39.13	pthread_rwlockattr_setkind_np.....	681
13.39.14	pthread_setaffinity_np.....	682
13.39.15	pthread_setattr_default_np.....	682
13.39.16	pthread_setname_np.....	682
13.39.17	pthread_sigqueue.....	683
13.39.18	pthread_timedjoin_np.....	683

13.39.19	pthread_tryjoin_np	683
13.40	Glibc <pty.h>	683
13.40.1	forkpty	684
13.40.2	openpty	684
13.41	Glibc Extensions to <pwd.h>	685
13.41.1	fgetpwent	685
13.41.2	fgetpwent_r	685
13.41.3	getpw	685
13.41.4	getpwent_r	685
13.41.5	putpwent	686
13.42	Glibc Extensions to <regex.h>	686
13.42.1	re_comp	686
13.42.2	re_compile_fastmap	686
13.42.3	re_compile_pattern	687
13.42.4	re_exec	687
13.42.5	re_match	687
13.42.6	re_match_2	687
13.42.7	re_search	687
13.42.8	re_search_2	688
13.42.9	re_set_registers	688
13.42.10	re_set_syntax	688
13.42.11	re_syntax_options	688
13.43	Glibc <regexp.h>	688
13.43.1	advance	688
13.43.2	loc1	689
13.43.3	loc2	689
13.43.4	locs	689
13.43.5	step	689
13.44	Glibc <resolv.h>	689
13.44.1	dn_comp	689
13.44.2	dn_expand	690
13.44.3	dn_skipname	690
13.44.4	res_dnok	690
13.44.5	res_hnok	690
13.44.6	res_init	690
13.44.7	res_mailok	691
13.44.8	res_mkquery	691
13.44.9	res_nmkquery	691
13.44.10	res_nquery	691
13.44.11	res_nquerydomain	692
13.44.12	res_nsearch	692
13.44.13	res_nsend	692
13.44.14	res_ownok	692
13.44.15	res_query	693
13.44.16	res_querydomain	693
13.44.17	res_search	693
13.44.18	res_send	693

13.45	Glibc <rpc/auth.h>	693
13.45.1	authdes_create	693
13.45.2	authdes_pk_create	694
13.45.3	authnone_create	694
13.45.4	authunix_create	694
13.45.5	authunix_create_default	694
13.45.6	getnetname	694
13.45.7	host2netname	695
13.45.8	key_decryptsession	695
13.45.9	key_decryptsession_pk	695
13.45.10	key_encryptsession	695
13.45.11	key_encryptsession_pk	695
13.45.12	key_gendes	696
13.45.13	key_get_conv	696
13.45.14	key_secretkey_is_set	696
13.45.15	key_setsecret	696
13.45.16	netname2host	696
13.45.17	netname2user	697
13.45.18	user2netname	697
13.45.19	xdr_des_block	697
13.45.20	xdr_opaque_auth	697
13.46	Glibc <rpc/auth_des.h>	697
13.46.1	authdes_getucred	697
13.46.2	getpublickey	698
13.46.3	getsecretkey	698
13.46.4	rtime	698
13.47	Glibc <rpc/auth_unix.h>	698
13.47.1	xdr_authunix_parms	698
13.48	Glibc <rpc/clnt.h>	698
13.48.1	callrpc	698
13.48.2	clnt_create	699
13.48.3	clnt_pcreateerror	699
13.48.4	clnt_perrno	699
13.48.5	clnt_perror	699
13.48.6	clnt_screateerror	700
13.48.7	clnt_sperrno	700
13.48.8	clnt_sperror	700
13.48.9	clntraw_create	700
13.48.10	clnttcp_create	700
13.48.11	clntudp_bufcreate	701
13.48.12	clntudp_create	701
13.48.13	clntunix_create	701
13.48.14	get_myaddress	701
13.48.15	getrpcport	702
13.48.16	rpc_createerr	702
13.49	Glibc <rpc/key_prot.h>	702
13.49.1	xdr_cryptkeyarg	702

13.49.2	xdr_cryptkeyarg2.....	702
13.49.3	xdr_cryptkeyres.....	702
13.49.4	xdr_getcredres.....	703
13.49.5	xdr_key_netstarg.....	703
13.49.6	xdr_key_netstres.....	703
13.49.7	xdr_keybuf.....	703
13.49.8	xdr_keystatus.....	703
13.49.9	xdr_netnamestr.....	703
13.49.10	xdr_unixcred.....	704
13.50	Glibc <rpc/netdb.h>.....	704
13.50.1	endrpccent.....	704
13.50.2	getrpcbyname.....	704
13.50.3	getrpcbyname_r.....	704
13.50.4	getrpcbynumber.....	704
13.50.5	getrpcbynumber_r.....	705
13.50.6	getrpccent.....	705
13.50.7	getrpccent_r.....	705
13.50.8	setrpccent.....	705
13.51	Glibc <rpc/pmap_clnt.h>.....	706
13.51.1	clnt_broadcast.....	706
13.51.2	pmap_getmaps.....	706
13.51.3	pmap_getport.....	706
13.51.4	pmap_rmtcall.....	706
13.51.5	pmap_set.....	707
13.51.6	pmap_unset.....	707
13.52	Glibc <rpc/pmap_prot.h>.....	707
13.52.1	xdr_pmap.....	707
13.52.2	xdr_pmaplist.....	707
13.53	Glibc <rpc/pmap_rmt.h>.....	708
13.53.1	xdr_rmtcall_args.....	708
13.53.2	xdr_rmtcallres.....	708
13.54	Glibc <rpc/rpc_msg.h>.....	708
13.54.1	xdr_callhdr.....	708
13.54.2	xdr_callmsg.....	708
13.54.3	xdr_replymsg.....	709
13.55	Glibc <rpc/svc.h>.....	709
13.55.1	svc_exit.....	709
13.55.2	svc_fdset.....	709
13.55.3	svc_getreq.....	709
13.55.4	svc_getreq_common.....	709
13.55.5	svc_getreq_poll.....	710
13.55.6	svc_getreqset.....	710
13.55.7	svc_max_pollfd.....	710
13.55.8	svc_pollfd.....	710
13.55.9	svc_register.....	710
13.55.10	svc_run.....	711
13.55.11	svc_sendreply.....	711

13.55.12	svc_unregister.....	711
13.55.13	svcerr_auth.....	711
13.55.14	svcerr_decode.....	712
13.55.15	svcerr_noproc.....	712
13.55.16	svcerr_noprogram.....	712
13.55.17	svcerr_progvers.....	712
13.55.18	svcerr_systemerr.....	712
13.55.19	svcerr_weakauth.....	713
13.55.20	svccraw_create.....	713
13.55.21	svctcp_create.....	713
13.55.22	svcudp_bufcreate.....	713
13.55.23	svcudp_create.....	714
13.55.24	svcunix_create.....	714
13.55.25	xprt_register.....	714
13.55.26	xprt_unregister.....	714
13.56	Glibc <rpc/xdr.h>.....	715
13.56.1	xdr_array.....	715
13.56.2	xdr_bool.....	715
13.56.3	xdr_bytes.....	715
13.56.4	xdr_char.....	715
13.56.5	xdr_double.....	716
13.56.6	xdr_enum.....	716
13.56.7	xdr_float.....	716
13.56.8	xdr_free.....	716
13.56.9	xdr_hyper.....	716
13.56.10	xdr_int.....	717
13.56.11	xdr_int16_t.....	717
13.56.12	xdr_int32_t.....	717
13.56.13	xdr_int64_t.....	717
13.56.14	xdr_int8_t.....	717
13.56.15	xdr_long.....	718
13.56.16	xdr_longlong_t.....	718
13.56.17	xdr_netobj.....	718
13.56.18	xdr_opaque.....	718
13.56.19	xdr_pointer.....	718
13.56.20	xdr_quad_t.....	719
13.56.21	xdr_reference.....	719
13.56.22	xdr_short.....	719
13.56.23	xdr_sizeof.....	719
13.56.24	xdr_string.....	719
13.56.25	xdr_u_char.....	720
13.56.26	xdr_u_hyper.....	720
13.56.27	xdr_u_int.....	720
13.56.28	xdr_u_long.....	720
13.56.29	xdr_u_longlong_t.....	721
13.56.30	xdr_u_quad_t.....	721
13.56.31	xdr_u_short.....	721

13.56.32	xdr_uint16_t	721
13.56.33	xdr_uint32_t	721
13.56.34	xdr_uint64_t	722
13.56.35	xdr_uint8_t	722
13.56.36	xdr_union	722
13.56.37	xdr_vector	722
13.56.38	xdr_void	722
13.56.39	xdr_wrapstring	723
13.56.40	xdrmem_create	723
13.56.41	xdrrec_create	723
13.56.42	xdrrec_endofrecord	723
13.56.43	xdrrec_eof	724
13.56.44	xdrrec_skiprecord	724
13.56.45	xdrstdio_create	724
13.57	Glibc <rpcsvc/nislib.h>	724
13.57.1	nis_add	724
13.57.2	nis_add_entry	725
13.57.3	nis_addmember	725
13.57.4	nis_checkpoint	725
13.57.5	nis_clone_object	725
13.57.6	nis_creategroup	725
13.57.7	nis_destroy_object	725
13.57.8	nis_destroygroup	726
13.57.9	nis_dir_cmp	726
13.57.10	nis_domain_of	726
13.57.11	nis_domain_of_r	726
13.57.12	nis_first_entry	726
13.57.13	nis_freenames	726
13.57.14	nis_freeresult	727
13.57.15	nis_freeservlist	727
13.57.16	nis_freetags	727
13.57.17	nis_getnames	727
13.57.18	nis_getservlist	727
13.57.19	nis_ismember	727
13.57.20	nis_leaf_of	728
13.57.21	nis_leaf_of_r	728
13.57.22	nis_lerror	728
13.57.23	nis_list	728
13.57.24	nis_local_directory	728
13.57.25	nis_local_group	728
13.57.26	nis_local_host	729
13.57.27	nis_local_principal	729
13.57.28	nis_lookup	729
13.57.29	nis_mkdir	729
13.57.30	nis_modify	729
13.57.31	nis_modify_entry	729
13.57.32	nis_name_of	730

13.57.33	nis_name_of_r	730
13.57.34	nis_next_entry	730
13.57.35	nis_perror	730
13.57.36	nis_ping	730
13.57.37	nis_print_directory	731
13.57.38	nis_print_entry	731
13.57.39	nis_print_group	731
13.57.40	nis_print_group_entry	731
13.57.41	nis_print_link	731
13.57.42	nis_print_object	732
13.57.43	nis_print_result	732
13.57.44	nis_print_rights	732
13.57.45	nis_print_table	732
13.57.46	nis_remove	732
13.57.47	nis_remove_entry	733
13.57.48	nis_removemember	733
13.57.49	nis_rmdir	733
13.57.50	nis_servstate	733
13.57.51	nis_sperrno	733
13.57.52	nis_sperror	733
13.57.53	nis_sperror_r	734
13.57.54	nis_stats	734
13.57.55	nis_verifygroup	734
13.58	Glibc <rpcsvc/nis_callback.h>	734
13.58.1	xdr_cback_data	734
13.58.2	xdr_obj_p	734
13.59	Glibc <rpcsvc/yp.h>	735
13.59.1	xdr_domainname	735
13.59.2	xdr_keydat	735
13.59.3	xdr_valdat	735
13.59.4	xdr_ypbind_resptype	735
13.59.5	xdr_ypmap_parms	735
13.59.6	xdr_ypmaplist	736
13.59.7	xdr_yppushresp_xfr	736
13.59.8	xdr_ypreq_key	736
13.59.9	xdr_ypreq_nokey	736
13.59.10	xdr_ypreq_xfr	736
13.59.11	xdr_ypresp_all	736
13.59.12	xdr_ypresp_key_val	737
13.59.13	xdr_ypresp_maplist	737
13.59.14	xdr_ypresp_master	737
13.59.15	xdr_ypresp_order	737
13.59.16	xdr_ypresp_val	737
13.59.17	xdr_ypresp_xfr	737
13.59.18	xdr_ypstat	738
13.59.19	xdr_ypxfirstat	738
13.60	Glibc <rpcsvc/ypclnt.h>	738

13.60.1	yp_all	738
13.60.2	yp_bind	738
13.60.3	yp_first	738
13.60.4	yp_get_default_domain	739
13.60.5	yp_master	739
13.60.6	yp_match	739
13.60.7	yp_next	739
13.60.8	yp_order	739
13.60.9	yp_unbind	739
13.60.10	ypbinderr_string	740
13.60.11	yperr_string	740
13.60.12	ypprot_err	740
13.61	Glibc Extensions to <sched.h>	740
13.61.1	clone	740
13.61.2	getcpu	740
13.61.3	sched_getaffinity	741
13.61.4	sched_getattr	741
13.61.5	sched_getcpu	741
13.61.6	sched_setaffinity	742
13.61.7	sched_setattr	742
13.61.8	setns	742
13.62	Glibc Extensions to <search.h>	742
13.62.1	hcreate_r	742
13.62.2	hdestroy_r	743
13.62.3	hsearch_r	743
13.62.4	tdestroy	744
13.62.5	twalk_r	744
13.63	Glibc Extensions to <selinux/selinux.h>	744
13.63.1	fgetfilecon	744
13.63.2	getfilecon	745
13.63.3	lgetfilecon	745
13.64	Glibc <shadow.h>	746
13.64.1	endspent	746
13.64.2	fgetspent	746
13.64.3	fgetspent_r	746
13.64.4	getspent	746
13.64.5	getspent_r	747
13.64.6	getspnam	747
13.64.7	getspnam_r	747
13.64.8	lckpword	747
13.64.9	putspent	748
13.64.10	setspent	748
13.64.11	sgetspent	748
13.64.12	sgetspent_r	748
13.64.13	ulckpword	749
13.65	Glibc Extensions to <signal.h>	749
13.65.1	gsignal	749

13.65.2	sigandset.....	749
13.65.3	sigblock.....	749
13.65.4	siggetmask.....	750
13.65.5	sigisemptyset.....	750
13.65.6	sigorset.....	750
13.65.7	sigreturn.....	751
13.65.8	sigsetmask.....	751
13.65.9	sigstack.....	751
13.65.10	sigvec.....	751
13.65.11	ssignal.....	752
13.65.12	sys_siglist.....	752
13.65.13	sysv_signal.....	752
13.65.14	tgkill.....	752
13.66	Glibc Extensions to <spawn.h>.....	753
13.66.1	pidfd_spawn.....	753
13.66.2	pidfd_spawnnp.....	753
13.66.3	posix_spawn_file_actions_addchdir_np.....	753
13.66.4	posix_spawn_file_actions_addclosefrom_np.....	754
13.66.5	posix_spawn_file_actions_addfchdir_np.....	754
13.66.6	posix_spawn_file_actions_addtcsetpgrp_np.....	754
13.66.7	posix_spawnattr_getcgroup_np.....	754
13.66.8	posix_spawnattr_setcgroup_np.....	755
13.67	Glibc Extensions to <stdio.h>.....	755
13.67.1	cuserid.....	755
13.67.2	clearerr_unlocked.....	755
13.67.3	fcloseall.....	755
13.67.4	feof_unlocked.....	756
13.67.5	ferror_unlocked.....	756
13.67.6	fflush_unlocked.....	756
13.67.7	fgetc_unlocked.....	757
13.67.8	fgets_unlocked.....	757
13.67.9	fileno_unlocked.....	757
13.67.10	fopencookie.....	758
13.67.11	fputc_unlocked.....	758
13.67.12	fputs_unlocked.....	758
13.67.13	fread_unlocked.....	759
13.67.14	fwrite_unlocked.....	759
13.67.15	getw.....	759
13.67.16	putw.....	760
13.67.17	renameat2.....	760
13.67.18	setbuffer.....	760
13.67.19	setlinebuf.....	761
13.67.20	sys_errlist.....	761
13.67.21	sys_nerr.....	761
13.67.22	tmpnam_r.....	761
13.68	Glibc Extensions to <stdlib.h>.....	762
13.68.1	arc4random.....	762

13.68.2	<code>arc4random_buf</code>	762
13.68.3	<code>arc4random_uniform</code>	762
13.68.4	<code>canonicalize_file_name</code>	763
13.68.5	<code>cfree</code>	763
13.68.6	<code>clearenv</code>	763
13.68.7	<code>drand48_r</code>	763
13.68.8	<code>ecvt_r</code>	764
13.68.9	<code>erand48_r</code>	764
13.68.10	<code>fcvt_r</code>	764
13.68.11	<code>getloadavg</code>	765
13.68.12	<code>getpt</code>	765
13.68.13	<code>initstate_r</code>	765
13.68.14	<code>jrand48_r</code>	766
13.68.15	<code>lcong48_r</code>	766
13.68.16	<code>lrand48_r</code>	767
13.68.17	<code>mkostemps</code>	767
13.68.18	<code>mkstemps</code>	767
13.68.19	<code>mrnd48_r</code>	768
13.68.20	<code>nrnd48_r</code>	768
13.68.21	<code>on_exit</code>	768
13.68.22	<code>qecvt</code>	769
13.68.23	<code>qecvt_r</code>	769
13.68.24	<code>qfcvt</code>	769
13.68.25	<code>qfcvt_r</code>	769
13.68.26	<code>qgcvt</code>	770
13.68.27	<code>random_r</code>	770
13.68.28	<code>rpmatch</code>	771
13.68.29	<code>seed48_r</code>	771
13.68.30	<code>setstate_r</code>	771
13.68.31	<code>srand48_r</code>	772
13.68.32	<code>srandom_r</code>	772
13.68.33	<code>strtod_l</code>	772
13.68.34	<code>strtof_l</code>	772
13.68.35	<code>strtol_l</code>	773
13.68.36	<code>strtold_l</code>	773
13.68.37	<code>strtoll_l</code>	773
13.68.38	<code>strtoq</code>	773
13.68.39	<code>strtoul_l</code>	774
13.68.40	<code>strtoull_l</code>	774
13.68.41	<code>strtouq</code>	774
13.68.42	<code>uabs</code>	774
13.68.43	<code>ulabs</code>	775
13.68.44	<code>ullabs</code>	775
13.68.45	<code>valloc</code>	775
13.69	Glibc Extensions to <code><string.h></code>	775
13.69.1	<code>explicit_bzero</code>	775
13.69.2	<code>memfrob</code>	776

13.69.3	mempcpy	776
13.69.4	memrchr	777
13.69.5	rawmemchr	777
13.69.6	sigabbrev_np	777
13.69.7	sigdescr_np	778
13.69.8	strcasestr	778
13.69.9	strchrnul	778
13.69.10	strerrordesc_np	779
13.69.11	strerrorname_np	779
13.69.12	strfry	779
13.69.13	strsep	780
13.69.14	strverscmp	780
13.70	Glibc <sys/auxv.h>	780
13.70.1	getauxval	780
13.71	Glibc <sys/capability.h>	781
13.71.1	capget	781
13.71.2	capset	781
13.72	Glibc <sys/epoll.h>	781
13.72.1	epoll_create	781
13.72.2	epoll_create1	782
13.72.3	epoll_ctl	782
13.72.4	epoll_pwait	782
13.72.5	epoll_pwait2	782
13.72.6	epoll_wait	783
13.73	Glibc <sys/eventfd.h>	783
13.73.1	eventfd	783
13.73.2	eventfd_read	783
13.73.3	eventfd_write	783
13.74	Glibc <sys/fanotify.h>	784
13.74.1	fanotify_init	784
13.74.2	fanotify_mark	784
13.75	Glibc <sys/file.h>	784
13.75.1	flock	784
13.76	Glibc <sys/fsuid.h>	785
13.76.1	setfsuid	785
13.76.2	setfsgid	785
13.77	Glibc <sys/gmon.h>	785
13.77.1	monstartup	785
13.78	Glibc <sys/inotify.h>	785
13.78.1	inotify_add_watch	785
13.78.2	inotify_init	786
13.78.3	inotify_init1	786
13.78.4	inotify_rm_watch	786
13.79	Glibc <sys/io.h>, <sys/perm.h>	787
13.79.1	ioperm	787
13.79.2	iopl	787
13.80	Glibc <sys/kdaemon.h>	787

13.80.1	bdflush	787
13.81	Glibc <sys/klog.h>	787
13.81.1	klogctl	787
13.82	Glibc Extensions to <sys/mman.h>	788
13.82.1	madvise	788
13.82.2	memfd_create	788
13.82.3	mincore	788
13.82.4	mlock2	789
13.82.5	mremap	789
13.82.6	pkey_alloc	789
13.82.7	pkey_free	790
13.82.8	pkey_get	790
13.82.9	pkey_mprotect	790
13.82.10	pkey_set	790
13.82.11	process_madvise	791
13.82.12	process_mrelease	791
13.82.13	remap_file_pages	791
13.83	Glibc <sys/mount.h>	791
13.83.1	fsconfig	792
13.83.2	fsmount	792
13.83.3	fsopen	792
13.83.4	fspick	792
13.83.5	mount	793
13.83.6	mount_setattr	793
13.83.7	move_mount	793
13.83.8	open_tree	793
13.83.9	umount	794
13.83.10	umount2	794
13.84	Glibc <sys/personality.h>	794
13.84.1	personality	794
13.85	Glibc <sys/pidfd.h>	794
13.85.1	pidfd_getfd	795
13.85.2	pidfd_getpid	795
13.85.3	pidfd_open	795
13.85.4	pidfd_send_signal	795
13.86	Glibc <sys/prctl.h>	796
13.86.1	prctl	796
13.87	Glibc <sys/profil.h>	796
13.87.1	sprofil	796
13.88	Glibc <sys/ptrace.h>	796
13.88.1	ptrace	796
13.89	Glibc <sys/quot.h>	796
13.89.1	quotactl	796
13.90	Glibc <sys/random.h>	797
13.90.1	getrandom	797
13.91	Glibc <sys/reboot.h>	797
13.91.1	reboot	797

13.92	Glibc Extensions to <code><sys/resource.h></code>	798
13.92.1	<code>prlimit</code>	798
13.93	Glibc Extensions to <code><sys/sem.h></code>	798
13.93.1	<code>semtimedop</code>	798
13.94	Glibc <code><sys/sendfile.h></code>	798
13.94.1	<code>sendfile</code>	798
13.95	Glibc <code><sys/signalfd.h></code>	799
13.95.1	<code>signalfd</code>	799
13.96	Glibc <code><sys/single_threaded.h></code>	799
13.96.1	<code>__libc_single_threaded</code>	799
13.97	Glibc Extensions to <code><sys/socket.h></code>	799
13.97.1	<code>isfdtype</code>	799
13.97.2	<code>recvmsg</code>	799
13.97.3	<code>sendmsg</code>	800
13.98	Glibc Extensions to <code><sys/stat.h></code>	800
13.98.1	<code>getumask</code>	800
13.98.2	<code>lchmod</code>	800
13.98.3	<code>statx</code>	801
13.99	Glibc <code><sys/statfs.h></code>	801
13.99.1	<code>fstatfs</code>	801
13.99.2	<code>statfs</code>	801
13.100	Glibc <code><sys/swap.h></code>	802
13.100.1	<code>swapoff</code>	802
13.100.2	<code>swapon</code>	802
13.101	Glibc <code><sys/sysctl.h></code>	802
13.101.1	<code>sysctl</code>	802
13.102	Glibc <code><sys/sysinfo.h></code>	803
13.102.1	<code>get_avphys_pages</code>	803
13.102.2	<code>get_nprocs</code>	803
13.102.3	<code>get_nprocs_conf</code>	803
13.102.4	<code>get_phys_pages</code>	804
13.102.5	<code>sysinfo</code>	804
13.103	Glibc <code><sys/syslog.h></code>	804
13.103.1	<code>syslog</code>	804
13.104	Glibc <code><sys/sysmacros.h></code>	805
13.104.1	<code>gnu_dev_major</code>	805
13.104.2	<code>gnu_dev_makedev</code>	805
13.104.3	<code>gnu_dev_minor</code>	805
13.105	Glibc Extensions to <code><sys/time.h></code>	806
13.105.1	<code>adjtime</code>	806
13.105.2	<code>futimes</code>	806
13.105.3	<code>futimesat</code>	806
13.105.4	<code>lutimes</code>	807
13.105.5	<code>settimeofday</code>	807
13.106	Glibc <code><sys/timerfd.h></code>	807
13.106.1	<code>timerfd_create</code>	807
13.106.2	<code>timerfd_gettime</code>	808

13.106.3	timerfd_settime.....	808
13.107	Glibc <sys/timex.h>.....	808
13.107.1	adjtimex.....	808
13.107.2	ntp_adjtime.....	808
13.107.3	ntp_gettime.....	809
13.107.4	ntp_gettimex.....	809
13.108	Glibc Extensions to <sys/uio.h>.....	809
13.108.1	preadv.....	809
13.108.2	preadv2.....	810
13.108.3	process_vm_readv.....	810
13.108.4	process_vm_writev.....	810
13.108.5	pwritev.....	810
13.108.6	pwritev2.....	811
13.109	Glibc <sys/ustat.h>.....	811
13.109.1	ustat.....	811
13.110	Glibc <sys/vlimit.h>.....	811
13.110.1	vlimit.....	811
13.111	Glibc Extensions to <sys/wait.h>.....	812
13.111.1	wait3.....	812
13.111.2	wait4.....	812
13.112	Glibc <sys/xattr.h>.....	812
13.112.1	fgetxattr.....	812
13.112.2	flistxattr.....	813
13.112.3	fremovexattr.....	813
13.112.4	fsetxattr.....	813
13.112.5	getxattr.....	814
13.112.6	lgetxattr.....	814
13.112.7	listxattr.....	814
13.112.8	llistxattr.....	815
13.112.9	lremovexattr.....	815
13.112.10	lsetxattr.....	815
13.112.11	removexattr.....	815
13.112.12	setxattr.....	816
13.113	Glibc Extensions to <termios.h>.....	816
13.113.1	cfmakeraw.....	816
13.113.2	cfsetspeed.....	816
13.114	Glibc Extensions to <time.h>.....	817
13.114.1	clock_adjtime.....	817
13.114.2	dysize.....	817
13.114.3	getdate_r.....	817
13.114.4	stime.....	817
13.114.5	strptime_l.....	818
13.114.6	timelocal.....	818
13.115	Glibc <ttyent.h>.....	818
13.115.1	endttyent.....	818
13.115.2	getttyent.....	818
13.115.3	getttynam.....	819

13.115.4	setttyent.....	819
13.116	Glibc Extensions to <unistd.h>	819
13.116.1	acct.....	819
13.116.2	brk.....	819
13.116.3	chroot.....	820
13.116.4	closefrom.....	820
13.116.5	close_range.....	820
13.116.6	copy_file_range.....	821
13.116.7	daemon.....	821
13.116.8	eaccess.....	821
13.116.9	endusershell.....	822
13.116.10	euidaccess.....	822
13.116.11	execveat.....	822
13.116.12	execvpe.....	823
13.116.13	get_current_dir_name.....	823
13.116.14	getdomainname.....	823
13.116.15	getdtablesize.....	824
13.116.16	getpagesize.....	824
13.116.17	getpass.....	825
13.116.18	gettid.....	825
13.116.19	getusershell.....	826
13.116.20	group_member.....	826
13.116.21	profil.....	826
13.116.22	revoke.....	826
13.116.23	sbrk.....	827
13.116.24	setlogin.....	827
13.116.25	setdomainname.....	827
13.116.26	sethostid.....	827
13.116.27	sethostname.....	828
13.116.28	setusershell.....	828
13.116.29	syncfs.....	828
13.116.30	syscall.....	829
13.116.31	ttyslot.....	829
13.116.32	vhangup.....	829
13.117	Glibc <utmp.h>.....	829
13.117.1	endutent.....	829
13.117.2	getutent.....	830
13.117.3	getutent_r.....	830
13.117.4	getutid.....	830
13.117.5	getutid_r.....	831
13.117.6	getutline.....	831
13.117.7	getutline_r.....	831
13.117.8	pututline.....	831
13.117.9	setutent.....	832
13.117.10	updwtmp.....	832
13.117.11	utmpname.....	832
13.117.12	login.....	833

13.117.13	<code>login_tty</code>	833
13.118	Glibc Extensions to <code><utmpx.h></code>	833
13.118.1	<code>getutmp</code>	834
13.118.2	<code>getutmpx</code>	834
13.118.3	<code>updwtmpx</code>	834
13.118.4	<code>utmpxname</code>	835
13.119	Glibc Extensions to <code><wchar.h></code>	835
13.119.1	<code>fgetwc_unlocked</code>	835
13.119.2	<code>fgetws_unlocked</code>	835
13.119.3	<code>fputwc_unlocked</code>	836
13.119.4	<code>fputws_unlocked</code>	836
13.119.5	<code>getwc_unlocked</code>	836
13.119.6	<code>getwchar_unlocked</code>	837
13.119.7	<code>putwc_unlocked</code>	837
13.119.8	<code>putwchar_unlocked</code>	838
13.119.9	<code>wcschrnul</code>	838
13.119.10	<code>wcsftime_l</code>	838
13.119.11	<code>wcstod_l</code>	839
13.119.12	<code>wcstof_l</code>	839
13.119.13	<code>wcstol_l</code>	839
13.119.14	<code>wcstold_l</code>	839
13.119.15	<code>wcstoll_l</code>	840
13.119.16	<code>wcstoq</code>	840
13.119.17	<code>wcstoul_l</code>	840
13.119.18	<code>wcstoull_l</code>	840
13.119.19	<code>wcstouq</code>	841
13.119.20	<code>wmempcpy</code>	841
14	Native Windows Support	842
14.1	Libtool and Windows	842
14.2	Large File Support	842
14.3	Inode numbers on Windows	842
14.4	Precise file timestamps on Windows	843
14.5	Avoiding the year 2038 problem	843
14.6	Windows sockets	844
14.6.1	Getaddrinfo and WINVER	844
14.7	Native Windows Support without MSVC Support	845
14.8	Visual Studio Compatibility	845
15	Multithreading	846
15.1	The three multithreading APIs	846
15.2	Choosing the right multithreading API	847
15.3	The POSIX multithreading API	847
15.4	The ISO C multithreading API	848
15.5	The Gnulib multithreading API	848
15.6	Optimizations of multithreaded code	849

16	Strings and Characters	851
16.1	Strings	851
16.1.1	The C string representation	851
16.1.2	Iterating through strings	853
16.1.3	Strings with NUL characters	854
16.1.4	Character and String Functions in C Locale	854
16.1.4.1	c-ctype	854
16.1.4.2	c-strcaseeq	855
16.1.4.3	c-strcasecmp	855
16.1.4.4	c-strncasecmp	855
16.1.4.5	c-strcasestr	856
16.1.4.6	c-strstr	856
16.1.4.7	c-strtod	856
16.1.4.8	c-strtold	856
16.1.5	Comparison of string APIs	857
16.2	Characters	859
16.2.1	The <code>char</code> type	860
16.2.2	The <code>wchar_t</code> type	860
16.2.3	The <code>char32_t</code> type	861
16.2.4	The <code>mbchar_t</code> type	861
16.2.4.1	Reading multibyte strings	862
16.2.5	Comparison of character APIs	862
17	Particular Modules	864
17.1	<code>alloca</code>	864
17.2	<code>alloca-opt</code>	864
17.3	Safe Allocation Macros	865
17.4	Attributes	866
17.5	Compile-time Assertions	866
17.6	Non-returning Functions	868
17.7	Integer Properties	869
17.7.1	Arithmetic Type Properties	869
17.7.2	Arithmetic Type Conversion	870
17.7.3	Integer Bounds	870
17.7.4	Checking Integer Overflow	871
17.7.5	Wraparound Arithmetic with Integers	872
17.7.6	Integer Type Overflow	874
17.7.7	Integer Range Overflow	875
17.8	Static inline functions	877
17.9	Extern inline functions	877
17.10	Handling closed standard file descriptors	879
17.11	Handling strings with NUL characters	881
17.12	Container data types	882
17.12.1	Ordinary container data types	882
17.12.1.1	Sequential lists	883
17.12.1.2	Sets	885

17.12.1.3	Ordered sets	886
17.12.1.4	Maps	886
17.12.1.5	Ordered maps	886
17.12.1.6	C++ classes for container data types	887
17.12.2	Specialized container data types	887
17.13	Modernized printf	888
17.14	Advanced functions on FILE streams	890
17.14.1	Determining the state of a FILE stream	890
17.14.2	Setting an error in a FILE stream	891
17.14.3	Dropping the contents of the buffers of a FILE stream ..	891
17.14.4	Accessing the write buffer of a FILE stream	891
17.14.5	Accessing the read buffer of a FILE stream	891
17.15	libunistring	892
17.15.1	How to use libunistring	892
17.15.2	The libunistring modules	893
17.15.2.1	Declarations in <unitypes.h>	893
17.15.2.2	Declarations in <unistr.h>	893
17.15.2.3	Declarations in <uniconv.h>	896
17.15.2.4	Declarations in <unistdio.h>	896
17.15.2.5	Declarations in <uniname.h>	897
17.15.2.6	Declarations in <unictype.h>	897
17.15.2.7	Declarations in <uniwidth.h>	906
17.15.2.8	Declarations in <unigbrk.h>	906
17.15.2.9	Declarations in <uniwbrk.h>	906
17.15.2.10	Declarations in <unilbrk.h>	906
17.15.2.11	Declarations in <uninorm.h>	907
17.15.2.12	Declarations in <unicase.h>	907
17.16	Stack traces	909
17.17	Printing -version and bug-reporting information	910
17.18	Recognizing Option Arguments	911
17.19	Quoting	913
17.20	progname and getprogname	913
17.21	gcd: greatest common divisor	914
17.22	Profiling of program phases	914
17.23	Library version handling	916
17.24	Supporting Relocation	916
17.25	func	920
17.26	stat-size	920
18	Regular expressions	921
18.1	Overview	921
18.2	Regular Expression Syntax	921
18.2.1	Syntax Bits	922
18.2.2	Predefined Syntaxes	924
18.2.3	Collating Elements vs. Characters	926
18.2.4	The Backslash Character	926

18.3	Common Operators	927
18.3.1	The Match-self Operator (<i>ordinary character</i>)	927
18.3.2	The Match-any-character Operator (.)	927
18.3.3	The Concatenation Operator	927
18.3.4	Repetition Operators	928
18.3.4.1	The Match-zero-or-more Operator (*)	928
18.3.4.2	The Match-one-or-more Operator (+ or \+)	928
18.3.4.3	The Match-zero-or-one Operator (? or \?)	928
18.3.4.4	Interval Operators ({ ... } or \{ ... \})	929
18.3.5	The Alternation Operator (or \)	929
18.3.6	List Operators ([...] and [^ ...])	930
18.3.6.1	Collating Symbol Operators ([.])	931
18.3.6.2	Equivalence Class Operators ([= ... =])	931
18.3.6.3	Character Class Operators ([: ... :])	931
18.3.6.4	The Range Operator (-)	932
18.3.7	Grouping Operators ((...) or \ (... \))	932
18.3.8	The Back-reference Operator (\digit)	933
18.3.9	Anchoring Operators	934
18.3.9.1	The Match-beginning-of-line Operator (^)	934
18.3.9.2	The Match-end-of-line Operator (\$)	934
18.4	GNU Operators	935
18.4.1	Word Operators	935
18.4.1.1	The Match-word-boundary Operator (\b)	935
18.4.1.2	The Match-within-word Operator (\B)	935
18.4.1.3	The Match-beginning-of-word Operator (\<)	935
18.4.1.4	The Match-end-of-word Operator (\>)	935
18.4.1.5	The Match-word-constituent Operator (\w)	935
18.4.1.6	The Match-non-word-constituent Operator (\W)	935
18.4.2	Space Operators	935
18.4.2.1	The Match-space Operator (\s)	935
18.4.2.2	The Match-non-space Operator (\S)	935
18.4.3	Whole-string Operators	935
18.4.3.1	The Match-beginning-of-string Operator (\`)	935
18.4.3.2	The Match-end-of-string Operator (\')	936
18.5	What Gets Matched?	936
18.6	Programming with Regex	936
18.6.1	GNU Regex Functions	936
18.6.1.1	GNU Pattern Buffers	936
18.6.1.2	GNU Regular Expression Compiling	936
18.6.1.3	GNU Matching	937
18.6.1.4	GNU Searching	938
18.6.1.5	Matching and Searching with Split Data	939
18.6.1.6	Searching with Fastmaps	939
18.6.1.7	GNU Translate Tables	940
18.6.1.8	Using Registers	941
18.6.1.9	Freeing GNU Pattern Buffers	943
18.6.2	BSD Regex Functions	943

18.6.2.1	BSD Regular Expression Compiling	943
18.6.2.2	BSD Searching	944
18.7	Regular expression syntaxes	944
18.7.1	‘awk’ regular expression syntax	944
18.7.2	‘ed’ regular expression syntax	945
18.7.3	‘egrep’ regular expression syntax	946
18.7.4	‘emacs’ regular expression syntax	946
18.7.5	‘gnu-awk’ regular expression syntax	948
18.7.6	‘grep’ regular expression syntax	949
18.7.7	‘posix-awk’ regular expression syntax	950
18.7.8	‘posix-basic’ regular expression syntax	950
18.7.9	‘posix-egrep’ regular expression syntax	951
18.7.10	‘posix-extended’ regular expression syntax	951
18.7.11	‘posix-minimal-basic’ regular expression syntax	952
18.7.12	‘sed’ regular expression syntax	952
19	Build Infrastructure Modules	953
19.1	Searching for Libraries	953
Simple Library Tests		953
Locating Libraries		953
Example of using AC_LIB_LINKFLAGS		954
Dependencies		954
Static vs. shared		955
CPPFLAGS vs. LDFLAGS		955
Bi-arch systems		955
19.2	Controlling the Exported Symbols of Shared Libraries	956
19.3	LD Version Scripts	958
19.4	configmake	959
19.5	warnings	959
19.6	manywarnings	960
19.7	Running self-tests under valgrind	963
19.7.1	Using valgrind without developer intervention	963
19.7.2	Valgrind options	963
19.7.3	Using valgrind at the developer’s discretion	964
19.7.4	How to use Valgrind with shell scripts	964
19.8	Package version management	965
19.8.1	Setting the package version	965
19.8.2	Propagating the package version	966
19.8.3	Using the package version during the build	966
19.9	VCS To ChangeLog	967
19.10	gitlog-to-changelog	968
20	Build Infrastructure Files	970
20.1	Recognizing platforms	970
20.2	Utilities for Makefiles	970
20.3	Programs for developing in Git checkouts	971

20.4	Utilities for building documentation	972
20.5	Utilities for building libraries	972
20.6	Utilities for running test suites	973
21	Release Management Files	974
21.1	Tools for releasing packages with shared libraries	974
21.2	Tools for uploading release tarballs	974
Appendix A GNU Free Documentation License		975
Appendix B Gnulib history		983
Module Index		984
Undocumented Module Index		999
General Index		1004

1 Brief Overview

Gnulib is a source code library that provides basic functionality to programs and libraries. Many software packages make use of Gnulib to avoid reinventing the portability wheel.

Resources:

- Gnulib is hosted at Savannah: <https://savannah.gnu.org/projects/gnulib>. Get the sources through Git from there.
- The Gnulib home page: <https://www.gnu.org/software/gnulib/>.

1.1 Gnulib Basics

While portability across operating systems is not one of GNU’s primary goals, it has helped introduce many people to the GNU system, and is worthwhile when it can be achieved at a low cost. This collection helps lower that cost.

Gnulib is intended to be the canonical source for most of the important “portability” and/or common files for GNU projects. These are files intended to be shared at the source level; Gnulib is not a typical library meant to be installed and linked against. Thus, unlike most projects, Gnulib does not normally generate a source tarball distribution; instead, developers grab modules directly from the source repository.

The easiest, and recommended, way to do this is to use the `gnulib-tool` script. Since there is no installation procedure for Gnulib, `gnulib-tool` needs to be run directly in the directory that contains the Gnulib source code. You can do this either by specifying the absolute filename of `gnulib-tool`, or by using a symbolic link from a place inside your `PATH` to the `gnulib-tool` file of your preferred Gnulib checkout. For example:

```
$ ln -s $HOME/gnu/src/gnulib.git/gnulib-tool $HOME/bin/gnulib-tool
```

1.2 Git Checkout

Gnulib is available for anonymous checkout. In any Bourne-shell the following should work:

```
$ git clone https://git.savannah.gnu.org/git/gnulib.git
```

For a read-write checkout you need to have a login on ‘savannah.gnu.org’ and be a member of the Gnulib project at <https://savannah.gnu.org/projects/gnulib>. Then, instead of the URL <https://git.savannah.gnu.org/git/gnulib.git>, use the URL ‘`ssh://user@git.savannah.gnu.org/srv/git/gnulib`’ where *user* is your login name on savannah.gnu.org.

git resources:

Overview: [https://en.wikipedia.org/wiki/Git_\(software\)](https://en.wikipedia.org/wiki/Git_(software))

Homepage:

<https://git-scm.com/>

When you use `git annotate` or `git blame` with Gnulib, it’s recommended that you use the `-w` option, in order to ignore massive whitespace changes that happened in 2009.

1.3 Keeping Up-to-date

The best way to work with Gnulib is to check it out of git. To synchronize, you can use `git pull`.

Subscribing to the bug-gnulib@gnu.org mailing list will help you to plan when to update your local copy of Gnulib (which you use to maintain your software) from git. You can review the archives, subscribe, etc., via <https://lists.gnu.org/mailman/listinfo/bug-gnulib>.

Sometimes, using an updated version of Gnulib will require you to use newer versions of GNU Automake or Autoconf. You may find it helpful to join the autotools-announce mailing list to be advised of such changes.

1.4 Contributing to Gnulib

All software here is copyrighted by the Free Software Foundation—you need to have filled out an assignment form for a project that uses the module for that contribution to be accepted here.

If you have a piece of code that you would like to contribute, please email bug-gnulib@gnu.org.

Generally we are looking for files that fulfill at least one of the following requirements:

- If your `.c` and `.h` files define functions that are broken or missing on some other system, we should be able to include it.
- If your functions remove arbitrary limits from existing functions (either under the same name, or as a slightly different name), we should be able to include it.

If your functions define completely new but rarely used functionality, you should probably consider packaging it as a separate library.

1.4.1 Gnulib licensing

Gnulib contains code both under GPL and LGPL. Because several packages that use Gnulib are GPL, the files state they are licensed under GPL. However, to support LGPL projects as well, you may use some of the files under LGPL. The “License:” information in the files under `modules/` clarifies the real license that applies to the module source.

Keep in mind that if you submit patches to files in Gnulib, you should license them under a compatible license, which means that sometimes the contribution will have to be LGPL, if the original file is available under LGPL via a “License: LGPL” information in the projects’ `modules/` file.

1.4.2 Indent with spaces not TABs

We use space-only indentation in nearly all files. This includes all `*.h`, `*.c`, `*.y` files, except for the `regex` module. Makefile and ChangeLog files are excluded, since TAB characters are part of their format.

In order to tell your editor to produce space-only indentation, you can use these instructions.

- For Emacs: Add these lines to your Emacs initialization file (`$HOME/.emacs` or similar):

```
;; In Gnulib, indent with spaces everywhere (not TABs).
```

```
;; Exceptions: Makefile and ChangeLog modes.
(add-hook 'find-file-hook '(lambda ()
  (if (and buffer-file-name
        (string-match "/gnulib\\>" (buffer-file-name))
        (not (string-equal mode-name "Change Log"))
        (not (string-equal mode-name "Makefile"))))
      (setq indent-tabs-mode nil))))
```

- For vi (vim): Add these lines to your `$HOME/.vimrc` file:

```
" Don't use tabs for indentation. Spaces are nicer to work with.
set expandtab
```

For Makefile and ChangeLog files, compensate for this by adding this to your `$HOME/.vim/after/indent/make.vim` file, and similarly for your `$HOME/.vim/after/indent/changelog.vim` file:

```
" Use tabs for indentation, regardless of the global setting.
set noexpandtab
```

- For Eclipse: In the “Window|Preferences” dialog (or “Eclipse|Preferences” dialog on Mac OS),
 1. Under “General|Editors|Text Editors”, select the “Insert spaces for tabs” checkbox.
 2. Under “C/C++|Code Style”, select a code style profile that has the “Indentation|Tab policy” combobox set to “Spaces only”, such as the “GNU [built-in]” policy.

If you use the GNU indent program, pass it the option `--no-tabs`.

1.4.3 How to add a new module

- Add the header files and source files to `lib/`.
- If the module needs configure-time checks, write an Autoconf macro for it in `m4/module.m4`. See `m4/README` for details.
- Write a module description `modules/module`, based on `modules/TEMPLATE`.
- If the module contributes a section to the end-user documentation, put this documentation in `doc/module.texi` and add it to the “Files” section of `modules/module`. Most modules don’t do this; they have only documentation for the programmer (= Gnulib user). Such documentation usually goes into the `lib/` source files. It may also go into `doc/`; but don’t add it to the module description in this case.
- Add the module to the list in `MODULES.html.sh`.

You can test that a module builds correctly with:

```
$ ./gnulib-tool --create-testdir --dir=/tmp/testdir module1 ... moduleN
$ cd /tmp/testdir
$ ./configure && make
```

Other things:

- Check the license and copyright year of headers.
- Check that the source code follows the GNU coding standards; see <https://www.gnu.org/prep/standards>.

- Add source files to `config/srclist*` if they are identical to upstream and should be upgraded in Gnulib whenever the upstream source changes.
- Include header files in source files to verify the function prototypes.
- Make sure a replacement function doesn't cause warnings or clashes on systems that have the function.
- Autoconf functions can use `'gl_*` prefix. The `'AC_*` prefix is for autoconf internal functions.
- Build files only if they are needed on a platform. Look at the `alloca` and `fnmatch` modules for how to achieve this. If for some reason you cannot do this, and you have a `.c` file that leads to an empty `.o` file on some platforms (through some big `#if` around all the code), then ensure that the compilation unit is not empty after preprocessing. One way to do this is to `#include <stddef.h>` or `<stdio.h>` before the big `#if`.

1.5 Portability guidelines

Gnulib code is intended to be portable to a wide variety of platforms, not just GNU platforms. Gnulib typically attempts to support a platform as long as it is still supported by its provider, even if the platform is not the latest version. See [Section 2.4 \[Target Platforms\]](#), page 13.

Many Gnulib modules exist so that applications need not worry about undesirable variability in implementations. For example, an application that uses the `malloc` module need not worry about `malloc (0)` returning a null pointer on some Standard C platforms; and `glob` users need not worry about `glob` silently omitting symbolic links to nonexistent files on some platforms that do not conform to POSIX.

Gnulib code is intended to port without problem to new hosts, e.g., hosts conforming to recent C and POSIX standards. Hence Gnulib code should avoid using constructs that these newer standards no longer require, without first testing for the presence of these constructs. For example, because C11 made variable length arrays optional, Gnulib code should avoid them unless it first uses the `vararrays` module to check whether they are supported.

The following subsections discuss some exceptions and caveats to the general Gnulib portability guidelines.

1.5.1 C language versions

Currently Gnulib assumes at least a freestanding C99 compiler, possibly operating with a C library that predates C99; with time this assumption will likely be strengthened to later versions of the C standard. Old platforms currently supported include AIX 7.3 and Solaris 10, though these platforms are rarely tested. Gnulib itself is so old that it contains many fixes for obsolete platforms, fixes that may be removed in the future.

Because of the freestanding C99 assumption, Gnulib code can include `<float.h>`, `<limits.h>`, `<stdarg.h>`, `<stddef.h>`, and `<stdint.h>` unconditionally; `<stdbool.h>` is also in the C99 freestanding list but is obsolescent as of C23. Gnulib code can also assume the existence of `<ctype.h>`, `<errno.h>`, `<fcntl.h>`, `<locale.h>`, `<signal.h>`, `<stdio.h>`, `<stdlib.h>`, `<string.h>`, and `<time.h>`. Similarly, many modules include `<sys/types.h>` even though it's not in the C standard; that's OK since `<sys/types.h>` has been around since Seventh Edition Unix (1979).

Even if the include files exist, they may not conform to the C standard. However, GCC has a `fixincludes` script that attempts to fix most conformance problems. Gnulib currently assumes include files largely conform to C99 or better. People still using ancient hosts should use `fixincludes` or fix their include files manually.

Even if the include files conform, the library itself may not. For example, `strtod` and `mktime` have some bugs on some platforms. You can work around some of these problems by requiring the relevant modules, e.g., the Gnulib `mktime` module supplies a working and conforming `mktime`.

1.5.2 C99 features assumed by Gnulib

Although the C99 standard specifies many features, Gnulib code is conservative about using them, partly because Gnulib predates the widespread adoption of C99, and partly because many C99 features are not well-supported in practice. C99 features that are reasonably portable nowadays include:

- A declaration after a statement, or as the first clause in a `for` statement.
- `long long int`.
- `<stdbool.h>`, although Gnulib code no longer uses it directly, preferring plain `bool` via the `bool` module instead. See [Section 9.51 \[stdbool.h\]](#), page 79.
- `<stdint.h>`, assuming the `stdint-h` module is used. See [Section 9.54 \[stdint.h\]](#), page 80.
- Compound literals and designated initializers.
- Empty arguments in macro calls.
- Variadic macros.
Note: The handling of `__VA_ARGS__` in MSVC differs from the one in ISO C 99, see <https://stackoverflow.com/questions/5134523/>. But usually this matters only for macros that decompose `__VA_ARGS__`.
- `static inline` functions.
- `__func__`, assuming the `func` module is used. See [Section 17.25 \[func\]](#), page 920.
- The `restrict` qualifier, assuming `AC_REQUIRE([AC_C_RESTRICT])` is used. This qualifier is sometimes implemented via a macro, so C++ code that uses Gnulib should avoid using `restrict` as an identifier.
- Flexible array members (however, see the `flexmember` module).

1.5.3 C99 features avoided by Gnulib

Gnulib avoids some features even though they are standardized by C99, as they have portability problems in practice. Here is a partial list of avoided C99 features. Many other C99 features are portable only if their corresponding modules are used; Gnulib code that uses such a feature should require the corresponding module.

- Variable length arrays (VLAs) or variably modified types, without checking whether `__STDC_NO_VLA__` is defined. See the `vararrays` and `vla` modules.
- Block-scope variable length arrays, without checking whether either `GNULIB_NO_VLA` or `__STDC_NO_VLA__` is defined. This lets you define `GNULIB_NO_VLA` to pacify GCC when using its `-Wvla-larger-than` warnings option, and to avoid large stack usage that

may have security implications. `GNULIB_NO_VLA` does not affect Gnulib’s other uses of VLAs and variably modified types, such as array declarations in function prototype scope.

- Converting to pointers via integer types other than `intptr_t` or `uintptr_t`. Although the C standard says that values of these integer types, if they exist, should be convertible to and from `intmax_t` and `uintmax_t` without loss of information, on CHERI platforms such conversions result in integers that, if converted back to a pointer, cannot be dereferenced.
- `extern inline` functions, without checking whether they are supported. See [Section 17.9 \[extern inline\]](#), page 877.
- Type-generic math functions.
- Universal character names in source code.
- `<iso646.h>`, since GNU programs need not worry about deficient source-code encodings.
- Comments beginning with `‘//’`. This is mostly for style reasons.

1.5.4 Other portability assumptions made by Gnulib

Gnulib code makes the following assumptions that go beyond what C and POSIX require:

- Standard internal types like `ptrdiff_t` and `size_t` are no wider than `long`. The GNU coding standards allow code to make this assumption, POSIX requires implementations to support at least one programming environment where this is true, and such environments are recommended for Gnulib-using applications. When it is easy to port to non-POSIX platforms like MinGW where these types are wider than `long`, new Gnulib code should do so, e.g., by using `ptrdiff_t` instead of `long`. However, it is not always that easy, and no effort has been made to check that all Gnulib modules work on MinGW-like environments.
- `int` and `unsigned int` are at least 32 bits wide. POSIX and the GNU coding standards both require this.
- Signed integer arithmetic is two’s complement.

Previously, Gnulib code sometimes also assumed that signed integer arithmetic wraps around, but modern compiler optimizations sometimes do not guarantee this, and Gnulib code with this assumption is now considered to be questionable. See [Section 17.7 \[Integer Properties\]](#), page 869.

Although some Gnulib modules contain explicit support for ones’ complement and signed magnitude integer representations, which are allowed by C17 and earlier, these modules are the exception rather than the rule. All practical Gnulib targets use two’s complement, which is required by C23.

- There are no “holes” in integer values: all the bits of an integer contribute to its value in the usual way. In particular, an unsigned type and its signed counterpart have the same number of bits when you count the latter’s sign bit. (As an exception, Gnulib code is portable to CHERI platforms even though this assumption is false for CHERI.)
- Objects with all bits zero are treated as zero or as null pointers. For example, `memset(A, 0, sizeof A)` initializes an array `A` of pointers to null pointers.

- The types `intptr_t` and `uintptr_t` exist, and pointers can be converted to and from these types without loss of information.
- Addresses and sizes behave as if objects reside in a flat address space. In particular:
 - If two nonoverlapping objects have sizes S and T represented as `ptrdiff_t` or `size_t` values, then $S + T$ cannot overflow.
 - A pointer P points within an object O if and only if $(\text{char} *) \&O \leq (\text{char} *) P$ and $(\text{char} *) P < (\text{char} *) (\&O + 1)$.
 - Arithmetic on a valid pointer is equivalent to the same arithmetic on the pointer converted to `uintptr_t`, except that offsets are multiplied by the size of the pointed-to objects. For example, if $P + I$ is a valid expression involving a pointer P and an integer I , then $(\text{uintptr_t}) (P + I) == (\text{uintptr_t}) ((\text{uintptr_t}) P + I * \text{sizeof} *P)$. Similar arithmetic can be done with `intptr_t`, although more care must be taken in case of integer overflow or negative integers.
 - A pointer P has alignment A if and only if $(\text{uintptr_t}) P \% A$ is zero, and similarly for `intptr_t`. Although C23 says that portable code should instead test whether `memalignment(P) \% A` is zero, `memalignment` is missing in many non-C23 systems, is not specified to work portably on pointers to atomic or volatile memory, and is likely a bit slower than the cast.
 - If an existing object has size S , and if T is sufficiently small (e.g., 8 KiB), then $S + T$ cannot overflow. Overflow in this case would mean that the rest of your program fits into T bytes, which can't happen in realistic flat-address-space hosts.
 - Adding zero to a null pointer does not change the pointer. For example, $0 + (\text{char} *) \text{NULL} == (\text{char} *) \text{NULL}$. Similarly, subtracting zero does not change a null pointer, and subtracting two null pointers results in zero. C23 was the first C standard to require this behavior; see “[Allow zero length operations on null pointers](https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3322.pdf) (<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3322.pdf>)”, WG14 N3322 (2024-08-28); which was adopted retroactively for C23.

Some system platforms violate these assumptions and are therefore not Gnuilib porting targets. See [Section 2.4.3 \[Unsupported Platforms\]](#), page 14.

1.6 High Quality

We develop and maintain a testsuite for Gnuilib. The goal is to have a 100% firm interface so that maintainers can feel free to update to the code in git at *any* time and know that their application will not break. This means that before any change can be committed to the repository, a test suite program must be produced that exposes the bug for regression testing.

1.6.1 Stable Branches

In Gnuilib, we don't use topic branches for experimental work. Therefore, occasionally a broken commit may be pushed in Gnuilib. It does not happen often, but it does happen.

To compensate for this, Gnuilib offers “stable branches”. These are branches of the Gnuilib code that are maintained over some longer period (a year, for example) and include

- bug fixes,
- portability enhancements (to existing as well as to new platforms),

- updates to `config.guess` and `config.sub`.

Not included in the stable branches are:

- new features, such as new modules,
- optimizations,
- refactorings,
- complex or risky changes in general,
- updates to `texinfo.tex`,
- documentation updates.

So far, we have nine stable branches:

stable-202601

A stable branch that starts at the beginning of January 2026.

stable-202507

A stable branch that starts at the beginning of July 2025.

stable-202501

A stable branch that starts at the beginning of January 2025. It is no longer updated.

stable-202407

A stable branch that starts at the beginning of July 2024. It is no longer updated.

stable-202401

A stable branch that starts at the beginning of January 2024. It is no longer updated.

stable-202307

A stable branch that starts at the beginning of July 2023. It is no longer updated.

stable-202301

A stable branch that starts at the beginning of January 2023. It is no longer updated.

stable-202207

A stable branch that starts at the beginning of July 2022. It is no longer updated.

stable-202201

A stable branch that starts at the beginning of January 2022. It is no longer updated.

The two use-cases of stable branches are thus:

- You want to protect yourself from occasional breakage in Gnulib.
- When making a bug-fix release of your code, you can incorporate bug fixes in Gnulib, by pulling in the newest commits from the same stable branch that you were already using for the previous release.

1.6.2 Writing reliable code

When compiling and testing Gnulib and Gnulib-using programs, certain compiler options can help improve reliability. First of all, make it a habit to use ‘`-Wall`’ in all compilation commands. Beyond that, the `manywarnings` module enables several forms of static checking in GCC and related compilers (see [Section 19.6 \[manywarnings\]](#), page 960).

For dynamic checking, you can run `configure` with `CFLAGS` options appropriate for your compiler. For example:

```
./configure \
  CPPFLAGS='-Wall'\
  CFLAGS='-g3 -O2'\
  ' -D_FORTIFY_SOURCE=3'\
  ' -D_GLIBCXX_ASSERTIONS'\
  ' -fsanitize=undefined'\
  ' -fsanitize-trap=all'
```

Here:

- `-D_FORTIFY_SOURCE=3` and `-D_GLIBCXX_ASSERTIONS` enable extra security hardening checks in the GNU C library and GNU C++ library, respectively.
- `-fsanitize=undefined` enables GCC’s undefined behavior sanitizer (`ubsan`), and
- `-fsanitize-trap=all` causes `ubsan` to abort the program (through an “illegal instruction” signal). This measure stops exploit attempts and also allows you to debug the issue.

Without the `-fsanitize-trap=all` option, `-fsanitize=undefined` causes messages to be printed, and execution continues after an undefined behavior situation. The message printing causes GCC-like compilers to arrange for the program to dynamically link to libraries it might not otherwise need. With GCC, instead of `-fsanitize-trap=all` you can use the `-static-libubsan` option to arrange for two of the extra libraries (`libstdc++` and `libubsan`) to be linked statically rather than dynamically, though this typically bloats the executable and the remaining extra libraries are still linked dynamically.

It is also good to occasionally run the programs under `valgrind` (see [Section 19.7 \[Running self-tests under valgrind\]](#), page 963).

GCC’s `-fhardened` option can also be used, but with caution because it is designed for production more than testing, and therefore enables `-ftrivial-auto-var-init=zero` which can mask program errors.

1.7 Join the GNU Project

GNU Gnulib is part of the GNU Operating System, developed by the GNU Project.

If you are the author of an awesome program and want to join us in writing Free (libre) Software, please consider making it an official GNU program and become a GNU Maintainer. Instructions on how to do this are [here](https://www.gnu.org/help/evaluation) (<https://www.gnu.org/help/evaluation>). We are looking forward to hacking with you!

Don’t have a program to contribute? Look at all the [other ways to help](https://www.gnu.org/help/help.html) (<https://www.gnu.org/help/help.html>).

And to learn more about Free (libre) Software in general, please read and share [this page \(https://gnu.org/philosophy/free-sw.html\)](https://gnu.org/philosophy/free-sw.html).

2 Philosophy

Gnulib's design and development philosophy is organized around steady, collaborative, and open development of reusable modules that are suitable for a reasonably wide variety of platforms.

2.1 Benefits of using Gnulib

Gnulib is useful to enhance various aspects of a package:

- **Portability:** With Gnulib, a package maintainer can program against the POSIX and GNU libc APIs and nevertheless expect good portability to platforms that don't implement POSIX.
- **Maintainability:** When a package uses modules from Gnulib instead of code written specifically for that package, the maintainer has less code to maintain.
- **Security:** Gnulib provides functions that are immune against vulnerabilities that plague the uses of the corresponding commonplace functions. For example, `asprintf`, `canonicalize_file_name` are not affected by buffer sizing problems that affect `sprintf`, `realpath`. `openat` does not have the race conditions that `open` has. Etc.
- **Reliability:** Gnulib provides functions that combine a call to a system function with a check of the result. Examples are `xalloc`, `xprintf`, `xstrtod`, `xgetcwd`.
- **Structure:** Gnulib offers a way to structure code into modules, typically one include file, one source code file, and one autoconf macro for each functionality. Modularity helps maintainability.

2.2 Library vs. Reusable Code

Classical libraries are installed as binary object code. Gnulib is different: It is used as a source code library. Each package that uses Gnulib thus ships with part of the Gnulib source code. The used portion of Gnulib is tailored to the package: A build tool, called `gnulib-tool`, is provided that copies a tailored subset of Gnulib into the package.

2.2.1 Comparison with other source code libraries

Many programming languages nowadays have

- an ecosystem of reusable source code packages, available through a central site,
- together with a tool that downloads dependencies from this central site.

Here is a table of such source code libraries.

Language	Central site	Download tool
Python	pypi.org	'pip'
Java	mvnrepository.com	'maven', 'gradle'
C#	nuget.org	'nuget'
JavaScript, TypeScript	npmjs.com	'npm', 'yarn'
Common Lisp	quicklisp.org	'ql:quickload'
Emacs Lisp	elpa.gnu.org	'M-x package-install'
Rust	crates.io	'cargo'
Go	pkg.go.dev	'go mod'

Ruby	rubygems.org	<code>'gem'</code>
Lua	luarocks.org	<code>'luarocks'</code>
OCaml	ocaml.org/packages	<code>'opam'</code>
D	code.dlang.org	<code>'dub'</code>
Perl	cpan.org	<code>'cpan'</code>
PHP	packagist.org	<code>'composer'</code>
R	cran.r-project.org	<code>'R CMD INSTALL'</code>

Most of them are vulnerable to *supply chain attacks*. These are cyber-attacks to the central site (library) or to one of the packages in the library, with the goal of attacking the packages that use that specific package as a (direct or indirect) dependency.

Examples of such supply chain attacks:

- A disenchanted package maintainer may replace the code of their package with something that does not work any more.
- An attacker may plant a backdoor in a package.
- An attacker may register packages whose names only slightly differ from well-known and heavily used packages (“typo-squatting”).

While some mitigations exist, they are often cumbersome to put in place. For example, a developer could set up a partial mirror of the library, where adding a package to the mirror is a manual operation, together with a configuration or modification of the tool that redirects download or installation requests to the mirror.

Gnulib is not vulnerable to such attacks, because all of its code is managed in a single repository, with a limited set of committers and with established code review practices.

2.3 Portability and Application Code

One of the goals of Gnulib is to make portable programming easy, on the basis of the standards relevant for GNU (and Unix). The objective behind that is to avoid a fragmentation of the user community into disjoint user communities according to the operating system, and instead allow synergies between users on different operating systems.

Another goal of Gnulib is to provide application code that can be shared between several applications. Some people wonder: "What? glibc doesn't have a function to copy a file?" Indeed, the scope of a system's libc is to implement the relevant standards (ISO C, POSIX) and to provide access functions to the kernel's system calls, and little more.

There is no clear borderline between both areas.

For example, Gnulib has a facility for generating the name of backup files. While this task is entirely at the application level—no standard specifies an API for it—the naïve code has some portability problems because on some platforms the length of file name components is limited to 30 characters or so. Gnulib handles that.

Similarly, Gnulib has a facility for executing a command in a subprocess. It is at the same time a portability enhancement (it works on GNU, Unix, and Windows, compared to the classical `fork/exec` idiom which is not portable to Windows), as well as an application aid: it takes care of redirecting stdin and/or stdout if desired, and emits an error message if the subprocess failed.

2.4 Target Platforms

Gnulib supports a number of platforms that we call the “reasonable portability targets”. This class consists of widespread operating systems, for three years after their last availability, or—for proprietary operating systems—as long as the vendor provides commercial support for it. Already existing Gnulib code for older operating systems is usually left in place for longer than these three years. So it comes that programs that use Gnulib run pretty well also on these older operating systems.

Some operating systems are not very widespread, but are Free Software and are actively developed. Such platforms are also supported by Gnulib, if that OS’s developers community keeps in touch with the Gnulib developers, by providing bug reports, analyses, or patches. For such platforms, Gnulib supports only the versions of the last year or the last few months, depending on the maturity of said OS project, the number of its users, and how often these users upgrade.

Niche operating systems are generally unsupported by Gnulib, unless some of their developers or users contribute support to Gnulib.

The degree of support Gnulib guarantees for a platform depends on the amount of testing it gets from volunteers. Platforms on which Gnulib is frequently tested are the best supported. Then come platforms with occasional testing, then platforms which are rarely tested. Usually, we fix bugs when they are reported. Except that some rarely tested platforms are also low priority; bug fixes for these platforms can take longer.

2.4.1 Supported Platforms

As of 2024, the list of supported platforms is the following:

- glibc systems. With glibc 2.19 or newer, they are frequently tested. About the kernels:
 - glibc on Linux is frequently tested.
 - glibc on kFreeBSD is rarely tested.
 - musl libc on Linux is occasionally tested.
- macOS. In versions 12.5, it’s occasionally tested. In version 10.5, it’s rarely tested.
- FreeBSD 14.0 or newer is occasionally tested.
- OpenBSD 7.0 or newer is occasionally tested.
- NetBSD 10.0 or newer is occasionally tested.
- AIX 7.1 and 7.3 are occasionally tested.
- Solaris 10 and 11.4 are occasionally tested. Solaris 9 is rarely tested and low priority.
- Android is occasionally tested, through the Termux app on Android 11.
- Cygwin 2.9 and 3.4 are occasionally tested. Cygwin 1.7.x no longer tested.
- Native Windows:
 - mingw is occasionally tested. Only the latest version of mingw is tested; older versions are not supported.
 - MSVC 14 (Microsoft Visual Studio 2015 14.0) is occasionally tested. Only “release” builds (compiler option ‘-MD’) are supported, not “debug” builds (compiler option ‘-MDd’).

Note that some modules are currently unsupported on native Windows: `mgetgroups`, `getugroups`, `idcache`, `userspec`, `openpty`, `login_tty`, `forkpty`, `pt_chown`, `grantpt`, `pty-h`, `savewd`, `mkancedirs`, `mkdir-p`, `euidaccess`, `faccessat`. The versions of Windows that are supported are Windows 10 and newer.

- GNU Hurd 0.9 is rarely tested.
- Minix 3.3.0 is no longer tested.
- Haiku is no longer tested.
- uClibc on Linux is no longer tested.
- QNX is no longer tested.

2.4.2 Formerly Supported Platforms

The following platforms were supported in the past, but are no longer supported:

- glibc versions 2.1.x and older.
- Mac OS X 10.4 and older.
- AIX 6 and older.
- HP-UX 11.31.
- IRIX 6.5 and older.
- OSF/1 5.1.
- Solaris 8 and older.
- Interix.
- BeOS.

Gnulib supports these operating systems only in an unvirtualized environment. When you run an OS inside a virtual machine, you have to be aware that the virtual machine can bring in bugs of its own. For example, floating-point operations on Solaris can behave slightly differently in QEMU than on real hardware. And Haiku's `bash` program misbehaves in VirtualBox 3, whereas it behaves fine in VirtualBox 4.

Similarly, running native Windows binaries on GNU/Linux under WINE is rarely tested and low priority: WINE has a set of behaviours and bugs that is slightly different from native Windows.

2.4.3 Unsupported Platforms

Some platforms with C compilers are not supported by Gnulib because the platforms violate Gnulib's C portability assumptions. See [Section 1.5.4 \[Other portability assumptions\]](#), [page 6](#).

These assumptions are not required by the C or POSIX standards but hold on almost all practical porting targets. If you need to port Gnulib code to a platform where these assumptions are not true, we would appreciate hearing of any fixes. We need fixes that do not increase runtime overhead on standard hosts and that are relatively easy to maintain.

These platforms are listed below to illustrate problems that Gnulib and Gnulib-using code would have if it were intended to be portable to all practical POSIX or C platforms.

- In Clang 19 and earlier, the `-fsanitize=undefined` option causes the program to crash if it adds zero to a null pointer – behavior that is undefined in strict C17 and

earlier, but which yields a null pointer in C23 and later and on all practical porting targets, and which the Gnulib portability guidelines allow.

If you use Clang 19 or earlier with `-fsanitize=undefined`, you can work around the problem by also using `-fno-sanitize=pointer-overflow`, although this may also disable some unrelated and useful pointer checks. Perhaps someday the Clang developers will fix the infelicity.

- The IBM i's pointers are 128 bits wide and it lacks the two types `intptr_t` and `uintptr_t`, which are optional in the C and POSIX standards. However, these two types are required for the XSI extension to POSIX, and many Gnulib modules use them. To work around this compatibility problem, Gnulib-using applications can be run on the IBM i's PASE emulation environment. The IBM i's architecture descends from the System/38 (1978).
- The Unisys ClearPath Dorado's machine word is 36 bits. Its signed integers use a ones'-complement representation. On these machines, `CHAR_BIT == 9` and `INT_MIN == -INT_MAX`. By default `UINT_MAX` is $2^{36} - 2$, which does not conform to the C requirement that it be one less than a power of two. Although compiler options can raise `UINT_MAX` to be $2^{36} - 1$, this can break system code that uses `-0` as a flag value. This platform's architecture descends from the UNIVAC 1103 (1953).
- The Unisys ClearPath Libra's machine word is 48 bits with a 4-bit tag and a 4-bit data extension. Its `unsigned int` uses the low-order 40 bits of the word, and `int` uses the low-order 41 bits of the word with a signed-magnitude representation. On these machines, `INT_MAX == UINT_MAX`, `INT_MIN == -INT_MAX`, and `sizeof (int) == 6`. This platform's architecture descends from the Burroughs B5000 (1961).

The following platforms are not supported by Gnulib. The cost of supporting them would exceed the benefit because they are rarely used, or poorly documented, or have been supplanted by other platforms, or diverge too much from POSIX, or some combination of these and other factors. Please don't bother sending us patches for them.

- Windows 95/98/ME.
- DJGPP and EMX (the 32-bit operating systems running in DOS).
- MSDOS (the 16-bit operating system).
- Windows Mobile, Symbian OS, iOS.

2.5 Modules

Gnulib is divided into modules. Every module implements a single facility. Modules can depend on other modules.

A module consists of a number of files and a module description. The files are copied by `gnulib-tool` into the package that will use it, usually verbatim, without changes. Source code files (`.h`, `.c` files) reside in the `lib/` subdirectory. Autoconf macro files reside in the `m4/` subdirectory. Build scripts reside in the `build-aux/` subdirectory.

The module description contains the list of files; `gnulib-tool` copies these files. It contains the module's dependencies; `gnulib-tool` installs them as well. It also contains the autoconf macro invocation (usually a single line or nothing at all); `gnulib-tool` ensures this is invoked from the package's `configure.ac` file. And also a `Makefile.am` snippet; `gnulib-tool` collects these into a `Makefile.am` for the tailored Gnulib part. The module

description and include file specification are for documentation purposes; they are combined into `MODULES.html`.

The module system serves two purposes:

1. It ensures consistency of the used `autoconf` macros and `Makefile.am` rules with the source code. For example, source code which uses the `getopt_long` function—this is a common way to implement parsing of command line options in a way that complies with the GNU standards—needs the source code (`lib/getopt.c` and others), the `autoconf` macro which detects whether the system's `libc` already has this function (in `m4/getopt.m4`), and a few `Makefile.am` lines that create the substitute `getopt.h` if not. These three pieces belong together. They cannot be used without each other. The module description and `gnulib-tool` ensure that they are copied altogether into the destination package.
2. It allows for scalability. It is well-known since the inception of the MODULA-2 language around 1978 that dissection into modules with dependencies allows for building large sets of code in a maintainable way. The maintainability comes from the facts that:
 - Every module has a single purpose; you don't worry about other parts of the program while creating, reading or modifying the code of a module.
 - The code you have to read in order to understand a module is limited to the source of the module and the `.h` files of the modules listed as dependencies. It is for this reason also that we recommend to put the comments describing the functions exported by a module into its `.h` file.

In other words, the module is the elementary unit of code in Gnulib, comparable to a class in object-oriented languages like Java or C#.

The module system is the basis of `gnulib-tool`. When `gnulib-tool` copies a part of Gnulib into a package, it first compiles a module list, starting with the requested modules and adding all the dependencies, and then collects the files, `configure.ac` snippets and `Makefile.am` snippets.

2.6 Various Kinds of Modules

There are modules of various kinds in Gnulib. For a complete list of the modules, see in `MODULES.html`.

2.6.1 Support for ISO C or POSIX functions.

When a function is not implemented by a system, the Gnulib module provides an implementation under the same name. Examples are the `'snprintf'` and `'readlink'` modules.

Similarly, when a function is not correctly implemented by a system, Gnulib provides a replacement. For functions, we use the pattern

```
#if !HAVE_WORKING_FOO
# define foo rpl_foo
#endif
```

and implement the `foo` function under the name `rpl_foo`. This renaming is needed to avoid conflicts at compile time (in case the system header files declare `foo`) and at link/run time (because the code making use of `foo` could end up residing in a shared library, and the executable program using this library could be defining `foo` itself).

For header files, such as `stdint.h`, we provide the substitute only if the system doesn't provide a correct one. The template of this replacement is distributed in a slightly different name, with `.in` inserted before the `.h` extension, so that on systems which do provide a correct header file the system's one is used.

The modules in this category are supported in C++ mode as well. This means, while the autoconfiguration uses the C compiler, the resulting header files and function substitutes can be used with a matching C++ compiler as well.

2.6.2 Enhancements of ISO C or POSIX functions

These are sometimes POSIX functions with GNU extensions also found in glibc—examples: `'getopt'`, `'fnmatch'`—and often new APIs—for example, for all functions that allocate memory in one way or the other, we have variants which also include the error checking against the out-of-memory condition.

2.6.3 Portable general use facilities

Examples are a module for copying a file—the portability problems relate to the copying of the file's modification time, access rights, and extended attributes—or a module for extracting the tail component of a file name—here the portability to native Windows requires a different API than the classical POSIX `basename` function.

2.6.4 Reusable application code

Examples are an error reporting function, a module that allows output of numbers with K/M/G suffixes, or cryptographic facilities.

2.6.5 Object oriented classes

Examples are data structures like `'list'`, or abstract output stream classes that work around the fact that an application cannot implement an stdio `FILE` with its logic. Here, while staying in C, we use implementation techniques like tables of function pointers, known from the C++ language or from the Linux kernel.

2.6.6 Interfaces to external libraries

Examples are the `'iconv'` module, which interfaces to the `iconv` facility, regardless whether it is contained in `libc` or in an external `libiconv`. Or the `'readline'` module, which interfaces to the GNU readline library.

2.6.7 Build / maintenance infrastructure

An example is the `'maintainer-makefile'` module, which provides extra Makefile tags for maintaining a package.

2.7 Collaborative Development

Gnulib is maintained collaboratively. The mailing list is `<bug-gnulib at gnu dot org>`. Be warned that some people on the list may be very active at some times and unresponsive at other times.

Every module has one or more maintainers. While issues are discussed collaboratively on the list, the maintainer of a module nevertheless has a veto right regarding changes in his module.

All patches should be posted to the list, regardless whether they are proposed patches or whether they are committed immediately by the maintainer of the particular module. The purpose is not only to inform the other users of the module, but mainly to allow peer review. It is not uncommon that several people contribute comments or spot bugs after a patch was proposed.

Conversely, if you are using Gnulib, and a patch is posted that affects one of the modules that your package uses, you have an interest in proofreading the patch.

2.8 Copyright

Most modules are under the GPL. Some, mostly modules which can reasonably be used in libraries, are under LGPL. Few modules are under other licenses, such as LGPLv2+, unlimited, or public domain.

If the module description file says "GPL", it means "GPLv3+" (GPLv3 or newer, at the licensee's choice); if it says "LGPL", it means "LGPLv3+" (LGPLv3 or newer, at the licensee's choice).

The source files, more precisely the files in `lib/` and `build-aux/`, are under a license compatible with the module's license. Most often, they are under the same license. But files can be shared among several modules, and in these cases it can happen that a source file is under a weaker license than noted in the module description – namely under the weakest license among the licenses of the modules that contain the file.

Different licenses apply to files in special directories:

`modules/` Module description files are under this copyright:

Copyright © 20XX–20YY Free Software Foundation, Inc.
Copying and distribution of this file, with or without modification, in any medium, are permitted without royalty provided the copyright notice and this notice are preserved. This file is offered as-is, without any warranty.

`m4/` Autoconf macro files are under this copyright:

Copyright © 20XX–20YY Free Software Foundation, Inc.
This file is free software; the Free Software Foundation gives unlimited permission to copy and/or distribute it, with or without modifications, as long as this notice is preserved. This file is offered as-is, without any warranty.

or under this copyright:

Copyright © 20XX–20YY Free Software Foundation, Inc.
This file is free software, distributed under the terms of the GNU General Public License as published by the Free Software Foundation; either version 2 of the License, or (at your option) any later version. As a special exception to the GNU General Public License, this file may be distributed as part of a program that contains a configuration script generated by Autoconf, under the same distribution terms as the rest of that program.

We denote either of these license statements as “unlimited”.

- tests/** If a license statement is not present in a test module, the test files are under GPL. Even if the corresponding source module is under LGPL, this is not a problem, since compiled tests are not installed by “make install”.
- doc/** Documentation files are under this copyright:
- Copyright © 2004–20YY Free Software Foundation, Inc.
 Permission is granted to copy, distribute and/or modify this document under the terms of the GNU Free Documentation License, Version 1.3 or any later version published by the Free Software Foundation; with no Invariant Sections, no Front-Cover Texts, and no Back-Cover Texts. A copy of the license is at <https://www.gnu.org/licenses/fdl-1.3.en.html>.

If you want to use some Gnulib modules under LGPL, you can do so by passing the option ‘--lgpl’ to `gnulib-tool`. This will ensure that all imported modules can be used under the LGPL license. Similarly, if you want some Gnulib modules under LGPLv2+ (Lesser GPL version 2.1 or newer), you can do so by passing the option ‘--lgpl=2’ to `gnulib-tool`.

Keep in mind that when you submit patches to files in Gnulib, you should license them under a compatible license. This means that sometimes the contribution will have to be LGPL, if the original file is available under LGPL. You can find out about it by looking at the license header of the file.

2.9 Steady Development

Gnulib modules are continually adapted, to match new practices, to be consistent with newly added modules, or simply as a response to build failure reports.

If you are willing to report an occasional regression, we recommend to use the newest version from git always, except in periods of major changes. Most Gnulib users do this.

2.10 Openness

Gnulib is open in the sense that we gladly accept contributions if they are generally useful, well engineered, and if the contributors have signed the obligatory papers with the FSF.

The module system is open in the sense that a package using Gnulib can

1. locally patch or override files in Gnulib,
2. locally add modules that are treated like Gnulib modules by `gnulib-tool`.

This is achieved by the ‘--local-dir’ option of `gnulib-tool` (see [Chapter 5 \[Extending Gnulib\]](#), page 48).

3 Invoking gnulib-tool

The `gnulib-tool` command is the recommended way to import Gnulib modules. It is possible to borrow Gnulib modules in a package without using `gnulib-tool`, relying only on the meta-information stored in the `modules/*` files, but with a growing number of modules this becomes tedious. `gnulib-tool` simplifies the management of source files, `Makefile.ams` and `configure.ac` in packages incorporating Gnulib modules.

`gnulib-tool` is not installed in a standard directory that is contained in the `PATH` variable. It needs to be run directly in the directory that contains the Gnulib source code. You can do this either by specifying the absolute filename of `gnulib-tool`, or you can also use a symbolic link from a place inside your `PATH` to the `gnulib-tool` file of your preferred and most up-to-date Gnulib checkout, like this:

```
$ ln -s $HOME/gnu/src/gnulib.git/gnulib-tool $HOME/bin/gnulib-tool
```

Run `'gnulib-tool --help'` for information. To get familiar with `gnulib-tool` without affecting your sources, you can also try some commands with the option `'--dry-run'`; then `gnulib-tool` will only report which actions it would perform in a real run without changing anything.

3.1 Finding modules

There are four ways of finding the names of Gnulib modules that you can use in your package:

- You have the complete module list, sorted according to categories, in <https://www.gnu.org/software/gnulib/MODULES.html>.
- If you are looking for POSIX function replacements that you don't know about yet, follow the procedure described in section [Section 3.7 \[Finding POSIX substitutes\]](#), page 30.
- If you are looking for a particular POSIX header or function replacement, look in the chapters [Chapter 9 \[Header File Substitutes\]](#), page 59, and [Chapter 10 \[Function Substitutes\]](#), page 98. For headers and functions that are provided by Glibc but not standardized by POSIX, look in the chapters [Chapter 12 \[Glibc Header File Substitutes\]](#), page 584, and [Chapter 13 \[Glibc Function Substitutes\]](#), page 597.
- If you have already found the source file in Gnulib and are looking for the module that contains this source file, you can use the command `'gnulib-tool --find filename'`.

3.2 Initial import

Gnulib assumes that your project uses Autoconf. When using Gnulib, you will need to have Autoconf among your build tools.

Gnulib also assumes that your project's `configure.ac` contains the line

```
AC_CONFIG_HEADERS([config.h])
```

The `config.h` file gets generated with platform dependent C macro definitions, and the source files include it (see [Section 3.5 \[Source changes\]](#), page 27).

Unless you use `gnulib-tool`'s `--gnu-make` option, Gnulib also assumes that your project uses Automake at least in a subdirectory of your project. While the use of Automake in

your project's top level directory is an easy way to fulfil the Makefile conventions of the GNU coding standards, Gnuilib does not require it.

Invoking 'gnuilib-tool --import' will copy source files, create a `Makefile.am` to build them, generate a file `gnuilib-comp.m4` with Autoconf M4 macro declarations used by `configure.ac`, and generate a file `gnuilib-cache.m4` containing the cached specification of how Gnuilib is used.

Our example will be a library that uses Autoconf, Automake and Libtool. It calls `strdup`, and you wish to use gnuilib to make the package portable to C99 and C11 (which don't have `strdup`).

```
~/src/libfoo$ gnuilib-tool --import strdup
```

```
Module list with included dependencies:
```

```
absolute-header
```

```
extensions
```

```
strdup
```

```
string
```

```
File list:
```

```
lib/dummy.c
```

```
lib/strdup.c
```

```
lib/string.in.h
```

```
m4/absolute-header.m4
```

```
m4/extensions.m4
```

```
m4/gnuilib-common.m4
```

```
m4/strdup.m4
```

```
m4/string_h.m4
```

```
Creating directory ./lib
```

```
Creating directory ./m4
```

```
Copying file lib/dummy.c
```

```
Copying file lib/strdup.c
```

```
Copying file lib/string.in.h
```

```
Copying file m4/absolute-header.m4
```

```
Copying file m4/extensions.m4
```

```
Copying file m4/gnuilib-common.m4
```

```
Copying file m4/gnuilib-tool.m4
```

```
Copying file m4/strdup.m4
```

```
Copying file m4/string_h.m4
```

```
Creating lib/Makefile.am
```

```
Creating m4/gnuilib-cache.m4
```

```
Creating m4/gnuilib-comp.m4
```

```
Finished.
```

You may need to add `#include` directives for the following `.h` files.

```
#include <string.h>
```

Don't forget to

- add "lib/Makefile" to `AC_CONFIG_FILES` in `./configure.ac`,
- mention "lib" in `SUBDIRS` in `Makefile.am`,

```

- mention "-I m4" in ACLOCAL_AMFLAGS in Makefile.am,
  or add an AC_CONFIG_MACRO_DIRS([m4]) invocation in ./configure.ac,
- invoke gl_EARLY in ./configure.ac, right after AC_PROG_CC,
- invoke gl_INIT in ./configure.ac.
~/src/libfoo$

```

By default, the source code is copied into `lib/` and the M4 macros in `m4/`. You can override these paths by using `--source-base=DIRECTORY` and `--m4-base=DIRECTORY`. Some modules also provide other files necessary for building. These files are copied into the directory specified by `'AC_CONFIG_AUX_DIR'` in `configure.ac` or by the `--aux-dir=DIRECTORY` option. If neither is specified, the current directory is assumed.

`gnuilib-tool` can make symbolic links instead of copying the source files. The option to specify for this is `'--symlink'`, or `'-s'` for short. This can be useful to save a few kilobytes of disk space. But it is likely to introduce bugs when `gnuilib` is updated; it is more reliable to use `'gnuilib-tool --update'` (see below) to update to newer versions of `gnuilib`. Furthermore it requires extra effort to create self-contained tarballs, and it may disturb some mechanism the maintainer applies to the sources. For these reasons, this option is generally discouraged.

`gnuilib-tool` will overwrite any preexisting files, in particular `Makefile.am`. It is also possible to separate the generated `Makefile.am` content (for building the `gnuilib` library) into a separate file, say `gnuilib.mk`, that can be included by your handwritten `Makefile.am`, but this is a more advanced use of `gnuilib-tool`.

Consequently, it is a good idea to choose directories that are not already used by your projects, to separate `gnuilib` imported files from your own files. This approach is also useful if you want to avoid conflicts between other tools (e.g., `gettextize` that also copy M4 files into your package. Simon Josefsson successfully uses a source base of `gl/`, and a M4 base of `gl/m4/`, in several packages.

After the `'--import'` option on the command line comes the list of Gnuilib modules that you want to incorporate in your package. The names of the modules coincide with the filenames in Gnuilib's `modules/` directory.

Some Gnuilib modules depend on other Gnuilib modules. `gnuilib-tool` will automatically add the needed modules as well; you need not list them explicitly. `gnuilib-tool` will also memorize which dependent modules it has added, so that when someday a dependency is dropped, the implicitly added module is dropped as well (unless you have explicitly requested that module).

If you want to cut a dependency, i.e., not add a module although one of your requested modules depends on it, you may use the option `'--avoid=module'` to do so. Multiple uses of this option are possible. Of course, you will then need to implement the same interface as the removed module.

A few manual steps are required to finish the initial import. `gnuilib-tool` printed a summary of these steps.

First, you must ensure Autoconf can find the macro definitions in `gnuilib-comp.m4`. Use the `ACLOCAL_AMFLAGS` specifier in your top-level `Makefile.am` file, as in:

```
ACLOCAL_AMFLAGS = -I m4
```

Alternatively, add an `AC_CONFIG_MACRO_DIRS` invocation in your `configure.ac` file, as in:

```
AC_CONFIG_MACRO_DIRS([m4])
```

You are now ready to call the M4 macros in `gnuilib-comp.m4` from `configure.ac`. The macro `gl_EARLY` must be called as soon as possible after verifying that the C compiler is working. Typically, this is immediately after `AC_PROG_CC`, as in:

```
...
AC_PROG_CC
gl_EARLY
...
```

The core part of the gnuilib checks are done by the macro `gl_INIT`. Place it further down in the file, typically where you normally check for header files or functions. It must come after other checks which may affect the compiler invocation, such as `AC_MINIX`. For example:

```
...
# For gnuilib.
gl_INIT
...
```

`gl_INIT` will in turn call the macros related with the gnuilib functions, be it specific gnuilib macros, like `gl_FUNC_ALLOCA` or Autoconf or Automake macros like `AC_FUNC_ALLOCA` or `AM_FUNC_GETLINE`. So there is no need to call those macros yourself when you use the corresponding gnuilib modules.

You must also make sure that the gnuilib library is built. Add the `Makefile` in the gnuilib source base directory to `AC_CONFIG_FILES`, as in:

```
AC_CONFIG_FILES(... lib/Makefile ...)
```

You must also make sure that `make` will recurse into the gnuilib directory. To achieve this, add the gnuilib source base directory to a `SUBDIRS` `Makefile.am` statement, as in:

```
SUBDIRS = lib
```

or if you, more likely, already have a few entries in `SUBDIRS`, you can add something like:

```
SUBDIRS += lib
```

Finally, you have to add compiler and linker flags in the appropriate source directories, so that you can make use of the gnuilib library. Since some modules (`'getopt'`, for example) may copy files into the build directory, `top_builddir/lib` is needed as well as `top_srcdir/lib`. For example:

```
...
AM_CPPFLAGS = -I$(top_builddir)/lib -I$(top_srcdir)/lib
...
LDADD = lib/libgnu.a
...
```

Don't forget to `#include` the various header files. In this example, you would need to make sure that `'#include <string.h>'` is evaluated when compiling all source code files, that want to make use of `strdup`.

In the usual case where Autoconf is creating a `config.h` file, you should include `config.h` first, before any other include file. That way, for example, if `config.h` defines `'restrict'` to be the empty string on a non-C99 host, or a macro like `'_FILE_OFFSET_BITS'` that affects

the layout of data structures, the definition is consistent for all include files. Also, on some platforms macros like ‘_FILE_OFFSET_BITS’ and ‘_GNU_SOURCE’ may be ineffective, or may have only a limited effect, if defined after the first system header file is included.

Finally, note that you cannot use `AC_LIBOBJ` or `AC_REPLACE_FUNCS` in your `configure.ac` and expect the resulting object files to be automatically added to `lib/libgnu.a`. This is because your `AC_LIBOBJ` and `AC_REPLACE_FUNCS` invocations from `configure.ac` augment a variable `@LIBOBJS@` (and/or `@LTLIBOBJS@` if using Libtool), whereas `lib/libgnu.a` is built from the contents of a different variable, usually `@gl_LIBOBJS@` (or `@gl_LTLIBOBJS@` if using Libtool).

3.3 Modified imports

You can at any moment decide to use Gnuilib differently than the last time.

There are two ways to change how Gnuilib is used. Which one you’ll use, depends on where you keep track of options and module names that you pass to `gnuilib-tool`.

- If you store the options and module names in a file under your own control, such as `autogen.sh`, `bootstrap`, `bootstrap.conf`, or similar, simply invoke `gnuilib-tool` again, with modified options and more or fewer module names.
- `gnuilib-tool` remembers which modules were used last time. If you want to rely on `gnuilib-tool`’s own memory of the last used options and module names, you can use the commands `gnuilib-tool --add-import` and `gnuilib-tool --remove-import`.

So, if you only want to use more Gnuilib modules, simply invoke `gnuilib-tool --add-import new-modules`. The list of modules that you pass after ‘`--add-import`’ is *added* to the previous list of modules.

Similarly, if you want to use fewer Gnuilib modules, simply invoke `gnuilib-tool --remove-import unneeded-modules`. The list of modules that you pass after ‘`--remove-import`’ is *removed* from the previous list of modules. Note that if a module is then still needed as dependency of other modules, it will be used nevertheless. If you want to *really* not use a module any more, regardless of whether other modules may need it, you need to use the ‘`--avoid`’ option.

For other changes, such as different choices of ‘`--lib`’, ‘`--source-base`’ or ‘`--aux-dir`’, the normal way is to modify manually the file `gnuilib-cache.m4` in the M4 macros directory, then launch ‘`gnuilib-tool --add-import`’.

The only change for which this doesn’t work is a change of the ‘`--m4-base`’ directory. Because, when you pass a different value of ‘`--m4-base`’, `gnuilib-tool` will not find the previous `gnuilib-cache.m4` file any more. A possible solution is to manually copy the `gnuilib-cache.m4` into the new M4 macro directory.

In the `gnuilib-cache.m4` file, the macros have the following meaning:

`gl_LOCAL_DIR`

The argument is a colon separated list of local directories where `gnuilib-tool` will search before looking at gnuilib’s directory. Corresponds to the ‘`--local-dir`’ command line argument.

`gl_MODULES`

The argument is a space separated list of the requested modules, not including dependencies.

gl_WITH_OBSOLETE

The presence of this macro corresponds to the ‘**--with-obsolete**’ command line argument. It takes no arguments.

gl_WITH_CXX_TESTS

The presence of this macro corresponds to the ‘**--with-c++-tests**’ command line argument and to the absence of the ‘**--without-c++-tests**’ command line argument. It takes no arguments.

gl_WITH_LONGRUNNING_TESTS

The presence of this macro corresponds to the ‘**--with-longrunning-tests**’ command line argument and to the absence of the ‘**--without-longrunning-tests**’ command line argument. It takes no arguments.

gl_WITH_PRIVILEGED_TESTS

The presence of this macro corresponds to the ‘**--with-longrunning-tests**’ command line argument and to the absence of the ‘**--without-longrunning-tests**’ command line argument. It takes no arguments.

gl_WITH_UNPORTABLE_TESTS

The presence of this macro corresponds to the ‘**--with-unportable-tests**’ command line argument and to the absence of the ‘**--without-unportable-tests**’ command line argument. It takes no arguments.

gl_WITH_ALL_TESTS

The presence of this macro corresponds to the ‘**--with-all-tests**’ command line argument. It takes no arguments.

gl_AVOID The argument is a space separated list of modules that should not be used, even if they occur as dependencies. Corresponds to the ‘**--avoid**’ command line argument.

gl_SOURCE_BASE

The argument is the relative file name of the directory containing the gnulib source files (mostly *.c and *.h files). Corresponds to the ‘**--source-base**’ command line argument.

gl_M4_BASE

The argument is the relative file name of the directory containing the gnulib M4 macros (*.m4 files). Corresponds to the ‘**--m4-base**’ command line argument.

gl_PO_BASE

The argument is the relative file name of the directory containing *.po files. Corresponds to the ‘**--po-base**’ command line argument.

gl_DOC_BASE

The argument is the relative file name of the directory containing documentation files. Corresponds to the ‘**--doc-base**’ command line argument.

gl_TESTS_BASE

The argument is the relative file name of the directory containing the gnulib unit test files. Corresponds to the ‘**--tests-base**’ command line argument.

- gl_WITH_TESTS**
The presence of this macro corresponds to the ‘**--with-tests**’ command line argument. It takes no arguments.
- gl_LIB**
The argument is the name of the library to be created. Corresponds to the ‘**--lib**’ command line argument.
- gl_LGPL**
The presence of this macro without arguments corresponds to the ‘**--lgpl**’ command line argument. The presence of this macro with an argument (whose value must be 2 or 3) corresponds to the ‘**--lgpl=arg**’ command line argument.
- gl_GPL**
The presence of this macro with an argument (whose value must be 2 or 3) corresponds to the ‘**--gpl=arg**’ command line argument.
- gl_MAKEFILE_NAME**
The argument is the name of the makefile in the source-base and tests-base directories. Corresponds to the ‘**--makefile-name**’ command line argument.
- gl_TESTS_MAKEFILE_NAME**
The argument is the name of the makefile in the tests-base directory. Corresponds to the ‘**--tests-makefile-name**’ command line argument.
- gl_AUTOMAKE_SUBDIR**
The presence of this macro corresponds to the ‘**--automake-subdir**’ command line argument. It takes no arguments.
- gl_CONDITIONAL_DEPENDENCIES**
The presence of this macro corresponds to the ‘**--conditional-dependencies**’ command line argument and to the absence of the ‘**--no-conditional-dependencies**’ command line argument. It takes no arguments.
- gl_LIBTOOL**
The presence of this macro corresponds to the ‘**--libtool**’ command line argument and to the absence of the ‘**--no-libtool**’ command line argument. It takes no arguments.
- gl_MACRO_PREFIX**
The argument is the prefix to use for macros in the `gnulib-comp.m4` file. Corresponds to the ‘**--macro-prefix**’ command line argument.
- gl_PO_DOMAIN**
The argument is the prefix of the i18n domain, typically matching the package name. Corresponds to the ‘**--po-domain**’ command line argument.
- gl_WITNESS_C_MACRO**
The argument is the C macro that is defined when the sources in this directory are compiled or used. Corresponds to the ‘**--witness-c-macro**’ command line argument.
- gl_VC_FILES**
The argument to this macro is **true** or **false**. The former corresponds to the ‘**--vc-files**’ command line argument and the latter corresponds to the ‘**--no-vc-files**’ command line argument.

3.4 Simple update

When you want to update to a more recent version of Gnulib, without changing the list of modules or other parameters, a simple call does it:

```
$ gnulib-tool --add-import
```

This will create, update or remove files, as needed.

Note: From time to time, changes are made in Gnulib that are not backward compatible. When updating to a more recent Gnulib, you should consult Gnulib's `NEWS` file to check whether the incompatible changes affect your project.

3.5 Changing your sources for use with Gnulib

When you use Gnulib, you need to make some small changes to your source code base.

3.5.1 -I options

Gnulib contains some header file overrides. This means that when building on systems with deficient header files in `/usr/include/`, it may create files named `string.h`, `stdlib.h`, `stdint.h` or similar in the build directory. In the other source directories of your package you will usually pass `-I` options to the compiler, so that these Gnulib substitutes are visible and take precedence over the files in `/usr/include/`.

3.5.2 Include `<config.h>`

These Gnulib substitute header files rely on `<config.h>` being already included. Furthermore `<config.h>` must be the first include in every compilation unit. This means that to *all your source files* and likely also to *all your tests source files* you need to add an `#include <config.h>` at the top. Which source files are affected? Exactly those whose compilation includes a `-I` option that refers to the Gnulib library directory.

This is annoying, but inevitable: On many systems, `<config.h>` is used to set system dependent flags (such as `_GNU_SOURCE` on GNU systems), and these flags have no effect after any system header file has been included.

3.5.3 Style of `#include` statements

When including including specific header files, you need to use the `#include <...>` syntax, not the `#include "..."` syntax. This is true for the following POSIX or ISO C standardized header files:

```
arpa/inet.h
assert.h
ctype.h
dirent.h
endian.h
errno.h
fcntl.h
fenv.h
float.h
```

fnmatch.h
glob.h
iconv.h
inttypes.h
langinfo.h
limits.h
locale.h
math.h
monetary.h
net/if.h
netdb.h
netinet/in.h
poll.h
pthread.h
sched.h
search.h
signal.h
spawn.h
stdalign.h
stdarg.h
stdcountof.h
stddef.h
stdint.h
stdio.h
stdlib.h
string.h
strings.h
sys/msg.h
sys/resource.h
sys/select.h
sys/sem.h
sys/shm.h
sys/socket.h
sys/stat.h
sys/time.h
sys/times.h
sys/types.h
sys/uio.h
sys/un.h


```
sys/utsname.h
sys/wait.h
termios.h
threads.h
time.h
uchar.h
unistd.h
utime.h
wchar.h
wctype.h
```

as well as for the following header files that exist in the GNU C library but are not standardized:

```
alloca.h
byteswap.h
error.h
getopt.h
malloc.h
mntent.h
omp.h
pty.h
selinux/selinux.h
sys/file.h
sys/ioctl.h
sys/random.h
sysexits.h
utmp.h
```

The reason for this requirement is that for these header files, the Gnullib override uses `#include_next` to include the system-provided header of the same name, and `#include_next` may not work right with the `'#include "...'"` syntax.

3.6 Changing your link commands for use with Gnullib

When you use Gnullib, you need to augment the set of libraries against which your programs and libraries are linked. This is done by augmenting the Automake variable `LDADD` (for all programs) or `prog_LDADD` (for a single program *prog*) or `library_la_LIBADD` (for a single library *library*.la).

What do you need to add to this Automake variable?

1. The reference to the Gnullib library. In the example of section [Section 3.2 \[Initial import\]](#), [page 20](#), this would be `lib/libgnu.a` for source in the top-level directory, or `../lib/libgnu.a` for source in a sibling directory of `lib/`.

2. References to additional libraries, brought in by some of the Gnulib modules that you use (directly or indirectly). The complete list of such libraries is printed when you invoke `gnulib-tool`. Alternatively, you can retrieve the set of additional libraries required by a specific Gnulib module by running

```
./gnulib-tool --extract-recursive-link-directive module
```

Beware: By looking into the module description file `modules/module` or by running

```
./gnulib-tool --extract-link-directive module
```

you would miss the link dependencies of indirectly used modules.

3.7 Finding recommended ISO C and POSIX function substitutes

Gnulib contains a wealth of portability workarounds for ISO C and POSIX functions. They are listed in detail in the chapter [Chapter 10 \[Function Substitutes\], page 98](#). If you want to know which function substitutes are recommended for your package, you can search your source code for ISO C and POSIX functions that it uses and read the corresponding sections of said documentation chapter. But this is a tedious task. Here is an alternative approach that makes this task easier.

1. Add the Gnulib module ‘`posixcheck`’ to the Gnulib imports of your package, as described earlier in this chapter.
2. Do a `make distclean` if you previously built in the top-level directory. Then regenerate the Autotools-generated parts of the package.
3. On a glibc system, build your package. Pay attention to the compiler warnings. Warnings are generated for uses of ISO C and POSIX functions that have portability problems or other important pitfalls and for which you have not yet imported the corresponding Gnulib module. If you get, say, a warning “warning: call to ‘close’ declared with attribute warning: close does not portably work on sockets - use gnulib module close for portability”, put ‘`close`’ on your list of modules to import.
4. Add the modules you noted to the Gnulib imports of your package.
5. Optionally, you can do the same steps again, and make sure that there are no warnings left except those that you want to intentionally ignore.
6. Finally, remove the Gnulib module ‘`posixcheck`’ from the Gnulib imports, and run `make distclean`.

3.8 Modifying the build rules of a Gnulib import directory

In some cases, you may want to set additional compiler options for use within the Gnulib import directory. For example, the ‘`relocatable`’ module operates better if you define the C macros `ENABLE_COSTLY_RELOCATABLE` and `INSTALLDIR` during its compilation.

There are two ways to do so: Use of the `gnulib-tool` option `--makefile-name`, and a kitchen-sink module.

With the `gnulib-tool` option `--makefile-name`, you are telling `gnulib-tool` to generate an includable `Makefile.am` portion, rather than a self-contained `Makefile.am`. For example, when you use `--makefile-name=Makefile.gnulib`, `gnulib-tool` will

generate `Makefile.gnuilib`, and you will provide a hand-written `Makefile.am` that includes `Makefile.gnuilib` through a line such as

```
include Makefile.gnuilib
```

Before this include, you need to initialize a set of `Makefile.am` variables.

For the directory into which `gnuilib-tool` copies the gnuilib source files, the set of variables to initialize is:

- `AUTOMAKE_OPTIONS`
- `SUBDIRS`
- `noinst_HEADERS`
- `noinst_LIBRARIES`
- `noinst_LTLIBRARIES`
- `pkgdata_DATA` (only with Automake $\geq$ 1.11.4)
- `EXTRA_DIST`
- `BUILT_SOURCES`
- `SUFFIXES`
- `MOSTLYCLEANFILES`
- `MOSTLYCLEANDIRS`
- `CLEANFILES`
- `DISTCLEANFILES`
- `MAINTAINERCLEANFILES`
- `AM_CPPFLAGS`
- `AM_CFLAGS`

`AUTOMAKE_OPTIONS` should be initialized as described in [Section “Changing Automake’s Behavior” in *GNU Automake*](#). The other variables can be initialized to empty. However, you will most likely want to initialize some of them with non-empty values, in order to achieve the desired customization.

For the directory into which `gnuilib-tool` copies the gnuilib unit test files (when option ‘`--with-tests`’ is used), the set of variables to initialize is:

- `AM_CFLAGS`
- `AM_CXXFLAGS` (when this directory contains some C++ code)

The other approach, the kitchen-sink module, is more advanced. See chapter [Chapter 5 \[Extending Gnuilib\]](#), page 48.

3.9 Building directly from the top-level directory

By default, the Gnuilib import directory will contain a generated `Makefile.am` file. After configuring, this produces a generated `Makefile` in this directory. As a consequence, the build from the top-level directory will use a recursive `make` invocation for this directory.

Some people prefer a build system where the `Makefile` in the top-level directory directly builds the artifacts in the subdirectories, without an intermediate `make` invocation. This is called “non-recursive make” and is supported by Automake. For more details, see <https://autotools.io/automake/nonrecursive.html>.

Gnuilib supports this flavour of build system too. To use it, pass two options to `gnuilib-tool`: `--makefile-name` and `--automake-subdir`.

With the `gnuilib-tool` option `--makefile-name`, you are telling `gnuilib-tool` to generate an includable `Makefile.am` portion in the Gnuilib import directory, rather than a self-contained `Makefile.am`. For example, when you use `--makefile-name=Makefile.gnuilib`, `gnuilib-tool` will generate `Makefile.gnuilib`.

With the option `--automake-subdir`, you are telling `gnuilib-tool` that you will include the generated file from the `Makefile.am` in the top-level directory, rather than from a `Makefile.am` in the same directory. For example, the top-level `Makefile.am` might contain this directive:

```
include lib/Makefile.gnuilib
```

The option `--automake-subdir` is also supported in combination with `--with-tests` (see [Section 3.14 \[Unit tests\], page 37](#)). Note that in this case, however, the generated unit tests directory will contain a `Makefile.am` and thus use a recursive `make` invocation. This is not a problem, since the built artifacts of your package have no dependencies towards the Gnuilib unit tests, nor vice versa.

3.10 Using Gnuilib for both a library and a program

Your project might build both a library and some accompanying programs in the same source tree. In that case you might want to use different modules for the library than for the programs. Typically the programs might want to make use of `getopt-posix` or `version-etc`, while the library wants to stay clear of these modules for technical or licensing reasons.

Let's assume that your project contains a `lib` directory where the source of the library resides and a `src` directory for the sources of the programs as follows.

```
.
|-- configure.ac
|-- lib
|   |-- foo.c
|   `-- Makefile.am
|-- Makefile.am
`-- src
    |-- bar.c
    `-- Makefile.am
```

You can now add two instances of Gnuilib to your project in separate source trees:

```
~/src/libfoo$ gnuilib-tool --import --lib=libgnu --source-base=gnuilib \
               --m4-base=gnuilib/m4 --macro-prefix=gl strndup
~/src/libfoo$ gnuilib-tool --import --lib=libgnutools \
               --source-base=src/gnuilib --m4-base=src/gnuilib/m4 \
               --macro-prefix=gl_tools getopt-gnu
```

The first one will import the module `strndup` in `gnuilib` and the second one will import `getopt-gnu` in `src/gnuilib` and you will end up with the following source tree (many files omitted in the interest of brevity):

```
.
|-- configure.ac
```

```

|-- gnuilib
|   |-- m4
|   |-- strdup.c
|-- lib
|   |-- foo.c
|   `-- Makefile.am
|-- Makefile.am
`-- src
    |-- bar.c
    |-- gnuilib
    |   |-- getopt.c
    |   |-- getopt.in.h
    |   |-- m4
    `-- Makefile.am

```

As discussed in [Section 3.14 \[Unit tests\]](#), [page 37](#), you may not use ‘`--with-tests`’ for this project since the `configure.ac` is shared.

Integration with your code is basically the same as outlined in [Section 3.2 \[Initial import\]](#), [page 20](#), with the one exception that you have to add both the macro `gl_EARLY` and the macro `gl_tools_EARLY` to your `configure.ac` (and of course also both macros `gl_INIT` and `gl_tools_INIT`). Obviously the name of the second macro is dependent on the value of the `--macro-prefix` option in your `gnuilib-tool` invocation.

```

...
AC_PROG_CC
gl_EARLY
gl_tools_EARLY
...
# For gnuilib.
gl_INIT
gl_tools_INIT
...

```

Also as outlined in [Section 3.2 \[Initial import\]](#), [page 20](#), you will have to add compiler and linker flags. For the library you might have to add something along the line of the following to your `Makefile.am`:

```

...
AM_CPPFLAGS = -I$(top_srcdir)/gnuilib -I$(top_builddir)/gnuilib
...
libfoo_la_LIBADD = $(top_builddir)/gnuilib/libgnu.la
...

```

Correspondingly for the programs you will have to add something like this:

```

...
AM_CPPFLAGS = -I$(top_srcdir)/src/gnuilib -I$(top_builddir)/src/gnuilib
...
LDADD = $(top_builddir)/src/gnuilib/libgnutools.la
...

```

The name of the library that you have pass in the linker option depends on the `--lib` option in `gnullib-tool` invocation.

3.11 Caveat: gettextize and autopoint users

The programs `gettextize` and `autopoint`, part of GNU `gettext`, import or update the internationalization infrastructure. Some of this infrastructure, namely ca. 20 Autoconf macro files and the `config.rpath` file, is also contained in Gnullib and may be imported by `gnullib-tool`. The use of `gettextize` or `autopoint` will therefore overwrite some of the files that `gnullib-tool` has imported, and vice versa.

Avoiding to use `gettextize` (manually, as package maintainer) or `autopoint` (as part of a script like `autoreconf` or `autogen.sh`) is not the solution: These programs also import the infrastructure in the `po/` and optionally in the `intl/` directory.

The copies of the conflicting files in Gnullib are more up-to-date than the copies brought in by `gettextize` and `autopoint`. When a new `gettext` release is made, the copies of the files in Gnullib will be updated immediately.

The choice of which version of `gettext` to require depends on the needs of your package. For a package that wants to comply to GNU Coding Standards, the steps are:

1. When you run `gettextize`, always use the `gettextize` from the matching GNU `gettext` release. For the most recent Gnullib checkout, this is the newest release found on <https://ftp.gnu.org/gnu/gettext/>. For an older Gnullib snapshot, it is the release that was the most recent release at the time the Gnullib snapshot was taken.
2. After running `gettextize`, invoke `gnullib-tool` and import the `gettext` module. Also, copy the latest version of gnullib's `build-aux/po/Makefile.in.in` to your `po/` directory (this is done for you if you use gnullib's `autogen.sh` script).
3. If you get an error message like `*** error: gettext infrastructure mismatch: using a Makefile.in.in from gettext version ... but the Autoconf macros are from gettext version ...`, it means that a new GNU `gettext` release was made, and its Autoconf macros were integrated into Gnullib and now mismatch the `po/` infrastructure. In this case, fetch and install the new GNU `gettext` release and run `gettextize` followed by `gnullib-tool`.

On the other hand, if your package is not as concerned with compliance to the latest standards, but instead favors development on stable environments, the steps are:

1. Determine the oldest version of `gettext` that you intend to support during development (at this time, gnullib recommends going no older than version 0.17). Run `autopoint` (not `gettextize`) to copy infrastructure into place (newer versions of `gettext` will install the older infrastructure that you requested).
2. Invoke `gnullib-tool`, and import the `gettext-h` module.

Regardless of which approach you used to get the infrastructure in place, the following steps must then be used to preserve that infrastructure (gnullib's `autogen.sh` script follows these rules):

1. When a script of yours run `autopoint`, invoke `gnullib-tool` afterwards.
2. When you invoke `autoreconf` after `gnullib-tool`, make sure to not invoke `autopoint` a second time, by setting the `AUTOPOINT` environment variable, like this:

```
$ env AUTOPOINT=true autoreconf --install
```

3.12 Handling Gnullib’s own message translations

Gnullib provides some functions that emit translatable messages using GNU `gettext`. The ‘gnullib’ domain at the [Translation Project \(https://translationproject.org/\)](https://translationproject.org/) collects translations of these messages, which you should incorporate into your own programs.

The recommended way to achieve this is that your package depends on the package ‘gnullib-110n’. This package contains the Gnullib localizations as `.mo` files.

In order to use this method, you must – in each program that might use Gnullib code – add an extra line to the part of the program that initializes locale-dependent behavior. Where you would normally write something like:

```
setlocale (LC_ALL, "");
bindtextdomain (PACKAGE, LOCALEDIR);
textdomain (PACKAGE);
```

you should add an additional `bindtextdomain` call to inform `gettext` of where the MO files for the extra message domain may be found:

```
bindtextdomain ("gnullib", GNULIB_LOCALEDIR);
```

Since you do not change the `textdomain` call, the default message domain for your program remains the same and your own use of `gettext` functions will not be affected.

If your package has a `DEPENDENCIES` file, for the ease of installers and distributors, you should add the ‘gnullib-110n’ dependency in this file.

If your package does not support localization, you can simplify its build process by using `gnullib-tool`’s `--avoid=gnullib-i18n` option. The package need not call `bindtextdomain` or `textdomain`, and all Gnullib-generated messages will be in American English.

Two older methods to use Gnullib localizations are deprecated:

- The first, and older, method is to list all the source files you use from Gnullib in your own `po/POTFILES.in` file. This will cause all the relevant translatable strings to be included in your POT file. When you send this POT file to the Translation Project, translators will normally fill in the translations of the Gnullib strings from their “translation memory”, and send you back updated PO files. For this to work, your program should define the macro `GNULIB_TEXT_DOMAIN` to its own `textdomain` name early, before including `gettext.h`; you can do this by putting `AC_DEFINE([GNULIB_TEXT_DOMAIN], [PACKAGE])` in your `configure.ac`.

However, this process is error-prone: you might forget to list some source files, or the translator might not be using a translation memory and provide a different translation than another translator, or the translation might not be kept in sync between Gnullib and your package. It is also slow and causes substantial extra work, because a human translator must be in the loop for each language and you will need to incorporate their work on request.

- The second method is that you pass the `--po-base=directory` and `--po-domain=domain` options to `gnullib-tool`, then `gnullib-tool` creates a separate directory with its own `POTFILES.in`, and fetches current translations directly from the Translation Project (using `wget`). The POT file in this directory will be called `domain-gnullib.pot`, depending on the `domain` you gave to the `--po-domain` option (typically the same as the package name). This causes these translations to reside in a separate message domain, so that they do not clash either with the translations for

the main part of your package nor with those of other packages on the system that use possibly different versions of Gnuilib. When you use these options, the functions in Gnuilib are built in such a way that they will always use this domain regardless of the default domain set by `textdomain`.

This method is deprecated because it adds complexity to your package and, additionally, mixes the “autopull” and “autogen” phases of package source preparation.

3.13 Integration with Version Control Systems

If a project stores its source files in a version control system (VCS), such as CVS, Subversion, or Git, one needs to decide which files to commit.

In principle, all files created by `gnuilib-tool`, except `gnuilib-cache.m4`, can be treated like generated source files, like for example a `parser.c` file generated from `parser.y`. Alternatively, they can be considered source files and updated manually.

Here are the three different approaches in common use. Each has its place, and you should use whichever best suits your particular project and development methods.

1. In projects which commit all source files, whether generated or not, into their VCS, the `gnuilib-tool` generated files should all be committed. In this case, you should pass the option ‘`--no-vc-files`’ to `gnuilib-tool`, which avoids alteration of VCS-related files such as `.gitignore`.

Gnuilib also contains files generated by `make` (and removed by `make clean`), using information determined by `configure`. For a Gnuilib source file of the form `lib/foo.in.h`, the corresponding `lib/foo.h` is such a `make`-generated file. These should *not* be checked into the VCS, but instead added to `.gitignore` or equivalent.

2. In projects which customarily omit from their VCS all files that are generated from other source files, none of these files and directories are added into the VCS. As described in [Section 3.3 \[Modified imports\]](#), page 24, there are two ways to keep track of options and module names that are passed to `gnuilib-tool`. The command for restoring the omitted files depends on it:
 - If they are stored in a file other than `gnuilib-cache.m4`, such as `autogen.sh`, `bootstrap`, `bootstrap.conf`, or similar, the restoration command is the entire `gnuilib-tool ... --import ...` invocation with all options and module names.
 - If the project relies on `gnuilib-tool`’s memory of the last used options and module names, then the file `gnuilib-cache.m4` in the M4 macros directory must be added to the VCS, and the restoration command is:

```
$ gnuilib-tool --update
```

The ‘`--update`’ option operates much like the ‘`--add-import`’ option, but it does not offer the possibility to change the way Gnuilib is used. Also it does not report in the `ChangeLogs` the files that it had to add because they were missing.

Most packages nowadays use the first among these two approaches. Over time, three ways of handling version control have evolved.

In the cases (A) and (B), a “git submodule” is used to reference the precise commit of the gnuilib repository, so that each developer running ‘`./bootstrap --pull`’ or `autopull.sh` will get the same version of all gnuilib-provided files.

The alternative is to always follow the newest Gnulib automatically. Note that this can cause breakages at unexpected moments, namely when a broken commit is pushed in Gnulib. It does not happen often, but it does happen.

- (A) In this approach, the developers use a git submodule manually.

The location of the submodule can be chosen to fit the package’s needs; here’s how to initially create the submodule in the directory **gnulib**:

```
$ git submodule add -- https://git.savannah.gnu.org/git/gnulib.git gnulib
```

Thereafter, the developer will run this command to update the submodule to the recorded checkout level:

```
$ git submodule update --init gnulib
```

Use this sequence to update to a newer version of gnulib:

```
$ git submodule update --remote gnulib
$ git add gnulib
$ ./bootstrap --bootstrap-sync
```

If multiple submodules are used, the following may be useful:

```
$ git config alias.syncsub "submodule foreach git pull origin master"
$ git syncsub
```

- (B) In this approach, the **build-aux/bootstrap** or **autopull.sh** program (see [Section 20.3 \[Developer tools\], page 971](#)) is used to aid a developer in using this setup. You copy this program (and if it’s **autopull.sh**, its companion files) into your package and place the copy or copies under version control. The program can be customized using **bootstrap.conf** which you also put under version control.
- (C) In this approach, you write the **autopull.sh** and **autogen.sh** files by hand.

autopull.sh is most easily written as a script that invokes

```
./gitsub.sh pull || exit 1
```

where **gitsub.sh** is described in [Section 20.3 \[Developer tools\], page 971](#).

autogen.sh typically contains an explicit **gnulib-tool** invocation, followed by

```
aclocal -I m4 \
  && autoconf \
  && autoheader && touch config.h.in \
  && automake --add-missing --copy \
  && rm -rf autom4te.cache \
  || exit $?
```

3. Some projects take a “middle road”: they do commit Gnulib source files as in the first approach, but they do not commit other derived files, such as a **Makefile.in** generated by Automake. This increases the size and complexity of the repository, but can help occasional contributors by not requiring them to have a full Gnulib checkout to do a build, and all developers by ensuring that all developers are working with the same version of Gnulib in the repository. It also supports multiple Gnulib instances within a project. It remains important not to commit the **make**-generated files, as described above.

3.14 Bundling the unit tests of the Gnulib modules

You can bundle the unit tests of the Gnulib modules together with your package, through the ‘**--with-tests**’ option. Together with ‘**--with-tests**’, you also specify the directory

for these tests through the ‘`--tests-base`’ option. Of course, you need to add this directory to the `SUBDIRS` variable in the `Makefile.am` of the parent directory.

The advantage of having the unit tests bundled is that when your program has a problem on a particular platform, running the unit tests may help determine quickly if the problem is on Gnu_{lib}’s side or on your package’s side. Also, it helps verifying Gnu_{lib}’s portability, of course.

The unit tests will be compiled and run when the user runs ‘`make check`’. When the user runs only ‘`make`’, the unit tests will not be compiled.

In the `SUBDIRS` variable, it is useful to put the Gnu_{lib} tests directory after the directory containing the other tests, not before:

```
SUBDIRS = gnulib-lib src man tests gnulib-tests
```

This will ensure that on platforms where there are test failures in either directory, users will see and report the failures from the tests of your program.

Note: In packages which use more than one invocation of `gnulib-tool` in the scope of the same `configure.ac`, you cannot use ‘`--with-tests`’. You will have to use a separate `configure.ac` in this case.

3.15 Avoiding unnecessary checks and compilations

In some cases, a module is needed by another module only on specific platforms. But when a module is present, its Autoconf checks are always executed, and its `Makefile.am` additions are always enabled. So it can happen that some Autoconf checks are executed and some source files are compiled, although no other module needs them on this particular platform, just *in case* some other module would need them.

The option ‘`--conditional-dependencies`’ enables an optimization of configure checks and `Makefile.am` snippets that avoids this. With this option, whether a module is considered “present” is no longer decided when `gnulib-tool` is invoked, but later, when `configure` is run. This applies to modules that were added as dependencies while `gnulib-tool` was run; modules that were passed on the command line explicitly are always “present”.

For example, the `timegm` module needs, on platforms where the system’s `timegm` function is missing or buggy, a replacement that is based on a function `mktime_internal`. The module `mktime-internal` that provides this function provides it on all platforms. So, by default, the file `mktime-internal.c` will be compiled on all platforms, even on glibc and BSD systems which have a working `timegm` function. When the option ‘`--conditional-dependencies`’ is given, on the other hand, and if `mktime-internal` was not explicitly required on the command line, the file `mktime-internal.c` will only be compiled on the platforms where the `timegm` needs them.

Conditional dependencies are specified in the module description by putting the condition on the same line as the dependent module, enclosed in brackets. The condition is a boolean shell expression that can assume that the `configure.ac` snippet from the module description has already been executed. In the example above, the dependency from `timegm` to `mktime-internal` is written like this:

```
Depends-on:
...
mktime-internal [test $HAVE_TIMEGM = 0 || test $REPLACE_TIMEGM = 1]
```

...

Note: The option ‘`--conditional-dependencies`’ cannot be used together with the option ‘`--with-tests`’. It also cannot be used when a package uses `gnulib-tool` for several subdirectories, with different values of ‘`--source-base`’, in the scope of a single `configure.ac` file.

4 Writing modules

This chapter explains how to write modules of your own, either to extend Gnulib for your own package (see [Chapter 5 \[Extending Gnulib\], page 48](#)), or for inclusion in gnulib proper.

The guidelines in this chapter do not necessarily need to be followed for using `gnulib-tool`. They merely represent a set of good practices. Following them will result in a good structure of your modules and in consistency with gnulib.

4.1 Source code files

Every API (C functions or variables) provided should be declared in a header file (`.h` file) and implemented in one or more implementation files (`.c` files). The separation has the effect that users of your module need to read only the contents of the `.h` file and the module description in order to understand what the module is about and how to use it—not the entire implementation. Furthermore, users of your module don't need to repeat the declarations of the functions in their code, and are likely to receive notification through compiler errors if you make incompatible changes to the API (like, adding a parameter or changing the return type of a function).

4.2 Header files

The `.h` file should declare the C functions and variables that the module provides.

The `.h` file should be stand-alone. That is, it does not require other `.h` files to be included before. Rather, it includes all necessary `.h` files by itself.

It is a tradition to use CPP tricks to avoid parsing the same header file more than once, which might cause warnings. The trick is to wrap the content of the header file (say, `foo.h`) in a block, as in:

```
#ifndef FOO_H
# define FOO_H
...
body of header file goes here
...
#endif /* FOO_H */
```

Whether to use `FOO_H` or `_FOO_H` is a matter of taste and style. The C99 and C11 standards reserve all identifiers that begin with an underscore and either an uppercase letter or another underscore, for any use. Thus, in theory, an application might not safely assume that `_FOO_H` has not already been defined by a library. On the other hand, using `FOO_H` will likely lead the higher risk of collisions with other symbols (e.g., `KEY_H`, `XX_H`, `BPF_H`, which are CPP macro constants, or `COFF_LONG_H`, which is a CPP macro function). Your preference may depend on whether you consider the header file under discussion as part of the application (which has its own namespace for CPP symbols) or a supporting library (that shouldn't interfere with the application's CPP symbol namespace).

Adapting C header files for use in C++ applications can use another CPP trick, as in:

```
# ifdef __cplusplus
extern "C"
```

```

{
# endif
...
body of header file goes here
...
# ifdef __cplusplus
}
# endif

```

The idea here is that `__cplusplus` is defined only by C++ implementations, which will wrap the header file in an `'extern "C"'` block. Again, whether to use this trick is a matter of taste and style. While the above can be seen as harmless, it could be argued that the header file is written in C, and any C++ application using it should explicitly use the `'extern "C"'` block itself. Your preference might depend on whether you consider the API exported by your header file as something available for C programs only, or for C and C++ programs alike.

Note that putting a `#include` in an `extern "C" { ... }` block yields a syntax error in C++ mode on some platforms (e.g., glibc systems with g++ v3.3 to v4.2, AIX). For this reason, it is recommended to place the `#include` before the `extern "C"` block.

4.3 Implementation files

The `.c` file or files implement the functions and variables declared in the `.h` file.

Include ordering

Every implementation file must start with `'#include <config.h>'`. This is necessary for activating the preprocessor macros that are defined on behalf of the Autoconf macros. Some of these preprocessor macros, such as `_GNU_SOURCE`, would have no effect if defined after a system header file has already been included.

Then comes the `'#include "..."'` specifying the header file that is being implemented. Putting this right after `'#include <config.h>'` has the effect that it verifies that the header file is self-contained.

Then come the system and application headers. It is customary to put all the system headers before all application headers, so as to minimize the risk that a preprocessor macro defined in an application header confuses the system headers on some platforms.

In summary:

- First comes `#include <config.h>`.
- Second comes the `#include "..."` specifying the module being implemented.
- Then come all the `#include <...>` of system or system-replacement headers, in arbitrary order.
- Then come all the `#include "..."` of gnuilib and application headers, in arbitrary order.

4.4 Specification

The specification of a function should answer at least the following questions:

- What is the purpose of the function?

- What are the arguments?
- What is the return value?
- What happens in case of failure? (Exit? A specific return value? Errno set?)
- Memory allocation policy: If pointers to memory are returned, are they freshly allocated and supposed to be freed by the caller?

Where to put the specification describing exported functions? Three practices are used in gnuilib:

- The specification can be as comments in the header file, just above the function declaration.
- The specification can be as comments in the implementation file, just above the function definition.
- The specification can be in texinfo format, so that it gets included in the gnuilib manual.

In any case, the specification should appear in just one place, unless you can ensure that the multiple copies will always remain identical.

The advantage of putting it in the header file is that the user only has to read the include file normally never needs to peek into the implementation file(s).

The advantage of putting it in the implementation file is that when reviewing or changing the implementation, you have both elements side by side.

The advantage of texinfo formatted documentation is that it is easily published in HTML or Info format.

Currently (as of 2020), 70% of gnuilib uses the first practice, 25% of gnuilib uses the second practice, and a small minority uses the texinfo practice.

4.5 Module description

For the module description, you can start from an existing module's description, or from a blank one: `module/TEMPLATE` for a normal module, or `module/TEMPLATE-TESTS` for a unit test module. Some more fields are possible but rarely used. Use `module/TEMPLATE-EXTENDED` if you want to use one of them.

Module descriptions have the following fields. Absent fields are equivalent to fields with empty contents.

Description

This field should contain a concise description of the module's functionality. One sentence is enough. For example, if it defines a single function `'frob'`, the description can be `'frob() function: frobnication.'` Gnuilib's documentation generator will automatically convert the first part to a hyperlink when it has this form.

Status

This field is either empty/absent, or contains the word `'obsolete'`. In the latter case, `gnuilib-tool` will, unless the option `--with-obsolete` is given, omit it when it used as a dependency. It is good practice to also notify the user about an obsolete module. This is done by putting into the `'Notice'` section (see below) text like `'This module is obsolete.'`

Notice This field contains text that `gnulib-tool` will show to the user when the module is used. This can be a status indicator like ‘`This module is obsolete.`’ or additional advice. Do not abuse this field.

Applicability

This field is either empty/absent, or contains the word ‘`all`’. It describes to which `Makefile.am` the module is applied. By default, a normal module is applied to `source_base/Makefile.am` (normally `lib/Makefile.am`), whereas a module ending in `-tests` is applied to `tests_base/Makefile.am` (normally `tests/Makefile.am`). If this field is ‘`all`’, it is applied to both `Makefile.am`s. This is useful for modules which provide `Makefile.am` macros rather than compiled source code.

Usable-in-testdir

This field should contain ‘`no`’ if the module cannot be used in a test directory created with `gnulib-tool --create-testdir`.

Files

This field contains a newline separated list of the files that are part of the module. `gnulib-tool` copies these files into the package that uses the module. This list is typically ordered by importance: First comes the header file, then the implementation files, then other files.

It is possible to have the same file mentioned in multiple modules. That is, if the maintainers of that module agree on the purpose and future of said file.

Depends-on

This field contains a newline separated list of the modules that are required for the proper working of this module. `gnulib-tool` includes each required module automatically, unless it is specified with option `--avoid` or it is marked as obsolete and the option `--with-obsolete` is not given.

A test modules `foo-tests` implicitly depends on the corresponding non-test module `foo`. `foo` implicitly depends on `foo-tests` if the latter exists and if the option `--with-tests` has been given.

Tests modules can depend on non-tests modules. Non-tests modules should not depend on tests modules. (Recall that tests modules are built in a separate directory.)

Each listed required module may be declared a conditional dependency. This is indicated by placing the condition for the dependency on the same line, enclosed in brackets, after the name of the required module. The condition is a shell expression that is run after the module’s `configure.ac` statements. For example:

```
strtoull    [test $ac_cv_func_strtoumax = no]
```

Lines starting with `#` are recognized as comments and are ignored.

configure.ac-early

This field contains `configure.ac` stuff (Autoconf macro invocations and shell statements) that are logically placed early in the `configure.ac` file: right after the `AC_PROG_CC` invocation. This section is adequate for statements that modify `CPPFLAGS`, as these can affect the results of other Autoconf macros.

configure.ac

This field contains **configure.ac** stuff (Autoconf macro invocations and shell statements).

It is forbidden to add items to the **CPPFLAGS** variable here, other than temporarily, as these could affect the results of other Autoconf macros.

We avoid adding items to the **LIBS** variable, other than temporarily. Instead, the module can export an Autoconf-substituted variable that contains link options. The user of the module can then decide to which executables to apply which link options. Recall that a package can build executables of different kinds and purposes; having all executables link against all libraries is inappropriate.

If the statements in this section grow larger than a couple of lines, we recommend moving them to a **.m4** file of their own.

Makefile.am

This field contains **Makefile.am** statements. Variables like **lib_SOURCES** are transformed to match the name of the library being built in that directory. For example, **lib_SOURCES** may become **libgnu_a_SOURCES** (for a plain library) or **libgnu_la_SOURCES** (for a libtool library). Therefore, the normal way of having an implementation file **lib/foo.c** compiled unconditionally is to write

```
lib_SOURCES += foo.c
```

Include

This field contains the preprocessor statements that users of the module need to add to their source code files. Typically it's a single include statement. A shorthand is allowed: You don't need to write the word **"#include"**, just the name of the include file in the way it will appear in an include statement. Example:

```
"foo.h"
```

Link

This field contains the set of libraries that are needed when linking libraries or executables that use this module. Often this will be written as a reference to a Makefile variable. Please write them one per line, so that **gnulib-tool** can remove duplicates when presenting a summary to the user. Example:

```
$(POW_LIBM)
$(LTLIBICONV) when linking with libtool, $(LIBICONV) otherwise
```

When this field is omitted, it defaults to the union of the **Link** field of the dependencies.

License

This field specifies the license that governs the source code parts of this module. See [Section 2.8 \[Copyright\], page 18](#), for details. Be sure to place, in every source code file, a copyright notice and the appropriate license notice, taken from the **etc/license-notices/** directory.

Maintainer

This field specifies the persons who have a definitive say about proposed changes to this module. You don't need to mention email addresses here: they can be inferred from the **ChangeLog** file.

Please put at least one person here. We don't like unmaintained modules.

4.6 Autoconf macros

For a module `foo`, an Autoconf macro file `m4/foo.m4` is typically created when the Autoconf macro invocations for the module are longer than one or two lines.

The name of the main entry point into this Autoconf macro file is typically `gl_FOO`. For modules outside Gnulib that are not likely to be moved into Gnulib, please use a prefix specific to your package: `gt_` for GNU gettext, `cu_` for GNU coreutils, etc.

For modules that define a function `foo`, the entry point is called `gl_FUNC_FOO` instead of `gl_FOO`. For modules that provide a header file with multiple functions, say `foo.h`, the entry point is called `gl_FOO_H` or `gl_HEADER_FOO_H`. This convention is useful because sometimes a header and a function name coincide (for example, `fcntl` and `fcntl.h`).

For modules that provide a replacement, it is useful to split the Autoconf macro into two macro definitions: one that detects whether the replacement is needed and requests the replacement by setting a `HAVE_FOO` variable to 0 or a `REPLACE_FOO` variable to 1 (this is the entry point, say `gl_FUNC_FOO`), and one that arranges for the macros needed by the replacement code `lib/foo.c` (typically called `gl_PREREQ_FOO`). The reason of this separation is

1. to make it easy to update the Autoconf macros when you have modified the source code file: after changing `lib/foo.c`, all you have to review is the **Depends-on** section of the module description and the `gl_PREREQ_FOO` macro in the Autoconf macro file.
2. The Autoconf macros are often large enough that splitting them eases maintenance.

Autoconf macro files in Gnulib all start with a header consisting of

1. A comment line with the file name. This is useful because in some cases the generated `aclocal.m4` file does not contain a reference to the `.m4` file but its entire contents. The comment makes it clear which `.m4` file is where in the `aclocal.m4` file.
2. A comment line with a serial number. This is useful when people use the ‘`aclocal --install`’ command. `aclocal` will then refrain from copying a file with a smaller serial number onto a file with a larger serial number. The serial number should be a positive integer on the main branch, or a positive fractional number on a stable branch.
3. The copyright and license header, as usual.

4.7 Making proper use of AC_LIBOBJ

Source files that provide a replacement should be only compiled on the platforms that need this replacement. While it is actually possible to compile a `.c` file whose contents is entirely `#ifdef`’ed out on the platforms that don’t need the replacement, this practice is discouraged because

- It makes the build time longer than needed, by invoking the compiler for nothing.
- It produces a `.o` file that suggests that a replacement was needed.
- Empty object files produce a linker warning on some platforms: MSVC.

The typical idiom for invoking `AC_LIBOBJ` is thus the following, in the module description:

```
if test $HAVE_FOO = 0 || test $REPLACE_FOO = 1; then
  AC_LIBOBJ([foo])
  gl_PREREQ_FOO
fi
```

Important: Do not place `AC_LIBOBJ` invocations in the Autoconf macros in the `m4/` directory. The purpose of the Autoconf macros is to determine what features or bugs the platform has, and to make decisions about which replacements are needed. The purpose of the `configure.ac` and `Makefile.am` sections of the module descriptions is to arrange for the replacements to be compiled. **Source file names do not belong in the `m4/` directory.**

When an `AC_LIBOBJ` invocation is unconditional, it is simpler to just have the source file compiled through an Automake variable augmentation: In the `Makefile.am` section write

```
lib_SOURCES += foo.c
```

When a module description contains an `AC_LIBOBJ([foo])` invocation, you **must** list the source file `lib/foo.c` in the `Files` section. This is needed even if the module depends on another module that already lists `lib/foo.c` in its `Files` section – because your module might be used among the test modules (in the directory specified through `--tests-base`) and the other module among the main modules (in the directory specified through `--source-base`), and in this situation, the `AC_LIBOBJ([foo])` of your module can only be satisfied by having `foo.c` be present in the tests source directory as well.

4.8 Unit test modules

A unit test that is a simple C program usually has a module description as simple as this:

```
Files:
tests/test-foo.c
tests/macros.h

Depends-on:

configure.ac:

Makefile.am:
TESTS += test-foo
check_PROGRAMS += test-foo
```

The test program `tests/test-foo.c` often has the following structure:

- First comes the obligatory `#include <config.h>`.
- Second comes the include of the header file that declares the API being tested. Including it here verifies that said header file is self-contained.
- Then come other includes. In particular, the file `macros.h` is often used here. It contains a convenient `ASSERT` macro.

The body of the test, then, contains many `ASSERT` invocations. When a test fails, the `ASSERT` macro prints the line number of the failing statement, thus giving you, the developer, an idea of which part of the test failed, even when you don't have access to the machine where the test failed and the reporting user cannot run a debugger.

Sometimes it is convenient to write part of the test as a shell script. (For example, in areas related to process control or interprocess communication, or when different locales should be tried.) In these cases, the typical module description is like this:

```
Files:
tests/test-foo.sh
tests/test-foo.c
tests/macros.h
```

```

Depends-on:

configure.ac:

Makefile.am:
TESTS += test-foo.sh
TESTS_ENVIRONMENT += FOO_BAR='@FOO_BAR@'
check_PROGRAMS += test-foo

```

Here, the `TESTS_ENVIRONMENT` variable can be used to pass values determined by `configure` or by the `Makefile` to the shell script, as environment variables. The Autoconf values `EXEEXT` and `srcdir` are already provided as environment variables, through an initial value of `TESTS_ENVIRONMENT` that `gnulib-tool` puts in place.

Regardless of the specific form of the unit test, the following guidelines should be respected:

- A test indicates success by exiting with exit code 0. It should normally not produce output in this case. (Output to temporary files that are cleaned up at the end of the test are possible, of course.)
- A test indicates failure by exiting with an exit code different from 0 and 77, typically 1. It is useful to print a message about the failure in this case. The `ASSERT` macro already does so.
- A test indicates "skip", that is, that most of its interesting functionality could not be performed, through a return code of 77. A test should also print a message to stdout or stderr about the reason for skipping. For example:

```

    fputs ("Skipping test: multithreading not enabled\n", stderr);
    return 77;

```

Such a message helps detecting bugs in the autoconf macros: A simple message '`SKIP: test-foo`' does not sufficiently catch the attention of the user.

4.9 Incompatible changes

Incompatible changes to Gnulib modules should be mentioned in Gnulib's `NEWS` file. Incompatible changes here mean that existing source code may not compile or work any more.

We don't mean changes in the binary interface (ABI), since

1. Gnulib code is used in source-code form.
2. The user who distributes libraries that contain Gnulib code is supposed to bump the version number in the way described in the Libtool documentation before every release.

5 Extending Gnulib

Gnulib modules are intended to be suitable for widespread use. Most problems with Gnulib can and should be fixed in a generic way, so that all of Gnulib’s users can benefit from the change. But occasionally a problem arises that is difficult or undesirable to fix generically, or a project that uses Gnulib may need to work around an issue before the Gnulib maintainers commit a final fix. Maintainers may also want to add their own pools of modules to projects as Gnulib “staging areas.”

The obvious way to make local changes to Gnulib modules is to use `gnulib-tool` to check out pristine modules, then to modify the results in-place. This works well enough for short-lived experiments. It is harder to keep modified versions of Gnulib modules for a long time, even though Git (or another distributed version control systems) can help out a lot with this during the development process.

Git, however, doesn’t address the distribution issue. When a package “foobar” needs a modified version of, say, `stdint.in.h`, it either has to put a comment into `foobar/autogen.sh` saying “Attention! This doesn’t work with a pristine Gnulib, you need this and that patch after checking out Gnulib,” or it has to use the ‘`--avoid=stdint-h`’ option and provide the modified `stdint-h` module in a different directory.

The `--local-dir` option to `gnulib-tool` solves this problem. It allows the package to override or augment Gnulib. This means:

- You can store files that are to override Gnulib files or modules.
- You can store context diffs to be applied to Gnulib files.
- You can add modules of your own, that are not (yet) in Gnulib.
- You can also add unstructured amounts of code to the library, by grouping the non-Gnulib files of the library in a single kitchen-sink “module.” (This kind of kitchen-sink module is not needed when you use the `gnulib-tool` option `--makefile-name`.)

In a release tarball, you can distribute the contents of this `--local-dir` directory that will be combinable with newer versions of Gnulib, barring incompatible changes to Gnulib.

If the `--local-dir=directory` option is specified, then `gnulib-tool` looks in *directory* whenever it reads a file from the Gnulib directory. Suppose `gnulib-tool` is looking for *file*. Then:

- If *directory/file* exists, then `gnulib-tool` uses it instead of the file included in Gnulib.
- Otherwise, if *directory/file.diff* exists, then `gnulib-tool` uses the file from Gnulib after applying the diff using the `patch` program.
- Otherwise, `gnulib-tool` uses the file included in Gnulib.

You can specify the `--local-dir` multiple times. In this case, the first specified directory has the highest precedence. That is, a *file* found in one directory will shadow any *file* and *file.diff* in the later directories and in the Gnulib directory. And a file *file.diff* found in one directory will be applied on top of the combination of *file* and *file.diff* files found in the later directories and in the Gnulib directory.

Please make wise use of this option. It also allows you to easily hold back modifications you make to Gnulib macros in cases it may be better to share them.

6 Miscellaneous Notes

6.1 Out of memory handling

The glibc API does not have a standard error code for the out of memory error condition. Instead of adding a non-standard error code, glibc has chosen to adopt a different strategy. Out of memory handling happens in rare situations, but performing the out of memory error handling after almost all API function invocations pollute your source code and might make it harder to spot more serious problems. The strategy chosen improves code readability and robustness.

For most applications, aborting the application with an error message when the out of memory situation occurs is the best that can be wished for. This is how the library behaves by default (using the ‘`xalloc-die`’ module).

However, we realize that some applications may not want to abort execution in any situation. Glibc supports a hook to let the application regain control and perform its own cleanups when an out of memory situation has occurred. The application can define a function (having a `void` prototype, i.e., no return value and no parameters) and set the library variable `xalloc_die` to that function. The variable should be declared as follows.

```
extern void (*xalloc_die) (void);
```

Glibc will invoke this function if an out of memory error occurs. Note that the function should not return. Of course, care must be taken to not allocate more memory, as that will likely also fail.

6.2 Obsolete modules

Modules can be marked obsolete. This means that the problems they fix don’t occur any more on the platforms that are reasonable porting targets now. `glibc-tool` warns when obsolete modules are mentioned on the command line, and by default ignores dependencies from modules to obsolete modules. When you pass the option `--with-obsolete` to `glibc-tool`, dependencies to obsolete modules will be included, however, unless blocked through an `--avoid` option. This option is useful if your package should be portable even to very old platforms.

In order to mark a module obsolete, you need to add this to the module description:

```
Status:
obsolete
```

```
Notice:
This module is obsolete.
```

6.3 Extra tests modules

Test modules can be marked with some special status attributes. When a test module has such an attribute, `glibc-tool --import` will not include it by default.

The supported status attributes are:

c++-test Indicates that the test is testing C++ interoperability. Such a test is useful in a C++ or mixed C/C++ package, but is useless in a C package.

longrunning-test

Indicates that the test takes a long time to compile or execute (more than five minutes or so). Such a test is better avoided in a release that is made for the general public.

privileged-test

Indicates that the test will request special privileges, for example, ask for the superuser password. Such a test may hang when run non-interactively and is therefore better avoided in a release that is made for the general public.

unportable-test

Indicates that the test is known to fail on some systems, and that there is no workaround about it. Such a test is better avoided in a release that is made for the general public.

`gnulib-tool --import --with-tests` will not include tests marked with these attributes by default. When `gnulib-tool` is invoked with one of the options `--with-c++-tests`, `--with-longrunning-tests`, `--with-privileged-tests`, `--with-unportable-tests`, it will include tests despite the corresponding special status attribute. When `gnulib-tool` receives the option `--with-all-tests`, it will include all tests regardless of their status attributes.

`gnulib-tool --create-testdir --with-tests` and `gnulib-tool --create-megatestdir --with-tests` by default include all tests of modules specified on the command line, regardless of their status attributes. Tests of modules occurring as dependencies are not included by default if they have one of these status attributes. The options `--with-c++-tests`, `--with-longrunning-tests`, `--with-privileged-tests`, `--with-unportable-tests` are recognized here as well. Additionally, `gnulib-tool` also understands the options `--without-c++-tests`, `--without-longrunning-tests`, `--without-privileged-tests`, `--without-unportable-tests`.

In order to mark a module with a status attribute, you need to add it to the module description, like this:

Status:

longrunning-test

If only a part of a test deserves a particular status attribute, you can split the module into a primary and a secondary test module, say `foo-tests` and `foo-extra-tests`. Then add a dependency from `foo-tests` to `foo-extra-tests`, and mark the `foo-extra-tests` with the particular status attribute.

6.4 Modules that modify the way other modules work

The normal way to design modules is that each module has its own code, and the module dependencies provide the facilities on which this code can rely. But sometimes it is necessary to use more advanced techniques. For example:

- You may want to have optional module dependencies: Let module A use facilities provided by module B, if module B is present, but without requiring that module B is present.
- A module can indicate support for particular behaviours. For example, Gnulib has a module `'sigpipe'` that requests POSIX compatible SIGPIPE behaviour from all other

modules – something that is not enabled by default. Or consider the ‘nonblocking’ module, that is an indicator that all I/O functions should handle non-blocking file descriptors – something that, equally, is not enabled by default.

- A module can indicate to other modules that they can rely on certain guarantees, and thus omit specific code. For example, when Gnulib’s ‘malloc-gnu’ module is present, you can omit code that test `n` against zero when you call `malloc (n)`.

Be aware that these advanced techniques likely cause breakage in the situation of multiple `gnulib-tool` invocations in the scope of a single `configure` file. This is because the question “is module B present?” does not have a unique answer in such situations. `gnulib-tool` has support for these techniques in the situation of `--create-testdir` (without option `--two-configures`), which basically has two `gnulib-tool` invocations, one for a set of modules that end up in `gllib`, and one for the set of modules that end up in `gltests`. But you should be aware that this does not cover the general situation.

Which technique to use, depends on the answer to the question: “If my module occurs among the modules of `gltests`, should it have an effect on the modules in `gllib`?”

If the answer is “no”, your module description should invoke the Autoconf macro `gl_MODULE_INDICATOR`. This Autoconf macro takes one argument: the name of your module. The effect of `gl_MODULE_INDICATOR([my-module])` is to define, in `config.h`, a C macro `GNULIB_MY_MODULE` that indicates whether your macro is considered to be present. This works even when your macro is used in `gltests`: `GNULIB_MY_MODULE` will then evaluate to 1 in `gltests` but to 0 in `gllib`.

If the answer is “yes”, you have two techniques available. The first one is to invoke a similar Autoconf macro, named `gl_MODULE_INDICATOR_FOR_TESTS`. It works similarly. However, when your macro is used in `gltests`, `GNULIB_MY_MODULE` will evaluate to 1 both in `gltests` and in `gllib`.

The second one is to define a shell variable in the `configure` file that tells whether your module is present, through use of `m4_divert_text`. The Autoconf macros of a dependency module will initialize this shell variable, through ‘`m4_divert_text([DEFAULTS], [my_shell_var=no])`’. The Autoconf macros of your module will override this value, through ‘`m4_divert_text([INIT_PREPARE], [my_shell_var=yes])`’. Then you can use `my_shell_var` in the Autoconf macros of both modules. You can find more details about this technique in the Gnulib module `getopt-gnu`.

Reminder: These techniques are advanced. They have the potential to cause lots of headaches if you apply them incorrectly.

6.5 A C++ namespace for gnulib

The function definitions provided by Gnulib (`.c` code) are meant to be compiled by a C compiler. The header files (`.h` files), on the other hand, can be used in either C or C++.

By default, when used in a C++ compilation unit, the `.h` files declare the same symbols and overrides as in C mode, except that functions defined by Gnulib or by the system are declared as ‘`extern "C"`’.

It is also possible to indicate to Gnulib to provide many of its symbols in a dedicated C++ namespace. If you define the macro `GNULIB_NAMESPACE` to an identifier, many functions will

be defined in the namespace specified by the identifier instead of the global namespace. For example, after you have defined

```
#define GNULIB_NAMESPACE gnu
```

at the beginning of a compilation unit, GnuLib’s `<fcntl.h>` header file will make available the `open` function as `gnu::open`. The symbol `open` will still refer to the system’s `open` function, with its platform specific bugs and limitations.

The symbols provided in the GnuLib namespace are those for which the corresponding header file contains a `_GL_CXXALIAS_RPL` or `_GL_CXXALIAS_SYS` macro invocation.

The benefits of this namespace mode are:

- GnuLib defines fewer symbols as preprocessor macros. For example, on a platform where `open` has to be overridden, GnuLib normally does `#define open rpl_open`. If your package has a class with a member `open`, for example a class `foo` with a method `foo::open`, then if you define this member in a compilation unit that includes `<fcntl.h>` and use it in a compilation unit that does not include `<fcntl.h>`, or vice versa, you will get a link error. Worse: You will not notice this problem on the platform where the system’s `open` function works fine. This problem goes away in namespace mode.
- It provides a safety check whether the set of modules your package requests from GnuLib is sufficient. For example, if you use the function `gnu::open` in your code, and you forgot to request the module ‘`open`’ from GnuLib, you will get a compilation error (regardless of the platform).

The drawback of this namespace mode is that the system provided symbols in the global namespace are still present, even when they contain bugs that GnuLib fixes. For example, if you call `open ( . . . )` in your code, it will invoke the possibly buggy system function, even if you have requested the module ‘`open`’ from gnuLib-tool.

You can turn on the namespace mode in some compilation units and keep it turned off in others. This can be useful if your package consists of an application layer that does not need to invoke POSIX functions and an operating system interface layer that contains all the OS function calls. In such a situation, you will want to turn on the namespace mode for the application layer—to avoid many preprocessor macro definitions—and turn it off for the OS interface layer—to avoid the drawback of the namespace mode, mentioned above.

6.6 License Texinfo sources

GnuLib provides copies of the GNU GPL, GNU LGPL, GNU Affero GPL, and GNU FDL licenses in Texinfo form. (The master location is <https://www.gnu.org/licenses/>). These Texinfo documents do not have any node names and structures built into them; for your manual, you should `@include` them in an appropriate `@node`.

The conventional name for the GPL node is ‘`Copying`’ and for the FDL ‘`GNU Free Documentation License`’. The LGPL doesn’t seem to have a conventional node name.

Of course the license texts themselves should not be changed at all.

The recommended way to make use of these license files, consistently with current practice, is as follows:

- The code license (GNU GPL, GNU LGPL, or GNU Affero GPL) is usually present as a file in the top-level directory. This is true not only for the release tarballs,

but also in the VCS repository. The file is typically named ‘COPYING’ for the GNU GPL, or ‘COPYING.LIB’ or ‘COPYING.LESSER’ for the GNU LGPL. The presence of this file fulfills a legal obligation; see <https://www.gnu.org/licenses/gpl-faq.html#WhyMustIInclude>.

To make use of the code license in your documentation, you may request one of the modules `gpl-3.0`, `gpl-2.0`, `lgpl-3.0`, `lgpl-2.1`, `agpl-3.0`, through a `gnulib-tool` invocation. Or you may copy the relevant Texinfo file directly into your VCS repository. Both approaches are equally good. The Texinfo file changes very rarely.

- The documentation license file (GNU FDL) is usually not present as a file in the top-level directory, because that would be ambiguous: When you use the GNU FDL, you need to specify the Invariant Sections, the Front-Cover Texts, and the Back-Cover Texts.

To make use of this documentation license, copy the relevant Texinfo file (`doc/fdl-1.3.texi`) into your VCS repository. This makes sure that anyone who receives a copy of your VCS repository has also received a copy of the documentation license. In the documentation, also state what are the Invariant Sections, the Front-Cover Texts, and the Back-Cover Texts.

We recommend to place the licenses as appendices at the end of the manual, right before any indices. For the FDL, we suggest the following `@menu` entry:

```
* GNU Free Documentation License:: License for copying this manual
```

For any `@detailmenu` entries, we suggest the following:

```
Copying This Manual
```

```
* GNU Free Documentation License:: Copying and sharing this manual
```

And for actual inclusion of the FDL itself, we suggest the following:

```
@node GNU Free Documentation License
@appendix GNU Free Documentation License

@include fdl.texi
```

6.7 Building gnulib

If you wish to help the gnulib development effort with build logs for your favorite platform, you may perform these steps:

1. Prerequisites tools

Install the proper development tools. To build and test all of Gnulib, you will need development tools for the programming languages C, C++, Python, and Perl, along with standard POSIX utilities such as `awk`, `make`, and `sh`. You will also need development tools that include Autoconf, Automake, Bison, Gettext, Git, GNU M4, Gperf, Libtool, and Texinfo. Some of these tools are needed only by some modules. More details can be found in Gnulib’s `DEPENDENCIES` file.

2. Obtain Gnulib

See <https://www.gnu.org/software/gnulib/> for how to get the current Gnulib sources via Git.

3. Create gnulib directory

On a machine with GNU development tools installed and with a gnulib git checkout, use

```
gnulib-tool --create-testdir --with-tests --dir=...
```

Alternatively, pick a small set of modules and run

```
gnulib-tool --create-megatestdir --with-tests --dir=... autobuild MODULES
```

(You can't do this with all of Gnulib at once: `gnulib-tool` would run for a week and produce a directory that takes more than 100 GB, maybe even 1 TB, of disk space.)

4. Transfer gnulib directory

Transfer this directory to a build machine (HP-UX, Cygwin, or whatever). Often it is easier to transfer one file, and this can be achieved by running, inside the directory the following commands:

```
./configure
make dist
```

And then transferring the `dummy-0.tar.gz` file.

5. Build modules

On the build machine, run `./do-autobuild` (or `"nohup ./do-autobuild"`). It creates a directory `logs/` with a log file for each module.

6.8 Gnulib Git Bundle

To provide a serialized archival copy of the Gnulib Git repository we publish Git Bundles (<https://git-scm.com/docs/git-bundle>) of Gnulib at <https://ftp.gnu.org/gnu/gnulib/>. These may be useful if Savannah happens to be offline or if you want to have a GnuPG signed confirmation of the Gnulib content.

The files are named like `gnulib-YYYYMMDD.bundle`, for example `gnulib-20250729.bundle`, where `YYYYMMDD` corresponds to the Git commit date (in UTC0) of the last commit on the `master` branch in the bundle.

After downloading the Git bundle you may use it to create a local gnulib clone using normal Git commands:

```
wget -nv https://ftp.gnu.org/gnu/gnulib/gnulib-20250729.bundle
git clone gnulib-20250729.bundle gnulib
cd gnulib
```

Below are SHA-256 checksums of known releases:

```
9dae009ef9dd7cff17b74c0cda5d7a423e2ed98b4f5b7aa29a970565b0591c06 gnulib-20250303.bundle
f01e423a7ef6b48e947fabd24bb11744204f4549342416e15dc64f427caa32e2 gnulib-20250729.bundle
```

Next to the Git Bundle is a GnuPG signature on the file, named `gnulib-YYYYMMDD.bundle.sig`, which can be verified using GnuPG as usual:

```
gpg --verify gnulib-20250729.bundle.sig
```

Or using the simpler `gpgv` tool like this:

```
gpgv gnulib-20250729.bundle.sig gnulib-20250729.bundle
```

The following GnuPG keys have signed releases:

```
sec> ed25519 2019-03-20 [SC] https://josefsson.org/key-20190320.txt
```

B1D2BD1375BECB784CF4F8C4D73CF638C53C06BE

uid [ultimate] Simon Josefsson <simon@josefsson.org>

We desire that the GnuLib Git bundles will be forever bit-by-bit reproducible for others from the official git repository. Currently gnuLib maintainers may invoke the following commands to prepare and upload a GnuLib git bundle. We appreciate ideas on how to improve these set of commands (or the upstream Git tool) to make further supply-chain security related improvements.

```
cd $(mktemp -d)
REV=225973a89f50c2b494ad947399425182dd42618c # master branch commit to package
S1REV=475dd38289d33270d0080085084bf687ad77c74d # stable-202501 branch commit
S2REV=e8cc0791e6bb0814cf4e88395c06d5e06655d8b5 # stable-202507 branch commit
git clone https://git.savannah.gnu.org/git/gnuLib.git
cd gnuLib
git fsck # attempt to validate input
# Manually inspect that the new tree matches a trusted previous copy
git checkout -B master $REV # put $REV at master
# Add all stable-* branches locally:
for b in $(git branch -r | grep origin/stable- | sort --version-sort); do git checkout
git checkout -B stable-202501 $S1REV
git checkout -B stable-202507 $S2REV
git remote remove origin # drop some unrelated branches
git gc --prune=now # drop any unrelated commits, not clear this helps
git -c pack.threads=1 repack -adF
git -c 'pack.threads=1' bundle create gnuLib.bundle --all
V=$(env TZ=UTC0 git show -s --date=format:%Y%m%d --pretty=%cd master)
mv gnuLib.bundle gnuLib-$V.bundle
build-aux/gnupload --to ftp.gnu.org:gnuLib gnuLib-$V.bundle
```

7 Building the ISO C and POSIX Substitutes

This section shows a radically different way to use Gnulib.

You can extract the ISO C / POSIX substitutes part of gnulib by running the command

```
gnulib-tool --create-testdir --source-base=lib \  
            --dir=/tmp/posixlib `posix-modules`
```

The command ‘`posix-modules`’ is found in the same directory as `gnulib-tool`.

The resulting directory can be built on a particular platform, independently of the program being ported. Then you can configure and build any program, by setting `CPPFLAGS` and `LDFLAGS` at configure time accordingly: set `CPPFLAGS="-I.../posixlib/lib"`, plus any essential type definitions and flags that you find in `.../posixlib/config.h`, and set `LDFLAGS=".../posixlib/lib/libgnu.a"`.

This way of using Gnulib is useful when you don’t want to modify the program’s source code, or when the program uses a mix between C and C++ sources (requiring separate builds of the `posixlib` for the C compiler and for the C++ compiler).

8 ISO C Keyword Substitutes

This chapter describes which keywords specified by ISO C are substituted by Gnulib.

8.1 alignof and alignas

Gnulib module: `alignasof`

The `alignasof` module arranges for `alignas` and `alignof` to be more like standard C.

Portability problems fixed by Gnulib:

- On pre-C23 platforms, `<stdalign.h>` must be included before using `alignas` or `alignof`. See [Section 9.47 \[stdalign.h\]](#), page 76.

Portability problems not fixed by Gnulib:

- The set of alignments supported by `alignas` is implementation dependent, and implementations need not support alignments greater than `alignof (max_align_t)`.
- Some older platforms do not support `alignas`. To test this, use the `alignasof` module along with the `#if` conditional `(defined alignas || defined __alignas_is_defined || 202311 <= __STDC_VERSION__ || 201103 <= __cplusplus)`. Platforms where `alignas` does not work include: IBM XL C for AIX V16.1.
- On some platforms, the argument of `alignas` must be an integer constant like `'8'`, instead of an integer constant expression like `'4 + 4'` that evaluates to 8: GCC 4.9.
- On pre-C23 platforms, `alignas` and `alignof` are macros.

8.2 bool

Gnulib module: `bool`

Portability problems fixed by Gnulib:

- The keywords `bool`, `true`, and `false` are not available: gcc 12 and other compilers predating C23.

Portability problems not fixed by Gnulib:

- On pre-C23 platforms, the keyword substitutes are macros.
- On pre-C23 platforms, the keyword substitutes assume C99 or later.

8.3 nullptr

Gnulib module: `nullptr`

The `nullptr` module arranges for `nullptr` to act like standard C and C++.

The `nullptr` keyword yields a null pointer. It differs from the `NULL` macro, in that `NULL` might be an integer whereas `nullptr` is of a special `nullptr_t` type with only one value, namely `nullptr` itself. Using `nullptr` can help some compilers emit more sensible warnings, can avoid the need to cast a null pointer passed to a function prototyped with an ellipsis, and removes the need to include `<stddef.h>` merely to define `NULL`.

Portability problems fixed by Gnulib:

- Some platforms lack `nullptr`: For C: GCC 12, Clang 15, and other pre-2023 C compilers. For C++: pre-2011 C++ compilers.

- Some platforms incorrectly warn when `nullptr` is a sentinel argument: GCC 13.2 and 14.0.

Portability problems not fixed by Gnulib:

- On older platforms, `nullptr` is a macro instead of a keyword.
- On older platforms, `nullptr` does not have the type `nullptr_t`. In C, it has type `void *`; in C++ it has an integer type.
- On older platforms Gnulib cannot easily emulate `nullptr_t`, so null pointer type checking is more error prone. In C, `_Generic` expressions cannot reliably distinguish the type of `nullptr` from integer or `void *` types. C++ overloading has similar limitations.
- GCC 14 defines `nullptr_t` even when `<stddef.h>` is not included. This bug is fixed in GCC 15.

8.4 `static_assert`

Gnulib module: `assert-h`

The `assert-h` module arranges for both `static_assert` and `<assert.h>` to be like standard C. See [Section 9.3 \[assert.h\], page 59](#).

Portability problems fixed by Gnulib:

- Pre-C11 platforms lack `static_assert`.
- On pre-C23 platforms, `<assert.h>` must be included before using `static_assert`.

Portability problems not fixed by Gnulib:

- On pre-C23 platforms, `static_assert` is a macro.

9 ISO C and POSIX Header File Substitutes

This chapter describes which header files specified by ISO C or POSIX are substituted by Gnulib, which portability pitfalls are fixed by Gnulib, and which (known) portability problems are not worked around by Gnulib.

The notation “Gnulib module: —” means that Gnulib does not provide a module providing a substitute for the header file. When the list “Portability problems not fixed by Gnulib” is empty, such a module is not needed: No portability problems are known. Otherwise, it indicates that such a module would be useful but is not available: No one so far found this header file important enough to contribute a substitute for it. If you need this particular header file, you may write to <bug-gnulib at gnu dot org>.

9.1 aio.h

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/aio.h.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on some platforms: NetBSD 3.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.10.x, mingw, MSVC 14, Haiku, Android 9.0.
- The type `off64_t` is not defined on some platforms: macOS 14, FreeBSD 10.4, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.

9.2 arpa/inet.h

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/arpa_inet.h.html

Gnulib module: `arpa_inet-h`

Portability problems fixed by Gnulib:

- This header file is missing on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

9.3 assert.h

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/assert.h.html>

Gnulib module: `assert-h`

See also the Gnulib modules `assert` and `verify`.

Portability problems fixed by Gnulib:

- On older C platforms <assert.h> must be included before using `static_assert`. For example, GCC versions before 13 do not support the `static_assert` keyword that was standardized by C23.

- On older platforms `static_assert` does not allow the second string-literal argument to be omitted. For example, GCC versions before 9.1 do not support the single-argument `static_assert` that was standardized by C23 and C++17.
- Even-older platforms do not support `static_assert` at all. For example, GCC versions before 4.6 and G++ versions before 4.3 do not support the two-argument form, which was standardized by C11 and C++11.
- Older C platforms might not support the obsolescent `_Static_assert` keyword or macro. This portability problem should not matter with code using this module, as such code should use `static_assert` instead.

Portability problems not fixed by GnuLib:

- A `static_assert` can also be used within a `struct` or `union` specifier, in place of an ordinary declaration of a member of the struct or union. The GnuLib substitute can be used only as an ordinary declaration in code intended to be portable to C99 or earlier.
- In C23 and C++11 and later, `static_assert` is a keyword. In C11 and C17 it is a macro. Any GnuLib substitute is also a macro.
- In C99 and later, `assert` can be applied to any scalar expression. In C89, the argument to `assert` is of type `int`.

9.4 `complex.h`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/complex.h.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This header file is missing on some platforms: NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 1.7.7, mingw, MSVC 9.

9.5 `cpio.h`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/cpio.h.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This header file is missing on some platforms: Minix 3.1.8, Cygwin 2.4.x, mingw, MSVC 14.

9.6 `ctype.h`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/ctype.h.html>

GnuLib module: `ctype-h`

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

9.7 devctl.h

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/devctl.h.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on many platforms: glibc 2.42, macOS 14, FreeBSD 9.3, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.6.0, mingw, MSVC 14, Android 9.0.

9.8 dirent.h

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/dirent.h.html>

Gnulib module: dirent-h

Portability problems fixed by Gnulib:

- This header file is missing on some platforms: MSVC 14.
- Although many systems define directory entry type macros like `DT_DIR`, some do not: Minix 3.1.8, AIX 7.2, HP-UX 11, Solaris 11.4, mingw.
- The directory entry type macro `DT_WHT` is missing on many systems: All systems where `DT_DIR` is missing, plus OpenBSD 7.5.
- The conversion macros `DTTOIF` and `IFTODT` are missing on many systems (however, the Gnulib replacement macros may evaluate their arguments multiple times): All systems where `DT_DIR` is missing, plus OpenBSD 7.5.
- The type `ino_t` is missing on some platforms: glibc 2.23 and others.

Portability problems not fixed by Gnulib:

- Although many systems define a `struct dirent` member named `d_type`, some do not: Minix 3.1.8, AIX 7.2, HP-UX 11, Solaris 11.4, mingw.
- On systems with `d_type`, not every filesystem supports `d_type`, and those lacking support will set it to `DT_UNKNOWN`.
- The POSIX.1-2024 directory entry type macros `DT_MQ`, `DT_SEM`, `DT_SHM` and `DT_TMO` are missing on many platforms.
- Some systems define a `struct dirent` member named `d_namlen` containing the string length of `d_name`, but others do not: glibc 2.23 on Linux, Minix 3.1.8, Solaris 11.4, Cygwin. All of these, except Cygwin, have a member `d_reclen` instead, that has a different semantics.
- Some systems define a `struct dirent` member named `d_off` containing a magic cookie suitable as an argument to `seekdir`, but others do not: glibc 2.23 on Hurd, macOS 11.1, FreeBSD 11.0, NetBSD 10.0, OpenBSD 7.5, AIX 5.1, HP-UX 11, Cygwin, mingw.
- Some systems define a `struct dirent` member named `d_reclen` containing the number of bytes in the directory entry record, but others do not. This member has limited utility, as it is an implementation detail.

9.9 dlfcn.h

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/dlfcn.h.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

9.10 endian.h

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/endian.h.html>

Describes the platform's endianness (byte ordering of words stored in memory). Defines the macros `BYTE_ORDER`, `LITTLE_ENDIAN`, `BIG_ENDIAN`, `PDP_ENDIAN`.

Gnulib module: endian

Portability problems fixed by Gnulib:

- This header file is missing on some platforms: macOS 14, FreeBSD 13.0, NetBSD 7.1, OpenBSD 3.8, Minix 3.1.8, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, mingw, MSVC 14.

Portability problems not fixed by Gnulib:

9.11 errno.h

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/errno.h.html>

Gnulib module: errno-h

Portability problems fixed by Gnulib:

- The macro `Eoverflow` is not defined on some platforms: OpenBSD 4.0, mingw, MSVC 9.
- The macro `ENOLINK` is not defined on some platforms: OpenBSD 7.5, mingw, MSVC 9.
- The macro `EMULTIHOP` is not defined on some platforms: OpenBSD 7.5, mingw, MSVC 14.
- The macro `ECANCELED` is not defined on some platforms: OpenBSD 4.0, Cygwin, mingw, MSVC 9.
- The macros `ENOMSG`, `EIDRM`, `EPROTO`, `EBADMSG`, `ENOTSUP` are not defined on some platforms: OpenBSD 4.0, mingw, MSVC 9.
- The macro `ESTALE` is not defined on some platforms: mingw, MSVC 14.
- The macro `EDQUOT` is not defined on some platforms: NonStop Kernel, mingw, MSVC 14.
- The macros `ENETRESET`, `ECONNABORTED` are not defined on some platforms: Minix 3.1.8, mingw, MSVC 9.

- The macros `EWouldBlock`, `ETXTBSY`, `ELOOP`, `ENOTSOCK`, `EDESTADDRREQ`, `EMSGSIZE`, `EPROTOTYPE`, `ENOPROTOOPT`, `EPROTONOSUPPORT`, `EOPNOTSUPP`, `EAFNOSUPPORT`, `EADDRINUSE`, `EADDRNOTAVAIL`, `ENETDOWN`, `ENETUNREACH`, `ECONNRESET`, `ENOBUFS`, `EISCONN`, `ENOTCONN`, `ETIMEDOUT`, `ECONNREFUSED`, `EHOSTUNREACH`, `EALREADY`, `EINPROGRESS` are not defined on some platforms: mingw, MSVC 9.
- The macros `EOWNERDEAD`, `ENOTRECOVERABLE` are not defined on some platforms: glibc/Linux 2.3.6, glibc/Hurd 2.15, glibc/kFreeBSD 2.15, Mac OS X 10.5, FreeBSD 6.0, NetBSD 9.3, OpenBSD 6.0, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin, mingw without pthreads-win32, MSVC 9.
- The macro `ESOCKTNOSUPPORT` is not defined on some platforms: Haiku.
- The macro `EILSEQ` is not defined on some platforms: LynxOS 178 2.2.2.
- The macros `EEXIST` and `ENOTEMPTY` have the same value on some platforms: AIX 7.3.

Portability problems not fixed by Gnulib:

- Error numbers are in the range `INT_MIN` through `INT_MIN + 0xffff`, instead of being positive as POSIX requires: Haiku.

9.12 `fcntl.h`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/fcntl.h.html>

Gnulib module: `fcntl-h`

Portability problems fixed by Gnulib:

- The type `pid_t` is not defined on some platforms: MSVC 14.
- The type `mode_t` is not defined on some platforms: MSVC 14.
- The type `off64_t` is not defined on some platforms: macOS 14, FreeBSD 10.4, NetBSD 10.0, OpenBSD 7.5, MSVC 14, Cygwin, Haiku, Minix 3.3.
- `'O_CLOEXEC'` is not defined on some platforms: Mac OS X 10.6, FreeBSD 8.4, NetBSD 5.1, OpenBSD 4.9, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 10, Cygwin 1.7.1, mingw, MSVC 14.
- `'O_DSYNC'`, `'O_NOCTTY'`, `'O_NOFOLLOW'`, `'O_RSYNC'`, `'O_SYNC'`, and `'O_TTY_INIT'` are not defined on some platforms. When not otherwise defined, Gnulib defines these macros to 0, which is generally safe.
- `'O_NONBLOCK'` is not defined on some platforms. If the `'nonblocking'` module is in use, gnuilib guarantees a working non-zero value; otherwise, the gnuilib replacement is 0.
- `'O_EXEC'` is not defined on some platforms: glibc 2.42, macOS 12, FreeBSD 7.4, NetBSD 9.4, OpenBSD 7.5, HP-UX 11, Solaris 10, Cygwin 1.5.x, mingw, MSVC 14, Android. However, glibc does define `'O_EXEC'` on GNU/Hurd. When not otherwise defined, Gnulib defines this macro to `'O_RDONLY'`, which is typically 0.

Note: `'O_EXEC'` is not suitable for opening a file descriptor that you want to use with `fchmod` later, because musl libc defines `'O_EXEC'` to `O_PATH` and Linux `fchmod` fails with error `EBADF` when given such a file descriptor.

- `'O_SEARCH'` is not defined on some platforms: glibc 2.42, macOS 12, FreeBSD 12.4, NetBSD 6.1, OpenBSD 7.5, AIX 6.1, HP-UX 11, Solaris 10, Cygwin 1.5.x, mingw,

MSVC 14, Android. When not otherwise defined, Gnulib defines this macro to ‘O_RDONLY’, which is typically 0.

- ‘O_SEARCH’ is defined to O_PATH on some platforms: musl libc.
- ‘O_ACCMODE’ is not defined on some platforms: MSVC 14.
- The ‘O_ACCMODE’ mask mistakenly omits ‘O_SEARCH’ and ‘O_EXEC’ on some platforms: Cygwin.
- ‘O_BINARY’, ‘O_TEXT’ (not specified by POSIX, but essential for portability to native Windows platforms) are defined on some platforms but not on others. Gnulib defines these macros to 0 on GNU and other platforms that do not distinguish between text and binary I/O.
- ‘O_CLOEXEC’, ‘O_NOFOLLOW’, and ‘O_TTY_INIT’ are defined to values that are too large for an `int` on some platforms: AIX 7.1 with XL C 12.1.
- ‘O_DIRECT’, ‘O_IGNORE_CTTY’, ‘O_NDELAY’, ‘O_NOATIME’, ‘O_NOLINK’, ‘O_NOLINKS’, and ‘O_NOTRANS’ (not specified by POSIX) are defined on some platforms but not on others. When not otherwise defined, Gnulib defines these macros to 0, which is generally safe.
- ‘O_DIRECTORY’ is not defined on some platforms: glibc 2.0, Mac OS X 10.4, FreeBSD 7.4, NetBSD 4.0.1, OpenBSD 4.9, HP-UX 11, Solaris 10, Cygwin 1.5.x, mingw, MSVC 14.
- ‘FD_CLOEXEC’, ‘F_DUPFD’, and ‘F_GETFD’ are not defined on some platforms: mingw, MSVC 14.
- ‘F_DUPFD_CLOEXEC’ is not defined on some platforms: Mac OS X 10.5, FreeBSD 6.0, NetBSD 5.0, OpenBSD 7.5, AIX 5.1, HP-UX 11, Solaris 11 2010-11, Cygwin 1.7.1, mingw, MSVC 14.
- ‘AT_FDCWD’, ‘AT_EACCESS’, ‘AT_SYMLINK_NOFOLLOW’, ‘AT_SYMLINK_FOLLOW’, and ‘AT_REMOVEDIR’ are not defined on many platforms: glibc 2.3.6, Mac OS X 10.5, FreeBSD 6.0, NetBSD 5.0, OpenBSD 7.5, AIX 5.1, HP-UX 11, Cygwin 1.5.x, mingw, MSVC 14.
- ‘AT_FDCWD’ is defined with a value too large for an `int` on some platforms: Solaris 11.3.

Portability problems not fixed by Gnulib:

- On some platforms, ‘O_NOATIME’ and ‘O_NOFOLLOW’ are defined but do not work. To help work around the resulting portability problems, the `fcntl-h` module defines ‘HAVE_WORKING_O_NOATIME’ to 1 on platforms where ‘O_NOATIME’ is known to work, and to 0 otherwise; and similarly for ‘HAVE_WORKING_O_NOFOLLOW’ and ‘O_NOFOLLOW’: Mac OS X 10.3.
- ‘O_PATH’ is not defined on some platforms: glibc 2.13, macOS 13, FreeBSD 13.0, NetBSD 9.2, OpenBSD 7.1, Minix 3.3.0, AIX 7.3, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14.
- Although POSIX.1-2024 standardized O_CLOFORK, FD_CLOFORK and F_DUPFD_CLOFORK, they are not defined on most platforms: glibc 2.42, musl libc, macOS 15.5, FreeBSD 14.2, NetBSD 10.1, OpenBSD 7.7, Minix 3.3.0, AIX 7.3, Solaris 11.3, Cygwin 3.6.2, mingw, MSVC 14.
- ‘F_SETFD’, ‘F_GETFL’, ‘F_SETFL’, ‘F_GETLK’, ‘F_SETLK’, ‘F_SETLKW’, ‘F_GETOWN’, and ‘F_SETOWN’ are not defined on some platforms: mingw, MSVC 14.

- ‘POSIX_FADV_DONTNEED’, ‘POSIX_FADV_NOREUSE’, ‘POSIX_FADV_NORMAL’, ‘POSIX_FADV_RANDOM’, ‘POSIX_FADV_SEQUENTIAL’, and ‘POSIX_FADV_WILLNEED’ are not defined on some platforms.
- POSIX allows ‘AT_FDCWD’ to be `-1`, so it is unsafe to use ‘AT_FDCWD’ as an invalid directory file descriptor argument to functions like ‘`openat`’. Instead, you can use ‘`AT_FDCWD == -1 ? -2 : -1`’. ‘AT_FDCWD’ is `-1` on some platforms: Haiku R1/Beta4.

9.13 fenv.h

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/fenv.h.html>

Gnulib module: fenv-h

Portability problems fixed by Gnulib:

- This header file is missing on some platforms: FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, AIX 5.1, Cygwin 1.7.7, MSVC 9.

Portability problems not fixed by Gnulib:

- On MSVC, the values of the rounding direction macros have changed: In MSVC 14, `FE_UPWARD` is `0x100` and `FE_DOWNWARD` is `0x200`, whereas in MSVC 14.30, it’s the opposite.

9.14 float.h

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/float.h.html>

Gnulib module: float-h

Portability problems fixed by Gnulib:

- The conversion from `int` to `long double` is incorrect on some platforms: glibc 2.7 on Linux/SPARC64.
- The values of `LDBL_*` macros are incorrect on some platforms: On OpenBSD 4.0 and MirBSD 10, they are the same as the values of the `DBL_*` macros, although ‘`long double`’ is a larger type than ‘`double`’. On FreeBSD/x86 6.4, they represent the incorrect 53-bit precision assumptions in the compiler, not the real 64-bit precision at runtime. On PowerPC with gcc 8.3 and the default `-mabi=ibmlongdouble`, they don’t reflect the “double double” representation of `long double` correctly; the non-IEEE representation is tricky, as it is a pair of IEEE `double` values such that adding the two elements and rounding to even yields the high element of the pair, and arithmetic overflow has undefined behavior instead of reliably yielding infinity.
- The macros `FLT_HAS_SUBNORM`, `FLT_DECIMAL_DIG`, `FLT_TRUE_MIN`, `DBL_HAS_SUBNORM`, `DBL_DECIMAL_DIG`, `DBL_TRUE_MIN`, `LDBL_HAS_SUBNORM`, `LDBL_DECIMAL_DIG`, `LDBL_TRUE_MIN`, that were added in ISO C 11, are missing on some platforms: gcc 4.5.4, clang 7, Solaris 11.3 with Sun C.
- The macro `LDBL_DECIMAL_DIG` is missing on some platforms: MSVC 14.
- The value of macro `FLT_TRUE_MIN` is not a `float` on some platforms: AIX 7.1 with xlc.
- The macros `FLT_IS_IEC_60559`, `FLT_NORM_MAX`, `FLT_SNAN`, `DBL_IS_IEC_60559`, `DBL_NORM_MAX`, `DBL_SNAN`, `LDBL_IS_IEC_60559`, `LDBL_NORM_MAX`, `LDBL_SNAN`, that

were added in ISO C 23, are missing on some platforms: gcc 14 without `-std=c23` or `-std=gnu23`, clang 18, musl libc 1.2.5, AIX 7.3 with xlclang, Solaris 11.3 with Sun C, MSVC 14.

Portability problems not fixed by Gnulib:

- The macro `FLT_ROUNDS` is a constant expression and does not represent the current rounding mode on some platforms: glibc 2.11, HP-UX 11, mingw.
- The value of `FLT_TRUE_MIN` is not a constant on some platforms: FreeBSD/x86.
- The values of `FLT_SNAN`, `DBL_SNAN`, `LDBL_SNAN` are not constants on some platforms: AIX 7.3 with xlclang, and all platforms which use the gnulib replacements.

9.15 fmtmsg.h

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/fmtmsg.h.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on some platforms: OpenBSD 7.5, Minix 3.1.8, Cygwin, mingw, MSVC 14, Android 9.0.

9.16 fnmatch.h

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/fnmatch.h.html>

Gnulib module: `fnmatch-h`, `fnmatch-gnu`

Portability problems fixed by Gnulib module `fnmatch-h`:

- This header file is missing on some platforms: mingw, MSVC 14.

Portability problems fixed by Gnulib module `fnmatch-gnu`, together with module `fnmatch-h`:

- The macros `FNM_LEADING_DIR` and `FNM_CASEFOLD` are not defined on some platforms: AIX 7.2, HP-UX 11.31, Solaris 10.
- The macro `FNM_EXTMATCH` is not defined on all non-glibc platforms: musl libc, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.2, Minix 3.3, AIX 7.2, HP-UX 11.31, Solaris 11.4, Cygwin 3.4.6, Android 13.
- The macro `FNM_FILE_NAME` is not defined on some platforms: NetBSD 10.0, AIX 7.2, HP-UX 11.31, Solaris 10.

Portability problems not fixed by Gnulib:

9.17 `ftw.h`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/ftw.h.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on some platforms: FreeBSD 5.2.1, NetBSD 3.0, Minix 3.1.8, mingw, MSVC 14.

9.18 `glob.h`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/glob.h.html>

Gnulib module: `glob-h`

Portability problems fixed by Gnulib:

- This header file is missing on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

9.19 `grp.h`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/grp.h.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on some platforms: mingw, MSVC 14.

9.20 `iconv.h`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/iconv.h.html>

Gnulib module: `iconv`

Portability problems fixed by Gnulib:

- The `<iconv.h>` from GNU libiconv is not found if installed in `$PREFIX/include`.

Portability problems not fixed by Gnulib:

- This header file is missing on some platforms: FreeBSD 6.0, OpenBSD 7.5, Minix 3.1.8, mingw, MSVC 14, when GNU libiconv is not installed.

9.21 inttypes.h

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/inttypes.h.html>

Gnulib module: inttypes-h

Portability problems fixed by Gnulib:

- This header file is missing on some platforms: MSVC 9.
- This header file is very incomplete on some platforms.
- The declarations of `imaxabs` and `imaxdiv` are missing on some platforms: NetBSD 3.0, OpenBSD 3.8, AIX 5.1, HP-UX 11.
- The declarations of `strtoimax` and `strtoumax` are missing on some platforms: OpenBSD 3.8, AIX 5.1 (missing only `strtoumax`).
- On some hosts that predate C++11, when using C++ one must define `__STDC_FORMAT_MACROS` to make visible the declarations of format macros such as `PRIdMAX`.

Portability problems not fixed by Gnulib:

- The macros `SCN*8`, `SCN*LEAST8`, `SCN*FAST8` are missing on some platforms: mingw without `__USE_MINGW_ANSI_STDIO`.
- The macros `PRId*`, `PRIB*`, `SCNb*`, that were added in ISO C 23, are missing on many platforms: glibc 2.37 and many others.

9.22 iso646.h

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/iso646.h.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on some platforms: Minix 3.1.8, HP-UX 11.00, Cygwin, mingw.

9.23 langinfo.h

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/langinfo.h.html>

Gnulib module: langinfo-h

Portability problems fixed by Gnulib:

- This header file is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.
- The constant `CODESET` is not defined on some platforms: OpenBSD 3.8.
- The constants `ALTMON_1` to `ALTMON_12` are not defined on some platforms: glibc 2.26, musl libc, macOS 11.1, NetBSD 8.0, OpenBSD 7.5, AIX 7.2, HP-UX 11.31, Solaris 11.4, Haiku, Cygwin 2.9.
- The constants `ABALTMON_1` to `ABALTMON_12` are not defined on many platforms: glibc 2.42, musl libc, macOS 14, NetBSD 10.0, OpenBSD 7.6, AIX 7.3, HP-UX 11.31, Solaris 11.4, Haiku, Cygwin 3.5.

- The constants `ERA`, `ERA_D_FMT`, `ERA_D_T_FMT`, `ERA_T_FMT`, `ALT_DIGITS` are not defined on some platforms: OpenBSD 7.5.

Portability problems not fixed by GnuLib:

9.24 `libgen.h`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/libgen.h.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This header file is missing on some platforms: mingw, MSVC 14.

The GnuLib module `dirname` provides similar API, with functions `base_name` and `dir_name` that also work with Windows file names.

9.25 `libintl.h`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/libintl.h.html>

Defines the macros `__USE_GNU_GETTEXT`, `__GNU_GETTEXT_SUPPORTED_REVISION`, and declares the functions `gettext`, `dgettext`, `dcgettext`, `ngettext`, `dngettext`, `dcngettext`, `textdomain`, `bindtextdomain`, `bind_textdomain_codeset`.

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Message-catalogs-with-gettext.html,
- https://www.gnu.org/software/gettext/manual/html_node/gettext.html.

GnuLib module: `gettext`

Portability problems fixed by GnuLib, if GNU `gettext` is installed:

- This header file is missing on some platforms: macOS 14, FreeBSD 14.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, mingw, MSVC 14, Android 9.0.
- The functions cannot deal with GNU `.mo` files with system-dependent strings (of major version 1 or of minor version 1) on some non-glibc platforms: NetBSD 3.0, Solaris 10.

Portability problems not fixed by GnuLib:

9.26 `limits.h`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/limits.h.html>

GnuLib module: `limits-h` or `gethostname`

Portability problems fixed by GnuLib module `limits-h`:

- The macros `LLONG_MIN`, `LLONG_MAX`, `ULLONG_MAX` are not defined on some platforms: older glibc systems (e.g. Fedora 1), AIX 5.1, HP-UX 11, OpenVMS.

- The macro `MB_LEN_MAX` is not defined on some platforms: pcc 1.2.0.DEVEL 20220331.
- The macros `WORD_BIT`, `LONG_BIT` are not defined on some platforms: glibc 2.11 without `-D_GNU_SOURCE`, Cygwin, mingw, MSVC 14.
- Macros like `CHAR_WIDTH` are not defined on some platforms: glibc 2.24, NetBSD 10.0, many others.
- The macros `BOOL_MAX` and `BOOL_WIDTH` are not defined on some platforms: glibc 2.32, many others.
- The macro `BOOL_MAX` is not defined with some compilers: clang 15.0.6.
- The macro `SSIZE_MAX` is not defined on some platforms: MSVC 14.

Portability problems fixed by Gnulib module `gethostname`:

- The `HOST_NAME_MAX` macro is not defined on some platforms: macOS 11.1, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.5.x, mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- The macro `SSIZE_MAX` has the wrong type, albeit with the correct value: 32-bit glibc 2.24 (on some architectures), Cygwin 2.5.2.

For `PATH_MAX`, Gnulib provides a module `pathmax` with a header file `"pathmax.h"`. It defines `PATH_MAX` to a constant on platforms with a file name length limit.

9.27 locale.h

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/locale.h.html>

Gnulib module: `locale-h`

Portability problems fixed by Gnulib:

- The definition of `'LC_MESSAGES'` is missing on some platforms: mingw, MSVC 14.
- The `locale_t` type is not defined on some platforms: glibc 2.11, macOS 11.1, FreeBSD 9.0, NetBSD 6.1, OpenBSD 6.1, Solaris 11.3, mingw, MSVC 14, Android 4.4.
- The `struct lconv` type does not contain any members on some platforms: Android up to 2014.
- The `struct lconv` type does not contain the members `int_p_cs_precedes`, `int_p_sign_posn`, `int_p_sep_by_space`, `int_n_cs_precedes`, `int_n_sign_posn`, `int_n_sep_by_space` on some platforms: glibc, OpenBSD 4.9, HP-UX 11, Solaris 11.4, Cygwin 1.5.x, mingw, MSVC 14.
- Some platforms provide a `NULL` macro that cannot be used in arbitrary expressions: NetBSD 5.0

Portability problems not fixed by Gnulib:

9.28 `math.h`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/math.h.html>

Gnulib module: `math-h`

Portability problems fixed by Gnulib:

- The conversion from `int` to `long double` is incorrect on some platforms: glibc 2.7 on Linux/SPARC64.
- The macro `NAN` is not defined on some platforms: OpenBSD 4.0, AIX 5.1.
- The macro `NAN` is not exposed outside of C99 compilation on some platforms: glibc.
- The macros `NAN` and `HUGE_VAL` expand to a function address rather than a floating point constant on some platforms: Solaris 10.
- The macros `HUGE_VALF` and `HUGE_VALL` are not defined on some platforms: glibc/HPPA, glibc/SPARC, AIX 5.1, Solaris 9, MSVC 9.
- The macros `FP_ILOGB0` and `FP_ILOGBNAN` are not defined on some platforms: NetBSD 5.1, AIX 5.1, Solaris 9, MSVC 9.
- `HUGE_VAL`, `HUGE_VALF`, `HUGE_VALL` are not compile time constants on some platforms: FreeBSD 5.2, AIX 7.1.
- `INFINITY` and `NAN` are not compile time constants on some platforms: FreeBSD 5.2, AIX 7.1.
- `INFINITY` and `NAN` are of type `double` instead of `float` on some platforms: FreeBSD 7.1, mingw 5.0.
- The macros `NAN`, `HUGE_VALL`, and `INFINITY` are not defined on some platforms: OpenVMS.

Portability problems not fixed by Gnulib:

- `NAN` is not a compile time constant with some compilers: OpenVMS.
- The macro or variable `math_errhandling` is not defined on some platforms: glibc 2.11, OpenBSD 4.9, NetBSD 5.1, UP-UX 11, Cygwin 1.7.9, mingw, MSVC 9.
- The macros `FP_ILOGB0` and `FP_ILOGBNAN` have wrong values on some platforms: Haiku before 2023-04-08.

9.29 `monetary.h`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/monetary.h.html>

Gnulib module: `monetary-h`

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on some platforms: NetBSD 3.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 1.7.1, mingw, MSVC 14, Android 9.0.
- This header file has a syntax error in C++ mode on some platforms: NetBSD 8.0.

9.30 `mqueue.h`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/mqueue.h.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on some platforms: macOS 14, FreeBSD 6.0, NetBSD 3.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 1.5.x, mingw, MSVC 14, Android 9.0.

9.31 `ndbm.h`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/ndbm.h.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on some platforms: Cygwin 3.0.x, mingw, MSVC 14, Android 9.0.

9.32 `net/if.h`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/net_if.h.html

Gnulib module: `net_if-h`

Portability problems fixed by Gnulib:

- This header file is missing on some platforms: mingw, MSVC 14.
- This header file is not self-contained on some platforms (needing `<sys/socket.h>` to be included first): Mac OS X 10.5, FreeBSD 8.2, OpenBSD 5.2.

Portability problems not fixed by Gnulib:

9.33 `netdb.h`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/netdb.h.html>

Gnulib module: `netdb-h`

Portability problems fixed by Gnulib:

- This header file is missing on some platforms: mingw, MSVC 14.
- This header file is incomplete on some platforms: Cygwin 1.5.x, Haiku.
- This header file does not define `AI_ALL`, `AI_V4MAPPED` on some platforms: NetBSD 10.0.
- This header file does not define `AI_ADDRCONFIG` on some platforms: NetBSD 5.0.
- This header file does not define `AI_NUMERICHOST` on some platforms: HP-UX 11.11.

- This header file does not define `AI_NUMERICSERV` on some platforms: macOS 10.5, HP-UX 11.11, mingw.
- This header file incorrectly defines `NI_MAXHOST` on some platforms: musl libc 1.2.4, OpenBSD 7.5.

Portability problems not fixed by GnuLib:

9.34 `netinet/in.h`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/netinet_in.h.html

GnuLib module: `netinet_in-h`

Portability problems fixed by GnuLib:

- This header file is missing on some platforms: mingw, MSVC 14.
- This header file is not self-contained on some platforms (it requires `<sys/types.h>` to be included first): OpenBSD 4.6.

Portability problems not fixed by GnuLib:

9.35 `netinet/tcp.h`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/inet_tcp.h.html

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This header file is missing on some platforms: mingw, MSVC 14.

9.36 `nl_types.h`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/nl_types.h.html

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This header file is missing on some platforms: Minix 3.1.8, Cygwin 2.9.x, mingw, MSVC 14, Android 9.0.

9.37 `poll.h`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/poll.h.html>

GnuLib module: `poll-h`

Portability problems fixed by GnuLib:

- This header file is missing on some platforms: mingw, MSVC 14.

Portability problems not fixed by GnuLib:

9.38 pthread.h

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/pthread.h.html>

Gnulib module: pthread-h

Portability problems fixed by Gnulib:

- This header file is missing on some platforms. Minix 3.1.8, mingw 2.x, MSVC 14. But the provided replacement covers only the essential POSIX threads API. Furthermore it is just a dummy on some of these platforms: Minix 3.1.8.
- This header pollutes the namespace with some broken macro implementations for various functions such as `strtok_r` and `gmtime_r`: mingw 3.0.

Portability problems not fixed by Gnulib:

9.39 pwd.h

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/pwd.h.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on some platforms: mingw, MSVC 14.

9.40 regex.h

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/regex.h.html>

Gnulib module: regex

Portability problems fixed by Gnulib:

- This header file is missing on some platforms: mingw, MSVC 14.
- On some systems that have this header file, GNU extensions like `re_set_syntax` and `RE_SYNTAX_EMACS` are not declared or supported: FreeBSD 14.2, OpenBSD 7.6, NetBSD 10.1, macOS 15, Minix 3.3.0, AIX 7.3, HP-UX 11, Solaris 11.4.
- On some systems that support GNU extensions, `RE_SYNTAX_EMACS` is 0 even though it should be `(RE_CHAR_CLASSES | RE_INTERVALS)` to be compatible with Emacs 21 (2001) and later: glibc 2.42, Cygwin 2.6.x.

Portability problems not fixed by Gnulib:

- This header file is not self-contained on some platforms: it requires `<sys/types.h>` to be included first.

9.41 sched.h

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/sched.h.html>

Gnulib module: sched-h

Portability problems fixed by Gnulib:

- This header file is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.
- This header file does not define the type `pid_t` on some platforms: glibc 2.11, macOS 11.1.
- `struct sched_param` is not defined on some platforms: Haiku.
- `SCHED_FIFO`, `SCHED_RR`, `SCHED_OTHER` are not defined on some platforms: Haiku.

Portability problems not fixed by Gnulib:

9.42 search.h

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/search.h.html>

Gnulib module: search-h

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on some platforms: Minix 3.1.8.

9.43 semaphore.h

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/semaphore.h.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

9.44 setjmp.h

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/setjmp.h.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

9.45 `signal.h`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/signal.h.html>

Gnulib module: `signal-h`

Portability problems fixed by Gnulib:

- `volatile sig_atomic_t` is rejected by older compilers on some platforms: AIX.
- `sigset_t` is missing on some platforms: MSVC 14.
- `sigset_t` is only declared in `<sys/types.h>` on some platforms: mingw.
- `struct sigaction` and `siginfo_t` are missing on some platforms: mingw, MSVC 14.
- The type `pid_t` is not defined on some platforms: MSVC 14.
- The signal `SIGPIPE` is not defined on some platforms: mingw, MSVC 14.
- The macros `SA_RESETHAND` and `SA_RESTART` are not defined on some platforms: Non-Stop.
- The type `sighandler_t` (a GNU extension) is not defined on most non-glibc platforms: macOS 11.1, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin, mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- Many signals are not defined on some platforms: mingw, MSVC 14.
- The macro `SIGBUS` is set to the same value as `SIGSEGV`, rather than being a distinct signal, on some platforms: Haiku.

9.46 `spawn.h`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/spawn.h.html>

Gnulib module: `spawn-h`

Portability problems fixed by Gnulib:

- This header file is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 1.7.25, mingw, MSVC 14.

Portability problems not fixed by Gnulib:

9.47 `stdalign.h`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> sections 6.5.3.4, 6.7.5, 7.15.

C++11 specification:

<https://www.open-std.org/jtc1/sc22/wg21/docs/papers/2011/n3242.pdf> section 18.10.

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/stdalign.h.html>

Gnulib module: `alignasof`

Portability problems fixed by Gnulib:

- On older C platforms `<stdalign.h>` must be included before using `alignas` or `alignof`. For example, GCC versions before 13 do not support these keywords, which were standardized by C23. On C23 and later platforms, `<stdalign.h>` has no effect and need not be included. (Gnulib-using code should not include `<stdalign.h>` without also employing Gnulib's now-deprecated `stdalign-h` module.)
- This header file is missing on many platforms: FreeBSD 6.4, NetBSD 7.1, OpenBSD 7.5, Minix 3.3.0, AIX 7.1, HP-UX 11.31, Solaris 11.3, mingw, MSVC 14, Android 9.0.
- Clang 3.0's `<stdalign.h>` does not define `alignof`.
- The `alignof` macro returns too large values for the types `double` and `long long` in GCC 4.7.0.
- Older C platforms might not support the obsolescent `_Alignas` and `_Alignof` keywords or macros. This portability problem should not matter with code using this module, as such code should use `alignas` and `alignof` instead.
- In C11 and C17, `<stdalign.h>` defines the macros `__alignas_is_defined` and `__alignof_is_defined` to 1. In C23, these macros are not defined. This portability problem should not matter with code using Gnulib's `alignasof` module, as such code should use `alignas` and `alignof` without checking these two macros. (Gnulib's now-deprecated `stdalign-h` module defines these two macros.)
- Clang 17 does not support an `alignas` attribute in the specifier-qualifier position of a member declaration.

Portability problems not fixed by Gnulib:

- In C11 and later, the operand of `alignof` must be a parenthesized type. Recent versions of GCC support an extension in which the operand can also be a unary expression, as with `sizeof`. The Gnulib substitute does not support this extension.
- On most pre-C11 platforms, the operand of `alignof` cannot be a structure type containing a flexible array member.
- The `alignas` keyword or macro is not always supported. Supported compilers include any compiler supporting C11 or later, which includes GCC, IBM C, Sun C 5.9 and later, and MSVC 7.0 and later.
- Some compilers do not support alignment via `alignas` of `auto` variables (i.e., variables on the stack). They diagnose and ignore the alignment: Sun C 5.11.
- Some linkers do not support operands of `alignas` that are greater than 8: mingw.
- Some compilers require the operand of `alignas` to be a single integer constant, not an expression: MSVC 7.0 through at least 10.0.
- The Sun C 5.13 (2014) compiler sometimes mishandles the alignment of multiple external variables that are declared close together with `alignas`. The bug is fixed in Sun C 5.15, also known as Oracle Developer Studio 12.6 (2017).
- You cannot assume that `alignas` and `alignof` are reserved words; they might be macros.

9.48 stdarg.h

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/stdarg.h.html>

Gnulib module: stdarg-h

Portability problems fixed by Gnulib:

- Some compilers (e.g., AIX 5.3 cc) need to be in c99 mode for the builtin `va_copy` to work.

Portability problems not fixed by Gnulib:

9.49 stdatomic.h

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.17.

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/stdatomic.h.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on many platforms: glibc with GCC 4.8.x or clang 2.8, FreeBSD 6.4, NetBSD 7.0, OpenBSD 6.5, OpenBSD 6.9 with gcc, Minix 3.3.0, AIX 7.2 with xlc or xlclang, HP-UX 11.31, Solaris 11.3 with Oracle Studio 12.4 (Sun C 5.13), Cygwin 1.5.19, mingw, MSVC 14.
- Linus Torvalds believes that “no compiler writer will get the C/C++ memory model right”; see <https://lkml.org/lkml/2018/6/7/761>.

9.50 stdbit.h

POSIX :

Not in POSIX yet, but we expect it will be. ISO C23 (latest free draft <https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf>) section 7.18.

Gnulib module: stdbit

Portability problems fixed by Gnulib:

- This header file is missing on many non-C23 platforms: glibc 2.38, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 17, Android 15.

Portability problems not fixed by Gnulib:

- On older non-C23 platforms lacking `typeof` or equivalent, a call to `stdc_bit_floor` and `stdc_bit_ceil` may yield a type that is wider than its argument. Although C23 seems to allow this, it differs from GNU behavior.
- On non-C23 platforms, type-generic functions apply portably only to the standard unsigned integer types specified by C17 or earlier.

9.51 `stdbool.h`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/stdbool.h.html>

Gnulib module: `stdbool-h-c99`

The `stdbool-h-c99` module is present only for programs that formerly used the old `stdbool` module for C99 compatibility, and that for some reason cannot use the current `bool` module for C23 compatibility.

Portability problems fixed by Gnulib:

- This header file is missing on some platforms: AIX 5.1, HP-UX 11.
- This header file is not usable in C++ mode with the vendor compiler on Solaris 10.
- Some compilers have bugs relating to ‘`bool`’.
- This header file defines `true` incorrectly on some platforms: OpenBSD 4.7 with gcc 2.95.

Portability problems not fixed by Gnulib:

- ‘`_Bool`’ cannot be used before `<stdbool.h>` is included, or if the program is intended to be compiled by a C++ compiler. (With the advent of C23, ‘`_Bool`’ is obsolescent anyway.)
- You cannot assume that `_Bool` is a keyword or a typedef; it might be a macro. For example, C23 allows `_Bool` to be a macro.
- Bit-fields of type ‘`bool`’ are not supported. Portable code should use ‘`unsigned int foo : 1;`’ rather than ‘`bool foo : 1;`’.
- Casts and automatic conversions to ‘`bool`’ don’t test against the zero value or the null pointer, as they should. Such casts should only be used if the value is known to be equal to 0 or 1.
- You cannot assume that casting a floating point literal to ‘`bool`’ will result in a constant expression.

9.52 `stdckdint.h`

POSIX specification:

Not in POSIX yet, but we expect it will be. ISO draft C23 (<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf>) section 7.20.

Gnulib module: `stdckdint-h`

Portability problems fixed by Gnulib:

- This header file is missing on many platforms: glibc 2.29, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9.0, mingw, MSVC 14, Android 9.0.

Portability problems not fixed by Gnulib:

- In draft C23, arguments of `stdckdint.h` macros can have side effects.

9.53 `stddef.h`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/stddef.h.html>

Gnulib module: `stddef-h`

Portability problems fixed by Gnulib:

- Some platforms fail to provide `unreachable`, which was added in C23: GCC 13, clang 15, AIX with xlc 12.1, Solaris with Sun C 5.15, and others.
- Some platforms fail to provide `max_align_t`, which was added in C11: NetBSD 8.0, Solaris 11.0, and others.
- `max_align_t` does not have the expected alignment on some platforms: NetBSD 8.0/x86, AIX 7.2 with xlc in 64-bit mode.
- Some platforms provide a `NULL` macro that cannot be used in arbitrary expressions: NetBSD 5.0
- Some platforms provide a `NULL` macro whose value does not have the size of a pointer: AIX 7.2 with xlc in 64-bit mode.
- When this header file is provided by TinyCC 0.9.27 on glibc or macOS systems, it does not fulfil the expectations of other system header files.

Portability problems not fixed by Gnulib:

- Some platforms fail to provide `nullptr_t`, which Gnulib cannot usefully emulate: GCC 12, Clang 15, and other pre-2023 C compilers.
- Some platforms define `nullptr_t` even when `<stddef.h>` is not included: GCC 14
- Some platforms provide an `offsetof` macro that cannot be used in arbitrary expressions: Solaris 11.4 This problem can be worked around by parenthesizing the `offsetof` expression in the unlikely case you use it with `sizeof` or `[]`.

9.54 `stdint.h`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/stdint.h.html>

Gnulib module: `stdint-h`

Portability problems fixed by Gnulib:

- This header file is missing on some platforms: OpenBSD 3.8, AIX 5.1, HP-UX 11.11, MSVC 9.
- This header file is very incomplete on some platforms.
- The values of `SIG_ATOMIC_MIN` and `SIG_ATOMIC_MAX` are incorrect on some platforms: FreeBSD 6.2/ia64, FreeBSD 13.0/arm64.
- The value of `WINT_MAX` is incorrect on some platforms: mingw.
- The values of `INT8_MAX`, `UINT8_MAX` etc. are not usable in preprocessor expressions on some platforms: HP-UX 11.23.
- The values of `INTPTR_MAX` and `UINTPTR_MAX`, although correctly defined in `<stdint.h>`, are replaced by empty values when `<limits.h>` or `<inttypes.h>` gets included later on some platforms: Solaris 9 with GCC 4.5 or newer.

- The macros `WCHAR_MIN` and `WCHAR_MAX` are not defined in `<stdint.h>` (only in `<wchar.h>`) on some platforms: Dragonfly.
- On some hosts that predate C++11, when using C++ one must define `__STDC_CONSTANT_MACROS` to make visible the definitions of constant macros such as `INTMAX_C`, and one must define `__STDC_LIMIT_MACROS` to make visible the definitions of limit macros such as `INTMAX_MAX`.
- The macro `SIZE_MAX` has the wrong type, albeit with the correct value: 32-bit glibc 2.24 (on s390 architecture), Mac OS X 10.7.
- Macros like `INTMAX_WIDTH` are not defined on some platforms: glibc 2.24, NetBSD 10.0, many others.

Portability problems not fixed by Gnulib:

- `{uint,int}_fast{8,16,32,64}_t` may not correspond to the fastest types available on the system. Other `<stdint.h>` substitutes may define these types differently, so public header files should avoid these types.
- Macros are used instead of typedefs.
- Some C preprocessors mishandle constants that do not fit in `long int`. For example, as of 2007, Sun C mishandled `#if LLONG_MIN < 0` on a platform with 32-bit `long int` and 64-bit `long long int`; this bug was fixed on or before Oracle Developer Studio 12.6 (Sun C 5.15 SunOS_sparc 2017/05/30). Some older preprocessors mishandle constants ending in `LL`. To work around these problems, compute the value of expressions like `LONG_MAX < LLONG_MAX` at `configure`-time rather than at `#if`-time.

The `stdint-h` module uses `#include_next`. If you wish to install the generated `stdint.h` file under another name, typically in order to be able to use some of the types defined by `stdint.h` in your public header file, you could use the following Makefile.am-snippet:

```
BUILT_SOURCES += idn-int.h
DISTCLEANFILES += idn-int.h
nodist_include_HEADERS += idn-int.h

idn-int.h:
if test -n "$(STDINT_H)"; then \
sed -e s/include_next/include/ gl/stdint.h > idn-int.h; \
else \
echo '#include <stdint.h>' > idn-int.h; \
fi
```

9.55 stdio.h

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/stdio.h.html>

Gnulib module: `stdio-h`

Portability problems fixed by Gnulib:

- The type `off_t` is not defined on some platforms: glibc 2.8, eglibc 2.11.2 and others.

- The type `ssize_t` is not defined on some platforms: glibc 2.8, Mac OS X 10.5, Solaris 10, MSVC 14, and others.
- The type `va_list` is not defined on some platforms: glibc 2.8, OpenBSD 4.0, Solaris 11.4, and others.
- The type `off64_t` is not defined on some platforms: macOS 14, FreeBSD 10.4, NetBSD 10.0, OpenBSD 7.5, MSVC 14, Cygwin, Haiku, Minix 3.3.
- Some platforms provide a `NULL` macro that cannot be used in arbitrary expressions: NetBSD 5.0

Portability problems not fixed by Gnulib:

9.56 `stdlib.h`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/stdlib.h.html>

Gnulib module: `stdlib-h`, `system-posix`

Portability problems fixed by the Gnulib module `stdlib-h`:

- The macros `EXIT_SUCCESS` and `EXIT_FAILURE` are not defined on some platforms.
- Some platforms provide a `NULL` macro that cannot be used in arbitrary expressions: NetBSD 5.0
- Using the macro `MB_CUR_MAX` causes link errors on some platforms: Android NDK r16.
- The value of `MB_CUR_MAX` is too small (3 instead of 4) in UTF-8 locales on some platforms: Solaris 10.

Portability problems fixed by the Gnulib module `system-posix`:

- The macros `WIFSIGNALED`, `WIFEXITED`, `WIFSTOPPED`, `WTERMSIG`, `WEXITSTATUS`, `WNOHANG`, `WUNTRACED`, `WSTOPSIG` are not defined in this header file (only in `<sys/wait.h>`) on some platforms: MirBSD 10.
- The macro `WCOREDUMP` is not defined in this header file (only in `<sys/wait.h>`) on some platforms: glibc 2.42.

Portability problems not fixed by Gnulib:

- The definition of the type `once_flag`, of the macro `ONCE_FLAG_INIT`, and the declaration of the function `call_once`, that are required by ISO C 23, are not provided. To get them, import Gnulib module `call_once` and include `<threads.h>` rather than `<stdlib.h>`.
- System status macros such as `WEXITSTATUS` require an lvalue argument on some platforms. macOS 11.1.

9.57 `stdnoreturn.h`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.25.

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/stdnoreturn.h.html>

Gnulib module: `stdnoreturn-h`

Portability problems fixed by Gnulib:

- This header file is missing on many platforms: FreeBSD 6.4, NetBSD 7.1, OpenBSD 7.5, Minix 3.3.0, AIX 7.1, HP-UX 11.31, Solaris 11.3, Cygwin 2.9.0, mingw, MSVC 14, Android 9.0.
- This file conflicts with some system header files, such as `<stdlib.h>` and `<process.h>`, on some platforms: MSVC/clang.

Portability problems not fixed by Gnulib:

- `<stdnoreturn.h>` and the `noreturn` macro are obsolescent in C23.
- `<stdnoreturn.h>` cannot be `#included` in C++ mode on some platforms: FreeBSD 13.1.
- `<stdnoreturn.h>` should be `#included` before `'_Noreturn'` is used.
- You cannot assume that `_Noreturn` is a reserved word; it might be a macro.
- When the macro `lint` is defined, standard headers define `_Noreturn` (and therefore `noreturn`) to be a macro that expands to the empty token sequence on some platforms: Cygwin 2.5.1, FreeBSD 10.3.
- On Cygwin 1.7.30 and MSVC 14, `noreturn` expands to the empty token sequence, to avoid problems with standard headers that use `noreturn` in combination with `__attribute__` or `__declspec`. Although the resulting code operates correctly, the compiler is not informed whether `noreturn` functions do not return, so it may generate incorrect warnings at compile-time, or code that is slightly less optimized. This problem does not occur with `_Noreturn`.
- Circa 2012 bleeding-edge GCC with `-Werror=old-style-declaration` requires `_Noreturn` or `noreturn` before the returned type in a declaration, and therefore rejects valid but unusually-worded declarations such as `void _Noreturn foo (void);`.

9.58 `string.h`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/string.h.html>

Gnulib module: `string-h`

Portability problems fixed by Gnulib:

- Some platforms provide a `NULL` macro that cannot be used in arbitrary expressions: NetBSD 5.0

Portability problems not fixed by Gnulib:

9.59 `strings.h`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/strings.h.html>

Gnulib module: `strings-h`

Portability problems fixed by Gnulib:

- This header file is not self-contained on some platforms: Minix 3.1.8.

Portability problems not fixed by Gnulib:

- This header file is missing on some platforms: MSVC 14.
- This header file defines symbols, such as ‘`index`’, often used for variables, making debugging harder.

9.60 `stropts.h`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/stropts.h.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin, mingw, MSVC 14, Android 9.0.
- This header is removed in POSIX.1-2024.

9.61 `sys/ipc.h`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/sys_ipc.h.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on some platforms: mingw, MSVC 14.

9.62 `sys/mman.h`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/sys_mman.h.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on some platforms: mingw, MSVC 14.

9.63 `sys/msg.h`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/sys_msg.h.html

Gnulib module: `sys_msg-h`

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

9.64 sys/resource.h

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/sys_resource.h.html

Gnulib module: sys_resource-h

Portability problems fixed by Gnulib:

- This header file is missing on some platforms: mingw, MSVC 14.
- On some platforms, this header file requires that `<sys/types.h>` and `<sys/time.h>` already be included: FreeBSD 5.0.
- On some platforms, this header file does not define the `RUSAGE_SELF` and `RUSAGE_CHILDREN` constants: OpenVMS.

Portability problems not fixed by Gnulib:

- On some platforms, this header does not define some or all of the symbolic constants required by POSIX. For example, OpenVMS and Android do not define `RLIM_SAVED_CUR` or `RLIM_SAVED_MAX`.

9.65 sys/select.h

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/sys_select.h.html

Gnulib module: sys_select-h

Portability problems fixed by Gnulib:

- This header file is missing on some platforms: HP-UX 11.11, NonStop Kernel, mingw, MSVC 14.
- This header file is not self-contained on some platforms: it requires `<sys/types.h>` to be included first.
- This header file is not self-contained—it requires `<string.h>` before `FD_ZERO` can be used—on some platforms: AIX 7.1, Solaris 11.4.

Portability problems not fixed by Gnulib:

9.66 sys/sem.h

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/sys_sem.h.html

Gnulib module: sys_sem-h

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on some platforms: mingw, MSVC 14.

9.67 sys/shm.h

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/sys_shm.h.html

Gnulib module: sys_shm-h

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on some platforms: mingw, MSVC 14.

9.68 sys/socket.h

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/sys_socket.h.html

Gnulib module: sys_socket-h

Portability problems fixed by Gnulib:

- This header file is missing on some platforms: mingw, MSVC 14.
- This header file is not self-contained on some platforms: it requires `<sys/types.h>` to be included first.
- This header file does not define the type `struct iovec` on some platforms: OpenBSD 4.4.
- This header file is lacking the `SHUT_RD`, `SHUT_WR`, `SHUT_RDWR` macros on some platforms, despite having the `shutdown` functions: emx+gcc.
- The `struct sockaddr_storage` type does not have a member `ss_family` on some platforms: AIX 7.1.
- The `CMSG_SPACE` and `CMSG_LEN` macros are not provided on some platforms: OpenVMS.
- This header file does not define the `SO_REUSEPORT` macro on some platforms: Minix 3.1.8, Solaris 10, Cygwin, mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- This header file does not declare the `msg_control` and `msg_controllen` members of `struct msghdr` on some platforms. This can be detected by the absence of the `CMSG_FIRSTHDR` macro: gnulib replacement header, old BSD
- This header file does not define the macro `SOCK_NONBLOCK` on some platforms: macOS 15, AIX 7.3, Solaris 10, mingw, MSVC 14.
- This header file does not define the macros `SOCK_CLOEXEC` and `MSG_CMSG_CLOEXEC` on some platforms: macOS 15, AIX 7.3, Solaris 10, mingw, MSVC 14.
- This header file does not define the macro `MSG_CMSG_CLOEXEC` on some platforms: Solaris 11 OmniOS.
- Although POSIX.1-2024 standardized `SOCK_CLOFORK` and `MSG_CMSG_CLOFORK`, no known platform supports them.
- This header file does not define the macro `SO_DOMAIN` on some platforms: macOS 15, NetBSD 10.0, AIX 7.3, Cygwin 3.5, mingw, MSVC 14.
- This header file does not define the macro `SO_PROTOCOL` on some platforms: macOS 15, NetBSD 10.0, AIX 7.3, Solaris 11.4, Cygwin 3.5, mingw, MSVC 14.

- This header file does not define the macro `MSG_NOSIGNAL` on some platforms: Solaris 10, mingw, MSVC 14.
- This header file does not define the macro `MSG_EOR` on some platforms: mingw, MSVC 14.
- This header file does not define the macros `MSG_CTRUNC` and `MSG_TRUNC` on some platforms: mingw.
- This header file does not define the macro `MSG_WAITALL` on some platforms: MSVC 14.
- This header file does not define the type `struct cmsghdr` and the constant `SCM_RIGHTS` on some platforms: mingw, MSVC 14.
- This header file does not define the constant `SOCK_CLOFORK` on some platforms: glibc 2.42, musl libc, macOS, FreeBSD, NetBSD, OpenBSD, AIX, Solaris 11.3, Cygwin, mingw, MSVC, Android.
- This header file does not define the constant `MSG_CMSG_CLOFORK` on some platforms: glibc 2.42, musl libc, macOS, FreeBSD, NetBSD, OpenBSD, AIX, Solaris 11.4, Cygwin, mingw, MSVC, Android.

9.69 `sys/stat.h`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/sys_stat.h.html

Gnulib module: `sys_stat-h`

Portability problems fixed by Gnulib module `sys_stat-h`:

- The type `mode_t` is not defined on some platforms: MSVC 14.
- The type `blksize_t` is not defined on some platforms: FreeBSD 5.5, OpenBSD 5.6, HP-UX 11.23, mingw, MSVC 14.
- The type `blkcnt_t` is not defined on some platforms: FreeBSD 5.5, OpenBSD 5.6, mingw, MSVC 14.
- Some macros, such as `S_IFMT` or `S_IFIFO`, are missing on some platforms.
- The macros `S_ISBLK`, `S_ISCHR`, `S_ISDIR`, `S_ISFIFO`, `S_ISLNK`, `S_ISREG`, `S_ISSOCK` are broken on some platforms.
- Some platforms define macros, such as `S_ISDOOR`, that are not defined on other platforms.
- The functions `lstat` and `mkdir` are not declared on some platforms: mingw, MSVC 14.
- The macros `UTIME_NOW` and `UTIME_OMIT` are missing on some platforms.
- On some platforms, `struct stat` does not include `st_atim`, `st_mtim`, or `st_ctim` members. Use the gnulib module ‘`stat-time`’ for accessors to portably get at subsecond resolution.

Portability problems fixed by Gnulib module `sys_stat-h`, together with module `windows-stat-inodes`:

- On Windows platforms (excluding Cygwin), `st_ino` is always 0.

On platforms where `off_t` is a 32-bit type, `struct stat` cannot represent the size of files or block devices 2 GiB or larger and may not work correctly on directories 2 GiB or

larger. Also, on platforms where `ino_t` is a 32-bit type, `struct stat` cannot represent larger inode numbers. See [Section 14.2 \[Large File Support\]](#), page 842, for how to address these problems.

See [Section 14.5 \[Avoiding the year 2038 problem\]](#), page 843, for portability issues with the `time_t` components of `struct stat`.

Portability problems not fixed by Gnulib:

- On some platforms the types `blksize_t` and `blkcnt_t` are unsigned: QNX, Android, glibc/alpha.
- The macro `S_IFBLK` is missing on some platforms: MSVC 14.
- On OpenVMS, `st_ino` is an array of three `ino_t` values, not a single value. `b` are known to represent the same file.
- On some platforms, two different files may have the same `st_dev` and `st_ino` values, even when `st_ino` is nonzero:
 - GNU/Linux NFS servers that export all local file systems as a single NFS file system, if a local `st_dev` exceeds 255, or if a local `st_ino` exceeds 16777215.
 - Network Appliance NFS servers in snapshot directories; see Network Appliance bug #195.
 - ClearCase MVFS; see bug id ATRia04618.
- On some file systems, `st_size` contains bogus information for symlinks; use the Gnulib module `areadlink-with-size` for a better way to get symlink contents.

To partially work around porting problems with Microsoft Windows and OpenVMS, you can use the Gnulib `same-inode` module to test whether two `struct stat` objects are known to represent the same file. For example, `psame_inode (&a, &b)` is true if the `struct stat` objects `a` and `b` are known to represent the same file.

Another partial workaround is to compare other file metadata such as `st_mode` and `st_mtime` on platforms where `st_dev` and `st_ino` not uniquely identify a file. However, this does not work reliably on files whose metadata are being changed by other programs, or where the metadata happen to be equal.

9.70 sys/statvfs.h

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/sys_statvfs.h.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on some platforms: OpenBSD 3.8, mingw, MSVC 14.

9.71 `sys/time.h`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/sys_time.h.html

Gnulib module: `sys_time-h`

Portability problems fixed by Gnulib:

- This header file is missing on some platforms: MSVC 14.
- ‘`struct timeval`’ is not defined on some platforms.
- ‘`struct timeval`’ is defined with a `tv_sec` type that is narrower than `time_t` on some native Windows platforms: mingw64 in 64-bit mode, mingw64 in 32-bit mode when `__MINGW_USE_VC2005_COMPAT` is defined, MSVC 14 in 64-bit mode, MSVC 14 in 32-bit mode when `_USE_32BIT_TIME_T` is not defined.

See [Section 14.5 \[Avoiding the year 2038 problem\]](#), [page 843](#), for portability issues with `time_t` and the `time_t` component of `struct timeval`.

Portability problems not fixed by Gnulib:

- ‘`struct timeval`’ is defined with a `tv_sec` type that is wider than `time_t`: OpenBSD 5.1 in 64-bit mode.

9.72 `sys/timeb.h`

Removed in POSIX.1-2008. POSIX.1-2004 specification:

https://pubs.opengroup.org/onlinepubs/009695399/basedefs/sys_timeb.h.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on some platforms: OpenBSD 7.5, Android 9.0.

9.73 `sys/times.h`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/sys_times.h.html

Gnulib module: `sys_times-h`

Portability problems fixed by Gnulib:

- This header file is missing on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

9.74 `sys/types.h`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/sys_types.h.html

Gnulib module: `sys_types-h`

Portability problems fixed by Gnulib:

- The type `pid_t` is not defined on some platforms: MSVC 14.

- The type `size_t` is not defined in this file on some platforms: MSVC 14.
- The type `ssize_t` is not defined on some platforms: MSVC 14.
- The type `mode_t` is not defined on some platforms: MSVC 14.
- The type `off64_t` is not defined on some platforms: macOS 14, FreeBSD 10.4, NetBSD 10.0, OpenBSD 7.5, MSVC 14, Cygwin, Haiku, Minix 3.3.
- The type `blksize_t` is not defined on some platforms: FreeBSD 5.5, OpenBSD 5.6, HP-UX 11.23, mingw, MSVC 14.
- The type `blkcnt_t` is not defined on some platforms: FreeBSD 5.5, OpenBSD 5.6, mingw, MSVC 14.
- Some systems leak definitions of `major`, `minor`, and `makedev` through this header; however, when `sys/sysmacros.h` exists, that file should also be included to avoid deprecation warnings from the versions in this header: glibc 2.25.

See [Section 14.5 \[Avoiding the year 2038 problem\]](#), page 843, for portability issues with `time_t`.

Portability problems not fixed by GnuLib:

- On some platforms the types `blksize_t` and `blkcnt_t` are unsigned: QNX, Android, glibc/alpha.
- On some platforms the types `blksize_t` and `suseconds_t` are signed integer types that are wider than `long`: glibc x32

This module, together with the module `largefile`, also defines the type `off_t` to a 64-bit integer type on some platforms: mingw, MSVC 14.

9.75 sys/uio.h

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/sys_uio.h.html

GnuLib module: `sys_uio-h`

Portability problems fixed by GnuLib:

- This header file is missing on some platforms: mingw, MSVC 14.
- This header file is not self-contained (it requires `<sys/types.h>` to be included first) on some platforms: OpenBSD 4.4.

Portability problems not fixed by GnuLib:

9.76 sys/un.h

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/sys_un.h.html

GnuLib module: `sys_un-h`

Portability problems fixed by GnuLib:

- This header file is missing on some platforms: mingw, MSVC 14.
- This header file does not define the `sa_family_t` type on some platforms: Cygwin 3.5.5, mingw, MSVC 14.

- This header file causes compilation errors in C++ mode on some platforms: glibc 2.42.

Portability problems not fixed by GnuLib:

- This header requires `sys/socket.h` to be included first on some platforms: Cygwin 1.7.18.

9.77 `sys/utsname.h`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/sys_utsname.h.html

GnuLib module: `sys_utsname-h`

Portability problems fixed by GnuLib:

- This header file is missing on some platforms: mingw, MSVC 14.
- This header file is not self-contained on some platforms: Minix 3.1.8.

Portability problems not fixed by GnuLib:

9.78 `sys/wait.h`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/sys_wait.h.html

GnuLib module: `sys_wait-h`

Portability problems fixed by GnuLib:

- This header file is missing on some platforms: mingw, MSVC 14.
- This header file does not define the `WCOREDUMP` macro on some platforms: AIX 7.3, Haiku.

Portability problems not fixed by GnuLib:

- System status macros such as `WEXITSTATUS` require an `lvalue` argument on some platforms: macOS 11.1.

9.79 `syslog.h`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/syslog.h.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This header file is missing on some platforms: mingw, MSVC 14.

9.80 tar.h

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/tar.h.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on some platforms: Cygwin 1.5.x, mingw, MSVC 14.

9.81 termios.h

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/termios.h.html>

Gnulib module: termios-h

Portability problems fixed by Gnulib:

- This header file is missing on some platforms: mingw, MSVC 14.
- This header does not declare `pid_t` on all platforms: glibc on some architectures, FreeBSD 6.4, OpenBSD 4.9, Cygwin 1.7.11.

Portability problems not fixed by Gnulib:

- The types `struct termios`, `struct winsize`, `cc_t`, `speed_t`, `tcflag_t` are not defined on some platforms: mingw, MSVC 14.
- `struct winsize` is defined in `sys/ioctl.h` on some platforms: glibc, AIX.

9.82 tgmath.h

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/tgmath.h.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on some platforms: Mac OS X 10.5, FreeBSD 5.2.1, NetBSD 10.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 1.7.9, mingw, MSVC 14, Android 9.0.

9.83 threads.h

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.28.

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/threads.h.html>

Defines the multithreading facility of ISO C11.

Gnulib module: threads-h

Portability problems fixed by Gnulib:

- This header file is missing on many platforms: glibc 2.27, macOS 14, FreeBSD 9.3, NetBSD 8.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.3, Cygwin 2.2.x, mingw, MSVC 14, Android 9.0.
- This header file defines `thrd_start_t` incorrectly on some platforms: AIX 7.2.
- This header file does not define `TSS_DTOR_ITERATIONS` on some platforms: AIX 7.2.

Portability problems not fixed by Gnulib:

- There is no way to define a working `thread_local` macro on some platforms:
 - Mac OS X 10.5,
 - OpenBSD 6.5,
 - AIX 7.1 with gcc (but it works with `'xlc -qthreaded -qtls'`),
 - HP-UX 11.31 with cc (but it works with gcc),
 - Android 4.3.

9.84 `time.h`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/time.h.html>

Gnulib module: `time-h`

Portability problems fixed by Gnulib:

- `'struct timespec'` is not defined on some platforms.
- The macro `TIME_UTC` is not defined on many platforms: glibc 2.15, macOS 10.13, FreeBSD 11.0, NetBSD 7.1, OpenBSD 6.0, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.3, Cygwin 2.9, mingw, MSVC 14, Android 9.0.
- Some platforms provide a `NULL` macro that cannot be used in arbitrary expressions: NetBSD 5.0.

Portability problems fixed by the Gnulib module `year2038`:

- On some platforms where `time_t` defaults to 32-bit but can be changed to 64-bit, functions like `stat` can fail with `errno == EOVERFLOW` when a 32-bit timestamp is out of range, such as with a file timestamp in the far future or past: glibc 2.34+ atop 32-bit x86 or ARM Linux. See [Section 14.5 \[Avoiding the year 2038 problem\]](#), page 843.

Portability problems not fixed by Gnulib:

- On some platforms `struct tm` lacks the `tm_gmtoff` and `tm_zone` members: AIX 7.3, HP-UX 11, Solaris 11.4, mingw, MSVC 14.
- On some platforms the global state variables `daylight`, `timezone` and `tzname` are not available. Even on platforms where they are available, their contents are often unreliable, even in single-threaded programs. Portable code can instead use `struct tm`'s `tm_gmtoff` and `tm_zone` members when available, and the `strftime` function with `%z` or `%Z` conversion specifiers otherwise.
- On platforms where `time_t` is always 32-bit, functions like `stat` can fail with `errno == EOVERFLOW` when a timestamp is out of range, such as with a file timestamp in the far future or past; on other such platforms, the functions silently return the low-order

32 bits of the correct timestamp. These platforms will be obsolete when 32-bit `time_t` rolls around, which will occur in 2038 for the typical case when `time_t` is signed. See [Section 14.5 \[Avoiding the year 2038 problem\]](#), page 843.

- On some platforms the `tv_nsec` member of `struct timespec` is not of type `long`, but is of type `long long` instead: glibc x32

9.85 `trace.h`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/trace.h.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin, mingw, MSVC 14, Android 9.0.
- This header is removed in POSIX.1-2024.

9.86 `uchar.h`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.30.

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/uchar.h.html>

Defines the types `char16_t`, `char32_t` and declares the functions `mbrtoc16`, `c16rtomb`, `mbrtoc32`, `c32rtomb`.

Gnulib module: `uchar-h` or `uchar-h-c23`

Portability problems fixed by either Gnulib module `uchar-h` or `uchar-h-c23`:

- This header file is missing on many non-glibc platforms: glibc 2.15, macOS 14, FreeBSD 6.4, NetBSD 10.0, OpenBSD 7.3, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.3, Cygwin 3.4.x, mingw, MSVC 9.
- This file is not self-contained on some platforms: Haiku.
- This file produces compilation errors in C++ mode on some platforms: AIX 7.2 with `xlclang++`.

Portability problems fixed by Gnulib module `uchar-h-c23`:

- `char32_t` values may not be Unicode code points. This is the case in ISO C 11 compliant but not ISO C 23 compliant implementations.

Portability problems not fixed by Gnulib:

9.87 `ucontext.h`

Removed in POSIX.1-2008. POSIX.1-2004 specification:

<https://pubs.opengroup.org/onlinepubs/009695399/basedefs/ucontext.h.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on some platforms: OpenBSD 7.5, Cygwin 1.7.35, mingw, MSVC 14.

9.88 `ulimit.h`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/ulimit.h.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on some platforms: OpenBSD 7.5, Minix 3.1.8, Cygwin, mingw, MSVC 14, Android 9.0.
- This header is removed in POSIX.1-2024.

9.89 `unistd.h`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/unistd.h.html>

Gnulib module: `unistd-h`

Portability problems fixed by Gnulib:

- This header file is missing on some platforms: MSVC 14.
- The type `off64_t` is not defined on some platforms: macOS 14, FreeBSD 10.4, NetBSD 10.0, OpenBSD 7.5, MSVC 14, Cygwin, Haiku, Minix 3.3.
- The `SEEK_*` macros are not defined in this file on some platforms: mingw.
- The `*_FILENO` macros are not defined in this file on some platforms: OS/2 EMX, mingw.
- The `_exit` function is not declared in this file on some platforms: mingw.
- This header file does not define `_CLOEXEC` on some platforms: glibc 2.42, macOS 12, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4, Cygwin 3.5, mingw.
- Some platforms provide a `NULL` macro that cannot be used in arbitrary expressions: NetBSD 5.0

Portability problems not fixed by Gnulib:

9.90 `utime.h`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/utime.h.html>

Gnulib module: `utime-h`

Portability problems fixed by Gnulib:

- This header file is missing on some platforms: MSVC 14.

Portability problems not fixed by Gnulib:

- This header is removed in POSIX.1-2024.

9.91 `utmpx.h`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/utmpx.h.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on some platforms: FreeBSD 8.4, OpenBSD 7.2, Minix 3.1.8, mingw, MSVC 14, Android 13.0.
- While some platforms have the `struct utmpx` field `ut_user`, older platforms have the field `ut_name`.
- The `struct utmpx` field `ut_exit` does not exist on some platforms: macOS, FreeBSD, AIX, Cygwin.
- The `struct utmpx` field `ut_session` does not exist on some platforms: macOS, FreeBSD, AIX, HP-UX, Cygwin.
- The `struct utmpx` field `ut_addr` or `ut_addr_v6` or `ut_ss` does not exist on some platforms: macOS, FreeBSD, AIX, Solaris.
- On some platforms, the `struct utmpx` field `ut_tv` is not of type `struct timeval`. Instead, it is a different struct with `tv_sec` and `tv_usec` members that may have different types than the members of `struct timeval`: glibc 2.42 on platforms where `time_t` was historically 32 bits and where log file formats were not changed when 64-bit `time_t` was introduced.
- On some platforms, this API does not support timestamps past the year 2038: glibc 2.39 on 32-bit platforms like x86 and ARM where `time_t` was historically 32 bits; later glibc versions support timestamps up to the year 2106, by changing `ut_tv.tv_sec`'s type to be a 32-bit unsigned integer.
- On some platforms, this header misbehaves if the `year2038` or `year2038-recommended` modules are used and the program is configured without the `--disable-year2038` option. The `readutmp` module works around this problem: glibc 2.38 on 32-bit platforms like x86 and ARM where `time_t` was historically 32 bits. See [Section 14.5 \[Avoiding the year 2038 problem\]](#), page 843.

9.92 wchar.h

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/wchar.h.html>

Gnulib module: wchar-h

Portability problems fixed by Gnulib:

- This header file cannot be included on some platforms: Linux uClibc built without wide character support.
- The type `wint_t` is incorrect on some platforms: mingw, MSVC 14.
- Some platforms provide a `NULL` macro that cannot be used in arbitrary expressions: NetBSD 5.0

Portability problems not fixed by Gnulib:

- This header file leads to link errors and endless recursions or endless loops on some platforms: glibc version 2.5 or older, together with gcc version 4.3 or newer and the option `'-std=c99'` or `'-std=gnu99'`.

9.93 wctype.h

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/wctype.h.html>

Gnulib module: wctype-h

Portability problems fixed by Gnulib:

- This header file is missing on some platforms: HP-UX 11.00.
- The type `wint_t` is incorrect on some platforms: mingw, MSVC 14.
- The functions `isw*` are missing on some platforms: FreeBSD 4.11.

Portability problems not fixed by Gnulib:

9.94 wordexp.h

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/basedefs/wordexp.h.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on some platforms: OpenBSD 7.5, Minix 3.1.8, Cygwin 1.5.x, mingw, MSVC 14, Android 9.0.

10 ISO C and POSIX Function Substitutes

This chapter describes which functions and function-like macros specified by ISO C (including ISO TS 18661-1) or POSIX are substituted by Gnulib, which portability pitfalls are fixed by Gnulib, and which (known) portability problems are not worked around by Gnulib.

The notation “Gnulib module: —” means that Gnulib does not provide a module providing a substitute for the function. When the list “Portability problems not fixed by Gnulib” is empty, such a module is not needed: No portability problems are known. Otherwise, it indicates that such a module would be useful but is not available: No one so far found this function important enough to contribute a substitute for it. If you need this particular function, you may write to <bug-gnulib at gnu dot org>.

This list of functions is sorted according to the header that declares them.

10.1 Functions in <aio.h>

10.1.1 aio_cancel

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/aio_cancel.html

Gnulib module: —

Portability problems fixed by Gnulib:

- On platforms where `off_t` is a 32-bit type, this function may not work correctly on files 2 GiB and larger. See [Section 14.2 \[Large File Support\]](#), page 842.

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: NetBSD 3.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.10.x, mingw, MSVC 14, Android 9.0.

10.1.2 aio_error

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/aio_error.html

Gnulib module: —

Portability problems fixed by Gnulib:

- On platforms where `off_t` is a 32-bit type, this function may not work correctly on files 2 GiB and larger. See [Section 14.2 \[Large File Support\]](#), page 842.

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: NetBSD 3.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, Cygwin 2.10.x, mingw, MSVC 14, Android 9.0.

10.1.3 aio_fsync

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/aio_fsync.html

Gnulib module: —

Portability problems fixed by GnuLib:

- On platforms where `off_t` is a 32-bit type, this function may not work correctly on files 2 GiB and larger. See [Section 14.2 \[Large File Support\]](#), page 842.

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 3.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, Cygwin 2.10.x, mingw, MSVC 14, Android 9.0.

10.1.4 `aio_read`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/aio_read.html

GnuLib module: —

Portability problems fixed by GnuLib:

- On platforms where `off_t` is a 32-bit type, this function may not work correctly on files 2 GiB and larger. See [Section 14.2 \[Large File Support\]](#), page 842.

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: NetBSD 3.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.10.x, mingw, MSVC 14, Android 9.0.

10.1.5 `aio_return`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/aio_return.html

GnuLib module: —

Portability problems fixed by GnuLib:

- On platforms where `off_t` is a 32-bit type, this function may not work correctly on files 2 GiB and larger. See [Section 14.2 \[Large File Support\]](#), page 842.

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: NetBSD 3.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, Cygwin 2.10.x, mingw, MSVC 14, Android 9.0.

10.1.6 `aio_suspend`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/aio_suspend.html

GnuLib module: —

Portability problems fixed by GnuLib:

- On platforms where `off_t` is a 32-bit type, this function may not work correctly on files 2 GiB and larger. See [Section 14.2 \[Large File Support\]](#), page 842.

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: NetBSD 3.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.10.x, mingw, MSVC 14, Android 9.0.

10.1.7 aio_write

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/aio_write.html

Gnulib module: —

Portability problems fixed by Gnulib:

- On platforms where `off_t` is a 32-bit type, this function may not work correctly on files 2 GiB and larger. See [Section 14.2 \[Large File Support\]](#), page 842.

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: NetBSD 3.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.10.x, mingw, MSVC 14, Android 9.0.

10.1.8 lio_listio

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/lio_listio.html

Gnulib module: —

Portability problems fixed by Gnulib:

- On platforms where `off_t` is a 32-bit type, this function may not work correctly on files 2 GiB and larger. See [Section 14.2 \[Large File Support\]](#), page 842.

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: NetBSD 3.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.10.x, mingw, MSVC 14, Android 9.0.

10.2 Functions in `<arpa/inet.h>`

10.2.1 htonl

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/htonl.html>

Gnulib module: `htonl`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: HP-UX 11, Android API level 20.
- This function is declared in `<winsock2.h>` instead of `<arpa/inet.h>` on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

10.2.2 htons

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/htons.html>

Gnulib module: `htonl`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: HP-UX 11, Android API level 20.

- This function is declared in `<winsock2.h>` instead of `<arpa/inet.h>` on some platforms: mingw, MSVC 14.

Portability problems not fixed by GnuLib:

10.2.3 `inet_addr`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/inet_addr.html

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is declared in `<winsock2.h>` instead of `<arpa/inet.h>` on some platforms: mingw, MSVC 14.
- POSIX.1-2024 says this function is obsolescent and it is planned to be removed in a future version. Use the function `inet_pton` instead.
- On some old platforms, this function returns a `'struct in_addr'` rather than a scalar type such as `'unsigned int'` or `'unsigned long'`.

10.2.4 `inet_ntoa`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/inet_ntoa.html

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is declared in `<winsock2.h>` instead of `<arpa/inet.h>` on some platforms: mingw, MSVC 14.
- POSIX.1-2024 says this function is obsolescent and it is planned to be removed in a future version. Use the function `inet_ntop` instead.
- The `inet_ntoa` function need not be reentrant, and consequently is not required to be thread safe. Implementations of `inet_ntoa` typically write the timestamp into static buffer. If two threads call `inet_ntoa` at roughly the same time, you might end up with the wrong date in one of the threads, or some undefined string.

Note: `inet_ntoa` is specific for IPv4 addresses. A protocol independent function is `inet_ntop`.

10.2.5 `inet_ntop`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/inet_ntop.html

GnuLib module: `inet_ntop`

Portability problems fixed by GnuLib:

- This function is missing on some platforms: HP-UX 11.00, mingw, MSVC 14.
- This function is declared in `<netdb.h>` instead of `<arpa/inet.h>` on some platforms: NonStop Kernel.

- This function is declared in `<ws2tcpip.h>`, with a POSIX incompatible declaration, on some platforms: MSVC 14 on Windows $\geq$ Vista.

Portability problems not fixed by GnuLib:

- This function's fourth argument type is `size_t` instead of `socklen_t` on some platforms: Solaris 10.

10.2.6 `inet_pton`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/inet_pton.html

GnuLib module: `inet_pton`

Portability problems fixed by GnuLib:

- This function is missing on some platforms: HP-UX 11.00, mingw, MSVC 14.
- This function is declared in `<netdb.h>` instead of `<arpa/inet.h>` on some platforms: NonStop Kernel.
- This function is declared in `<ws2tcpip.h>`, with a POSIX incompatible declaration, on some platforms: MSVC 14 on Windows $\geq$ Vista.

Portability problems not fixed by GnuLib:

10.2.7 `ntohl`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/ntohl.html>

GnuLib module: `htonl`

Portability problems fixed by GnuLib:

- This function is missing on some platforms: HP-UX 11, Android API level 20.
- This function is declared in `<winsock2.h>` instead of `<arpa/inet.h>` on some platforms: mingw, MSVC 14.

Portability problems not fixed by GnuLib:

10.2.8 `ntohs`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/ntohs.html>

GnuLib module: `htonl`

Portability problems fixed by GnuLib:

- This function is missing on some platforms: HP-UX 11, Android API level 20.
- This function is declared in `<winsock2.h>` instead of `<arpa/inet.h>` on some platforms: mingw, MSVC 14.

Portability problems not fixed by GnuLib:

10.3 Functions in <assert.h>

10.3.1 assert

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/assert.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

Extension: Gnulib offers a module ‘assert’ that allows the installer to disable assertions through a ‘configure’ option: ‘--disable-assert’.

10.4 Functions in <complex.h>

10.4.1 CMPLX

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.3.9.3.

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/CMPLX.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This macro is missing on many platforms: glibc with GCC 4.9.x or clang 18, Mac OS X 10.5, FreeBSD 6.4, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.1, HP-UX 11.11, Solaris 11.0, Solaris 11 OpenIndiana, Cygwin, mingw, MSVC 14.

10.4.2 CMPLXF

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.3.9.3.

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/CMPLXF.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This macro is missing on many platforms: glibc with GCC 4.9.x or clang 18, Mac OS X 10.5, FreeBSD 6.4, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.1, HP-UX 11.11, Solaris 11.0, Solaris 11 OpenIndiana, Cygwin, mingw, MSVC 14.

10.4.3 CMPLXL

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.3.9.3.

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/CMPLXL.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This macro is missing on many platforms: glibc with GCC 4.9.x or clang 18, Mac OS X 10.5, FreeBSD 6.4, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.1, HP-UX 11.11, Solaris 11.0, Solaris 11 OpenIndiana, Cygwin, mingw, MSVC 14.

10.4.4 cabs

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/cabs.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, HP-UX 11, Solaris 9.

10.4.5 cabsf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/cabsf.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, mingw, MSVC 9.

10.4.6 cabsl

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/cabsl.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 10.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 9, Android 5.1.

10.4.7 cacos

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/cacos.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.7, mingw, MSVC 9, Android 5.1.

10.4.8 `cacosf`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/cacosf.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.7, mingw, MSVC 9, Android 5.1.

10.4.9 `cacosh`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/cacosh.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.7, mingw, MSVC 9, Android 5.1.

10.4.10 `cacoshf`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/cacoshf.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.7, mingw, MSVC 9, Android 5.1.

10.4.11 `cacoshl`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/cacoshl.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 11.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 9, Android 7.1.

10.4.12 `cacosl`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/cacosl.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 11.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 9, Android 7.1.

10.4.13 `carg`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/carg.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.7, mingw, MSVC 9.

10.4.14 `cargf`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/cargf.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.7, mingw, MSVC 9.

10.4.15 `cargl`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/cargl.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 9, Android 5.1.

10.4.16 `casin`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/casin.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.7, mingw, MSVC 9, Android 5.1.

10.4.17 `casinf`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/casinf.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.7, mingw, MSVC 9, Android 5.1.

10.4.18 `casinh`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/casinh.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.7, mingw, MSVC 9, Android 5.1.

10.4.19 `casinhf`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/casinhf.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.7, mingw, MSVC 9, Android 5.1.

10.4.20 `casinh1`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/casinh1.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 11.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 9, Android 7.1.

10.4.21 `casinl`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/casinl.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 11.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 9, Android 7.1.

10.4.22 `catan`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/catan.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.7, mingw, MSVC 9, Android 5.1.

10.4.23 `catanf`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/catanf.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.7, mingw, MSVC 9, Android 5.1.

10.4.24 catanh

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/catanh.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.7, mingw, MSVC 9, Android 5.1.

10.4.25 catanhf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/catanhf.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.7, mingw, MSVC 9, Android 5.1.

10.4.26 catanhl

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/catanhl.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 11.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 9, Android 7.1.

10.4.27 catanl

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/catanl.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 11.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 9, Android 7.1.

10.4.28 `ccos`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/ccos.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.7, mingw, MSVC 9.

10.4.29 `ccosf`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/ccosf.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.7, mingw, MSVC 9.

10.4.30 `ccosh`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/ccosh.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.7, mingw, MSVC 9.

10.4.31 `ccoshf`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/ccoshf.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.7, mingw, MSVC 9.

10.4.32 `ccoshl`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/ccoshl.html>

Gnulib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: FreeBSD 14.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 9, Android 7.1.

10.4.33 `ccosl`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/ccosl.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: FreeBSD 14.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 9, Android 7.1.

10.4.34 `cexp`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/cexp.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.7, mingw, MSVC 9.

10.4.35 `cexpf`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/cexpf.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.7, mingw, MSVC 9.

10.4.36 `cexpl`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/cexpl.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: FreeBSD 13.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 9, Android 7.1.

10.4.37 cimag

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/cimag.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.7, mingw, MSVC 9.

10.4.38 cimagf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/cimagf.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.7, mingw, MSVC 9.

10.4.39 cimagl

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/cimagl.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 9, Android 5.1.

10.4.40 clog

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/clog.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 11.0, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.7, mingw, MSVC 9, Android 7.1.

10.4.41 clogf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/clogf.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 11.0, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.7, mingw, MSVC 9, Android 7.1.

10.4.42 clogl

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/clogl.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 11.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 9, Android 7.1.

10.4.43 conj

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/conj.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.7, mingw, MSVC 9.

10.4.44 conjf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/conjf.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.7, mingw, MSVC 9.

10.4.45 `conj1`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/conj1.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 9, Android 5.1.

10.4.46 `cpow`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/cpow.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 11.0, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.7, mingw, MSVC 9, Android 7.1.

10.4.47 `cpowf`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/cpowf.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 11.0, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.7, mingw, MSVC 9, Android 7.1.

10.4.48 `cpowl`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/cpowl.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 11.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 9, Android 7.1.

10.4.49 cproj

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/cproj.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.7, mingw, MSVC 9.
- The glibc implementation is or was broken.

10.4.50 cprojf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/cprojf.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.7, mingw, MSVC 9.
- The glibc implementation is or was broken.

10.4.51 cprojl

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/cprojl.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 9, Android 5.1.
- The glibc implementation is or was broken.

10.4.52 creal

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/creal.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.7, mingw, MSVC 9.

10.4.53 crealf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/crealf.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.7, mingw, MSVC 9.

10.4.54 creall

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/creall.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 9, Android 5.1.

10.4.55 csin

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/csin.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.7, mingw, MSVC 9.

10.4.56 csinf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/csinf.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.7, mingw, MSVC 9.

10.4.57 csinh

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/csinh.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.7, mingw, MSVC 9.

10.4.58 csinhf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/csinhf.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.7, mingw, MSVC 9.

10.4.59 csinh1

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/csinh1.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 14.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 9, Android 7.1.

10.4.60 csinl

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/csinl.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 14.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 9, Android 7.1.

10.4.61 csqrt

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/csqrt.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.7, mingw, MSVC 9.

10.4.62 csqrtf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/csqrtf.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.7, mingw, MSVC 9.

10.4.63 csqrtl

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/csqrtl.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 9, Android 5.1.

10.4.64 ctan

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/ctan.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.7, mingw, MSVC 9.

10.4.65 `ctanf`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/ctanf.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.7, mingw, MSVC 9.

10.4.66 `ctanh`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/ctanh.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.7, mingw, MSVC 9.

10.4.67 `ctanhf`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/ctanhf.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.7, mingw, MSVC 9.

10.4.68 `ctanh1`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/ctanh1.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 14.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 9, Android 7.1.

10.4.69 `ctanl`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/ctanl.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 14.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 9, Android 7.1.

10.5 Functions in `<ctype.h>`

10.5.1 `isalnum`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/isalnum.html>

Gnulib module: `ctype`

Portability problems fixed by Gnulib:

- This function cannot be called from plain inline or extern inline functions on some platforms: macOS 10.8.

Portability problems not fixed by Gnulib:

Note: This function's behaviour depends on the locale, but does not support the multi-byte characters that occur in strings in locales with `MB_CUR_MAX > 1` (this includes all the common UTF-8 locales). There are five alternative APIs:

`c_isalnum`

This function operates in a locale independent way and returns true only for ASCII characters. It is provided by the Gnulib module '`c-ctype`'.

`iswalnum` This function operates in a locale dependent way, on wide characters. In order to use it, you first have to convert from multibyte to wide characters, using the `mbrtowc` function. It is provided by the Gnulib module '`wctype`'.

`c32isalnum`

This function operates in a locale dependent way, on 32-bit wide characters. In order to use it, you first have to convert from multibyte to 32-bit wide characters, using the `mbrtoc32` function. It is provided by the Gnulib module '`c32isalnum`'.

`mb_isalnum`

This function operates in a locale dependent way, on multibyte characters. It is provided by the Gnulib module '`mbchar`'.

`uc_is_alnum`

This function operates in a locale independent way, on Unicode characters. It is provided by the Gnulib module '`unictype/ctype-alnum`'.

10.5.2 `isalnum_l`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/isalnum_l.html

Gnulib module: `isalnum_l`

Portability problems fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 9.0, NetBSD 6.1, OpenBSD 6.1, Minix 3.1.8, AIX 6.1, HP-UX 11, Solaris 11.3, Cygwin 2.5.x, mingw, MSVC 14, Android 4.4.

Portability problems not fixed by Gnulib:

10.5.3 `isalpha`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/isalpha.html>

Gnulib module: `ctype`

Portability problems fixed by Gnulib:

- This function cannot be called from plain inline or extern inline functions on some platforms: macOS 10.8.

Portability problems not fixed by Gnulib:

Note: This function's behaviour depends on the locale, but does not support the multi-byte characters that occur in strings in locales with `MB_CUR_MAX > 1` (this includes all the common UTF-8 locales). There are five alternative APIs:

`c_isalpha`

This function operates in a locale independent way and returns true only for ASCII characters. It is provided by the Gnulib module '`c-ctype`'.

`iswalpha` This function operates in a locale dependent way, on wide characters. In order to use it, you first have to convert from multibyte to wide characters, using the `mbrtowc` function. It is provided by the Gnulib module '`wctype`'.

`c32isalpha`

This function operates in a locale dependent way, on 32-bit wide characters. In order to use it, you first have to convert from multibyte to 32-bit wide characters, using the `mbrtoc32` function. It is provided by the Gnulib module '`c32isalpha`'.

`mb_isalpha`

This function operates in a locale dependent way, on multibyte characters. It is provided by the Gnulib module '`mbchar`'.

`uc_is_alpha`

This function operates in a locale independent way, on Unicode characters. It is provided by the Gnulib module '`unictype/ctype-alpha`'.

10.5.4 `isalpha_l`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/isalpha_l.html

Gnulib module: `isalpha_l`

Portability problems fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 9.0, NetBSD 6.1, OpenBSD 6.1, Minix 3.1.8, AIX 6.1, HP-UX 11, Solaris 11.3, Cygwin 2.5.x, mingw, MSVC 14, Android 4.4.

Portability problems not fixed by Gnulib:

10.5.5 `isblank`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/isblank.html>

Gnulib module: `isblank`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Solaris 9, mingw, MSVC 9.
- This function cannot be called from plain inline or extern inline functions on some platforms: macOS 10.8.

Portability problems not fixed by Gnulib:

Note: This function's behaviour depends on the locale, but does not support the multi-byte characters that occur in strings in locales with `MB_CUR_MAX > 1` (this includes all the common UTF-8 locales). There are five alternative APIs:

`c_isblank`

This function operates in a locale independent way and returns true only for ASCII characters. It is provided by the Gnulib module '`c-ctype`'.

`iswblank` This function operates in a locale dependent way, on wide characters. In order to use it, you first have to convert from multibyte to wide characters, using the `mbrtowc` function. It is provided by the Gnulib module '`wctype`'.

`c32isblank`

This function operates in a locale dependent way, on 32-bit wide characters. In order to use it, you first have to convert from multibyte to 32-bit wide characters, using the `mbrtoc32` function. It is provided by the Gnulib module '`c32isblank`'.

`mb_isblank`

This function operates in a locale dependent way, on multibyte characters. It is provided by the Gnulib module '`mbchar`'.

`uc_is_blank`

This function operates in a locale independent way, on Unicode characters. It is provided by the Gnulib module '`unictype/ctype-blank`'.

10.5.6 isblank_l

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/isblank_l.html

Gnulib module: isblank_l

Portability problems fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 9.0, NetBSD 6.1, OpenBSD 6.1, Minix 3.1.8, AIX 6.1, HP-UX 11, Solaris 11.3, Cygwin 2.5.x, mingw, MSVC 14, Android 4.4.

Portability problems not fixed by Gnulib:

10.5.7 iscntrl

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/iscntrl.html>

Gnulib module: ctype

Portability problems fixed by Gnulib:

- This function cannot be called from plain inline or extern inline functions on some platforms: macOS 10.8.

Portability problems not fixed by Gnulib:

Note: This function's behaviour depends on the locale, but does not support the multi-byte characters that occur in strings in locales with `MB_CUR_MAX > 1` (this includes all the common UTF-8 locales). There are five alternative APIs:

`c_iscntrl`

This function operates in a locale independent way and returns true only for ASCII characters. It is provided by the Gnulib module 'c-ctype'.

`iswcntrl` This function operates in a locale dependent way, on wide characters. In order to use it, you first have to convert from multibyte to wide characters, using the `mbrtowc` function. It is provided by the Gnulib module 'wctype'.

`c32iscntrl`

This function operates in a locale dependent way, on 32-bit wide characters. In order to use it, you first have to convert from multibyte to 32-bit wide characters, using the `mbrtoc32` function. It is provided by the Gnulib module 'c32iscntrl'.

`mb_iscntrl`

This function operates in a locale dependent way, on multibyte characters. It is provided by the Gnulib module 'mbchar'.

`uc_is_cntrl`

This function operates in a locale independent way, on Unicode characters. It is provided by the Gnulib module 'unictype/ctype-cntrl'.

10.5.8 `iscntrl_1`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/iscntrl_1.html

Gnulib module: `iscntrl_1`

Portability problems fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 9.0, NetBSD 6.1, OpenBSD 6.1, Minix 3.1.8, AIX 6.1, HP-UX 11, Solaris 11.3, Cygwin 2.5.x, mingw, MSVC 14, Android 4.4.

Portability problems not fixed by Gnulib:

10.5.9 `isdigit`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/isdigit.html>

Gnulib module: `ctype`

Portability problems fixed by Gnulib:

- This function cannot be called from plain inline or extern inline functions on some platforms: macOS 10.8.

Portability problems not fixed by Gnulib:

- This function's behavior depends on locale, and it can return nonzero on arguments other than the usual decimal digits: mingw, MSVC.

Note: This function does not support the multibyte characters that occur in strings in locales with `MB_CUR_MAX > 1` (this includes all the common UTF-8 locales). There are five alternative APIs:

`c_isdigit`

This function operates in a locale independent way and returns true only for ASCII characters. It is provided by the Gnulib module '`c-ctype`'.

`iswdigit` This function operates in a locale dependent way, on wide characters. In order to use it, you first have to convert from multibyte to wide characters, using the `mbrtowc` function. It is provided by the Gnulib module '`wctype`'.

`c32isdigit`

This function operates in a locale dependent way, on 32-bit wide characters. In order to use it, you first have to convert from multibyte to 32-bit wide characters, using the `mbrtoc32` function. It is provided by the Gnulib module '`c32isdigit`'.

`mb_isdigit`

This function operates in a locale dependent way, on multibyte characters. It is provided by the Gnulib module '`mbchar`'.

`uc_is_digit`

This function operates in a locale independent way, on Unicode characters. It is provided by the Gnulib module '`unictype/ctype-digit`'.

10.5.10 `isdigit_l`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/isdigit_l.html

Gnulib module: `isdigit_l`

Portability problems fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 9.0, NetBSD 6.1, OpenBSD 6.1, Minix 3.1.8, AIX 6.1, HP-UX 11, Solaris 11.3, Cygwin 2.5.x, mingw, MSVC 14, Android 4.4.

Portability problems not fixed by Gnulib:

10.5.11 `isgraph`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/isgraph.html>

Gnulib module: `ctype`

Portability problems fixed by Gnulib:

- This function cannot be called from plain inline or extern inline functions on some platforms: macOS 10.8.

Portability problems not fixed by Gnulib:

Note: This function's behaviour depends on the locale, but does not support the multi-byte characters that occur in strings in locales with `MB_CUR_MAX > 1` (this includes all the common UTF-8 locales). There are five alternative APIs:

`c_isgraph`

This function operates in a locale independent way and returns true only for ASCII characters. It is provided by the Gnulib module '`c-ctype`'.

`iswgraph` This function operates in a locale dependent way, on wide characters. In order to use it, you first have to convert from multibyte to wide characters, using the `mbrtowc` function. It is provided by the Gnulib module '`wctype`'.

`c32isgraph`

This function operates in a locale dependent way, on 32-bit wide characters. In order to use it, you first have to convert from multibyte to 32-bit wide characters, using the `mbrtoc32` function. It is provided by the Gnulib module '`c32isgraph`'.

`mb_isgraph`

This function operates in a locale dependent way, on multibyte characters. It is provided by the Gnulib module '`mbchar`'.

`uc_is_graph`

This function operates in a locale independent way, on Unicode characters. It is provided by the Gnulib module '`unictype/ctype-graph`'.

10.5.12 `isgraph_l`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/isgraph_l.html

Gnulib module: `isgraph_l`

Portability problems fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 9.0, NetBSD 6.1, OpenBSD 6.1, Minix 3.1.8, AIX 6.1, HP-UX 11, Solaris 11.3, Cygwin 2.5.x, mingw, MSVC 14, Android 4.4.

Portability problems not fixed by Gnulib:

10.5.13 `islower`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/islower.html>

Gnulib module: `ctype`

Portability problems fixed by Gnulib:

- This function cannot be called from plain inline or extern inline functions on some platforms: macOS 10.8.

Portability problems not fixed by Gnulib:

Note: This function's behaviour depends on the locale, but does not support the multi-byte characters that occur in strings in locales with `MB_CUR_MAX > 1` (this includes all the common UTF-8 locales). There are five alternative APIs:

`c_islower`

This function operates in a locale independent way and returns true only for ASCII characters. It is provided by the Gnulib module '`c-ctype`'.

`iswlower` This function operates in a locale dependent way, on wide characters. In order to use it, you first have to convert from multibyte to wide characters, using the `mbrtowc` function. It is provided by the Gnulib module '`wctype`'.

`c32islower`

This function operates in a locale dependent way, on 32-bit wide characters. In order to use it, you first have to convert from multibyte to 32-bit wide characters, using the `mbrtoc32` function. It is provided by the Gnulib module '`c32islower`'.

`mb_islower`

This function operates in a locale dependent way, on multibyte characters. It is provided by the Gnulib module '`mbchar`'.

`uc_is_lower`

This function operates in a locale independent way, on Unicode characters. It is provided by the Gnulib module '`unictype/ctype-lower`'.

10.5.14 `islower_l`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/islower_l.html

Gnulib module: `islower_l`

Portability problems fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 9.0, NetBSD 6.1, OpenBSD 6.1, Minix 3.1.8, AIX 6.1, HP-UX 11, Solaris 11.3, Cygwin 2.5.x, mingw, MSVC 14, Android 4.4.

Portability problems not fixed by Gnulib:

10.5.15 `isprint`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/isprint.html>

Gnulib module: `ctype`

Portability problems fixed by Gnulib:

- This function cannot be called from plain inline or extern inline functions on some platforms: macOS 10.8.

Portability problems not fixed by Gnulib:

Note: This function's behaviour depends on the locale, but does not support the multi-byte characters that occur in strings in locales with `MB_CUR_MAX > 1` (this includes all the common UTF-8 locales). There are five alternative APIs:

`c_isprint`

This function operates in a locale independent way and returns true only for ASCII characters. It is provided by the Gnulib module '`c-ctype`'.

`iswprint` This function operates in a locale dependent way, on wide characters. In order to use it, you first have to convert from multibyte to wide characters, using the `mbrtowc` function. It is provided by the Gnulib module '`wctype`'.

`c32isprint`

This function operates in a locale dependent way, on 32-bit wide characters. In order to use it, you first have to convert from multibyte to 32-bit wide characters, using the `mbrtoc32` function. It is provided by the Gnulib module '`c32isprint`'.

`mb_isprint`

This function operates in a locale dependent way, on multibyte characters. It is provided by the Gnulib module '`mbchar`'.

`uc_is_print`

This function operates in a locale independent way, on Unicode characters. It is provided by the Gnulib module '`unictype/ctype-print`'.

10.5.16 isprint_l

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/isprint_l.html

Gnulib module: isprint_l

Portability problems fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 9.0, NetBSD 6.1, OpenBSD 6.1, Minix 3.1.8, AIX 6.1, HP-UX 11, Solaris 11.3, Cygwin 2.5.x, mingw, MSVC 14, Android 4.4.

Portability problems not fixed by Gnulib:

10.5.17 ispunct

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/ispunct.html>

Gnulib module: ctype

Portability problems fixed by Gnulib:

- This function cannot be called from plain inline or extern inline functions on some platforms: macOS 10.8.

Portability problems not fixed by Gnulib:

Note: This function's behaviour depends on the locale, but does not support the multi-byte characters that occur in strings in locales with `MB_CUR_MAX > 1` (this includes all the common UTF-8 locales). There are five alternative APIs:

`c_ispunct`

This function operates in a locale independent way and returns true only for ASCII characters. It is provided by the Gnulib module 'c-ctype'.

`iswpunct` This function operates in a locale dependent way, on wide characters. In order to use it, you first have to convert from multibyte to wide characters, using the `mbrtowc` function. It is provided by the Gnulib module 'wctype'.

`c32ispunct`

This function operates in a locale dependent way, on 32-bit wide characters. In order to use it, you first have to convert from multibyte to 32-bit wide characters, using the `mbrtoc32` function. It is provided by the Gnulib module 'c32ispunct'.

`mb_ispunct`

This function operates in a locale dependent way, on multibyte characters. It is provided by the Gnulib module 'mbchar'.

`uc_is_punct`

This function operates in a locale independent way, on Unicode characters. It is provided by the Gnulib module 'unictype/ctype-punct'.

10.5.18 `ispunct_l`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/ispunct_l.html

Gnulib module: `ispunct_l`

Portability problems fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 9.0, NetBSD 6.1, OpenBSD 6.1, Minix 3.1.8, AIX 6.1, HP-UX 11, Solaris 11.3, Cygwin 2.5.x, mingw, MSVC 14, Android 4.4.

Portability problems not fixed by Gnulib:

10.5.19 `isspace`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/isspace.html>

Gnulib module: `ctype`

Portability problems fixed by Gnulib:

- This function cannot be called from plain inline or extern inline functions on some platforms: macOS 10.8.

Portability problems not fixed by Gnulib:

Note: This function's behaviour depends on the locale, but does not support the multi-byte characters that occur in strings in locales with `MB_CUR_MAX > 1` (this includes all the common UTF-8 locales). There are five alternative APIs:

`c_isspace`

This function operates in a locale independent way and returns true only for ASCII characters. It is provided by the Gnulib module '`c-ctype`'.

`iswspace` This function operates in a locale dependent way, on wide characters. In order to use it, you first have to convert from multibyte to wide characters, using the `mbrtowc` function. It is provided by the Gnulib module '`wctype`'.

`c32isspace`

This function operates in a locale dependent way, on 32-bit wide characters. In order to use it, you first have to convert from multibyte to 32-bit wide characters, using the `mbrtoc32` function. It is provided by the Gnulib module '`c32isspace`'.

`mb_isspace`

This function operates in a locale dependent way, on multibyte characters. It is provided by the Gnulib module '`mbchar`'.

`uc_is_space`

This function operates in a locale independent way, on Unicode characters. It is provided by the Gnulib module '`unictype/ctype-space`'.

10.5.20 `isspace_l`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/isspace_l.html

Gnulib module: `isspace_l`

Portability problems fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 9.0, NetBSD 6.1, OpenBSD 6.1, Minix 3.1.8, AIX 6.1, HP-UX 11, Solaris 11.3, Cygwin 2.5.x, mingw, MSVC 14, Android 4.4.

Portability problems not fixed by Gnulib:

10.5.21 `isupper`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/isupper.html>

Gnulib module: `ctype`

Portability problems fixed by Gnulib:

- This function cannot be called from plain inline or extern inline functions on some platforms: macOS 10.8.

Portability problems not fixed by Gnulib:

Note: This function's behaviour depends on the locale, but does not support the multi-byte characters that occur in strings in locales with `MB_CUR_MAX > 1` (this includes all the common UTF-8 locales). There are five alternative APIs:

`c_isupper`

This function operates in a locale independent way and returns true only for ASCII characters. It is provided by the Gnulib module '`c-ctype`'.

`iswupper` This function operates in a locale dependent way, on wide characters. In order to use it, you first have to convert from multibyte to wide characters, using the `mbrtowc` function. It is provided by the Gnulib module '`wctype`'.

`c32isupper`

This function operates in a locale dependent way, on 32-bit wide characters. In order to use it, you first have to convert from multibyte to 32-bit wide characters, using the `mbrtoc32` function. It is provided by the Gnulib module '`c32isupper`'.

`mb_isupper`

This function operates in a locale dependent way, on multibyte characters. It is provided by the Gnulib module '`mbchar`'.

`uc_is_upper`

This function operates in a locale independent way, on Unicode characters. It is provided by the Gnulib module '`unictype/ctype-upper`'.

10.5.22 `isupper_l`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/isupper_l.html

Gnulib module: `isupper_l`

Portability problems fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 9.0, NetBSD 6.1, OpenBSD 6.1, Minix 3.1.8, AIX 6.1, HP-UX 11, Solaris 11.3, Cygwin 2.5.x, mingw, MSVC 14, Android 4.4.

Portability problems not fixed by Gnulib:

10.5.23 `isxdigit`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/isxdigit.html>

Gnulib module: `ctype`

Portability problems fixed by Gnulib:

- This function cannot be called from plain inline or extern inline functions on some platforms: macOS 10.8.

Portability problems not fixed by Gnulib:

- This function's behavior depends on locale, and it can return nonzero on arguments other than the usual hexadecimal digits: mingw, MSVC.

Note: This function does not support the multibyte characters that occur in strings in locales with `MB_CUR_MAX > 1` (this includes all the common UTF-8 locales). There are five alternative APIs:

`c_isxdigit`

This function operates in a locale independent way and returns true only for ASCII characters. It is provided by the Gnulib module '`c-ctype`'.

`iswxdigit`

This function operates in a locale dependent way, on wide characters. In order to use it, you first have to convert from multibyte to wide characters, using the `mbrtowc` function. It is provided by the Gnulib module '`wctype`'.

`c32isxdigit`

This function operates in a locale dependent way, on 32-bit wide characters. In order to use it, you first have to convert from multibyte to 32-bit wide characters, using the `mbrtoc32` function. It is provided by the Gnulib module '`c32isxdigit`'.

`mb_isxdigit`

This function operates in a locale dependent way, on multibyte characters. It is provided by the Gnulib module '`mbchar`'.

`uc_is_xdigit`

This function operates in a locale independent way, on Unicode characters. It is provided by the Gnulib module '`unictype/ctype-xdigit`'.

10.5.24 isxdigit_l

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/isxdigit_l.html

Gnulib module: isxdigit_l

Portability problems fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 9.0, NetBSD 6.1, OpenBSD 6.1, Minix 3.1.8, AIX 6.1, HP-UX 11, Solaris 11.3, Cygwin 2.5.x, mingw, MSVC 14, Android 4.4.

Portability problems not fixed by Gnulib:

10.5.25 tolower

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/tolower.html>

Gnulib module: ctype

Portability problems fixed by Gnulib:

- This function cannot be called from plain inline or extern inline functions on some platforms: macOS 10.8.

Portability problems not fixed by Gnulib:

Note: This function's behaviour depends on the locale, but does not support the multi-byte characters that occur in strings in locales with `MB_CUR_MAX > 1` (this includes all the common UTF-8 locales). There are four alternative APIs:

`c_tolower`

This function operates in a locale independent way and returns a different value than the argument only for uppercase ASCII characters. It is provided by the Gnulib module `'c-ctype'`.

`tolower` This function operates in a locale dependent way, on wide characters. In order to use it, you first have to convert from multibyte to wide characters, using the `mbrtowc` function. It is provided by the Gnulib module `'wctype'`.

`c32tolower`

This function operates in a locale dependent way, on 32-bit wide characters. In order to use it, you first have to convert from multibyte to 32-bit wide characters, using the `mbrtoc32` function. It is provided by the Gnulib module `'c32tolower'`.

`uc_tolower`

This function operates in a locale independent way, on Unicode characters. It is provided by the Gnulib module `'unicase/tolower'`.

10.5.26 tolower_l

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/tolower_l.html

Gnulib module: tolower_l

Portability problems fixed by GnuLib:

- This function is missing on many platforms: FreeBSD 9.0, NetBSD 6.1, OpenBSD 6.1, Minix 3.1.8, AIX 6.1, HP-UX 11, Solaris 11.3, Cygwin 2.5.x, mingw, MSVC 14, Android 4.4.

Portability problems not fixed by GnuLib:

10.5.27 toupper

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/toupper.html>

GnuLib module: ctype

Portability problems fixed by GnuLib:

- This function cannot be called from plain inline or extern inline functions on some platforms: macOS 10.8.

Portability problems not fixed by GnuLib:

Note: This function's behaviour depends on the locale, but does not support the multi-byte characters that occur in strings in locales with `MB_CUR_MAX > 1` (this includes all the common UTF-8 locales). There are four alternative APIs:

`c_toupper`

This function operates in a locale independent way and returns a different value than the argument only for lowercase ASCII characters. It is provided by the GnuLib module 'c-ctype'.

`toupper` This function operates in a locale dependent way, on wide characters. In order to use it, you first have to convert from multibyte to wide characters, using the `mbrtowc` function. It is provided by the GnuLib module 'wctype'.

`c32toupper`

This function operates in a locale dependent way, on 32-bit wide characters. In order to use it, you first have to convert from multibyte to 32-bit wide characters, using the `mbrtoc32` function. It is provided by the GnuLib module 'c32toupper'.

`uc_toupper`

This function operates in a locale independent way, on Unicode characters. It is provided by the GnuLib module 'unicase/toupper'.

10.5.28 toupper_l

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/toupper_l.html

GnuLib module: toupper_l

Portability problems fixed by GnuLib:

- This function is missing on many platforms: FreeBSD 9.0, NetBSD 6.1, OpenBSD 6.1, Minix 3.1.8, AIX 6.1, HP-UX 11, Solaris 11.3, Cygwin 2.5.x, mingw, MSVC 14, Android 4.4.

Portability problems not fixed by GnuLib:

10.6 Functions in <devctl.h>

10.6.1 posix_devctl

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/posix_devctl.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc 2.42, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android 9.0.

10.7 Functions in <dirent.h>

10.7.1 alphasort

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/alphasort.html>

Gnulib module: alphasort

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Solaris 9, mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- The parameters of this function are declared as `const void *` on some platforms: glibc 2.3.6, macOS 10.7, FreeBSD 6.0, NetBSD 7.1, OpenBSD 7.5.
- The parameters of this function are declared as `void *` on some platforms: AIX 5.1.

10.7.2 closedir

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/closedir.html>

Gnulib module: closedir

Portability problems fixed by Gnulib:

- This function is missing on some platforms: MSVC 14.

Portability problems not fixed by Gnulib:

10.7.3 dirfd

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/dirfd.html>

Gnulib module: dirfd

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 7.1, HP-UX 11, Solaris 10, mingw, MSVC 14.

Portability problems not fixed by GnuLib:

- This function always fails on some platforms: mingw.
- There is a `dirfd` macro but no function, and the macro does not work with an argument of type `void *`, as a function would: NetBSD 9.2.

10.7.4 fdopendir

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fdopendir.html>

GnuLib module: `fdopendir`

Portability problems fixed by GnuLib:

- This function is missing on some platforms: glibc 2.3.6, Mac OS X 10.5, FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 1.5.x, mingw, MSVC 14. But the replacement function is not safe to be used in libraries and is not multithread-safe. Also, the replacement does not guarantee that `'dirfd(fdopendir(n))==n'` (`dirfd` might fail, or return a different file descriptor than `n`).
- This function exists but is not declared on some platforms: FreeBSD 7.3.
- This function does not reject non-directory file descriptors on some platforms: GNU/Hurd.
- This function mistakenly closes non-directory file descriptors on some platforms: FreeBSD 8.1.

Portability problems not fixed by GnuLib:

10.7.5 opendir

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/opendir.html>

GnuLib module: `opendir`

Portability problems fixed by GnuLib:

- This function is missing on some platforms: MSVC 14.
- On platforms where `off_t` is a 32-bit type, this function may not work correctly on huge directories 2 GiB and larger. Also, on platforms where `ino_t` is a 32-bit type, this function may report inode numbers incorrectly. See [Section 14.2 \[Large File Support\]](#), [page 842](#).

Portability problems not fixed by GnuLib:

10.7.6 posix_getdents

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/posix_getdents.html

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on many platforms: glibc 2.42, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.x, mingw, MSVC 14, Android 9.0.

10.7.7 readdir

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/readdir.html>

Gnulib module: readdir

Portability problems fixed by Gnulib:

- This function is missing on some platforms: MSVC 14.
- On platforms where `off_t` is a 32-bit type, this function may not work correctly on huge directories 2 GiB and larger. Also, on platforms where `ino_t` is a 32-bit type, this function may report inode numbers incorrectly. This can occur with file systems such as XFS (typically on large disks) and NFS. See [Section 14.2 \[Large File Support\]](#), [page 842](#).

Portability problems not fixed by Gnulib:

- Although POSIX places no restrictions on `d_ino` values, some older systems are rumored to return `d_ino` values equal to zero for directory entries that do not really exist. Although Gnulib formerly attempted to cater to these older systems, this caused misbehavior on standard systems and so Gnulib does not attempt to cater to them any more. If you know of any problems caused by this, please send a bug report.
- When reading a directory that has been removed, this function sets `errno` to `ENOENT` instead of leaving `errno` alone to indicate EOF: glibc 2.2.5.

10.7.8 readdir_r

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/readdir_r.html

Gnulib module: extensions

Portability problems fixed by Gnulib:

- This function is planned to be removed from POSIX and to be deprecated in glibc. Portable applications should use `readdir`.
- This function has an incompatible declaration on some platforms: Solaris 11.4 (when `_POSIX_PTHREAD_SEMANTICS` is not defined).
- On platforms where `off_t` is a 32-bit type, this function may not work correctly on huge directories 2 GiB and larger. Also, on platforms where `ino_t` is a 32-bit type, this function may report inode numbers incorrectly. This can occur with file systems such as XFS (typically on large disks) and NFS. See [Section 14.2 \[Large File Support\]](#), [page 842](#).

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.
- When reading a directory that has been removed, this function sets `errno` to `ENOENT` instead of leaving `errno` alone to indicate EOF: glibc 2.2.5.

10.7.9 rewinddir

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/rewinddir.html>

Gnulib module: `rewinddir`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: MSVC 14.
- On Mac OS X platforms where `long int` is a 32-bit type, this function may not work correctly on huge directories 2 GiB and larger. See [Section 14.2 \[Large File Support\]](#), [page 842](#).

Portability problems not fixed by Gnulib:

10.7.10 scandir

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/scandir.html>

Gnulib module: `scandir`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Solaris 9, mingw, MSVC 14.
- On platforms where `off_t` is a 32-bit type, this function may not work correctly on huge directories 2 GiB and larger. Also, on platforms where `ino_t` is a 32-bit type, this function may report inode numbers incorrectly. See [Section 14.2 \[Large File Support\]](#), [page 842](#).

Portability problems not fixed by Gnulib:

- The fourth parameter of this function is declared as `int (*) (const void *, const void *)` on some platforms: glibc 2.3.6, macOS 10.7, FreeBSD 6.0, NetBSD 7.1, OpenBSD 7.5.
- The fourth parameter of this function is declared as `int (*) (void *, void *)` on some platforms: AIX 5.1.

10.7.11 seekdir

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/seekdir.html>

Gnulib module: —

Portability problems fixed by Gnulib:

- On platforms where `long int` is a 32-bit type, this function may not work correctly on huge directories 2 GiB and larger. See [Section 14.2 \[Large File Support\]](#), [page 842](#).

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: MSVC 14, Android 5.1.

10.7.12 telldir

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/telldir.html>

Gnulib module: —

Portability problems fixed by Gnulib:

- On platforms where `long int` is a 32-bit type, this function may not work correctly on huge directories 2 GiB and larger. See [Section 14.2 \[Large File Support\]](#), page 842.

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: MSVC 14, Android 5.1.

10.8 Functions in <dlfcn.h>

10.8.1 dladdr

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/dladdr.html>

Documentation:

[man dladdr](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, Cygwin 1.7.x, mingw, MSVC 14.

10.8.2 dlclose

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/dlclose.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

10.8.3 dlerror

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/dlerror.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

10.8.4 dlopen

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/dlopen.html>

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-dlopen-1.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.
- If the file name argument is not absolute, the file is searched for. The search algorithm is system specific.

10.8.5 dlsym

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/dlsym.html>

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-dlsym-1.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.
- The visibility of symbols loaded in dependent shared libraries or present in the main executable is system dependent.

10.9 Functions in <endian.h>

10.9.1 be16toh

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/be16toh.html>

Documentation:

`man be16toh`

Gnulib module: endian

Portability problems fixed by Gnulib:

- This function is missing on many platforms: glibc 2.8, macOS 14, FreeBSD 13.0, NetBSD 7.2, OpenBSD 5.5, Minix 3.1.8, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14.

Portability problems not fixed by Gnulib:

10.9.2 be32toh

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/be32toh.html>

Documentation:

`man be32toh`

Gnulib module: endian

Portability problems fixed by Gnulib:

- This function is missing on many platforms: glibc 2.8, macOS 14, FreeBSD 13.0, NetBSD 7.2, OpenBSD 5.5, Minix 3.1.8, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14.

Portability problems not fixed by Gnulib:

10.9.3 be64toh

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/be64toh.html>

Documentation:

`man be64toh`

Gnulib module: endian

Portability problems fixed by Gnulib:

- This function is missing on many platforms: glibc 2.8, macOS 14, FreeBSD 13.0, NetBSD 7.2, OpenBSD 5.5, Minix 3.1.8, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14.

Portability problems not fixed by Gnulib:

10.9.4 htobe16

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/htobe16.html>

Documentation:

`man htobe16`

Gnulib module: endian

Portability problems fixed by Gnulib:

- This function is missing on many platforms: glibc 2.8, macOS 14, FreeBSD 13.0, NetBSD 7.2, OpenBSD 5.5, Minix 3.1.8, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14.

Portability problems not fixed by Gnulib:

10.9.5 htobe32

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/htobe32.html>

Documentation:

`man htobe32`

Gnulib module: endian

Portability problems fixed by Gnulib:

- This function is missing on many platforms: glibc 2.8, macOS 14, FreeBSD 13.0, NetBSD 7.2, OpenBSD 5.5, Minix 3.1.8, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14.

Portability problems not fixed by Gnulib:

10.9.6 htobe64

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/htobe64.html>

Documentation:

`man htobe64`

Gnulib module: endian

Portability problems fixed by Gnulib:

- This function is missing on many platforms: glibc 2.8, macOS 14, FreeBSD 13.0, NetBSD 7.2, OpenBSD 5.5, Minix 3.1.8, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14.

Portability problems not fixed by Gnulib:

10.9.7 htole16

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/htole16.html>

Documentation:

`man htole16`

Gnulib module: endian

Portability problems fixed by Gnulib:

- This function is missing on many platforms: glibc 2.8, macOS 14, FreeBSD 13.0, NetBSD 7.2, OpenBSD 5.5, Minix 3.1.8, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14.

Portability problems not fixed by Gnulib:

10.9.8 htole32

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/htole32.html>

Documentation:

`man htole32`

Gnulib module: endian

Portability problems fixed by Gnulib:

- This function is missing on many platforms: glibc 2.8, macOS 14, FreeBSD 13.0, NetBSD 7.2, OpenBSD 5.5, Minix 3.1.8, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14.

Portability problems not fixed by Gnulib:

10.9.9 htole64

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/htole64.html>

Documentation:

`man htole64`

Gnulib module: endian

Portability problems fixed by Gnulib:

- This function is missing on many platforms: glibc 2.8, macOS 14, FreeBSD 13.0, NetBSD 7.2, OpenBSD 5.5, Minix 3.1.8, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14.

Portability problems not fixed by Gnulib:

10.9.10 le16toh

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/le16toh.html>

Documentation:

`man le16toh`

Gnulib module: endian

Portability problems fixed by Gnulib:

- This function is missing on many platforms: glibc 2.8, macOS 14, FreeBSD 13.0, NetBSD 7.2, OpenBSD 5.5, Minix 3.1.8, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14.

Portability problems not fixed by Gnulib:

10.9.11 le32toh

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/le32toh.html>

Documentation:

`man le32toh`

Gnulib module: endian

Portability problems fixed by Gnulib:

- This function is missing on many platforms: glibc 2.8, macOS 14, FreeBSD 13.0, NetBSD 7.2, OpenBSD 5.5, Minix 3.1.8, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14.

Portability problems not fixed by Gnulib:

10.9.12 le64toh

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/le64toh.html>

Documentation:

`man le64toh`

Gnulib module: endian

Portability problems fixed by Gnulib:

- This function is missing on many platforms: glibc 2.8, macOS 14, FreeBSD 13.0, NetBSD 7.2, OpenBSD 5.5, Minix 3.1.8, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14.

Portability problems not fixed by Gnulib:

10.10 Functions in `<errno.h>`

10.10.1 `errno`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/errno.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- On Windows, the socket functions don't set `errno`; their error code is available through `WSAGetLastError()` instead.

10.11 Functions in `<fcntl.h>`

10.11.1 `creat`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/creat.html>

Gnulib module: `creat`

Portability problems fixed by Gnulib:

- This function does not support modes with execution bits (such as `0700`) on some platforms: MSVC 14.
- On platforms where `off_t` is a 32-bit type, `creat` may not work correctly with files 2 GiB and larger. See [Section 14.2 \[Large File Support\]](#), page 842.
- This function does not fail when the file name argument ends in a slash and (without the slash) names a nonexistent file, on some platforms: FreeBSD 7.2, AIX 7.1, HP-UX 11.31, Solaris 9.

Portability problems not fixed by Gnulib:

- On Windows, this function returns a file handle in `O_TEXT` mode. If you need a file handle in `O_BINARY` mode, you need to use the function `open` instead.

10.11.2 `fcntl`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fcntl.html>

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-fcntl-3.html

Gnulib module: `fcntl`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.
- This function does not support `F_DUPFD_CLOEXEC` on some platforms: glibc with Linux kernels before 2.6.24, Mac OS X 10.5, FreeBSD 6.0, NetBSD 5.0, OpenBSD 7.5, AIX 7.1, HP-UX 11, Solaris 11 2010-11, Cygwin 1.7.1. Note that the gnulib replacement code is functional but not atomic.
- The `F_DUPFD_CLOEXEC` action of this function does not set the `FD_CLOEXEC` flag on some platforms: NetBSD 10.0.
- The `F_DUPFD_CLOEXEC` action of this function sets the `FD_CLOEXEC` flag on the wrong file descriptor on some platforms: Haiku.
- The `F_DUPFD` action of this function does not reject out-of-range targets properly on some platforms: AIX 7.1, Cygwin 1.5.x, Haiku.
- The `F_DUPFD` action of this function mistakenly clears `FD_CLOEXEC` on the source descriptor on some platforms: Haiku.

Portability problems not fixed by Gnulib:

- The replacement function does not support `F_SETFD`, `F_GETFL`, `F_SETFL`, `F_GETOWN`, `F_SETOWN`, `F_GETLK`, `F_SETLK`, and `F_SETLKW` on some platforms: mingw, MSVC 14.
- When a file does not support locking (such as on an NFS file system that does not support file locking), calls with `F_SETLK` and `F_SETLKW` fail with `errno` set to different values on different systems: `EINVAL` on OpenIndiana (as suggested by the POSIX 1003.1-2017 `fcntl` specification), `ENOLCK` on GNU/Linux, and `EOPNOTSUPP` on FreeBSD.

10.11.3 open

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/open.html>

Gnulib module: `open`, `fchdir`

Portability problems fixed by the Gnulib module `open`:

- Some platforms do not support `O_CLOEXEC`: Mac OS X 10.6, FreeBSD 8.4, NetBSD 5.1, OpenBSD 4.9, Minix 3.1.8, AIX 7.1, HP-UX 11, Solaris 10, Cygwin 1.7.x, mingw, MSVC 14.
- Some platforms do not support `O_DIRECTORY`: glibc 2.0, Mac OS X 10.4, FreeBSD 7.4, NetBSD 4.0.1, OpenBSD 4.9, HP-UX 11, Solaris 10, Cygwin 1.5.x, mingw, MSVC 14.
- On platforms where `off_t` is a 32-bit type, `open` may not work correctly with files 2 GiB and larger. See [Section 14.2 \[Large File Support\]](#), page 842.
- This function does not fail when the file name argument ends in a slash and (without the slash) names a nonexistent file or a file that is not a directory, on some platforms: macOS 14, FreeBSD 7.2, AIX 7.1, HP-UX 11.00, Solaris 9.
- This function does not support the `O_NONBLOCK` flag when it is defined by the gnulib module `nonblocking` on some platforms: mingw, MSVC 14.
- On Windows platforms (excluding Cygwin), this function does usually not recognize the `/dev/null` filename.

Portability problems fixed by the Gnulib module `fchdir`:

- On Windows platforms (excluding Cygwin), this function fails to open a read-only descriptor for directories.

Portability problems not fixed by Gnulib:

- The Gnulib replacement for `O_CLOEXEC` is not atomic, and so is not safe in the presence of multiple threads or signal handlers.
- The Gnulib replacement for `O_SEARCH` may require read access instead of search/execute access.
- On some platforms, `O_SEARCH` may reject directories where you have search access even though `O_SEARCH` is not replaced: macOS 12.6.
- When the file does not exist, POSIX says the behavior of `O_CREAT | O_DIRECTORY` is unspecified. Most platforms fail with `EINVAL` in this case. However, in Linux kernels 6.3 and earlier, this function creates a regular file and then either returns successfully (Linux kernel 5.6 and earlier) or fails with `ENOTDIR` (Linux kernels 5.7–6.3).
- `open ("symlink", O_CREAT ...)` fails when the argument points to a nonexistent file in an existing directory on some platforms: Haiku.
- `open ("symlink", O_NOFOLLOW ...)` fails with `errno` set to `EMLINK` instead of the POSIX-required `ELoop` on some platforms: FreeBSD 14.2.
- `open ("symlink", O_NOFOLLOW ...)` fails with `errno` set to `EFTYPE` instead of the POSIX-required `ELoop` on some platforms: NetBSD 10.0.
- On Windows, this function returns a file handle in `O_TEXT` mode by default; this means that it translates `'\n'` to CR/LF by default. Use the `O_BINARY` flag if you need reliable binary I/O.

10.11.4 `openat`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/openat.html>

Gnulib module: `openat`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, Mac OS X 10.5, FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 1.5.x, mingw, MSVC 14. But the replacement function is not safe to be used in libraries and is not multithread-safe.
- Some platforms do not support `O_CLOEXEC`: AIX 7.1, Solaris 10.
- Some platforms do not support `O_DIRECTORY`: glibc 2.0, Mac OS X 10.4, FreeBSD 7.4, NetBSD 4.0.1, OpenBSD 4.9, HP-UX 11, Solaris 10, Cygwin 1.5.x, mingw, MSVC 14.
- On platforms where `off_t` is a 32-bit type, `open` may not work correctly with files 2 GiB and larger. See [Section 14.2 \[Large File Support\]](#), page 842.
- This function does not fail when the file name argument ends in a slash and (without the slash) names a nonexistent file or a file that is not a directory, on some platforms: Solaris 9.

Portability problems not fixed by Glibc:

- The Glibc replacement for `O_CLOEXEC` is not atomic, and so is not safe in the presence of multiple threads or signal handlers.
- The Glibc replacement for `O_SEARCH` may require read access instead of search/execute access.
- On some platforms, `O_SEARCH` may reject directories where you have search access even though `O_SEARCH` is not replaced: macOS 12.6.
- When the file does not exist, POSIX says the behavior of `O_CREAT | O_DIRECTORY` is unspecified. Most platforms fail with `EINVAL` in this case. However, in Linux kernels 6.3 and earlier, this function creates a regular file and then either returns successfully (Linux kernel 5.6 and earlier) or fails with `ENOTDIR` (Linux kernels 5.7–6.3).
- `openat (fd, "symlink", O_NOFOLLOW ...)` fails with `errno` set to `EMLINK` instead of the POSIX-required `ELoop` on some platforms: FreeBSD 10.1.
- `openat (fd, "symlink", O_NOFOLLOW ...)` fails with `errno` set to `EFTYPE` instead of the POSIX-required `ELoop` on some platforms: NetBSD 6.1.

10.11.5 `posix_fadvise`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/posix_fadvise.html

Glibc module: —

Portability problems fixed by Glibc:

- On platforms where `off_t` is a 32-bit type, this function may not work correctly across the entire data range of files 2 GiB and larger. See [Section 14.2 \[Large File Support\]](#), [page 842](#).

Portability problems not fixed by Glibc:

- This function is missing on some platforms: macOS 14, FreeBSD 6.0, NetBSD 3.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11.23, Solaris 10, Cygwin 1.5.x, mingw, MSVC 14, Android 4.4.

10.11.6 `posix_fallocate`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/posix_fallocate.html

Glibc module: —

Portability problems fixed by Glibc:

- On platforms where `off_t` is a 32-bit type, this function may not work correctly across the entire data range of files 2 GiB and larger. See [Section 14.2 \[Large File Support\]](#), [page 842](#).

Portability problems not fixed by Glibc:

- This function is missing on some platforms: macOS 14, FreeBSD 6.0, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 10, Cygwin 1.5.x, mingw, MSVC 14, Android 4.4.

- POSIX specifies that `EINVAL` should be returned when the file system doesn't support the allocation operation directly. glibc however emulates the file system allocation with writes where unsupported, and apps have depended on that long term implementation. This is both inefficient, and as of glibc 2.21 at least, buggy on certain NFS setups.

10.12 Functions in `<fenv.h>`

10.12.1 `feclearexcept`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/feclearexcept.html>

Gnulib module: `fenv-exceptions-tracking-c99`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, Solaris 9, Cygwin 1.7.7, MSVC 9, Android 4.4.
- This function is broken on some platforms: OpenBSD 7.4/mips64, Minix 3.3/i386.

Portability problems not fixed by Gnulib:

10.12.2 `fegetenv`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fegetenv.html>

Gnulib module: `fenv-environment`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, Solaris 9, Cygwin 1.7.7, MSVC 9, Android 4.4.
- This function clears all exception trap bits on some platforms: glibc 2.19/x86_64, Mac OS X 10.5/i386, Mac OS X 10.5/x86_64.
- This function does not save the exception trap bits on some platforms: AIX 7.3.

Portability problems not fixed by Gnulib:

10.12.3 `fegetexceptflag`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fegetexceptflag.html>

Gnulib module: `fenv-exceptions-state-c99`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, Solaris 9, Cygwin 1.7.7, MSVC 9, Android 4.4.

Portability problems not fixed by Gnulib:

10.12.4 fegetmode

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Control-Functions.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.12.5 fegetround

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fegetround.html>

Gnulib module: fenv-rounding

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, AIX 5.1, Solaris 9, Cygwin 1.7.7, MSVC 9, Android 4.4.

Portability problems not fixed by Gnulib:

10.12.6 feholdexcept

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/feholdexcept.html>

Gnulib module: fenv-environment

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, AIX 5.1, Solaris 9, Cygwin 1.7.7, MSVC 9, Android 4.4.
- This function does not work on some platforms: glibc 2.5, FreeBSD 12.2/arm64.
- This function does not save the exception trap bits on some platforms: AIX 7.3.
- This function does not clear the exception trap bits on some platforms: musl libc, mingw.
- This function may fail on some platforms: MSVC 14.
-

Portability problems not fixed by Gnulib:

10.12.7 feraiseexcept

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/feraiseexcept.html>

Gnulib module: fenv-exceptions-tracking-c99

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, Solaris 9, Cygwin 1.7.7, MSVC 9, Android 4.4.
- This function does not detect failures on glibc 2.19/arm.
- This function does not trigger traps on Cygwin 3.4.x/x86_64.

Portability problems not fixed by Gnulib:

10.12.8 fesetenv

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fesetenv.html>

Gnulib module: fenv-environment

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, Solaris 9, Cygwin 1.7.7, MSVC 9, Android 4.4.
- This function does not work on some platforms: FreeBSD 12.2/arm64, mingw 13.
- This function does not restore the floating-point exception trap bits on some platforms: musl libc/i386, musl libc/x86_64, AIX 7.3, Solaris 10 and 11, MSVC 14.
- The macro `FE_DFL_ENV` cannot be used because it leads to a link error on some platforms: NetBSD 10.0/hppa, NetBSD 10.0/sparc, Cygwin 2.9.0.
- This function, when called with `FE_DFL_ENV` argument, has no effect on the x86 `mxcsr` register and thus on floating-point operations performed in the SSE unit on some platforms: mingw 10.

Portability problems not fixed by Gnulib:

10.12.9 fesetexcept

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Status-bit-operations.html.

Gnulib module: fenv-exceptions-tracking-c23

Portability problems fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.
- This function triggers floating-point exception traps although it shouldn't, on glibc 2.37/i386, glibc 2.37/powerpc.

Portability problems not fixed by Gnulib:

10.12.10 fesetexceptflag

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fesetexceptflag.html>

Gnulib module: fenv-exceptions-state-c99

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, Solaris 9, Cygwin 1.7.7, MSVC 9, Android 4.4.
- This function triggers floating-point exception traps although it shouldn't, on glibc 2.37/i386, glibc 2.37/x86_64, glibc 2.37/powerpc, musl libc, Mac OS X 10.5, Minix 3.3, mingw, Haiku.
- This function clears too many floating-point exception flags on glibc 2.37/alpha.
- This function loses the `FE_INVALID` flag on FreeBSD 14.0/powerpc64, NetBSD 9.3/powerpc.
- This function is also buggy on AIX 7.3.1.

Portability problems not fixed by Gnulib:

10.12.11 fesetmode

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Control-Functions.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.12.12 fesetround

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fesetround.html>

Gnulib module: `fenv-rounding`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, AIX 5.1, Solaris 9, Cygwin 1.7.7, MSVC 9, Android 4.4.
- This function does not work on some platforms: MSVC 14.

Portability problems not fixed by Gnulib:

- The rounding mode has no effect on `long double` operations on some platforms: Android/arm64, FreeBSD/arm64, NetBSD/sparc64, OpenBSD/arm64, OpenBSD/mips64.

10.12.13 fetestexcept

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fetestexcept.html>

Gnulib module: `fenv-exceptions-tracking-c99`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, Solaris 9, Cygwin 1.7.7, MSVC 9, Android 4.4.

- This function clears the floating-point exception trap bits on some platforms: NetBSD 9.3/x86_64.
- This function is broken on some platforms: Minix 3.3/i386.

Portability problems not fixed by GnuLib:

10.12.14 `fetestexceptflag`

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Status-bit-operations.html.

GnuLib module: `fenv-exceptions-state-c23`

Portability problems fixed by GnuLib:

- This function is missing on all non-glibc platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

Portability problems not fixed by GnuLib:

10.12.15 `feupdateenv`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/feupdateenv.html>

GnuLib module: `fenv-environment`

Portability problems fixed by GnuLib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, Solaris 9, Cygwin 1.7.7, MSVC 14, Android 4.4.
- This function crashes when given the argument `FE_DFL_ENV` on some platforms: glibc 2.37/riscv64.
- This function does not work on some platforms: glibc 2.5/ia64, FreeBSD 12.2/arm64, mingw 13.
- This function forgets about the currently set floating-point exception flags on some platforms: Mac OS X 10.5/i386, Mac OS X 10.5/x86_64.
- This function does not restore the floating-point exception trap bits on some platforms: musl libc/i386, musl libc/x86_64, AIX 7.3, Solaris 10 and 11, mingw 10.
- This function does not trigger traps on glibc 2.37/hppa, musl libc/s390x, NetBSD 10.0/powerpc.
- This function may fail on some platforms: MSVC 14.

Portability problems not fixed by GnuLib:

10.13 Functions in `<fmtmsg.h>`

10.13.1 `fmtmsg`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fmtmsg.html>

GnuLib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.14 Functions in <fnmatch.h>

10.14.1 fnmatch

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fnmatch.html>

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-fnmatch-3.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Wildcard-Matching.html,
- `man fnmatch`.

Gnulib module: `fnmatch` or `fnmatch-gnu`

Portability problems fixed by either Gnulib module `fnmatch` or `fnmatch-gnu`:

- This function is missing on some platforms: mingw, MSVC 14.
- The "?" pattern character fails to match characters outside the single-byte range on some platforms: NetBSD 10.0, Android 13.
- The "?" pattern character fails to match characters outside the Unicode BMP on some platforms: Solaris 10, Cygwin 3.4.6, MSVC, 32-bit AIX.
- In the pattern, negated character ranges (such as `[!a-z]`) are not supported on some platforms: Solaris 11.4.
- In the pattern, character classes (such as `[:alnum:]`) inside bracket expressions are not supported on some platforms: FreeBSD 13.2, NetBSD 10.0, Solaris 11 OpenIndiana, Cygwin 3.4.6.
- In the pattern, character classes (such as `[:alnum:]`) inside bracket expressions fail to match characters outside the single-byte range on some platforms: Android 13.
- The character class `[:cntrl:]` matches the empty string on some platforms: Solaris 11.4.
- Ranges that start or end with a backslash don't work right on some platforms: glibc 2.3.3.
- In the pattern, an opening bracket without closing bracket does not match a literal '[' on some platforms: glibc 2.12, macOS 12.5, NetBSD 10.0.

Portability problems fixed by Gnulib module `fnmatch-gnu`:

- This function does not support the flags `FNM_LEADING_DIR` and `FNM_CASEFOLD` on some platforms: AIX 7.2, HP-UX 11.31, Solaris 10.
- The flag `FNM_CASEFOLD` does not work in many situations on some platforms: NetBSD 10.0.

- The flag `FNM_CASEFOLD` does not work for multibyte characters consisting of more than one byte on some platforms: Android 13.
- This function does not support the flag `FNM_EXTMATCH` on all non-glibc platforms: musl libc, macOS 12.5, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.2, Minix 3.3, AIX 7.2, HP-UX 11.31, Solaris 11.4, Cygwin 3.4.6, Android 13.
- This function does not support the flag `FNM_FILE_NAME` as an alias of `FNM_PATHNAME` on some platforms: NetBSD 10.0, AIX 7.2, HP-UX 11.31, Solaris 10.

Portability problems not fixed by Gnulib:

Note: Gnulib's replacement function has some limitations:

- It does not implement patterns with collating elements (such as "`[[.ch.]]`") or equivalence classes (such as "`[[=a=]]`").

10.15 Functions in `<ftw.h>`

10.15.1 `nftw`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/nftw.html>

Gnulib module: —

Portability problems fixed by Gnulib:

- On platforms where `off_t` is a 32-bit type, this function may not correctly report the size of files or block devices 2 GiB and larger. See [Section 14.2 \[Large File Support\]](#), [page 842](#).

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 3.0, Minix 3.1.8, mingw, MSVC 14, Android 4.1.

10.16 Functions in `<glob.h>`

10.16.1 `glob`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/glob.html>

Gnulib module: `glob`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14, Android 8.1.
- This function does not list symbolic links to nonexistent files among the results, on some platforms: glibc 2.26, AIX 7.2, HP-UX 11, Solaris 11.4.
- On platforms where `off_t` is a 32-bit type, this function may not work correctly on huge directories 2 GiB and larger. See [Section 14.2 \[Large File Support\]](#), [page 842](#).
- This function makes a recursive call for every trailing `/` and every `/` following a wildcard character on some platforms, which can cause the stack to overflow: glibc 2.42, Cygwin, Android.

Portability problems not fixed by GnuLib:

- Some platforms may store additional flags in the `gl_flags` field.

10.16.2 globfree

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/globfree.html>

GnuLib module: `glob`

Portability problems fixed by GnuLib:

- This function is missing on some platforms: mingw, MSVC 14, Android 8.1.

Portability problems not fixed by GnuLib:

10.17 Functions in `<grp.h>`

10.17.1 endgrent

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/endgrent.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: mingw, MSVC 14, Android 7.1.

10.17.2 getgrent

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getgrent.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: mingw, MSVC 14, Android 7.1.

10.17.3 getgrgid

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getgrgid.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: mingw, MSVC 14.

10.17.4 `getgrgid_r`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/getgrgid_r.html

Gnulib module: extensions

Portability problems fixed by Gnulib:

- This function has an incompatible declaration on some platforms: Solaris 11.3 (when `_POSIX_PTHREAD_SEMANTICS` is not defined).

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14, Android 6.0.

10.17.5 `getgrnam`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getgrnam.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

10.17.6 `getgrnam_r`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/getgrnam_r.html

Gnulib module: extensions

Portability problems fixed by Gnulib:

- This function has an incompatible declaration on some platforms: Solaris 11.3 (when `_POSIX_PTHREAD_SEMANTICS` is not defined).

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14, Android 6.0.

10.17.7 `setgrent`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/setgrent.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14, Android 7.1.

10.18 Functions in <iconv.h>

10.18.1 iconv

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/iconv.html>

Gnulib module: iconv

Portability problems fixed by Gnulib:

- GNU libiconv is not found if installed in \$PREFIX/lib.

Portability problems handled by Gnulib (in the sense that HAVE_ICONV does not get defined if the system's iconv function has this problem):

- Failures are not distinguishable from successful returns on some platforms: macOS 14.4, AIX 5.1..7.2, Solaris 10.
- A buffer overrun can occur on some platforms: AIX 6.1..7.1.

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 8.1, when GNU libiconv is not installed.
- This function was not correctly implemented in glibc versions before 2.2.
- When iconv encounters an input character that is valid but that cannot be converted to the output character set, glibc's and GNU libiconv's iconv stop the conversion. Some other implementations put an implementation-defined character into the output buffer. Gnulib provides higher-level facilities striconv and striconveh (wrappers around iconv) that deal with conversion errors in a platform independent way.
- This function returns a positive return value, instead of zero, when converting from ISO-8859-1 to UTF-8 on HP-UX 11.

10.18.2 iconv_close

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/iconv_close.html

Gnulib module: iconv

Portability problems fixed by Gnulib:

- GNU libiconv is not found if installed in \$PREFIX/lib.

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 8.1, when GNU libiconv is not installed.

10.18.3 iconv_open

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/iconv_open.html

Gnulib module: iconv, iconv_open, iconv_open-utf

Portability problems fixed by either Gnulib module iconv or iconv_open:

- GNU libiconv is not found if installed in \$PREFIX/lib.

Portability problems handled by either Gnulib module `iconv` or `iconv_open` (in the sense that `HAVE_ICONV` does not get defined if the system's `iconv_open` function has this problem):

- No converter from EUC-JP to UTF-8 is provided on some platforms: HP-UX 11.

Portability problems fixed by Gnulib module `iconv_open`:

- This function recognizes only non-standard aliases for many encodings (not the IANA registered encoding names) on many platforms: AIX 5.1, HP-UX 11, Solaris 11 2010-11, z/OS.

Portability problems fixed by Gnulib module `iconv_open-utf`:

- This function does not support the encodings UTF-16BE, UTF-16LE, UTF-32BE, UTF-32LE on many platforms: AIX 5.1, HP-UX 11.

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 8.1, when GNU libiconv is not installed.
- For some encodings A and B, this function cannot convert directly from A to B, although an indirect conversion from A through UTF-8 to B is possible. This occurs on some platforms: Solaris 11 2010-11. Gnulib provides a higher-level facility `striconveh` (a wrapper around `iconv`) that deals with this problem.
- The set of supported encodings and conversions is system dependent.

10.19 Functions in `<inttypes.h>`

10.19.1 `imaxabs`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/imaxabs.html>

Gnulib module: `imaxabs`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11.23, Solaris 9, MSVC 9, Android 4.3.

Portability problems not fixed by Gnulib:

10.19.2 `imaxdiv`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/imaxdiv.html>

Gnulib module: `imaxdiv`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11.23, Solaris 9, MSVC 9, Android 4.3.

Portability problems not fixed by Gnulib:

10.19.3 strtoumax

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/strtoumax.html>

Gnulib module: strtoumax

Portability problems fixed by Gnulib:

- This function is missing on some platforms: OpenBSD 3.8, Minix 3.1.8, Solaris 9, MSVC 9.
- This function fails on valid input strings on some platforms: AIX 5.1.

Portability problems not fixed by Gnulib:

- This function is only defined as a macro on some platforms: HP-UX 11.11.
- If no conversion could be performed, this function sets `errno` to `EINVAL` on some platforms. Portable programs should check for no conversion by comparing `str` to `*endptr`, not by inspecting `errno`.

10.19.4 strtoumax

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/strtoumax.html>

Gnulib module: strtoumax

Portability problems fixed by Gnulib:

- This function is missing on some platforms: OpenBSD 3.8, Minix 3.1.8, AIX 5.1, Solaris 9, MSVC 9.

Portability problems not fixed by Gnulib:

- This function is only defined as a macro on some platforms: HP-UX 11.11.
- If no conversion could be performed, this function sets `errno` to `EINVAL` on some platforms. Portable programs should check for no conversion by comparing `str` to `*endptr`, not by inspecting `errno`.

10.19.5 wcstoumax

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wcstoumax.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11.11, Solaris 9, Cygwin 1.5.x, MSVC 9, Android 4.4.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.19.6 wcstoumax

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wcstoumax.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11.11, Solaris 9, Cygwin 1.5.x, MSVC 9, Android 4.4.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.20 Functions in <langinfo.h>

10.20.1 nl_langinfo

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/nl_langinfo.html

Gnulib module: `nl_langinfo`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14, Android 7.1.
- The constant `CODESET` is not supported on some platforms: OpenBSD 3.8.
- The constants `ALTMON_1` to `ALTMON_12` are not supported on some platforms: glibc 2.26, musl libc, macOS 14, NetBSD 8.0, OpenBSD 7.5, AIX 7.2, HP-UX 11.31, Solaris 11.4, Haiku, Cygwin 2.9.
- The constants `ABALTMON_1` to `ABALTMON_12` are not supported on many platforms: glibc 2.42, musl libc, macOS 14, NetBSD 10.0, OpenBSD 7.6, AIX 7.3, HP-UX 11.31, Solaris 11.4, Haiku, Cygwin 3.5.
- The constants `ERA`, `ERA_D_FMT`, `ERA_D_T_FMT`, `ERA_T_FMT`, `ALT_DIGITS` are not supported on some platforms: OpenBSD 7.5.
- This function is not multithread-safe on some platforms: macOS 26, Solaris 11.3.

Portability problems not fixed by Gnulib:

- On Cygwin 1.5.x, which doesn't have locales, `nl_langinfo(CODESET)` always returns `"US-ASCII"`.
- On Cygwin 1.7.0, only the charset portion of a locale designation is honored.
- On NetBSD 5.0, in some locales, `nl_langinfo(CRNCYSTR)` returns the empty string, although the local currency symbol, as returned by `localeconv()->currency_symbol`, is non-empty.
- On NetBSD 5.1, in the `"C"` locale, the results of `nl_langinfo(ABMON_1) ... nl_langinfo(ABMON_12)` are full month names, not abbreviated month names.

10.20.2 nl_langinfo_l

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/nl_langinfo_l.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 6.0, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.3, Cygwin 1.7.x, mingw, MSVC 14, Android 7.1.

10.21 Functions in <libgen.h>

10.21.1 basename

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/basename.html>

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-basename-3.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.
- glibc and Android have two different functions **basename**: the POSIX version and the GNU version.
- **basename** assumes file names in POSIX syntax; it does not work with file names in Windows syntax.

The Gnulib module **basename-lgpl** provides similar API, with a function **last_component**, that also works with Windows file names.

10.21.2 dirname

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/dirname.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.
- **dirname** assumes file names in POSIX syntax; it does not work with file names in Windows syntax.

The Gnulib module **dirname** provides similar API, with functions **dir_name** and **mdir_name**, that also works with Windows file names.

10.22 Functions in <libintl.h>

10.22.1 bind_textdomain_codeset

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/bind_textdomain_codeset.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Charset-conversion-in-gettext.html,
- `man bind_textdomain_codeset`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.22.2 bindtextdomain

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/bindtextdomain.html>

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Locating-gettext-catalog.html,
- `man bindtextdomain`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.22.3 dcgettext

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/dcgettext.html>

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Translation-with-gettext.html,
- `man dcgettext`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

- This function does not treat a `LANGUAGE` environment variable with an empty value like an unset `LANGUAGE` environment variable on some platforms: NetBSD 10.0.

10.22.4 `dcgettext_l`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/dcgettext_l.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc 2.42, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android 9.0.

10.22.5 `dcngettext`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/dcngettext.html>

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Advanced-gettext-functions.html,
- `man dcngettext`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.
- This function does not treat a `LANGUAGE` environment variable with an empty value like an unset `LANGUAGE` environment variable on some platforms: NetBSD 10.0.

10.22.6 `dcngettext_l`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/dcngettext_l.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc 2.41, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android 9.0.

10.22.7 dgettext

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/dgettext.html>

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Translation-with-gettext.html,
- `man dgettext`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.
- This function does not treat a `LANGUAGE` environment variable with an empty value like an unset `LANGUAGE` environment variable on some platforms: NetBSD 10.0.

10.22.8 dgettext_l

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/dgettext_l.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc 2.42, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android 9.0.

10.22.9 dngettext

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/dngettext.html>

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Advanced-gettext-functions.html,
- `man dngettext`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.
- This function does not treat a `LANGUAGE` environment variable with an empty value like an unset `LANGUAGE` environment variable on some platforms: NetBSD 10.0.

10.22.10 `dngettext_1`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/dngettext_1.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc 2.42, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android 9.0.

10.22.11 `gettext`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/gettext.html>

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Translation-with-gettext.html,
- `man gettext`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.
- This function does not treat a `LANGUAGE` environment variable with an empty value like an unset `LANGUAGE` environment variable on some platforms: NetBSD 10.0.

10.22.12 `gettext_1`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/gettext_1.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc 2.42, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android 9.0.

10.22.13 `ngettext`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/ngettext.html>

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Advanced-gettext-functions.html,

- `man ngettext`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.
- This function does not treat a `LANGUAGE` environment variable with an empty value like an unset `LANGUAGE` environment variable on some platforms: NetBSD 10.0.

10.22.14 `ngettext_1`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/ngettext_1.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc 2.42, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android 9.0.

10.22.15 `textdomain`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/textdomain.html>

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Locating-gettext-catalog.html,
- `man textdomain`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.23 Functions in `<locale.h>`

10.23.1 `duplocale`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/duplocale.html>

Gnulib module: `duplocale`

Portability problems fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 9.0, NetBSD 6.1, OpenBSD 6.1, Minix 3.1.8, AIX 6.1, HP-UX 11, Solaris 11.3, Cygwin 2.5.x, mingw, MSVC 14, Android 4.4.

- This function is useless because the `locale_t` type is not defined on some platforms: z/OS.
- The argument `LC_GLOBAL_LOCALE` is not supported on some platforms: glibc 2.11, AIX 7.1.
- With the argument `LC_GLOBAL_LOCALE`, this function returns a wrong result on some platforms: NetBSD 7.1.

Portability problems not fixed by Gnuilib:

- With the argument `LC_GLOBAL_LOCALE`, this function returns a wrong result on some platforms: Haiku.

10.23.2 freelocale

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/freelocale.html>

Gnuilib module: `freelocale`

Portability problems fixed by Gnuilib:

- This function is missing on many platforms: FreeBSD 9.0, NetBSD 6.1, OpenBSD 6.1, Minix 3.1.8, AIX 6.1, HP-UX 11, Solaris 11.3, Cygwin 2.5.x, mingw, MSVC 14, Android 4.4.
- This function is useless because the `locale_t` type is not defined on some platforms: z/OS.

Portability problems not fixed by Gnuilib:

- This function's return type is `int` instead of `void` on macOS.
- This function may cause crashes in subsequent `newlocale` invocations on some platforms: Haiku.

10.23.3 getlocalename_l

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/getlocalename_l.html

Gnuilib module: `getlocalename_l-simple`

Portability problems fixed by Gnuilib:

- This function is missing on many platforms: glibc 2.42, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.x, mingw, MSVC 14, Android 9.0. But the gnuilib replacement supports only a single locale category: the argument `LC_ALL` is unsupported.
- This function returns an invalid pointer when the second argument is `LC_GLOBAL_LOCALE`, on some platforms: Cygwin 3.6.0.

Portability problems not fixed by Gnuilib:

- This function supports only a single locale category, i.e. the argument `LC_ALL` is unsupported, on some platforms: Cygwin 3.6.0.

10.23.4 localeconv

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/localeconv.html>

Gnulib module: localeconv

Portability problems fixed by Gnulib:

- The `struct lconv` type does not contain any members on some platforms: Android up to 2014.
- The `struct lconv` type does not contain the members `int_p_cs_precedes`, `int_p_sign_posn`, `int_p_sep_by_space`, `int_n_cs_precedes`, `int_n_sign_posn`, `int_n_sep_by_space` on some platforms: glibc, OpenBSD 4.9, HP-UX 11, Solaris 11.4, Cygwin 1.5.x, mingw, MSVC 14.
- The values of fields of `struct lconv` of type `char` are -1 instead of `CHAR_MAX` on some platforms: mingw.

Portability problems not fixed by Gnulib:

10.23.5 newlocale

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/newlocale.html>

Gnulib module: newlocale

Portability problems fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 9.0, NetBSD 6.1, OpenBSD 6.1, Minix 3.1.8, AIX 6.1, HP-UX 11, Solaris 11.3, Cygwin 2.5.x, mingw, MSVC 14, Android 4.4.
- This function is useless because the `locale_t` type is not defined on some platforms: z/OS.
- When the third argument is `NULL`, this function uses locale category data from the current locale instead of from the "C" locale on some platforms: macOS, NetBSD 10.0, Solaris 11 OpenIndiana.
- When the second argument is an invalid or unsupported locale name, this function uses the "C" locale instead of failing on some platforms: NetBSD 10.0.

Portability problems not fixed by Gnulib:

- This function is useless because the `locale_t` type contains basically no information on some platforms: OpenBSD 6.3.

10.23.6 setlocale

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/setlocale.html>

Gnulib module: setlocale, setlocale-null

Portability problems fixed by Gnulib module `setlocale`:

- On Windows platforms (excluding Cygwin), `setlocale(category, "")` and `setlocale(category, NULL)` both ignore the environment variables `LC_ALL`, `category`, and `LANG`.

- On Windows platforms (excluding Cygwin) and Cygwin 1.5.x, `setlocale(LC_ALL, name)` succeeds and sets the `LC_CTYPE` category to 'C' when it does not support the encoding, instead of failing.
- On Windows platforms (excluding Cygwin), `setlocale` understands different locale names, that are not based on ISO 639 language names and ISO 3166 country names.
- On Android 5.0 and newer, setting any category other than `LC_CTYPE` of the global locale affects the `LC_CTYPE` category as well.
- On Android < 5.0, which doesn't have locales, the `setlocale` function always fails. The replacement, however, supports only the locale names "C" and "POSIX".

Portability problems fixed by Gnuilib module `setlocale` or `setlocale-null`:

- Invocations of `setlocale(..., NULL)` are not multithread-safe on some platforms: `musl libc`, `macOS`, `FreeBSD`, `NetBSD`, `OpenBSD`, `AIX`, `Haiku`, `Cygwin`. To make these invocations multithread-safe, you need the Gnuilib module `setlocale`, or you need to change the code to invoke `setlocale_null` or `setlocale_null_r` instead.

Portability problems not fixed by Gnuilib:

- On Cygwin 1.5.x, which doesn't have locales, `setlocale(LC_ALL, NULL)` always returns "C".
- On Cygwin 1.7.0, only the charset portion of a locale designation is honored.
- On OpenBSD, `setlocale(LC_ALL, "")` will only update categories that are deemed appropriate for the `LC_ALL` environment value, even if there are other categories set to different values in the environment. In addition any value is accepted for `LC_CTYPE`, and so `NULL` is never returned to indicate a failure to set locale. To verify category values, each category must be set individually with `setlocale(LC_COLLATE, "")` etc.
- On Android 5.0 and newer, the default locale (i.e. the locale in use when `setlocale` was not called) is the "C.UTF-8" locale, not the "C" locale. Additionally, a `setlocale` call that is meant to set the "C" or "POSIX" locale actually sets an equivalent of the "C.UTF-8" locale.

Note: The names of locales with UTF-8 encoding are platform dependent:

- On most platforms, they follow the naming scheme '`ll_CC.UTF-8`'. Here '`ll`' is an ISO 639 two-letter language code, and '`CC`' is an ISO 3166 two-letter country code. For example, '`en_US.UTF-8`' (this locale exists on nearly all systems), '`fr_FR.UTF-8`', and similar.
- On native Windows with mingw, UTF-8 locales are not supported.
- On native Windows with MSVC (version 14 or newer, and with the `-MD` option in `CFLAGS`), they follow the naming scheme '`language_country.65001`' or, equivalently, '`language_country.utf8`', where '`language`' is a language name in English and '`country`' is a country name in English. For example '`English_United States.65001`' (yes, with a space in it!) or '`French_France.65001`'.

10.23.7 uselocale

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/uselocale.html>

Gnuilib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on many platforms: FreeBSD 9.0, NetBSD 10.0, OpenBSD 6.1, Minix 3.1.8, AIX 6.1, HP-UX 11, Solaris 11.3, Cygwin 2.5.x, mingw, MSVC 14, Android API level 20.
- This function is not documented and leads to crashes in subsequent `setlocale` invocations on some platforms: AIX 7.2.
- This function is useless because the `locale_t` type is not defined on some platforms: z/OS.
- This function is useless because the `locale_t` type contains basically no information on some platforms: OpenBSD 7.4.

10.24 Functions in `<math.h>`

10.24.1 `acos`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/acos.html>

GnuLib module: `acos`

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

10.24.2 `acosf`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/acosf.html>

GnuLib module: `acosf`

Portability problems fixed by GnuLib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, Solaris 9.
- This function is only defined as a macro with arguments on some platforms: MSVC 14.

Portability problems not fixed by GnuLib:

10.24.3 `acosh`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/acosh.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 9.

10.24.4 acoshf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/acoshf.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, mingw, MSVC 9.

10.24.5 acoshl

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/acoshl.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 9, Android 4.4.
- This function produces results which are accurate to only 16 digits on some platforms: NetBSD 10.0.

10.24.6 acosl

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/acosl.html>

Gnulib module: `acosl`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, HP-UX 11, Solaris 9, Cygwin 1.7.x, Android 4.4.
- This function is only defined as a macro with arguments on some platforms: MSVC 14.

Portability problems not fixed by Gnulib:

- This function produces results which are accurate to only 16 digits on some platforms: NetBSD 10.0.

10.24.7 acospi

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.4.8

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-C23 platforms: glibc 2.40, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.8 `acospi`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.4.8

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-C23 platforms: glibc 2.40, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.9 `acospi`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.4.8

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-C23 platforms: glibc 2.40, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.10 `asin`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/asin.html>

Gnulib module: `asin`

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

10.24.11 `asinf`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/asinf.html>

Gnulib module: `asinf`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, Solaris 9.
- This function is only defined as a macro with arguments on some platforms: MSVC 14.

Portability problems not fixed by Gnulib:

10.24.12 asinh

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/asinh.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 9.

10.24.13 asinhf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/asinhf.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, mingw, MSVC 9.

10.24.14 asinhl

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/asinhl.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 9, Android 4.4.
- This function produces results which are accurate to only 16 digits on some platforms: NetBSD 10.0.

10.24.15 asinl

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/asinl.html>

Gnulib module: asinl

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, HP-UX 11, Solaris 9, Cygwin 1.7.x, Android 4.4.
- This function is only defined as a macro with arguments on some platforms: MSVC 14.

Portability problems not fixed by Gnulib:

- This function produces results which are accurate to only 16 digits on some platforms: NetBSD 10.0.

10.24.16 asinpi

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.4.9

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-C23 platforms: glibc 2.40, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.17 asinpif

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.4.9

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-C23 platforms: glibc 2.40, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.18 asinpil

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.4.9

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-C23 platforms: glibc 2.40, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.19 atan

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/atan.html>

Gnulib module: atan

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

10.24.20 atan2

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/atan2.html>

Gnulib module: atan2

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

10.24.21 atan2f

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/atan2f.html>

GnuLib module: atan2f

Portability problems fixed by GnuLib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, Solaris 9.
- This function is only defined as a macro with arguments on some platforms: MSVC 14.

Portability problems not fixed by GnuLib:

10.24.22 atan2l

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/atan2l.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, HP-UX 11, Solaris 9, Cygwin 1.7.x, Android 4.4.
- This function is only defined as a macro with arguments on some platforms: MSVC 14.
- This function produces results which are accurate to only 16 digits on some platforms: NetBSD 10.0.

10.24.23 atan2pi

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.4.11

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on non-C23 platforms: glibc 2.40, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.24 atan2pif

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.4.11

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on many non-C23 platforms: glibc 2.40, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.25 atan2pil

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.4.11

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on many non-C23 platforms: glibc 2.40, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.26 atanf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/atanf.html>

GnuLib module: atanf

Portability problems fixed by GnuLib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, Solaris 9.
- This function is only defined as a macro with arguments on some platforms: MSVC 14.

Portability problems not fixed by GnuLib:

10.24.27 atanh

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/atanh.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 9.

10.24.28 atanhf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/atanhf.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, mingw, MSVC 9.

10.24.29 `atanhl`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/atanhl.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 9, Android 4.4.
- This function produces results which are accurate to only 16 digits on some platforms: NetBSD 10.0.

10.24.30 `atanl`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/atanl.html>

Gnulib module: `atanl`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, HP-UX 11, Solaris 9, Cygwin 1.7.x, Android 4.4.
- This function is only defined as a macro with arguments on some platforms: MSVC 14.

Portability problems not fixed by Gnulib:

- This function produces results which are accurate to only 16 digits on some platforms: NetBSD 10.0.

10.24.31 `atanpi`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.4.10

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-C23 platforms: glibc 2.40, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.32 `atanpif`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.4.10

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-C23 platforms: glibc 2.40, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.33 atanpil

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.4.10

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on non-C23 platforms: glibc 2.40, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.34 canonicalize

Documentation:

https://www.gnu.org/software/libc/manual/html_node/FP-Bit-Twiddling.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.24.35 canonicalizef

Documentation:

https://www.gnu.org/software/libc/manual/html_node/FP-Bit-Twiddling.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.24.36 canonicalizel

Documentation:

https://www.gnu.org/software/libc/manual/html_node/FP-Bit-Twiddling.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.24.37 `cbirt`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/cbirt.html>

Gnulib module: `cbirt` or `cbirt-ieee`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, MSVC 9.

Portability problems not fixed by Gnulib:

10.24.38 `cbirtf`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/cbirtf.html>

Gnulib module: `cbirtf` or `cbirtf-ieee`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, Solaris 9, MSVC 9.

Portability problems not fixed by Gnulib:

10.24.39 `cbirtl`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/cbirtl.html>

Gnulib module: `cbirtl` or `cbirtl-ieee`

Portability problems fixed by either Gnulib module `cbirtl` or `cbirtl-ieee`

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, MSVC 9, Android 4.4.
- This function produces grossly wrong results on some platforms: OpenBSD 5.1/SPARC.

Portability problems fixed by Gnulib module `cbirtl-ieee`:

Portability problems not fixed by Gnulib:

10.24.40 `ceil`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/ceil.html>

Gnulib module: `ceil` or `ceil-ieee`

Portability problems fixed by either Gnulib module `ceil` or `ceil-ieee`:

Portability problems fixed by Gnulib module `ceil-ieee`:

- This function returns a positive zero for an argument between -1 and 0 on some platforms: AIX 7.1.

Portability problems not fixed by Gnulib:

10.24.41 `ceilf`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/ceilf.html>

Gnulib module: `ceilf` or `ceilf-ieee`

Portability problems fixed by either Gnulib module `ceilf` or `ceilf-ieee`:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9.
- This function is only defined as a macro with arguments on some platforms: MSVC 14.

Portability problems fixed by Gnulib module `ceilf-ieee`:

- This function returns a positive zero for an argument between -1 and 0 on some platforms: macOS 10.5, AIX 7.1.

Portability problems not fixed by Gnulib:

10.24.42 `ceil`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/ceil.html>

Gnulib module: `ceil` or `ceil-ieee`

Portability problems fixed by either Gnulib module `ceil` or `ceil-ieee`:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, HP-UX 11, Solaris 9, Cygwin 1.7.x.
- This function is only defined as a macro with arguments on some platforms: MSVC 14.
- This function returns a wrong result for small arguments on some platforms: OpenBSD 5.6.

Portability problems fixed by Gnulib module `ceil-ieee`:

Portability problems not fixed by Gnulib:

10.24.43 `compoundn`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.7.2

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-C23 platforms: glibc 2.41, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.44 compoundnf

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.7.2

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-C23 platforms: glibc 2.41, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.45 compoundnl

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.7.2

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-C23 platforms: glibc 2.41, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.46 copysign

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/copysign.html>

Gnulib module: copysign

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, MSVC 9.

Portability problems not fixed by Gnulib:

10.24.47 copysignf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/copysignf.html>

Gnulib module: copysignf

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, Solaris 9, MSVC 9.

Portability problems not fixed by Gnulib:

10.24.48 copysignl

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/copysignl.html>

Gnulib module: copysignl

Portability problems fixed by GnuLib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, MSVC 9.

Portability problems not fixed by GnuLib:

10.24.49 cos

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/cos.html>

GnuLib module: cos

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

10.24.50 cosf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/cosf.html>

GnuLib module: cosf

Portability problems fixed by GnuLib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, Solaris 9.
- This function is only defined as a macro with arguments on some platforms: MSVC 14.

Portability problems not fixed by GnuLib:

10.24.51 cosh

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/cosh.html>

GnuLib module: cosh

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

10.24.52 coshf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/coshf.html>

GnuLib module: coshf

Portability problems fixed by GnuLib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, Solaris 9.
- This function is only defined as a macro with arguments on some platforms: MSVC 14.

Portability problems not fixed by GnuLib:

10.24.53 coshl

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/cosh1.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, HP-UX 11, Solaris 9, Cygwin 1.7.x, Android 4.4.
- This function is only defined as a macro with arguments on some platforms: MSVC 14.
- This function produces results which are accurate to only 16 digits on some platforms: NetBSD 10.0.

10.24.54 cosl

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/cosl.html>

Gnulib module: cosl

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, HP-UX 11, Solaris 9, Cygwin 1.7.x, Android 4.4.
- This function is only defined as a macro with arguments on some platforms: MSVC 14.

Portability problems not fixed by Gnulib:

- This function produces results which are accurate to only 16 digits on some platforms: NetBSD 10.0.

10.24.55 cospi

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.4.12

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-C23 platforms: glibc 2.40, macOS 14, FreeBSD 13.4, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.56 cospif

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.4.12

Gnulib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on many non-C23 platforms: glibc 2.40, macOS 14, FreeBSD 13.4, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.57 `cospil`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.4.12

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on many non-C23 platforms: glibc 2.40, macOS 14, FreeBSD 13.4, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.58 `daddl`

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on all non-glibc platforms: glibc 2.27, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.24.59 `ddivl`

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on all non-glibc platforms: glibc 2.27, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.24.60 `dfmal`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.14.5

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on all non-glibc platforms: glibc 2.34, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.61 `dmull`

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on all non-glibc platforms: glibc 2.27, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.24.62 `dsqrtl`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.14.6

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on all non-glibc platforms: glibc 2.34, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.63 `dsubl`

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on all non-glibc platforms: glibc 2.27, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.24.64 erf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/erf.html>

Gnulib module: erf

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, MSVC 9.

10.24.65 erfc

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/erfc.html>

Gnulib module: erfc

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, MSVC 9.

10.24.66 erfcf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/erfcf.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, MSVC 9.

10.24.67 erfcl

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/erfcl.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 7.1, OpenBSD 3.8, Minix 3.1.8, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 9, Android 4.4.
- This function produces results which are accurate to only 16 digits on some platforms: NetBSD 10.0.

10.24.68 erff

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/erff.html>

Gnulib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, MSVC 9.

10.24.69 `erfl`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/erfl.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 7.1, OpenBSD 3.8, Minix 3.1.8, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 9, Android 4.4.
- This function produces results which are accurate to only 16 digits on some platforms: NetBSD 10.0.

10.24.70 `exp`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/exp.html>

GnuLib module: `exp` or `exp-ieee`

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

10.24.71 `exp10`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.6.2

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Exponents-and-Logarithms.html,
- `man exp10`.

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on many platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 1.5.x, mingw, MSVC 14, Android.

10.24.72 `exp10f`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.6.2

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Exponents-and-Logarithms.html,
- `man exp10f`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 1.5.x, mingw, MSVC 14, Android.

10.24.73 `exp10l`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.6.2

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Exponents-and-Logarithms.html,
- `man exp10l`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 1.5.x, mingw, MSVC 14, Android.

10.24.74 `exp10m1`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.6.3

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.39, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.75 exp10m1f

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.6.3

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.39, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.76 exp10m1l

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.6.3

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.39, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.77 exp2

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/exp2.html>

Gnulib module: exp2 or exp2-ieee

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, Solaris 9, MSVC 9.
- This function returns grossly wrong results on some platforms: OpenBSD 4.9.

Portability problems not fixed by Gnulib:

10.24.78 exp2f

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/exp2f.html>

Gnulib module: exp2f or exp2f-ieee

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, MSVC 9.

Portability problems not fixed by Gnulib:

10.24.79 exp2l

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/exp2l.html>

Gnulib module: `exp2l` or `exp2l-ieee`

Portability problems fixed by either Gnulib module `exp2l` or `exp2l-ieee`:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, MSVC 9, Android 4.4.
- This function returns Infinity instead of large finite values on some platforms: NetBSD 10.0/i386.
- This function produces results which are accurate to only 16 digits on some platforms: NetBSD 10.0/x86_64.

Portability problems fixed by Gnulib module `exp2l-ieee`:

- This function returns a wrong value for a NaN argument on some platforms: OpenBSD 4.9.

Portability problems not fixed by Gnulib:

10.24.80 exp2m1

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.6.5

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.39, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.81 exp2m1f

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.6.5

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.39, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.82 exp2m1l

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.6.5

Gnulib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on all non-glibc platforms: glibc 2.39, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.83 expf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/expf.html>

GnuLib module: expf or expf-ieee

Portability problems fixed by GnuLib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, Solaris 9.
- This function is only defined as a macro with arguments on some platforms: MSVC 14.

Portability problems not fixed by GnuLib:

10.24.84 expl

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/expl.html>

GnuLib module: expl or expl-ieee

Portability problems fixed by GnuLib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, HP-UX 11, Solaris 9, Cygwin 1.7.x, Android 4.4.
- This function returns 0.0 for all arguments on some platforms: Haiku 2017.
- This function returns NaN for small operands on some platforms: OpenBSD 5.4.
- This function is only defined as a macro with arguments on some platforms: MSVC 14.
- This function produces results which are accurate to only 16 digits on some platforms: musl libc 1.2.2/arm64, musl libc 1.2.2/s390x, NetBSD 10.0.

Portability problems not fixed by GnuLib:

10.24.85 expm1

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/expm1.html>

GnuLib module: expm1 or expm1-ieee

Portability problems fixed by either GnuLib module `expm1` or `expm1-ieee`:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 9.

Portability problems fixed by GnuLib module `expm1-ieee`:

- This function has problems when the argument is minus zero on some platforms: AIX 7.1.

Portability problems not fixed by GnuLib:

10.24.86 expm1f

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/expm1f.html>

Gnulib module: expm1f or expm1f-ieee

Portability problems fixed by either Gnulib module `expm1f` or `expm1f-ieee`:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, mingw, MSVC 9.

Portability problems fixed by Gnulib module `expm1f-ieee`:

- This function returns a positive zero for a minus zero argument on some platforms: AIX 7.2.

Portability problems not fixed by Gnulib:

10.24.87 expm1l

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/expm1l.html>

Gnulib module: expm1l or expm1l-ieee

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 9.3, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 9, Android 4.4.
- This function returns 0 instead of small positive values on some platforms: NetBSD 10.0/i386.
- This function produces results which are accurate to only 16 digits on some platforms: musl libc 1.2.2/arm64, musl libc 1.2.2/s390x, Mac OS X 10.5, NetBSD 10.0/x86_64.

Portability problems not fixed by Gnulib:

10.24.88 fabs

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fabs.html>

Gnulib module: fabs or fabs-ieee

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

10.24.89 fabsf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fabsf.html>

Gnulib module: fabsf or fabsf-ieee

Portability problems fixed by Gnulib:

- This function is missing on some platforms: AIX 5.1, Solaris 9.
- This function is only defined as a macro with arguments on some platforms: MSVC 14.

Portability problems not fixed by Gnulib:

10.24.90 fabs1

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fabs1.html>

Gnulib module: fabs1 or fabs1-ieee

Portability problems fixed by Gnulib:

- This function is missing on some platforms: NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, HP-UX 11, Solaris 9, Cygwin 1.7.x.
- This function is only defined as a macro with arguments on some platforms: MSVC 14.

Portability problems not fixed by Gnulib:

10.24.91 fadd

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.27, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.24.92 faddl

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.27, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.24.93 fdim

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fdim.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, Solaris 9, MSVC 9.

10.24.94 fdimf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fdimf.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, MSVC 9.

10.24.95 fdiml

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fdiml.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, MSVC 9.

10.24.96 fdiv

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.27, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.24.97 fdivl

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.27, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.24.98 ffma

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.14.5

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.34, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.99 ffmt

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.14.5

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.34, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.100 floor

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/floor.html>

Gnulib module: floor or floor-ieee

Portability problems fixed by either Gnulib module `floor` or `floor-ieee`:

Portability problems fixed by Gnulib module `floor-ieee`:

Portability problems not fixed by Gnulib:

10.24.101 floorf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/floorf.html>

Gnulib module: floorf or floorf-ieee

Portability problems fixed by either Gnulib module `floorf` or `floorf-ieee`:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9.
- This function is only defined as a macro with arguments on some platforms: MSVC 14.

Portability problems fixed by Gnulib module `floorf-ieee`:

Portability problems not fixed by Gnulib:

10.24.102 `floorl`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/floorl.html>

Gnulib module: `floorl` or `floorl-ieee`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, HP-UX 11, Solaris 9, Cygwin 1.7.x.
- This function is only defined as a macro with arguments on some platforms: MSVC 14.

Portability problems not fixed by Gnulib:

10.24.103 `fma`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fma.html>

Gnulib module: `fma` or `fma-ieee`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, MSVC 9.
- This function produces wrong results on some platforms: glibc 2.11, Mac OS X 10.5, FreeBSD 6.4/x86, NetBSD 8.0, OpenBSD 7.4/arm64, Cygwin 1.5, mingw.

Portability problems not fixed by Gnulib:

10.24.104 `fmaf`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fmaf.html>

Gnulib module: `fmaf` or `fmaf-ieee`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, MSVC 9.
- This function produces wrong results on some platforms: glibc 2.11, Mac OS X 10.5, FreeBSD 6.4/x86, FreeBSD 12.2/arm, OpenBSD 7.4/arm64, Cygwin 1.5, mingw.

Portability problems not fixed by Gnulib:

- This function produces wrong results on some platforms: musl libc/powerpc64le when emulated by QEMU 5.0.0.

10.24.105 fmal

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fmal.html>

Gnulib module: fmal or fmal-ieee

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, MSVC 9, Android 4.4.
- This function produces wrong results on some platforms: glibc 2.17, macOS 13/x86_64, FreeBSD 6.4/x86, OpenBSD 7.4/arm64, mingw.

Portability problems not fixed by Gnulib:

10.24.106 fmax

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fmax.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, Solaris 9, MSVC 9.

10.24.107 fmaxf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fmaxf.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, MSVC 9.

10.24.108 fmaximum

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.12.4

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.34, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.109 fmaximum_mag

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.12.6

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.34, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.110 fmaximum_magf

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.12.6

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.34, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.111 fmaximum_magl

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.12.6

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.34, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.112 fmaximum_mag_num

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.12.10

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

Gnulib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on all non-glibc platforms: glibc 2.34, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.113 `fmaximum_mag_numf`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.12.10

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on all non-glibc platforms: glibc 2.34, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.114 `fmaximum_mag_numl`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.12.10

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on all non-glibc platforms: glibc 2.34, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.115 `fmaximum_num`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.12.8

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on all non-glibc platforms: glibc 2.34, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.116 fmaximum_numf

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.12.8

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.34, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.117 fmaximum_numl

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.12.8

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.34, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.118 fmaximumf

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.12.4

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.34, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.119 fmaximuml

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.12.4

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.34, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.120 `fmaxl`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fmaxl.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, MSVC 9.

10.24.121 `fmaxmag`

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.24.122 `fmaxmagf`

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.24.123 `fmaxmagl`

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

Gnulib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on all non-glibc platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.24.124 fmin

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fmin.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, Solaris 9, MSVC 9.

10.24.125 fminf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fminf.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, MSVC 9.

10.24.126 fminimum

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.12.5

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on all non-glibc platforms: glibc 2.34, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.127 fminimum_mag

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.12.7

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on all non-glibc platforms: glibc 2.34, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.128 `fminimum_magf`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.12.7

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on all non-glibc platforms: glibc 2.34, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.129 `fminimum_magl`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.12.7

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on all non-glibc platforms: glibc 2.34, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.130 `fminimum_mag_num`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.12.11

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on all non-glibc platforms: glibc 2.34, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.131 fminimum_mag_numf

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.12.11

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.34, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.132 fminimum_mag_numl

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.12.11

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.34, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.133 fminimum_num

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.12.9

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.34, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.134 fminimum_numf

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.12.9

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

Gnulib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on all non-glibc platforms: glibc 2.34, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.135 `fminimum_num1`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.12.9

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on all non-glibc platforms: glibc 2.34, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.136 `fminimumf`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.12.5

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on all non-glibc platforms: glibc 2.34, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.137 `fminimuml`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.12.5

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on all non-glibc platforms: glibc 2.34, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.138 fminl

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fminl.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, MSVC 9.

10.24.139 fminmag

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.24.140 fminmagf

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.24.141 fminmagl

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.24.142 fmod

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fmod.html>

Gnulib module: `fmod` or `fmod-ieee`

Portability problems fixed by either Gnulib module `fmod` or `fmod-ieee`:

Portability problems fixed by Gnulib module `fmod-ieee`:

- This function has problems when the first argument is minus zero on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

10.24.143 fmodf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fmodf.html>

Gnulib module: `fmodf` or `fmodf-ieee`

Portability problems fixed by either Gnulib module `fmodf` or `fmodf-ieee`:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, Solaris 9.
- This function is only defined as a macro with arguments on some platforms: MSVC 14.

Portability problems fixed by Gnulib module `fmodf-ieee`:

- This function has problems when the first argument is minus zero on some platforms: MSVC 14.

Portability problems not fixed by Gnulib:

10.24.144 fmodl

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fmodl.html>

Gnulib module: `fmodl` or `fmodl-ieee`

Portability problems fixed by either Gnulib module `fmodl` or `fmodl-ieee`:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, HP-UX 11, Solaris 9, Cygwin 1.7.x, Android 4.4.
- This function is only defined as a macro with arguments on some platforms: MSVC 14.
- This function is not declared and does not work on some platforms: AIX 5.1.

Portability problems fixed by Gnulib module `fmodl-ieee`:

- This function has problems when the first argument is minus zero on some platforms: MSVC 14.

Portability problems not fixed by Gnulib:

10.24.145 fmul

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.27, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.24.146 fmull

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.27, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.24.147 fpclassify

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fpclassify.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, OpenBSD 7.5, AIX 5.1, Solaris 11.4, Android 9.0.

10.24.148 frexp

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/frexp.html>

Gnulib module: frexp or frexp-ieee

Portability problems fixed by Gnulib:

- This function does not work on denormalized numbers on some platforms: NetBSD 3.0.
- This function does not work on negative zero on some platforms: NetBSD 4.99, MSVC 14.
- This function does not work on infinite numbers on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

10.24.149 frexpf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/frexpf.html>

Gnulib module: frexpf or frexpf-ieee

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9.
- This function is only defined as a macro with arguments on some platforms: MSVC 14.
- This function does not work on negative zero on some platforms: mingw.
- This function does not work on infinite numbers on some platforms: mingw.

Portability problems not fixed by Gnulib:

10.24.150 frexpl

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/frexpl.html>

Gnulib module: frexpl or frexpl-ieee

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, HP-UX 11, Solaris 9, Cygwin 1.7.x, Android 4.4.
- This function is only defined as a macro with arguments on some platforms: MSVC 14.
- This function does not work on finite numbers on some platforms: AIX 5.1, MSVC 14.
- This function does not work on denormalized numbers on some platforms: macOS 10.13/i386.
- This function does not work on infinite numbers on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

10.24.151 fromfp

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Rounding-Functions.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.24.152 fromfpf

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Rounding-Functions.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.24.153 fromfpl

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Rounding-Functions.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.24.154 fromfpx

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Rounding-Functions.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.24.155 fromfpxf

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Rounding-Functions.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.24.156 fromfpxl

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Rounding-Functions.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.24.157 fsqrt

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.14.6

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.34, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.158 fsqrtl

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.14.6

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.34, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.159 fsub

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.27, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.24.160 fsubl

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Misc-FP-Arithmetic.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.27, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.24.161 getpayload

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section F.10.13.1

Documentation:

https://www.gnu.org/software/libc/manual/html_node/FP-Bit-Twiddling.html.

Gnulib module: getpayload

Portability problems fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.
- This function returns a wrong result for non-NaN arguments on some platforms: glibc 2.31.

Portability problems not fixed by Gnulib:

10.24.162 getpayloadf

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section F.10.13.1

Documentation:

https://www.gnu.org/software/libc/manual/html_node/FP-Bit-Twiddling.html.

Gnulib module: getpayloadf

Portability problems fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.
- This function returns a wrong result for non-NaN arguments on some platforms: glibc 2.31.

Portability problems not fixed by Gnulib:

10.24.163 `getpayloadl`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section F.10.13.1

Documentation:

https://www.gnu.org/software/libc/manual/html_node/FP-Bit-Twiddling.html.

Gnulib module: `getpayloadl`

Portability problems fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.
- This function returns a wrong result for non-NaN arguments on some platforms: glibc 2.31.

Portability problems not fixed by Gnulib:

10.24.164 `hypot`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/hypot.html>

Gnulib module: `hypot` or `hypot-ieee`

Portability problems fixed by either Gnulib module `hypot` or `hypot-ieee`:

Portability problems fixed by Gnulib module `hypot-ieee`:

- When the arguments are mixed NaN and Infinity, this function returns a wrong value on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

10.24.165 `hypotf`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/hypotf.html>

Gnulib module: `hypotf` or `hypotf-ieee`

Portability problems fixed by either Gnulib module `hypotf` or `hypotf-ieee`:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, MSVC 14.
- This function produces wrong values on some platforms: NetBSD 5.1, OpenBSD 4.9.

Portability problems fixed by Gnulib module `hypot-ieee`:

- When the arguments are mixed NaN and Infinity, this function returns a wrong value on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

10.24.166 hypotl

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/hypotl.html>

Gnulib module: `hypotl` or `hypotl-ieee`

Portability problems fixed by either Gnulib module `hypotl` or `hypotl-ieee`:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, MSVC 14, Android 4.4.
- This function produces very imprecise results on some platforms: NetBSD 10.0, OpenBSD 5.1/SPARC.

Portability problems fixed by Gnulib module `hypotl-ieee`:

- When the arguments are mixed NaN and Infinity, this function returns a wrong value on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

10.24.167 ilogb

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/ilogb.html>

Gnulib module: `ilogb`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, MSVC 9.
- This function returns a wrong result for a zero argument on some platforms: OpenBSD 6.7, AIX 5.1.
- This function returns a wrong result for denormalized arguments on some platforms: Mac OS X 10.5 64-bit, AIX 7.1 64-bit.
- This function returns a wrong result for an infinite argument on some platforms: NetBSD 7.1, OpenBSD 6.7.
- This function returns a wrong result for a NaN argument on some platforms: NetBSD 7.1, OpenBSD 6.7.

Portability problems not fixed by Gnulib:

10.24.168 ilogbf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/ilogbf.html>

Gnulib module: `ilogbf`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, MSVC 9.
- This function returns a wrong result for a zero argument on some platforms: OpenBSD 6.7.
- This function returns a wrong result for an infinite argument on some platforms: NetBSD 7.1, OpenBSD 6.7.

Portability problems not fixed by Gnulib:

10.24.169 `ilogbl`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/ilogbl.html>

Gnulib module: `ilogbl`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 7.1, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, MSVC 9.
- This function returns a wrong result for a zero argument on some platforms: Cygwin 2.9.
- This function returns a wrong result for a NaN argument on some platforms: Cygwin 3.4.x.
- This function returns a wrong result for denormalized arguments on some platforms: AIX 7.1 64-bit, Haiku 2017.

Portability problems not fixed by Gnulib:

10.24.170 `isfinite`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/isfinite.html>

Gnulib module: `isfinite`

Portability problems fixed by Gnulib:

- This macro is missing on some platforms: macOS 14, OpenBSD 7.5, AIX 5.1, Solaris 11.4.
- This macro incorrectly yields true for some ‘double’ and for some ‘long double’ arguments, on some platforms: Linux/ia64 (NaNs).

Portability problems not fixed by Gnulib:

- It is implementation-dependent whether `isfinite` raises an exception given a signaling NaN operand.
- This macro returns an unspecified result when given noncanonical values such as unnormalized numbers, pseudo-denormals, pseudo-NaNs, pseudo-Infinities, and pseudo-zeroes.

10.24.171 `isgreater`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/isgreater.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, NetBSD 10.0, OpenBSD 7.5, AIX 5.1, Solaris 11.4, Android 9.0.

10.24.172 isgreaterequal

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/isgreaterequal.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, NetBSD 10.0, OpenBSD 7.5, AIX 5.1, Solaris 11.4, Android 9.0.

10.24.173 isinf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/isinf.html>

Gnulib module: isinf

Portability problems fixed by Gnulib:

- This macro is missing on some platforms: AIX 5.1, Solaris 11.4.

Portability problems not fixed by Gnulib:

- This macro returns an unspecified result when given noncanonical values such as unnormalized numbers, pseudo-denormals, pseudo-NaNs, pseudo-Infinities, and pseudo-zeroes.

10.24.174 isless

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/isless.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, NetBSD 10.0, OpenBSD 7.5, AIX 5.1, Solaris 11.4, Android 9.0.

10.24.175 islessequal

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/islessequal.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, NetBSD 10.0, OpenBSD 7.5, AIX 5.1, Solaris 11.4, Android 9.0.

10.24.176 islessgreater

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/islessgreater.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, NetBSD 10.0, OpenBSD 7.5, AIX 5.1, Solaris 11.4, Android 9.0.

10.24.177 isnan

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/isnan.html>

Gnulib module: isnan

Portability problems fixed by Gnulib:

- This macro is missing on some platforms: MSVC 9.
- `isnan` was introduced with C99 and is thus commonly not present on pre-C99 systems.
- `isnan` is not a macro on some platforms: Solaris 11.4.
- This macro does not work on some platforms: MSVC 14.

Portability problems not fixed by Gnulib:

- This macro returns an unspecified result when given noncanonical values such as unnormalized numbers, pseudo-denormals, pseudo-NaNs, pseudo-Infinities, and pseudo-zeroes.

10.24.178 isnormal

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/isnormal.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, OpenBSD 7.5, AIX 5.1, Solaris 11.4, Android 4.4.

10.24.179 isunordered

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/isunordered.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, NetBSD 10.0, OpenBSD 7.5, AIX 5.1, Solaris 11.4, Android 9.0.

10.24.180 j0

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/j0.html>

Gnulib module: j0

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8.

10.24.181 j1

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/j1.html>

Gnulib module: j1

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8.

10.24.182 jn

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/jn.html>

Gnulib module: jn

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8.

10.24.183 ldexp

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/ldexp.html>

Gnulib module: ldexp or ldexp-ieee

Portability problems fixed by Gnulib:

- This function produces wrong results on some platforms: OpenBSD 7.3/mips64.

Portability problems not fixed by Gnulib:

10.24.184 ldexpf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/ldexpf.html>

Gnulib module: ldexpf or ldexpf-ieee

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9.
- This function is only defined as a macro with arguments on some platforms: MSVC 14.

Portability problems not fixed by Gnulib:

10.24.185 ldexpl

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/ldexpl.html>

Gnulib module: ldexpl or ldexpl-ieee

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, HP-UX 11, Solaris 9, Cygwin 1.7.x.
- This function is only defined as a macro with arguments on some platforms: MSVC 14.
- This function has no prototype in `<math.h>` on some platforms: Mac OS X.
- This function does not work on finite numbers on some platforms: AIX 5.1.

Portability problems not fixed by Gnulib:

10.24.186 lgamma

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/lgamma.html>

Gnulib module: lgamma

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, MSVC 9.

10.24.187 lgammaf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/lgammaf.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, MSVC 9.

10.24.188 lgammal

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/lgammal.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 9.3, OpenBSD 3.8, Minix 3.1.8, HP-UX 11, Solaris 9, Cygwin 1.7.x, MSVC 9, Android 4.4.

10.24.189 llogb

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Exponents-and-Logarithms.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.24.190 llogbf

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Exponents-and-Logarithms.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.24.191 llogbl

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Exponents-and-Logarithms.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.24.192 llrint

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/llrint.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, Solaris 9, Cygwin 1.5.x, MSVC 9.

10.24.193 llrintf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/llrintf.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.5.x, MSVC 9.

10.24.194 llrintl

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/llrintl.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 9.3, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.5.x, MSVC 9, Android 4.4.

10.24.195 llround

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/llround.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, Solaris 9, Cygwin 1.7.7, MSVC 9.

10.24.196 llroundf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/llroundf.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.7, MSVC 9.

10.24.197 llroundl

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/llroundl.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 7.1, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, MSVC 9.

10.24.198 `log`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/log.html>

Gnulib module: `log` or `log-ieee`

Portability problems fixed by either Gnulib module `log` or `log-ieee`:

Portability problems fixed by Gnulib module `log-ieee`:

- This function returns a wrong value for a negative argument on some platforms: NetBSD 5.1, Solaris 11.4.

Portability problems not fixed by Gnulib:

10.24.199 `log10`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/log10.html>

Gnulib module: `log10` or `log10-ieee`

Portability problems fixed by either Gnulib module `log10` or `log10-ieee`:

Portability problems fixed by Gnulib module `log10-ieee`:

- This function returns a wrong value for a negative argument on some platforms: NetBSD 5.1, Solaris 11.4.

Portability problems not fixed by Gnulib:

10.24.200 `log10f`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/log10f.html>

Gnulib module: `log10f` or `log10f-ieee`

Portability problems fixed by either Gnulib module `log10f` or `log10f-ieee`:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, Solaris 9.
- This function is only defined as a macro with arguments on some platforms: MSVC 14.

Portability problems fixed by Gnulib module `log10f-ieee`:

- This function returns a wrong value for a negative argument on some platforms: NetBSD 5.1.

Portability problems not fixed by Gnulib:

10.24.201 log10l

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/log10l.html>

Gnulib module: log10l or log10l-ieee

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, HP-UX 11, Solaris 9, Cygwin 1.7.x, Android 4.4.
- This function is only defined as a macro with arguments on some platforms: MSVC 14.
- This function is not declared and does not work on some platforms: AIX 5.1.
- This function produces results which are accurate to only 16 digits on some platforms: musl libc 1.2.2/arm64, musl libc 1.2.2/s390x, NetBSD 10.0.

Portability problems not fixed by Gnulib:

10.24.202 log10p1

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.6.13

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.39, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.203 log10p1f

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.6.13

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.39, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.204 log10p11

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.6.13

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.39, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.205 `log1p`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/log1p.html>

Gnulib module: `log1p` or `log1p-ieee`

Portability problems fixed by either Gnulib module `log1p` or `log1p-ieee`:

- This function is missing on some platforms: Minix 3.1.8, MSVC 9.

Portability problems fixed by Gnulib module `log1p-ieee`:

- This function has problems when the argument is minus zero on some platforms: AIX 7.1, HP-UX 11.

Portability problems not fixed by Gnulib:

10.24.206 `log1pf`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/log1pf.html>

Gnulib module: `log1pf` or `log1pf-ieee`

Portability problems fixed by either Gnulib module `log1pf` or `log1pf-ieee`:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, MSVC 9.

Portability problems fixed by Gnulib module `log1pf-ieee`:

- This function has problems when the argument is minus zero on some platforms: OpenBSD 4.9, AIX 7.1.

Portability problems not fixed by Gnulib:

10.24.207 `log1pl`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/log1pl.html>

Gnulib module: `log1pl` or `log1pl-ieee`

Portability problems fixed by either Gnulib module `log1pl` or `log1pl-ieee`:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 9.3, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, MSVC 9, Android 4.4.
- This function produces results which are accurate to only 16 digits on some platforms: musl libc 1.2.2/arm64, musl libc 1.2.2/s390x, NetBSD 10.0.

Portability problems fixed by Gnulib module `log1pl-ieee`:

- This function has problems when the argument is minus zero on some platforms: AIX 7.1.

Portability problems not fixed by Gnulib:

10.24.208 `log2`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/log2.html>

Gnulib module: `log2` or `log2-ieee`

Portability problems fixed by either Gnulib module `log2` or `log2-ieee`:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, Solaris 9, MSVC 9, Android 4.2.
- This function is only provided as a macro on some platforms: Cygwin 1.5.x.
- This function returns slightly wrong values for exact powers of 2 on some platforms: Cygwin 1.7.9.

Portability problems fixed by Gnulib module `log2-ieee`:

- This function returns a wrong value for a negative argument on some platforms: NetBSD 5.1, Solaris 10/x86_64.

Portability problems not fixed by Gnulib:

10.24.209 `log2f`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/log2f.html>

Gnulib module: `log2f` or `log2f-ieee`

Portability problems fixed by either Gnulib module `log2f` or `log2f-ieee`:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, Solaris 9, MSVC 9, Android 4.2.
- This function is only provided as a macro on some platforms: Cygwin 1.5.x.
- This function returns slightly wrong values for exact powers of 2 on some platforms: Cygwin 1.7.9.

Portability problems fixed by Gnulib module `log2f-ieee`:

- This function returns a wrong value for a negative argument on some platforms: NetBSD 5.1, Solaris 10/x86_64.

Portability problems not fixed by Gnulib:

10.24.210 `log2l`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/log2l.html>

Gnulib module: `log2l` or `log2l-ieee`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 9.3, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, MSVC 9, Android 4.2.
- This function produces results which are accurate to only 16 digits on some platforms: musl libc 1.2.2/arm64, musl libc 1.2.2/s390x.
- This function returns Infinity for some large finite arguments on some platforms: musl libc 1.2.2/arm64, musl libc 1.2.2/s390x, NetBSD 10.0.

Portability problems not fixed by Gnulib:

10.24.211 log2p1

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.6.16

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.39, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.212 log2p1f

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.6.16

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.39, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.213 log2p1l

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.6.16

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.39, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.214 logb

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/logb.html>

Gnulib module: logb or logb-ieee

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, MSVC 9.
- This function is missing a declaration on some platforms: Cygwin 1.5.x.
- This function produces wrong results for subnormal numbers on some platforms: glibc 2.17/ppc, glibc 2.7/sparc, glibc 2.7/hppa, Solaris 11.4, Cygwin 1.5.x.

Portability problems not fixed by Gnulib:

10.24.215 logbf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/logbf.html>

Gnulib module: `logbf` or `logbf-ieee`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, MSVC 9.
- This function produces wrong results for subnormal numbers on some platforms: glibc 2.11/ppc, glibc 2.7/sparc, glibc 2.7/hppa, Solaris 11.4.

Portability problems not fixed by Gnulib:

10.24.216 logbl

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/logbl.html>

Gnulib module: `logbl` or `logbl-ieee`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, MSVC 9, Android 4.2.
- This function goes into an endless loop for arguments such as 1.6810515715560467531313389086608763014932L on some platforms: OpenBSD 7.3/mips64.
- This function produces wrong results for subnormal numbers on some platforms: glibc 2.11/powerpc, glibc 2.7/sparc, glibc 2.7/hppa, Solaris 11.4, glibc 2.23/powerpc64le.

Portability problems not fixed by Gnulib:

10.24.217 logf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/logf.html>

Gnulib module: `logf` or `logf-ieee`

Portability problems fixed by either Gnulib module `logf` or `logf-ieee`:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, Solaris 9.
- This function is only defined as a macro with arguments on some platforms: MSVC 14.

Portability problems fixed by Gnulib module `logf-ieee`:

- This function returns a wrong value for a negative argument on some platforms: NetBSD 5.1.

Portability problems not fixed by Gnulib:

10.24.218 logl

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/logl.html>

Gnulib module: logl or logl-ieee

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, HP-UX 11, Solaris 9, Cygwin 1.7.x, Android 4.4.
- This function is only defined as a macro with arguments on some platforms: MSVC 14.
- This function returns wrong results on some platforms: glibc 2.7 on Linux/SPARC64.
- This function produces results which are accurate to only 16 digits on some platforms: musl libc 1.2.2/arm64, musl libc 1.2.2/s390x, NetBSD 10.0.

Portability problems not fixed by Gnulib:

10.24.219 logp1

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.6.14

Gnulib module: logp1

Portability problems fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.39, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

Portability problems not fixed by Gnulib:

10.24.220 logp1f

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.6.14

Gnulib module: logp1f

Portability problems fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.39, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

Portability problems not fixed by Gnulib:

10.24.221 logp1l

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.6.14

Gnulib module: logp1l

Portability problems fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.39, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

Portability problems not fixed by GnuLib:

10.24.222 `lrint`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/lrint.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: FreeBSD 5.2.1, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, Solaris 9, MSVC 9.

10.24.223 `lrintf`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/lrintf.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: FreeBSD 5.2.1, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, MSVC 9.

10.24.224 `lrintl`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/lrintl.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 9.3, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.5.x, MSVC 9, Android 4.4.

10.24.225 `lround`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/lround.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: FreeBSD 5.2.1, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, Solaris 9, MSVC 9.

10.24.226 lroundf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/lroundf.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, MSVC 9.

10.24.227 lroundl

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/lroundl.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 7.1, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, MSVC 9.

10.24.228 math_errhandling

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.1

Documentation:

`man math_error`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This macro is missing on some platforms: NetBSD 10.0, mingw.
- This macro does not describe the error behaviour of elementary arithmetic operations (+, -, *, /) and of mathematical operations for which the compiler emits inline code (such as `sqrt` on some CPUs).
- This macro does not describe the error behaviour of functions such as `strtod`.
- For mathematical operations in general, it is a safer bet to look at the exceptions set in the floating-point environment (by calling `feclearexcept` (`FE_ALL_EXCEPT`) before the operation and `fetestexcept` after the operation), regardless of `math_errhandling`.

10.24.229 modf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/modf.html>

Gnulib module: `modf` or `modf-ieee`

Portability problems fixed by either Gnulib module `modf` or `modf-ieee`:

Portability problems fixed by Gnulib module `modf-ieee`:

- This function has problems with a NaN argument on some platforms: NetBSD 5.1, Cygwin.

- This function has problems with infinite arguments on some platforms: FreeBSD 6.4, OpenBSD 7.3, mingw with MSVCRT.

Portability problems not fixed by Gnulib:

10.24.230 `modff`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/modff.html>

Gnulib module: `modff` or `modff-ieee`

Portability problems fixed by either Gnulib module `modff` or `modff-ieee`:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, HP-UX 11.
- This function is only defined as a buggy macro with arguments on some platforms: MSVC 14.

Portability problems fixed by Gnulib module `modff-ieee`:

- This function has problems with a NaN argument on some platforms: NetBSD 5.1, OpenBSD 7.5, Solaris 9, Cygwin.
- This function has problems with infinite arguments on some platforms: mingw with MSVCRT.

Portability problems not fixed by Gnulib:

10.24.231 `modfl`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/modfl.html>

Gnulib module: `modfl` or `modfl-ieee`

Portability problems fixed by either Gnulib module `modfl` or `modfl-ieee`:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, HP-UX 11, Solaris 9, Cygwin 1.7.x, Android 4.4.
- This function is only defined as a macro with arguments on some platforms: MSVC 14.
- This function produces wrong results on some platforms: NetBSD 10.0/arm64.

Portability problems fixed by Gnulib module `modfl-ieee`:

- This function has problems with infinite arguments on some platforms: mingw.

Portability problems not fixed by Gnulib:

10.24.232 `nan`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/nan.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, Solaris 9, MSVC 9.

10.24.233 nanf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/nanf.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, MSVC 9.

10.24.234 nanl

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/nanl.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, MSVC 9, Android 3.1.

10.24.235 nearbyint

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/nearbyint.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, AIX 5.1, Solaris 9, MSVC 9.

10.24.236 nearbyintf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/nearbyintf.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, MSVC 9.

10.24.237 nearbyintl

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/nearbyintl.html>

Gnulib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, MSVC 9, Android 4.4.

10.24.238 nextafter

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/nextafter.html>

GnuLib module: nextafter

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: Minix 3.1.8, MSVC 9.

10.24.239 nextafterf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/nextafterf.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, Solaris 9, MSVC 9.

10.24.240 nextafterl

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/nextafterl.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 9, Android 4.4.

10.24.241 nextdown

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/FP-Bit-Twiddling.html,
- `man nextdown`.

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on all non-glibc platforms: glibc 2.23, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.24.242 nextdownf

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/FP-Bit-Twiddling.html,
- `man nextdownf`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.23, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.24.243 nextdownl

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/FP-Bit-Twiddling.html,
- `man nextdownl`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.23, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.24.244 nexttoward

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/nexttoward.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 9, Android 4.2.

10.24.245 nexttowardf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/nexttowardf.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 9.

10.24.246 nexttowardl

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/nexttowardl.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 8.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 9, Android 4.2.

10.24.247 nextup

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/FP-Bit-Twiddling.html,
- `man nextup`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.23, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.24.248 nextupf

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/FP-Bit-Twiddling.html,
- `man nextupf`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.23, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.24.249 nextupl

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/FP-Bit-Twiddling.html,
- `man nextupl`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.23, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.24.250 pow

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/pow.html>

Gnulib module: pow

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

10.24.251 powf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/powf.html>

Gnulib module: powf

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, Solaris 9.
- This function is only defined as a macro with arguments on some platforms: MSVC 9.

Portability problems not fixed by Gnulib:

10.24.252 powl

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/powl.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, HP-UX 11, Solaris 9, Cygwin 1.7.x, Android 4.4.
- This function is only defined as a macro with arguments on some platforms: MSVC 14.
- This function produces results which are accurate to only 16 digits on some platforms: NetBSD 10.0.

10.24.253 pown

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.7.6

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all platforms: glibc 2.41, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.254 pownf

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.7.6

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all platforms: glibc 2.41, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.255 pownl

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.7.6

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all platforms: glibc 2.41, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.256 powr

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.7.7

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all platforms: glibc 2.41, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.257 powrf

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.7.7

Gnulib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on all platforms: glibc 2.41, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.258 `powrl`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.7.7

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on all platforms: glibc 2.41, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.259 `remainder`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/remainder.html>

GnuLib module: `remainder` or `remainder-ieee`

Portability problems fixed by either GnuLib module `remainder` or `remainder-ieee`:

- This function is missing on some platforms: MSVC 9.

Portability problems fixed by GnuLib module `remainder-ieee`:

Portability problems not fixed by GnuLib:

10.24.260 `remainderf`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/remainderf.html>

GnuLib module: `remainderf` or `remainderf-ieee`

Portability problems fixed by either GnuLib module `remainderf` or `remainderf-ieee`:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, MSVC 9.

Portability problems fixed by GnuLib module `remainderf-ieee`:

Portability problems not fixed by GnuLib:

10.24.261 `remainderl`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/remainderl.html>

GnuLib module: `remainderl` or `remainderl-ieee`

Portability problems fixed by either GnuLib module `remainderl` or `remainderl-ieee`:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 9.3, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, MSVC 9, Android 4.4.

- This function returns completely wrong values on some platforms: OpenBSD 5.1/SPARC.
- This function produces results which are accurate to only 16 digits on some platforms: musl libc 1.2.2/arm64, musl libc 1.2.2/s390x, NetBSD 10.0.

Portability problems fixed by Gnulib module `remainder1-ieee`:

Portability problems not fixed by Gnulib:

10.24.262 `remquo`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/remquo.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, Solaris 9, MSVC 9.

10.24.263 `remquof`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/remquof.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, MSVC 9.

10.24.264 `remquol`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/remquol.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 9.3, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, MSVC 9, Android 4.4.

10.24.265 `rint`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/rint.html>

Gnulib module: `rint` or `rint-ieee`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: MSVC 9.

Portability problems not fixed by Gnulib:

10.24.266 rintf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/rintf.html>

Gnulib module: rintf or rintf-ieee

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, MSVC 9.

Portability problems not fixed by Gnulib:

10.24.267 rintl

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/rintl.html>

Gnulib module: rintl or rintl-ieee

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.5.x, MSVC 9, Android 4.4.
- This function produces wrong results for negative numbers on some platforms: NetBSD 8.0.

Portability problems not fixed by Gnulib:

10.24.268 rootn

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.7.8

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all platforms: glibc 2.41, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.269 rootnf

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.7.8

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all platforms: glibc 2.41, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.270 rootnl

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.7.8

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all platforms: glibc 2.41, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.271 round

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/round.html>

Gnulib module: round or round-ieee

Portability problems fixed by either Gnulib module **round** or **round-ieee**:

- This function is missing on some platforms: FreeBSD 5.2.1, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, Solaris 9, MSVC 9.
- This function is not declared on some platforms: glibc 2.8.
- This function returns a wrong result for $x = 1/2 - 2^{-54}$ on some platforms: NetBSD 3.0, AIX 7.1, mingw with MSVCRT.

Portability problems fixed by Gnulib module **round-ieee**:

- This function returns a positive zero for an argument between -0.5 and 0 on some platforms: AIX 7.1.

Portability problems not fixed by Gnulib:

10.24.272 roundeven

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Rounding-Functions.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.24.273 roundevenf

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Rounding-Functions.html.

Gnulib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on all non-glibc platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.24.274 `roundevenl`

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Rounding-Functions.html.

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on all non-glibc platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.24.275 `roundf`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/roundf.html>

GnuLib module: `roundf` or `roundf-ieee`

Portability problems fixed by either GnuLib module `roundf` or `roundf-ieee`:

- This function is missing on some platforms: FreeBSD 5.2.1, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, MSVC 9.
- This function is not declared on some platforms: glibc 2.8.
- This function returns a wrong result for $x = 1/2 - 2^{-25}$ on some platforms: mingw with MSVCRT.

Portability problems fixed by GnuLib module `roundf-ieee`:

- This function returns a positive zero for an argument between -0.5 and 0 on some platforms: AIX 7.1.

Portability problems not fixed by GnuLib:

10.24.276 `roundl`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/roundl.html>

GnuLib module: `roundl` or `roundl-ieee`

Portability problems fixed by either GnuLib module `roundl` or `roundl-ieee`:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, MSVC 9.
- This function is not declared on some platforms: glibc 2.8.

Portability problems fixed by GnuLib module `roundl-ieee`:

- This function returns a positive zero for an argument between -0.5 and 0 on some platforms: AIX 7.1.

Portability problems not fixed by GnuLib:

10.24.277 `rsqrt`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.7.9

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on many platforms: glibc 2.41, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.278 `rsqrtf`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.7.9

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on many platforms: glibc 2.41, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.279 `rsqrtl`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.7.9

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on many platforms: glibc 2.41, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.280 `scalbln`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/scalbln.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, AIX 5.1, HP-UX 11, Solaris 9, MSVC 9, Android 9.0.

10.24.281 scalblnf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/scalblnf.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, AIX 5.1, HP-UX 11, Solaris 9, MSVC 9, Android 9.0.

10.24.282 scalblnl

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/scalblnl.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, MSVC 9, Android 9.0.

10.24.283 scalbn

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/scalbn.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: AIX 5.1, MSVC 9.

10.24.284 scalbnf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/scalbnf.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: AIX 5.1, HP-UX 11, Solaris 9, MSVC 9.

10.24.285 scalbnl

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/scalbnl.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, MSVC 9.

10.24.286 setpayload

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section F.10.13.2

Documentation:

https://www.gnu.org/software/libc/manual/html_node/FP-Bit-Twiddling.html.

Gnulib module: setpayload

Portability problems fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

Portability problems not fixed by Gnulib:

10.24.287 setpayloadf

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section F.10.13.2

Documentation:

https://www.gnu.org/software/libc/manual/html_node/FP-Bit-Twiddling.html.

Gnulib module: setpayloadf

Portability problems fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

Portability problems not fixed by Gnulib:

10.24.288 setpayloadl

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section F.10.13.2

Documentation:

https://www.gnu.org/software/libc/manual/html_node/FP-Bit-Twiddling.html.

Gnulib module: setpayloadl

Portability problems fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

Portability problems not fixed by Gnulib:

10.24.289 setpayloadsig

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section F.10.13.3

Documentation:

https://www.gnu.org/software/libc/manual/html_node/FP-Bit-Twiddling.html.

Gnulib module: setpayloadsig

Portability problems fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

Portability problems not fixed by Gnulib:

10.24.290 setpayloadsigf

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section F.10.13.3

Documentation:

https://www.gnu.org/software/libc/manual/html_node/FP-Bit-Twiddling.html.

Gnulib module: setpayloadsigf

Portability problems fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

Portability problems not fixed by Gnulib:

10.24.291 setpayloadsigl

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section F.10.13.3

Documentation:

https://www.gnu.org/software/libc/manual/html_node/FP-Bit-Twiddling.html.

Gnulib module: setpayloadsigl

Portability problems fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

Portability problems not fixed by Gnulib:

10.24.292 signbit

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/signbit.html>

Gnulib module: signbit

Portability problems fixed by Gnulib:

- This function is missing on some platforms: macOS 14, OpenBSD 7.5, AIX 5.1, Solaris 11.0, MSVC 9, Android 9.0.

Portability problems not fixed by Gnulib:

10.24.293 `signgam`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/signgam.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This variable is missing on some platforms: macOS 10.13, Minix 3.1.8, AIX 5.1, mingw, MSVC 14.

10.24.294 `sin`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/sin.html>

Gnulib module: `sin`

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

10.24.295 `sinf`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/sinf.html>

Gnulib module: `sinf`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, Solaris 9.
- This function is only defined as a macro with arguments on some platforms: MSVC 14.

Portability problems not fixed by Gnulib:

10.24.296 `sinh`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/sinh.html>

Gnulib module: `sinh`

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

10.24.297 `sinhf`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/sinhf.html>

Gnulib module: `sinhf`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, Solaris 9.
- This function is only defined as a macro with arguments on some platforms: MSVC 14.

Portability problems not fixed by Gnulib:

10.24.298 sinh1

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/sinh1.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, HP-UX 11, Solaris 9, Cygwin 1.7.x, Android 4.4.
- This function is only defined as a macro with arguments on some platforms: MSVC 14.
- This function produces results which are accurate to only 16 digits on some platforms: NetBSD 10.0.

10.24.299 sinl

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/sinl.html>

Gnulib module: sinl

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, HP-UX 11, Solaris 9, Cygwin 1.7.x, Android 4.4.
- This function is only defined as a macro with arguments on some platforms: MSVC 14.
- This function returns a value that is larger than the argument, for some positive arguments, on some platforms: NetBSD 10.0.

Portability problems not fixed by Gnulib:

- This function produces results which are accurate to only 16 digits on some platforms: NetBSD 10.0.

10.24.300 sinpi

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.4.13

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-C23 platforms: glibc 2.40, macOS 14, FreeBSD 13.4, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.301 sinpif

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.4.13

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-C23 platforms: glibc 2.40, macOS 14, FreeBSD 13.4, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.302 sinpil

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.4.13

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-C23 platforms: glibc 2.40, macOS 14, FreeBSD 13.4, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.303 sqrt

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/sqrt.html>

Gnulib module: sqrt or sqrt-ieee

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

10.24.304 sqrtf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/sqrtf.html>

Gnulib module: sqrtf or sqrtf-ieee

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, Solaris 9.
- This function is only defined as a macro with arguments on some platforms: MSVC 14.

Portability problems not fixed by Gnulib:

10.24.305 sqrtl

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/sqrtl.html>

Gnulib module: sqrtl or sqrtl-ieee

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, HP-UX 11, Solaris 9, Cygwin 1.7.x, Android 4.4.
- This function is only defined as a macro with arguments on some platforms: MSVC 14.
- This function produces very imprecise results on some platforms: OpenBSD 5.1/SPARC.

Portability problems not fixed by Gnulib:

10.24.306 tan

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/tan.html>

Gnulib module: tan

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

10.24.307 tanf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/tanf.html>

Gnulib module: tanf

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, Solaris 9.
- This function is only defined as a macro with arguments on some platforms: MSVC 14.

Portability problems not fixed by Gnulib:

10.24.308 tanh

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/tanh.html>

Gnulib module: tanh

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

10.24.309 tanhf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/tanhf.html>

Gnulib module: tanhf

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, Solaris 9.
- This function is only defined as a macro with arguments on some platforms: MSVC 14.

Portability problems not fixed by Gnulib:

10.24.310 tanhl

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/tanhl.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, HP-UX 11, Solaris 9, Cygwin 1.7.x, Android 4.4.
- This function is only defined as a macro with arguments on some platforms: MSVC 14.
- This function produces results which are accurate to only 16 digits on some platforms: NetBSD 10.0.

10.24.311 tanl

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/tanl.html>

Gnulib module: tanl

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, HP-UX 11, Solaris 9, Cygwin 1.7.x, Android 4.4.
- This function is only defined as a macro with arguments on some platforms: MSVC 14.

Portability problems not fixed by Gnulib:

- This function produces results which are accurate to only 16 digits on some platforms: NetBSD 10.0.

10.24.312 tanpi

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.4.14

Gnulib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on many non-C23 platforms: glibc 2.40, macOS 14, FreeBSD 13.4, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.313 tanpif

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.4.14

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on many non-C23 platforms: glibc 2.40, macOS 14, FreeBSD 13.4, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.314 tanpil

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.12.4.14

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on many non-C23 platforms: glibc 2.40, macOS 14, FreeBSD 13.4, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

10.24.315 tgamma

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/tgamma.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, MSVC 9.

10.24.316 tgammaf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/tgammaf.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, MSVC 9.

10.24.317 tgamma1

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/tgamma1.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 9.3, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, MSVC 9, Android 4.4.

10.24.318 totalorder

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section F.10.12.1

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/FP-Comparison-Functions.html.
- <https://lists.gnu.org/archive/html/bug-gnulib/2023-10/msg00037.html>

Gnulib module: totalorder

Portability problems fixed by Gnulib:

- This function is missing on many platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.
- This function has a different signature on some platforms: glibc 2.30.

Portability problems not fixed by Gnulib:

- This function treats signalling NaNs incorrectly on some platforms: glibc 2.35/sh4.

10.24.319 totalorderf

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section F.10.12.1

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/FP-Comparison-Functions.html.
- <https://lists.gnu.org/archive/html/bug-gnulib/2023-10/msg00037.html>

Gnulib module: totalorderf

Portability problems fixed by Gnulib:

- This function is missing on many platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.
- This function has a different signature on some platforms: glibc 2.30.

Portability problems not fixed by Gnulib:

- This function treats signalling NaNs incorrectly on some platforms: glibc 2.35/sh4.

10.24.320 totalorderl

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section F.10.12.1

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/FP-Comparison-Functions.html.
- <https://lists.gnu.org/archive/html/bug-gnulib/2023-10/msg00037.html>

Gnulib module: totalorderl

Portability problems fixed by Gnulib:

- This function is missing on many platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.
- This function has a different signature on some platforms: glibc 2.30.

Portability problems not fixed by Gnulib:

- This function treats signalling NaNs incorrectly on some platforms: glibc 2.35/sh4.

10.24.321 totalordermag

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section F.10.12.2

Documentation:

https://www.gnu.org/software/libc/manual/html_node/FP-Comparison-Functions.html.

Gnulib module: totalordermag

Portability problems fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.
- This function has a different signature on some platforms: glibc 2.30.

Portability problems not fixed by Gnulib:

10.24.322 totalordermagf

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section F.10.12.2

Documentation:

https://www.gnu.org/software/libc/manual/html_node/FP-Comparison-Functions.html.

Gnulib module: totalordermagf

Portability problems fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

- This function has a different signature on some platforms: glibc 2.30.

Portability problems not fixed by Gnulib:

10.24.323 `totalordermagl`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section F.10.12.2

Documentation:

https://www.gnu.org/software/libc/manual/html_node/FP-Comparison-Functions.html.

Gnulib module: `totalordermagl`

Portability problems fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.
- This function has a different signature on some platforms: glibc 2.30.

Portability problems not fixed by Gnulib:

10.24.324 `trunc`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/trunc.html>

Gnulib module: `trunc` or `trunc-ieee`

Portability problems fixed by either Gnulib module `trunc` or `trunc-ieee`:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 3.0, OpenBSD 3.8, Solaris 9, MSVC 9.
- This function is not declared (without `-D_GNU_SOURCE`) on some platforms: glibc 2.8.

Portability problems fixed by Gnulib module `trunc-ieee`:

Portability problems not fixed by Gnulib:

10.24.325 `truncf`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/truncf.html>

Gnulib module: `truncf` or `truncf-ieee`

Portability problems fixed by either Gnulib module `truncf` or `truncf-ieee`:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, MSVC 9.
- This function is not declared (without `-D_GNU_SOURCE`) on some platforms: glibc 2.8.

Portability problems fixed by Gnulib module `truncf-ieee`:

Portability problems not fixed by Gnulib:

10.24.326 trunc1

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/trunc1.html>

Gnulib module: `trunc1` or `trunc1-ieee`

Portability problems fixed by either Gnulib module `trunc1` or `trunc1-ieee`:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, MSVC 9.
- This function is not declared (without `-D_GNU_SOURCE`) on some platforms: glibc 2.8.

Portability problems fixed by Gnulib module `trunc1-ieee`:

Portability problems not fixed by Gnulib:

10.24.327 ufromfp

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Rounding-Functions.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.24.328 ufromfpf

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Rounding-Functions.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.24.329 ufromfpl

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Rounding-Functions.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.24.330 `ufromfpx`

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Rounding-Functions.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.24.331 `ufromfpxf`

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Rounding-Functions.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.24.332 `ufromfpxl`

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Rounding-Functions.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.24.333 `y0`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/y0.html>

Gnulib module: `y0`

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8.

10.24.334 y1

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/y1.html>

Gnulib module: y1

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8.

10.24.335 yn

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/yn.html>

Gnulib module: yn

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8.

10.25 Functions in <monetary.h>

10.25.1 strfmon

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/strfmon.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: NetBSD 3.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 1.7.1, mingw, MSVC 14, Android 9.0.

10.25.2 strfmon_l

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/strfmon_l.html

Gnulib module: strfmon_l

Portability problems fixed by Gnulib:

- This function ignores the locale argument on some platforms: FreeBSD 13.0, Cygwin 2.9.
- This function uses a wrong locale for the numbers on some platforms: glibc 2.23.

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 6.0, NetBSD 7.1, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.3, Cygwin 1.7.x, mingw, MSVC 14, Android 9.0.

10.26 Functions in <mqueue.h>

10.26.1 mq_close

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/mq_close.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 6.0, NetBSD 3.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 1.5.x, mingw, MSVC 14, Android 9.0.

10.26.2 mq_getattr

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/mq_getattr.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 6.0, NetBSD 3.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 1.5.x, mingw, MSVC 14, Android 9.0.

10.26.3 mq_notify

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/mq_notify.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 6.0, NetBSD 3.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 1.5.x, mingw, MSVC 14, Android 9.0.

10.26.4 mq_open

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/mq_open.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 6.0, NetBSD 3.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 1.5.x, mingw, MSVC 14, Android 9.0.

10.26.5 mq_receive

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/mq_receive.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 6.0, NetBSD 3.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 1.5.x, mingw, MSVC 14, Android 9.0.

10.26.6 mq_send

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/mq_send.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 6.0, NetBSD 3.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 1.5.x, mingw, MSVC 14, Android 9.0.

10.26.7 mq_setattr

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/mq_setattr.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 6.0, NetBSD 3.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 1.5.x, mingw, MSVC 14, Android 9.0.

10.26.8 mq_timedreceive

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/mq_timedreceive.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 6.0, NetBSD 3.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.5.x, mingw, MSVC 14, Android 9.0.

10.26.9 mq_timedsend

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/mq_timedsend.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 6.0, NetBSD 3.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.5.x, mingw, MSVC 14, Android 9.0.

10.26.10 mq_unlink

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/mq_unlink.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 6.0, NetBSD 3.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 1.5.x, mingw, MSVC 14, Android 9.0.

10.27 Functions in <ndbm.h>

10.27.1 dbm_clearerr

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/dbm_clearerr.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, HP-UX 11.11, Cygwin 3.0.x, mingw, MSVC 14, Android 9.0.

10.27.2 dbm_close

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/dbm_close.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, Cygwin 3.0.x, mingw, MSVC 14, Android 9.0.

10.27.3 dbm_delete

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/dbm_delete.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, Cygwin 3.0.x, mingw, MSVC 14, Android 9.0.

10.27.4 dbm_error

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/dbm_error.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, HP-UX 11.11, Cygwin 3.0.x, mingw, MSVC 14, Android 9.0.

10.27.5 dbm_fetch

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/dbm_fetch.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, Cygwin 3.0.x, mingw, MSVC 14, Android 9.0.

10.27.6 dbm_firstkey

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/dbm_firstkey.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, Cygwin 3.0.x, mingw, MSVC 14, Android 9.0.

10.27.7 dbm_nextkey

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/dbm_nextkey.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, Cygwin 3.0.x, mingw, MSVC 14, Android 9.0.

10.27.8 dbm_open

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/dbm_open.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, Cygwin 3.0.x, mingw, MSVC 14, Android 9.0.

10.27.9 dbm_store

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/dbm_store.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, Cygwin 3.0.x, mingw, MSVC 14, Android 9.0.

10.28 Functions in <net/if.h>

10.28.1 if_freenameindex

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/if_freenameindex.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: OpenBSD 3.8, Minix 3.1.8, HP-UX 11.23, Cygwin 1.5.x, mingw, MSVC 14, Android 6.0.
- This is available only as a macro, rather than a function, on some platforms: OpenBSD 4.6, MirBSD 10.

10.28.2 `if_indextoname`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/if_indextoname.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, HP-UX 11.23, Cygwin 1.5.x, mingw, MSVC 14.

10.28.3 `if_nameindex`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/if_nameindex.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, HP-UX 11.23, Cygwin 1.5.x, mingw, MSVC 14, Android 6.0.

10.28.4 `if_nametoindex`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/if_nametoindex.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, HP-UX 11.23, Cygwin 1.5.x, mingw, MSVC 14.

10.29 Functions in `<netdb.h>`

10.29.1 `endhostent`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/endhostent.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14, Android 8.1.

10.29.2 endnetent

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/endnetent.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Cygwin 2.9, mingw, MSVC 14, Android 8.1.

10.29.3 endprotoent

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/endprotoent.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14, Android 8.1.

10.29.4 endservent

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/endservent.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

10.29.5 freeaddrinfo

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/freeaddrinfo.html>

Gnulib module: getaddrinfo

Portability problems fixed by Gnulib:

- This function is missing on some platforms: HP-UX 11.11, Cygwin 1.5.x, mingw, MSVC 14.
- On Windows, this function is declared in `<ws2tcpip.h>` rather than in `<netdb.h>`.
- On Windows, in 32-bit mode, this function is defined with a calling convention that is different from `cdecl`.

Portability problems not fixed by Gnulib:

10.29.6 `gai_strerror`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/gai_strerror.html

Gnulib module: `getaddrinfo`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: HP-UX 11.11, Cygwin 1.5.x.
- This function is only available in `<ws2tcpip.h>` on some platforms: mingw, MSVC 14.
- This function's return type is `char *` instead of `const char *` on some platforms: AIX 7.1, HP-UX 11, Solaris 9, mingw, MSVC 14.

Portability problems not fixed by Gnulib:

10.29.7 `getaddrinfo`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getaddrinfo.html>

Gnulib module: `getaddrinfo`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: HP-UX 11.11, Cygwin 1.5.x, mingw, MSVC 14.
- On native Windows, this function is declared in `<ws2tcpip.h>` rather than in `<netdb.h>`.
- On native Windows, this function does not support the `AI_NUMERICSERV` hints flag.
- On native Windows, in 32-bit mode, this function is defined with a calling convention that is different from `cdecl`.

Portability problems not fixed by Gnulib:

- Unlike glibc's implementation, gnulib's replacement does not support internationalized domain names (IDN) encoding.

10.29.8 `gethostent`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/gethostent.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14.

10.29.9 `getnameinfo`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getnameinfo.html>

Gnulib module: `getaddrinfo`

Portability problems fixed by GnuLib:

- This function is missing on some platforms: HP-UX 11.11, Cygwin 1.5.x, mingw, MSVC 14.

Portability problems not fixed by GnuLib:

10.29.10 `getnetbyaddr`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getnetbyaddr.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: Cygwin 2.9, mingw, MSVC 14.

10.29.11 `getnetbyname`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getnetbyname.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: Cygwin 2.9, mingw, MSVC 14.

10.29.12 `getnetent`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getnetent.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: Cygwin 2.9, mingw, MSVC 14, Android 8.1.

10.29.13 `getprotobyname`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getprotobyname.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is declared in `<winsock2.h>` instead of `<netdb.h>` on some platforms: mingw, MSVC 14.

10.29.14 getprotobynumber

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getprotobynumber.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is declared in `<winsock2.h>` instead of `<netdb.h>` on some platforms: mingw, MSVC 14.

10.29.15 getprotoent

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getprotoent.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14, Android 8.1.

10.29.16 getservbyname

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getservbyname.html>

Gnulib module: `servent`

Portability problems fixed by Gnulib:

- This function is declared in `<winsock2.h>` instead of `<netdb.h>` on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- On Windows, in 32-bit mode, this function is defined with a calling convention that is different from `cdecl`.

10.29.17 getservbyport

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getservbyport.html>

Gnulib module: `servent`

Portability problems fixed by Gnulib:

- This function is declared in `<winsock2.h>` instead of `<netdb.h>` on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- On Windows, in 32-bit mode, this function is defined with a calling convention that is different from `cdecl`.

10.29.18 getservent

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getservent.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

10.29.19 sethostent

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/sethostent.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14, Android 8.1.

10.29.20 setnetent

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/setnetent.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Cygwin 2.9, mingw, MSVC 14, Android 8.1.

10.29.21 setprotoent

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/setprotoent.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14, Android 8.1.

10.29.22 setservent

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/setservent.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

10.30 Functions in <netinet/in.h>

10.30.1 in6addr_any

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/in6addr_any.html

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Host-Address-Data-Type.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This constant is missing on some platforms: Minix 3.1.8, HP-UX 11.00, Cygwin 1.5.x, mingw, MSVC 14, Android 6.0.

10.30.2 in6addr_loopback

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/in6addr_loopback.html

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Host-Address-Data-Type.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This constant is missing on some platforms: Minix 3.1.8, HP-UX 11.00, Cygwin 1.5.x, mingw, MSVC 14, Android 6.0.

10.31 Functions in <nl_types.h>

10.31.1 catclose

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/catclose.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9.x, mingw, MSVC 14, Android 7.1.

10.31.2 catgets

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/catgets.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9.x, mingw, MSVC 14, Android 7.1.

10.31.3 catopen

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/catopen.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9.x, mingw, MSVC 14, Android 7.1.

10.32 Functions in <poll.h>

10.32.1 poll

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/poll.html>

Gnulib module: poll

Portability problems fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14, HP NonStop.
- This function doesn't work on special files like `/dev/null` and ttys like `/dev/tty` on some platforms: macOS 14, AIX 5.3.

Portability problems not fixed by Gnulib:

- Under Windows, when passing a pipe, Gnulib's `poll` replacement might return 0 even before the timeout has passed. Programs using it with pipes can thus busy wait.
- On some platforms, file descriptors other than sockets do not support `POLLHUP`; they will return a "readable" or "writable" status instead: AIX 7.2, Cygwin 3.4.6, HP NonStop, mingw, MSVC.

10.32.2 ppoll

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/ppoll.html>

Documentation:

`man ppoll`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 6.4, NetBSD 9.3, OpenBSD 3.8, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 10, Cygwin 1.7.x, mingw, MSVC 14, Android 4.3.

10.33 Functions in <pthread.h>

10.33.1 pthread_atfork

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_atfork.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, OpenBSD 3.8, Minix 3.1.8, mingw, MSVC 14, Android 3.0.

10.33.2 pthread_attr_destroy

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_attr_destroy.html

Gnulib module: pthread-thread

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.

Portability problems not fixed by Gnulib:

10.33.3 pthread_attr_getdetachstate

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_attr_getdetachstate.html

Gnulib module: pthread-thread

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.

Portability problems not fixed by Gnulib:

10.33.4 pthread_attr_getguardsize

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_attr_getguardsize.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: OpenBSD 3.8, Minix 3.1.8, Cygwin 1.7.9, mingw, MSVC 14.

10.33.5 pthread_attr_getinheritsched

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_attr_getinheritsched.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14, Android 8.1.

10.33.6 pthread_attr_getschedparam

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_attr_getschedparam.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

10.33.7 pthread_attr_getschedpolicy

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_attr_getschedpolicy.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

10.33.8 pthread_attr_getscope

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_attr_getscope.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

10.33.9 pthread_attr_getstack

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_attr_getstack.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, HP-UX 11.11, Solaris 9, Cygwin 1.7.9, mingw, MSVC 14.

10.33.10 pthread_attr_getstacksize

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_attr_getstacksize.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

10.33.11 pthread_attr_init

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_attr_init.html

Gnulib module: pthread-thread

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.
- This function is only defined as an inline function on some platforms: HP-UX 11.11.

Portability problems not fixed by Gnulib:

10.33.12 pthread_attr_setdetachstate

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_attr_setdetachstate.html

Gnulib module: pthread-thread

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.

Portability problems not fixed by Gnulib:

10.33.13 pthread_attr_setguardsize

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_attr_setguardsize.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: OpenBSD 3.8, Minix 3.1.8, Cygwin 1.7.9, mingw, MSVC 14.

10.33.14 pthread_attr_setinheritsched

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_attr_setinheritsched.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14, Android 8.1.

10.33.15 pthread_attr_setschedparam

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_attr_setschedparam.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

10.33.16 pthread_attr_setschedpolicy

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_attr_setschedpolicy.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

10.33.17 pthread_attr_setscope

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_attr_setscope.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

10.33.18 pthread_attr_setstack

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_attr_setstack.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, HP-UX 11.11, Solaris 9, Cygwin 1.7.9, mingw, MSVC 14.

10.33.19 pthread_attr_setstacksize

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_attr_setstacksize.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

10.33.20 pthread_barrier_destroy

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_barrier_destroy.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 5.2.1, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 14, Android 6.0.

10.33.21 pthread_barrier_init

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_barrier_init.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 5.2.1, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 14, Android 6.0.

10.33.22 pthread_barrier_wait

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_barrier_wait.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 5.2.1, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 14, Android 6.0.

10.33.23 pthread_barrierattr_destroy

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_barrierattr_destroy.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 5.2.1, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 14, Android 6.0.

10.33.24 pthread_barrierattr_getpshared

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_barrierattr_getpshared.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.2, macOS 14, FreeBSD 5.2.1, NetBSD 10.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 14, Android 6.0.

10.33.25 pthread_barrierattr_init

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_barrierattr_init.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 5.2.1, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 14, Android 6.0.

10.33.26 pthread_barrierattr_setpshared

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_barrierattr_setpshared.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 5.2.1, NetBSD 10.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 14, Android 6.0.

10.33.27 pthread_cancel

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_cancel.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14, Android 9.0.

10.33.28 pthread_cleanup_pop

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_cleanup_pop.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, NetBSD 10.0, Minix 3.1.8, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.33.29 pthread_cleanup_push

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_cleanup_push.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, NetBSD 10.0, Minix 3.1.8, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.33.30 pthread_cond_broadcast

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_cond_broadcast.html

Gnulib module: pthread-cond

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.

Portability problems not fixed by Gnulib:

10.33.31 pthread_cond_clockwait

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_cond_clockwait.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Non_002dPOSIX-Extensions.html,

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 3.2.x, mingw, MSVC 14, Android API level 29.

10.33.32 pthread_cond_destroy

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_cond_destroy.html

Gnulib module: pthread-cond

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.

Portability problems not fixed by Gnulib:

10.33.33 pthread_cond_init

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_cond_init.html

Gnulib module: pthread-cond

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.

Portability problems not fixed by Gnulib:

10.33.34 pthread_cond_signal

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_cond_signal.html

Gnulib module: pthread-cond

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.

Portability problems not fixed by Gnulib:

10.33.35 pthread_cond_timedwait

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_cond_timedwait.html

Gnulib module: pthread-cond

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.

Portability problems not fixed by Gnulib:

- This function sometimes waits indefinitely instead of up to the specified time point, on some platforms (glibc/Linux, musl libc) in some hypervisors: VirtualBox 6.1.50-Ubuntu with paravirtualization set to "Default" or "KVM".

10.33.36 pthread_cond_wait

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_cond_wait.html

Gnulib module: pthread-cond

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.

Portability problems not fixed by Gnulib:

10.33.37 pthread_condattr_destroy

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_condattr_destroy.html

Gnulib module: pthread-cond

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.

Portability problems not fixed by Gnulib:

10.33.38 pthread_condattr_getclock

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_condattr_getclock.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.2, macOS 14, FreeBSD 5.2.1, NetBSD 7.1, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.9, mingw, MSVC 14, Android 4.4.

10.33.39 pthread_condattr_getpshared

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_condattr_getpshared.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, mingw, MSVC 14.

10.33.40 pthread_condattr_init

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_condattr_init.html

Gnulib module: pthread-cond

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.

Portability problems not fixed by Gnulib:

10.33.41 pthread_condattr_setclock

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_condattr_setclock.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.2, macOS 14, FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.9, mingw, MSVC 14, Android 4.4.

10.33.42 pthread_condattr_setpshared

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_condattr_setpshared.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, mingw, MSVC 14.

10.33.43 pthread_create

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_create.html

Gnulib module: pthread-thread

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.
- This function is only defined as an inline function on some platforms: HP-UX 11.11.

Portability problems not fixed by GnuLib:

- On Linux/glibc platforms before the advent of NPTL in 2003, signals could only be sent to one particular thread. In POSIX, signals are sent to the entire process and executed by any thread of the process that happens to have the particular signal currently unblocked.

10.33.44 pthread_detach

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_detach.html

GnuLib module: pthread-thread

Portability problems fixed by GnuLib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.

Portability problems not fixed by GnuLib:

10.33.45 pthread_equal

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_equal.html

GnuLib module: pthread-thread

Portability problems fixed by GnuLib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.

Portability problems not fixed by GnuLib:

10.33.46 pthread_exit

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_exit.html

GnuLib module: pthread-thread

Portability problems fixed by GnuLib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.

Portability problems not fixed by GnuLib:

10.33.47 pthread_getcpuclockid

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_getcpuclockid.html

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: macOS 14, FreeBSD 6.0, NetBSD 7.1, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.9, mingw, MSVC 14.

10.33.48 pthread_getschedparam

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_getschedparam.html

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

10.33.49 pthread_getspecific

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_getspecific.html

GnuLib module: pthread-tss

Portability problems fixed by GnuLib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.

Portability problems not fixed by GnuLib:

10.33.50 pthread_join

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_join.html

GnuLib module: pthread-thread

Portability problems fixed by GnuLib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.

Portability problems not fixed by GnuLib:

10.33.51 pthread_key_create

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_key_create.html

GnuLib module: pthread-tss

Portability problems fixed by GnuLib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.

Portability problems not fixed by Gnulib:

10.33.52 pthread_key_delete

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_key_delete.html

Gnulib module: pthread-tss

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.

Portability problems not fixed by Gnulib:

10.33.53 pthread_mutex_clocklock

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_mutex_clocklock.html

Documentation: —

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 3.2.x, mingw, MSVC 14, Android API level 29.

10.33.54 pthread_mutex_consistent

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_mutex_consistent.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.11, macOS 14, FreeBSD 6.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 10, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.33.55 pthread_mutex_destroy

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_mutex_destroy.html

Gnulib module: pthread-mutex

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.

Portability problems not fixed by Gnulib:

10.33.56 pthread_mutex_getprioceiling

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_mutex_getprioceiling.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, NetBSD 7.1, Minix 3.1.8, mingw, MSVC 14, Android 9.0.

10.33.57 pthread_mutex_init

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_mutex_init.html

Gnulib module: pthread-mutex

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14, Android 4.3. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.

Portability problems not fixed by Gnulib:

10.33.58 pthread_mutex_lock

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_mutex_lock.html

Gnulib module: pthread-mutex

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.

Portability problems not fixed by Gnulib:

- This function does not behave as required by POSIX for mutexes of type NORMAL and ERRORCHECK, by allowing RECURSIVE locking, in programs linked without `-lpthread` on some platforms: FreeBSD 14.0, NetBSD 10.0.

10.33.59 pthread_mutex_setprioceiling

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_mutex_setprioceiling.html

Gnulib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: glibc 2.3.6, NetBSD 7.1, Minix 3.1.8, mingw, MSVC 14, Android 9.0.

10.33.60 pthread_mutex_timedlock

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_mutex_timedlock.html

GnuLib module: pthread_mutex_timedlock

Portability problems fixed by GnuLib:

- This function is missing on some platforms: macOS 14, FreeBSD 5.2.1, NetBSD 7.1, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 14, Android 4.4. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.

Portability problems not fixed by GnuLib:

10.33.61 pthread_mutex_trylock

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_mutex_trylock.html

GnuLib module: pthread-mutex

Portability problems fixed by GnuLib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.

Portability problems not fixed by GnuLib:

- This function does not behave as required by POSIX for mutexes of type NORMAL and ERRORCHECK, by allowing RECURSIVE locking, in programs linked without `-lpthread` on some platforms: FreeBSD 14.0, NetBSD 10.0.

10.33.62 pthread_mutex_unlock

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_mutex_unlock.html

GnuLib module: pthread-mutex

Portability problems fixed by GnuLib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.

Portability problems not fixed by GnuLib:

10.33.63 pthread_mutexattr_destroy

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_mutexattr_destroy.html

Gnulib module: pthread-mutex

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.

Portability problems not fixed by Gnulib:

10.33.64 pthread_mutexattr_getprioceiling

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_mutexattr_getprioceiling.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, NetBSD 7.1, Minix 3.1.8, mingw, MSVC 14, Android 9.0.

10.33.65 pthread_mutexattr_getprotocol

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_mutexattr_getprotocol.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, NetBSD 7.1, Minix 3.1.8, mingw, MSVC 14, Android 8.1.

10.33.66 pthread_mutexattr_getpshared

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_mutexattr_getpshared.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, mingw, MSVC 14.

10.33.67 pthread_mutexattr_getrobust

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_mutexattr_getrobust.html

Gnulib module: pthread-mutex

Portability problems fixed by Gnulib:

- This function is missing on some platforms: glibc 2.11, macOS 14, FreeBSD 6.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 10, Cygwin 2.9, mingw, MSVC 14, Android 9.0. But the provided replacement is just a dummy.

Portability problems not fixed by Gnulib:

10.33.68 pthread_mutexattr_gettype

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_mutexattr_gettype.html

Gnulib module: pthread-mutex

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.

Portability problems not fixed by Gnulib:

10.33.69 pthread_mutexattr_init

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_mutexattr_init.html

Gnulib module: pthread-mutex

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.

Portability problems not fixed by Gnulib:

10.33.70 pthread_mutexattr_setprioceiling

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_mutexattr_setprioceiling.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, NetBSD 7.1, Minix 3.1.8, mingw, MSVC 14, Android 9.0.

10.33.71 pthread_mutexattr_setprotocol

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_mutexattr_setprotocol.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, NetBSD 7.1, Minix 3.1.8, mingw, MSVC 14, Android 8.1.

10.33.72 pthread_mutexattr_setpshared

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_mutexattr_setpshared.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, mingw, MSVC 14.

10.33.73 pthread_mutexattr_setrobust

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_mutexattr_setrobust.html

Gnulib module: pthread-mutex

Portability problems fixed by Gnulib:

- This function is missing on some platforms: glibc 2.11, macOS 14, FreeBSD 6.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 10, Cygwin 2.9, mingw, MSVC 14, Android 9.0. But the provided replacement is just a dummy.

Portability problems not fixed by Gnulib:

10.33.74 pthread_mutexattr_settype

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_mutexattr_settype.html

Gnulib module: pthread-mutex

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.

Portability problems not fixed by Gnulib:

10.33.75 pthread_once

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_once.html

Gnulib module: pthread-once

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.
- This function makes applications hang forever on some platforms: Cygwin 3.5.3.

Portability problems not fixed by Gnulib:

10.33.76 pthread_rwlock_clockrdlock

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_rwlock_clockrdlock.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Non_002dPOSIX-Extensions.html,

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 3.2.x, mingw, MSVC 14, Android API level 29.

10.33.77 pthread_rwlock_clockwrlock

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_rwlock_clockwrlock.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Non_002dPOSIX-Extensions.html,

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 3.2.x, mingw, MSVC 14, Android API level 29.

10.33.78 pthread_rwlock_destroy

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_rwlock_destroy.html

Gnulib module: pthread-rwlock

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14, Android 4.3. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.

Portability problems not fixed by Gnulib:

10.33.79 pthread_rwlock_init

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_rwlock_init.html

Gnulib module: pthread-rwlock

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14, Android 4.3. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.

Portability problems not fixed by Gnulib:

10.33.80 pthread_rwlock_rdlock

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_rwlock_rdlock.html

Gnulib module: pthread-rwlock

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14, Android 4.3. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.
- This function prefers readers to writers (meaning, when this function is called on an rwlock that is already taken by one or more readers, and another writer is already waiting to take it, this function may return successfully immediately) – a behaviour that may lead to writer starvation – on some platforms: glibc 2.42.

Portability problems not fixed by Gnulib:

10.33.81 pthread_rwlock_timedrdlock

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_rwlock_timedrdlock.html

Gnulib module: pthread-rwlock

Portability problems fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 5.2.1, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 14, Android 4.3. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.
- This function prefers readers to writers (meaning, when this function is called on an `rwlock` that is already taken by one or more readers, and another writer is already waiting to take it, this function may return successfully immediately) – a behaviour that may lead to writer starvation – on some platforms: glibc 2.42.

Portability problems not fixed by Gnulib:

10.33.82 `pthread_rwlock_timedwrlock`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_rwlock_timedwrlock.html

Gnulib module: `pthread-rwlock`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 5.2.1, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 14, Android 4.3. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.

Portability problems not fixed by Gnulib:

10.33.83 `pthread_rwlock_tryrdlock`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_rwlock_tryrdlock.html

Gnulib module: `pthread-rwlock`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14, Android 4.3. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.
- This function prefers readers to writers (meaning, when this function is called on an `rwlock` that is already taken by one or more readers, and another writer is already waiting to take it, this function may return successfully immediately) – a behaviour that may lead to writer starvation – on some platforms: glibc 2.42.

Portability problems not fixed by Gnulib:

10.33.84 `pthread_rwlock_trywrlock`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_rwlock_trywrlock.html

Gnulib module: `pthread-rwlock`

Portability problems fixed by GnuLib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14, Android 4.3. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.

Portability problems not fixed by GnuLib:

- This function returns EDEADLK instead of EBUSY when the current thread already owns the lock for writing on some platforms: macOS 12.

10.33.85 pthread_rwlock_unlock

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_rwlock_unlock.html

GnuLib module: pthread-rwlock

Portability problems fixed by GnuLib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14, Android 4.3. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.

Portability problems not fixed by GnuLib:

10.33.86 pthread_rwlock_wrlock

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_rwlock_wrlock.html

GnuLib module: pthread-rwlock

Portability problems fixed by GnuLib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14, Android 4.3. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.

Portability problems not fixed by GnuLib:

10.33.87 pthread_rwlockattr_destroy

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_rwlockattr_destroy.html

GnuLib module: pthread-rwlock

Portability problems fixed by GnuLib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.

Portability problems not fixed by GnuLib:

10.33.88 pthread_rwlockattr_getpshared

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_rwlockattr_getpshared.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: NetBSD 10.0, Minix 3.1.8, mingw, MSVC 14.

10.33.89 pthread_rwlockattr_init

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_rwlockattr_init.html

Gnulib module: pthread-rwlock

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.

Portability problems not fixed by Gnulib:

10.33.90 pthread_rwlockattr_setpshared

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_rwlockattr_setpshared.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: NetBSD 10.0, Minix 3.1.8, mingw, MSVC 14.

10.33.91 pthread_self

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_self.html

Gnulib module: pthread-thread

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.

Portability problems not fixed by Gnulib:

10.33.92 pthread_setcancelstate

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_setcancelstate.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14, Android 9.0.

10.33.93 pthread_setcanceltype

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_setcanceltype.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14, Android 9.0.

10.33.94 pthread_setschedparam

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_setschedparam.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

10.33.95 pthread_setschedprio

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_setschedprio.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.3, macOS 14, FreeBSD 14.0, NetBSD 3.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.9, mingw, MSVC 14, Android 8.1.

10.33.96 pthread_setspecific

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_setspecific.html

Gnulib module: pthread-tss

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.

Portability problems not fixed by Gnulib:

10.33.97 pthread_spin_destroy

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_spin_destroy.html

Gnulib module: pthread-spin

Portability problems fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 5.2.1, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.9, mingw, MSVC 14, Android 6.0. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.

Portability problems not fixed by Gnulib:

10.33.98 pthread_spin_init

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_spin_init.html

Gnulib module: pthread-spin

Portability problems fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 5.2.1, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.9, mingw, MSVC 14, Android 6.0. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.

Portability problems not fixed by Gnulib:

10.33.99 pthread_spin_lock

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_spin_lock.html

Gnulib module: pthread-spin

Portability problems fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 5.2.1, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.9, mingw, MSVC 14, Android

6.0. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.

Portability problems not fixed by Gnulib:

10.33.100 pthread_spin_trylock

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_spin_trylock.html

Gnulib module: pthread-spin

Portability problems fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 5.2.1, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.9, mingw, MSVC 14, Android 6.0. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.

Portability problems not fixed by Gnulib:

10.33.101 pthread_spin_unlock

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_spin_unlock.html

Gnulib module: pthread-spin

Portability problems fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 5.2.1, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.9, mingw, MSVC 14, Android 6.0. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.

Portability problems not fixed by Gnulib:

10.33.102 pthread_testcancel

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_testcancel.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14, Android 9.0.

10.34 Functions in <pwd.h>

10.34.1 `endpwent`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/endpwent.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

10.34.2 `getpwent`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getpwent.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14, Android 7.1.

10.34.3 `getpwnam`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getpwnam.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

10.34.4 `getpwnam_r`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/getpwnam_r.html

Gnulib module: extensions

Portability problems fixed by Gnulib:

- This function has an incompatible declaration on some platforms: Solaris 11.3 (when `_POSIX_PTHREAD_SEMANTICS` is not defined).

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: OpenBSD 3.8, Minix 3.1.8, mingw, MSVC 14, Android 3.0.

10.34.5 `getpwuid`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getpwuid.html>

Gnulib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is unsafe to call between `fork` and `exec` if the parent process is multi-threaded. Instead, use `getpwuid_r` prior to forking.
- This function is missing on some platforms: mingw, MSVC 14.

10.34.6 `getpwuid_r`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/getpwuid_r.html

GnuLib module: extensions

Portability problems fixed by GnuLib:

- This function has an incompatible declaration on some platforms: Solaris 11.3 (when `_POSIX_PTHREAD_SEMANTICS` is not defined).

Portability problems not fixed by GnuLib:

- This function is unsafe to call between `fork` and `exec` if the parent process is multi-threaded. Use it prior to forking.
- This function is missing on some platforms: OpenBSD 3.8, Minix 3.1.8, mingw, MSVC 14, Android 3.0.

10.34.7 `setpwent`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/setpwent.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: mingw, MSVC 14, Android 7.1.

10.35 Functions in `<regex.h>`

10.35.1 `regcomp`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/regcomp.html>

GnuLib module: `regex`

Portability problems fixed by GnuLib:

- This function is missing on some platforms: mingw, MSVC 14.
- Many regular expression implementations have bugs.

Portability problems not fixed by GnuLib:

10.35.2 regerror

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/regerror.html>

Gnulib module: regex

Portability problems fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

10.35.3 regexec

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/regexec.html>

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-regexec-2.html

Gnulib module: regex

Portability problems fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.
- Many regular expression implementations have bugs.

Portability problems not fixed by Gnulib:

10.35.4 regfree

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/regfree.html>

Gnulib module: regex

Portability problems fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

10.36 Functions in <sched.h>

10.36.1 sched_get_priority_max

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/sched_get_priority_max.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: OpenBSD 3.8, Minix 3.1.8, HP-UX 11, mingw, MSVC 14.

10.36.2 sched_get_priority_min

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/sched_get_priority_min.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: OpenBSD 3.8, Minix 3.1.8, HP-UX 11, mingw, MSVC 14.

10.36.3 sched_getparam

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/sched_getparam.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, OpenBSD 7.5, Minix 3.1.8, mingw, MSVC 14.
- This function always fails with error ENOSYS on some platforms: musl libc.

10.36.4 sched_getscheduler

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/sched_getscheduler.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, OpenBSD 7.5, Minix 3.1.8, mingw, MSVC 14.
- This function always fails with error ENOSYS on some platforms: musl libc.

10.36.5 sched_rr_get_interval

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/sched_rr_get_interval.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, mingw, MSVC 14.

10.36.6 sched_setparam

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/sched_setparam.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, OpenBSD 7.5, Minix 3.1.8, mingw, MSVC 14.
- This function always fails with error ENOSYS on some platforms: musl libc.

10.36.7 sched_setscheduler

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/sched_setscheduler.html

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-sched-setscheduler.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, OpenBSD 7.5, Minix 3.1.8, mingw, MSVC 14.
- This function always fails with error ENOSYS on some platforms: musl libc.

10.36.8 sched_yield

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/sched_yield.html

Gnulib module: sched_yield

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14. But the provided replacement is just a dummy on some of these platforms: Minix 3.1.8.

Portability problems not fixed by Gnulib:

10.37 Functions in <search.h>

10.37.1 hcreate

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/hcreate.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14, Android 8.1.

10.37.2 hdestroy

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/hdestroy.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14, Android 8.1.

10.37.3 hsearch

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/hsearch.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14, Android 8.1.

10.37.4 insque

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/insque.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.5.x, mingw, MSVC 14, Android 4.4.

10.37.5 lfind

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/lfind.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Android 4.4.

10.37.6 lsearch

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/lsearch.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Android 4.4.

10.37.7 remque

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/remque.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.5.x, mingw, MSVC 14, Android 4.4.

10.37.8 tdelete

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/tdelete.html>

Gnulib module: tsearch

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14, Android 4.0.4.
- `tdelete` returns `NULL` when removing the last element of a tree on some platforms: OpenBSD 4.0.

Portability problems not fixed by Gnulib:

10.37.9 tfind

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/tfind.html>

Gnulib module: tsearch

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14, Android 4.0.4.

Portability problems not fixed by Gnulib:

- This function uses `void **` instead of `void *const *` for the second parameter on some platforms: Cygwin.

10.37.10 tsearch

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/tsearch.html>

Gnulib module: tsearch

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14, Android 4.0.4.

Portability problems not fixed by Gnulib:

10.37.11 twalk

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/twalk.html>

Gnulib module: tsearch

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14, Android 4.4.

Portability problems not fixed by Gnulib:

10.38 Functions in <semaphore.h>

10.38.1 sem_clockwait

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/sem_clockwait.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Non_002dPOSIX-Extensions.html,

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 3.2.x, mingw, MSVC 14, Android API level 29.

10.38.2 sem_close

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/sem_close.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

10.38.3 `sem_destroy`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/sem_destroy.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.
- This function is not implemented on some platforms: Mac OS X 10.11.

10.38.4 `sem_getvalue`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/sem_getvalue.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.
- This function is not implemented on some platforms: Mac OS X 10.11.

10.38.5 `sem_init`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/sem_init.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.
- This function is not implemented on some platforms: Mac OS X 10.11.

10.38.6 `sem_open`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/sem_open.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

10.38.7 `sem_post`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/sem_post.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

10.38.8 `sem_timedwait`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/sem_timedwait.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, mingw, MSVC 14.

10.38.9 `sem_trywait`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/sem_trywait.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

10.38.10 `sem_unlink`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/sem_unlink.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.5.x, mingw, MSVC 14.

10.38.11 `sem_wait`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/sem_wait.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

10.39 Functions in <setjmp.h>

10.39.1 longjmp

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/longjmp.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- The effects of this call are system and compiler optimization dependent, since it restores the contents of register-allocated variables but not the contents of stack-allocated variables.
- When longjumping out of a signal handler that was being executed on an alternate stack (installed through `sigaltstack`), on Hurd, FreeBSD, NetBSD, OpenBSD, you need to clear the `SS_ONSTACK` flag in the `stack_t` structure managed by the kernel.
- This function may crash when the invoker code was compiled with option `_FORTIFY_SOURCE >= 2` on Hurd.

10.39.2 setjmp

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/setjmp.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- POSIX does not specify whether `setjmp` saves the signal mask in the `jmp_buf`. It does on BSD platforms, and on glibc platforms when `_BSD_SOURCE` (and/or `_DEFAULT_SOURCE` on glibc `>= 2.19`) is defined; in this case `setjmp` behaves like `sigsetjmp(. , 1)`, and functions `_setjmp` and `_longjmp` are available that don't save or restore the signal mask. On System V platforms (excluding HP-UX), and on glibc platforms by default, `setjmp` doesn't save the signal mask.

10.39.3 siglongjmp

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/siglongjmp.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.
- This is only provided as a macro on some platforms: Cygwin.

10.39.4 sigsetjmp

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/sigsetjmp.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

10.40 Functions in <signal.h>

10.40.1 kill

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/kill.html>

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-kill-3.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

10.40.2 killpg

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/killpg.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

10.40.3 psiginfo

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/psiginfo.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.9, macOS 14, FreeBSD 14.0, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 1.7.9, mingw, MSVC 14, Android 4.1.

10.40.4 psignal

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/psignal.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, HP-UX 11, Cygwin 1.7.9, mingw, MSVC 14, Android API level 16.
- This function is declared in `<siginfo.h>`, not `<signal.h>`, on some platforms: Solaris 11.3.
- The first parameter is of type `unsigned int`, not `int`, on some platforms: macOS 13, FreeBSD 10.4, NetBSD 5.1, OpenBSD 7.5.

10.40.5 pthread_kill

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_kill.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

10.40.6 pthread_sigmask

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/pthread_sigmask.html

Gnulib module: `pthread_sigmask`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.
- This function is declared in `<pthread.h>` instead of `<signal.h>` on some platforms: FreeBSD 6.4, OpenBSD 3.8.
- This function does nothing and always returns 0 in programs that are not linked with `-lpthread` on some platforms: FreeBSD 14.0, MidnightBSD 1.1, HP-UX 11.31, Solaris 9.
- When it fails, this functions returns `-1` instead of the error number on some platforms: Cygwin 1.7.5.

Portability problems not fixed by Gnulib:

- On platforms that do not natively support this function, it has unspecified behavior in a multi-threaded process.
- In case of failure, the return value is wrong on some platforms: NetBSD 9.3 when `libpthread` is not in use.

10.40.7 raise

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/raise.html>

Gnulib module: raise

Portability problems fixed by Gnulib:

- This function is missing on some old platforms.
- This function crashes when invoked with invalid arguments on some platforms: MSVC 14.

Portability problems not fixed by Gnulib:

10.40.8 sig2str

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/sig2str.html>

Gnulib module: sig2str

Portability problems fixed by Gnulib:

- This function is missing on many platforms: glibc 2.42, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Cygwin 3.2.x, mingw, MSVC 14, Android API level 35.

Portability problems not fixed by Gnulib:

- This function does not map the argument 0 to "EXIT" on some platforms: FreeBSD 15.0.

10.40.9 sigaction

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/sigaction.html>

Gnulib module: sigaction

Portability problems fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- POSIX recommends that when specifying SA_RESETHAND, SA_NODEFER must also be specified.
- Support for SA_ONSTACK is missing on some platforms: mingw, MSVC 14, cygwin.
- Support for SA_SIGINFO is missing on some platforms: mingw, MSVC 14.
- Support for SIGCHLD, and thus for SA_NOCLDSTOP and SA_NOCLDWAIT, is missing on some platforms: mingw, MSVC 14.
- Support for SA_RESETHAND is missing on some platforms: NonStop.
- Support for SA_RESTART is missing on some platforms: mingw, MSVC 14, NonStop.
- The symbolic value SIG_IGN for the SIGCHLD signal is equivalent to a signal handler

```
void handle_child (int sigchld)
{
    while (waitpid (-1, NULL, WNOHANG) > 0)
```

```
    }
    ;
```

except that `SIG_IGN` for `SIGCHLD` has the effect that the children execution times are not accounted in the `times` function. On some platforms (BSD? SystemV? Linux?), you need to use the `sigaction` flag `SA_NOCLDWAIT` in order to obtain this behavior.

10.40.10 sigaddset

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/sigaddset.html>

Gnulib module: `sigprocmask`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14, Android 4.4.
- This function cannot be called from plain inline or extern inline functions on some platforms: macOS 10.8.

Portability problems not fixed by Gnulib:

10.40.11 sigaltstack

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/sigaltstack.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.7.x, mingw, MSVC 14.
- `sigaltstack` doesn't work on HP-UX 11/IA-64 and OpenBSD 3.6/Sparc64.

10.40.12 sigdelset

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/sigdelset.html>

Gnulib module: `sigprocmask`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14, Android 4.4.
- This function cannot be called from plain inline or extern inline functions on some platforms: macOS 10.8.

Portability problems not fixed by Gnulib:

10.40.13 sigemptyset

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/sigemptyset.html>

Gnulib module: `sigprocmask`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14, Android 4.4.

- This function cannot be called from plain inline or extern inline functions on some platforms: macOS 10.8.

Portability problems not fixed by GnuLib:

10.40.14 sigfillset

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/sigfillset.html>

GnuLib module: sigprocmask

Portability problems fixed by GnuLib:

- This function is missing on some platforms: mingw, MSVC 14, Android 4.4.
- This function cannot be called from plain inline or extern inline functions on some platforms: macOS 10.8.

Portability problems not fixed by GnuLib:

10.40.15 sigismember

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/sigismember.html>

GnuLib module: sigprocmask

Portability problems fixed by GnuLib:

- This function is missing on some platforms: mingw, MSVC 14, Android 4.4.
- This function cannot be called from plain inline or extern inline functions on some platforms: macOS 10.8.

Portability problems not fixed by GnuLib:

10.40.16 signal

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/signal.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function crashes when invoked with invalid arguments on some platforms: MSVC 14.
- On System V platforms, when the signal is triggered, the kernel uninstalls the handler (i.e. resets the signal's action to SIG_DFL) before invoking the handler. This opens the door to race conditions: undesired things happen if the signal is triggered twice and the signal handler was not quick enough reinstalling itself as a handler. On BSD platforms and glibc platforms, on the other hand, when the signal is triggered, the kernel blocks the signal before invoking the handler. This is saner, but POSIX still allows either behavior. To avoid this problem, use `sigaction` instead of `signal`.

10.40.17 sigpending

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/sigpending.html>

Gnulib module: sigprocmask

Portability problems fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

10.40.18 sigprocmask

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/sigprocmask.html>

Gnulib module: sigprocmask

Portability problems fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- In case of failure, the return value is wrong on some platforms: NetBSD 10.0 when libpthread is in use.

Note: Although **sigprocmask** officially has undefined behaviour in multi-threaded programs, in practice it is essentially equivalent to **pthread_sigmask**, with only a difference regarding the error return convention. It's simpler to use **sigprocmask**, since it does not require linking with **-lpthread** on some platforms: glibc, NetBSD, OpenBSD, AIX.

Note: While on POSIX platforms, **sigprocmask** is multithread-safe and async-signal safe (cf. POSIX section “Signal Actions” https://pubs.opengroup.org/onlinepubs/9799919799/functions/V2_chap02.html#tag_16_04_03), the gnulib replacement on native Windows is only multithread-safe, not async-signal safe.

10.40.19 sigqueue

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/sigqueue.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 6.0, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, mingw, MSVC 14, Android 5.1.

10.40.20 sigsuspend

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/sigsuspend.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

10.40.21 sigtimedwait

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/sigtimedwait.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9.x, mingw, MSVC 14, Android 5.1.
- Linux implements the meaning of NULL timeout by doing what `sigwaitinfo` does; other platforms may not do the same.

10.40.22 sigwait

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/sigwait.html>

Gnulib module: extensions

Portability problems fixed by Gnulib:

- This function has an incompatible declaration on some platforms: Solaris 11.4 (when `_POSIX_PTHREAD_SEMANTICS` is not defined).

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.
- On Linux/glibc platforms before the advent of NPTL, signals could only be sent to one particular thread. In POSIX, signals are sent to the entire process and executed by any thread of the process that happens to have the particular signal currently unblocked.

10.40.23 sigwaitinfo

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/sigwaitinfo.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, OpenBSD 7.5, Minix 3.1.8, mingw, MSVC 14, Android 5.1.

10.40.24 str2sig

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/str2sig.html>

Gnulib module: sig2str

Portability problems fixed by Gnulib:

- This function is missing on many platforms: glibc 2.42, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Cygwin 3.2.x, mingw, MSVC 14, Android API level 35.

Portability problems not fixed by Gnulib:

10.41 Functions in <spawn.h>

10.41.1 posix_spawn

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/posix_spawn.html

Gnulib module: posix_spawn

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.3, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 14, Android 8.1.
- When this function fails, it causes the stdio buffer contents to be output twice on some platforms: AIX 6.1.
- When the program to be invoked is an executable script without a ‘#!’ marker in the first line, this function executes the script as if it were a shell script, on some platforms: GNU/Hurd.

Portability problems not fixed by Gnulib:

- This function does not work on some platforms: AIX 6.1 (under particular circumstances).

The Gnulib modules `posix_spawn_file_actions_addchdir` and `posix_spawn_file_actions_addfchdir` provide additional actions, that consist in changing the current directory of the child process before starting the specified program.

10.41.2 posix_spawn_file_actions_addchdir

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/posix_spawn_file_actions_addchdir.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc 2.42, macOS 14, FreeBSD 14.0, NetBSD 9.3, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android 9.0.

10.41.3 posix_spawn_file_actions_addclose

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/posix_spawn_file_actions_addclose.html

Gnulib module: posix_spawn_file_actions_addclose

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 14, Android 8.1.
- This function does not reject a negative file descriptor on some platforms: musl libc.

Portability problems not fixed by Gnulib:

- This function does not reject an out-of-range file descriptor on some platforms: musl libc, NetBSD, Cygwin.

10.41.4 `posix_spawn_file_actions_adddup2`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/posix_spawn_file_actions_adddup2.html

Gnulib module: `posix_spawn_file_actions_adddup2`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 14, Android 8.1.
- This function does not reject a too large file descriptor on some platforms: musl libc, Solaris 11.4.

Portability problems not fixed by Gnulib:

10.41.5 `posix_spawn_file_actions_addfchdir`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/posix_spawn_file_actions_addfchdir.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc 2.42, macOS 14, FreeBSD 14.0, NetBSD 9.3, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android 9.0.

10.41.6 `posix_spawn_file_actions_addopen`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/posix_spawn_file_actions_addopen.html

Gnulib module: `posix_spawn_file_actions_addopen`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 14, Android 8.1.

- This function does not reject a too large file descriptor on some platforms: musl libc, Solaris 11.4.

Portability problems not fixed by GnuLib:

10.41.7 `posix_spawn_file_actions_destroy`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/posix_spawn_file_actions_destroy.html

GnuLib module: `posix_spawn_file_actions_destroy`

Portability problems fixed by GnuLib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 14, Android 8.1.

Portability problems not fixed by GnuLib:

10.41.8 `posix_spawn_file_actions_init`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/posix_spawn_file_actions_init.html

GnuLib module: `posix_spawn_file_actions_init`

Portability problems fixed by GnuLib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 14, Android 8.1.

Portability problems not fixed by GnuLib:

10.41.9 `posix_spawnattr_destroy`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/posix_spawnattr_destroy.html

GnuLib module: `posix_spawnattr_destroy`

Portability problems fixed by GnuLib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 14, Android 8.1.

Portability problems not fixed by GnuLib:

10.41.10 `posix_spawnattr_getflags`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/posix_spawnattr_getflags.html

GnuLib module: `posix_spawnattr_getflags`

Portability problems fixed by GnuLib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 14, Android 8.1.

Portability problems not fixed by GnuLib:

10.41.11 `posix_spawnattr_getpgroup`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/posix_spawnattr_getpgroup.html

GnuLib module: `posix_spawnattr_getpgroup`

Portability problems fixed by GnuLib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 14, Android 8.1.

Portability problems not fixed by GnuLib:

10.41.12 `posix_spawnattr_getschedparam`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/posix_spawnattr_getschedparam.html

GnuLib module: `posix_spawnattr_getschedparam`

Portability problems fixed by GnuLib:

- This function is missing on some platforms: macOS 14, FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 14, Android 8.1.

Portability problems not fixed by GnuLib:

- This function always fails with error `ENOSYS` on some platforms: musl libc.

10.41.13 `posix_spawnattr_getschedpolicy`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/posix_spawnattr_getschedpolicy.html

GnuLib module: `posix_spawnattr_getschedpolicy`

Portability problems fixed by GnuLib:

- This function is missing on some platforms: macOS 14, FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 14, Android 8.1.

Portability problems not fixed by GnuLib:

- This function always fails with error `ENOSYS` on some platforms: musl libc.

10.41.14 `posix_spawnattr_getsigdefault`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/posix_spawnattr_getsigdefault.html

Gnulib module: `posix_spawnattr_getsigdefault`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 14, Android 8.1.

Portability problems not fixed by Gnulib:

10.41.15 `posix_spawnattr_getsigmask`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/posix_spawnattr_getsigmask.html

Gnulib module: `posix_spawnattr_getsigmask`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 14, Android 8.1.

Portability problems not fixed by Gnulib:

10.41.16 `posix_spawnattr_init`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/posix_spawnattr_init.html

Gnulib module: `posix_spawnattr_init`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 14, Android 8.1.

Portability problems not fixed by Gnulib:

10.41.17 `posix_spawnattr_setflags`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/posix_spawnattr_setflags.html

Gnulib module: `posix_spawnattr_setflags`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 14, Android 8.1.

Portability problems not fixed by Gnulib:

10.41.18 `posix_spawnattr_setpgroup`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/posix_spawnattr_setpgroup.html

Gnulib module: `posix_spawnattr_setpgroup`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 14, Android 8.1.

Portability problems not fixed by Gnulib:

10.41.19 `posix_spawnattr_setschedparam`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/posix_spawnattr_setschedparam.html

Gnulib module: `posix_spawnattr_setschedparam`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 14, Android 8.1.

Portability problems not fixed by Gnulib:

- This function always fails with error `ENOSYS` on some platforms: musl libc.

10.41.20 `posix_spawnattr_setschedpolicy`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/posix_spawnattr_setschedpolicy.html

Gnulib module: `posix_spawnattr_setschedpolicy`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 14, Android 8.1.

Portability problems not fixed by Gnulib:

- This function always fails with error `ENOSYS` on some platforms: musl libc.

10.41.21 `posix_spawnattr_setsigdefault`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/posix_spawnattr_setsigdefault.html

Gnulib module: `posix_spawnattr_setsigdefault`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 14, Android 8.1.

Portability problems not fixed by Gnulib:

10.41.22 `posix_spawnattr_setsigmask`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/posix_spawnattr_setsigmask.html

Gnulib module: `posix_spawnattr_setsigmask`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 14, Android 8.1.

Portability problems not fixed by Gnulib:

10.41.23 `posix_spawn`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/posix_spawn.html

Gnulib module: `posix_spawn`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.3, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.x, mingw, MSVC 14, Android 8.1.
- When this function fails, it causes the stdio buffer contents to be output twice on some platforms: AIX 6.1.
- When the program to be invoked is an executable script without a ‘#!’ marker in the first line, this function executes the script as if it were a shell script, on some platforms: glibc 2.14/Linux, glibc 2.32/Hurd, macOS 14, FreeBSD 14.0, OpenBSD 7.2, AIX 7.2, Solaris 11.4, Cygwin 3.4.6.

Portability problems not fixed by Gnulib:

- This function does not work on some platforms: AIX 6.1 (under particular circumstances).

The Gnulib modules `posix_spawn_file_actions_addchdir` and `posix_spawn_file_actions_addfchdir` provide additional actions, that consist in changing the current directory of the child process before starting the specified program.

10.42 Functions in `<stdarg.h>`

10.42.1 `va_arg`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/va_arg.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- The second argument of `va_arg` must be a type that is invariant under the “default argument promotions” (ISO C 99 6.5.2.2 paragraph 6). This means that the following are not valid here:

`'float'` Use `'double'` instead.

`'bool'` Use `'int'` instead.

Integer types smaller than `'int'`.

Use `'int'` or `'unsigned int'` instead.

This is a portability problem because you don't know the width of some abstract types like `uid_t`, `gid_t`, `mode_t`. So, instead of

```
mode = va_arg (ap, mode_t);
```

you have to write

```
mode = (sizeof (mode_t) < sizeof (int)
        ? va_arg (ap, int)
        : va_arg (ap, mode_t));
```

10.42.2 `va_copy`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/va_copy.html

Gnulib module: `stdarg-h`

Portability problems fixed by Gnulib:

- This macro is missing on some platforms: AIX 5.1 with `cc` or `xlc`, HP-UX 11 with `cc`.

Portability problems not fixed by Gnulib:

10.42.3 `va_end`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/va_end.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

10.42.4 `va_start`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/va_start.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

10.43 Functions in `<stdatomic.h>`

10.43.1 `atomic_compare_exchange_strong`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.17.7.4.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/atomic_compare_exchange_strong.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc with GCC 4.8.x or clang 2.8, FreeBSD 6.4, NetBSD 7.0, OpenBSD 6.5, OpenBSD 6.9 with gcc, Minix 3.3.0, AIX 7.2 with xlc or xlclang, HP-UX 11.31, Solaris 11.3 with Oracle Studio 12.4 (Sun C 5.13), Cygwin 1.5.19, mingw, MSVC 14..

10.43.2 `atomic_compare_exchange_strong_explicit`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.17.7.4.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/atomic_compare_exchange_strong_explicit.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc with GCC 4.8.x or clang 2.8, FreeBSD 6.4, NetBSD 7.0, OpenBSD 6.5, OpenBSD 6.9 with gcc, Minix 3.3.0, AIX 7.2 with xlc or xlclang, HP-UX 11.31, Solaris 11.3 with Oracle Studio 12.4 (Sun C 5.13), Cygwin 1.5.19, mingw, MSVC 14..

10.43.3 `atomic_compare_exchange_weak`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.17.7.4.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/atomic_compare_exchange_weak.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc with GCC 4.8.x or clang 2.8, FreeBSD 6.4, NetBSD 7.0, OpenBSD 6.5, OpenBSD 6.9 with gcc, Minix 3.3.0, AIX 7.2 with xlc or xlclang, HP-UX 11.31, Solaris 11.3 with Oracle Studio 12.4 (Sun C 5.13), Cygwin 1.5.19, mingw, MSVC 14..

10.43.4 `atomic_compare_exchange_weak_explicit`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.17.7.4.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/atomic_compare_exchange_weak_explicit.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc with GCC 4.8.x or clang 2.8, FreeBSD 6.4, NetBSD 7.0, OpenBSD 6.5, OpenBSD 6.9 with gcc, Minix 3.3.0, AIX 7.2 with xlc or xlclang, HP-UX 11.31, Solaris 11.3 with Oracle Studio 12.4 (Sun C 5.13), Cygwin 1.5.19, mingw, MSVC 14..

10.43.5 `atomic_exchange`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.17.7.3.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/atomic_exchange.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc with GCC 4.8.x or clang 2.8, FreeBSD 6.4, NetBSD 7.0, OpenBSD 6.5, OpenBSD 6.9 with gcc, Minix 3.3.0, AIX 7.2 with xlc or xlclang, HP-UX 11.31, Solaris 11.3 with Oracle Studio 12.4 (Sun C 5.13), Cygwin 1.5.19, mingw, MSVC 14..

10.43.6 `atomic_exchange_explicit`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.17.7.3.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/atomic_exchange_explicit.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc with GCC 4.8.x or clang 2.8, FreeBSD 6.4, NetBSD 7.0, OpenBSD 6.5, OpenBSD 6.9 with gcc, Minix 3.3.0, AIX 7.2 with xlc or xlclang, HP-UX 11.31, Solaris 11.3 with Oracle Studio 12.4 (Sun C 5.13), Cygwin 1.5.19, mingw, MSVC 14..

10.43.7 `atomic_fetch_add`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.17.7.5.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/atomic_fetch_add.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc with GCC 4.8.x or clang 2.8, FreeBSD 6.4, NetBSD 7.0, OpenBSD 6.5, OpenBSD 6.9 with gcc, Minix 3.3.0, AIX 7.2 with xlc or xlclang, HP-UX 11.31, Solaris 11.3 with Oracle Studio 12.4 (Sun C 5.13), Cygwin 1.5.19, mingw, MSVC 14..

10.43.8 `atomic_fetch_add_explicit`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.17.7.5.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/atomic_fetch_add_explicit.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc with GCC 4.8.x or clang 2.8, FreeBSD 6.4, NetBSD 7.0, OpenBSD 6.5, OpenBSD 6.9 with gcc, Minix 3.3.0, AIX 7.2 with xlc or xlclang, HP-UX 11.31, Solaris 11.3 with Oracle Studio 12.4 (Sun C 5.13), Cygwin 1.5.19, mingw, MSVC 14..

10.43.9 `atomic_fetch_and`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.17.7.5.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/atomic_fetch_and.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc with GCC 4.8.x or clang 2.8, FreeBSD 6.4, NetBSD 7.0, OpenBSD 6.5, OpenBSD 6.9 with gcc, Minix 3.3.0, AIX 7.2 with xlc or xlclang, HP-UX 11.31, Solaris 11.3 with Oracle Studio 12.4 (Sun C 5.13), Cygwin 1.5.19, mingw, MSVC 14..

10.43.10 `atomic_fetch_and_explicit`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.17.7.5.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/atomic_fetch_and_explicit.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc with GCC 4.8.x or clang 2.8, FreeBSD 6.4, NetBSD 7.0, OpenBSD 6.5, OpenBSD 6.9 with gcc, Minix 3.3.0, AIX 7.2 with xlc or xlclang, HP-UX 11.31, Solaris 11.3 with Oracle Studio 12.4 (Sun C 5.13), Cygwin 1.5.19, mingw, MSVC 14..

10.43.11 `atomic_fetch_or`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.17.7.5.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/atomic_fetch_or.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc with GCC 4.8.x or clang 2.8, FreeBSD 6.4, NetBSD 7.0, OpenBSD 6.5, OpenBSD 6.9 with gcc, Minix 3.3.0, AIX 7.2 with xlc or xlclang, HP-UX 11.31, Solaris 11.3 with Oracle Studio 12.4 (Sun C 5.13), Cygwin 1.5.19, mingw, MSVC 14..

10.43.12 `atomic_fetch_or_explicit`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.17.7.5.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/atomic_fetch_or_explicit.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc with GCC 4.8.x or clang 2.8, FreeBSD 6.4, NetBSD 7.0, OpenBSD 6.5, OpenBSD 6.9 with gcc, Minix 3.3.0, AIX 7.2 with xlc or xlclang, HP-UX 11.31, Solaris 11.3 with Oracle Studio 12.4 (Sun C 5.13), Cygwin 1.5.19, mingw, MSVC 14..

10.43.13 `atomic_fetch_sub`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.17.7.5.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/atomic_fetch_sub.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc with GCC 4.8.x or clang 2.8, FreeBSD 6.4, NetBSD 7.0, OpenBSD 6.5, OpenBSD 6.9 with gcc, Minix 3.3.0, AIX 7.2 with xlc or xlclang, HP-UX 11.31, Solaris 11.3 with Oracle Studio 12.4 (Sun C 5.13), Cygwin 1.5.19, mingw, MSVC 14..

10.43.14 `atomic_fetch_sub_explicit`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.17.7.5.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/atomic_fetch_sub_explicit.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc with GCC 4.8.x or clang 2.8, FreeBSD 6.4, NetBSD 7.0, OpenBSD 6.5, OpenBSD 6.9 with gcc, Minix 3.3.0, AIX 7.2 with xlc or xlclang, HP-UX 11.31, Solaris 11.3 with Oracle Studio 12.4 (Sun C 5.13), Cygwin 1.5.19, mingw, MSVC 14..

10.43.15 `atomic_fetch_xor`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.17.7.5.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/atomic_fetch_xor.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc with GCC 4.8.x or clang 2.8, FreeBSD 6.4, NetBSD 7.0, OpenBSD 6.5, OpenBSD 6.9 with gcc, Minix 3.3.0, AIX 7.2 with xlc or xlclang, HP-UX 11.31, Solaris 11.3 with Oracle Studio 12.4 (Sun C 5.13), Cygwin 1.5.19, mingw, MSVC 14..

10.43.16 `atomic_fetch_xor_explicit`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.17.7.5.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/atomic_fetch_xor_explicit.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc with GCC 4.8.x or clang 2.8, FreeBSD 6.4, NetBSD 7.0, OpenBSD 6.5, OpenBSD 6.9 with gcc, Minix 3.3.0, AIX 7.2 with xlc or xlclang, HP-UX 11.31, Solaris 11.3 with Oracle Studio 12.4 (Sun C 5.13), Cygwin 1.5.19, mingw, MSVC 14..

10.43.17 `atomic_flag_clear`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.17.8.2.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/atomic_flag_clear.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc with GCC 4.8.x or clang 2.8, FreeBSD 6.4, NetBSD 7.0, OpenBSD 6.5, OpenBSD 6.9 with gcc, Minix 3.3.0, AIX 7.2 with xlc or xlclang, HP-UX 11.31, Solaris 11.3 with Oracle Studio 12.4 (Sun C 5.13), Cygwin 1.5.19, mingw, MSVC 14..

10.43.18 `atomic_flag_clear_explicit`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.17.8.2.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/atomic_flag_clear_explicit.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc with GCC 4.8.x or clang 2.8, FreeBSD 6.4, NetBSD 7.0, OpenBSD 6.5, OpenBSD 6.9 with gcc, Minix 3.3.0, AIX 7.2 with xlc or xlclang, HP-UX 11.31, Solaris 11.3 with Oracle Studio 12.4 (Sun C 5.13), Cygwin 1.5.19, mingw, MSVC 14..

10.43.19 `atomic_flag_test_and_set`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.17.8.1.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/atomic_flag_test_and_set.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc with GCC 4.8.x or clang 2.8, FreeBSD 6.4, NetBSD 7.0, OpenBSD 6.5, OpenBSD 6.9 with gcc, Minix 3.3.0, AIX 7.2 with xlc or xlclang, HP-UX 11.31, Solaris 11.3 with Oracle Studio 12.4 (Sun C 5.13), Cygwin 1.5.19, mingw, MSVC 14..

10.43.20 `atomic_flag_test_and_set_explicit`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.17.8.1.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/atomic_flag_test_and_set_explicit.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc with GCC 4.8.x or clang 2.8, FreeBSD 6.4, NetBSD 7.0, OpenBSD 6.5, OpenBSD 6.9 with gcc, Minix 3.3.0, AIX 7.2 with xlc or xlclang, HP-UX 11.31, Solaris 11.3 with Oracle Studio 12.4 (Sun C 5.13), Cygwin 1.5.19, mingw, MSVC 14..

10.43.21 `atomic_init`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.17.2.1.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/atomic_init.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc with GCC 4.8.x or clang 2.8, FreeBSD 6.4, NetBSD 7.0, OpenBSD 6.5, OpenBSD 6.9 with gcc, Minix 3.3.0, AIX 7.2 with xlc or xlclang, HP-UX 11.31, Solaris 11.3 with Oracle Studio 12.4 (Sun C 5.13), Cygwin 1.5.19, mingw, MSVC 14..

10.43.22 `atomic_is_lock_free`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.17.5.1.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/atomic_is_lock_free.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc with GCC 4.8.x or clang 2.8, FreeBSD 6.4, NetBSD 7.0, OpenBSD 6.5, OpenBSD 6.9 with gcc, Minix 3.3.0, AIX 7.2 with xlc or xlclang, HP-UX 11.31, Solaris 11.3 with Oracle Studio 12.4 (Sun C 5.13), Cygwin 1.5.19, mingw, MSVC 14..

10.43.23 `atomic_load`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.17.7.2.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/atomic_load.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc with GCC 4.8.x or clang 2.8, FreeBSD 6.4, NetBSD 7.0, OpenBSD 6.5, OpenBSD 6.9 with gcc, Minix 3.3.0, AIX 7.2 with xlc or xlclang, HP-UX 11.31, Solaris 11.3 with Oracle Studio 12.4 (Sun C 5.13), Cygwin 1.5.19, mingw, MSVC 14..

10.43.24 `atomic_load_explicit`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.17.7.2.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/atomic_load_explicit.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc with GCC 4.8.x or clang 2.8, FreeBSD 6.4, NetBSD 7.0, OpenBSD 6.5, OpenBSD 6.9 with gcc, Minix 3.3.0, AIX 7.2 with xlc or xlclang, HP-UX 11.31, Solaris 11.3 with Oracle Studio 12.4 (Sun C 5.13), Cygwin 1.5.19, mingw, MSVC 14..

10.43.25 `atomic_signal_fence`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.17.4.2.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/atomic_signal_fence.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc with GCC 4.8.x or clang 2.8, FreeBSD 6.4, NetBSD 7.0, OpenBSD 6.5, OpenBSD 6.9 with gcc, Minix 3.3.0, AIX 7.2 with xlc or xlclang, HP-UX 11.31, Solaris 11.3 with Oracle Studio 12.4 (Sun C 5.13), Cygwin 1.5.19, mingw, MSVC 14..

10.43.26 `atomic_store`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.17.7.1.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/atomic_store.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc with GCC 4.8.x or clang 2.8, FreeBSD 6.4, NetBSD 7.0, OpenBSD 6.5, OpenBSD 6.9 with gcc, Minix 3.3.0, AIX 7.2 with xlc or xlclang, HP-UX 11.31, Solaris 11.3 with Oracle Studio 12.4 (Sun C 5.13), Cygwin 1.5.19, mingw, MSVC 14..

10.43.27 `atomic_store_explicit`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.17.7.1.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/atomic_store_explicit.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc with GCC 4.8.x or clang 2.8, FreeBSD 6.4, NetBSD 7.0, OpenBSD 6.5, OpenBSD 6.9 with gcc, Minix 3.3.0, AIX 7.2 with xlc or xlclang, HP-UX 11.31, Solaris 11.3 with Oracle Studio 12.4 (Sun C 5.13), Cygwin 1.5.19, mingw, MSVC 14..

10.43.28 `atomic_thread_fence`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.17.4.1.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/atomic_thread_fence.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc with GCC 4.8.x or clang 2.8, FreeBSD 6.4, NetBSD 7.0, OpenBSD 6.5, OpenBSD 6.9 with gcc, Minix 3.3.0, AIX 7.2 with xlc or xlclang, HP-UX 11.31, Solaris 11.3 with Oracle Studio 12.4 (Sun C 5.13), Cygwin 1.5.19, mingw, MSVC 14..

10.43.29 `kill_dependency`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.17.3.1.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/kill_dependency.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc with GCC 4.8.x or clang 2.8, FreeBSD 6.4, NetBSD 7.0, OpenBSD 6.5, OpenBSD 6.9 with gcc, Minix 3.3.0, AIX 7.2 with xlc or xlclang, HP-UX 11.31, Solaris 11.3 with Oracle Studio 12.4 (Sun C 5.13), Cygwin 1.5.19, mingw, MSVC 14..

10.44 Functions in `<stdbit.h>`

10.44.1 `stdc_bit_ceil`

Documentation:

ISO C23 (latest free draft <https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf>) section 7.18.16.

Gnulib module: `stdc_bit_ceil`

Portability problems fixed by Gnulib:

- This macro is missing on many platforms: glibc 2.38, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 17, Android 15.

Portability problems not fixed by Gnulib:

10.44.2 stdc_bit_floor

Documentation:

ISO C23 (latest free draft <https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf>) section 7.18.15.

Gnulib module: stdc_bit_floor

Portability problems fixed by Gnulib:

- This macro is missing on many platforms: glibc 2.38, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 17, Android 15.

Portability problems not fixed by Gnulib:

10.44.3 stdc_bit_width

Documentation:

ISO C23 (latest free draft <https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf>) section 7.18.14.

Gnulib module: stdc_bit_width

Portability problems fixed by Gnulib:

- This macro is missing on many platforms: glibc 2.38, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 17, Android 15.

Portability problems not fixed by Gnulib:

10.44.4 stdc_count_ones

Documentation:

ISO C23 (latest free draft <https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf>) section 7.18.12.

Gnulib module: stdc_count_ones

Portability problems fixed by Gnulib:

- This macro is missing on many platforms: glibc 2.38, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 17, Android 15.

Portability problems not fixed by Gnulib:

10.44.5 stdc_count_zeros

Documentation:

ISO C23 (latest free draft <https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf>) section 7.18.11.

Gnulib module: stdc_count_zeros

Portability problems fixed by Gnulib:

- This macro is missing on many platforms: glibc 2.38, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 17, Android 15.

Portability problems not fixed by Gnulib:

10.44.6 `stdc_first_leading_one`

Documentation:

ISO C23 (latest free draft <https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf>) section 7.18.8.

Gnulib module: `stdc_first_leading_one`

Portability problems fixed by Gnulib:

- This macro is missing on many platforms: glibc 2.38, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 17, Android 15.

Portability problems not fixed by Gnulib:

10.44.7 `stdc_first_leading_zero`

Documentation:

ISO C23 (latest free draft <https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf>) section 7.18.7.

Gnulib module: `stdc_first_leading_zero`

Portability problems fixed by Gnulib:

- This macro is missing on many platforms: glibc 2.38, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 17, Android 15.

Portability problems not fixed by Gnulib:

10.44.8 `stdc_first_trailing_one`

Documentation:

ISO C23 (latest free draft <https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf>) section 7.18.10.

Gnulib module: `stdc_first_trailing_one`

Portability problems fixed by Gnulib:

- This macro is missing on many platforms: glibc 2.38, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 17, Android 15.

Portability problems not fixed by Gnulib:

10.44.9 `stdc_first_trailing_zero`

Documentation:

ISO C23 (latest free draft <https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf>) section 7.18.9.

Gnulib module: `stdc_first_trailing_zero`

Portability problems fixed by Gnulib:

- This macro is missing on many platforms: glibc 2.38, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 17, Android 15.

Portability problems not fixed by Gnulib:

10.44.10 `stdc_has_single_bit`

Documentation:

ISO C23 (latest free draft <https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf>) section 7.18.13.

Gnulib module: `stdc_has_single_bit`

Portability problems fixed by Gnulib:

- This macro is missing on many platforms: glibc 2.38, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 17, Android 15.

Portability problems not fixed by Gnulib:

10.44.11 `stdc_leading_ones`

Documentation:

ISO C23 (latest free draft <https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf>) section 7.18.4.

Gnulib module: `stdc_leading_ones`

Portability problems fixed by Gnulib:

- This macro is missing on many platforms: glibc 2.38, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 17, Android 15.

Portability problems not fixed by Gnulib:

10.44.12 `stdc_leading_zeros`

Documentation:

ISO C23 (latest free draft <https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf>) section 7.18.3.

Gnulib module: `stdc_leading_zeros`

Portability problems fixed by Gnulib:

- This macro is missing on many platforms: glibc 2.38, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 17, Android 15.

Portability problems not fixed by Gnulib:

10.44.13 `stdc_trailing_ones`

Documentation:

ISO C23 (latest free draft <https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf>) section 7.18.6.

Gnulib module: `stdc_trailing_ones`

Portability problems fixed by Gnulib:

- This macro is missing on many platforms: glibc 2.38, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 17, Android 15.

Portability problems not fixed by Gnulib:

10.44.14 stdc_trailing_zeros

Documentation:

ISO C23 (latest free draft <https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf>) section 7.18.5.

Gnulib module: `stdc_trailing_zeros`

Portability problems fixed by Gnulib:

- This macro is missing on many platforms: glibc 2.38, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 17, Android 15.

Portability problems not fixed by Gnulib:

10.45 Functions in `<stdio.h>`

10.45.1 asprintf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/asprintf.html>

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Dynamic-Output.html,
- `man asprintf`.

Gnulib module: `vasprintf` or `vasprintf-posix` or `vasprintf-gnu`

Portability problems fixed by either Gnulib module `vasprintf` or `vasprintf-posix` or `vasprintf-gnu`:

- This function is missing on some platforms: AIX 5.1, HP-UX 11, Solaris 10, mingw, MSVC 14.

Portability problems fixed by either Gnulib module `vasprintf-posix` or `vasprintf-gnu`:

- This function does not support size specifiers as in C99 (`hh`, `ll`, `j`, `t`, `z`) on some platforms: Cygwin 1.5.24.
- This function does not support size specifiers as in C23 (`w8`, `w16`, `w32`, `w64`, `wf8`, `wf16`, `wf32`, `wf64`) on some platforms: glibc 2.37, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC, Android 9.0.
- `printf "%f", "%e", "%g"` of Infinity and NaN yields an incorrect result on some platforms: AIX 7.3, Solaris 11.4.
- This function does not support the `'a'` and `'A'` directives on some platforms: FreeBSD 5.2.1, OpenBSD 4.0, Solaris 11.4, Cygwin 1.5.x.
- This function does not support the `'b'` directive, required by ISO C23, on some platforms: glibc 2.34, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, HP-UX 11, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android 9.0.
- This function does not support the `'F'` directive on some platforms: NetBSD 3.0, Cygwin 1.5.x.

- This function does not support the ‘**ls**’ directive on some platforms: OpenBSD 4.0, Cygwin 1.5.x, Haiku.
- This function does not support precisions in the ‘**ls**’ directive correctly on some platforms: Solaris 11.4.
- This function does not support format directives that access arguments in an arbitrary order, such as “%2\$s”, on some platforms: NetBSD 3.0.
- This function doesn’t support the ‘**l**’ flag at all on some platforms: mingw without `__USE_MINGW_ANSI_STDIO`, MSVC 14.
- This function doesn’t support the ‘**l**’ flag correctly for integers (it uses the decimal point instead of the thousands separator!) on some platforms: mingw with `__USE_MINGW_ANSI_STDIO`.
- This function doesn’t support the ‘**l**’ flag on some platforms: NetBSD 3.0, Cygwin 1.5.24.
- This function doesn’t support the ‘**l**’ flag together with a precision on an integer argument correctly on some platforms: glibc 2.36, FreeBSD 14.0, NetBSD 10.0, Solaris 11, Cygwin 3.6.0, Haiku.
- This function doesn’t support the ‘**l**’ flag correctly in locales where the thousands separator is a multibyte character on some platforms: NetBSD 10.0, Solaris 11 OpenIndiana.
- This function does not round the argument of the ‘**a**’ directive correctly on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4.
- `printf "%010f"` of NaN and Infinity yields an incorrect result (padded with zeroes) on some platforms: Mac OS X 10.5, FreeBSD 6.0, NetBSD 5.0, AIX 7.3, Solaris 11.4, Cygwin 1.5.x.
- `printf "%#.0x"` or `printf "%#.0X"` with a zero argument yields an incorrect result (non-empty) on some platforms: Mac OS X 10.6.
- This function produces wrong output for the ‘**lc**’ directive with a NUL wide character argument on some platforms: musl libc 1.2.4.
- This function can crash in out-of-memory conditions on some platforms: FreeBSD 14.0, NetBSD 5.0.

Portability problems fixed by Gnulib module `vasprintf-gnu`:

- This function does not support the ‘**B**’ directive on some platforms: glibc 2.34, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC, Android 9.0.

Portability problems not fixed by Gnulib:

- The `%m` directive is not portable, use `%s` mapped to an argument of `strerror(errno)` (or a version of `strerror_r`) instead.

10.45.2 clearerr

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/clearerr.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

10.45.3 ctermid

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/ctermid.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14, Android 7.1.

10.45.4 dprintf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/dprintf.html>

Gnulib module: dprintf or dprintf-posix or dprintf-gnu

Portability problems fixed by either Gnulib module `dprintf` or `dprintf-posix` or `dprintf-gnu`:

- This function is missing on many non-glibc platforms: Mac OS X 10.5, FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.3, Cygwin 1.5.x, mingw, MSVC 14.

Portability problems fixed by either Gnulib module `dprintf-posix` or `dprintf-gnu`:

- This function does not support size specifiers as in C23 (`w8`, `w16`, `w32`, `w64`, `wf8`, `wf16`, `wf32`, `wf64`) on some platforms: glibc 2.37, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC, Android 9.0.
- `printf "%f", "%e", "%g"` of Infinity and NaN yields an incorrect result on some platforms: AIX 7.3, Solaris 11.4.
- This function doesn't support the `'` flag at all on some platforms: mingw without `__USE_MINGW_ANSI_STDIO`, MSVC 14.
- This function doesn't support the `'` flag correctly for integers (it uses the decimal point instead of the thousands separator!) on some platforms: mingw with `__USE_MINGW_ANSI_STDIO`.
- This function doesn't support the `'` flag together with a precision on an integer argument correctly on some platforms: glibc 2.36, FreeBSD 14.0, NetBSD 10.0, Solaris 11, Cygwin 3.6.0, Haiku.
- This function doesn't support the `'` flag correctly in locales where the thousands separator is a multibyte character on some platforms: NetBSD 10.0, Solaris 11 OpenIndiana.
- This function does not support the `'a'` and `'A'` directives on some platforms: Solaris 11.4.
- This function does not support the `'b'` directive, required by ISO C23, on some platforms: glibc 2.34, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, HP-UX 11, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android 9.0.
- This function does not support precisions in the `'ls'` directive correctly on some platforms: Solaris 11.4.

- This function does not round the argument of the ‘a’ directive correctly on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4.
- `printf "%010f"` of NaN and Infinity yields an incorrect result (padded with zeroes, or wrong capitalization) on some platforms: AIX 7.3, Solaris 11.4.
- `printf "%#.0x"` or `"%#.0X"` with a zero argument yields an incorrect result (non-empty) on some platforms: Mac OS X 10.6.
- This function does not support precisions larger than 512 or 1024 in integer, floating-point and pointer output on some platforms: AIX 7.1.
- This function produces wrong output for the ‘lc’ directive with a NUL wide character argument on some platforms: musl libc 1.2.4.

Portability problems fixed by Gnulib module `dprintf-gnu`:

- This function does not support the ‘B’ directive on some platforms: glibc 2.34, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC, Android 9.0.

Portability problems fixed by either Gnulib module `dprintf-posix` or `dprintf-gnu` and additionally the Gnulib module `printf-with-n-directive`:

- This function does not support the ‘n’ directive on some platforms: glibc when used with `_FORTIFY_SOURCE >= 2` (set by default on Ubuntu), macOS 14, OpenBSD, Android.

Portability problems not fixed by Gnulib:

- The `%m` directive is not portable, use `%s` mapped to an argument of `strerror(errno)` (or a version of `strerror_r`) instead.
- Formatting noncanonical ‘long double’ numbers produces nonmeaningful results on some platforms: glibc and others, on x86, x86_64, IA-64 CPUs.
- When formatting an integer with grouping flag, this function inserts thousands separators even in the "C" locale on some platforms: NetBSD 5.1.
- On some platforms, this function does not set `errno` or the stream error indicator on attempts to write to a read-only stream: Cygwin 1.7.9.

10.45.5 `fclose`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fclose.html>

Gnulib module: `fclose`

Portability problems fixed by Gnulib:

- On some platforms, this function fails to set the file position of a seekable input stream to the byte after the last one actually read: glibc 2.34, FreeBSD, AIX 7.2.
- This function crashes if the stream’s file descriptor has already been closed on some platforms: MSVC 14.
- On Windows platforms (excluding Cygwin), `socket` and `accept` followed by `fdopen` do not return streams that can be closed by `fclose`.

Portability problems not fixed by Gnulib:

- On Windows platforms (excluding Cygwin), this function does not set `errno` upon failure.

10.45.6 fdopen

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fdopen.html>

Gnulib module: fdopen

Portability problems fixed by Gnulib:

- This function crashes when invoked with invalid arguments on some platforms: MSVC 14.
- On Windows platforms (excluding Cygwin), this function does not set `errno` upon failure.

Portability problems not fixed by Gnulib:

10.45.7 feof

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/feof.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

10.45.8 ferror

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/ferror.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

10.45.9 fflush

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fflush.html>

Gnulib module: fflush

Portability problems fixed by Gnulib:

- `fflush` followed by `fseek` or `fseeko`, applied to an input stream, should have the effect of positioning the underlying file descriptor. It doesn't do this on some platforms: glibc 2.34, FreeBSD 14.0, and others.
- `fflush` on an input stream changes the position of the stream to the end of the previous buffer, on some platforms: mingw, MSVC 14.
- `fflush` on an input stream right after `ungetc` does not discard the `ungetc` buffer, on some platforms: macOS 14, FreeBSD 6.0, NetBSD 10.0, OpenBSD 7.5, Cygwin 1.5.25-10.
- `fflush` fails with `EBADF` if the stream is not open for writing: NetBSD 10.0, OpenBSD 7.5.

Portability problems not fixed by GnuLib:

- `fflush`, `ftell`, `ftello`, `fgetpos` behave incorrectly on input streams that are opened in `O_TEXT` mode and whose contents contains Unix line terminators (LF), on some platforms: mingw, MSVC 14.
- On Windows platforms (excluding Cygwin), this function does not set `errno` upon failure.
- This function crashes if the stream's file descriptor has already been closed, if `MSVC_INVALID_PARAMETER_HANDLING` is `HAIRY_LIBRARY_HANDLING` or `SANE_LIBRARY_HANDLING`, on some platforms: MSVC 14.
- `fflush` on an input stream right after `ungetc` does not discard the `ungetc` buffer, on some platforms: AIX 5.1, HP-UX 11, Solaris 11 2010-11, mingw, MSVC 14.

10.45.10 `fgetc`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fgetc.html>

GnuLib module: `stdio-h`, nonblocking

Portability problems fixed by GnuLib module `stdio-h`, together with module `nonblocking`:

- When reading from a non-blocking pipe whose buffer is empty, this function fails with `errno` being set to `EINVAL` instead of `EAGAIN` on some platforms: mingw, MSVC 14.

Portability problems not fixed by GnuLib:

- C99 and POSIX.1-2001 and later require end-of-file to be sticky, that is, they require this function to act as if it reads end-of-file if `feof` would return nonzero. However, on some systems this function attempts to read from the underlying file descriptor even if the stream's end-of-file indicator is set. These systems include glibc and default Solaris.
- On Windows platforms (excluding Cygwin), this function does not set `errno` upon failure.
- This function crashes if the stream's file descriptor has already been closed, if `MSVC_INVALID_PARAMETER_HANDLING` is `HAIRY_LIBRARY_HANDLING` or `SANE_LIBRARY_HANDLING`, on some platforms: MSVC 14.

10.45.11 `fgetpos`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fgetpos.html>

GnuLib module: —

Portability problems fixed by GnuLib:

- On platforms where `off_t` is a 32-bit type, this function may not work correctly on files 2 GiB and larger. See [Section 14.2 \[Large File Support\]](#), page 842.

Portability problems not fixed by GnuLib:

- `fflush`, `ftell`, `ftello`, `fgetpos` behave incorrectly on input streams that are opened in `O_TEXT` mode and whose contents contains Unix line terminators (LF), on some platforms: mingw, MSVC 14.
- This function mistakenly succeeds on pipes on some platforms: mingw 10.

10.45.12 `fgets`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fgets.html>

Gnulib module: `stdio-h`, `nonblocking`

Portability problems fixed by Gnulib module `stdio-h`, together with module `nonblocking`:

- When reading from a non-blocking pipe whose buffer is empty, this function fails with `errno` being set to `EINVAL` instead of `EAGAIN` on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- C99 and POSIX.1-2001 and later require end-of-file to be sticky, that is, they require this function to act as if it reads end-of-file if `feof` would return nonzero. However, on some systems this function attempts to read from the underlying file descriptor even if the stream's end-of-file indicator is set. These systems include glibc and default Solaris.
- On Windows platforms (excluding Cygwin), this function does not set `errno` upon failure.

10.45.13 `fileno`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fileno.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

10.45.14 `flockfile`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/flockfile.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

10.45.15 `fmemopen`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fmemopen.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: Mac OS X 10.5, FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.3, Cygwin 1.5.x, mingw, MSVC 14, Android 5.1.

An alternative to the `fmemopen` function is the Gnulib module `sf-istream`.

10.45.16 fopen

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fopen.html>

Gnulib module: `fopen` or `fopen-gnu`

Portability problems fixed by either Gnulib module `fopen` or `fopen-gnu`:

- This function does not fail when the file name argument ends in a slash and (without the slash) names a nonexistent file or a file that is not a directory, on some platforms: HP-UX 11.00, AIX 7.1, Solaris 9.
- On platforms where `off_t` is a 32-bit type, `fopen` may not work correctly with files 2 GiB and larger. See [Section 14.2 \[Large File Support\]](#), page 842.
- On Windows platforms (excluding Cygwin), this function does usually not recognize the `/dev/null` filename.

Portability problems fixed by Gnulib module `fopen-gnu`:

- This function does not support the mode character ‘x’ (corresponding to `O_EXCL`), introduced in ISO C11, on some platforms: FreeBSD 8.2, NetBSD 6.1, OpenBSD 5.6, Minix 3.2, AIX 6.1, HP-UX 11.31, Solaris 11.3, Cygwin 1.7.16 (2012), mingw with MSVCRT, MSVC 14.
- This function does not support the mode character ‘e’ (corresponding to `O_CLOEXEC`), introduced into a future POSIX revision through <https://www.austingroupbugs.net/view.php?id=411>, on some platforms: glibc 2.6, macOS 14, FreeBSD 9.0, NetBSD 5.1, OpenBSD 5.6, Minix 3.2, AIX 7.2, HP-UX 11.31, Solaris 11.3, Cygwin 1.7.16 (2012), mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- On Windows platforms (excluding Cygwin), this function does not set `errno` upon failure.
- On Windows, this function returns a file stream in “text” mode by default; this means that it translates ‘\n’ to CR/LF by default. Use the “b” flag if you need reliable binary I/O.
- On Windows platforms (excluding Cygwin), this function fails to open directories for reading. Such streams have implementation-defined semantics on other platforms. To avoid directory streams with a consistent error message, use `fstat` after `open` and `fdopen`, rather than `fopen` and `fileno`.

10.45.17 fprintf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fprintf.html>

Gnulib module: `fprintf-posix` or `fprintf-gnu` or `stdio-h`, nonblocking, sigpipe

Portability problems fixed by either Gnulib module `stdio-h` or `fprintf-posix` or `fprintf-gnu`:

- This function does not support non-ASCII characters in double-byte encoding, corresponding to the locale, on some platforms: mingw with `__USE_MINGW_ANSI_STDIO` in combination with `msvcrt`, when the output goes to a Windows console.

Portability problems fixed by either Gnulib module `fprintf-posix` or `fprintf-gnu`:

- This function does not support size specifiers as in C99 (`hh`, `ll`, `j`, `t`, `z`) on some platforms: AIX 5.1, HP-UX 11.23, Solaris 9, Cygwin 1.5.24, old mingw, MSVC 9.
- This function does not support size specifiers as in C23 (`w8`, `w16`, `w32`, `w64`, `wf8`, `wf16`, `wf32`, `wf64`) on some platforms: glibc 2.37, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC, Android 9.0.
- `printf` of ‘long double’ numbers is unsupported on some platforms: mingw, MSVC 14.
- `printf "%f", "%e", "%g"` of Infinity and NaN yields an incorrect result on some platforms: AIX 7.3, Solaris 11.4, mingw, MSVC 14.
- This function does not support the ‘a’ and ‘A’ directives on some platforms: FreeBSD 5.2.1, OpenBSD 4.0, AIX 5.2, HP-UX 11, Solaris 11.4, Cygwin 1.5.x, mingw, MSVC 14.
- This function does not support the ‘b’ directive, required by ISO C23, on some platforms: glibc 2.34, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, HP-UX 11, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android 9.0.
- This function does not support the ‘F’ directive on some platforms: NetBSD 3.0, AIX 5.1, HP-UX 11.23, Solaris 9, Cygwin 1.5.x, mingw, MSVC 14.
- This function does not support the ‘ls’ directive on some platforms: OpenBSD 4.0, Cygwin 1.5.x, Haiku.
- This function does not support precisions in the ‘ls’ directive correctly on some platforms: Solaris 11.4.
- This function does not support format directives that access arguments in an arbitrary order, such as `"%2$s"`, on some platforms: NetBSD 3.0, mingw, MSVC 14.
- This function doesn’t support the ‘l’ flag at all on some platforms: mingw without `__USE_MINGW_ANSI_STDIO`, MSVC 14.
- This function doesn’t support the ‘l’ flag correctly for integers (it uses the decimal point instead of the thousands separator!) on some platforms: mingw with `__USE_MINGW_ANSI_STDIO`.
- This function doesn’t support the ‘l’ flag on some platforms: NetBSD 3.0, Cygwin 1.5.24, mingw, MSVC 14.
- This function doesn’t support the ‘l’ flag together with a precision on an integer argument correctly on some platforms: glibc 2.36, FreeBSD 14.0, NetBSD 10.0, Solaris 11, Cygwin 3.6.0, Haiku.
- This function doesn’t support the ‘l’ flag correctly in locales where the thousands separator is a multibyte character on some platforms: NetBSD 10.0, Solaris 11 OpenIndiana.
- This function does not round the argument of the ‘a’ directive correctly on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4.
- `printf "%010f"` of NaN and Infinity yields an incorrect result (padded with zeroes, or wrong capitalization) on some platforms: Mac OS X 10.5, FreeBSD 6.0, NetBSD 5.0, AIX 7.3, Solaris 11.4, Cygwin 1.5.x, mingw, MSVC/clang.
- `printf "%#.0x" or "%#.0X"` with a zero argument yields an incorrect result (non-empty) on some platforms: Mac OS X 10.6.

- This function does not support precisions larger than 512 or 1024 in integer, floating-point and pointer output on some platforms: AIX 7.1, Solaris 10/x86, mingw, MSVC/clang.
- This function mishandles large floating point precisions (for example, formatting 1.0 with `"%.511f"`) on some platforms: Solaris 10.
- This function produces wrong output for the `'lc'` directive with a NUL wide character argument on some platforms: musl libc 1.2.4.
- This function can crash in out-of-memory conditions on some platforms: FreeBSD 14.0, NetBSD 5.0.

Portability problems fixed by Gnulib module `fprintf-gnu`:

- This function does not support the `'B'` directive on some platforms: glibc 2.34, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC, Android 9.0.

Portability problems fixed by either Gnulib module `fprintf-posix` or `fprintf-gnu` and additionally the Gnulib module `printf-with-n-directive`:

- This function does not support the `'n'` directive on some platforms: glibc when used with `_FORTIFY_SOURCE >= 2` (set by default on Ubuntu), macOS 14, OpenBSD, MSVC 14, Android.

Portability problems fixed by Gnulib module `stdio-h` or `fprintf-posix` or `fprintf-gnu`, together with module `nonblocking`:

- When writing to a non-blocking pipe whose buffer is full, this function fails with `errno` being set to `ENOSPC` instead of `EAGAIN` on some platforms: mingw, MSVC 14.

Portability problems fixed by Gnulib module `stdio-h` or `fprintf-posix` or `fprintf-gnu`, together with module `sigpipe`:

- When writing to a pipe with no readers, this function fails, instead of obeying the current `SIGPIPE` handler, on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- The `%m` directive is not portable, use `%s` mapped to an argument of `strerror(errno)` (or a version of `strerror_r`) instead.
- Formatting noncanonical `'long double'` numbers produces nonmeaningful results on some platforms: glibc and others, on x86, x86_64, IA-64 CPUs.
- When formatting an integer with grouping flag, this function inserts thousands separators even in the `"C"` locale on some platforms: NetBSD 5.1.
- Attempting to write to a read-only stream fails with `EOF` but does not set the error flag for `ferror` on some platforms: glibc 2.13, cygwin 1.7.9.

10.45.18 fputc

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fputc.html>

Gnulib module: `stdio-h`, `nonblocking`, `sigpipe`

Portability problems fixed by Gnulib module `stdio-h`, together with module `nonblocking`:

- When writing to a non-blocking pipe whose buffer is full, this function fails with `errno` being set to `ENOSPC` instead of `EAGAIN` on some platforms: mingw, MSVC 14.

Portability problems fixed by Gnulib module `stdio-h`, together with module `sigpipe`:

- When writing to a pipe with no readers, this function fails, instead of obeying the current `SIGPIPE` handler, on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- On Windows platforms (excluding Cygwin), this function does not set `errno` upon failure.
- This function fails and produces garbled output when invoked twice, for outputting a non-ASCII character in double-byte encoding, corresponding to the locale, on some platforms: mingw in combination with `msvcrt`, when the output goes to a Windows console.
- On some platforms, this function does not set `errno` or the stream error indicator on attempts to write to a read-only stream: Cygwin 1.7.9.
- This function crashes if the stream's file descriptor has already been closed, if `MSVC_INVALID_PARAMETER_HANDLING` is `HAIRY_LIBRARY_HANDLING` or `SANE_LIBRARY_HANDLING`, on some platforms: MSVC 14.

10.45.19 `fputs`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fputs.html>

Gnulib module: `stdio-h`, `nonblocking`, `sigpipe`

Portability problems fixed by Gnulib module `stdio-h`, together with module `nonblocking`:

- When writing to a non-blocking pipe whose buffer is full, this function fails with `errno` being set to `ENOSPC` instead of `EAGAIN` on some platforms: mingw, MSVC 14.

Portability problems fixed by Gnulib module `stdio-h`, together with module `sigpipe`:

- When writing to a pipe with no readers, this function fails, instead of obeying the current `SIGPIPE` handler, on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- On Windows platforms (excluding Cygwin), this function does not set `errno` upon failure.
- On some platforms, this function does not set `errno` or the stream error indicator on attempts to write to a read-only stream: Cygwin 1.7.9.

10.45.20 `fread`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fread.html>

Gnulib module: `stdio-h`, `nonblocking`

Portability problems fixed by Gnulib module `stdio-h`, together with module `nonblocking`:

- When reading from a non-blocking pipe whose buffer is empty, this function fails with `errno` being set to `EINVAL` instead of `EAGAIN` on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- C99 and POSIX.1-2001 and later require end-of-file to be sticky, that is, they require this function to act as if it reads end-of-file if `feof` would return nonzero. However, on some systems this function attempts to read from the underlying file descriptor even if the stream's end-of-file indicator is set. These systems include glibc and default Solaris.
- On Windows platforms (excluding Cygwin), this function does not set `errno` upon failure.
- This function crashes if the stream's file descriptor has already been closed, if `MSVC_INVALID_PARAMETER_HANDLING` is `HAIRY_LIBRARY_HANDLING` or `SANE_LIBRARY_HANDLING`, on some platforms: MSVC 14.

10.45.21 `freopen`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/freopen.html>

Gnulib module: `freopen`

Portability problems fixed by Gnulib:

- On some platforms, if `stream` does not already have an open file descriptor, `freopen` returns the stream without opening the file: glibc 2.24.
- On platforms where `off_t` is a 32-bit type, `freopen` may not work correctly with files 2 GiB and larger. See [Section 14.2 \[Large File Support\]](#), page 842.
- On Windows platforms (excluding Cygwin), this function does usually not recognize the `/dev/null` filename.

Portability problems not fixed by Gnulib:

- On Windows platforms (excluding Cygwin), this function does not set `errno` upon failure.
- This function does not support a `NULL` file name argument on some platforms: OpenBSD 4.9, AIX 7.1, HP-UX 11.23, Solaris 10, mingw, MSVC 14.
- This function does not fail when the file name argument ends in a slash and (without the slash) names a nonexistent file or a file that is not a directory, on some platforms: HP-UX 11.00, Solaris 9.
- Applications should not assume that `fileno(f)` will be the same before and after a call to `freopen(name, mode, f)`. However, the module `freopen-safer` can at least protect `stdin`, `stdout`, and `stderr`.

10.45.22 `fscanf`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fscanf.html>

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-fscanf.html

Gnulib module: `stdio-h`, `nonblocking`

Portability problems fixed by Gnulib module `stdio-h`, together with module `nonblocking`:

- When reading from a non-blocking pipe whose buffer is empty, this function fails with `errno` being set to `EINVAL` instead of `EAGAIN` on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- C99 and POSIX.1-2001 and later require end-of-file to be sticky, that is, they require this function to act as if it reads end-of-file if `feof` would return nonzero. However, on some systems this function attempts to read from the underlying file descriptor even if the stream's end-of-file indicator is set. These systems include glibc and default Solaris.
- This function does not support size specifiers as in C23 (`w8`, `w16`, `w32`, `w64`, `wf8`, `wf16`, `wf32`, `wf64`) on some platforms: glibc 2.38, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.2, AIX 7.2, HP-UX 11, Solaris 11.4, Cygwin 2.9.0, mingw, MSVC 14.
- On Windows platforms (excluding Cygwin), this function does not set `errno` upon failure.
- On Windows, this function doesn't support the `hh`, `ll`, `j`, `t`, `z` size specifiers.

10.45.23 `fseek`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fseek.html>

Gnulib module: `fseek`

Portability problems fixed by Gnulib:

- This function mistakenly succeeds on pipes on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- On Windows platforms (excluding Cygwin), this function does not set `errno` upon failure.
- On platforms where `long` is a 32-bit type, `fseek` does not work correctly with files 2 GiB and larger, even when the `AC_SYS_LARGEFILE` macro is used. The fix is to use `fseeko` instead.

10.45.24 `fseeko`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fseeko.html>

Gnulib module: `fseeko`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.
- The declaration of `fseeko` in `<stdio.h>` is not enabled by default on some platforms: glibc 2.3.6.

- This function fails on seekable stdin, stdout, and stderr: cygwin <= 1.5.24.
- On platforms where `off_t` is a 32-bit type, `fseeko` does not work correctly with files 2 GiB and larger. See [Section 14.2 \[Large File Support\]](#), page 842.

Portability problems not fixed by GnuLib:

10.45.25 `fsetpos`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fsetpos.html>

GnuLib module: —

Portability problems fixed by GnuLib:

- On platforms where `off_t` is a 32-bit type, this function may not work correctly on files 2 GiB and larger. See [Section 14.2 \[Large File Support\]](#), page 842.

Portability problems not fixed by GnuLib:

10.45.26 `ftell`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/ftell.html>

GnuLib module: `ftell`

Portability problems fixed by GnuLib:

- This function mistakenly succeeds on pipes on some platforms: mingw, MSVC 14.
- This function produces incorrect results after `putc` that followed a `getc` call that reached EOF on some platforms: Solaris 11 2010-11.
- This function, when invoked after `ungetc`, throws away the `ungetc` buffer, changes the stream's file position, and returns the wrong position on some platforms: macOS 10.15, 11, 12.
- This function mistakenly succeeds on pipes on some platforms: mingw 10.

Portability problems not fixed by GnuLib:

- This function produces incorrect results immediately after `fseek` on some platforms: HP-UX 11.
- `fflush`, `ftell`, `ftello`, `fgetpos` behave incorrectly on input streams that are opened in `O_TEXT` mode and whose contents contains Unix line terminators (LF), on some platforms: mingw, MSVC 14.
- On platforms where `long` is a 32-bit type, `ftell` does not work correctly with files 2 GiB and larger, even when the `AC_SYS_LARGEFILE` macro is used. The fix is to use `ftello` instead.

10.45.27 `ftello`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/ftello.html>

GnuLib module: `ftello`

Portability problems fixed by GnuLib:

- This function is missing on some platforms: mingw, MSVC 14.

- The declaration of `ftello` in `<stdio.h>` is not enabled by default on some platforms: glibc 2.3.6.
- This function produces incorrect results after `putc` that followed a `getc` call that reached EOF on some platforms: Solaris 11 2010-11.
- This function, when invoked after `ungetc`, throws away the `ungetc` buffer, changes the stream's file position, and returns the wrong position on some platforms: macOS 10.15, 11, 12.
- This function fails on seekable stdin, stdout, and stderr: cygwin <= 1.5.24.
- This function mistakenly succeeds on pipes on some platforms: mingw 10.
- On platforms where `off_t` is a 32-bit type, `ftello` does not work correctly with files 2 GiB and larger. See [Section 14.2 \[Large File Support\]](#), page 842.

Portability problems not fixed by Gnuilib:

- This function produces incorrect results immediately after `fseek` on some platforms: HP-UX 11.
- `fflush`, `ftell`, `ftello`, `fgetpos` behave incorrectly on input streams that are opened in `O_TEXT` mode and whose contents contains Unix line terminators (LF), on some platforms: mingw, MSVC 14.

10.45.28 `ftrylockfile`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/ftrylockfile.html>

Gnuilib module: —

Portability problems fixed by Gnuilib:

Portability problems not fixed by Gnuilib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

10.45.29 `funlockfile`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/funlockfile.html>

Gnuilib module: —

Portability problems fixed by Gnuilib:

Portability problems not fixed by Gnuilib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

10.45.30 `fwrite`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fwrite.html>

Gnuilib module: `stdio-h`, `nonblocking`, `sigpipe`

Portability problems fixed by Gnuilib module `stdio-h`:

- This function fails and produces no output when the argument string starts with a non-ASCII character in double-byte encoding, corresponding to the locale, on some

platforms: mingw in combination with msvcrt, when the output goes to a Windows console.

Portability problems fixed by Gnulib module `stdio-h`, together with module `nonblocking`:

- When writing to a non-blocking pipe whose buffer is full, this function fails with `errno` being set to `ENOSPC` instead of `EAGAIN` on some platforms: mingw, MSVC 14.

Portability problems fixed by Gnulib module `stdio-h`, together with module `sigpipe`:

- When writing to a pipe with no readers, this function fails, instead of obeying the current `SIGPIPE` handler, on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- On Windows platforms (excluding Cygwin), this function does not set `errno` upon failure.
- On some platforms, this function does not set `errno` or the stream error indicator on attempts to write to a read-only stream: Cygwin 1.7.9.
- This function crashes if the stream's file descriptor has already been closed, if `MSVC_INVALID_PARAMETER_HANDLING` is `HAIRY_LIBRARY_HANDLING` or `SANE_LIBRARY_HANDLING`, on some platforms: MSVC 14.

10.45.31 `getc`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getc.html>

Gnulib module: `stdio-h`, `nonblocking`

Portability problems fixed by Gnulib module `stdio-h`, together with module `nonblocking`:

- When reading from a non-blocking pipe whose buffer is empty, this function fails with `errno` being set to `EINVAL` instead of `EAGAIN` on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- C99 and POSIX.1-2001 and later require end-of-file to be sticky, that is, they require this function to act as if it reads end-of-file if `fEOF` would return nonzero. However, on some systems this function attempts to read from the underlying file descriptor even if the stream's end-of-file indicator is set. These systems include glibc and default Solaris.
- On Windows platforms (excluding Cygwin), this function does not set `errno` upon failure.

10.45.32 `getc_unlocked`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/getc_unlocked.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

10.45.33 `getchar`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getchar.html>

Gnulib module: `stdio-h`, nonblocking

Portability problems fixed by Gnulib module `stdio-h`, together with module `nonblocking`:

- When reading from a non-blocking pipe whose buffer is empty, this function fails with `errno` being set to `EINVAL` instead of `EAGAIN` on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- C99 and POSIX.1-2001 and later require end-of-file to be sticky, that is, they require this function to act as if it reads end-of-file if `feof` would return nonzero. However, on some systems this function attempts to read from the underlying file descriptor even if the stream's end-of-file indicator is set. These systems include glibc and default Solaris.
- On Windows platforms (excluding Cygwin), this function does not set `errno` upon failure.

10.45.34 `getchar_unlocked`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/getchar_unlocked.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

10.45.35 `getdelim`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getdelim.html>

Gnulib module: `getdelim`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Mac OS X 10.5, FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 10, mingw, MSVC 14, Android 4.2.
- This function makes out-of-bounds reads on some platforms: macOS 10.13.
- This function crashes when passed a pointer to a NULL buffer together with a pointer to a non-zero buffer size on some platforms: FreeBSD 8.0.
- This function does not NUL terminate the buffer when the first character read is EOF on some platforms: glibc 2.42.

Portability problems not fixed by Gnulib:

- This function has quadratic running time for long lines on some platforms: uClibc 0.9.31.

10.45.36 `getline`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getline.html>

Gnulib module: `getline`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Mac OS X 10.5, FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 10, mingw, MSVC 14, Android 4.2.
- This function is missing a declaration on some platforms: AIX 7.1.
- Some platforms provide a function by this name but with the wrong signature, for example in `-linet`.
- This function crashes when passed a pointer to a NULL buffer together with a pointer to a non-zero buffer size on some platforms: FreeBSD 8.0.
- This function does not NUL terminate the buffer when the first character read is EOF on some platforms: glibc 2.42.

Portability problems not fixed by Gnulib:

- This function has quadratic running time for long lines on some platforms: uClibc 0.9.31.

10.45.37 `open_memstream`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/open_memstream.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Mac OS X 10.5, FreeBSD 6.0, NetBSD 7.1, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.3, Cygwin 1.5.x, mingw, MSVC 14, Android 5.1.

An alternative to the `open_memstream` function is the Gnulib module `string-buffer`.

10.45.38 `pclose`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/pclose.html>

Gnulib module: `pclose`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: MSVC 14.

Portability problems not fixed by Gnulib:

10.45.39 perror

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/perror.html>

Gnulib module: perror

Portability problems fixed by Gnulib:

- This function does not support the error values that are specified by POSIX but not defined by the system, on some platforms: OpenBSD 4.0, Cygwin 1.5.x, mingw, MSVC 14.
- This function treats `errno` of 0 like failure, although POSIX requires that the message declare it as a success, on some platforms: FreeBSD 8.2, OpenBSD 4.7, macOS 14.
- This function clobbers the `strerror` buffer on some platforms: Cygwin 1.7.9, Android 11.
- This function fails to print a useful a string for out-of-range integers on some platforms: HP-UX 11.

Portability problems not fixed by Gnulib:

- POSIX requires that this function set the stream error bit (detected by `ferror`) on write failure, but not all platforms do this: glibc 2.13, cygwin 1.7.9.
- POSIX requires that this function not alter stream orientation, but the gnulib replacement locks in byte orientation and fails on wide character streams.

10.45.40 popen

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/popen.html>

Gnulib module: popen

Portability problems fixed by Gnulib:

- This function is missing on some platforms: MSVC 14.
- Some platforms start the child with closed stdin or stdout if the standard descriptors were closed in the parent: Cygwin 1.5.x.
- This function hangs if called with closed stdin or stdout on some platforms: OS/2 kLIBC.

Portability problems not fixed by Gnulib:

- On native Windows platforms, this functions terminates the current process with exit code 127 if the environment variable `COMSPEC` is not set.
- Some platforms mistakenly set the close-on-exec bit, then if it is cleared by the application, the platform then leaks file descriptors from earlier `popen` calls into subsequent `popen` children: Cygwin 1.5.x.

10.45.41 printf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/printf.html>

Gnulib module: printf-posix or printf-gnu or stdio-h, nonblocking, sigpipe

Portability problems fixed by either Gnulib module `stdio-h` or `fprintf-posix` or `fprintf-gnu`:

- This function does not support non-ASCII characters in double-byte encoding, corresponding to the locale, on some platforms: mingw with `__USE_MINGW_ANSI_STDIO` in combination with `msvcrt`, when the output goes to a Windows console.

Portability problems fixed by either Gnulib module `printf-posix` or `printf-gnu`:

- This function does not support size specifiers as in C99 (`hh`, `ll`, `j`, `t`, `z`) on some platforms: AIX 5.1, HP-UX 11.23, Solaris 9, Cygwin 1.5.24, old mingw, MSVC 9.
- This function does not support size specifiers as in C23 (`w8`, `w16`, `w32`, `w64`, `wf8`, `wf16`, `wf32`, `wf64`) on some platforms: glibc 2.37, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC, Android 9.0.
- `printf` of ‘long double’ numbers is unsupported on some platforms: mingw, MSVC 14.
- `printf "%f", "%e", "%g"` of Infinity and NaN yields an incorrect result on some platforms: AIX 7.3, Solaris 11.4, mingw, MSVC 14.
- This function doesn’t support the ‘`'`’ flag at all on some platforms: mingw without `__USE_MINGW_ANSI_STDIO`, MSVC 14.
- This function doesn’t support the ‘`'`’ flag correctly for integers (it uses the decimal point instead of the thousands separator!) on some platforms: mingw with `__USE_MINGW_ANSI_STDIO`.
- This function doesn’t support the ‘`'`’ flag together with a precision on an integer argument correctly on some platforms: glibc 2.36, FreeBSD 14.0, NetBSD 10.0, Solaris 11, Cygwin 3.6.0, Haiku.
- This function doesn’t support the ‘`'`’ flag correctly in locales where the thousands separator is a multibyte character on some platforms: NetBSD 10.0, Solaris 11 OpenIndiana.
- This function does not support the ‘`a`’ and ‘`A`’ directives on some platforms: FreeBSD 5.2.1, OpenBSD 4.0, AIX 5.2, HP-UX 11, Solaris 11.4, Cygwin 1.5.x, mingw, MSVC 14.
- This function does not support the ‘`b`’ directive, required by ISO C23, on some platforms: glibc 2.34, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, HP-UX 11, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android 9.0.
- This function does not support the ‘`F`’ directive on some platforms: NetBSD 3.0, AIX 5.1, HP-UX 11.23, Solaris 9, Cygwin 1.5.x, mingw, MSVC 14.
- This function does not support the ‘`ls`’ directive on some platforms: OpenBSD 4.0, Cygwin 1.5.x, Haiku.
- This function does not support precisions in the ‘`ls`’ directive correctly on some platforms: Solaris 11.4.
- This function does not support format directives that access arguments in an arbitrary order, such as `"%2$s"`, on some platforms: NetBSD 3.0, mingw, MSVC 14.
- This function doesn’t support the ‘`'`’ flag on some platforms: NetBSD 3.0, Cygwin 1.5.24, mingw, MSVC 14.
- This function does not round the argument of the ‘`a`’ directive correctly on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4.

- `printf "%010f"` of NaN and Infinity yields an incorrect result (padded with zeroes, or wrong capitalization) on some platforms: Mac OS X 10.5, FreeBSD 6.0, NetBSD 5.0, AIX 7.3, Solaris 11.4, Cygwin 1.5.x, mingw, MSVC/clang.
- `printf "%#.0x" or "%#.0X"` with a zero argument yields an incorrect result (non-empty) on some platforms: Mac OS X 10.6.
- This function does not support precisions larger than 512 or 1024 in integer, floating-point and pointer output on some platforms: AIX 7.1, Solaris 10/x86, mingw, MSVC/clang.
- This function mishandles large floating point precisions (for example, formatting 1.0 with `"%.511f"`) on some platforms: Solaris 10.
- This function produces wrong output for the `'lc'` directive with a NUL wide character argument on some platforms: musl libc 1.2.4.
- This function can crash in out-of-memory conditions on some platforms: FreeBSD 14.0, NetBSD 5.0.

Portability problems fixed by Gnulib module `printf-gnu`:

- This function does not support the `'B'` directive on some platforms: glibc 2.34, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC, Android 9.0.

Portability problems fixed by either Gnulib module `printf-posix` or `printf-gnu` and additionally the Gnulib module `printf-with-n-directive`:

- This function does not support the `'n'` directive on some platforms: glibc when used with `_FORTIFY_SOURCE >= 2` (set by default on Ubuntu), macOS 14, OpenBSD, MSVC 14, Android.

Portability problems fixed by Gnulib module `stdio-h` or `printf-posix` or `printf-gnu`, together with module `nonblocking`:

- When writing to a non-blocking pipe whose buffer is full, this function fails with `errno` being set to `ENOSPC` instead of `EAGAIN` on some platforms: mingw, MSVC 14.

Portability problems fixed by Gnulib module `stdio-h` or `printf-posix` or `printf-gnu`, together with module `sigpipe`:

- When writing to a pipe with no readers, this function fails, instead of obeying the current `SIGPIPE` handler, on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- The `%m` directive is not portable, use `%s` mapped to an argument of `strerror(errno)` (or a version of `strerror_r`) instead.
- Formatting noncanonical `'long double'` numbers produces nonmeaningful results on some platforms: glibc and others, on x86, x86_64, IA-64 CPUs.
- When formatting an integer with grouping flag, this function inserts thousands separators even in the `"C"` locale on some platforms: NetBSD 5.1.
- Attempting to write to a read-only stream fails with `EOF` but does not set the error flag for `ferror` on some platforms: glibc 2.13, cygwin 1.7.9.

10.45.42 `putc`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/putc.html>

Gnulib module: `stdio-h`, `nonblocking`, `sigpipe`

Portability problems fixed by Gnulib module `stdio-h`, together with module `nonblocking`:

- When writing to a non-blocking pipe whose buffer is full, this function fails with `errno` being set to `ENOSPC` instead of `EAGAIN` on some platforms: mingw, MSVC 14.

Portability problems fixed by Gnulib module `stdio-h`, together with module `sigpipe`:

- When writing to a pipe with no readers, this function fails, instead of obeying the current `SIGPIPE` handler, on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- On Windows platforms (excluding Cygwin), this function does not set `errno` upon failure.
- This function fails and produces garbled output when invoked twice, for outputting a non-ASCII character in double-byte encoding, corresponding to the locale, on some platforms: mingw in combination with `msvcrt`, when the output goes to a Windows console.
- On some platforms, this function does not set `errno` or the stream error indicator on attempts to write to a read-only stream: Cygwin 1.7.9.

10.45.43 `putc_unlocked`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/putc_unlocked.html

Gnulib module: —

Portability problems fixed by Gnulib:

- This function cannot be called from plain inline or extern inline functions on some platforms: macOS 10.8.

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.
- On some platforms, this function does not set `errno` or the stream error indicator on attempts to write to a read-only stream: Cygwin 1.7.9.

10.45.44 `putchar`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/putchar.html>

Gnulib module: `stdio-h`, `nonblocking`, `sigpipe`

Portability problems fixed by Gnulib module `stdio-h`, together with module `nonblocking`:

- When writing to a non-blocking pipe whose buffer is full, this function fails with `errno` being set to `ENOSPC` instead of `EAGAIN` on some platforms: mingw, MSVC 14.

Portability problems fixed by Gnulib module `stdio-h`, together with module `sigpipe`:

- When writing to a pipe with no readers, this function fails, instead of obeying the current `SIGPIPE` handler, on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- On Windows platforms (excluding Cygwin), this function does not set `errno` upon failure.
- On some platforms, this function does not set `errno` or the stream error indicator on attempts to write to a read-only stream: Cygwin 1.7.9.

10.45.45 `putchar_unlocked`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/putchar_unlocked.html

Gnulib module: —

Portability problems fixed by Gnulib:

- This function cannot be called from plain inline or extern inline functions on some platforms: macOS 10.8.

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.
- On some platforms, this function does not set `errno` or the stream error indicator on attempts to write to a read-only stream: Cygwin 1.7.9.

10.45.46 `puts`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/puts.html>

Gnulib module: `stdio-h`, `nonblocking`, `sigpipe`

Portability problems fixed by Gnulib module `stdio-h`, together with module `nonblocking`:

- When writing to a non-blocking pipe whose buffer is full, this function fails with `errno` being set to `ENOSPC` instead of `EAGAIN` on some platforms: mingw, MSVC 14.

Portability problems fixed by Gnulib module `stdio-h`, together with module `sigpipe`:

- When writing to a pipe with no readers, this function fails, instead of obeying the current `SIGPIPE` handler, on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- On Windows platforms (excluding Cygwin), this function does not set `errno` upon failure.
- On some platforms, this function does not set `errno` or the stream error indicator on attempts to write to a read-only stream: Cygwin 1.7.9.

10.45.47 remove

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/remove.html>

Gnulib module: remove

Portability problems fixed by Gnulib:

- This function fails to reject trailing slashes on non-directories on some platforms: FreeBSD 7.2, Solaris 9.
- This function mistakenly removes a directory with `remove("dir/.")` on some platforms: Cygwin 1.5.x.
- This function does not remove empty directories on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

10.45.48 rename

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/rename.html>

Gnulib module: rename

Portability problems fixed by Gnulib:

- This function does not allow trailing slashes when creating a destination directory, as in `rename("dir","new/")`: NetBSD 1.6.
- This function does not reject trailing slashes on the destination for non-directories on some platforms, as in `rename("file","new/")`: AIX 7.1, Solaris 11.3, Cygwin 1.5.x, mingw, MSVC 14.
- This function does not reject trailing slashes on symlinks to non-directories on some platforms, as in `rename("link-to-file/","f")`: FreeBSD 7.2.
- This function ignores trailing slashes on symlinks on some platforms, such that `rename("link/","new")` corrupts link: Solaris 9.
- This function incorrectly reduces the link count when comparing two spellings of a hard link on some platforms: NetBSD 1.6, Cygwin 1.5.x.
- This function will not always replace an existing destination on some platforms: Cygwin 1.5.x, mingw, MSVC 14. However, the replacement is not atomic for directories, and may end up losing the empty destination if the source could not be renamed.
- This function mistakenly allows names ending in `'.'` or `'..'` on some platforms: Cygwin 1.5.x, mingw, MSVC 14.
- This function does not reject attempts to rename existing directories and non-directories onto one another on some platforms: Cygwin 1.5.x, mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- POSIX requires that `rename("symlink-to-dir/","dir2")` rename `dir` and leave `symlink-to-dir` dangling; likewise, it requires that `rename("dir","dangling/")` rename `dir` so that `dangling` is no longer a dangling symlink. This behavior is counter-intuitive, so on some systems, `rename` fails with `ENOTDIR` if either argument is a symlink with a trailing slash: glibc, OpenBSD, Cygwin 1.7.

- POSIX requires that `rename` do nothing and return 0 if the source and destination are hard links to the same file. This behavior is counterintuitive, and on some systems `renameat` is a no-op in this way only if the source and destination identify the same directory entry. On these systems, for example, although renaming `./f` to `f` is a no-op, renaming `f` to `g` deletes `f` when `f` and `g` are hard links to the same file: NetBSD 7.0.
- After renaming a non-empty directory over an existing empty directory, the old directory name is still visible through the `stat` function for 30 seconds after the rename, on NFS file systems, on some platforms: Linux 2.6.18.
- This function will not rename a source that is currently opened by any process: mingw, MSVC 14.

10.45.49 `renameat`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/renameat.html>

Gnulib module: `renameat`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, Mac OS X 10.5, FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 1.5.x, mingw, MSVC 14. But the replacement function is not safe to be used in libraries and is not multithread-safe.
- This function is declared in `<unistd.h>`, not in `<stdio.h>`, on some platforms: NetBSD 7.0, Solaris 11.4.
- This function is declared in `<sys/stat.h>`, not in `<stdio.h>`, on some platforms: Android 4.3.
- This function does not reject trailing slashes on non-directories on some platforms, as in `renameat(fd,"file",fd,"new/")`: Solaris 11.4.
- This function ignores trailing slashes on symlinks on some platforms, such that `renameat(fd,"link/",fd,"new")` corrupts `link`: Solaris 9.
- `renameat(AT_FDCWD,"file",AT_FDCWD,"new/")` succeeds instead of failing and setting `errno` to `ENOTDIR`, on some platforms: GNU/Hurd with glibc 2.41.
- `renameat(AT_FDCWD,"dir",AT_FDCWD,"new/")` fails with `errno` set to `ENOENT`, on some platforms: GNU/Hurd with glibc 2.42.
- `renameat(AT_FDCWD,"dir/",AT_FDCWD,"new")` fails with `errno` set to `ENOTDIR`, on some platforms: GNU/Hurd with glibc 2.42.

Portability problems not fixed by Gnulib:

- POSIX requires that `renameat(fd,"symlink-to-dir/",fd,"dir2")` rename `dir` and leave `symlink-to-dir` dangling; likewise, it requires that `renameat(fd,"dir",fd,"dangling/")` rename `dir` so that `dangling` is no longer a dangling symlink. This behavior is counter-intuitive, so on some systems, `renameat` fails with `ENOTDIR` if either argument is a symlink with a trailing slash: glibc, OpenBSD, Cygwin 1.7.
- POSIX requires that `renameat` do nothing and return 0 if the source and destination are hard links to the same file. This behavior is counterintuitive, and on some systems

`renameat` is a no-op in this way only if the source and destination identify the same directory entry. On these systems, for example, although renaming `./f` to `f` is a no-op, renaming `f` to `g` deletes `f` when `f` and `g` are hard links to the same file: NetBSD 7.0.

- After renaming a non-empty directory over an existing empty directory, the old directory name is still visible through the `stat` function for 30 seconds after the rename, on NFS file systems, on some platforms: Linux 2.6.18.
- This function will not rename a source that is currently opened by any process: mingw, MSVC 14.

10.45.50 `rewind`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/rewind.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- On Windows platforms (excluding Cygwin), this function does not set `errno` upon failure.

10.45.51 `scanf`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/scanf.html>

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-scanf.html

Gnulib module: `stdio-h`, `nonblocking`

Portability problems fixed by Gnulib module `stdio-h`, together with module `nonblocking`:

- When reading from a non-blocking pipe whose buffer is empty, this function fails with `errno` being set to `EINVAL` instead of `EAGAIN` on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- C99 and POSIX.1-2001 and later require end-of-file to be sticky, that is, they require this function to act as if it reads end-of-file if `feof` would return nonzero. However, on some systems this function attempts to read from the underlying file descriptor even if the stream's end-of-file indicator is set. These systems include glibc and default Solaris.
- This function does not support size specifiers as in C23 (`w8`, `w16`, `w32`, `w64`, `wf8`, `wf16`, `wf32`, `wf64`) on some platforms: glibc 2.38, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.2, AIX 7.2, HP-UX 11, Solaris 11.4, Cygwin 2.9.0, mingw, MSVC 14.
- On Windows platforms (excluding Cygwin), this function does not set `errno` upon failure.
- On Windows, this function doesn't support the `hh`, `ll`, `j`, `t`, `z` size specifiers.

10.45.52 setbuf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/setbuf.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

10.45.53 setvbuf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/setvbuf.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- On Windows platforms (excluding Cygwin), this function does not set `errno` upon failure.
- On Windows platforms (excluding Cygwin), `_IOLBF` (line-buffering) is treated as if it were `_IOFBF` (full buffering), possibly with a warning generated.

10.45.54 snprintf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/snprintf.html>

Gnulib module: `snprintf` or `snprintf-posix` or `snprintf-gnu`

Portability problems fixed by either Gnulib module `snprintf` or `snprintf-posix` or `snprintf-gnu`:

- This function is missing on some platforms: MSVC 14.
- This function does not support format directives that access arguments in an arbitrary order, such as `"%2$s"`, on some platforms: NetBSD 3.0, mingw.
- This function does not return a byte count as specified in C99 on some platforms: HP-UX 11, Solaris 9, mingw.

Portability problems fixed by either Gnulib module `snprintf-posix` or `snprintf-gnu`:

- This function does not support size specifiers as in C99 (`hh`, `ll`, `j`, `t`, `z`) on some platforms: AIX 5.1, HP-UX 11.23, Solaris 9, Cygwin 1.5.24, old mingw, MSVC 9.
- This function does not support size specifiers as in C23 (`w8`, `w16`, `w32`, `w64`, `wf8`, `wf16`, `wf32`, `wf64`) on some platforms: glibc 2.37, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC, Android 9.0.
- printf of 'long double' numbers is unsupported on some platforms: mingw, MSVC 14.
- printf `"%f"`, `"%e"`, `"%g"` of Infinity and NaN yields an incorrect result on some platforms: AIX 7.3, Solaris 11.4, mingw, MSVC 14.
- This function doesn't support the `'` flag at all on some platforms: mingw without `__USE_MINGW_ANSI_STDIO`, MSVC 14.

- This function doesn't support the ' flag correctly for integers (it uses the decimal point instead of the thousands separator!) on some platforms: mingw with `__USE_MINGW_ANSI_STDIO`.
- This function doesn't support the ' flag together with a precision on an integer argument correctly on some platforms: glibc 2.36, FreeBSD 14.0, NetBSD 10.0, Solaris 11, Cygwin 3.6.0, Haiku.
- This function doesn't support the ' flag correctly in locales where the thousands separator is a multibyte character on some platforms: NetBSD 10.0, Solaris 11 OpenIndiana.
- This function does not support the 'a' and 'A' directives on some platforms: FreeBSD 5.2.1, OpenBSD 4.0, AIX 5.2, HP-UX 11, Solaris 11.4, Cygwin 1.5.x, mingw, MSVC 14.
- This function does not support the 'b' directive, required by ISO C23, on some platforms: glibc 2.34, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, HP-UX 11, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android 9.0.
- This function does not support the 'F' directive on some platforms: NetBSD 3.0, AIX 5.1, HP-UX 11.23, Solaris 9, Cygwin 1.5.x, mingw, MSVC 14.
- This function does not support the 'ls' directive on some platforms: OpenBSD 4.0, Cygwin 1.5.x, Haiku.
- This function does not support precisions in the 'ls' directive correctly on some platforms: Solaris 11.4.
- This function doesn't support the ' flag on some platforms: NetBSD 3.0, Cygwin 1.5.24, mingw, MSVC 14.
- This function does not round the argument of the 'a' directive correctly on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4.
- `printf "%010f"` of NaN and Infinity yields an incorrect result (padded with zeroes, or wrong capitalization) on some platforms: Mac OS X 10.5, FreeBSD 6.0, NetBSD 5.0, AIX 7.3, Solaris 11.4, Cygwin 1.5.x, mingw, MSVC/clang.
- `printf "%#.0x"` or `printf "%#.0X"` with a zero argument yields an incorrect result (non-empty) on some platforms: Mac OS X 10.6.
- This function does not support precisions larger than 512 or 1024 in integer, floating-point and pointer output on some platforms: AIX 7.1, Solaris 10/x86, mingw, MSVC/clang.
- This function mishandles large floating point precisions (for example, formatting 1.0 with `("%.511f")`) on some platforms: Solaris 10.
- This function produces wrong output for the 'lc' directive with a NUL wide character argument on some platforms: musl libc 1.2.4.
- This function can crash in out-of-memory conditions on some platforms: FreeBSD 14.0, NetBSD 5.0.
- This function does not truncate the result as specified in C99 on some platforms: mingw, MSVC 14.
- This function does not fully support the 'n' directive on some platforms: HP-UX 11, mingw.

Portability problems fixed by Gnulib module `snprintf-gnu`:

- This function does not support the ‘B’ directive on some platforms: glibc 2.34, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC, Android 9.0.

Portability problems fixed by either Gnulib module `snprintf-posix` or `snprintf-gnu` and additionally the Gnulib module `printf-with-n-directive`:

- This function does not support the ‘n’ directive on some platforms: glibc when used with `_FORTIFY_SOURCE >= 2` (set by default on Ubuntu), macOS 14, OpenBSD, MSVC 14, Android.

Portability problems not fixed by Gnulib:

- The `%m` directive is not portable, use `%s` mapped to an argument of `strerror(errno)` (or a version of `strerror_r`) instead.
- Formatting noncanonical ‘long double’ numbers produces nonmeaningful results on some platforms: glibc and others, on x86, x86_64, IA-64 CPUs.
- When formatting an integer with grouping flag, this function inserts thousands separators even in the “C” locale on some platforms: NetBSD 5.1.

10.45.55 sprintf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/sprintf.html>

Gnulib module: `sprintf-posix` or `sprintf-gnu`

Portability problems fixed by either Gnulib module `sprintf-posix` or `sprintf-gnu`:

- This function does not support size specifiers as in C99 (`hh`, `ll`, `j`, `t`, `z`) on some platforms: AIX 5.1, HP-UX 11.23, Solaris 9, Cygwin 1.5.24, old mingw, MSVC 9.
- This function does not support size specifiers as in C23 (`w8`, `w16`, `w32`, `w64`, `wf8`, `wf16`, `wf32`, `wf64`) on some platforms: glibc 2.37, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC, Android 9.0.
- `printf` of ‘long double’ numbers is unsupported on some platforms: mingw, MSVC 14.
- `printf "%f", "%e", "%g"` of Infinity and NaN yields an incorrect result on some platforms: AIX 7.3, Solaris 11.4, mingw, MSVC 14.
- This function doesn’t support the ‘l’ flag at all on some platforms: mingw without `__USE_MINGW_ANSI_STDIO`, MSVC 14.
- This function doesn’t support the ‘l’ flag correctly for integers (it uses the decimal point instead of the thousands separator!) on some platforms: mingw with `__USE_MINGW_ANSI_STDIO`.
- This function doesn’t support the ‘l’ flag together with a precision on an integer argument correctly on some platforms: glibc 2.36, FreeBSD 14.0, NetBSD 10.0, Solaris 11, Cygwin 3.6.0, Haiku.
- This function doesn’t support the ‘l’ flag correctly in locales where the thousands separator is a multibyte character on some platforms: NetBSD 10.0, Solaris 11 OpenIndiana.

- This function does not support the ‘a’ and ‘A’ directives on some platforms: FreeBSD 5.2.1, OpenBSD 4.0, AIX 5.2, HP-UX 11, Solaris 11.4, Cygwin 1.5.x, mingw, MSVC 14.
- This function does not support the ‘b’ directive, required by ISO C23, on some platforms: glibc 2.34, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, HP-UX 11, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android 9.0.
- This function does not support the ‘F’ directive on some platforms: NetBSD 3.0, AIX 5.1, HP-UX 11.23, Solaris 9, Cygwin 1.5.x, mingw, MSVC 14.
- This function does not support the ‘ls’ directive on some platforms: OpenBSD 4.0, Cygwin 1.5.x, Haiku.
- This function does not support precisions in the ‘ls’ directive correctly on some platforms: Solaris 11.4.
- This function does not support format directives that access arguments in an arbitrary order, such as “%2\$s”, on some platforms: NetBSD 3.0, mingw, MSVC 14.
- This function doesn’t support the ‘ ’ flag on some platforms: NetBSD 3.0, Cygwin 1.5.24, mingw, MSVC 14.
- This function does not round the argument of the ‘a’ directive correctly on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4.
- printf “%010f” of NaN and Infinity yields an incorrect result (padded with zeroes, or wrong capitalization) on some platforms: Mac OS X 10.5, FreeBSD 6.0, NetBSD 5.0, AIX 7.3, Solaris 11.4, Cygwin 1.5.x, mingw, MSVC/clang.
- printf “%#.0x” or “%#.0X” with a zero argument yields an incorrect result (non-empty) on some platforms: Mac OS X 10.6.
- This function does not support precisions larger than 512 or 1024 in integer, floating-point and pointer output on some platforms: AIX 7.1, Solaris 10/x86, mingw, MSVC/clang.
- This function mishandles large floating point precisions (for example, formatting 1.0 with “%.511f”) on some platforms: Solaris 10.
- This function produces wrong output for the ‘lc’ directive with a NUL wide character argument on some platforms: musl libc 1.2.4.
- This function can crash in out-of-memory conditions on some platforms: FreeBSD 14.0, NetBSD 5.0.
- The compiler warns that this function is deprecated: macOS 14.

Portability problems fixed by Gnulib module **sprintf-gnu**:

- This function does not support the ‘B’ directive on some platforms: glibc 2.34, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC, Android 9.0.

Portability problems fixed by either Gnulib module **sprintf-posix** or **sprintf-gnu** and additionally the Gnulib module **printf-with-n-directive**:

- This function does not support the ‘n’ directive on some platforms: glibc when used with `_FORTIFY_SOURCE >= 2` (set by default on Ubuntu), macOS 14, OpenBSD, MSVC 14, Android.

Portability problems not fixed by GnuLib:

- The `%m` directive is not portable, use `%s` mapped to an argument of `strerror(errno)` (or a version of `strerror_r`) instead.
- Formatting noncanonical ‘long double’ numbers produces nonmeaningful results on some platforms: glibc and others, on x86, x86_64, IA-64 CPUs.
- When formatting an integer with grouping flag, this function inserts thousands separators even in the "C" locale on some platforms: NetBSD 5.1.

10.45.56 `sscanf`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/sscanf.html>

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-sscanf.html

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function does not support size specifiers as in C23 (`w8`, `w16`, `w32`, `w64`, `wf8`, `wf16`, `wf32`, `wf64`) on some platforms: glibc 2.38, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.2, AIX 7.2, HP-UX 11, Solaris 11.4, Cygwin 2.9.0, mingw, MSVC 14.
- On Windows platforms (excluding Cygwin), this function does not set `errno` upon failure.
- On Windows, this function doesn’t support the `hh`, `ll`, `j`, `t`, `z` size specifiers.

10.45.57 `stderr`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/stderr.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- `stderr` is created in 32-bit mode instead of 64-bit mode: Cygwin 1.5.x. One workaround is to use `freopen(NULL, "r+", stderr)` on Cygwin 1.5.21 or newer. Another is to use the gnuLib `ftello` module and do `ftello(stderr)`.
- POSIX states that a setuid application can guarantee that fd 2 is open, but some systems guarantee this even for non-setuid programs. If an application is executed with fd 2 closed, use of `stderr` can affect an unrelated file that happened to be assigned to fd 2. The gnuLib `xstdopen` or the *-safer modules may be used to guarantee that fd 2 stays reserved for `stderr`.

10.45.58 `stdin`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/stdin.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- `stdin` is created in 32-bit mode instead of 64-bit mode: Cygwin 1.5.x. One workaround is to use `freopen(NULL, "r", stdin)` on Cygwin 1.5.21 or newer. Another is to use the gnulib `ftello` module and do `ftello(stdin)`.
- POSIX states that a setuid application can guarantee that fd 0 is open, but some systems guarantee this even for non-setuid programs. If an application is executed with fd 0 closed, use of `stdin` can affect an unrelated file that happened to be assigned to fd 0. The gnulib `xstdopen` or the *-safer modules may be used to guarantee that fd 0 stays reserved for `stdin`.

10.45.59 `stdout`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/stdout.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- `stdout` is created in 32-bit mode instead of 64-bit mode: Cygwin 1.5.x. One workaround is to use `freopen(NULL, "w", stdout)` on Cygwin 1.5.21 or newer. Another is to use the gnulib `ftello` module and do `ftello(stdout)`.
- POSIX states that a setuid application can guarantee that fd 1 is open, but some systems guarantee this even for non-setuid programs. If an application is executed with fd 1 closed, use of `stdout` can affect an unrelated file that happened to be assigned to fd 1. The gnulib `xstdopen` or the *-safer modules may be used to guarantee that fd 1 stays reserved for `stdout`.

10.45.60 `tmpfile`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/tmpfile.html>

Gnulib module: `tmpfile`

Portability problems fixed by Gnulib:

- This function always fails on some platforms: Android 4.3.
- This function often fails for trivial reasons on some platforms: mingw, MSVC 14.
- On platforms where `off_t` is a 32-bit type, `tmpfile` may not work correctly to create files 2 GiB and larger. See [Section 14.2 \[Large File Support\]](#), page 842.

Portability problems not fixed by Gnulib:

10.45.61 tmpnam

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/tmpnam.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is not appropriate for creating temporary files. (It has security risks.) Better use `mkstemp` instead.

10.45.62 ungetc

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/ungetc.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- On Windows platforms (excluding Cygwin), this function does not set `errno` upon failure.

10.45.63 vasprintf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/vasprintf.html>

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Variable-Arguments-Output.html,
- `man vasprintf`.

Gnulib module: `vasprintf` or `vasprintf-posix` or `vasprintf-gnu`

Portability problems fixed by either Gnulib module `vasprintf` or `vasprintf-posix` or `vasprintf-gnu`:

- This function is missing on some platforms: AIX 5.1, HP-UX 11, Solaris 10, mingw, MSVC 14.

Portability problems fixed by either Gnulib module `vasprintf-posix` or `vasprintf-gnu`:

- This function does not support size specifiers as in C99 (`hh`, `ll`, `j`, `t`, `z`) on some platforms: Cygwin 1.5.24.
- This function does not support size specifiers as in C23 (`w8`, `w16`, `w32`, `w64`, `wf8`, `wf16`, `wf32`, `wf64`) on some platforms: glibc 2.37, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC, Android 9.0.
- `printf "%f", "%e", "%g"` of Infinity and NaN yields an incorrect result on some platforms: AIX 7.3, Solaris 11.4.

- This function doesn't support the ' flag at all on some platforms: mingw without `__USE_MINGW_ANSI_STDIO`, MSVC 14.
- This function doesn't support the ' flag correctly for integers (it uses the decimal point instead of the thousands separator!) on some platforms: mingw with `__USE_MINGW_ANSI_STDIO`.
- This function doesn't support the ' flag together with a precision on an integer argument correctly on some platforms: glibc 2.36, FreeBSD 14.0, NetBSD 10.0, Solaris 11, Cygwin 3.6.0, Haiku.
- This function doesn't support the ' flag correctly in locales where the thousands separator is a multibyte character on some platforms: NetBSD 10.0, Solaris 11 OpenIndiana.
- This function does not support the 'a' and 'A' directives on some platforms: FreeBSD 5.2.1, OpenBSD 4.0, Solaris 11.4, Cygwin 1.5.x.
- This function does not support the 'b' directive, required by ISO C23, on some platforms: glibc 2.34, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, HP-UX 11, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android 9.0.
- This function does not support the 'F' directive on some platforms: NetBSD 3.0, Cygwin 1.5.x.
- This function does not support the 'ls' directive on some platforms: OpenBSD 4.0, Cygwin 1.5.x, Haiku.
- This function does not support precisions in the 'ls' directive correctly on some platforms: Solaris 11.4.
- This function does not support format directives that access arguments in an arbitrary order, such as "%2\$s", on some platforms: NetBSD 3.0.
- This function doesn't support the ' flag on some platforms: NetBSD 3.0, Cygwin 1.5.24.
- This function does not round the argument of the 'a' directive correctly on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4.
- printf "%010f" of NaN and Infinity yields an incorrect result (padded with zeroes) on some platforms: Mac OS X 10.5, FreeBSD 6.0, NetBSD 5.0, AIX 7.3, Solaris 11.4, Cygwin 1.5.x.
- printf "%#.0x" or "%#.0X" with a zero argument yields an incorrect result (non-empty) on some platforms: Mac OS X 10.6.
- This function produces wrong output for the 'lc' directive with a NUL wide character argument on some platforms: musl libc 1.2.4.
- This function can crash in out-of-memory conditions on some platforms: FreeBSD 14.0, NetBSD 5.0.

Portability problems fixed by Gnulib module `vasprintf-gnu`:

- This function does not support the 'B' directive on some platforms: glibc 2.34, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC, Android 9.0.

Portability problems not fixed by Gnulib:

- The %m directive is not portable, use %s mapped to an argument of `strerror(errno)` (or a version of `strerror_r`) instead.

10.45.64 `vdprintf`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/vdprintf.html>

Gnulib module: `vdprintf` or `vdprintf-posix` or `vdprintf-gnu`

Portability problems fixed by either Gnulib module `vdprintf` or `vdprintf-posix` or `vdprintf-gnu`:

- This function is missing on some platforms: Mac OS X 10.5, FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.3, Cygwin 1.5.x, mingw, MSVC 14.

Portability problems fixed by either Gnulib module `vdprintf-posix` or `vdprintf-gnu`:

- This function does not support size specifiers as in C23 (`w8`, `w16`, `w32`, `w64`, `wf8`, `wf16`, `wf32`, `wf64`) on some platforms: glibc 2.37, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC, Android 9.0.
- `printf "%f", "%e", "%g"` of Infinity and NaN yields an incorrect result on some platforms: AIX 7.3, Solaris 11.4.
- This function doesn't support the `'` flag at all on some platforms: mingw without `__USE_MINGW_ANSI_STDIO`, MSVC 14.
- This function doesn't support the `'` flag correctly for integers (it uses the decimal point instead of the thousands separator!) on some platforms: mingw with `__USE_MINGW_ANSI_STDIO`.
- This function doesn't support the `'` flag together with a precision on an integer argument correctly on some platforms: glibc 2.36, FreeBSD 14.0, NetBSD 10.0, Solaris 11, Cygwin 3.6.0, Haiku.
- This function doesn't support the `'` flag correctly in locales where the thousands separator is a multibyte character on some platforms: NetBSD 10.0, Solaris 11 OpenIndiana.
- This function does not support the `'a'` and `'A'` directives on some platforms: Solaris 11.4.
- This function does not support the `'b'` directive, required by ISO C23, on some platforms: glibc 2.34, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, HP-UX 11, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android 9.0.
- This function does not support precisions in the `'ls'` directive correctly on some platforms: Solaris 11.4.
- This function does not round the argument of the `'a'` directive correctly on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4.
- `printf "%010f"` of NaN and Infinity yields an incorrect result (padded with zeroes, or wrong capitalization) on some platforms: AIX 7.3, Solaris 11.4.
- `printf "%#.0x"` or `"%#.0X"` with a zero argument yields an incorrect result (non-empty) on some platforms: Mac OS X 10.6.
- This function does not support precisions larger than 512 or 1024 in integer, floating-point and pointer output on some platforms: AIX 7.1.
- This function produces wrong output for the `'lc'` directive with a NUL wide character argument on some platforms: musl libc 1.2.4.

Portability problems fixed by Gnulib module `vdprintf-gnu`:

- This function does not support the ‘B’ directive on some platforms: glibc 2.34, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC, Android 9.0.

Portability problems fixed by either Gnulib module `vdprintf-posix` or `vdprintf-gnu` and additionally the Gnulib module `printf-with-n-directive`:

- This function does not support the ‘n’ directive on some platforms: glibc when used with `_FORTIFY_SOURCE >= 2` (set by default on Ubuntu), macOS 14, OpenBSD, Android.

Portability problems not fixed by Gnulib:

- The `%m` directive is not portable, use `%s` mapped to an argument of `strerror(errno)` (or a version of `strerror_r`) instead.
- Formatting noncanonical ‘long double’ numbers produces nonmeaningful results on some platforms: glibc and others, on x86, x86_64, IA-64 CPUs.
- When formatting an integer with grouping flag, this function inserts thousands separators even in the “C” locale on some platforms: NetBSD 5.1.
- On some platforms, this function does not set `errno` or the stream error indicator on attempts to write to a read-only stream: Cygwin 1.7.9.

10.45.65 `vfprintf`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/vfprintf.html>

Gnulib module: `vfprintf-posix` or `vfprintf-gnu` or `stdio-h`, nonblocking, sigpipe

Portability problems fixed by either Gnulib module `stdio-h` or `fprintf-posix` or `fprintf-gnu`:

- This function does not support non-ASCII characters in double-byte encoding, corresponding to the locale, on some platforms: mingw with `__USE_MINGW_ANSI_STDIO` in combination with `msvcrt`, when the output goes to a Windows console.

Portability problems fixed by either Gnulib module `vfprintf-posix` or `vfprintf-gnu`:

- This function does not support size specifiers as in C99 (`hh`, `ll`, `j`, `t`, `z`) on some platforms: AIX 5.1, HP-UX 11.23, Solaris 9, Cygwin 1.5.24, old mingw, MSVC 9.
- This function does not support size specifiers as in C23 (`w8`, `w16`, `w32`, `w64`, `wf8`, `wf16`, `wf32`, `wf64`) on some platforms: glibc 2.37, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC, Android 9.0.
- printf of ‘long double’ numbers is unsupported on some platforms: mingw, MSVC 14.
- printf “%f”, “%e”, “%g” of Infinity and NaN yields an incorrect result on some platforms: AIX 7.3, Solaris 11.4, mingw, MSVC 14.
- This function doesn’t support the ‘l’ flag at all on some platforms: mingw without `__USE_MINGW_ANSI_STDIO`, MSVC 14.
- This function doesn’t support the ‘l’ flag correctly for integers (it uses the decimal point instead of the thousands separator!) on some platforms: mingw with `__USE_MINGW_ANSI_STDIO`.

- This function doesn't support the ' flag together with a precision on an integer argument correctly on some platforms: glibc 2.36, FreeBSD 14.0, NetBSD 10.0, Solaris 11, Cygwin 3.6.0, Haiku.
- This function doesn't support the ' flag correctly in locales where the thousands separator is a multibyte character on some platforms: NetBSD 10.0, Solaris 11 OpenIndiana.
- This function does not support the 'a' and 'A' directives on some platforms: FreeBSD 5.2.1, OpenBSD 4.0, AIX 5.2, HP-UX 11, Solaris 11.4, Cygwin 1.5.x, mingw, MSVC 14.
- This function does not support the 'b' directive, required by ISO C23, on some platforms: glibc 2.34, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, HP-UX 11, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android 9.0.
- This function does not support the 'F' directive on some platforms: NetBSD 3.0, AIX 5.1, HP-UX 11.23, Solaris 9, Cygwin 1.5.x, mingw, MSVC 14.
- This function does not support the 'ls' directive on some platforms: OpenBSD 4.0, Cygwin 1.5.x, Haiku.
- This function does not support precisions in the 'ls' directive correctly on some platforms: Solaris 11.4.
- This function does not support format directives that access arguments in an arbitrary order, such as "%2\$s", on some platforms: NetBSD 3.0, mingw, MSVC 14.
- This function doesn't support the ' flag on some platforms: NetBSD 3.0, Cygwin 1.5.24, mingw, MSVC 14.
- This function does not round the argument of the 'a' directive correctly on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4.
- printf "%010f" of NaN and Infinity yields an incorrect result (padded with zeroes, or wrong capitalization) on some platforms: Mac OS X 10.5, FreeBSD 6.0, NetBSD 5.0, AIX 7.3, Solaris 11.4, Cygwin 1.5.x, mingw, MSVC/clang.
- printf "%#.0x" or "%#.0X" with a zero argument yields an incorrect result (non-empty) on some platforms: Mac OS X 10.6.
- This function does not support precisions larger than 512 or 1024 in integer, floating-point and pointer output on some platforms: AIX 7.1, Solaris 10/x86, mingw, MSVC/clang.
- This function mishandles large floating point precisions (for example, formatting 1.0 with "%.511f") on some platforms: Solaris 10.
- This function produces wrong output for the 'lc' directive with a NUL wide character argument on some platforms: musl libc 1.2.4.
- This function can crash in out-of-memory conditions on some platforms: FreeBSD 14.0, NetBSD 5.0.

Portability problems fixed by Gnulib module **vfprintf-gnu**:

- This function does not support the 'B' directive on some platforms: glibc 2.34, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC, Android 9.0.

Portability problems fixed by either Gnulib module **vfprintf-posix** or **vfprintf-gnu** and additionally the Gnulib module **printf-with-n-directive**:

- This function does not support the ‘n’ directive on some platforms: glibc when used with `_FORTIFY_SOURCE >= 2` (set by default on Ubuntu), macOS 14, OpenBSD, MSVC 14, Android.

Portability problems fixed by Gnulib module `stdio-h` or `vfprintf-posix` or `vfprintf-gnu`, together with module `nonblocking`:

- When writing to a non-blocking pipe whose buffer is full, this function fails with `errno` being set to `ENOSPC` instead of `EAGAIN` on some platforms: mingw, MSVC 14.

Portability problems fixed by Gnulib module `stdio-h` or `vfprintf-posix` or `vfprintf-gnu`, together with module `sigpipe`:

- When writing to a pipe with no readers, this function fails, instead of obeying the current `SIGPIPE` handler, on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- The `%m` directive is not portable, use `%s` mapped to an argument of `strerror(errno)` (or a version of `strerror_r`) instead.
- Formatting noncanonical ‘long double’ numbers produces nonmeaningful results on some platforms: glibc and others, on x86, x86_64, IA-64 CPUs.
- When formatting an integer with grouping flag, this function inserts thousands separators even in the “C” locale on some platforms: NetBSD 5.1.
- Attempting to write to a read-only stream fails with EOF but does not set the error flag for `ferror` on some platforms: glibc 2.13, cygwin 1.7.9.

10.45.66 vfscanf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/vfscanf.html>

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-vfscanf.html

Gnulib module: `vfscanf`, `nonblocking`

Portability problems fixed by Gnulib module `vfscanf`, together with module `nonblocking`:

- When reading from a non-blocking pipe whose buffer is empty, this function fails with `errno` being set to `EINVAL` instead of `EAGAIN` on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8.
- This function does not support size specifiers as in C23 (`w8`, `w16`, `w32`, `w64`, `wf8`, `wf16`, `wf32`, `wf64`) on some platforms: glibc 2.38, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.2, AIX 7.2, HP-UX 11, Solaris 11.4, Cygwin 2.9.0, mingw, MSVC 14.
- On Windows platforms (excluding Cygwin), this function does not set `errno` upon failure.
- On Windows, this function doesn’t support the `hh`, `ll`, `j`, `t`, `z` size specifiers.

10.45.67 `vprintf`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/vprintf.html>

Gnulib module: `vprintf-posix` or `vprintf-gnu` or `stdio-h`, `nonblocking`, `sigpipe`

Portability problems fixed by either Gnulib module `stdio-h` or `fprintf-posix` or `fprintf-gnu`:

- This function does not support non-ASCII characters in double-byte encoding, corresponding to the locale, on some platforms: mingw with `__USE_MINGW_ANSI_STDIO` in combination with `msvcrt`, when the output goes to a Windows console.

Portability problems fixed by either Gnulib module `vprintf-posix` or `vprintf-gnu`:

- This function does not support size specifiers as in C99 (`hh`, `ll`, `j`, `t`, `z`) on some platforms: AIX 5.1, HP-UX 11.23, Solaris 9, Cygwin 1.5.24, old mingw, MSVC 9.
- This function does not support size specifiers as in C23 (`w8`, `w16`, `w32`, `w64`, `wf8`, `wf16`, `wf32`, `wf64`) on some platforms: glibc 2.37, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC, Android 9.0.
- printf of ‘long double’ numbers is unsupported on some platforms: mingw, MSVC 14.
- printf “%f”, “%e”, “%g” of Infinity and NaN yields an incorrect result on some platforms: AIX 7.3, Solaris 11.4, mingw, MSVC 14.
- This function doesn’t support the ‘`'`’ flag at all on some platforms: mingw without `__USE_MINGW_ANSI_STDIO`, MSVC 14.
- This function doesn’t support the ‘`'`’ flag correctly for integers (it uses the decimal point instead of the thousands separator!) on some platforms: mingw with `__USE_MINGW_ANSI_STDIO`.
- This function doesn’t support the ‘`'`’ flag together with a precision on an integer argument correctly on some platforms: glibc 2.36, FreeBSD 14.0, NetBSD 10.0, Solaris 11, Cygwin 3.6.0, Haiku.
- This function doesn’t support the ‘`'`’ flag correctly in locales where the thousands separator is a multibyte character on some platforms: NetBSD 10.0, Solaris 11 OpenIndiana.
- This function does not support the ‘`a`’ and ‘`A`’ directives on some platforms: FreeBSD 5.2.1, OpenBSD 4.0, AIX 5.2, HP-UX 11, Solaris 11.4, Cygwin 1.5.x, mingw, MSVC 14.
- This function does not support the ‘`b`’ directive, required by ISO C23, on some platforms: glibc 2.34, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, HP-UX 11, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android 9.0.
- This function does not support the ‘`F`’ directive on some platforms: NetBSD 3.0, AIX 5.1, HP-UX 11.23, Solaris 9, Cygwin 1.5.x, mingw, MSVC 14.
- This function does not support the ‘`ls`’ directive on some platforms: OpenBSD 4.0, Cygwin 1.5.x, Haiku.
- This function does not support precisions in the ‘`ls`’ directive correctly on some platforms: Solaris 11.4.
- This function does not support format directives that access arguments in an arbitrary order, such as “%2\$s”, on some platforms: NetBSD 3.0, mingw, MSVC 14.

- This function doesn't support the ' flag on some platforms: NetBSD 3.0, Cygwin 1.5.24, mingw, MSVC 14.
- This function does not round the argument of the 'a' directive correctly on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4.
- printf "%010f" of NaN and Infinity yields an incorrect result (padded with zeroes, or wrong capitalization) on some platforms: Mac OS X 10.5, FreeBSD 6.0, NetBSD 5.0, AIX 7.3, Solaris 11.4, Cygwin 1.5.x, mingw, MSVC/clang.
- printf "%#.0x" or "%#.0X" with a zero argument yields an incorrect result (non-empty) on some platforms: Mac OS X 10.6.
- This function does not support precisions larger than 512 or 1024 in integer, floating-point and pointer output on some platforms: AIX 7.1, Solaris 10/x86, mingw, MSVC/clang.
- This function mishandles large floating point precisions (for example, formatting 1.0 with "%.511f") on some platforms: Solaris 10.
- This function produces wrong output for the 'lc' directive with a NUL wide character argument on some platforms: musl libc 1.2.4.
- This function can crash in out-of-memory conditions on some platforms: FreeBSD 14.0, NetBSD 5.0.

Portability problems fixed by Gnulib module `vprintf-gnu`:

- This function does not support the 'B' directive on some platforms: glibc 2.34, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC, Android 9.0.

Portability problems fixed by either Gnulib module `vprintf-posix` or `vprintf-gnu` and additionally the Gnulib module `printf-with-n-directive`:

- This function does not support the 'n' directive on some platforms: glibc when used with `_FORTIFY_SOURCE >= 2` (set by default on Ubuntu), macOS 14, OpenBSD, MSVC 14, Android.

Portability problems fixed by Gnulib module `stdio-h` or `vprintf-posix` or `vprintf-gnu`, together with module `nonblocking`:

- When writing to a non-blocking pipe whose buffer is full, this function fails with `errno` being set to `ENOSPC` instead of `EAGAIN` on some platforms: mingw, MSVC 14.

Portability problems fixed by Gnulib module `stdio-h` or `vprintf-posix` or `vprintf-gnu`, together with module `sigpipe`:

- When writing to a pipe with no readers, this function fails, instead of obeying the current `SIGPIPE` handler, on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- The `%m` directive is not portable, use `%s` mapped to an argument of `strerror(errno)` (or a version of `strerror_r`) instead.
- Formatting noncanonical 'long double' numbers produces nonmeaningful results on some platforms: glibc and others, on x86, x86_64, IA-64 CPUs.
- When formatting an integer with grouping flag, this function inserts thousands separators even in the "C" locale on some platforms: NetBSD 5.1.

- Attempting to write to a read-only stream fails with `EOF` but does not set the error flag for `ferror` on some platforms: glibc 2.13, cygwin 1.7.9.

10.45.68 vscanf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/vscanf.html>

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-vscanf.html

Gnulib module: `vscanf`, `nonblocking`

Portability problems fixed by Gnulib module `vscanf`, together with module `nonblocking`:

- When reading from a non-blocking pipe whose buffer is empty, this function fails with `errno` being set to `EINVAL` instead of `EAGAIN` on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- This function does not support size specifiers as in C23 (`w8`, `w16`, `w32`, `w64`, `wf8`, `wf16`, `wf32`, `wf64`) on some platforms: glibc 2.38, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.2, AIX 7.2, HP-UX 11, Solaris 11.4, Cygwin 2.9.0, mingw, MSVC 14.
- On Windows platforms (excluding Cygwin), this function does not set `errno` upon failure.
- On Windows, these functions don't support the `hh`, `ll`, `j`, `t`, `z` size specifiers.

10.45.69 vsnprintf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/vsnprintf.html>

Gnulib module: `vsnprintf` or `vsnprintf-posix` or `vsnprintf-gnu`

Portability problems fixed by either Gnulib module `vsnprintf` or `vsnprintf-posix` or `vsnprintf-gnu`:

- This function does not support format directives that access arguments in an arbitrary order, such as `"%2$s"`, on some platforms: NetBSD 3.0, mingw, MSVC 14.
- This function does not return a byte count as specified in C99 on some platforms: HP-UX 11, Solaris 9, mingw, MSVC 14.

Portability problems fixed by either Gnulib module `vsnprintf-posix` or `vsnprintf-gnu`:

- This function does not support size specifiers as in C99 (`hh`, `ll`, `j`, `t`, `z`) on some platforms: AIX 5.1, HP-UX 11.23, Solaris 9, Cygwin 1.5.24, old mingw, MSVC 9.
- This function does not support size specifiers as in C23 (`w8`, `w16`, `w32`, `w64`, `wf8`, `wf16`, `wf32`, `wf64`) on some platforms: glibc 2.37, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC, Android 9.0.
- printf of 'long double' numbers is unsupported on some platforms: mingw, MSVC 14.

- `printf "%f", "%e", "%g"` of Infinity and NaN yields an incorrect result on some platforms: AIX 7.3, Solaris 11.4, mingw, MSVC 14.
- This function doesn't support the `'` flag at all on some platforms: mingw without `__USE_MINGW_ANSI_STDIO`, MSVC 14.
- This function doesn't support the `'` flag correctly for integers (it uses the decimal point instead of the thousands separator!) on some platforms: mingw with `__USE_MINGW_ANSI_STDIO`.
- This function doesn't support the `'` flag together with a precision on an integer argument correctly on some platforms: glibc 2.36, FreeBSD 14.0, NetBSD 10.0, Solaris 11, Cygwin 3.6.0, Haiku.
- This function doesn't support the `'` flag correctly in locales where the thousands separator is a multibyte character on some platforms: NetBSD 10.0, Solaris 11 OpenIndiana.
- This function does not support the `'a'` and `'A'` directives on some platforms: FreeBSD 5.2.1, OpenBSD 4.0, AIX 5.2, HP-UX 11, Solaris 11.4, Cygwin 1.5.x, mingw, MSVC 14.
- This function does not support the `'b'` directive, required by ISO C23, on some platforms: glibc 2.34, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, HP-UX 11, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android 9.0.
- This function does not support the `'F'` directive on some platforms: NetBSD 3.0, AIX 5.1, HP-UX 11.23, Solaris 9, Cygwin 1.5.x, mingw, MSVC 14.
- This function does not support the `'ls'` directive on some platforms: OpenBSD 4.0, Cygwin 1.5.x, Haiku.
- This function does not support precisions in the `'ls'` directive correctly on some platforms: Solaris 11.4.
- This function doesn't support the `'` flag on some platforms: NetBSD 3.0, Cygwin 1.5.24, mingw, MSVC 14.
- This function does not round the argument of the `'a'` directive correctly on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4.
- `printf "%010f"` of NaN and Infinity yields an incorrect result (padded with zeroes, or wrong capitalization) on some platforms: Mac OS X 10.5, FreeBSD 6.0, NetBSD 5.0, AIX 7.3, Solaris 11.4, Cygwin 1.5.x, mingw, MSVC/clang.
- `printf "%#.0x" or "%#.0X"` with a zero argument yields an incorrect result (non-empty) on some platforms: Mac OS X 10.6.
- This function does not support precisions larger than 512 or 1024 in integer, floating-point and pointer output on some platforms: AIX 7.1, Solaris 10/x86, mingw, MSVC/clang.
- This function mishandles large floating point precisions (for example, formatting 1.0 with `"%.511f"`) on some platforms: Solaris 10.
- This function produces wrong output for the `'lc'` directive with a NUL wide character argument on some platforms: musl libc 1.2.4.
- This function can crash in out-of-memory conditions on some platforms: FreeBSD 14.0, NetBSD 5.0.
- This function does not truncate the result as specified in C99 on some platforms: mingw, MSVC 14.

- This function overwrites memory even when a zero size argument is passed on some platforms: HP-UX 11.

Portability problems fixed by Gnulib module `vsnprintf-gnu`:

- This function does not support the ‘B’ directive on some platforms: glibc 2.34, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC, Android 9.0.

Portability problems fixed by either Gnulib module `vsnprintf-posix` or `vsnprintf-gnu` and additionally the Gnulib module `printf-with-n-directive`:

- This function does not fully support the ‘n’ directive on some platforms: glibc when used with `_FORTIFY_SOURCE >= 2` (set by default on Ubuntu), macOS 14, OpenBSD, HP-UX 11, mingw, MSVC 14, Android.

Portability problems not fixed by Gnulib:

- The `%m` directive is not portable, use `%s` mapped to an argument of `strerror(errno)` (or a version of `strerror_r`) instead.
- Formatting noncanonical ‘long double’ numbers produces nonmeaningful results on some platforms: glibc and others, on x86, x86-64, IA-64 CPUs.
- When formatting an integer with grouping flag, this function inserts thousands separators even in the “C” locale on some platforms: NetBSD 5.1.

10.45.70 vsprintf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/vsprintf.html>

Gnulib module: `vsprintf-posix` or `vsprintf-gnu`

Portability problems fixed by either Gnulib module `vsprintf-posix` or `vsprintf-gnu`:

- This function does not support size specifiers as in C99 (`hh`, `ll`, `j`, `t`, `z`) on some platforms: AIX 5.1, HP-UX 11.23, Solaris 9, Cygwin 1.5.24, old mingw, MSVC 9.
- This function does not support size specifiers as in C23 (`w8`, `w16`, `w32`, `w64`, `wf8`, `wf16`, `wf32`, `wf64`) on some platforms: glibc 2.37, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC, Android 9.0.
- `printf` of ‘long double’ numbers is unsupported on some platforms: mingw, MSVC 14.
- `printf` “%f”, “%e”, “%g” of Infinity and NaN yields an incorrect result on some platforms: AIX 7.3, Solaris 11.4, mingw, MSVC 14.
- This function doesn’t support the ‘l’ flag at all on some platforms: mingw without `__USE_MINGW_ANSI_STDIO`, MSVC 14.
- This function doesn’t support the ‘l’ flag correctly for integers (it uses the decimal point instead of the thousands separator!) on some platforms: mingw with `__USE_MINGW_ANSI_STDIO`.
- This function doesn’t support the ‘l’ flag together with a precision on an integer argument correctly on some platforms: glibc 2.36, FreeBSD 14.0, NetBSD 10.0, Solaris 11, Cygwin 3.6.0, Haiku.
- This function doesn’t support the ‘l’ flag correctly in locales where the thousands separator is a multibyte character on some platforms: NetBSD 10.0, Solaris 11 OpenIndiana.

- This function does not support the ‘a’ and ‘A’ directives on some platforms: FreeBSD 5.2.1, OpenBSD 4.0, AIX 5.2, HP-UX 11, Solaris 11.4, Cygwin 1.5.x, mingw, MSVC 14.
- This function does not support the ‘b’ directive, required by ISO C23, on some platforms: glibc 2.34, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, HP-UX 11, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android 9.0.
- This function does not support the ‘F’ directive on some platforms: NetBSD 3.0, AIX 5.1, HP-UX 11.23, Solaris 9, Cygwin 1.5.x, mingw, MSVC 14.
- This function does not support the ‘ls’ directive on some platforms: OpenBSD 4.0, Cygwin 1.5.x, Haiku.
- This function does not support precisions in the ‘ls’ directive correctly on some platforms: Solaris 11.4.
- This function does not support format directives that access arguments in an arbitrary order, such as “%2\$s”, on some platforms: NetBSD 3.0, mingw, MSVC 14.
- This function doesn’t support the ‘ ’ flag on some platforms: NetBSD 3.0, Cygwin 1.5.24, mingw, MSVC 14.
- This function does not round the argument of the ‘a’ directive correctly on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4.
- printf “%010f” of NaN and Infinity yields an incorrect result (padded with zeroes, or wrong capitalization) on some platforms: Mac OS X 10.5, FreeBSD 6.0, NetBSD 5.0, AIX 7.3, Solaris 11.4, Cygwin 1.5.x, mingw, MSVC/clang.
- printf “%#.0x” or “%#.0X” with a zero argument yields an incorrect result (non-empty) on some platforms: Mac OS X 10.6.
- This function does not support precisions larger than 512 or 1024 in integer, floating-point and pointer output on some platforms: AIX 7.1, Solaris 10/x86, mingw, MSVC/clang.
- This function mishandles large floating point precisions (for example, formatting 1.0 with “%.511f”) on some platforms: Solaris 10.
- This function produces wrong output for the ‘lc’ directive with a NUL wide character argument on some platforms: musl libc 1.2.4.
- This function can crash in out-of-memory conditions on some platforms: FreeBSD 14.0, NetBSD 5.0.
- The compiler warns that this function is deprecated: macOS 14.

Portability problems fixed by Gnulib module **vsprintf-gnu**:

- This function does not support the ‘B’ directive on some platforms: glibc 2.34, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC, Android 9.0.

Portability problems fixed by either Gnulib module **vsprintf-posix** or **vsprintf-gnu** and additionally the Gnulib module **printf-with-n-directive**:

- This function does not support the ‘n’ directive on some platforms: glibc when used with `_FORTIFY_SOURCE >= 2` (set by default on Ubuntu), macOS 14, OpenBSD, MSVC 14, Android.

Portability problems not fixed by GnuLib:

- The `%m` directive is not portable, use `%s` mapped to an argument of `strerror(errno)` (or a version of `strerror_r`) instead.
- Formatting noncanonical ‘long double’ numbers produces nonmeaningful results on some platforms: glibc and others, on x86, x86_64, IA-64 CPUs.
- When formatting an integer with grouping flag, this function inserts thousands separators even in the "C" locale on some platforms: NetBSD 5.1.

10.45.71 vsscanf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/vsscanf.html>

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-vsscanf.html

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: MSVC 14.
- This function does not support size specifiers as in C23 (`w8`, `w16`, `w32`, `w64`, `wf8`, `wf16`, `wf32`, `wf64`) on some platforms: glibc 2.38, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.2, AIX 7.2, HP-UX 11, Solaris 11.4, Cygwin 2.9.0, mingw, MSVC 14.
- On Windows platforms (excluding Cygwin), this function does not set `errno` upon failure.
- On Windows, these functions don’t support the `hh`, `ll`, `j`, `t`, `z` size specifiers.

10.46 Functions in <stdlib.h>

10.46.1 _Exit

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/_Exit.html

GnuLib module: `_Exit`

Portability problems fixed by GnuLib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.5.x, Android 4.4.

Portability problems not fixed by GnuLib:

10.46.2 a64l

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/a64l.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: FreeBSD 6.0, Minix 3.1.8, mingw, MSVC 14, Android 9.0.
- This function was not correctly implemented in glibc versions before 2.2.5.

10.46.3 abort

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/abort.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- Some platforms mistakenly close all stdio streams prior to raising SIGABRT: Cygwin 1.5.x.
- Some platforms always print a message to stderr, even if a SIGABRT handler uses longjmp to resume execution at a safe point: mingw, MSVC 14.

10.46.4 abs

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/abs.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: Android 4.3.

10.46.5 aligned_alloc

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.24.3.1.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/aligned_alloc.html

Documentation:

`man aligned_alloc`

GnuLib module: `aligned_alloc`

Portability problems fixed by GnuLib:

- This function fails if the alignment argument is smaller than `sizeof (void *)` on some platforms: macOS 14, AIX 7.3.1.
- This function returns a null pointer if the requested size is not a multiple of the alignment: Solaris 11.4.
- This function returns a null pointer if the size argument is zero: macOS 15, AIX 7.3, Solaris 11.4.

- On some platforms, `aligned_alloc` crashes if the requested size is not a multiple of the alignment: AddressSanitizer (gcc 11.2 or clang 13).

Portability problems not fixed by Glibc:

- This function is missing on many older platforms: glibc 2.15, macOS 10.14, FreeBSD 6.4, NetBSD 7.1, OpenBSD 6.0, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.3, Cygwin 1.7.x, mingw, MSVC 14, Android 8.1.
- If the alignment and size are absurdly large, this function crashes: glibc 2.42.

Glibc has partial substitutes for `aligned_alloc` that do not crash even if the AddressSanitizer bug is present:

- The Glibc module `alignalloc` provides a portable function `alignalloc` that is a near-substitute for glibc `aligned_alloc`, except that the result must be freed with `alignfree` rather than plain `free`.
- The Glibc module `aligned-malloc` provides functions for allocating and freeing blocks of suitably aligned memory.
- The Glibc module `pagealign_alloc` provides a similar API for allocating and freeing blocks of memory aligned on a system page boundary.

10.46.6 `at_quick_exit`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.24.4.3.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/at_quick_exit.html

Glibc module: —

Portability problems fixed by Glibc:

Portability problems not fixed by Glibc:

- This function is missing on many platforms: glibc 2.9, macOS 14, FreeBSD 6.4, NetBSD 5.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.1, HP-UX 11.31, Solaris 11.3, Cygwin 1.7.x, mingw, MSVC 14, Android 4.3.

10.46.7 `atexit`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/atexit.html>

Glibc module: `atexit`

Portability problems fixed by Glibc:

- This function is missing on some old platforms.

Portability problems not fixed by Glibc:

- The exit status can be incorrect for a process terminated by a signal during a call to a function registered by `atexit`. This is because if a signal terminates a process after `exit` is called or `main` returns, the process's exit status is that specified by `exit`'s argument or by `main`'s return value, instead of being the exit status corresponding to the signal: z/OS 3.1.

10.46.8 atof

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/atof.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Android 4.4.
- This function mis-parses strings with leading ‘+’ on some old platforms: Old versions of Linux.
- This function returns a positive value for negative underflow on some platforms: glibc 2.4, Mingw, Cygwin.
- This function fails to do a valid parse of ‘-0x’ on some platforms: glibc 2.4, Cygwin < 1.5.25-11.
- This function fails to parse Infinities and plain NaNs on some platforms: Mingw, OpenBSD 4.0.
- This function fails to parse NaN() on some platforms: Mingw, OpenBSD 4.0, Cygwin < 1.5.25-11.
- This function fails to parse NaN(n-char-sequence) on some platforms: Mingw, OpenBSD 4.0.
- This function fails to parse C99 hexadecimal floating point on some platforms: Mingw, OpenBSD 4.0.
- This function fails to correctly parse very long strings on some platforms: Mingw, Cygwin.

10.46.9 atoi

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/atoi.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

10.46.10 atol

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/atol.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

10.46.11 `atoll`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/atoll.html>

Gnulib module: `atoll`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, HP-UX 11.23, MSVC 9.

Portability problems not fixed by Gnulib:

10.46.12 `bsearch`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/bsearch.html>

Gnulib module: `bsearch`

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

10.46.13 `calloc`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/calloc.html>

Gnulib module: `calloc-posix`

Portability problems fixed by Gnulib:

- On some platforms, when the function fails it does not set `errno`: mingw, MSVC 14.
- On some platforms, when the function fails it might set `errno` to `EAGAIN` instead of to `ENOMEM`. Although POSIX allows `EAGAIN`, the `calloc-posix` module insists on `ENOMEM` which also conforms to POSIX and is GNU-compatible: Solaris 11.4.
- On some platforms, `calloc (n, s)` can succeed even if multiplying `n` by `s` would exceed `PTRDIFF_MAX` or `SIZE_MAX`. Although failing to check for exceeding `PTRDIFF_MAX` is arguably allowed by POSIX it can lead to undefined behavior later, so `calloc-posix` does not allow going over the limit.

Extension: Gnulib provides a module ‘`calloc-gnu`’ that substitutes a `calloc` implementation that behaves more like the glibc implementation. It fixes this portability problem:

- `calloc (0, s)` and `calloc (n, 0)` return `NULL` on success on some platforms: AIX 7.3.

10.46.14 `div`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/div.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

10.46.15 drand48

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/drand48.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

10.46.16 erand48

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/erand48.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

10.46.17 exit

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/exit.html>

Gnulib module: stdlib-h

Portability problems fixed by Gnulib:

- Some problems with the macros `EXIT_SUCCESS` and `EXIT_FAILURE`, see [Section 9.56 \[stdlib.h\]](#), page 82.

Portability problems not fixed by Gnulib:

10.46.18 free

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/free.html>

Gnulib module: free-posix

Portability problems fixed by Gnulib:

- This function may clobber `errno` on many platforms: glibc 2.32, Mac OS X, FreeBSD, NetBSD, OpenBSD 4.4, Minix, AIX, HP-UX, Cygwin, mingw, MSVC.

Portability problems not fixed by Gnulib:

10.46.19 free_aligned_sized

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.24.3.5

Gnulib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on most non-C23 platforms: glibc 2.42, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android 9.0.

10.46.20 `free_sized`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.24.3.4

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on most non-C23 platforms: glibc 2.42, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android 9.0.

10.46.21 `getenv`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getenv.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

10.46.22 `getsubopt`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getsubopt.html>

GnuLib module: `getsubopt`

Portability problems fixed by GnuLib:

- This function is missing on some platforms: mingw, MSVC 14, Android 7.1.
- This function is declared in `unistd.h` instead of `stdlib.h` on some platforms: Cygwin 1.7.1.

Portability problems not fixed by GnuLib:

10.46.23 `grantpt`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/grantpt.html>

GnuLib module: `grantpt`

Portability problems fixed by GnuLib:

- This function is missing on some platforms: OpenBSD 3.8, Minix 3.1.8, mingw, MSVC 14, Android 4.4.

Portability problems not fixed by GnuLib:

- This function reports success for invalid file descriptors on some platforms: OpenBSD, Cygwin 1.7.9, musl libc.

10.46.24 `initstate`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/initstate.html>

Gnulib module: `random`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14, Android 4.4.
- This function is not declared on some platforms: Cygwin 1.5.25.

Portability problems not fixed by Gnulib:

- This function crashes if the state buffer is unaligned on some platforms: glibc 2.40/sparc.
- The first parameter is `unsigned long` instead of `unsigned int` on some platforms: MidnightBSD 2.0.
- The third parameter is `long` instead of `size_t` on some platforms: MidnightBSD 2.0.
- This function makes out-of-bounds writes on some platforms: Haiku.

10.46.25 `jrand48`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/jrand48.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

10.46.26 `l64a`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/l64a.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, Minix 3.1.8, mingw, MSVC 14, Android 9.0.
- This function was not correctly implemented in glibc versions before 2.2.5.

10.46.27 `labs`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/labs.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Android 4.3.

10.46.28 lcong48

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/lcong48.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14, Android 5.1.

10.46.29 ldiv

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/ldiv.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

10.46.30 llabs

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/llabs.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, HP-UX 11.23, MSVC 9, Android 4.3.

10.46.31 lldiv

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/lldiv.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: OpenBSD 3.8, Minix 3.1.8, HP-UX 11.23, MSVC 9.

10.46.32 lrand48

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/lrand48.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

10.46.33 malloc

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/malloc.html>

Gnulib module: malloc-posix

Portability problems fixed by Gnulib:

- On some platforms, when the function fails it does not set `errno`: mingw, MSVC 14.
- On some platforms, when the function fails it might set `errno` to `EAGAIN` instead of to `ENOMEM`. Although POSIX allows `EAGAIN`, the `malloc-posix` module insists on `ENOMEM` which also conforms to POSIX and is GNU-compatible: Solaris 11.4.
- On some platforms, `malloc (n)` can succeed even if `n` exceeds `PTRDIFF_MAX`. Although this behavior is arguably allowed by POSIX it can lead to behavior not defined by POSIX later, so `malloc-posix` does not allow going over the limit.

Extension: Gnulib provides a module ‘`malloc-gnu`’ that substitutes a `malloc` implementation that behaves more like the glibc implementation, by fixing this portability problem:

- `malloc (0)` returns `NULL` on success on some platforms: AIX 7.3.

10.46.34 mblen

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/mblen.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Android 7.1.

10.46.35 mbstowcs

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/mbstowcs.html>

Gnulib module: mbstowcs

Portability problems fixed by Gnulib:

- In the C or POSIX locales, this function can return `(size_t) -1` and set `errno` to `EILSEQ`: glibc 2.35.

Portability problems not fixed by Gnulib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters. However, the Gnulib function `mbstoc32s`, provided by Gnulib module `mbstoc32s`, operates on 32-bit wide characters and therefore does not have this limitation.

10.46.36 mbtowc

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/mbtowc.html>

Gnulib module: mbtowc

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Android 4.4.

Portability problems not fixed by Gnulib:

- In UTF-8 locales, this function may return wide characters up to 0x7FFFFFFF (that is, beyond 0x0010FFFF) on some platforms: glibc 2.34.
- This function accumulates hidden state on some platforms: glibc 2.8 (see <https://sourceware.org/PR9674>).
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.46.37 memalignment

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.24.9.1

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on most non-C23 platforms: glibc 2.42, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android 9.0.

To check whether a pointer P has alignment A , Gnulib code typically tests `(uintptr_t) P % A` instead of `memalignment (P) % A`. See [Section 1.5.4 \[Other portability assumptions\]](#), [page 6](#).

10.46.38 mkdtemp

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/mkdtemp.html>

Gnulib module: `mkdtemp`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: AIX 5.1, HP-UX 11, Solaris 10, mingw, MSVC 14.

Portability problems not fixed by Gnulib:

10.46.39 mkostemp

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/mkostemp.html>

Documentation:

`man mkostemp`

Gnulib module: `mkostemp`

Portability problems fixed by Gnulib:

- This function is missing on many non-glibc platforms: glibc 2.6, Mac OS X 10.5, FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.5, mingw, MSVC 14, Android 5.1.

- This function is declared in `<unistd.h>` instead of `<stdlib.h>` on some platforms: macOS 14.
- On platforms where `off_t` is a 32-bit type, `mkostemp` may not work correctly to create files larger than 2 GB. (Cf. `AC_SYS_LARGEFILE`.)

Portability problems not fixed by Gnulib:

The gnulib module `clean-temp` can create temporary files that will not be left behind after signals such as `SIGINT`.

10.46.40 `mkstemp`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/mkstemp.html>

Gnulib module: `mkstemp`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.
- On platforms where `off_t` is a 32-bit type, `mkstemp` may not work correctly to create files 2 GiB and larger. See [Section 14.2 \[Large File Support\]](#), page 842.
- On some older platforms, `mkstemp` can create a world or group writable or readable file, if you haven't set the process `umask` to 077. This is a security risk.

Portability problems not fixed by Gnulib:

The gnulib module `clean-temp` can create temporary files that will not be left behind after signals such as `SIGINT`.

10.46.41 `rand48`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/rand48.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

10.46.42 `rand48`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/rand48.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

10.46.43 `posix_memalign`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/posix_memalign.html

Gnulib module: `posix_memalign`

Although this function is superseded by `aligned_alloc`, it is more portable to older systems that do not support C11. See [Section 10.46.5 \[aligned_alloc\]](#), page 382.

Portability problems fixed by Gnulib:

- This function returns a null pointer if the size argument is zero: AIX 7.3.
- This function produces misaligned results on some platforms: OpenBSD 6.1.

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Mac OS X 10.5, FreeBSD 6.0, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 10, Cygwin 1.5.x, mingw, MSVC 14, Android 4.1.
- If the alignment and size are absurdly large, this function crashes: glibc 2.42.

The Gnulib module `aligned-malloc` provides functions for allocating and freeing blocks of suitably aligned memory.

The Gnulib module `pagealign_alloc` provides a similar API for allocating and freeing blocks of memory aligned on a system page boundary.

10.46.44 `posix_openpt`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/posix_openpt.html

Gnulib module: `posix_openpt`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11.23, Solaris 9, Cygwin 1.5.x, mingw, MSVC 14, Android 4.4. However, the replacement may fail with `ENOSYS` or `ENOENT` on some platforms.

Portability problems not fixed by Gnulib:

Note that when using this function to open the master side of a pseudo-terminal, you still need platform dependent code to open the corresponding slave side. The Gnulib module `openpty` provides an easy-to-use API that does both at once.

10.46.45 `ptsname`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/ptsname.html>

Gnulib module: `ptsname`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: OpenBSD 3.8, Minix 3.1.8, mingw, MSVC 14.

- This function fails to set `errno` on failure on some platforms: FreeBSD 8.2.

Portability problems not fixed by GnuLib:

- On Solaris 11 2010-11, this function fails on all BSD-style `/dev/pty*` device files.
- This function is not thread-safe on some platforms: Cygwin 1.7.9. Likewise, the gnuLib replacement is not thread-safe.

Note that the GnuLib module `ptsname_r` is a version of this function that is more likely to be thread-safe.

10.46.46 `ptsname_r`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/ptsname_r.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Allocation.html,
- `man ptsname_r`.

GnuLib module: `ptsname_r`

Portability problems fixed by GnuLib:

- This function is missing on some platforms: Mac OS X 10.12, FreeBSD 12.0, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.9, mingw, MSVC 14.
- When this function fails, it returns -1 instead of the error code on some platforms: macOS 14, FreeBSD 14.0, Android 4.3.
- When this function fails with `ERANGE`, it stores an empty string as result on some platforms: NetBSD 10.0, Solaris 11 OmniOS.

Portability problems not fixed by GnuLib:

- When this function fails, it returns 0 and stores an empty string as result on some platforms: Cygwin 2.9.

Note: Portable programs should expect to find the error code as the return value of this function, not as the value of `errno`. This is needed for compatibility with musl libc and with the forthcoming POSIX Issue 8.

10.46.47 `putenv`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/putenv.html>, amended through <https://www.austingroupbugs.net/view.php?id=1598>.

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Environment-Access.html.

Note: POSIX `putenv` supports adding or changing the value of an environment variable, while glibc also supports removing an environment variable (as if by `unsetenv`).

GnuLib module: `putenv-gnu`

Portability problems fixed by Gnulib:

- This function does not support removing an environment variable on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, mingw, MSVC 14, Android 11.0.

Portability problems not fixed by Gnulib:

10.46.48 `qsort`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/qsort.html>

Gnulib module: `qsort`

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

10.46.49 `qsort_r`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/qsort_r.html

Documentation:

`man qsort_r`

Gnulib module: `qsort_r`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: glibc 2.7, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android API level 35.
- This function has an incompatible API on some platforms: FreeBSD 13.4.

Portability problems not fixed by Gnulib:

10.46.50 `quick_exit`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.24.4.7.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/quick_exit.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: glibc 2.9, macOS 14, FreeBSD 6.4, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.3, Cygwin 1.7.x, mingw, MSVC 9, Android 4.4.

10.46.51 rand

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/rand.html>

Gnulib module: rand

Portability problems fixed by Gnulib:

- This function crashes when used in multithreaded programs on some platforms: CheriBSD.

Portability problems not fixed by Gnulib:

- This function is only defined as an inline function on some platforms: Android 4.4.
- This function is not multithread-safe on some platforms: musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, AIX 7.1, Solaris 11.4.

10.46.52 rand_r

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/rand_r.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14, Android 4.4.
- This function is removed in POSIX.1-2024. Use the function `random_r` from Gnulib module `random_r` instead.

10.46.53 random

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/random.html>

Gnulib module: random

Portability problems fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.
- This function is only defined as an inline function on some platforms: Android 4.4.
- This function is not multithread-safe on some platforms: macOS 14, FreeBSD 13.2, Solaris 11.4, Cygwin 3.4.6, Haiku.

Portability problems not fixed by Gnulib:

- This function has a slightly incompatible declaration (the return type being ‘`int`’ instead of ‘`long`’) on some platforms: Cygwin 1.5.25, Haiku.
- When ‘`srandom`’ was not called, this function returns a non-deterministic sequence rather than a deterministic sequence of numbers on some platforms: OpenBSD 7.4.

10.46.54 `realloc`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/realloc.html>

Gnulib module: `realloc-posix`

Portability problems fixed by Gnulib:

- On some platforms, when the function fails it does not set `errno`: FreeBSD 15 (for a size > 0x7000000000000000), mingw, MSVC 14.
- On some platforms, when the function fails it might set `errno` to `EAGAIN` instead of to `ENOMEM`. Although POSIX allows `EAGAIN`, the `realloc-posix` module insists on `ENOMEM` which also conforms to POSIX and is GNU-compatible: Solaris 11.4.
- On some platforms, `realloc(p, n)` can succeed even if `n` exceeds `PTRDIFF_MAX`. Although this behavior is arguably allowed by POSIX it is not compatible with GNU and can lead to behavior not defined by POSIX later, so `realloc-posix` does not allow going over the limit.
- It is not portable to call `realloc` with a size of 0.

With a null pointer argument, this is the same ambiguity as `malloc(0)` as to whether a successful call returns a null pointer or a pointer to a new zero-sized memory region. Although both behaviors conform to POSIX, `realloc-posix` insists on the latter behavior, which is compatible with GNU.

With non-null `p`, behavior is a real mess for `realloc(p, 0)`. C23 says behavior is undefined. C89 through C17 say a successful call returns either a null pointer or a pointer to a new zero-sized memory region. POSIX.1-2017 extends C99 by saying that if a successful call returns a null pointer it must also set `errno` to an implementation-defined value, and POSIX.1-2024 extends C17 a bit further by saying that such a call must set `errno` to `EINVAL`. It is not known what POSIX.1-2024's successor will say, though presumably it will extend C23. In practice, successful `realloc(p, 0)` calls have one of the following behaviors:

1. Free `p`, do not change `errno`, and return a null pointer: glibc 2.33–2.41 by default, Android.
2. Free `p`, possibly set `errno`, and return a null pointer: glibc 1–2.1 if specially built with `REALLOC_ZERO_BYTES_FREES=1`, glibc 2.1.1–2.32 by default, mingw, MSVC 14.
3. Free `p`, set `errno` to `EINVAL`, and return a null pointer: AIX 7.3 without `_LINUX_SOURCE_COMPAT`.
4. Free `p` and return a pointer to a new region of size zero: AIX 7.3 with `_LINUX_SOURCE_COMPAT`, glibc 1–2.1 by default, glibc 2.1.1–2.41 if specially built with `REALLOC_ZERO_BYTES_FREES=0`, musl libc, macOS, FreeBSD, NetBSD, OpenBSD, Solaris, Cygwin.

Behaviors (3) and (4) conform to POSIX; behaviors (1) and (2) do not. The `realloc-posix` module insists on behavior (4) as it is more popular than (3) and is more useful in practice. Behavior (4) is the only one of the four in which returning a null pointer means failure, which is what non-expert programmers typically expect.

A program not suspecting these variations in semantics will either:

- Leak memory (the still-valid `p`) if it unwisely does not check for `realloc` failure, when it assumes behavior (1), (2) or (3) but the system implements behavior (4).
- Falsely respond to memory exhaustion (if it wisely checks for `realloc` failure), or have double-free bugs (if it unwisely does not check), when it assumes behavior (4) but the system implements (1), (2) or (3).
- When `realloc (p, 0)` sets `errno` to `EINVAL` and returns null, a POSIX.1-2024 application cannot tell whether the call has succeeded and freed `p`, or has failed without freeing `p`. This is only a theoretical problem, though, as the only practical implementation with this behavior is AIX without `_LINUX_SOURCE_COMPAT`, which behaves this way only when the call succeeds.

Portability problems not fixed by Gnulib:

- When not growing an already-allocated region, i.e., when `p` points to a region of size `psize` and `n <= psize`, `realloc (p, n)` can fail and return a null pointer: glibc 2.42 and probably other platforms.

10.46.55 reallocarray

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/reallocarray.html>

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Changing-Block-Size.html,
- `man reallocarray`.

Gnulib module: `reallocarray`

Portability problems fixed by Gnulib:

- This function is missing on many platforms: glibc 2.25, macOS 14, FreeBSD 10, NetBSD 7, OpenBSD 5.5, Minix 3.3.0, AIX 7.2, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android 9.0.
- On some platforms, when the function fails it might set `errno` to `EAGAIN` instead of to `ENOMEM`. Although POSIX allows `EAGAIN`, the `reallocarray` module insists on `ENOMEM` which also conforms to POSIX and is GNU-compatible: Solaris 11.4.
- On some platforms, `reallocarray (p, n, s)` can succeed even if multiplying `n` by `s` would exceed `PTRDIFF_MAX`, which can lead to undefined behavior later: FreeBSD 13, NetBSD 9, OpenBSD 6, musl 1.2.
- It is not portable to call `reallocarray (p, n, s)` when either `n` or `s` is zero, as `reallocarray` has the same issues with zero sizes that `realloc` does. See [Section 10.46.54 \[realloc\]](#), page 397.

Portability problems not fixed by Gnulib:

- When not growing an already-allocated region, `reallocarray`, like `realloc`, can fail and return a null pointer: glibc 2.42 and probably other platforms.

10.46.56 realpath

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/realpath.html>

Gnulib module: canonicalize-lgpl

Portability problems fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.
- This function does not allow for a NULL ‘resolved’ parameter on some platforms: Mac OS X 10.5, FreeBSD 6.4, OpenBSD 4.4, Solaris 10.
- This function does not always return an absolute path on some platforms: Solaris 10.
- This function fails to detect trailing slashes on non-directories on some platforms: glibc 2.3.5, macOS 14, OpenBSD 6.0.
- This function fails to recognize non-directories or symlinks to non-directories followed by ‘.’ on some platforms: macOS 14, Cygwin 2.9.
- This function misbehaves on consecutive slashes on some platforms: musl libc 1.2.2, AIX 7.

Portability problems not fixed by Gnulib:

- This function does not allow to determine the required size of output buffer; the use of a non-NULL ‘resolved’ buffer is non-portable, since PATH_MAX, if it is defined, is nothing more than a guess.

10.46.57 secure_getenv

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/secure_getenv.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Environment-Access.html,
- `man secure_getenv`.

Gnulib module: secure_getenv

Portability problems fixed by Gnulib:

- This function is missing on many non-glibc platforms: glibc 2.16, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 5.2, Minix 3.2.0, AIX 7.1, HP-UX 11, Solaris 11.3, Cygwin 3.0, mingw, MSVC 14, Android 9.0.

Portability problems not fixed by Gnulib:

10.46.58 seed48

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/seed48.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

10.46.59 `setenv`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/setenv.html>

Gnulib module: `setenv`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: HP-UX 11.23, Solaris 9, mingw, MSVC 14.
- On some platforms, this function does not fail with `EINVAL` when passed an empty string or a string containing '=': Mac OS X 10.5, FreeBSD 6.0, NetBSD 1.6, OpenBSD 7.5, Cygwin 1.5.x.
- On some platforms, this function removes a leading '=' from the value argument: Cygwin 1.5.x.

Portability problems not fixed by Gnulib:

- Older versions of POSIX required that `setenv(NULL, "", 0)` gracefully fail with `EINVAL`, but not all implementations guarantee this, and the requirement was removed.

10.46.60 `setkey`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/setkey.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.34, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.
- This function is not declared in `<stdlib.h>` (without `-D_GNU_SOURCE`) on some platforms: glibc (at least 2.11–2.13).
- POSIX.1-2024 says this function is obsolescent and it is planned to be removed in a future version.

10.46.61 `setstate`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/setstate.html>

Gnulib module: `random`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14, Android 4.4.
- This function is not declared on some platforms: Cygwin 1.5.25.

Portability problems not fixed by Gnulib:

- This function has a slightly incompatible declaration (the argument type being '`const char *`' instead of '`char *`') on some platforms: macOS 14, AIX 7.1, HP-UX 11.31, Solaris 11.4, Solaris 11 OpenIndiana.

10.46.62 srand

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/srand.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is only defined as an inline function on some platforms: Android 4.4.
- This function has no effect on `rand` invocations in other threads on some platforms: Cygwin 3.4.6.

10.46.63 srand48

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/srand48.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

10.46.64 srandom

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/srandom.html>

Gnulib module: random

Portability problems fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.
- This function is only defined as an inline function on some platforms: Android 4.4.

Portability problems not fixed by Gnulib:

- This function has a slightly incompatible declaration (the return type being ‘`long`’ instead of ‘`void`’) on some platforms: Cygwin 1.5.25.
- The parameter is `unsigned long` instead of `unsigned int` on some platforms: MidnightBSD 2.0.
- This function has no effect on some platforms: OpenBSD 7.4. This platform has, instead, a function `srandom_deterministic`.

10.46.65 strfromd

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Printing-of-Floats.html,
- `man strfromd`.

Gnulib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on all non-glibc platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.46.66 `strfromf`

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Printing-of-Floats.html,
- `man strfromf`.

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on all non-glibc platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.46.67 `strfroml`

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Printing-of-Floats.html,
- `man strfroml`.

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on all non-glibc platforms: glibc 2.24, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.46.68 `strtod`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/strtod.html>

GnuLib module: `strtod` or `strtod-obsolete`

Portability problems fixed by either GnuLib module `strtod` or `strtod-obsolete`:

- This function mis-parses strings with leading ‘+’ on some old platforms: Old versions of Linux.
- This function returns a wrong end pointer on some old platforms.
- This function allows whitespace between ‘e’ and the exponent on some platforms: HP-UX 11.11.

- This function returns the wrong end pointer for ‘-0x’ on some platforms: glibc 2.4, Mac OS X 10.5, FreeBSD 6.2, AIX 7.1, Cygwin < 1.5.25-11.
- This function fails to parse Infinities and plain NaNs on some platforms: OpenBSD 4.0, HP-UX 11.11, Solaris 9, mingw, MSVC 14.
- This function fails to parse ‘NaN()’ on some platforms: glibc 2.3.6, Mac OS X 10.5, FreeBSD 6.2, OpenBSD 4.0, AIX 7.1, HP-UX 11.11, Cygwin < 1.5.25-11, mingw, MSVC 14.
- This function fails to parse ‘NaN(*n-char-sequence*)’ on some platforms: OpenBSD 4.0, HP-UX 11.11, mingw, MSVC 14.
- This function parses ‘NaN(*n-char-sequence*)’, but returns the wrong end pointer on some platforms: glibc 2.4, AIX 7.1.
- This function misparses ‘nan(’ on some platforms: macOS 10.6.6.
- This function fails to parse C99 hexadecimal floating point on some platforms: NetBSD 5.0, OpenBSD 4.0, AIX 5.1, HP-UX 11.11, Solaris 11.4, mingw, MSVC 14.
- In hexadecimal floats, this function allows whitespace between ‘p’ and the exponent on some platforms: HP-UX 11.31/ia64.
- This function returns the wrong end pointer for ‘0x1p’ on some platforms: AIX 7.1.
- This function fails to set `errno` upon gradual underflow (resulting in a denormalized number) on some platforms: Cygwin 2.9.

Portability problems fixed by Gnulib module `strtod-obsolete`:

- This function is missing on some old platforms.

Portability problems not fixed by Gnulib:

- This function returns +0.0 (not -0.0) for negative underflow on some platforms: glibc 2.7, Cygwin 1.5.x, mingw, MSVC 14.
- This function cannot distinguish between “nan” and “-nan” on some platforms: glibc 2.7, mingw, MSVC 14.
- This function fails to correctly parse very long strings on some platforms: Mac OS X 10.5, FreeBSD 6.2, NetBSD 5.0, OpenBSD 4.0, Cygwin, mingw, MSVC 14.
- This function fails to set `errno` upon gradual underflow (resulting in a denormalized number) on some platforms: MSVC 14.
- The replacement function does not always return correctly rounded results.
- If no conversion could be performed, this function sets `errno` to `EINVAL` on some platforms. Portable programs should check for no conversion by comparing `str` to `*endptr`, not by inspecting `errno`.

10.46.69 `strtof`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/strtof.html>

Gnulib module: `strtof`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, HP-UX 11, Solaris 9, MSVC 9, Android 4.4.

- This function returns the wrong end pointer for ‘-0x’ on some platforms: glibc 2.4, Mac OS X 10.5, FreeBSD 6.2.
- This function fails to parse ‘NaN()’ on some platforms: glibc 2.5, FreeBSD 6.2.
- This function misparses ‘nan()’ on some platforms: macOS 10.6.6.
- This function fails to set `errno` upon overflow on some platforms: mingw 9.0.
- This function fails to set `errno` upon underflow on some platforms: Cygwin 2.9, mingw 5.0.

Portability problems not fixed by Gnulib:

- This function returns +0.0 (not -0.0) for negative underflow on some platforms: glibc 2.7, mingw, MSVC 14.
- This function cannot distinguish between “nan” and “-nan” on some platforms: glibc 2.7, mingw, MSVC 14.
- This function fails to correctly parse very long strings on some platforms: Mac OS X 10.5, FreeBSD 6.2, NetBSD 5.0, Cygwin, mingw, MSVC 14.
- This function fails to set `errno` upon gradual underflow (resulting in a denormalized number) on some platforms: MSVC 14.
- The replacement function does not always return correctly rounded results.
- If no conversion could be performed, this function sets `errno` to `EINVAL` on some platforms. Portable programs should check for no conversion by comparing `str` to `*endptr`, not by inspecting `errno`.

10.46.70 `strtol`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/strtol.html>

Gnulib module: `strtol`

Portability problems fixed by Gnulib:

- This function is missing on some old platforms.
- This function does not parse the leading ‘0’ when the input string is “0x” and the base is 16 or 0 on some platforms: Minix 3.3, mingw, MSVC 14.
- This function does not parse binary integers (with a ‘0b’ or ‘0B’ prefix) when the base is 2 or 0 on many platforms.

Portability problems not fixed by Gnulib:

- If no conversion could be performed, this function sets `errno` to `EINVAL` on some platforms. Portable programs should check for no conversion by comparing `str` to `*endptr`, not by inspecting `errno`.

10.46.71 `strtold`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/strtold.html>

Gnulib module: `strtold`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, Solaris 9, older Cygwin 1.7.x, MSVC 9, Android 4.4.

- This function returns a struct, not a `long double`, on some platforms: HP-UX 11.31/hppa.
- This function always returns a wrong value on some platforms: mingw 5.0.
- This function allows whitespace between ‘e’ and the exponent on some platforms: HP-UX 11.31/ia64.
- This function returns the wrong end pointer for ‘-0x’ on some platforms: glibc-2.3.2, Mac OS X 10.5, Haiku.
- This function fails to parse Infinities and plain NaNs on some platforms: HP-UX 11.31/ia64.
- This function fails to parse ‘NaN()’ on some platforms: glibc-2.3.2, mingw, Haiku.
- This function parses ‘NaN(*n-char-sequence*)’, but returns the wrong end pointer on some platforms: glibc-2.3.2, mingw, Haiku.
- This function fails to parse C99 hexadecimal floating point on some platforms: mingw.
- In hexadecimal floats, this function allows whitespace between ‘p’ and the exponent on some platforms: HP-UX 11.31/ia64.
- This function fails to set `errno` upon underflow on some platforms: Cygwin 2.9.
- This function maps denormalized numbers to zero on some platforms: Haiku 2024.
- This function leaks memory on mingw 5.0 and allocates an unbounded amount of stack on mingw 9.0.

Portability problems not fixed by GnuLib:

- This function fails to set `errno` upon gradual underflow (resulting in a denormalized number) on some platforms: MSVC 14.
- The replacement function does not always return correctly rounded results.
- If no conversion could be performed, this function sets `errno` to `EINVAL` on some platforms. Portable programs should check for no conversion by comparing `str` to `*endptr`, not by inspecting `errno`.

10.46.72 strtoll

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/strtoll.html>

GnuLib module: strtoll

Portability problems fixed by GnuLib:

- This function is missing on some platforms: HP-UX 11.11, MSVC 9.
- This function does not parse the leading ‘0’ when the input string is “0x” and the base is 16 or 0 on some platforms: Minix 3.3, mingw, MSVC 14.
- This function does not parse binary integers (with a ‘0b’ or ‘0B’ prefix) when the base is 2 or 0 on many platforms.

Portability problems not fixed by GnuLib:

- If no conversion could be performed, this function sets `errno` to `EINVAL` on some platforms. Portable programs should check for no conversion by comparing `str` to `*endptr`, not by inspecting `errno`.

10.46.73 strtoul

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/strtoul.html>

Gnulib module: strtoul

Portability problems fixed by Gnulib:

- This function is missing on some old platforms.
- This function does not parse the leading ‘0’ when the input string is "0x" and the base is 16 or 0 on some platforms: Minix 3.3, mingw, MSVC 14.
- This function does not parse binary integers (with a ‘0b’ or ‘0B’ prefix) when the base is 2 or 0 on many platforms.

Portability problems not fixed by Gnulib:

- If no conversion could be performed, this function sets `errno` to `EINVAL` on some platforms. Portable programs should check for no conversion by comparing `str` to `*endptr`, not by inspecting `errno`.

10.46.74 strtoull

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/strtoull.html>

Gnulib module: strtoull

Portability problems fixed by Gnulib:

- This function is missing on some platforms: HP-UX 11.11, MSVC 9.
- This function does not parse the leading ‘0’ when the input string is "0x" and the base is 16 or 0 on some platforms: Minix 3.3, mingw, MSVC 14.
- This function does not parse binary integers (with a ‘0b’ or ‘0B’ prefix) when the base is 2 or 0 on many platforms.

Portability problems not fixed by Gnulib:

- If no conversion could be performed, this function sets `errno` to `EINVAL` on some platforms. Portable programs should check for no conversion by comparing `str` to `*endptr`, not by inspecting `errno`.

10.46.75 system

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/system.html>

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-system-3.html

Gnulib module: system-posix

Portability problems fixed by Gnulib:

- The macros `WIFSIGNALED`, `WIFEXITED`, `WIFSTOPPED`, `WTERMSIG`, `WEXITSTATUS`, `WNOHANG`, `WUNTRACED`, `WSTOPSIG` are not defined in `<stdlib.h>` (only in `<sys/wait.h>`) on some platforms: MirBSD 10.

Portability problems not fixed by Gnulib:

- On Windows platforms (excluding Cygwin), the command processor used by the `system` function is `cmd.exe`, not `/bin/sh`. Accordingly, the rules for quoting shell arguments containing spaces, quote or other special characters are different.

10.46.76 `unlockpt`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/unlockpt.html>

Gnulib module: `unlockpt`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: OpenBSD 3.8, Minix 3.1.8, mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- This function reports success for invalid file descriptors on some platforms: NetBSD 5.1, Cygwin 1.7.9, Haiku.

10.46.77 `unsetenv`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/unsetenv.html>

Gnulib module: `unsetenv`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: AIX 5.1, HP-UX 11.23, Solaris 9, mingw, MSVC 14.
- This function has the return type `'void'` instead of `'int'` on some platforms: FreeBSD 6.0, NetBSD 1.6, OpenBSD 3.8.
- On some platforms, this function does not fail with `EINVAL` when passed an empty string or a string containing `'='`: FreeBSD 6.0, NetBSD 1.6, OpenBSD 4.7.
- This function removes only the first value association for the given environment variable, not all of them, on some platforms: Solaris 11.4, Haiku.

Portability problems not fixed by Gnulib:

- Older versions of POSIX required that `unsetenv(NULL)` gracefully fail with `EINVAL`, but not all implementations guarantee this, and the requirement was removed.

10.46.78 `wcstombs`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wcstombs.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters. However, the Gnulib function `c32stombs`, provided by Gnulib module `c32stombs`, operates on 32-bit wide characters and therefore does not have this limitation.

10.46.79 wctomb

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wctomb.html>

Gnulib module: wctomb

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Android 4.4.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.47 Functions in <string.h>

10.47.1 memccpy

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/memccpy.html>

Gnulib module: memccpy

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

10.47.2 memchr

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/memchr.html>

Gnulib module: memchr

Portability problems fixed by Gnulib:

- This function dereferences too much memory on some platforms: glibc 2.10 on x86_64, IA-64; glibc 2.11 on Alpha, AIX 7.2.
- This function returns NULL if the character argument is not in the range of an `unsigned char` on some platforms: Android 5.0.

Portability problems not fixed by Gnulib:

10.47.3 memcmp

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/memcmp.html>

Gnulib module: memcmp

Portability problems fixed by Gnulib:

- This function fails when comparing 16 bytes or more and with at least one buffer not starting on a 4-byte boundary on some older platforms: NeXTstep/x86.

Portability problems not fixed by Gnulib:

- This function produces wrong results when the arguments are constant strings and the compiler is clang 6.0.1.

10.47.4 memcpy

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/memcpy.html>

Gnulib module: memcpy

Portability problems fixed by Gnulib:

- This function cannot be called from plain inline or extern inline functions on some platforms: macOS 10.8.

Portability problems not fixed by Gnulib:

10.47.5 memmem

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/memmem.html>

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Search-Functions.html,
- `man memmem`.

Gnulib module: memmem or memmem-simple

Both modules implement the same replacement for the `memmem` function with the `memmem` module providing a replacement on more platforms where the existing `memmem` function has a quadratic worst-case complexity.

Portability problems fixed by either Gnulib module `memmem-simple` or `memmem`:

- This function is missing on some platforms: Mac OS X 10.5, FreeBSD 5.2.1, OpenBSD 4.0, Minix 3.1.8, HP-UX 11, Solaris 10, mingw, MSVC 14.
- This function has reversed arguments on some older platforms: Linux libc 5.0.9
- This function can trigger false positives for long periodic needles on some platforms: glibc 2.12, Cygwin 1.7.7.
- This function returns incorrect values in some cases, such as when given an empty needle: glibc <= 2.0, macOS 14, AIX 7.2, Solaris 11.3, Cygwin 1.5.x.

Performance problems fixed by Gnulib module `memmem`:

- This function has quadratic instead of linear worst-case complexity on some platforms: glibc 2.8, macOS 14, FreeBSD 11.4, NetBSD 8.2, OpenBSD 6.6, AIX 7.2, Solaris 11.4, Cygwin 1.5.x. Note for small needles the replacement may be slower.

Portability problems not fixed by Gnulib:

10.47.6 memmove

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/memmove.html>

Gnulib module: memmove

Portability problems fixed by Gnulib:

- This function cannot be called from plain inline or extern inline functions on some platforms: macOS 10.8.

Portability problems not fixed by Gnulib:

10.47.7 `memset`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/memset.html>

Gnulib module: `memset`

Portability problems fixed by Gnulib:

- This function cannot be called from plain inline or extern inline functions on some platforms: macOS 10.8.

Portability problems not fixed by Gnulib:

10.47.8 `memset_explicit`

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Erasing-Sensitive-Data.html,

Gnulib module: `memset_explicit`

The `memset_explicit` function is an approximation to what is needed, and does not suffice in general to erase information. Although calling `memset_explicit` should clear the memory in question, the information that was in memory may still be available elsewhere on the machine. Proper implementation of information erasure requires support from levels below C code.

Portability problems fixed by Gnulib:

- This function is missing on some platforms: glibc 2.36, FreeBSD 13.1, NetBSD 10.0, OpenBSD 7.2, macOS 13, Solaris 11.4, Android API level 33, and many other systems.

Portability problems not fixed by Gnulib:

- Although the module's implementation should set the memory on platforms compatible with GCC and on platforms using traditional linkers, it may not set the memory on non-GCC platforms that use whole-program optimization.

10.47.9 `stpcpy`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/stpcpy.html>

Gnulib module: `stpcpy`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, HP-UX 11, Solaris 10, Cygwin 1.5.x, mingw, MSVC 14, Android 4.4.
- This function cannot be called from plain inline or extern inline functions on some platforms: macOS 10.8.

Portability problems not fixed by Gnulib:

10.47.10 stpncpy

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/stpncpy.html>

Gnulib module: stpncpy

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Mac OS X 10.5, FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, HP-UX 11, Solaris 10, Cygwin 1.5.x, mingw, MSVC 14, Android 4.4.
- This function has an incompatible return value on some platforms: AIX 5.1.
- This function cannot be called from plain inline or extern inline functions on some platforms: macOS 10.8.
- This function looks at more bytes of the source string than allowed on some platforms: FreeBSD 15.0/x86_64.

Portability problems not fixed by Gnulib:

10.47.11 strcat

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/strcat.html>

Gnulib module: string-h

Portability problems fixed by Gnulib:

- This function cannot be called from plain inline or extern inline functions on some platforms: macOS 10.8.

Portability problems not fixed by Gnulib:

10.47.12 strchr

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/strchr.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function cannot work correctly on character strings in some multibyte locales. Gnulib provides an alternative function `mbschr` that works on character strings in all locales.

10.47.13 strcmp

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/strcmp.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

10.47.14 `strcoll`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/strcoll.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function fails, setting `errno` to `EILSEQ`, on Solaris 10, in UTF-8 locales, when at least one of the strings contains a Unicode character in a block that was not assigned in Unicode 4.0.

10.47.15 `strcoll_l`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/strcoll_l.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 6.0, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.3, Cygwin 1.7.x, mingw, MSVC 14, Android 4.4.

10.47.16 `strcpy`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/strcpy.html>

Gnulib module: `string-h`

Portability problems fixed by Gnulib:

- This function cannot be called from plain inline or extern inline functions on some platforms: macOS 10.8.

Portability problems not fixed by Gnulib:

Note: `strcpy(dst, src)` is only safe to use when you can guarantee that there are at least `strlen(src) + 1` bytes allocated at `dst`.

10.47.17 `strcspn`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/strcspn.html>

Gnulib module: `strcspn`

Portability problems fixed by Gnulib:

- This function is missing on some old platforms.

Portability problems not fixed by Gnulib:

- This function cannot work correctly on character strings in multibyte locales. Gnulib provides an alternative function `mbcspn` that works on character strings in all locales.

10.47.18 strdup

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/strdup.html>

Gnulib module: strdup or strdup-posix

Portability problems fixed by either Gnulib module **strdup** or **strdup-posix**:

- This function has no prototype in `<string.h>` on some old platforms.

Portability problems fixed by Gnulib module **strdup-posix**:

- On some platforms, when the function fails it does not set **errno**: mingw, MSVC 14.
- On some platforms, when the function fails it might set **errno** to **EAGAIN** instead of to **ENOMEM**. Although POSIX allows **EAGAIN**, the **strdup-posix** module insists on **ENOMEM** which also conforms to POSIX and is GNU-compatible: Solaris 11.4.

Portability problems not fixed by Gnulib:

10.47.19 strerror

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/strerror.html>

Gnulib module: strerror

Portability problems fixed by Gnulib:

- This function does not support the error values that are specified by POSIX but not defined by the system, on some platforms: OpenBSD 4.0, NonStop Kernel, Cygwin 1.5.x, mingw, MSVC 14.
- This function reports failure for **strerror(0)** (by setting **errno** or using a string similar to out-of-range values), although POSIX requires this to leave **errno** unchanged and report success, on some platforms: FreeBSD 8.2, NetBSD 10.0, OpenBSD 4.7, macOS 14.
- This function returns an empty string for out-of-range integers on some platforms: HP-UX 11.

Portability problems not fixed by Gnulib:

10.47.20 strerror_l

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/strerror_l.html

Gnulib module: strerror_l

Portability problems fixed by Gnulib:

- This function is missing on some platforms: glibc 2.5, macOS 14, FreeBSD 12.4, NetBSD 6.1, OpenBSD 6.1, Minix 3.1.8, AIX 7.1, HP-UX 11, Solaris 11.3, Cygwin 2.5.x, mingw, MSVC 14, Android 5.1.

Portability problems not fixed by Gnulib:

10.47.21 `strerror_r`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/strerror_r.html

LSB specification:

- https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-strerror-r.html
- https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib---xpg-strerror-r.html

Gnulib module: `strerror_r-posix`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: NetBSD 3.0, Minix 3.1.8, HP-UX 11.23, Solaris 9, mingw, MSVC 14.
- glibc, Cygwin, Android, and AIX+_LINUX_SOURCE_COMPAT have an incompatible version of this function. The POSIX compliant code

```
char *s = (strerror_r (err, buf, buflen) == 0 ? buf : NULL);
```

is essentially equivalent to this code using the glibc function:

```
char *s = strerror_r (err, buf, buflen);
```

- This function clobbers the `strerror` buffer on some platforms: Cygwin 1.7.9.
- This function is sometimes not declared in `<string.h>` on some platforms: glibc 2.8.
- The third argument is of type `int` instead of `size_t` on some platforms: AIX 5.1.
- When this function fails, it returns `-1` and sets `errno`, instead of returning the error number, on some platforms: glibc 2.12 with `-D_POSIX_C_SOURCE=200112L`, AIX 6.1.
- When this function fails, it corrupts `errno`, on some platforms: Solaris 10.
- This function does not support many error values defined in `<errno.h>` on some platforms: MSVC 14.
- This function does not support the error values that are specified by POSIX but not defined by the system, on some platforms: OpenBSD 4.0, NonStop Kernel, Cygwin 1.5.x.
- This function reports failure for `strerror_r(0, buf, len)`, although POSIX requires this to succeed, on some platforms: FreeBSD 8.2.
- This function produces a different string for 0 than `strerror` on some platforms: macOS 14.
- This function always fails when the third argument is less than 80 on some platforms: HP-UX 11.31.
- When the buffer is too small and the value is in range, this function does not fail, but instead truncates the result and returns 0 on some platforms: AIX 6.1, Haiku 2017.
- When the value is not in range or the buffer is too small, this function fails to leave a NUL-terminated string in the buffer on some platforms: glibc 2.13, FreeBSD 8.2, Solaris 10.
- When the value is out of range but the buffer is too small, this function does not always return the longest possible string on some platforms: OpenBSD 4.7.

Portability problems not fixed by GnuLib:

Note: GnuLib has a module `xstrerror`, with the property that `xstrerror (NULL, errnum)` returns the value of `strerror_r` as a freshly allocated string. (Recall that the expression `strerror (errnum)` is not multithread-safe.)

10.47.22 `strlcat`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/strlcat.html>

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Truncating-Strings.html.

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: glibc 2.37, AIX 7.3.1, HP-UX 11.31, mingw, MSVC 14.

10.47.23 `strncpy`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/strncpy.html>

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Truncating-Strings.html.

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: glibc 2.37, AIX 7.3.1, HP-UX 11.31, mingw, MSVC 14.

10.47.24 `strlen`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/strlen.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

10.47.25 `strncat`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/strncat.html>

GnuLib module: `strncat`

Portability problems fixed by GnuLib:

- This function dereferences too much memory on some platforms: Solaris 11.4 on SPARC, Solaris 11.0 on x86.

- This function cannot be called from plain inline or extern inline functions on some platforms: macOS 10.8.

Portability problems not fixed by GnuLib:

10.47.26 `strncmp`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/strncmp.html>

GnuLib module: `strncmp`

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

10.47.27 `strncpy`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/strncpy.html>

GnuLib module: `string-h` or `strncpy`

Portability problems fixed by GnuLib module `string-h` or GnuLib module `strncpy`:

- This function cannot be called from plain inline or extern inline functions on some platforms: macOS 10.8.

Portability problems fixed by GnuLib module `strncpy`:

- This function looks at more bytes of the source string than allowed on some platforms: FreeBSD 15.0/x86_64.

Portability problems not fixed by GnuLib:

Note: This function was designed for the use-case of filling a fixed-size record with a string, before writing it to a file. This function is **not** appropriate for copying a string into a bounded memory area, because you have no guarantee that the result will be NUL-terminated. Even if you add the NUL byte at the end yourself, this function is inefficient (as it spends time clearing unused memory) and will allow silent truncation to occur, which is not a good behavior for GNU programs. For more details, see <https://meyering.net/crusade-to-eliminate-strncpy/>.

10.47.28 `strndup`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/strndup.html>

GnuLib module: `strndup`

Portability problems fixed by GnuLib:

- This function is missing on some platforms: Mac OS X 10.5, FreeBSD 6.0, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, HP-UX 11, Solaris 10, mingw, MSVC 14.
- This function does not NUL-terminate the result on some platforms: AIX 5.1.
- This function does not support zero-length operations on NULL pointers on some platforms: AIX 7.3.

Portability problems not fixed by GnuLib:

10.47.29 `strnlen`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/strnlen.html>

Gnulib module: `strnlen`

Portability problems fixed by Gnulib:

- On some platforms, calls like `strnlen(s, maxlen)` can crash if `s` is null-terminated but address arithmetic overflows (i.e., `s + maxlen < s`): Android 5.0.
- This function is missing on some platforms: Mac OS X 10.5, FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, HP-UX 11, Solaris 10, mingw.

Portability problems not fixed by Gnulib:

10.47.30 `strpbrk`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/strpbrk.html>

Gnulib module: `strpbrk`

Portability problems fixed by Gnulib:

- This function is missing on some old platforms.

Portability problems not fixed by Gnulib:

- This function cannot work correctly on character strings in multibyte locales. Gnulib provides an alternative function `mbsspbrk` that works on character strings in all locales.

10.47.31 `strrchr`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/strrchr.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function cannot work correctly on character strings in some multibyte locales. Gnulib provides an alternative function `mbsrchr` that works on character strings in all locales.

10.47.32 `strsignal`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/strsignal.html>

Gnulib module: `strsignal`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, HP-UX 11, mingw, MSVC 14.
- This function does not return a string for out-of-range numbers on some platforms: Solaris, AIX 5.1.
- This function is declared in `unistd.h` instead of `string.h` on some platforms: NetBSD 5.0.

Portability problems not fixed by Gnulib:

- This function returns `const char *` instead of `char *` on some platforms: cygwin 1.5.25, Haiku.

10.47.33 `strspn`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/strspn.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function cannot work correctly on character strings in multibyte locales. Gnulib provides an alternative function `mbsspn` that works on character strings in all locales.

10.47.34 `strstr`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/strstr.html>

Gnulib module: `strstr` or `strstr-simple`

Portability problems fixed by either Gnulib module `strstr-simple` or `strstr`:

- This function can trigger `memchr` bugs on some platforms: glibc 2.10.
- This function can trigger false positives for long periodic needles on some platforms: glibc 2.12, Cygwin 1.7.7.
- This function may fail to find matches on some platforms: glibc 2.28.

Portability problems fixed by Gnulib module `strstr`:

- This function has quadratic instead of linear worst-case complexity on some platforms: glibc 2.8, macOS 14, FreeBSD 11.4, NetBSD 10.0, OpenBSD 6.1, AIX 7.2, HP-UX 11, Solaris 11.4, Cygwin 1.5.x, mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- This function cannot work correctly on character strings in most multibyte locales. Gnulib provides an alternative function `mbsstr` that works on character strings in all locales.

10.47.35 `strtok`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/strtok.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

10.47.36 strtok_r

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/strtok_r.html

Gnulib module: strtok_r

Portability problems fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.
- This function crashes when invoked from code compiled with optimization enabled on some platforms: glibc 2.7.

Portability problems not fixed by Gnulib:

- This function cannot work correctly on character strings in multibyte locales. Gnulib provides an alternative function `mbstok_r` that works on character strings in all locales.

10.47.37 strxfrm

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/strxfrm.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function has a wrong return value on some platforms: Haiku.

10.47.38 strxfrm_l

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/strxfrm_l.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 6.0, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.3, Cygwin 1.7.x, mingw, MSVC 14, Android 4.4.

10.48 Functions in <strings.h>

10.48.1 ffs

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/ffs.html>

Gnulib module: ffs

Portability problems fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- This function is only defined as an inline function on some platforms: Android 13.

10.48.2 ffs1

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/ffs1.html>

Documentation:

`man ffs1`

Gnulib module: ffs1

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 10, Cygwin 1.7.x, mingw, MSVC 14, Android 9.0.
- This function is declared in `<strings.h>` instead of `<string.h>` on some platforms: AIX 7.2, Android 13.

Portability problems not fixed by Gnulib:

- This function is only defined as an inline function on some platforms: Android 13.

10.48.3 ffs11

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/ffs11.html>

Documentation:

`man ffs11`

Gnulib module: ffs11

Portability problems fixed by Gnulib:

- This function is missing on many non-glibc platforms: Mac OS X 10.5, FreeBSD 6.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 10, Cygwin 1.7.x, mingw, MSVC 14, Android 9.0.
- This function is declared in `<strings.h>` instead of `<string.h>` on some platforms: AIX 7.2, Android 13.
- This function returns completely wrong values on some platforms: AIX 7.2 in 32-bit mode.

Portability problems not fixed by Gnulib:

- This function is only defined as an inline function on some platforms: Android 13.

10.48.4 strcasecmp

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/strcasecmp.html>

Gnulib module: strcasecmp

Portability problems fixed by Gnulib:

- This function is missing on some platforms: MSVC 14.
- This function uses the case mappings of a wrong locale on some platforms: Solaris 11.4, Cygwin 3.5.6.

Portability problems not fixed by Gnuilib:

- This function cannot work correctly on character strings in multibyte locales. Gnuilib provides an alternative function `mbscasecmp` that does a case insensitive comparison of character strings and that works in all locales.

10.48.5 `strcasecmp_l`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/strcasecmp_l.html

Gnuilib module: `strcasecmp_l`

Portability problems fixed by Gnuilib:

- This function is missing on many platforms: FreeBSD 9.0, NetBSD 10.0, OpenBSD 6.1, Minix 3.3.0, AIX 6.1, HP-UX 11, Solaris 11.3, Cygwin 2.5.x, mingw, MSVC 14, Android 5.1.
- This function is declared in `<string.h>` instead of `<strings.h>` on some platforms: glibc 2.9, macOS 15.
- This function uses the case mappings of a wrong locale on some platforms: Solaris 11.4, Cygwin 3.5.6.

Portability problems not fixed by Gnuilib:

10.48.6 `strncasecmp`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/strncasecmp.html>

Gnuilib module: `strncasecmp`

Portability problems fixed by Gnuilib:

- This function is missing on some platforms: MSVC 14.
- This function uses the case mappings of a wrong locale on some platforms: Solaris 11.4, Cygwin 3.5.6.

Portability problems not fixed by Gnuilib:

- This function cannot work correctly on character strings in multibyte locales. Gnuilib provides alternative functions `mbsnccasecmp` and `mbspccasecmp` that do a case insensitive comparison of character strings and that work in all locales.

10.48.7 `strncasecmp_l`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/strncasecmp_l.html

Gnuilib module: `strncasecmp_l`

Portability problems fixed by Gnuilib:

- This function is missing on many platforms: FreeBSD 9.0, NetBSD 10.0, OpenBSD 6.1, Minix 3.3.0, AIX 6.1, HP-UX 11, Solaris 11.3, Cygwin 2.5.x, mingw, MSVC 14, Android 5.1.

- This function is declared in `<string.h>` instead of `<strings.h>` on some platforms: glibc 2.9, macOS 15.
- This function uses the case mappings of a wrong locale on some platforms: Solaris 11.4, Cygwin 3.5.6.

Portability problems not fixed by GnuLib:

10.49 Functions in `<sys/ipc.h>`

10.49.1 `ftok`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/ftok.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: mingw, MSVC 14.

10.50 Functions in `<sys/mman.h>`

10.50.1 `mlock`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/mlock.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

10.50.2 `mlockall`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/mlockall.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 4.1.

10.50.3 `mmap`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/mmap.html>

GnuLib module: —

Portability problems fixed by GnuLib:

- On platforms where `off_t` is a 32-bit type, this function may not work correctly across the entire data range of files 2 GiB and larger. See [Section 14.2 \[Large File Support\]](#), page 842.

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: mingw, MSVC 14.
- To get anonymous memory, on some platforms, you can use the flags `MAP_ANONYMOUS` | `MAP_PRIVATE` and `-1` instead of a file descriptor; on others you have to use a read-only file descriptor of `/dev/zero`.
- On HP-UX, passing a non-NULL first argument, as a hint for the address (even without `MAP_FIXED`), often causes `mmap` to fail. Better pass NULL in this case.
- On HP-UX, `MAP_FIXED` basically never works. On other platforms, it depends on the circumstances whether memory can be returned at a given address.

10.50.4 `mprotect`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/mprotect.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.
- This function does not set `errno` on some platforms: mingw.
- On AIX, it is not possible to use `mprotect` on memory regions allocated with `malloc`.

10.50.5 `msync`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/msync.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.
- On NetBSD, `msync` takes only two arguments.

10.50.6 `munlock`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/munlock.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

10.50.7 munlockall

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/munlockall.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 4.1.

10.50.8 munmap

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/munmap.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

10.50.9 posix_madvise

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/posix_madvise.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 10, Cygwin 1.5.x, mingw, MSVC 14, Android 5.1.

10.50.10 posix_mem_offset

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/posix_mem_offset.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.50.11 `posix_typed_mem_get_info`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/posix_typed_mem_get_info.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.50.12 `posix_typed_mem_open`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/posix_typed_mem_open.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.50.13 `shm_open`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/shm_open.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, Cygwin 1.5.x, mingw, MSVC 14, Android 9.0.

10.50.14 `shm_unlink`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/shm_unlink.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, Cygwin 1.5.x, mingw, MSVC 14, Android 9.0.

10.51 Functions in <sys/msg.h>

10.51.1 msgctl

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/msgctl.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14, Android 7.1.

10.51.2 msgget

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/msgget.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14, Android 7.1.

10.51.3 msgrcv

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/msgrcv.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14, Android 7.1.

10.51.4 msgsnd

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/msgsnd.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14, Android 7.1.

10.52 Functions in <sys/resource.h>

10.52.1 getpriority

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getpriority.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

10.52.2 getrlimit

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getrlimit.html>

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-getrlimit-1.html

Gnulib module: —

Portability problems fixed by Gnulib:

- On platforms where `rlim_t` is a 32-bit type, this function does not allow to retrieve limits larger than 4 GiB and larger, such as for `RLIMIT_FSIZE`. See [Section 14.2 \[Large File Support\]](#), page 842.

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

10.52.3 getrusage

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getrusage.html>

Gnulib module: `getrusage`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- Many platforms don't fill in all the fields of `struct rusage` with meaningful values.

10.52.4 setpriority

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/setpriority.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

10.52.5 setrlimit

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/setrlimit.html>

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-getrlimit-1.html

Gnulib module: —

Portability problems fixed by Gnulib:

- On platforms where `rlim_t` is a 32-bit type, this function does not allow to set limits 4 GiB and larger, such as for `RLIMIT_FSIZE`. See [Section 14.2 \[Large File Support\]](#), [page 842](#).

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

10.53 Functions in `<sys/select.h>`

10.53.1 FD_CLR

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/FD_CLR.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

10.53.2 FD_ISSET

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/FD_ISSET.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

10.53.3 FD_SET

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/FD_SET.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

10.53.4 FD_ZERO

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/FD_ZERO.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

10.53.5 pselect

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/pselect.html>

Gnulib module: pselect

Portability problems fixed by Gnulib:

- This function is missing on some platforms: OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11.23, Solaris 9, mingw, MSVC 14.
- On some platforms, this function fails to detect invalid fds with EBADF, but only if they lie beyond the current maximum open fd: FreeBSD 8.2.

Portability problems not fixed by Gnulib:

- This function has a slightly incompatible declaration on some platforms: AIX 7.1 (when `_ALL_SOURCE` is defined).
- When the *sigmask* argument is nonnull on platforms that do not natively support this function, race conditions are possible when its gnulib implementation temporarily modifies the signal mask, and the behavior is unspecified in a multi-threaded process.

10.53.6 select

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/select.html>

Gnulib module: select

Portability problems fixed by Gnulib:

- This function is declared in `<winsock2.h>` instead of `<sys/select.h>` on some platforms: mingw, MSVC 14.
- On Windows platforms (excluding Cygwin), `select` can only be called on descriptors created by the `socket` function, not on regular file descriptors.
- On Windows platforms (excluding Cygwin), error codes from this function are not placed in `errno`, and `WSAGetLastError` must be used instead.
- On some platforms, this function fails to detect invalid fds with EBADF, but only if they lie beyond the current maximum open fd: FreeBSD 8.2.

Portability problems not fixed by Gnulib:

- When you call `select` with a timeout, some implementations modify the timeout parameter so that upon return from the function, it contains the amount of time not slept. Other implementations leave the timeout parameter unmodified.
- Under Windows, when passing a pipe, Gnulib's `select` replacement might return 0 even before the timeout has passed. Programs using it with pipes can thus busy wait.

- On Linux, when some file descriptor refers to a regular file, `select` may fail, setting `errno` to `EBADF`.
- On Haiku, the maximum allowed value for the first argument is not `FD_SETSIZE` but `FD_SETSIZE / 2`.

10.54 Functions in `<sys/sem.h>`

10.54.1 `semctl`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/semctl.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14, Android 7.1.

10.54.2 `semget`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/semget.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14, Android 7.1.

10.54.3 `semop`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/semop.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14, Android 7.1.

10.55 Functions in `<sys/shm.h>`

10.55.1 `shmat`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/shmat.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14, Android 7.1.
- On Linux, the flag `SHM_REMAP` is needed in order to force `shmat` to replace existing memory mappings in the specify address range. On other platforms, it is not needed.

10.55.2 shmctl

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/shmctl.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14, Android 7.1.

10.55.3 shmdt

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/shmdt.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14, Android 7.1.

10.55.4 shmget

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/shmget.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14, Android 7.1.
- On many platforms (not Linux), SHMMAX is so small that it is unusable for reasonable applications, and/or `shmget` requires superuser privileges.

10.56 Functions in <sys/socket.h>

10.56.1 accept

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/accept.html>

Gnulib module: `accept`

Portability problems fixed by Gnulib:

- This function is declared in `<winsock2.h>` instead of `<sys/socket.h>` on some platforms: mingw, MSVC 14.
- On Windows platforms (excluding Cygwin), the descriptors returned by the `accept` function cannot be used in calls to `read`, `write`, and `close`; you have to use `recv`, `send`, `closesocket` in these cases instead.
- On Windows platforms (excluding Cygwin), error codes from this function are not placed in `errno`, and `WSAGetLastError` must be used instead.

- On HP-UX 11, in 64-bit mode, when the macro `_HPUX_ALT_XOPEN_SOCKET_API` is not defined, this function behaves incorrectly because it is declared to take a pointer to a 64-bit wide `socklen_t` entity but in fact considers it as a pointer to a 32-bit wide `unsigned int` entity.

Portability problems not fixed by GnuLib:

- Some platforms don't have a `socklen_t` type; in this case this function's third argument type is `'int *'`.
- On some platforms, this function's third argument type is `'void *'`, not `'socklen_t *'`: Solaris 10, Solaris 11 OmniOS.

10.56.2 accept4

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/accept4.html>

Documentation:

`man accept4`

GnuLib module: `accept4`

Portability problems fixed by GnuLib:

- This function is missing on many non-glibc platforms: glibc 2.9, macOS 14, FreeBSD 6.0, NetBSD 7.1, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.3, Cygwin 1.7.1, mingw, MSVC 14, Android 4.4. But the replacement function is not atomic; this matters in multi-threaded programs that spawn child processes.

Portability problems not fixed by GnuLib:

- `SOCK_CLOEXEC` and `SOCK_NONBLOCK` may not be defined as they're also significant to the `socket()` function.

10.56.3 bind

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/bind.html>

GnuLib module: `bind`

Portability problems fixed by GnuLib:

- This function is declared in `<winsock2.h>` instead of `<sys/socket.h>` on some platforms: mingw, MSVC 14.
- On Windows platforms (excluding Cygwin), error codes from this function are not placed in `errno`, and `WSAGetLastError` must be used instead.

Portability problems not fixed by GnuLib:

10.56.4 connect

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/connect.html>

GnuLib module: `connect`

Portability problems fixed by GnuLib:

- This function is declared in `<winsock2.h>` instead of `<sys/socket.h>` on some platforms: mingw, MSVC 14.

- On Windows platforms (excluding Cygwin), error codes from this function are not placed in `errno`, and `WSAGetLastError` must be used instead.

Portability problems not fixed by Gnulib:

10.56.5 `getpeername`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getpeername.html>

Gnulib module: `getpeername`

Portability problems fixed by Gnulib:

- This function is declared in `<winsock2.h>` instead of `<sys/socket.h>` on some platforms: mingw, MSVC 14.
- On Windows platforms (excluding Cygwin), error codes from this function are not placed in `errno`, and `WSAGetLastError` must be used instead.
- On HP-UX 11, in 64-bit mode, when the macro `_HPUX_ALT_XOPEN_SOCKET_API` is not defined, this function behaves incorrectly because it is declared to take a pointer to a 64-bit wide `socklen_t` entity but in fact considers it as a pointer to a 32-bit wide `unsigned int` entity.

Portability problems not fixed by Gnulib:

- Some platforms don't have a `socklen_t` type; in this case this function's third argument type is `'int *'`.
- On some platforms, this function's third argument type is `'void *'`, not `'socklen_t *'`: Solaris 11 OmniOS.

10.56.6 `getsockname`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getsockname.html>

Gnulib module: `getsockname`

Portability problems fixed by Gnulib:

- This function is declared in `<winsock2.h>` instead of `<sys/socket.h>` on some platforms: mingw, MSVC 14.
- On Windows platforms (excluding Cygwin), error codes from this function are not placed in `errno`, and `WSAGetLastError` must be used instead.
- On HP-UX 11, in 64-bit mode, when the macro `_HPUX_ALT_XOPEN_SOCKET_API` is not defined, this function behaves incorrectly because it is declared to take a pointer to a 64-bit wide `socklen_t` entity but in fact considers it as a pointer to a 32-bit wide `unsigned int` entity.

Portability problems not fixed by Gnulib:

- Some platforms don't have a `socklen_t` type; in this case this function's third argument type is `'int *'`.
- On some platforms, this function's third argument type is `'void *'`, not `'socklen_t *'`: Solaris 11 OmniOS.

10.56.7 getsockopt

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getsockopt.html>

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-getsockopt-1.html

Gnulib module: getsockopt

Portability problems fixed by Gnulib:

- This function is declared in `<winsock2.h>` instead of `<sys/socket.h>` on some platforms: mingw, MSVC 14.
- On Windows platforms (excluding Cygwin), error codes from this function are not placed in `errno`, and `WSAGetLastError` must be used instead.
- On HP-UX 11, in 64-bit mode, when the macro `_HPUX_ALT_XOPEN_SOCKET_API` is not defined, this function behaves incorrectly because it is declared to take a pointer to a 64-bit wide `socklen_t` entity but in fact considers it as a pointer to a 32-bit wide `unsigned int` entity.

Portability problems not fixed by Gnulib:

- Some platforms don't have a `socklen_t` type; in this case this function's fifth argument type is `'int *'`.
- On some platforms, this function's fifth argument type is `'void *'`, not `'socklen_t *'`: Solaris 11 OmniOS.
- Many socket options are not available on all platforms.

10.56.8 listen

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/listen.html>

Gnulib module: listen

Portability problems fixed by Gnulib:

- This function is declared in `<winsock2.h>` instead of `<sys/socket.h>` on some platforms: mingw, MSVC 14.
- On Windows platforms (excluding Cygwin), error codes from this function are not placed in `errno`, and `WSAGetLastError` must be used instead.

Portability problems not fixed by Gnulib:

10.56.9 recv

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/recv.html>

Gnulib module: recv

Portability problems fixed by Gnulib:

- This function is declared in `<winsock2.h>` instead of `<sys/socket.h>` on some platforms: mingw, MSVC 14.

- On Windows platforms (excluding Cygwin), error codes from this function are not placed in `errno`, and `WSAGetLastError` must be used instead.

Portability problems not fixed by GnuLib:

10.56.10 `recvfrom`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/recvfrom.html>

GnuLib module: `recvfrom`

Portability problems fixed by GnuLib:

- This function is declared in `<winsock2.h>` instead of `<sys/socket.h>` on some platforms: mingw, MSVC 14.
- On Windows platforms (excluding Cygwin), error codes from this function are not placed in `errno`, and `WSAGetLastError` must be used instead.
- On HP-UX 11, in 64-bit mode, when the macro `_HPUX_ALT_XOPEN_SOCKET_API` is not defined, this function behaves incorrectly because it is declared to take a pointer to a 64-bit wide `socklen_t` entity but in fact considers it as a pointer to a 32-bit wide `unsigned int` entity.

Portability problems not fixed by GnuLib:

- Some platforms don't have a `socklen_t` type; in this case this function's sixth argument type is `'int *'`.
- On some platforms, this function's sixth argument type is `'void *'`, not `'socklen_t *'`: Solaris 11 OmniOS.

10.56.11 `recvmsg`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/recvmsg.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: mingw, MSVC 14.

10.56.12 `send`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/send.html>

GnuLib module: `send`

Portability problems fixed by GnuLib:

- This function is declared in `<winsock2.h>` instead of `<sys/socket.h>` on some platforms: mingw, MSVC 14.
- On Windows platforms (excluding Cygwin), error codes from this function are not placed in `errno`, and `WSAGetLastError` must be used instead.

Portability problems not fixed by GnuLib:

10.56.13 sendmsg

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/sendmsg.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

10.56.14 sendto

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/sendto.html>

Gnulib module: sendto

Portability problems fixed by Gnulib:

- This function is declared in `<winsock2.h>` instead of `<sys/socket.h>` on some platforms: mingw, MSVC 14.
- On Windows platforms (excluding Cygwin), error codes from this function are not placed in `errno`, and `WSAGetLastError` must be used instead.

Portability problems not fixed by Gnulib:

10.56.15 setsockopt

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/setsockopt.html>

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-setsockopt-2.html

Gnulib module: setsockopt

Portability problems fixed by Gnulib:

- This function is declared in `<winsock2.h>` instead of `<sys/socket.h>` on some platforms: mingw, MSVC 14.
- On Windows platforms (excluding Cygwin), error codes from this function are not placed in `errno`, and `WSAGetLastError` must be used instead.

Portability problems not fixed by Gnulib:

- Many socket options are not available on all platforms.

10.56.16 shutdown

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/shutdown.html>

Gnulib module: shutdown

Portability problems fixed by Gnulib:

- This function is declared in `<winsock2.h>` instead of `<sys/socket.h>` on some platforms: mingw, MSVC 14.

- On Windows platforms (excluding Cygwin), error codes from this function are not placed in `errno`, and `WSAGetLastError` must be used instead.

Portability problems not fixed by Gnulib:

10.56.17 `socketatmark`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/socketatmark.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11.23, Solaris 9, Cygwin 1.7.x, mingw, MSVC 14, Android 9.0.

10.56.18 `socket`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/socket.html>

Gnulib module: `socket`

Portability problems fixed by Gnulib:

- This function is declared in `<winsock2.h>` instead of `<sys/socket.h>` on some platforms: mingw, MSVC 14.
- On Windows platforms (excluding Cygwin), the descriptors returned by the `socket` function cannot be used in calls to `read`, `write`, and `close`; you have to use `recv`, `send`, `closesocket` in these cases instead.
- On Windows platforms (excluding Cygwin), error codes from this function are not placed in `errno`, and `WSAGetLastError` must be used instead.

Portability problems not fixed by Gnulib:

10.56.19 `socketpair`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/socketpair.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

10.57 Functions in `<sys/stat.h>`

10.57.1 `chmod`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/chmod.html>

Gnulib module: `chmod`

Portability problems fixed by Gnulib:

- This function does not fail when the file name argument ends in a slash and (without the slash) names a non-directory, on some platforms: AIX 7.2.
- This function fails with a wrong error code (`EINVAL` instead of `ENOTDIR`) when the file name argument ends in a slash and (without the slash) names a non-directory, on some platforms: mingw, MSVC.

Portability problems not fixed by Gnulib:

- This function sometimes fails with `EACCES` when the failure is due to lack of appropriate privileges (`EPERM`), not to search permission denied on the file name prefix (`EACCES`): Linux kernel 5.15 with glibc 2.35 and a CIFS v1 file system (see <https://bugs.gnu.org/65599>).

10.57.2 fchmod

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fchmod.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.
- This function sometimes fails with `EACCES` when the failure is due to lack of appropriate privileges (`EPERM`), not to search permission denied on the file name prefix (`EACCES`): Linux kernel 5.15 with glibc 2.35 and a CIFS v1 file system (see <https://bugs.gnu.org/65599>).

10.57.3 fchmodat

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fchmodat.html>

Gnulib module: fchmodat

Portability problems fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, Mac OS X 10.5, FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 10, Cygwin 1.5.x, mingw, MSVC 14. But the replacement function is not safe to be used in libraries and is not multithread-safe.
- This function does not fail when the file name argument ends in a slash and (without the slash) names a non-directory, on some platforms: AIX 7.2.
- When given the `AT_SYMLINK_NOFOLLOW` flag, this function fails with `errno` set to `ENOTSUP`, even when the file is not a symbolic link: GNU/Linux with glibc 2.31, Cygwin 2.9.

Portability problems not fixed by Gnulib:

- Some platforms do not allow changing the access bits on symbolic links.
- If the `AT_SYMLINK_NOFOLLOW` flag is specified, this function can fail with `errno` set to `EMFILE` or `ENFILE`, and it fails with `errno` set to `EOPNOTSUPP` if the `/proc` file system is not mounted: GNU/Linux with glibc 2.34.

- This function sometimes fails with **EACCES** when the failure is due to lack of appropriate privileges (**EPERM**), not to search permission denied on the file name prefix (**EACCES**): Linux kernel 5.15 with glibc 2.35 and a CIFS v1 file system (see <https://bugs.gnu.org/65599>).

10.57.4 fstat

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fstat.html>

Gnulib module: `fstat`

Portability problems fixed by Gnulib:

- This function crashes when invoked with invalid arguments on some platforms: MSVC 14.
- On platforms where `off_t` is a 32-bit type, `fstat` may not correctly report the size of files or block devices 2 GiB and larger.. See [Section 14.2 \[Large File Support\]](#), page 842.
- On Linux/x86 and Linux/x86_64, applications compiled in 32-bit mode cannot access files that happen to have a 64-bit inode number. This can occur with file systems such as XFS (typically on large disks) and NFS. See [Section 14.2 \[Large File Support\]](#), page 842.
- On macOS 12.6, when this function yields a timestamp with a nonpositive `tv_sec` value, `tv_nsec` might be in the range `-999999999..-1`, representing a negative nanoseconds offset from `tv_sec`. Solaris 11.4 is similar, except that `tv_sec` might also be `-1000000000`.
- The `st_atime`, `st_ctime`, `st_mtime` fields are affected by the current time zone and by the DST flag of the current time zone on some platforms: mingw, MSVC 14 (when the environment variable `TZ` is set).

Portability problems not fixed by Gnulib:

- See [Section 9.69 \[sys/stat.h\]](#), page 87, for general portability problems with `struct stat`.
- This function fails if the file descriptor was opened with `O_PATH`: Linux kernel 3.5.
- On Cygwin, `fstat` applied to the file descriptors 0 and 1, returns different `st_ino` values, even if standard input and standard output are not redirected and refer to the same terminal.

10.57.5 fstatat

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fstatat.html>

Gnulib module: `fstatat`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, Mac OS X 10.5, FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 1.5.x, mingw, MSVC 14. But the replacement function is not safe to be used in libraries and is not multithread-safe.

- On platforms where `off_t` is a 32-bit type, `fstatat` may not correctly report the size of files or block devices larger than 2 GB. See [Section 14.2 \[Large File Support\]](#), page 842.
- On some platforms, `fstatat(fd, "file/", buf, flag)` succeeds instead of failing with `ENOTDIR`. Solaris 9.
- For symlinks, when the argument ends in a slash, some platforms don't dereference the argument: Solaris 9.
- On macOS 12.6, when this function yields a timestamp with a nonpositive `tv_sec` value, `tv_nsec` might be in the range `-999999999..-1`, representing a negative nanoseconds offset from `tv_sec`. Solaris 11.4 is similar, except that `tv_sec` might also be `-1000000000`.

Portability problems not fixed by Gnulib:

- This function does not fail when the second argument is an empty string on some platforms, even when `AT_EMPTY_PATH` is not used: glibc 2.7, Linux 2.6.38.
- This function sets `st_ino` only to the low-order 32 bits of the inode number of a socket or pipe, which thus can disagree with the `st_ino` obtained by `fstat`: macOS 14.
- See [Section 9.69 \[sys/stat.h\]](#), page 87, for general portability problems with `struct stat`.

10.57.6 futimens

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/futimens.html>

Gnulib module: futimens

Portability problems fixed by Gnulib:

- This function is missing on some platforms: glibc 2.5, Mac OS X 10.5, FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 10, Cygwin 1.5.x, mingw, MSVC 14, Android 4.3. However, the replacement function may end up truncating timestamps to worse resolution than supported by the file system.
- This function returns a bogus value instead of failing with `ENOSYS` on some platforms: Linux kernel 2.6.21.
- When using `UTIME_OMIT` or `UTIME_NOW`, some systems require the `tv_sec` argument to be 0, and don't necessarily handle all file permissions in the manner required by POSIX: Linux kernel 2.6.25.
- When using `UTIME_OMIT` for the modification time, but specifying an access time, some systems fail to update the change time: Linux kernel 2.6.32, macOS 14, NetBSD 10.0, Solaris 11.1.
- Passing `AT_FDCWD` as the `fd` argument does not properly fail with `EBADF` on some systems: glibc 2.11, musl libc, Solaris 11.
- Out-of-range values of `tv_nsec` do not lead to a failure on some platforms: NetBSD 10.0, GNU/Hurd with glibc 2.41.

Portability problems not fixed by Gnulib:

- On file systems mounted with the `noatime` attribute, this function might not modify the access time as requested: Linux kernel 6.9.

- This function sets a slightly different time stamp than the requested one: Haiku.
- Some platforms lack the ability to change the timestamps of a file descriptor, so the replacement can fail with `ENOSYS`; the glibc module ‘`utimens`’ provides a more reliable interface `fdutimens`.
- The mere act of using `stat` modifies the access time of directories on some platforms, so `utimensat` can only effectively change directory modification time: Cygwin 1.5.x.

10.57.7 `lstat`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/lstat.html>

Glibc module: `lstat`

Portability problems fixed by Glibc:

- On platforms where `off_t` is a 32-bit type, `lstat` may not correctly report the size of files or block devices 2 GiB and larger. See [Section 14.2 \[Large File Support\]](#), page 842.
- On Linux/x86 and Linux/x86_64, applications compiled in 32-bit mode cannot access files that happen to have a 64-bit inode number. This can occur with file systems such as XFS (typically on large disks) and NFS. See [Section 14.2 \[Large File Support\]](#), page 842.
- For symlinks, when the argument ends in a slash, some platforms don’t dereference the argument: Solaris 9.
- On some platforms, `lstat("file/",buf)` succeeds instead of failing with `ENOTDIR`. macOS 14, Solaris 9.
- On macOS 12.6, when this function yields a timestamp with a nonpositive `tv_sec` value, `tv_nsec` might be in the range `−999999999..−1`, representing a negative nanoseconds offset from `tv_sec`. Solaris 11.4 is similar, except that `tv_sec` might also be `−1000000000`.
- On Windows platforms (excluding Cygwin), a different name has to be used for the null device, namely `"NUL"` instead of `"/dev/null"`.
- On Windows platforms (excluding Cygwin), symlinks are not supported, so `lstat` does not exist.

Portability problems not fixed by Glibc:

- See [Section 9.69 \[sys/stat.h\]](#), page 87, for general portability problems with `struct stat`.

10.57.8 `mkdir`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/mkdir.html>

Glibc module: `sys_stat-h` or `mkdir`

Portability problems fixed by either Glibc module `sys_stat-h` or `mkdir`:

- This function is declared in different header files (namely, `<io.h>` or `<direct.h>`) on some platforms: mingw, MSVC 14.

- On Windows platforms (excluding Cygwin), this function is called `_mkdir` and takes only one argument. The fix (without Gnulib) is to define a macro like this:

```
#define mkdir ((int (*)( )) _mkdir)
```

or

```
#define mkdir(path,mode) _mkdir (path)
```

Portability problems fixed by Gnulib module `mkdir`:

- When the argument ends in a slash, the function call fails on some platforms.
- This function mistakenly succeeds on `'mkdir("d/./",mode)'` on some platforms: Cygwin 1.5.x, mingw, MSVC 14.

Portability problems not fixed by Gnulib:

10.57.9 mkdirat

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/mkdirat.html>

Gnulib module: `mkdirat`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, Mac OS X 10.5, FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 10, Cygwin 1.5.x, mingw, MSVC 14. But the replacement function is not safe to be used in libraries and is not multithread-safe.

Portability problems not fixed by Gnulib:

10.57.10 mkfifo

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/mkfifo.html>

Gnulib module: `mkfifo`

Portability problems fixed by Gnulib:

- This function mishandles trailing slash on some platforms: FreeBSD 7.2, Solaris 9.
- This function is missing on some platforms; however, the replacement always fails with `ENOSYS`: mingw, MSVC 14, Android 4.4.

Portability problems not fixed by Gnulib:

- If the argument exists and is a directory, this function fails with `EISDIR` instead of the correct `EEXIST`: HP-UX 11.11.

10.57.11 mkfifoat

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/mkfifoat.html>

Gnulib module: `mkfifoat`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 12, FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 10, Cygwin 1.5.x, mingw, MSVC 14, Android 5.1. But the replacement function is not safe to be used in libraries and is not multithread-safe.

- This function does not fail when the file name argument ends in a slash and (without the slash) names a nonexistent file, on some platforms: AIX 7.2.
- This function does not fail when the file name argument ends in a slash and (without the slash) names a symbolic link, on some platforms: macOS 14.

Portability problems not fixed by GnuLib:

- The gnuLib replacement function always fails with `ENOSYS` on some platforms: mingw, MSVC 14.
- If the argument exists and is a directory, the gnuLib replacement function fails with `EISDIR` instead of the correct `EEXIST`: HP-UX 11.11.

10.57.12 `mknod`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/mknod.html>

GnuLib module: `mknod`

Portability problems fixed by GnuLib:

- This function requires super-user privileges to create a fifo: FreeBSD 7.2, NetBSD 8.0, OpenBSD 7.5.
- This function mishandles trailing slash on some platforms: FreeBSD 7.2, Solaris 9.
- This function does not handle the `S_IFIFO` flag on some platforms: macOS 14, DragonFly 6.0, Haiku.
- This function is missing on some platforms; however, the replacement always fails with `ENOSYS`: mingw, MSVC 14.

Portability problems not fixed by GnuLib:

- Use of this function for anything except fifos is not portable, generally requiring super-user privileges and knowledge of supported device numbers.
- If the argument exists and is a directory, this function fails with `EISDIR` instead of the correct `EEXIST`: HP-UX 11.11.

10.57.13 `mknodat`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/mknodat.html>

GnuLib module: `mkfifoat`

Portability problems fixed by GnuLib:

- This function is missing on some platforms: glibc 2.3.6, macOS 12, FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 10, Cygwin 1.5.x, mingw, MSVC 14, Android 4.4. But the replacement function is not safe to be used in libraries and is not multithread-safe.
- This function does not fail when the file name argument ends in a slash and (without the slash) names a nonexistent file, on some platforms: AIX 7.2.
- This function does not handle the `S_IFIFO` flag on some platforms: Haiku.

Portability problems not fixed by GnuLib:

- The gnuLib replacement function always fails with `ENOSYS` on some platforms: mingw, MSVC 14.

- If the argument exists and is a directory, the glibc replacement function fails with `EISDIR` instead of the correct `EEXIST`: HP-UX 11.11.

10.57.14 `stat`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/stat.html>

Glibc module: `stat`

Portability problems fixed by Glibc:

- On platforms where `off_t` is a 32-bit type, `stat` may not correctly report the size of files or block devices 2 GiB and larger. See [Section 14.2 \[Large File Support\]](#), page 842.
- On Linux/x86 and Linux/x86_64, applications compiled in 32-bit mode cannot access files that happen to have a 64-bit inode number. This can occur with file systems such as XFS (typically on large disks) and NFS. See [Section 14.2 \[Large File Support\]](#), page 842.
- The `st_atime`, `st_ctime`, `st_mtime` fields are affected by the current time zone and by the DST flag of the current time zone on some platforms: mingw, MSVC 14 (when the environment variable `TZ` is set).
- On MSVC 14, this function fails with error `ENOENT` on files such as `'C:\pagefile.sys'` and on directories such as `'C:\System Volume Information'`.
- On some platforms, `stat("link-to-file/",buf)` succeeds instead of failing with `ENOTDIR`. macOS 14, FreeBSD 7.2, AIX 7.1, Solaris 9, mingw64.
- On some platforms, `stat(".",buf)` and `stat("./",buf)` give different results: mingw, MSVC 14.
- On mingw, the `st_mode` field of `stat("NUL",buf)` is bogus, suggesting that `NUL` is a regular file.
- On macOS 12.6, when this function yields a timestamp with a nonpositive `tv_sec` value, `tv_nsec` might be in the range `-999999999..-1`, representing a negative nanoseconds offset from `tv_sec`. Solaris 11.4 is similar, except that `tv_sec` might also be `-1000000000`.
- On Windows platforms (excluding Cygwin), a different name has to be used for the null device, namely `"NUL"` instead of `"/dev/null"`.

Portability problems not fixed by Glibc:

- See [Section 9.69 \[sys/stat.h\]](#), page 87, for general portability problems with `struct stat`.
- Cygwin's `stat` function sometimes sets `errno` to `EACCES` when `ENOENT` would be more appropriate.
- This function sets `st_ino` only to the low-order 32 bits of the inode number of a socket or pipe, which thus can disagree with the `st_ino` obtained by `fstat`: macOS 14.
- Because of the definition of `struct stat`, it is not possible to portably replace `stat` via an object-like macro. Therefore, expressions such as `(islnk ? lstat : stat) (name, buf)` are not portable, and should instead be written `islnk ? lstat (name, buf) : stat (name, buf)`.

10.57.15 umask

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/umask.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

10.57.16 utimensat

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/utimensat.html>

Gnulib module: utimensat

Portability problems fixed by Gnulib:

- This function is missing on some platforms: glibc 2.5, Mac OS X 10.5, FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 10, Cygwin 1.5.x, mingw, MSVC 14, Android 3.0. However, the replacement function may end up truncating timestamps to worse resolution than supported by the file system. Furthermore, the replacement function is not safe to be used in libraries and is not multithread-safe.
- This function returns a bogus value instead of failing with `ENOSYS` on some platforms: Linux kernel 2.6.21.
- This function fails with `ENOSYS` if passed the flag `AT_SYMLINK_NOFOLLOW` on a regular file: Linux kernel 2.6.22.
- When using `UTIME_OMIT` or `UTIME_NOW`, some systems require the `tv_sec` argument to be 0, and don't necessarily handle all file permissions in the manner required by POSIX: Linux kernel 2.6.25.
- When using `UTIME_OMIT` for the modification time, but specifying an access time, some systems fail to update the change time: Linux kernel 2.6.32, macOS 14, NetBSD 10.0, Solaris 11.1.
- Out-of-range values of `tv_nsec` do not lead to a failure on some platforms: Linux kernel 2.6.22.19 on hppa, NetBSD 10.0, GNU/Hurd with glibc 2.41.
- On some platforms, this function mis-handles a trailing slash: AIX 7.2.

Portability problems not fixed by Gnulib:

- On file systems mounted with the `noatime` attribute, this function might not modify the access time as requested: Linux kernel 6.9.
- This function ignores the flag `AT_SYMLINK_NOFOLLOW`: Linux kernel 6.8 on a CIFS file system.
- This function sets a slightly different time stamp than the requested one: Haiku.
- On some platforms, timestamps of symbolic links cannot be modified, so the replacement fails with `ENOSYS` if passed the flag `AT_SYMLINK_NOFOLLOW` on a symlink.
- The mere act of using `lstat` modifies the access time of symlinks on some platforms, so `utimensat` with `AT_SYMLINK_NOFOLLOW` can only effectively change modification time: Cygwin.

- The mere act of using `stat` modifies the access time of directories on some platforms, so `utimensat` can only effectively change directory modification time: Cygwin 1.5.x.

The `gnulib` module `fdutimensat` provides a similar interface.

10.58 Functions in `<sys/statvfs.h>`

10.58.1 `fstatvfs`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fstatvfs.html>

Gnulib module: —

Portability problems fixed by Gnulib:

- On platforms where `f_blocks` in `'struct statvfs'` is a 32-bit value, this function may not work correctly on files systems larger than 4 TiB. See [Section 14.2 \[Large File Support\]](#), page 842. This affects `glibc/Hurd`, `HP-UX 11`, `Solaris`.

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: `OpenBSD 3.8`, `mingw`, `MSVC 14`, `Android 4.3`.
- This function fails if the file descriptor was opened with `O_PATH`: `Linux kernel 3.11`.

10.58.2 `statvfs`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/statvfs.html>

Gnulib module: —

Portability problems fixed by Gnulib:

- On platforms where `f_blocks` in `'struct statvfs'` is a 32-bit value, this function may not work correctly on files systems larger than 4 TiB. See [Section 14.2 \[Large File Support\]](#), page 842. This affects `glibc/Hurd`, `HP-UX 11`, `Solaris`.

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: `OpenBSD 3.8`, `mingw`, `MSVC 14`, `Android 4.3`.
- This function can hang if it stats all preceding entries in `/proc/mounts`, and any of those file systems are hard-mounted and not available. This affects `Linux < 2.6.36`.

Gnulib provides a module `fsusage` that provides similar information as `statvfs`.

10.59 Functions in `<sys/time.h>`

10.59.1 `utimes`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/utimes.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.
- On some platforms, this function mis-handles a trailing slash: FreeBSD 7.2, Solaris 9.
- This function cannot set full timestamp resolution. In particular, some platforms incorrectly round rather than truncate. Use `utimensat(AT_FDCWD,file,times,0)`, or the gnulib module `utimens`, instead.
- On file systems mounted with the `noatime` attribute, this function might not modify the access time as requested: Linux kernel 6.9.
- On some platforms, `utimes (file, NULL)` fails to set the file's timestamp to the current time: glibc 2.3.3.
- On some platforms, `utimes` failed on read-only files when `utime` worked fine. glibc 2.2.5.
- This function sets a slightly different time stamp than the requested one: Haiku.
- On OS/2, this function cannot set the timestamp to earlier than the year 1980 in local time.
- On OS/2, this function cannot set the timestamp to an odd number of seconds.
- On OS/2, this function does not work on an opened file.

Extension: Gnulib provides a module 'utimens' that works around these problems and can set the time with nanosecond resolution (as far as supported by the file system).

10.60 Functions in <sys/times.h>

10.60.1 times

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/times.html>

Gnulib module: times

Portability problems fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- There is no function on Windows to measure consumed process child times, thus the `tms_cutime` and `tms_cstime` will always be 0 when the module is used.

10.61 Functions in <sys/uio.h>

10.61.1 readv

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/readv.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

10.61.2 writev

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/writev.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

10.62 Functions in <sys/utsname.h>

10.62.1 uname

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/uname.html>

Gnulib module: uname

Portability problems fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

10.63 Functions in <sys/wait.h>

10.63.1 wait

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wait.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

10.63.2 waitid

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/waitid.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 7.1, OpenBSD 7.2, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14.
- As of 2005, no system is known on which `waitid` with flag `WNOWAIT` works correctly.

10.63.3 waitpid

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/waitpid.html>

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-waitpid-3.html

Gnulib module: waitpid

Portability problems fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

10.64 Functions in <syslog.h>

10.64.1 closelog

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/closelog.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

10.64.2 openlog

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/openlog.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

10.64.3 setlogmask

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/setlogmask.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

10.64.4 syslog

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/syslog.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

10.65 Functions in <termios.h>

10.65.1 cfgetispeed

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/cfgetispeed.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14, Android 4.4.

10.65.2 cfgetospeed

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/cfgetospeed.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14, Android 4.4.

10.65.3 cfsetispeed

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/cfsetispeed.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14, Android 4.4.

10.65.4 cfsetospeed

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/cfsetospeed.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14, Android 4.4.

10.65.5 `tcdrain`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/tcdrain.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14, Android 4.4.
- On some platforms, `tcdrain` on a non-tty fails with `errno` set to `EINVAL` or, on Mac OS X, also `EOPNOTSUPP` or `ENODEV`, rather than `ENOTTY`.

10.65.6 `tcflow`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/tcflow.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14, Android 4.4.

10.65.7 `tcflush`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/tcflush.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14, Android 4.4.
- On some platforms, `tcflush` of `TCIFLUSH` on a non-tty fails with `errno` set to `EINVAL` rather than `ENOTTY`.
- On some platforms, `tcflush` of `TCOFLUSH` on a non-tty fails with `errno` set to `EINVAL` or, on Mac OS X, also `EOPNOTSUPP` or `ENODEV`, rather than `ENOTTY`.

10.65.8 `tcgetattr`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/tcgetattr.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14, Android 4.4.

10.65.9 tcgetsid

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/tcgetsid.html>

Gnulib module: tcgetsid

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, OpenBSD 4.5, Minix 3.1.8, Cygwin 1.7.9, mingw, MSVC 14, Android 4.4.
- The declaration of this function in C++ compilation units has C++ linkage, not C linkage, on some platforms: HP-UX 11.00.

Portability problems not fixed by Gnulib:

- This function always fails on some platforms: FreeBSD 6.0, Cygwin 1.7.8, mingw, MSVC 14.
- This function returns int instead of pid_t on some platforms: Cygwin 1.7.11.

10.65.10 tcgetwinsize

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/tcgetwinsize.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc 2.42, macOS 14, FreeBSD 12.0, NetBSD 8.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android 9.0.

10.65.11 tcsendbreak

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/tcsendbreak.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14, Android 4.4.

10.65.12 tcsetattr

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/tcsetattr.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14, Android 4.4.

10.65.13 tcsetwinsize

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/tcsetwinsize.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc 2.42, macOS 14, FreeBSD 12.0, NetBSD 8.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android 9.0.

10.66 Functions in <threads.h>

10.66.1 call_once

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.28.2.1.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/call_once.html

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Call-Once.html.

Gnulib module: call_once

Portability problems fixed by Gnulib:

- This function is missing on many platforms: glibc 2.27, macOS 14, FreeBSD 9.3, NetBSD 8.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.3, Cygwin 3.1.x, mingw, MSVC 14, Android API level 29.

Portability problems not fixed by Gnulib:

- This function makes applications hang forever on some platforms: Cygwin 3.5.3.
- This function does not work on some platforms: Haiku.

10.66.2 cnd_broadcast

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.28.3.1.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/cnd_broadcast.html

Documentation:

https://www.gnu.org/software/libc/manual/html_node/ISO-C-Condition-Variables.html.

Gnulib module: cnd

Portability problems fixed by Gnulib:

- This function is missing on many platforms: glibc 2.27, macOS 14, FreeBSD 9.3, NetBSD 8.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.3, Cygwin 3.1.x, mingw, MSVC 14, Android API level 29.

Portability problems not fixed by Gnulib:

10.66.3 `cnd_destroy`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.28.3.2.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/cnd_destroy.html

Documentation:

https://www.gnu.org/software/libc/manual/html_node/ISO-C-Condition-Variables.html.

Gnulib module: `cnd`

Portability problems fixed by Gnulib:

- This function is missing on many platforms: glibc 2.27, macOS 14, FreeBSD 9.3, NetBSD 8.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.3, Cygwin 3.1.x, mingw, MSVC 14, Android API level 29.

Portability problems not fixed by Gnulib:

10.66.4 `cnd_init`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.28.3.3.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/cnd_init.html

Documentation:

https://www.gnu.org/software/libc/manual/html_node/ISO-C-Condition-Variables.html.

Gnulib module: `cnd`

Portability problems fixed by Gnulib:

- This function is missing on many platforms: glibc 2.27, macOS 14, FreeBSD 9.3, NetBSD 8.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.3, Cygwin 3.1.x, mingw, MSVC 14, Android API level 29.

Portability problems not fixed by Gnulib:

10.66.5 `cnd_signal`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.28.3.4.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/cnd_signal.html

Documentation:

https://www.gnu.org/software/libc/manual/html_node/ISO-C-Condition-Variables.html.

Gnulib module: `cnd`

Portability problems fixed by Gnulib:

- This function is missing on many platforms: glibc 2.27, macOS 14, FreeBSD 9.3, NetBSD 8.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.3, Cygwin 3.1.x, mingw, MSVC 14, Android API level 29.

Portability problems not fixed by Gnulib:

10.66.6 `cnd_timedwait`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.28.3.5.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/cnd_timedwait.html

Documentation:

https://www.gnu.org/software/libc/manual/html_node/ISO-C-Condition-Variables.html.

Gnulib module: `cnd`

Portability problems fixed by Gnulib:

- This function is missing on many platforms: glibc 2.27, macOS 14, FreeBSD 9.3, NetBSD 8.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.3, Cygwin 3.1.x, mingw, MSVC 14, Android API level 29.

Portability problems not fixed by Gnulib:

- This function sometimes waits indefinitely instead of up to the specified time point, on some platforms (glibc/Linux, musl libc) in some hypervisors: VirtualBox 6.1.50-Ubuntu with paravirtualization set to "Default" or "KVM".

10.66.7 `cnd_wait`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.28.3.6.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/cnd_wait.html

Documentation:

https://www.gnu.org/software/libc/manual/html_node/ISO-C-Condition-Variables.html.

Gnulib module: `cnd`

Portability problems fixed by Gnulib:

- This function is missing on many platforms: glibc 2.27, macOS 14, FreeBSD 9.3, NetBSD 8.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.3, Cygwin 3.1.x, mingw, MSVC 14, Android API level 29.

Portability problems not fixed by Gnulib:

10.66.8 `mtx_destroy`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.28.4.1.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/mtx_destroy.html

Documentation:

https://www.gnu.org/software/libc/manual/html_node/ISO-C-Mutexes.html.

Gnulib module: `mtx`

Portability problems fixed by Gnulib:

- This function is missing on many platforms: glibc 2.27, macOS 14, FreeBSD 9.3, NetBSD 8.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.3, Cygwin 3.1.x, mingw, MSVC 14, Android API level 29.

Portability problems not fixed by Gnulib:

10.66.9 `mtx_init`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.28.4.2.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/mtx_init.html

Documentation:

https://www.gnu.org/software/libc/manual/html_node/ISO-C-Mutexes.html.

Gnulib module: `mtx`

Portability problems fixed by Gnulib:

- This function is missing on many platforms: glibc 2.27, macOS 14, FreeBSD 9.3, NetBSD 8.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.3, Cygwin 3.1.x, mingw, MSVC 14, Android API level 29.

Portability problems not fixed by Gnulib:

10.66.10 `mtx_lock`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.28.4.3.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/mtx_lock.html

Documentation:

https://www.gnu.org/software/libc/manual/html_node/ISO-C-Mutexes.html.

Gnulib module: `mtx`

Portability problems fixed by Gnulib:

- This function is missing on many platforms: glibc 2.27, macOS 14, FreeBSD 9.3, NetBSD 8.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.3, Cygwin 3.1.x, mingw, MSVC 14, Android API level 29.

Portability problems not fixed by Gnulib:

10.66.11 `mtx_timedlock`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.28.4.4.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/mtx_timedlock.html

Documentation:

https://www.gnu.org/software/libc/manual/html_node/ISO-C-Mutexes.html.

Gnulib module: `mtx`

Portability problems fixed by Gnulib:

- This function is missing on many platforms: glibc 2.27, macOS 14, FreeBSD 9.3, NetBSD 8.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.3, Cygwin 3.1.x, mingw, MSVC 14, Android API level 29.

Portability problems not fixed by Gnulib:

10.66.12 `mtx_trylock`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.28.4.5.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/mtx_trylock.html

Documentation:

https://www.gnu.org/software/libc/manual/html_node/ISO-C-Mutexes.html.

Gnulib module: `mtx`

Portability problems fixed by Gnulib:

- This function is missing on many platforms: glibc 2.27, macOS 14, FreeBSD 9.3, NetBSD 8.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.3, Cygwin 3.1.x, mingw, MSVC 14, Android API level 29.

Portability problems not fixed by Gnulib:

10.66.13 `mtx_unlock`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.28.4.6.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/mtx_unlock.html

Documentation:

https://www.gnu.org/software/libc/manual/html_node/ISO-C-Mutexes.html.

Gnulib module: `mtx`

Portability problems fixed by Gnulib:

- This function is missing on many platforms: glibc 2.27, macOS 14, FreeBSD 9.3, NetBSD 8.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.3, Cygwin 3.1.x, mingw, MSVC 14, Android API level 29.

Portability problems not fixed by Gnulib:

10.66.14 `thrd_create`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.28.5.1.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/thrd_create.html

Documentation:

https://www.gnu.org/software/libc/manual/html_node/ISO-C-Thread-Management.html.

Gnulib module: `thrd`

Portability problems fixed by Gnulib:

- This function is missing on many platforms: glibc 2.27, macOS 14, FreeBSD 9.3, NetBSD 8.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.3, Cygwin 3.1.x, mingw, MSVC 14, Android API level 29.
- This function uses an incorrectly defined `thrd_start_t` on some platforms: AIX 7.2.

Portability problems not fixed by Gnulib:

10.66.15 `thrd_current`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.28.5.2.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/thrd_current.html

Documentation:

https://www.gnu.org/software/libc/manual/html_node/ISO-C-Thread-Management.html.

Gnulib module: `thrd`

Portability problems fixed by Gnulib:

- This function is missing on many platforms: glibc 2.27, macOS 14, FreeBSD 9.3, NetBSD 8.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.3, Cygwin 3.1.x, mingw, MSVC 14, Android API level 29.

Portability problems not fixed by Gnulib:

10.66.16 `thrd_detach`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.28.5.3.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/thrd_detach.html

Documentation:

https://www.gnu.org/software/libc/manual/html_node/ISO-C-Thread-Management.html.

Gnulib module: `thrd`

Portability problems fixed by GnuLib:

- This function is missing on many platforms: glibc 2.27, macOS 14, FreeBSD 9.3, NetBSD 8.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.3, Cygwin 3.1.x, mingw, MSVC 14, Android API level 29.

Portability problems not fixed by GnuLib:

10.66.17 `thrd_equal`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.28.5.4.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/thrd_equal.html

Documentation:

https://www.gnu.org/software/libc/manual/html_node/ISO-C-Thread-Management.html.

GnuLib module: `thrd`

Portability problems fixed by GnuLib:

- This function is missing on many platforms: glibc 2.27, macOS 14, FreeBSD 9.3, NetBSD 8.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.3, Cygwin 3.1.x, mingw, MSVC 14, Android API level 29.

Portability problems not fixed by GnuLib:

10.66.18 `thrd_exit`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.28.5.5.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/thrd_exit.html

Documentation:

https://www.gnu.org/software/libc/manual/html_node/ISO-C-Thread-Management.html.

GnuLib module: `thrd`

Portability problems fixed by GnuLib:

- This function is missing on many platforms: glibc 2.27, macOS 14, FreeBSD 9.3, NetBSD 8.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.3, Cygwin 3.1.x, mingw, MSVC 14, Android API level 29.
- The exit code provided to this function is discarded on some platforms: AIX 7.3.1.

Portability problems not fixed by GnuLib:

10.66.19 `thrd_join`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.28.5.6.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/thrd_join.html

Documentation:

https://www.gnu.org/software/libc/manual/html_node/ISO-C-Thread-Management.html.

Gnulib module: `thrd`

Portability problems fixed by Gnulib:

- This function is missing on many platforms: glibc 2.27, macOS 14, FreeBSD 9.3, NetBSD 8.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.3, Cygwin 3.1.x, mingw, MSVC 14, Android API level 29.
- This function never stores an exit code on some platforms: AIX 7.3.1.
- This function crashes when the second argument is NULL on some platforms: Solaris 11.4.

Portability problems not fixed by Gnulib:

10.66.20 `thrd_sleep`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.28.5.7.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/thrd_sleep.html

Documentation:

https://www.gnu.org/software/libc/manual/html_node/ISO-C-Thread-Management.html.

Gnulib module: `thrd`

Portability problems fixed by Gnulib:

- This function is missing on many platforms: glibc 2.27, macOS 14, FreeBSD 9.3, NetBSD 8.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.3, Cygwin 3.1.x, mingw, MSVC 14, Android API level 29.

Portability problems not fixed by Gnulib:

10.66.21 `thrd_yield`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.28.5.8.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/thrd_yield.html

Documentation:

https://www.gnu.org/software/libc/manual/html_node/ISO-C-Thread-Management.html.

Gnulib module: `thrd`

Portability problems fixed by Gnulib:

- This function is missing on many platforms: glibc 2.27, macOS 14, FreeBSD 9.3, NetBSD 8.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.3, Cygwin 3.1.x, mingw, MSVC 14, Android API level 29.

Portability problems not fixed by Gnulib:

10.66.22 tss_create

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.28.6.1.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/tss_create.html

Documentation:

https://www.gnu.org/software/libc/manual/html_node/ISO-C-Thread_002dlocal-Storage.html.

Gnulib module: tss

Portability problems fixed by Gnulib:

- This function is missing on many platforms: glibc 2.27, macOS 14, FreeBSD 9.3, NetBSD 8.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.3, Cygwin 3.1.x, mingw, MSVC 14, Android API level 29.

Portability problems not fixed by Gnulib:

10.66.23 tss_delete

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.28.6.2.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/tss_delete.html

Documentation:

https://www.gnu.org/software/libc/manual/html_node/ISO-C-Thread_002dlocal-Storage.html.

Gnulib module: tss

Portability problems fixed by Gnulib:

- This function is missing on many platforms: glibc 2.27, macOS 14, FreeBSD 9.3, NetBSD 8.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.3, Cygwin 3.1.x, mingw, MSVC 14, Android API level 29.

Portability problems not fixed by Gnulib:

10.66.24 tss_get

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.28.6.3.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/tss_get.html

Documentation:

https://www.gnu.org/software/libc/manual/html_node/ISO-C-Thread_002dlocal-Storage.html.

Gnulib module: tss

Portability problems fixed by Gnulib:

- This function is missing on many platforms: glibc 2.27, macOS 14, FreeBSD 9.3, NetBSD 8.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.3, Cygwin 3.1.x, mingw, MSVC 14, Android API level 29.

Portability problems not fixed by Gnulib:

10.66.25 `tss_set`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.28.6.4.

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/tss_set.html

Documentation:

https://www.gnu.org/software/libc/manual/html_node/ISO-C-Thread_002dlocal-Storage.html.

Gnulib module: `tss`

Portability problems fixed by Gnulib:

- This function is missing on many platforms: glibc 2.27, macOS 14, FreeBSD 9.3, NetBSD 8.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.3, Cygwin 3.1.x, mingw, MSVC 14, Android API level 29.

Portability problems not fixed by Gnulib:

10.67 Functions in `<time.h>`

10.67.1 `asctime`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/asctime.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is deprecated in C23. Likewise, POSIX says this function is obsolescent and it is planned to be removed in a future version. Portable applications can use `strftime` (or even `sprintf`) instead. However, `strftime` is locale dependent.
- This function may overflow its internal buffer if its argument specifies a year before 1000 or after 9999. See [Section 10.67.9 \[ctime\]](#), page 465.
- Although it is tempting to call this function on the value that functions like `localtime` return, this is typically a mistake. On most current platforms, these functions return a null pointer for timestamps out of range, and this function has undefined behavior in that case.

10.67.2 `asctime_r`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/asctime_r.html

Gnulib module: `extensions`

Portability problems fixed by Gnulib:

- This function has an incompatible declaration on some platforms: Solaris 11.4 (when `_POSIX_PTHREAD_SEMANTICS` is not defined).

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: mingw, MSVC 14.
- This function is removed in POSIX.1-2024. Use the function `strftime` (or even `sprintf`) instead. However, `strftime` is locale dependent.
- This function may overflow its output buffer if its argument specifies a year before 1000 or after 9999. See [Section 10.67.9 \[ctime\]](#), page 465.
- Although it is tempting to call this function on the value that functions like `localtime` return, this is typically a mistake. On most current platforms, these functions return a null pointer for timestamps out of range, and this function has undefined behavior in that case.

10.67.3 clock

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/clock.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

10.67.4 clock_gettime

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/clock_gettime.html

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: Mac OS X 10.11, FreeBSD 6.0, NetBSD 7.1, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.9, mingw, MSVC 14, Android 5.1.

10.67.5 clock_getres

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/clock_getres.html

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: Mac OS X 10.11, Minix 3.1.8, mingw, MSVC 14.
- On many platforms, this function returns a value other than the clock resolution of `clock_gettime`, i.e., the minimum distance between differing timestamps. For example, on AIX 7.2 it returns 10 milliseconds even though the clock resolution is 1 microsecond. Conversely, on GNU/Linux it typically returns 1 nanosecond even though the clock resolution may be greater.

The Glibc module `gettime-res` is a partial substitute; it implements the `CLOCK_REALTIME` functionality of `clock_getres`, and fixes the too-high resolution bug of platforms like AIX 7.2.

10.67.6 `clock_gettime`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/clock_gettime.html

Glibc module: —

Portability problems fixed by Glibc:

Portability problems not fixed by Glibc:

- This function is missing on some platforms: Mac OS X 10.11, Minix 3.1.8, mingw, MSVC 14.
- This function leaves the upper 32 bits of the `tv_sec` field of the result uninitialized on some platforms: mingw in 32-bit mode.

The Glibc modules `gettime` and `timespec_get` are partial substitutes; they implement the `CLOCK_REALTIME` functionality of `clock_gettime`.

10.67.7 `clock_nanosleep`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/clock_nanosleep.html

Glibc module: —

Portability problems fixed by Glibc:

Portability problems not fixed by Glibc:

- This function is missing on some platforms: Mac OS X 10.11, FreeBSD 11.0, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 9, Cygwin 1.7.9, mingw, MSVC 14.

10.67.8 `clock_settime`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/clock_settime.html

Glibc module: —

Portability problems fixed by Glibc:

Portability problems not fixed by Glibc:

- This function is missing on some platforms: Mac OS X 10.11, Minix 3.1.8, Cygwin 1.7.9, mingw, MSVC 14.

10.67.9 `ctime`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/ctime.html>

Gnulib module: `ctime`

Portability problems fixed by Gnulib:

- On native Windows platforms (mingw, MSVC), this function works incorrectly when the environment variable `TZ` has been set by Cygwin.

Portability problems not fixed by Gnulib:

- This function is deprecated in C23. Likewise, POSIX says this function is obsolescent and it is planned to be removed in a future version. Portable applications can use `localtime_r` and `strftime` (or even `sprintf`) instead. However, `localtime_r` can fail and `strftime` is locale dependent.
- This function may overflow its internal buffer if its argument specifies a time before the year 1000 or after the year 9999.
- This function may dereference an internal null pointer if its argument specifies a time before the year `INT_MIN+1900` or after the year `INT_MAX+1900`.
- The `ctime` function need not be reentrant, and consequently is not required to be thread safe. Implementations of `ctime` typically write the timestamp into static buffer. If two threads call `ctime` at roughly the same time, you might end up with the wrong date in one of the threads, or some undefined string.
- Native Windows platforms (mingw, MSVC) support only a subset of time zones supported by GNU or specified by POSIX. See [Section 10.67.35 \[tzset\]](#), page 474.

Although `ctime` and related functions `asctime`, `asctime_r` and `ctime_r` formerly were plausible to use, they are now unsafe in general, and should be avoided.

Decades ago when `time_t` was narrow and there was no `strftime` or internationalization, code could call these functions and then select the parts needed. For example, in Unix 7th Edition `/usr/src/cmd/ls.c` (1979):

```
cp = ctime(&p->lmtime);
if(p->lmtime < year)
    printf(" %-7.7s %-4.4s ", cp+4, cp+20); else
    printf(" %-12.12s ", cp+4);
```

This had well-defined behavior when `time_t` was only 32 bits and so was OK for circa 1979 platforms.

However, today's platforms have a `time_t` so wide that the year might not be in the range [1000, 9999]. In this case the behavior of `ctime` is undefined and some platforms behave badly, overrunning a buffer or dereferencing an internal null pointer; and even on platforms where no undefined behavior occurs, the 7th Edition code generates wrong output for out-of-range years, because it incorrectly assumes that every year is represented by exactly four digits.

10.67.10 `ctime_r`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/ctime_r.html

Gnulib module: extensions

Portability problems fixed by Gnulib:

- This function has an incompatible declaration on some platforms: Solaris 11.4 (when `_POSIX_PTHREAD_SEMANTICS` is not defined).

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.
- This function is removed in POSIX.1-2024. Use the functions `localtime_r` and `strftime` (or even `sprintf`) instead. However, `localtime_r` can fail and `strftime` is locale dependent.
- This function may overflow its output buffer if its argument specifies a time before the year 1000 or after the year 9999. See [Section 10.67.9 \[ctime\]](#), page 465.
- This function may dereference an internal null pointer if its argument specifies a time before the year `INT_MIN+1900` or after the year `INT_MAX+1900`.

10.67.11 `daylight`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/daylight.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This variable's contents are unreliable if you use a geographical TZ setting like `TZ="America/Los_Angeles"`.
- This variable is missing on some platforms: FreeBSD 14.0, OpenBSD 5.7.
- This variable is called `_daylight` on some platforms: mingw with `-DNO_OLDNAMES`, MSVC.
- The address of this variable is not a compile-time constant on some platforms: mingw, MSVC.
- Native Windows platforms (mingw, MSVC) support only a subset of time zones supported by GNU or specified by POSIX. See [Section 10.67.35 \[tzset\]](#), page 474.

This variable is vestigial: it is not needed and does not suffice in general. It is planned to be removed from a future version of POSIX. A more portable way of getting the UTC offset is to use `strftime` with the `%z` format. See [Section 10.67.21 \[strftime\]](#), page 469.

10.67.12 `difftime`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/difftime.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

10.67.13 `getdate`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getdate.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 14.0, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

Gnulib provides a module `parse-datetime` that contains a function `parse_datetime` that has similar functionality as the `getdate` function.

10.67.14 `getdate_err`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/getdate_err.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This variable is missing on some platforms: FreeBSD 14.0, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

10.67.15 `gmtime`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/gmtime.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- On some platforms, this function yields incorrect values for timestamps before the year 1: MacOS X 10.5, Solaris 11.3.
- On some platforms, this function returns NULL for arguments < -43200, that is, for timestamps before 1969-12-31 12:00:00 UTC: mingw, MSVC.

10.67.16 `gmtime_r`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/gmtime_r.html

Gnulib module: `time_r`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.
- This function is not declared unless `_REENTRANT` is defined, on some platforms: HP-UX 11.

Portability problems not fixed by GnuLib:

- On some platforms, this function yields incorrect values for timestamps before the year 1: MacOS X 10.5, Solaris 11.3.
- This function sets the field `tm_zone` to "GMT" instead of "UTC" on some platforms: glibc, NetBSD, OpenBSD 7.7, Minix, Cygwin, Android.

10.67.17 localtime

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/localtime.html>

GnuLib module: localtime

Portability problems fixed by GnuLib:

- On native Windows platforms (mingw, MSVC), this function works incorrectly when the environment variable TZ has been set by Cygwin.

Portability problems not fixed by GnuLib:

- On some platforms, this function loops forever for values near extrema (such as the year $-2 * 31$): Mac OS X 10.6. You can use the `time_rz` module to work around the problem.
- On some platforms, this function returns nonsense values for unsupported arguments (like 2^{56}), rather than failing: FreeBSD 10.
- On some platforms, this function yields incorrect values for timestamps before the year 1: MacOS X 10.5, Solaris 11.3.
- On some platforms, this function returns NULL for negative arguments, that is, for timestamps before 1970-01-01 00:00:00 local time: mingw, MSVC.
- Native Windows platforms (mingw, MSVC) support only a subset of time zones supported by GNU or specified by POSIX. See [Section 10.67.35 \[tzset\]](#), page 474.

10.67.18 localtime_r

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/localtime_r.html

GnuLib module: time_r

Portability problems fixed by GnuLib:

- This function is missing on some platforms: mingw, MSVC 14.
- This function is not declared unless `_REENTRANT` is defined, on some platforms: HP-UX 11.

Portability problems not fixed by GnuLib:

- On some platforms, this function loops forever for values near extrema (such as the year $-2 * 31$): Mac OS X 10.6. You can use the `time_rz` module to work around the problem.
- On some platforms, this function returns nonsense values for unsupported arguments (like 2^{56}), rather than failing: FreeBSD 10.
- On some platforms, this function yields incorrect values for timestamps before the year 1: MacOS X 10.5, Solaris 11.3.

10.67.19 mktime

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/mktime.html>

Gnulib module: mktime

Portability problems fixed by Gnulib:

- On native Windows platforms (mingw, MSVC), this function works incorrectly when the environment variable TZ has been set by Cygwin.
- mktime may go into an endless loop on some platforms.
- mktime may occasionally return wrong results on some platforms.

Portability problems not fixed by Gnulib:

- On some platforms, this function yields incorrect values for timestamps before the year 1: MacOS X 10.5, Solaris 11.3.
- Native Windows platforms (mingw, MSVC) support only a subset of time zones supported by GNU or specified by POSIX. See [Section 10.67.35 \[tzset\]](#), page 474.

10.67.20 nanosleep

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/nanosleep.html>

Gnulib module: nanosleep

Portability problems fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.
- This function mishandles large arguments when interrupted by a signal on some platforms: Linux 64-bit, Solaris 64-bit.
- This function cannot sleep longer than 49.7 days on some platforms: Cygwin 1.5.x.
- This function does not fail when passed a negative nanosecond value on some platforms: newer 32-bit mingw.

Portability problems not fixed by Gnulib:

10.67.21 strftime

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/strftime.html>

Gnulib module: strftime-fixes

Portability problems fixed by Gnulib:

- On native Windows platforms (mingw, MSVC), this function works incorrectly when the environment variable TZ has been set by Cygwin.

Portability problems not fixed by Gnulib:

- The %c specifier does not work with a time retrieved through gmtime or gmtime_r, since it outputs also the current time zone, on some platforms: glibc 2.30, NetBSD 10.0, Solaris 11.4.
- The %c specifier produces no weekday name and no month name, only a potentially ambiguous numerical output, on some platforms: mingw, MSVC.

- The %h specifier produces empty output on some platforms: mingw, MSVC.
- The %r specifier produces empty output on some platforms: mingw, MSVC.
- The %r specifier produces empty output, at least in a French locale, on some platforms: macOS 12.5, FreeBSD 14.0.
- The %r specifier includes an AM/PM indicator, at least in a French locale, on some platforms: NetBSD 10.0, Solaris 11.4.
- The %p specifier produces non-empty output, at least in a French locale, on some platforms: NetBSD 10.0, Solaris 11.4.
- The Windows C runtime library (which is used by MinGW) does not support the %e specifier (and possibly the other more recent SUS specifiers too, i.e., %C, %D, %h, %n, %r, %R, %t, and %T).
- Native Windows platforms (mingw, MSVC) support only a subset of time zones supported by GNU or specified by POSIX. See [Section 10.67.35 \[tzset\]](#), page 474.

Extension: Gnulib offers a module ‘nstrftime’ that provides an `nstrftime` function with various GNU extensions.

10.67.22 strftime_l

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/strftime_l.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 6.0, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.3, Cygwin 1.7.x, mingw, MSVC 14, Android 4.4.

10.67.23 strptime

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/strptime.html>

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-strptime-3.html

Gnulib module: `strptime`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

10.67.24 time

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/time.html>

Gnulib module: `time`

Portability problems fixed by GnuLib:

- This function is not consistent with `gettimeofday` and `timespec_get` on some platforms: glibc 2.31 or newer on Linux, FreeBSD 14.0, AIX 7.2, native Windows. Namely, in the first 1 to 2.5 milliseconds of every second (or, on AIX and Windows, in the first 5 milliseconds of every second), `time` returns a value that is one less than the `tv_sec` part of the return value of `gettimeofday` or `timespec_get`.

Portability problems not fixed by GnuLib:

10.67.25 `timegm`

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Broken_002ddown-Time.html,
- `man timegm`.

GnuLib module: `timegm`

Portability problems fixed by GnuLib:

- This function is missing on some platforms: AIX 5.1, HP-UX 11, Solaris 11.3, mingw, MSVC 14, Android 3.0.

Portability problems not fixed by GnuLib:

10.67.26 `timer_create`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/timer_create.html

GnuLib module: `timer-time`

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: Mac OS X 10.11, FreeBSD 6.4, Minix 3.1.8, mingw, MSVC 14.
- This function fails with `ENOSYS` on some platforms: OpenBSD 4.9.

10.67.27 `timer_delete`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/timer_delete.html

GnuLib module: `timer-time`

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: Mac OS X 10.11, FreeBSD 6.4, Minix 3.1.8, mingw, MSVC 14.
- This function fails with `ENOSYS` on some platforms: OpenBSD 4.9.

10.67.28 timer_getoverrun

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/timer_getoverrun.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Mac OS X 10.11, FreeBSD 6.0, Minix 3.1.8, Cygwin 2.11.x, mingw, MSVC 14.

10.67.29 timer_gettime

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/timer_gettime.html

Gnulib module: timer-time

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Mac OS X 10.11, FreeBSD 6.4, Minix 3.1.8, mingw, MSVC 14.
- This function fails with `ENOSYS` on some platforms: OpenBSD 4.9.

10.67.30 timer_settime

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/timer_settime.html

Gnulib module: timer-time

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Mac OS X 10.11, FreeBSD 6.4, Minix 3.1.8, mingw, MSVC 14.
- This function fails with `ENOSYS` on some platforms: OpenBSD 4.9.

10.67.31 timespec_get

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.29.2.6

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/timespec_get.html

Gnulib module: timespec_get

Portability problems fixed by Gnulib:

- This function is missing on many platforms: glibc 2.15, macOS 10.14, FreeBSD 11.0, NetBSD 7.1, OpenBSD 6.0, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.3, Cygwin 3.5.x, mingw with MSVCRT, MSVC 14, Android 9.0.

Portability problems not fixed by Gnulib:

10.67.32 `timespec_getres`

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.29.2.7

Gnulib module: `timespec_getres`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: glibc 2.33, macOS 12, FreeBSD 14.0, NetBSD 9.2, OpenBSD 7.0, Minix 3.3.0, AIX 7.1, HP-UX 11.31, Solaris 11.3, Cygwin 2.9, mingw, MSVC 14, Android API level 34.

Portability problems not fixed by Gnulib:

- On some platforms, this function returns a value other than the clock resolution of `timespec_get`, i.e., the minimum distance between differing timestamps. For example, on GNU/Linux it typically returns 1 nanosecond regardless of the actual clock resolution.

The Gnulib module `gettime-res` is a partial substitute; it implements the `TIME_UTC` functionality of `timespec_getres`.

10.67.33 `timezone`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/timezone.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This variable's contents are unreliable if you use a geographical TZ setting like `TZ="America/Los_Angeles"`.
- This variable is missing (and instead, a function named `timezone` exists) on some platforms: FreeBSD 14.0, OpenBSD 5.7.
- This variable is called `_timezone` on some platforms: mingw with `-DNO_OLDNAMES`, MSVC.
- The address of this variable is not a compile-time constant on some platforms: mingw, MSVC.
- Native Windows platforms (mingw, MSVC) support only a subset of time zones supported by GNU or specified by POSIX. See [Section 10.67.35 \[tzset\]](#), page 474.

This variable is vestigial: it is not needed and does not suffice in general. It is planned to be removed from a future version of POSIX. A more portable way of getting the UTC offset is to use `strftime` with the `%z` format. See [Section 10.67.21 \[strftime\]](#), page 469.

10.67.34 `tzname`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/tzname.html>

Gnulib module: `tzname`

Portability problems fixed by Gnulib:

- This variable is called `_tzname` on some platforms: mingw with `-DNO_OLDNAMES`, MSVC.

Portability problems not fixed by Gnulib:

- This variable's contents are unreliable if you use a geographical TZ setting like `TZ="America/Los_Angeles"`.
- The address of this variable is not a compile-time constant on some platforms: Cygwin, mingw, MSVC.
- Native Windows platforms (mingw, MSVC) support only a subset of time zones supported by GNU or specified by POSIX. See [Section 10.67.35 \[tzset\]](#), page 474.

This variable is vestigial: it is not needed and does not suffice in general. It is planned to be removed from a future version of POSIX. A more portable way of getting the time zone abbreviation is to use `strftime` with the `%Z` format. See [Section 10.67.21 \[strftime\]](#), page 469.

10.67.35 `tzset`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/tzset.html>

Gnulib module: `tzset`

Portability problems fixed by Gnulib:

- On native Windows platforms (mingw, MSVC), this function works incorrectly when the environment variable `TZ` has been set by Cygwin.

Portability problems not fixed by Gnulib:

- Native Windows platforms (mingw, MSVC) support only a subset of POSIX-specified values for the `TZ` environment variable, consisting of a time zone abbreviation containing exactly three ASCII letters with no daylight saving time or angle brackets, and with no support for `tz` database settings like `TZ='America/New_York'`. Even this subset does not work on applications built via the Universal Windows Platform, as it does not make environment variables like `TZ` available to applications.
- Older POSIX platforms do not support angle brackets in `TZ` values, as this feature was added in IEEE Std 1003.1-2001.

10.68 Functions in `<uchar.h>`

10.68.1 `c8rtomb`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on most platforms: glibc 2.35, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin, mingw, MSVC 14, Android 9.0.

10.68.2 c16rtomb

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.30.1.4.

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/c16rtomb.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on most non-glibc platforms: glibc 2.15, macOS 14, FreeBSD 6.4, NetBSD 10.0, OpenBSD 7.3, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.3, Cygwin 3.4.x, mingw, MSVC 9, Android 4.4.

10.68.3 c32rtomb

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.30.1.6.

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/c32rtomb.html>

Gnulib module: c32rtomb

Portability problems fixed by Gnulib:

- This function is missing on most non-glibc platforms: glibc 2.15, macOS 14, FreeBSD 6.4, NetBSD 10.0, OpenBSD 7.3, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.3, Cygwin 3.4.x, mingw, MSVC 9, Android 4.4.
- This function returns 0 when the first argument is NULL in some locales on some platforms: AIX 7.2.

Portability problems not fixed by Gnulib:

- This function is only defined as an inline function on some platforms: Haiku 2020.

10.68.4 mbrtoc8

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on most platforms: glibc 2.35, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin, mingw, MSVC 14, Android 9.0.

10.68.5 mbrtoc16

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.30.1.3.

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/mbrtoc16.html>

Gnulib module: mbrtoc16

Portability problems fixed by Gnulib:

- This function is missing on most non-glibc platforms: glibc 2.15, macOS 14, FreeBSD 6.4, NetBSD 10.0, OpenBSD 7.3, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.3, Cygwin 3.4.x, mingw, MSVC 9, Android 4.4.
- This function may crash when the first argument is NULL on some platforms: glibc 2.36.
- In the C or POSIX locales, this function can return `(size_t) -1` and set `errno` to `EILSEQ`: glibc 2.36.
- This function returns 0 instead of `(size_t) -2` when the input is empty: glibc 2.19, Android 11.
- This function returns the total number of bytes that make up the multibyte character, not the number of bytes that were needed to complete the multibyte character, on some platforms: mingw.
- This function returns `(size_t) -3` instead of a byte count when it has stored a high surrogate, and returns a byte count instead of `(size_t) -3` when it has stored a low surrogate, on some platforms: Android.
- This function does not recognize multibyte sequences that `mbrtowc` recognizes on some platforms: FreeBSD 13.2, Solaris 11.4, MSVC 14.

Portability problems not fixed by Gnulib:

- After `mbrtoc16` returns a `char16_t` value, `mbsinit` cannot be used to determine whether the function is ready to return another `char16_t` value. To do so, instead call `mbrtoc16` again, with an appropriately incremented `const char *` argument and an appropriately decremented `size_t` argument.

10.68.6 mbrtoc32

ISO C23 specification:

<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3220.pdf> section 7.30.1.5.

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/mbrtoc32.html>

Gnulib module: `mbrtoc32` or `mbrtoc32-regular`

Portability problems fixed by either Gnulib module `mbrtoc32` or `mbrtoc32-regular`:

- This function is missing on most non-glibc platforms: glibc 2.15, macOS 14, FreeBSD 6.4, NetBSD 10.0, OpenBSD 7.3, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.3, Cygwin 3.4.x, mingw, MSVC 9, Android 4.4.
- In the C or POSIX locales, this function can return `(size_t) -1` and set `errno` to `EILSEQ`: glibc 2.35.
- This function returns 0 instead of `(size_t) -2` when the input is empty: glibc 2.19, mingw, Android 11, Haiku.
- This function does not recognize multibyte sequences that `mbrtowc` recognizes on some platforms: FreeBSD 13.2, Solaris 11.4, mingw, MSVC 14.
- This function does not work when it's fed the input bytes one-by-one on some platforms: Cygwin 3.5.3.

Portability problems fixed by Gnuilib module `mbrtoc32-regular`:

- This function can map some multibyte characters to a sequence of two or more Unicode characters, and may thus return (`size_t`) `-3`. No known implementation currently (2023) behaves that way, but it may theoretically happen. With the `mbrtoc32-regular` module, you have the guarantee that the Gnuilib-provided `mbrtoc32` function maps each multibyte character to exactly one Unicode character and thus never returns (`size_t`) `-3`.
- This function behaves incorrectly when converting precomposed characters from the BIG5-HKSCS encoding: glibc 2.36.

Portability problems not fixed by Gnuilib:

- This function is only defined as an inline function on some platforms: Haiku 2020.

Note: If you want the guarantee that the `char32_t` values returned by this function are Unicode code points, you also need to request the `uchar-h-c23` module.

10.69 Functions in `<unistd.h>`

10.69.1 `_exit`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/_exit.html

Gnuilib module: `unistd-h`

Portability problems fixed by Gnuilib:

- This function is declared in a different header file (namely, `<stdlib.h>`) on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnuilib:

10.69.2 `_Fork`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/_Fork.html

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Creating-a-Process.html

Gnuilib module: `—`

Portability problems fixed by Gnuilib:

Portability problems not fixed by Gnuilib:

- This function is missing on many platforms: glibc 2.33, macOS 14, FreeBSD 13.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android API level 34.

10.69.3 `access`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/access.html>

Gnuilib module: `access`

Portability problems fixed by Gnuilib:

- This function does not support the `X_OK` mode on some platforms: MSVC 14.

- This function does not reject trailing slashes on symlinks to non-directories on some platforms: Mac OS X 10.5.

Portability problems not fixed by GnuLib:

- This function uses the effective id instead of the real id on some platforms: Cygwin 1.5.x.

Other problems of this function:

- There is an inherent race between calling this function and performing some action based on the results; you should think twice before trusting this function, especially in a set-uid or set-gid program.
- This function does not have an option for not following symbolic links (like `stat` versus `lstat`). If you need this option, use the GnuLib module `faccessat` with the `AT_EACCESS` flag.
- On Solaris, for the root user, any file is `X_OK` even if the file does not have the `x` permission bit set.
- On native Windows, files whose basename does not contain a `'.'` cannot be executed through `execlp` or `execvp`. Nevertheless, this function may return true for such files.
- On Windows, different facilities for executing a program have different ways of finding an executable file, by trying various suffixes. For example, `execlp` and `execvp` search for files with the suffixes `.com`, `.exe`, `.bat`, `.cmd`, when the file with the given file name does not exist. Whereas `cmd.exe` searches according to the `PATHEXT` environment variable. This function does not perform any search; it merely looks at the file with the given file name.

10.69.4 alarm

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/alarm.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function has no impact if `SIGALRM` is inherited as ignored; programs should use `signal` (`SIGALRM`, `SIG_DFL`) if it is important to ensure the alarm will fire.
- Use of this function in multi-threaded applications is not advised.
- This function is missing on some platforms: mingw (2011), MSVC 14.
- This function is conditionally declared in the non-standard `<io.h>` header on some platforms: mingw (2012 or newer).

10.69.5 chdir

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/chdir.html>

GnuLib module: `chdir`

Portability problems fixed by GnuLib:

- This function is declared in different header files (namely, `<io.h>` or `<direct.h>`) on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

10.69.6 `chown`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/chown.html>

Gnulib module: `chown`

Portability problems fixed by Gnulib:

- Some platforms fail to detect trailing slash on non-directories, as in `chown("link-to-file/",uid,gid)`: macOS 14, FreeBSD 7.2, AIX 7.3.1, Solaris 9.
- Some platforms fail to update the change time when at least one argument was not `-1`, but no ownership changes resulted: OpenBSD 7.7, Cygwin 2.9.0.
- When passed an argument of `-1`, some implementations really set the owner user/group id of the file to this value, rather than leaving that id of the file alone: some very old platforms.
- When applied to a symbolic link, some implementations don't dereference the symlink, i.e. they behave like `lchown`: some very old platforms.
- This function is missing on some platforms; however, the replacement always fails with `ENOSYS`: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- This function sometimes fails with `EACCES` when the failure is due to lack of appropriate privileges (`EPERM`), not to search permission denied on the file name prefix (`EACCES`): Linux kernel 5.15 with glibc 2.35 and a CIFS v1 file system (see <https://bugs.gnu.org/65599>).

10.69.7 `close`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/close.html>

Gnulib module: `close`

Portability problems fixed by Gnulib:

- This function is declared in a different header file (namely, `<io.h>`) on some platforms: MSVC 14.
- This function crashes when invoked with invalid arguments on some platforms: MSVC 14.
- On Windows platforms (excluding Cygwin), `socket` and `accept` do not return file descriptors that can be closed by `close`. Instead, `closesocket` must be used.

Portability problems not fixed by Gnulib:

10.69.8 `confstr`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/confstr.html>

Gnulib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.5.x, mingw, MSVC 14, Android 9.0.

10.69.9 crypt

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/crypt.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: glibc 2.34, FreeBSD 6.0, NetBSD 5.0, Cygwin 2.9, mingw, MSVC 14, Android 9.0.
- This function is not declared in `<unistd.h>` (without `-D_GNU_SOURCE`) on some platforms: glibc (at least 2.11–2.13).

10.69.10 dup

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/dup.html>

GnuLib module: dup

Portability problems fixed by GnuLib:

- This function is declared in a different header file (namely, `<io.h>`) on some platforms: MSVC 14.
- This function crashes when invoked with invalid arguments on some platforms: MSVC 14.

Portability problems not fixed by GnuLib:

10.69.11 dup2

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/dup2.html>

GnuLib module: dup2

Portability problems fixed by GnuLib:

- This function is declared in a different header file (namely, `<io.h>`) on some platforms: MSVC 14.
- This function always returns 0 for success on some platforms: mingw, MSVC 14.
- This function can hang when duplicating an fd to itself on some platforms: mingw, MSVC 14.
- This function crashes when invoked with invalid arguments on some platforms: Cygwin 1.7.17, MSVC 14.
- This function crashes when invoked with valid arguments on some platforms: Cygwin 1.7.25.

- This function fails with `EINVAL` when duplicating an fd to itself: Android.
- This function resets the `FD_CLOEXEC` flag when duplicating an fd to itself on some platforms: Haiku.
- This function returns 0 for `dup2 (1, 1)` on some platforms: Cygwin 1.5.x.
- This function may return `-EBADF` instead of `-1` on some platforms: Linux releases between July 2008 and May 2009 (versions 2.6.27 to 2.6.29).
- This function returns `EMFILE` instead of `EBADF` for large targets, which interferes with using `dup2(fd,fd)==fd` as the minimal `EBADF` filter: AIX 7.1, FreeBSD 6.1, Cygwin 1.5.

Portability problems not fixed by Gnulib:

10.69.12 dup3

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/dup3.html>

Documentation:

`man dup3`

Gnulib module: `dup3`

Portability problems fixed by Gnulib:

- This function is missing on many non-glibc platforms: glibc 2.8, macOS 14, FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.3, Cygwin 1.7.1, mingw, MSVC 14, Android 4.4.
- This function can crash on some platforms: Cygwin 1.7.25.
- This function mistakenly succeeds when given two equal file descriptors on some platforms: NetBSD 10.0, Haiku.

Portability problems not fixed by Gnulib:

10.69.13 encrypt

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/encrypt.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.
- This function is not declared in `<unistd.h>` (without `-D_GNU_SOURCE`) on some platforms: glibc (at least 2.11–2.13).
- POSIX.1-2024 says this function is obsolescent and it is planned to be removed in a future version.

10.69.14 environ

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/environ.html>

Gnulib module: environ

Portability problems fixed by Gnulib:

- POSIX does not require this variable to be declared, and it is indeed not declared on some platforms: macOS 14, FreeBSD 14.0, NetBSD 5.0, OpenBSD 3.8, Solaris 11.4.
- On macOS, this variable is not declared. Up to Mac OS X 10.4, one can use

```
extern char **environ;
```

to get the variable declared. This does not work any more, however, in shared libraries on macOS 10.5. Here is a workaround: Instead, one can use

```
#include <crt_externs.h>
#define environ (*_NSGetEnviron())
```

This works at all versions of macOS.

- On Cygwin in 64-bit mode, references to this variable cause a link error when the option `-Wl,--disable-auto-import` is in use.

Portability problems not fixed by Gnulib:

- The address of this variable is not a compile-time constant on some platforms: mingw.
- Assigning NULL to `environ` to clear all variables is not portable; better is to assign `environ` to one-element array containing a NULL pointer. That said, an empty environment is not portable either, as some systems may require particular environment variables (such as PATH) to be present in order to operate consistently.

10.69.15 execl

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/execl.html>

Gnulib module: execl

Portability problems fixed by Gnulib:

- On Windows platforms (excluding Cygwin), this function does not pass command-line arguments correctly if they contain space, tab, backslash, or double-quote characters.
- On Windows platforms (excluding Cygwin), this function spawns an asynchronous child process and then exits the current process immediately. As a consequence, the parent of the current process 1. may incorrectly proceed as if its child had exited, and 2. will never see the child's exit status.
- On Windows platforms (excluding Cygwin), the return type of this function is `intptr_t`, not `int`.

Note: The Gnulib replacement for this function is not async-safe, that is, it must not be invoked from a signal handler.

Portability problems not fixed by Gnulib:

- On some platforms, a script without executable permission is still run: Cygwin 1.5.x.

10.69.16 `execle`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/execle.html>

Gnulib module: `execle`

Portability problems fixed by Gnulib:

- On Windows platforms (excluding Cygwin), this function does not pass command-line arguments correctly if they contain space, tab, backslash, or double-quote characters.
- On Windows platforms (excluding Cygwin), this function spawns an asynchronous child process and then exits the current process immediately. As a consequence, the parent of the current process 1. may incorrectly proceed as if its child had exited, and 2. will never see the child's exit status.
- On Windows platforms (excluding Cygwin), the return type of this function is `intptr_t`, not `int`.

Note: The Gnulib replacement for this function is not async-safe, that is, it must not be invoked from a signal handler.

Portability problems not fixed by Gnulib:

- On some platforms, a script without executable permission is still run: Cygwin 1.5.x.

10.69.17 `execlp`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/execlp.html>

Gnulib module: `execlp`

Portability problems fixed by Gnulib:

- On Windows platforms (excluding Cygwin), this function does not pass command-line arguments correctly if they contain space, tab, backslash, or double-quote characters.
- On Windows platforms (excluding Cygwin), this function spawns an asynchronous child process and then exits the current process immediately. As a consequence, the parent of the current process 1. may incorrectly proceed as if its child had exited, and 2. will never see the child's exit status.
- On Windows platforms (excluding Cygwin), the return type of this function is `intptr_t`, not `int`.

Portability problems not fixed by Gnulib:

- On some platforms, a script without executable permission is still run: Cygwin 1.5.x.

10.69.18 `execv`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/execv.html>

Gnulib module: `execv`

Portability problems fixed by Gnulib:

- On Windows platforms (excluding Cygwin), this function does not pass command-line arguments correctly if they contain space, tab, backslash, or double-quote characters.

- On Windows platforms (excluding Cygwin), this function spawns an asynchronous child process and then exits the current process immediately. As a consequence, the parent of the current process 1. may incorrectly proceed as if its child had exited, and 2. will never see the child's exit status.
- On Windows platforms (excluding Cygwin), the return type of this function is `intptr_t`, not `int`.

Note: The Gnuilib replacement for this function is not async-safe, that is, it must not be invoked from a signal handler.

Portability problems not fixed by Gnuilib:

- On some platforms, a script without executable permission is still run: Cygwin 1.5.x.

10.69.19 `execve`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/execve.html>

Gnuilib module: `execve`

Portability problems fixed by Gnuilib:

- On Windows platforms (excluding Cygwin), this function does not pass command-line arguments correctly if they contain space, tab, backslash, or double-quote characters.
- On Windows platforms (excluding Cygwin), this function spawns an asynchronous child process and then exits the current process immediately. As a consequence, the parent of the current process 1. may incorrectly proceed as if its child had exited, and 2. will never see the child's exit status.
- On Windows platforms (excluding Cygwin), the return type of this function is `intptr_t`, not `int`.

Note: The Gnuilib replacement for this function is not async-safe, that is, it must not be invoked from a signal handler.

Portability problems not fixed by Gnuilib:

- On some platforms, a script without executable permission is still run: Cygwin 1.5.x.

10.69.20 `execvp`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/execvp.html>

Gnuilib module: `execvp`

Portability problems fixed by Gnuilib:

- On Windows platforms (excluding Cygwin), this function does not pass command-line arguments correctly if they contain space, tab, backslash, or double-quote characters.
- On Windows platforms (excluding Cygwin), this function spawns an asynchronous child process and then exits the current process immediately. As a consequence, the parent of the current process 1. may incorrectly proceed as if its child had exited, and 2. will never see the child's exit status.
- On Windows platforms (excluding Cygwin), the return type of this function is `intptr_t`, not `int`.

Portability problems not fixed by Gnulib:

- On some platforms, a script without executable permission is still run: Cygwin 1.5.x.

10.69.21 faccessat

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/faccessat.html>

Gnulib module: faccessat

Portability problems fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 10.9, FreeBSD 7.4, NetBSD 6.1.5, OpenBSD 4.9, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 10, Cygwin 1.5.x, mingw, MSVC 14, Android 4.0.4.
- On some platforms, `faccessat (dfd, "file/", amode, flag)` succeeds instead of failing when `file` is not a directory. macOS 14.

Portability problems not fixed by Gnulib:

- The replacement does not always take ACLs into account.
- The replacement is not safe to be used in libraries.
- The replacement is not multithread-safe.
- The replacement does not support the `AT_SYMLINK_NOFOLLOW` flag, which is supported by GNU `faccessat`.
- On some platforms, `faccessat` can mishandle `AT_EACCESS` after a process starts as root and then becomes non-root: GNU/Linux with glibc 2.32.

Other problems of this function:

- There is an inherent race between calling this function and performing some action based on the results; you should think twice before trusting this function, especially in a set-uid or set-gid program.

10.69.22 fchdir

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fchdir.html>

Gnulib module: fchdir

Portability problems fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14. But the replacement function is not safe to be used in libraries and is not multithread-safe.

Portability problems not fixed by Gnulib:

- This function fails if the file descriptor was opened with `O_PATH`: Linux kernel 3.4.

10.69.23 fchown

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fchown.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.
- This function sometimes fails with `EACCES` when the failure is due to lack of appropriate privileges (`EPERM`), not to search permission denied on the file name prefix (`EACCES`): Linux kernel 5.15 with glibc 2.35 and a CIFS v1 file system (see <https://bugs.gnu.org/65599>).

10.69.24 `fchownat`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fchownat.html>

Gnulib module: `fchownat`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, Mac OS X 10.5, FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 1.5.x, mingw, MSVC 14. But the replacement function is not safe to be used in libraries and is not multithread-safe. Also, the replacement may fail to change symlinks if `lchown` is unsupported, or fail altogether if `chown` is unsupported.
- This function is declared in `<sys/stat.h>`, not in `<unistd.h>`, on some platforms: Android 4.3.
- Some platforms fail to detect trailing slash on non-directories, as in `fchown(dir, "link-to-file/", uid, gid, flag)`: Solaris 9.
- Some platforms fail to update the change time when at least one argument was not `-1`, but no ownership changes resulted: OpenBSD 7.7.
- Some platforms mistakenly dereference symlinks when using `AT_SYMLINK_NOFOLLOW`: Linux kernel 2.6.17.
- This function does not fail for an empty filename on some platforms: Linux with glibc < 2.11.

Portability problems not fixed by Gnulib:

- This function sometimes fails with `EACCES` when the failure is due to lack of appropriate privileges (`EPERM`), not to search permission denied on the file name prefix (`EACCES`): Linux kernel 5.15 with glibc 2.35 and a CIFS v1 file system (see <https://bugs.gnu.org/65599>).

10.69.25 `fdatasync`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fdatasync.html>

Gnulib module: `fdatasync`

Portability problems fixed by Gnulib:

- This function is present but not declared on some platforms: Mac OS X 10.7.
- This function is missing on some platforms: Mac OS X 10.5, FreeBSD 11.0, OpenBSD 3.8, Minix 3.1.8, mingw, MSVC 14, Android 2.2.

Portability problems not fixed by GnuLib:

- If the argument is a file descriptor that lacks write access, such as a directory file descriptor, this function fails with `EBADF` on some platforms: AIX 7.3, Cygwin 3.6.x.

10.69.26 `fexecve`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fexecve.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 6.0, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11 2010-11, Cygwin 1.5.x, mingw, MSVC 14, Android 8.1.

10.69.27 `fork`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fork.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: mingw, MSVC 14.
- On some platforms, `fork` followed by a call of the `exec` family (`execl`, `execvp`, `execle`, `execv`, `execvp`, or `execve`) is less efficient than `vfork` followed by the same call. `vfork` is a variant of `fork` that has been introduced to optimize the `fork/exec` pattern.
- On Windows platforms (excluding Cygwin), this function is not implemented; use `_spawnvp` instead.

10.69.28 `fpathconf`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fpathconf.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: mingw, MSVC 14.

10.69.29 `fsync`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fsync.html>

GnuLib module: `fsync`

Portability problems fixed by GnuLib:

- This function is missing on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- If the argument is a file descriptor that lacks write access, such as a directory file descriptor, this function fails with `EBADF` on some platforms: AIX 7.3, Cygwin 3.6.x.

10.69.30 `ftruncate`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/ftruncate.html>

Gnulib module: `ftruncate`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: MSVC 14.
- On platforms where `off_t` is a 32-bit type, this function may not work correctly on files 2 GiB and larger. See [Section 14.2 \[Large File Support\]](#), page 842.

Portability problems not fixed by Gnulib:

10.69.31 `getcwd`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getcwd.html>

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-getcwd.html

Gnulib module: `getcwd` or `getcwd-lgpl`

Portability problems fixed by either Gnulib module `getcwd` or `getcwd-lgpl`:

- This function is declared in different header files (namely, `<io.h>` or `<direct.h>`) on some platforms: mingw, MSVC 14.
- On glibc and Solaris 11.4 platforms, `getcwd (NULL, n)` allocates memory for the result. On some other platforms, this call is not allowed.
- On some platforms, the prototype for `getcwd` uses `int` instead of `size_t` for the size argument when using non-standard headers, and the declaration is missing from `<unistd.h>`: mingw, MSVC 14.
- On some platforms, `getcwd (buf, 0)` crashes: MSVC 14.
- On some platforms, `getcwd (buf, 0)` fails with `ERANGE` instead of the required `EINVAL`: mingw.

Portability problems fixed by Gnulib module `getcwd`:

- This function does not handle long file names (greater than `PATH_MAX`) correctly on some platforms: glibc on Linux 2.4.20, musl libc 1.2.2/powerpc64le, macOS 13, FreeBSD 6.4, NetBSD 10.0, OpenBSD 7.5, AIX 7.1.

Portability problems not fixed by Gnulib:

- When using `getcwd (NULL, nonzero)`, some platforms, such as glibc or cygwin, allocate exactly `nonzero` bytes and fail with `ERANGE` if it was not big enough, while other platforms, such as FreeBSD, mingw, or MSVC 14, ignore the size argument and allocate whatever size is necessary. If this call succeeds, an application cannot portably access beyond the string length of the result.

10.69.32 getegid

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getegid.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

10.69.33 getentropy

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getentropy.html>

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Unpredictable-Bytes.html,
- `man getentropy`.

Gnulib module: getentropy

Portability problems fixed by Gnulib:

- This function is missing on some platforms: glibc 2.24, Mac OS X 10.11, FreeBSD 11.0, NetBSD 9.3, OpenBSD 5.5, Minix 3.3, AIX 7.1, HP-UX 11.31, Solaris 11.0, Cygwin 1.7.x, mingw, MSVC 14, Android 8.1.
- This function is declared in `<sys/random.h>`, not in `<unistd.h>`, on some platforms: macOS 14, Solaris 11.4, Android 9.0.

Portability problems not fixed by Gnulib:

Although this function is intended to produce random data, the data’s security properties may not be appropriate for your application. For example, identical “random” data streams might be produced by rebooted virtual machines. If this is of concern you may need to use additional techniques such as hedging.¹

Related modules include `getrandom`, which has a more-flexible but more-complex API, and `crypto/gc-random`, which is likely a better match for code already using the other `crypto` APIs.

10.69.34 geteuid

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/geteuid.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

¹ Ristenpart T, Yilek S. When good randomness goes bad: virtual machine vulnerabilities and hedging deployed cryptography (<https://pages.cs.wisc.edu/~rist/papers/sslhedge.pdf>). NDSS 2010.

10.69.35 `getgid`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getgid.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

10.69.36 `getgroups`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getgroups.html>

Gnulib module: `getgroups`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.
- On some platforms, this function fails to reject a negative count, even though that is less than the size that would be returned: macOS 14, FreeBSD 13.0.
- On NeXTstep 3.2, `getgroups(0, NULL)` always fails. See macro `'AC_FUNC_GETGROUPS'`.
- On very old systems, this function operated on an array of `'int'`, even though that was a different size than an array of `'gid_t'`.

Portability problems not fixed by Gnulib:

- This function is unsafe to call between `fork` and `exec` if the parent process is multi-threaded.
- It is unspecified whether the effective group id will be included in the returned list, nor whether the list will be sorted in any particular order. For that matter, some platforms include the effective group id twice, if it is also a member of the current supplemental group ids.

The Gnulib module `mgetgroups` provides a similar API.

10.69.37 `gethostid`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/gethostid.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14, Android 9.0.

10.69.38 gethostname

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/gethostname.html>

Gnulib module: gethostname

Portability problems fixed by Gnulib:

- This function is declared in `<winsock2.h>` instead of `<unistd.h>` on some platforms: MSVC 14.
- On mingw and MSVC 14, this function has a prototype that differs from that specified by POSIX, and it is defined only in the `ws2_32` library.

Portability problems not fixed by Gnulib:

- This function's second argument type is `int` instead of `size_t` on some platforms: Solaris 10, Solaris 11 OmniOS.
- If the given buffer is too small for the host name, some implementations fail with `EINVAL`, instead of returning a truncated host name.

10.69.39 getlogin

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getlogin.html>

Gnulib module: getlogin

Portability problems fixed by Gnulib:

- This function is missing on some platforms: older mingw, MSVC 14.
- This function is not declared unless `_POSIX` is defined on some platforms: mingw.
- This function returns the value of the `LOGNAME` environment variable and is therefore arbitrarily fakeable: musl libc 1.2.5.

Portability problems not fixed by Gnulib:

- This function returns an empty string even when standard input is a tty on some platforms: HP-UX 11.11.
- When there are several user names with the same user ID, this function may return, instead of the user name that logged in, another user name that has the same user ID, on some platforms: Linux and others.

10.69.40 getlogin_r

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/getlogin_r.html

Gnulib module: getlogin_r

Portability problems fixed by Gnulib:

- This function is missing on some platforms: NetBSD 3.0, Minix 3.1.8, mingw, MSVC 14, Android 8.1.
- This function is not declared unless `_REENTRANT` is defined, on some platforms: HP-UX 11.
- This function returns a truncated result, instead of failing with error code `ERANGE`, when the buffer is not large enough, on some platforms: macOS 14.

- This function returns the value of the `LOGNAME` environment variable and is therefore arbitrarily fakeable: musl libc 1.2.5.

Portability problems not fixed by Gnulib:

- This function has an incompatible declaration on some platforms: FreeBSD 12.0, MidnightBSD 2.0, Solaris 11.4 (when `_POSIX_PTHREAD_SEMANTICS` is not defined).
- This function fails even when standard input is a tty on some platforms: HP-UX 11.11.
- This function fails with error code `EINVAL` instead of `ERANGE` when the second argument is zero on some platforms: HP-UX 11.31.
- This function fails with error code `ENOMEM` instead of `ERANGE` on some platforms: Haiku.
- When there are several user names with the same user ID, this function may return, instead of the user name that logged in, another user name that has the same user ID, on some platforms: Linux and others.

10.69.41 getopt

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getopt.html>

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/libutil-getopt-3.html

Gnulib module: `getopt-posix` or `getopt-gnu`

The module `getopt-gnu` has support for “long options” and for “options that take optional arguments”. Compared to the API defined by POSIX, it adds a header file `<getopt.h>` and a function `getopt_long`.

Portability problems fixed by either Gnulib module `getopt-posix` or `getopt-gnu`:

- This function is missing on some platforms: MSVC 14.
- The value of `optind` after a missing required argument is wrong on some platforms: macOS 14, AIX 7.1, mingw.

Portability problems fixed by Gnulib module `getopt-gnu`:

- The function `getopt` does not support the ‘+’ flag in the options string on some platforms: macOS 14, AIX 5.2, HP-UX 11, Solaris 11 2010-11.
- The function `getopt` does not obey the combination of ‘+’ and ‘:’ flags in the options string on some platforms: glibc 2.11.
- The function `getopt` does not obey the ‘-’ flag in the options string when `POSIXLY_CORRECT` is set on some platforms: Cygwin 1.7.0.
- The function `getopt` does not support options with optional arguments on some platforms: macOS 14, OpenBSD 4.0, AIX 5.2, HP-UX 11, Solaris 11 2010-11, Cygwin 1.5.x.
- The function `getopt_long` is missing on some platforms: AIX 5.1, HP-UX 11, Solaris 9, MSVC 14.
- The function `getopt_long` does not support abbreviated long options where all disambiguations are equivalent on some platforms: OpenBSD 5.0.

- The function `getopt_long_only` is missing on some platforms: FreeBSD 5.2.1, NetBSD 10.0, AIX 5.1, HP-UX 11, Solaris 9, mingw, MSVC 14.
- This function crashes if the option string includes `W`; on some platforms: glibc 2.14.

Portability problems not fixed by Gnulib:

- The default behavior of the glibc implementation of `getopt` allows mixing option and non-option arguments on the command line in any order. Other implementations, such as the one in Cygwin, enforce strict POSIX compliance: they require that the option arguments precede the non-option arguments. This is something to watch out in your program's testsuite.
- The glibc implementation allows a complete reset of the environment, including re-checking for `POSIXLY_CORRECT`, by setting `optind` to 0. Several BSD implementations provide `optreset`, causing a reset by setting it non-zero, although it does not necessarily re-read `POSIXLY_CORRECT`. Solaris `getopt` does not support either reset method, but does not maintain state that needs the extra level of reset.
- On some platforms, this function does not set the stream error indicator on attempts to write to a read-only stream: glibc 2.13, Cygwin 1.7.9.

10.69.42 `getpgid`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getpgid.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

10.69.43 `getpgrp`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getpgrp.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

10.69.44 `getpid`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getpid.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

10.69.45 getppid

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getppid.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

10.69.46 getresgid

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getresgid.html>

Documentation:

`man getresgid`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, NetBSD 10.0, Minix 3.1.8, AIX 5.1, HP-UX 11.00, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14.

10.69.47 getresuid

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getresuid.html>

Documentation:

`man getresuid`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, NetBSD 10.0, Minix 3.1.8, AIX 5.1, HP-UX 11.00, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14.

10.69.48 getsid

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getsid.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14, Android 4.1.

10.69.49 `getuid`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getuid.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

10.69.50 `isatty`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/isatty.html>

Gnulib module: `isatty`

Portability problems fixed by Gnulib:

- This function is declared in a different header file (namely, `<io.h>`) on some platforms: MSVC 14.
- On native Windows, this function also returns true for character devices such as NUL.
- On native Windows, this function returns false for Cygwin consoles.
- This function crashes when invoked with invalid arguments on some platforms: MSVC 14.

Portability problems not fixed by Gnulib:

10.69.51 `lchown`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/lchown.html>

Gnulib module: `lchown`

Portability problems fixed by Gnulib:

- This function is missing on some platforms; however, the replacement fails with `ENOSYS`: mingw, MSVC 14.
- This function is missing on some platforms; however, the replacement fails on symlinks: Minix 3.2.1.
- Some platforms fail to detect trailing slash on non-directories, as in `lchown("link-to-file/",uid,gid)`: FreeBSD 7.2, Solaris 9.
- Some platforms fail to update the change time when at least one argument was not `-1`, but no ownership changes resulted: OpenBSD 7.7.

Portability problems not fixed by Gnulib:

- This function sometimes fails with `EACCES` when the failure is due to lack of appropriate privileges (`EPERM`), not to search permission denied on the file name prefix (`EACCES`): Linux kernel 5.15 with glibc 2.35 and a CIFS v1 file system (see <https://bugs.gnu.org/65599>).

10.69.52 link

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/link.html>

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-link-1.html

Gnulib module: link

Portability problems fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.
- This function fails to reject trailing slashes on non-directories on some platforms: FreeBSD 7.2, Solaris 11.3, Cygwin 1.5.x.

Portability problems not fixed by Gnulib:

- When the first argument is a symlink, some platforms create a hard-link to what the symlink referenced, rather than to the symlink itself. Use ‘linkat’ to force a particular behavior.

10.69.53 linkat

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/linkat.html>

Gnulib module: linkat

Portability problems fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, Mac OS X 10.9, FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 10, Cygwin 1.5.x, mingw, MSVC 14, Android 4.4. But the replacement function is not safe to be used in libraries and is not multithread-safe.
- This function fails to directly hardlink symlinks on some platforms: Mac OS X 10.10.
- This function fails to reject trailing slashes on non-directories on some platforms: macOS 14, AIX 7.1, Solaris 11.3.
- This functions does not support AT_SYMLINK_FOLLOW on some platforms: Linux kernel 2.6.17.

Portability problems not fixed by Gnulib:

- linkat (fd, "directory", fd, "symlink/" ...) sets `errno` to `EINVAL` instead of `EEXIST` or `ENOTDIR` on some platforms: GNU/Hurd with glibc 2.41.

10.69.54 lockf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/lockf.html>

Gnulib module: —

Portability problems fixed by Gnulib:

- On platforms where `off_t` is a 32-bit type, this function may not work correctly across the entire data range of files 2 GiB and larger. See [Section 14.2 \[Large File Support\]](#), [page 842](#).

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.5.x, mingw, MSVC 14, Android 6.0.

10.69.55 `lseek`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/lseek.html>

Gnulib module: `lseek`

Portability problems fixed by Gnulib:

- On some platforms, `lseek (fd, offset, SEEK_DATA)` returns a value greater than `offset` even when `offset` addresses data: macOS 14.
- This function is declared in a different header file (namely, `<io.h>`) on some platforms: MSVC 14.
- On platforms where `off_t` is a 32-bit type, `lseek` does not work correctly with files 2 GiB and larger. See [Section 14.2 \[Large File Support\]](#), page 842.
- This function mistakenly succeeds on pipes on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- POSIX does not specify which file descriptors support seeking and which don't. In practice, regular files and block devices support seeking, and ttys, pipes, and most character devices don't support it.
- Some systems do not support `SEEK_DATA` and `SEEK_HOLE`: AIX, HP-UX, Microsoft Windows, NetBSD, OpenBSD, Cygwin 3.4.6.
- Some systems have a buggy `SEEK_DATA` and `SEEK_HOLE`, and Gnulib works around the problem via `#undef SEEK_DATA` and `#undef SEEK_HOLE`: FreeBSD 13, macOS 12.

10.69.56 `nice`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/nice.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.
- In glibc before glibc 2.2.4, `nice` returned 0 upon success.

10.69.57 `optarg`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/optarg.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This variable is missing on some platforms: Mac OS X 10.5, MSVC 14.

10.69.58 opterr

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/opterr.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This variable is missing on some platforms: MSVC 14.

10.69.59 optind

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/optind.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This variable is missing on some platforms: MSVC 14.

10.69.60 optopt

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/optopt.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This variable is missing on some platforms: Mac OS X 10.5, MSVC 14.

10.69.61 pathconf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/pathconf.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

10.69.62 pause

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/pause.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

10.69.63 pipe

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/pipe.html>

Gnulib module: pipe-posix

Portability problems fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- This function crashes rather than failing with `EMFILE` if no resources are left on some platforms: Cygwin 1.7.9.

10.69.64 pipe2

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/pipe2.html>

Documentation:

`man pipe2`

Gnulib module: pipe2

Portability problems fixed by Gnulib:

- This function is missing on many non-glibc platforms: glibc 2.8, macOS 14, FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.3, Cygwin 1.7.1, mingw, MSVC 14, Android 2.2. But the replacement function is not atomic; this matters in multi-threaded programs that spawn child processes.

Portability problems not fixed by Gnulib:

- This function crashes rather than failing with `EMFILE` if no resources are left on some platforms: Cygwin 1.7.9.

Note: This function portably supports the `O_NONBLOCK` flag only if the gnulib module `nonblocking` is also used.

10.69.65 posix_close

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/posix_close.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc 2.42, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android 9.0.

10.69.66 pread

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/pread.html>

Gnulib module: pread

Portability problems fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.
- On platforms where `off_t` is a 32-bit type, this function may not work correctly on files 2 GiB and larger. See [Section 14.2 \[Large File Support\]](#), page 842.
- This function returns zero instead of positive values when large file support is enabled on some platforms: HP-UX 11.11.
- This function does not fail on pipes on some platforms: HP-UX 11.31.

Portability problems not fixed by Gnulib:

10.69.67 `pwrite`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/pwrite.html>

Gnulib module: `pwrite`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.
- On platforms where `off_t` is a 32-bit type, this function may not work correctly on files 2 GiB and larger. See [Section 14.2 \[Large File Support\]](#), page 842.
- This function does not fail when an invalid (negative) offset is passed when large file support is enabled on some platforms: HP-UX 11.11.
- This function uses an arbitrary offset instead of the `off_t` argument when large file support is enabled on some platforms: HP-UX 11.11.

Portability problems not fixed by Gnulib:

10.69.68 `read`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/read.html>

Gnulib module: `read`, `stdio-h`, `nonblocking`

Portability problems fixed by Gnulib module `read`:

- This function is declared in a different header file (namely, `<io.h>`) on some platforms: mingw, MSVC 14.
- This function crashes when invoked with invalid arguments on some platforms: MSVC 14.

Portability problems fixed by Gnulib module `stdio-h`, together with module `nonblocking`:

- When reading from a non-blocking pipe whose buffer is empty, this function fails with `errno` being set to `EINVAL` instead of `EAGAIN` on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- This function may fail with error `EINTR`, even in programs that don't install any signal handlers, on some platforms: macOS 14.

For handling `EINTR`, Gnulib provides a module `'safe-read'` with a function `safe_read`.

10.69.69 readlink

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/readlink.html>

Gnulib module: readlink

Portability problems fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.
- Some platforms mistakenly succeed on file names ending in `/`: FreeBSD 7.2, Solaris 9, macOS 14.
- On some platforms, this function returns `int` instead of `ssize_t`: glibc 2.4, FreeBSD 6.0, OpenBSD 7.5, Cygwin 1.5.x, AIX 7.1.
- For the file name `/dev/null`, this function returns `\Device\Null`, which is unusable, on some platforms: Cygwin 3.3.6.

Portability problems mostly fixed by Gnulib:

- On some platforms, this function fails and sets `errno` to `ERANGE` rather than returning truncated contents: AIX 7.2, HP-UX 11. The Gnulib replacement normally works as POSIX requires by returning the truncated contents. However, if the full link contents are unreasonably large (more than 4000 bytes) the replacement clears the entire buffer and returns the buffer size; although this is not a complete fix, it suffices for typical callers, which ignore the buffer contents anyway.

Portability problems not fixed by Gnulib:

- This function always fails on platforms that don't support symlinks: mingw, MSVC 14.
- When this function is called on a directory: In the case of NFS mounted directories, Cygwin sets `errno` to `ENOENT` or `EIO` instead of `EINVAL`. To avoid this problem, check for a directory before calling this function.
- When this function is called on a file that is not a symbolic link: Cygwin may set `errno` to `EACCES` instead of `EINVAL`.
- When this function fails because it is called on an existing non-directory's name concatenated to `/`, it sets `errno` to `EINVAL`: AIX 7.2.
- Symlink contents do not always have a trailing null byte, and there is no indication if symlink contents were truncated if the return value matches the length. Furthermore, Linux sets `errno` to `EINVAL` if the requested length is zero. Use the gnulib module `areadlink` for improved ability to read symlink contents.

10.69.70 readlinkat

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/readlinkat.html>

Gnulib module: readlinkat

Portability problems fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, Mac OS X 10.9, FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 10, Cygwin 1.5.x, mingw, MSVC 14, Android 4.4. But the replacement function is not safe to be used in libraries and is not multithread-safe.

- Some platforms mistakenly succeed on file names ending in `/`: macOS 14.
- On some platforms, this function returns `int` instead of `ssize_t`: AIX 7.1.
- For the file name `/dev/null`, this function returns `\Device\Null`, which is unusable, on some platforms: Cygwin 3.3.6.

Portability problems mostly fixed by Gnulib:

- On some platforms, this function fails and sets `errno` to `ERANGE` rather than returning truncated contents: AIX 7.2. The Gnulib replacement normally works as POSIX requires by returning the truncated contents. However, if the full link contents are unreasonably large (more than 4000 bytes) the replacement clears the entire buffer and returns the buffer size; although this is not a complete fix, it suffices for typical callers, which ignore the buffer contents anyway.

Portability problems not fixed by Gnulib:

- This function always fails on platforms that don't support symlinks: mingw, MSVC 14.
- When this function is called on a directory: In the case of NFS mounted directories, Cygwin sets `errno` to `ENOENT` or `EIO` instead of `EINVAL`. To avoid this problem, check for a directory before calling this function.
- When this function is called on a file that is not a symbolic link: Cygwin may set `errno` to `EACCES` instead of `EINVAL`.
- When this function fails because it is called on an existing non-directory's name concatenated to `/`, it sets `errno` to `EINVAL`: AIX 7.2.
- When this function is called on an empty file name, it fails with error `EBADF` instead of `ENOENT`: Cygwin 3.4.6.
- Symlink contents do not always have a trailing null byte, and there is no indication if symlink contents were truncated if the return value matches the length. Furthermore, Linux sets `errno` to `EINVAL` if the requested length is zero. Use the gnu lib module `areadlink` for improved ability to read symlink contents.

10.69.71 `rmdir`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/rmdir.html>

Gnulib module: `rmdir`

Portability problems fixed by Gnulib:

- This function is declared in different header files (namely, `<io.h>` or `<direct.h>`) on some platforms: mingw, MSVC 14.
- This function mistakenly removes a directory with `rmdir("dir/.")` on some platforms: Cygwin 1.5.x.
- This function fails with `EINVAL` instead of the expected `ENOTDIR` for `rmdir("file/")` on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- When `rmdir` fails because the specified directory is not empty, the `errno` value is system dependent.

- POSIX requires that `rmdir("link-to-empty/")` remove `empty` and leave `link-to-empty` as a dangling symlink. This is counter-intuitive, so some systems fail with `ENOTDIR` instead: glibc

10.69.72 `setegid`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/setegid.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: HP-UX 11.23, mingw, MSVC 14.

10.69.73 `seteuid`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/seteuid.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: HP-UX 11.23, mingw, MSVC 14.

10.69.74 `setgid`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/setgid.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

10.69.75 `setpgid`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/setpgid.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

10.69.76 `setregid`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/setregid.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

10.69.77 setresgid

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/setresgid.html>

Documentation:

`man setresgid`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, NetBSD 10.0, Minix 3.1.8, AIX 5.1, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14.

10.69.78 setresuid

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/setresuid.html>

Documentation:

`man setresuid`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, NetBSD 10.0, Minix 3.1.8, AIX 5.1, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14.

10.69.79 setreuid

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/setreuid.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

10.69.80 setsid

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/setsid.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

10.69.81 `setuid`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/setuid.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

10.69.82 `sleep`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/sleep.html>

Gnulib module: `sleep`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: mingw (2005 or newer), MSVC 14.
- This function takes milliseconds as argument and returns `void` on some platforms: mingw (2005 and older).
- This function cannot sleep longer than 49.7 days on some platforms: Cygwin 1.5.x.

Portability problems not fixed by Gnulib:

- According to POSIX, the `sleep` function may interfere with the program's use of the `SIGALRM` signal. On Linux, it doesn't; on other platforms, it may.

10.69.83 `swab`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/swab.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Android 8.1.

10.69.84 `symlink`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/symlink.html>

Gnulib module: `symlink`

Portability problems fixed by Gnulib:

- On some systems, `symlink(value, "name/")` mistakenly creates a symlink: macOS 14, FreeBSD 7.2, AIX 7.1, Solaris 9.
- This function is missing on some platforms; however, the replacement always fails with `EPERM`: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- Some file systems do not support symbolic links.

- If the target is the empty string, this function can create an empty symbolic link instead of failing with `ENOENT` as GNU/Linux does: FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.8, Solaris 11.4. On Solaris, the empty symbolic link `""` resolves like `."` does; on the other operating systems, resolution fails with `ENOENT`.

10.69.85 `symlinkat`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/symlinkat.html>

Gnulib module: `symlinkat`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, Mac OS X 10.5, FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 10, Cygwin 1.5.x, mingw, MSVC 14, Android 4.4. But the replacement function is not safe to be used in libraries and is not multithread-safe.
- Some platforms declare this function in `stdio.h` instead of `unistd.h`: Cygwin 1.7.1.
- On some systems, `symlinkat(value, fd, "name/")` mistakenly creates a symlink: macOS 14, AIX 7.1.

Portability problems not fixed by Gnulib:

- This function always fails with `ENOSYS` on platforms that don't support symlinks: mingw, MSVC 14.
- If the target is the empty string, this function can create an empty symbolic link instead of failing with `ENOENT` as GNU/Linux does: FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.8, Solaris 11.4. On Solaris, the empty symbolic link `""` resolves like `."` does; on the BSDs, resolution fails with `ENOENT`.

10.69.86 `sync`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/sync.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

10.69.87 `sysconf`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/sysconf.html>

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-sysconf.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

10.69.88 `tcgetpgrp`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/tcgetpgrp.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

10.69.89 `tcsetpgrp`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/tcsetpgrp.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

10.69.90 `truncate`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/truncate.html>

Gnulib module: `truncate`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14, Android 4.4 with `AC_SYS_LARGEFILE` in effect.
- On platforms where `off_t` is a 32-bit type, this function is not applicable to arbitrary lengths for files 2 GiB and larger. See [Section 14.2 \[Large File Support\]](#), page 842.
- This function does not fail when the file name argument ends in a slash and (without the slash) names a non-directory, on some platforms: AIX 7.2.

Portability problems not fixed by Gnulib:

10.69.91 `ttyname`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/ttyname.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.
- This function is just a stub that produces an error message on standard error on some platforms: Android 4.3.

10.69.92 `ttynam_r`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/ttynam_r.html

Gnulib module: `ttynam_r`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: NetBSD 3.0, Minix 3.1.8, mingw, MSVC 14.
- This function is not declared unless `_REENTRANT` is defined, on some platforms: HP-UX 11.
- This function has an incompatible declaration on some platforms: Solaris 11.4 (when `_POSIX_PTHREAD_SEMANTICS` is not defined).
- This function refuses to do anything when the output buffer is less than 128 bytes large, on some platforms: Solaris 11 2010-11.
- This function is just a stub that produces an error message on standard error on some platforms: Android 4.3.

Portability problems not fixed by Gnulib:

10.69.93 `unlink`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/unlink.html>

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-unlink-3.html

Gnulib module: `unlink`

Portability problems fixed by Gnulib:

- This function is declared in a different header file (namely, `<stdio.h>`) on some platforms: MSVC 14.
- Some systems mistakenly succeed on `unlink("link-to-file/")`: GNU/Hurd, FreeBSD 7.2, AIX 7.1, Solaris 9.
- On Mac OS X 10.10, in a writable HFS mount, `unlink("../")` succeeds without doing anything.

Portability problems not fixed by Gnulib:

- Some systems allow a superuser to unlink directories, even though this can cause file system corruption. The error given if a process is not permitted to unlink directories varies across implementations; it is not always the POSIX value of `EPERM`. Meanwhile, if a process has the ability to unlink directories, POSIX requires that `unlink("symlink-to-dir/")` remove `dir` and leave `symlink-to-dir` dangling; this behavior is counter-intuitive. The gnulib module `unlinkdir` can help determine whether code must be cautious of unlinking directories.
- Removing an open file is non-portable: On Unix this allows the programs that have the file already open to continue working with it; the file's storage is only freed when the no process has the file open any more. On Windows, the attempt to remove an open file fails.

10.69.94 unlinkat

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/unlinkat.html>

Gnulib module: `unlinkat`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, Mac OS X 10.5, FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 1.5.x, mingw, MSVC 14. But the replacement function is not safe to be used in libraries and is not multithread-safe.
- This function is declared in `<fcntl.h>`, not in `<unistd.h>`, on some platforms: Cygwin 1.7.1, Android 4.3.
- On Mac OS X 10.10, in a writable HFS mount, `unlinkat(fd, "..", 0)` succeeds without doing anything.
- Some systems mistakenly succeed on `unlinkat(fd, "file/", flag)`: GNU/Hurd, Solaris 9.

Portability problems not fixed by Gnulib:

- When `unlinkat(fd, name, AT_REMOVEDIR)` fails because the specified directory is not empty, the `errno` value is system dependent.
- POSIX requires that `unlinkdir(fd, "link-to-empty/", AT_REMOVEDIR)` remove `empty` and leave `link-to-empty` as a dangling symlink. This is counter-intuitive, so some systems fail with `ENOTDIR` instead: glibc
- Some systems allow a superuser to unlink directories, even though this can cause file system corruption. The error given if a process is not permitted to unlink directories varies across implementations; it is not always the POSIX value of `EPERM`. Meanwhile, if a process has the ability to unlink directories, POSIX requires that `unlinkat(fd, "symlink-to-dir/", 0)` remove `dir` and leave `symlink-to-dir` dangling; this behavior is counter-intuitive. The gnulib module `unlinkdir` can help determine whether code must be cautious of unlinking directories.
- Removing an open file is non-portable: On Unix this allows the programs that have the file already open to continue working with it; the file's storage is only freed when the no process has the file open any more. On Windows, the attempt to remove an open file fails.

10.69.95 write

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/write.html>

Gnulib module: `write`, `nonblocking`, `sigpipe`

Portability problems fixed by Gnulib module `write`:

- This function is declared in a different header file (namely, `<io.h>`) on some platforms: mingw, MSVC 14.
- This function crashes when invoked with invalid arguments on some platforms: MSVC 14.

Portability problems fixed by Gnulib module `stdio-h`, together with module `nonblocking`:

- When writing to a non-blocking pipe whose buffer is full, this function fails with `errno` being set to `ENOSPC` instead of `EAGAIN` on some platforms: mingw, MSVC 14.
- When writing to a non-blocking pipe on which no reader is currently waiting an amount of bytes that exceeds the pipe buffer's size, then—even if the pipe's buffer is empty—this function fails, instead of performing a partial write into the pipe buffer, on some platforms: mingw, MSVC 14.

Portability problems fixed by Gnulib module `stdio-h`, together with module `sigpipe`:

- When writing to a pipe with no readers, this function fails with error `EINVAL`, instead of obeying the current `SIGPIPE` handler, on some platforms: mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- This function may fail with error `EINTR`, even in programs that don't install any signal handlers, on some platforms: macOS 14.

For handling `EINTR`, Gnulib provides a module `'safe-write'` with a function `safe_write`.

10.70 Functions in `<utmpx.h>`

10.70.1 `endutxent`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/endutxent.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, OpenBSD 7.5, Minix 3.1.8, mingw, MSVC 14, Android 9.0.

10.70.2 `getutxent`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getutxent.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, OpenBSD 7.5, Minix 3.1.8, mingw, MSVC 14, Android 9.0.

10.70.3 `getutxid`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getutxid.html>

Gnulib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: FreeBSD 6.0, OpenBSD 7.5, Minix 3.1.8, mingw, MSVC 14, Android 9.0.

10.70.4 `getutxline`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getutxline.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: FreeBSD 6.0, OpenBSD 7.5, Minix 3.1.8, mingw, MSVC 14, Android 9.0.

10.70.5 `pututxline`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/pututxline.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: FreeBSD 6.0, OpenBSD 7.5, Minix 3.1.8, mingw, MSVC 14, Android 9.0.

10.70.6 `setutxent`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/setutxent.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: FreeBSD 6.0, OpenBSD 7.5, Minix 3.1.8, mingw, MSVC 14, Android 9.0.

10.71 Functions in `<wchar.h>`

10.71.1 `btowc`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/btowc.html>

GnuLib module: `btowc`

Portability problems fixed by GnuLib:

- This function is missing on some platforms: Minix 3.1.8, HP-UX 11.00, mingw.
- This function returns WEOF for a NUL argument on some platforms: Cygwin 1.7.2.

- In the C or POSIX locales, this function is not consistent with Gnulib's `mbrtowc` and can return `WEOF`: glibc 2.35, MirOS BSD #10.
- In the C or POSIX locales, this function is not consistent with `mbrtowc` on some platforms: Cygwin 3.4.x, mingw with MSVCRT.

Portability problems not fixed by Gnulib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters. However, the Gnulib function `btoc32`, provided by Gnulib module `btoc32`, operates on 32-bit wide characters and therefore does not have this limitation.

10.71.2 `fgetwc`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fgetwc.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.5.x.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.3 `fgetws`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fgetws.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.5.x.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.4 `fputwc`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fputwc.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.5.x.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.5 fputws

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fputws.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.5.x.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.
- On some platforms, this function does not set `errno` or the stream error indicator on attempts to write to a read-only stream: Cygwin 1.7.9.

10.71.6 fwide

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fwide.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, HP-UX 11.00, Cygwin 1.5.x, MSVC 14.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.
- `fwide` is not guaranteed to be able to change a file stream's mode to a different mode than the current one.

10.71.7 fwprintf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fwprintf.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, HP-UX 11.00, Cygwin 1.5.x.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.
- This function does not support size specifiers as in C23 (`w8`, `w16`, `w32`, `w64`, `wf8`, `wf16`, `wf32`, `wf64`) on some platforms: glibc 2.37, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC, Android 9.0.
- This function doesn't support the `'` flag at all on some platforms: mingw without `__USE_MINGW_ANSI_STDIO`, MSVC 14, Haiku.

- This function doesn't support the ' flag correctly for integers (it uses the decimal point instead of the thousands separator!) on some platforms: mingw with `__USE_MINGW_ANSI_STDIO`.
- This function doesn't support the ' flag together with a precision on an integer argument correctly on some platforms: glibc 2.36, FreeBSD 14.0, NetBSD 10.0, Solaris 11, Cygwin 3.6.0, Haiku.
- This function doesn't support the ' flag correctly in locales where the thousands separator is a multibyte character on some platforms: NetBSD 10.0, Solaris 11 OpenIndiana.
- This function does not support the 'b' directive, required by ISO C23, on some platforms: glibc 2.34, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, HP-UX 11, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android 9.0.
- This function does not support the 'B' directive on some platforms: glibc 2.34, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC, Android 9.0.
- `printf "%#.0x" or "%#.0X"` with a zero argument yields an incorrect result (non-empty) on some platforms: Mac OS X 10.6.
- The `%m` directive is not portable, use `%s` mapped to an argument of `strerror(errno)` (or a version of `strerror_r`) instead.
- In the C or POSIX locales, the `%c` and `%s` conversions may fail on some platforms: glibc 2.35.
- When formatting an integer with grouping flag, this function inserts thousands separators even in the "C" locale on some platforms: NetBSD 5.1.
- On some platforms, this function does not set `errno` or the stream error indicator on attempts to write to a read-only stream: Cygwin 1.7.9.

10.71.8 fwscanf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/fwscanf.html>

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-fwscanf.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, HP-UX 11.00, Cygwin 1.5.x.
- This function does not support size specifiers as in C23 (`w8`, `w16`, `w32`, `w64`, `wf8`, `wf16`, `wf32`, `wf64`) on some platforms: glibc 2.38, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.2, AIX 7.2, HP-UX 11, Solaris 11.4, Cygwin 2.9.0, mingw, MSVC 14.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.9 `getwc`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getwc.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.5.x.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.10 `getwchar`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/getwchar.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.5.x.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.11 `mbrlen`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/mbrlen.html>

Gnulib module: `mbrlen`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, HP-UX 11.00, mingw.
- In the C or POSIX locales, this function can return `(size_t) -1` and set `errno` to `EILSEQ`: glibc 2.35.
- This function returns 0 instead of `(size_t) -2` when the input is empty: glibc 2.19.
- This function returns `(size_t) -1` instead of `(size_t) -2` when the input is empty: AIX 5.1.
- This function does not put the state into non-initial state when parsing an incomplete multibyte character on some platforms: AIX 5.1.
- This function returns the total number of bytes that make up the multibyte character, not the number of bytes that were needed to complete the multibyte character, on some platforms: HP-UX 11.11, Solaris 11 2010-11.
- This function may not return 0 when parsing the NUL character on some platforms: Solaris 9.

Portability problems not fixed by Gnulib:

10.71.12 `mbrtowc`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/mbrtowc.html>

Gnulib module: `mbrtowc`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, HP-UX 11.00, mingw.
- In the C or POSIX locales, this function can return `(size_t) -1` and set `errno` to `EILSEQ`: glibc 2.35.
- This function returns 0 instead of `(size_t) -2` when the input is empty: glibc 2.19, mingw with UCRT, MSVC 14, Android 11.
- This function returns `(size_t) -1` instead of `(size_t) -2` when the input is empty: AIX 7.2.
- This function does not put the state into non-initial state when parsing an incomplete multibyte character on some platforms: AIX 7.2.
- This function stores a wide character when parsing an incomplete multibyte character on some platforms: mingw with UCRT, MSVC 14.
- This function returns the total number of bytes that make up the multibyte character, not the number of bytes that were needed to complete the multibyte character, on some platforms: HP-UX 11.11, Solaris 11 2010-11, mingw, MSVC 14.
- This function may not return 0 when parsing the NUL character on some platforms: Solaris 9.

Portability problems not fixed by Gnulib:

- In UTF-8 locales, this function may return wide characters up to `0x7FFFFFFF` (that is, beyond `0x0010FFFF`) on some platforms: glibc 2.34.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters. However, the ISO C11 function `mbrtoc32`, provided by Gnulib module `mbrtoc32`, operates on 32-bit wide characters and therefore does not have this limitation.

10.71.13 `mbsinit`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/mbsinit.html>

Gnulib module: `mbsinit`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, HP-UX 11.00, mingw with UCRT, MSVC 14.
- This function always returns 1, even in multibyte locales, on some platforms: mingw.

Portability problems not fixed by Gnulib:

- This function is not useful after calls to `mbrtoc16` or `mbrtoc8`.

10.71.14 mbsnrtowcs

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/mbsnrtowcs.html>

Gnulib module: mbsnrtowcs

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, HP-UX 11, Solaris 11.3, Cygwin 1.5.x, mingw, MSVC 14, Android 4.4.
- This function produces invalid wide characters on some platforms: Solaris 11.4.
- In the C or POSIX locales, this function can return `(size_t) -1` and set `errno` to `EILSEQ`: glibc 2.35.

Portability problems not fixed by Gnulib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters. However, the Gnulib function `mbsnrtoc32s`, provided by Gnulib module `mbsnrtoc32s`, operates on 32-bit wide characters and therefore does not have this limitation.
- The specification is not clear about whether this function should update the conversion state when the first argument (the destination pointer) is `NULL`. The glibc, Mac OS X, FreeBSD implementations do update the state in this case. For portability, when passing a `NULL` destination argument, it is best to pass a pointer to a temporary copy of the conversion state.

10.71.15 mbsrtowcs

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/mbsrtowcs.html>

Gnulib module: mbsrtowcs

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, HP-UX 11.00, mingw.
- This function does not work on some platforms: HP-UX 11, Solaris 11 2010-11.
- In the C or POSIX locales, this function can return `(size_t) -1` and set `errno` to `EILSEQ`: glibc 2.35.
- This function does not work when the first argument is `NULL` on some platforms: mingw.

Portability problems not fixed by Gnulib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters. However, the Gnulib function `mbsrtoc32s`, provided by Gnulib module `mbsrtoc32s`, operates on 32-bit wide characters and therefore does not have this limitation.
- The specification is not clear about whether this function should update the conversion state when the first argument (the destination pointer) is `NULL`. The glibc implementation does not update the state in this case; the macOS and FreeBSD implementations do. For portability, when passing a `NULL` destination argument, it is best to pass a pointer to a temporary copy of the conversion state.

10.71.16 `open_wmemstream`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/open_wmemstream.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, Mac OS X 10.5, FreeBSD 6.0, NetBSD 7.1, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.3, Cygwin 1.5.x, mingw, MSVC 14, Android 5.1.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.17 `putwc`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/putwc.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.5.x.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.
- On some platforms, this function does not set `errno` or the stream error indicator on attempts to write to a read-only stream: Cygwin 1.7.9.

10.71.18 `putwchar`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/putwchar.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.5.x.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.
- On some platforms, this function does not set `errno` or the stream error indicator on attempts to write to a read-only stream: Cygwin 1.7.9.

10.71.19 `swprintf`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/swprintf.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, HP-UX 11.00, Cygwin 1.5.x, mingw with UCRT.
- This function has a different signature on some platforms: MSVC 9.
- This function is only defined as a macro on some platforms: MSVC 14.
- This function does not support the ‘n’ directive on some platforms: glibc when used with `_FORTIFY_SOURCE >= 2` (set by default on Ubuntu), macOS 14, OpenBSD, MSVC 14, Android.
- This function sometimes returns a wrong value through the ‘n’ directive on some platforms: musl libc 1.2.3.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.
- On Windows, this function does not take a buffer size as second argument.
- This function produces wrong values for the ‘La’ directive on some platforms: glibc 2.15, Haiku, NetBSD 10.0.
- This function does not support size specifiers as in C23 (`w8`, `w16`, `w32`, `w64`, `wf8`, `wf16`, `wf32`, `wf64`) on some platforms: glibc 2.37, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC, Android 9.0.
- This function ignores the minimum field width in the ‘lc’ directive on some platforms: musl libc 1.2.3.
- This function is broken when it produces output that contains null wide characters on some platforms: musl libc 1.2.3, macOS 14, FreeBSD 13.1, NetBSD 10.0, OpenBSD 7.2, AIX 7.2, mingw.
- This function doesn’t support the ‘f’ flag at all on some platforms: mingw without `__USE_MINGW_ANSI_STDIO`, MSVC 14, Haiku.
- This function doesn’t support the ‘f’ flag correctly for integers (it uses the decimal point instead of the thousands separator!) on some platforms: mingw with `__USE_MINGW_ANSI_STDIO`.
- This function doesn’t support the ‘f’ flag together with a precision on an integer argument correctly on some platforms: glibc 2.36, FreeBSD 14.0, NetBSD 10.0, Solaris 11, Cygwin 3.6.0, Haiku.
- This function doesn’t support the ‘f’ flag correctly in locales where the thousands separator is a multibyte character on some platforms: NetBSD 10.0, Solaris 11 OpenIndiana.
- This function does not support the ‘b’ directive, required by ISO C23, on some platforms: glibc 2.34, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, HP-UX 11, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android 9.0.
- This function does not support the ‘B’ directive on some platforms: glibc 2.34, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC, Android 9.0.
- `printf "%#.0x" or "%#.0X"` with a zero argument yields an incorrect result (non-empty) on some platforms: Mac OS X 10.6.

- The `%m` directive is not portable, use `%s` mapped to an argument of `strerror(errno)` (or a version of `strerror_r`) instead.
- In the C or POSIX locales, the `%c` and `%s` conversions may fail on some platforms: glibc 2.35, musl libc 1.2.4, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.2, Cygwin 2.9.0.
- The `%lc` directive may fail on some platforms: musl libc 1.2.4, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.2.
- When formatting an integer with grouping flag, this function inserts thousands separators even in the "C" locale on some platforms: NetBSD 5.1.

10.71.20 swscanf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/swscanf.html>

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-swscanf.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, HP-UX 11.00, Cygwin 1.5.x.
- This function does not support size specifiers as in C23 (`w8`, `w16`, `w32`, `w64`, `wf8`, `wf16`, `wf32`, `wf64`) on some platforms: glibc 2.38, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.2, AIX 7.2, HP-UX 11, Solaris 11.4, Cygwin 2.9.0, mingw, MSVC 14.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.21 ungetwc

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/ungetwc.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.5.x.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.22 vfwprintf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/vfwprintf.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, HP-UX 11.00, Cygwin 1.5.x.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.
- This function does not support size specifiers as in C23 (`w8`, `w16`, `w32`, `w64`, `wf8`, `wf16`, `wf32`, `wf64`) on some platforms: glibc 2.37, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC, Android 9.0.
- This function doesn't support the ' flag at all on some platforms: mingw without `__USE_MINGW_ANSI_STDIO`, MSVC 14, Haiku.
- This function doesn't support the ' flag correctly for integers (it uses the decimal point instead of the thousands separator!) on some platforms: mingw with `__USE_MINGW_ANSI_STDIO`.
- This function doesn't support the ' flag together with a precision on an integer argument correctly on some platforms: glibc 2.36, FreeBSD 14.0, NetBSD 10.0, Solaris 11, Cygwin 3.6.0, Haiku.
- This function doesn't support the ' flag correctly in locales where the thousands separator is a multibyte character on some platforms: NetBSD 10.0, Solaris 11 OpenIndiana.
- This function does not support the 'b' directive, required by ISO C23, on some platforms: glibc 2.34, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, HP-UX 11, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android 9.0.
- This function does not support the 'B' directive on some platforms: glibc 2.34, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC, Android 9.0.
- `printf "%#.0x" or "%#.0X"` with a zero argument yields an incorrect result (non-empty) on some platforms: Mac OS X 10.6.
- The `%m` directive is not portable, use `%s` mapped to an argument of `strerror(errno)` (or a version of `strerror_r`) instead.
- In the C or POSIX locales, the `%c` and `%s` conversions may fail on some platforms: glibc 2.35.
- When formatting an integer with grouping flag, this function inserts thousands separators even in the "C" locale on some platforms: NetBSD 5.1.
- On some platforms, this function does not set `errno` or the stream error indicator on attempts to write to a read-only stream: Cygwin 1.7.9.

10.71.23 vfwscanf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/vfwscanf.html>

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-vfwscanf.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11.23, Cygwin 1.5.x, Android 4.4.
- This function does not support size specifiers as in C23 (`w8`, `w16`, `w32`, `w64`, `wf8`, `wf16`, `wf32`, `wf64`) on some platforms: glibc 2.38, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.2, AIX 7.2, HP-UX 11, Solaris 11.4, Cygwin 2.9.0, mingw, MSVC 14.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.24 vswprintf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/vswprintf.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, HP-UX 11.00, Cygwin 1.5.x.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.
- On Windows, this function does not take a buffer size as second argument.
- This function does not support size specifiers as in C23 (`w8`, `w16`, `w32`, `w64`, `wf8`, `wf16`, `wf32`, `wf64`) on some platforms: glibc 2.37, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC, Android 9.0.
- This function doesn't support the ' flag at all on some platforms: mingw without `__USE_MINGW_ANSI_STDIO`, MSVC 14, Haiku.
- This function doesn't support the ' flag correctly for integers (it uses the decimal point instead of the thousands separator!) on some platforms: mingw with `__USE_MINGW_ANSI_STDIO`.
- This function doesn't support the ' flag together with a precision on an integer argument correctly on some platforms: glibc 2.36, FreeBSD 14.0, NetBSD 10.0, Solaris 11, Cygwin 3.6.0, Haiku.
- This function doesn't support the ' flag correctly in locales where the thousands separator is a multibyte character on some platforms: NetBSD 10.0, Solaris 11 OpenIndiana.
- This function does not support the 'b' directive, required by ISO C23, on some platforms: glibc 2.34, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, HP-UX 11, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android 9.0.
- This function does not support the 'B' directive on some platforms: glibc 2.34, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC, Android 9.0.

- `printf "%#.0x" or "%#.0X"` with a zero argument yields an incorrect result (non-empty) on some platforms: Mac OS X 10.6.
- The `%m` directive is not portable, use `%s` mapped to an argument of `strerror(errno)` (or a version of `strerror_r`) instead.
- In the C or POSIX locales, the `%c` and `%s` conversions may fail on some platforms: glibc 2.35.
- When formatting an integer with grouping flag, this function inserts thousands separators even in the "C" locale on some platforms: NetBSD 5.1.

10.71.25 vswscanf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/vswscanf.html>

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-vswscanf.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11.23, Cygwin 1.5.x, MSVC 14, Android 4.4.
- This function does not support size specifiers as in C23 (`w8`, `w16`, `w32`, `w64`, `wf8`, `wf16`, `wf32`, `wf64`) on some platforms: glibc 2.38, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.2, AIX 7.2, HP-UX 11, Solaris 11.4, Cygwin 2.9.0, mingw, MSVC 14.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.26 vwprintf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/vwprintf.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, HP-UX 11.00, Cygwin 1.7.
- This function exists, but does not support wide arguments on some platforms: Cygwin 1.5.x.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.
- This function does not support size specifiers as in C23 (`w8`, `w16`, `w32`, `w64`, `wf8`, `wf16`, `wf32`, `wf64`) on some platforms: glibc 2.37, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC, Android 9.0.

- This function doesn't support the ' flag at all on some platforms: mingw without `__USE_MINGW_ANSI_STDIO`, MSVC 14, Haiku.
- This function doesn't support the ' flag correctly for integers (it uses the decimal point instead of the thousands separator!) on some platforms: mingw with `__USE_MINGW_ANSI_STDIO`.
- This function doesn't support the ' flag together with a precision on an integer argument correctly on some platforms: glibc 2.36, FreeBSD 14.0, NetBSD 10.0, Solaris 11, Cygwin 3.6.0, Haiku.
- This function doesn't support the ' flag correctly in locales where the thousands separator is a multibyte character on some platforms: NetBSD 10.0, Solaris 11 OpenIndiana.
- This function does not support the 'b' directive, required by ISO C23, on some platforms: glibc 2.34, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, HP-UX 11, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android 9.0.
- This function does not support the 'B' directive on some platforms: glibc 2.34, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC, Android 9.0.
- `printf "%#.0x" or "%#.0X"` with a zero argument yields an incorrect result (non-empty) on some platforms: Mac OS X 10.6.
- The `%m` directive is not portable, use `%s` mapped to an argument of `strerror(errno)` (or a version of `strerror_r`) instead.
- In the C or POSIX locales, the `%c` and `%s` conversions may fail on some platforms: glibc 2.35.
- When formatting an integer with grouping flag, this function inserts thousands separators even in the "C" locale on some platforms: NetBSD 5.1.
- On some platforms, this function does not set `errno` or the stream error indicator on attempts to write to a read-only stream: Cygwin 1.7.9.

10.71.27 vwscanf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/vwscanf.html>

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-vwscanf.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11.23, Cygwin 1.5.x, Android 4.4.
- This function does not support size specifiers as in C23 (`w8`, `w16`, `w32`, `w64`, `wf8`, `wf16`, `wf32`, `wf64`) on some platforms: glibc 2.38, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.2, AIX 7.2, HP-UX 11, Solaris 11.4, Cygwin 2.9.0, mingw, MSVC 14.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.28 `wcpcpy`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wcpcpy.html>

Gnulib module: `wcpcpy`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Mac OS X 10.5, FreeBSD 6.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Solaris 10, Cygwin 1.5.x, mingw, MSVC 14.
- This function is not declared (without `-D_GNU_SOURCE`) on some platforms: glibc 2.13.

Portability problems not fixed by Gnulib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.29 `wcpncpy`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wcpncpy.html>

Gnulib module: `wcpncpy`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Mac OS X 10.5, FreeBSD 6.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Solaris 10, Cygwin 1.5.x, mingw, MSVC 14.
- This function is not declared (without `-D_GNU_SOURCE`) on some platforms: glibc 2.13.

Portability problems not fixed by Gnulib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.30 `wcrtomb`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wcrtomb.html>

Gnulib module: `wcrtomb`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, HP-UX 11.00, mingw.
- This function produces wrong characters in the C locale on some platforms: Android 4.3.
- This function returns 0 when the first argument is NULL in some locales on some platforms: Solaris 11.3.
- This function does not ignore the second argument when the first argument is NULL on some platforms: mingw with UCRT, MSVC 14.

Portability problems not fixed by Gnulib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters. However, the ISO C11 function `c32rtomb`, provided by Gnulib module `c32rtomb`, operates on 32-bit wide characters and therefore does not have this limitation.

10.71.31 `wscasecmp`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wscasecmp.html>

Gnulib module: `wscasecmp`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Mac OS X 10.5, FreeBSD 6.0, NetBSD 3.0, OpenBSD 3.8, HP-UX 11, Solaris 10, Cygwin 1.5.x, mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.32 `wscasecmp_l`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/wscasecmp_l.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: Mac OS X 10.5, FreeBSD 14.0, NetBSD 5.0, OpenBSD 6.0, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.3, Cygwin 1.7.x, mingw, MSVC 14, Android 5.1.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.33 `wscat`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wscat.html>

Gnulib module: `wscat`

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.34 `wcschr`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wcschr.html>

Gnulib module: `wcschr`

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.35 `wscmp`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wscmp.html>

Gnulib module: `wscmp`

Portability problems fixed by Gnulib:

- This function compares the wide characters as if they were unsigned, although `wchar_t` is signed, on some platforms: glibc 2.14.1 on x86 or x86_64, musl libc 1.2.3, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.2, Solaris 11.4.
- This function may return a wrong result if the two arguments are of different length, on some platforms: glibc 2.37 on arm and arm64 CPUs, AIX 7.2 in 64-bit mode.

Portability problems not fixed by Gnulib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.36 `wscoll`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wscoll.html>

Gnulib module: `wscoll`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8.

Portability problems not fixed by Gnulib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.37 `wscoll_l`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/wscoll_l.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 6.0, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.3, Cygwin 1.7.x, mingw, MSVC 14, Android 4.4.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.38 `wscpy`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wscpy.html>

Gnulib module: `wscpy`

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

Note: `wscpy (dst, src)` is only safe to use when you can guarantee that there are at least `wcslen (src) + 1` wide characters allocated at `dst`.

10.71.39 `wscspn`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wscspn.html>

GnuLib module: `wscspn`

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.40 `wcsdup`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wcsdup.html>

GnuLib module: `wcsdup`

Portability problems fixed by GnuLib:

- This function is missing on some platforms: Mac OS X 10.5, FreeBSD 6.0, NetBSD 3.0, OpenBSD 3.8, HP-UX 11, Solaris 10, Cygwin 1.5.x.

Portability problems not fixed by GnuLib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.41 `wcsftime`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wcsftime.html>

GnuLib module: `wcsftime`

Portability problems fixed by GnuLib:

- On native Windows platforms (mingw, MSVC), this function works incorrectly when the environment variable `TZ` has been set by Cygwin.

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: OpenBSD 3.8, Minix 3.1.8, Cygwin 1.5.x.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.
- Native Windows platforms (mingw, MSVC) support only a subset of time zones specified by POSIX. See [Section 10.67.35 \[tzset\]](#), page 474.

10.71.42 `wcslcat`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wcslcat.html>

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Truncating-Strings.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc 2.37, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, mingw, MSVC 14.

10.71.43 `wcsncpy`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wcsncpy.html>

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Truncating-Strings.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc 2.37, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, mingw, MSVC 14.

10.71.44 `wcslen`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wcslen.html>

Gnulib module: `wcslen`

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.45 `wcsncasecmp`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wcsncasecmp.html>

Gnulib module: `wcsncasecmp`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Mac OS X 10.5, FreeBSD 6.0, NetBSD 3.0, OpenBSD 3.8, HP-UX 11, Solaris 10, Cygwin 1.5.x, mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.46 `wcsncasecmp_l`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/wcsncasecmp_l.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: Mac OS X 10.5, FreeBSD 14.0, NetBSD 5.0, OpenBSD 6.0, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.3, Cygwin 1.7.x, mingw, MSVC 14, Android 5.1.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.47 `wcsncat`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wcsncat.html>

Gnulib module: `wcsncat`

Portability problems fixed by Gnulib:

- This function does not support zero-length operations on NULL pointers on some platforms: macOS 15, NetBSD 10.0, OpenBSD 7.6, Solaris 11.4, Cygwin 3.5.4.

Portability problems not fixed by Gnulib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.48 `wcsncmp`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wcsncmp.html>

Gnulib module: `wcsncmp`

Portability problems fixed by Gnulib:

- This function compares the wide characters as if they were unsigned, although `wchar_t` is signed, on some platforms: glibc 2.14.1 on x86 or x86_64, musl libc 1.2.3, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.2, Solaris 11.4.
- This function may return a wrong result if the two arguments are of different length, on some platforms: AIX 7.2 in 64-bit mode.

Portability problems not fixed by Gnulib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.49 `wcsncpy`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wcsncpy.html>

Gnulib module: `wcsncpy`

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

Note: This function has no real use: It cannot be used for filling a fixed-size record with a wide string, before writing it to a file, because the wide string encoding is platform dependent and, on some platforms, also locale dependent. And this function is **not** appropriate for copying a wide string into a bounded memory area, because you have no guarantee that the result will be null-terminated. Even if you add the null character at the end yourself, this function is inefficient (as it spends time clearing unused memory) and will allow silent truncation to occur, which is not a good behavior for GNU programs.

10.71.50 `wcsnlen`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wcsnlen.html>

Gnulib module: `wcsnlen`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Mac OS X 10.5, FreeBSD 6.0, NetBSD 7.1, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Solaris 10, Cygwin 1.5.x, mingw.

Portability problems not fixed by Gnulib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.51 `wcsnrtombs`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wcsnrtombs.html>

Gnulib module: `wcsnrtombs`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, HP-UX 11, Solaris 11.3, Cygwin 1.5.x, mingw, MSVC 14, Android 4.4.
- This function cannot consume valid sequences of wide characters on some platforms: Solaris 11.4.
- In C++ mode, the system's `<wchar.h>` defines `std::wcsnrtombs` but not `::wcsnrtombs` on some platforms: Solaris 11 OpenIndiana.

Portability problems not fixed by Gnulib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters. However, the Gnulib function `c32snrtombs`, provided by Gnulib module `c32snrtombs`, operates on 32-bit wide characters and therefore does not have this limitation.

10.71.52 wbspbrk

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wbspbrk.html>

Gnulib module: wbspbrk

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.53 wcsrchr

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wcsrchr.html>

Gnulib module: wcsrchr

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.54 wcsrtombs

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wcsrtombs.html>

Gnulib module: wcsrtombs

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, HP-UX 11.00, mingw.
- This function does not ignore the length argument if the destination argument is NULL on some platforms: mingw.
- This function updates the source pointer also if the destination argument is NULL on some platforms: HP-UX 11.

Portability problems not fixed by Gnulib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters. However, the Gnulib function `c32srtombs`, provided by Gnulib module `c32srtombs`, operates on 32-bit wide characters and therefore does not have this limitation.

10.71.55 wcssp

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wcssp.html>

Gnulib module: wcssp

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.56 wcsstr

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wcsstr.html>

Gnulib module: `wcsstr` or `wcsstr-simple`

Portability problems fixed by either Gnulib module `wcsstr-simple` or `wcsstr`:

- This function is missing on some platforms: HP-UX 11.00.

Portability problems fixed by Gnulib module `wcsstr`:

- This function has quadratic instead of linear worst-case complexity on some platforms: glibc 2.39, macOS 14, FreeBSD 13.1, NetBSD 10.0, OpenBSD 7.2, AIX 7.2, HP-UX 11, Solaris 11.4, Cygwin 2.9.0, mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.57 wcstod

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wcstod.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.5.x.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.58 wcstof

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wcstof.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, HP-UX 11, Solaris 9, Cygwin 1.5.x, MSVC 9, Android 4.4.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.59 wcstok

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wcstok.html>

Gnulib module: `wcstok`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Cygwin 1.5.x.

- This function takes only two arguments on some platforms: mingw, older MSVC.
- This function may use hidden state, ignoring the third argument, and thus exhibit a bug when two or more `wcstok` iteration loops are being performed in the same thread, on some platforms: HP-UX 11.31.

Portability problems not fixed by GnuLib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.60 `wcstol`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wcstol.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.5.x.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.61 `wcstold`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wcstold.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, HP-UX 11, Solaris 9, Cygwin 1.7.x, MSVC 9, Android 4.4.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.62 `wcstoll`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wcstoll.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: Minix 3.1.8, HP-UX 11.11, Solaris 9, Cygwin 1.5.x, MSVC 9, Android 4.4.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.63 `wcstoul`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wcstoul.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.5.x.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.64 `wcstoull`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wcstoull.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, HP-UX 11.11, Solaris 9, Cygwin 1.5.x, MSVC 9, Android 4.4.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.65 `wcswidth`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wcswidth.html>

Gnulib module: `wcswidth`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: OpenBSD 3.8, Minix 3.1.8, mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters. However, the Gnulib function `c32swidth`, provided by Gnulib module `c32swidth`, operates on 32-bit wide characters and therefore does not have this limitation.

10.71.66 `wcsxfrm`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wcsxfrm.html>

Gnulib module: `wcsxfrm`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.5.x.

Portability problems not fixed by Gnulib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.67 `wcsxfrm_l`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/wcsxfrm_l.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 6.0, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.3, Cygwin 1.7.x, mingw, MSVC 14, Android 4.4.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.68 `wctob`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wctob.html>

Gnulib module: `wctob`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, HP-UX 11.00, mingw.
- This function clobbers caller-owned registers on some platforms: Cygwin 1.7.5.
- This function does not work on some platforms: Solaris 9.

Portability problems not fixed by Gnulib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters. However, the Gnulib function `c32tob`, provided by Gnulib module `c32tob`, operates on 32-bit wide characters and therefore does not have this limitation.

10.71.69 `wcwidth`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wcwidth.html>

Gnulib module: `wcwidth`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.
- This function is not declared (without `-D_GNU_SOURCE`) on some platforms: glibc 2.8.
- This function handles combining characters in UTF-8 locales incorrectly on some platforms: macOS 14, NetBSD 9.3, OpenBSD 5.8, MidnightBSD 1.1.
- This function returns 2 for characters with ambiguous east asian width, even in Western locales, on some platforms: Solaris 11.4.

Portability problems not fixed by Gnulib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters. However, the Gnulib function `c32width`, provided by Gnulib module `c32width`, operates on 32-bit wide characters and therefore does not have this limitation.

- This function treats zero-width spaces like control characters on some platforms: AIX 7.2.

10.71.70 `wmemchr`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wmemchr.html>

Gnulib module: `wmemchr`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: HP-UX 11.00, MSVC 14.

Portability problems not fixed by Gnulib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.71 `wmemcmp`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wmemcmp.html>

Gnulib module: `wmemcmp`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: HP-UX 11.00, MSVC 14.
- This function compares the wide characters as if they were unsigned, although `wchar_t` is signed, on some platforms: glibc 2.14.1 on x86 or x86_64, musl libc 1.2.3, NetBSD 10.0, OpenBSD 7.2, Solaris 11.4.

Portability problems not fixed by Gnulib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.72 `wmemcpy`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wmemcpy.html>

Gnulib module: `wmemcpy`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: HP-UX 11.00, MSVC 9.

Portability problems not fixed by Gnulib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.73 `wmemmove`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wmemmove.html>

Gnulib module: `wmemmove`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: HP-UX 11.00, MSVC 9.

Portability problems not fixed by GnuLib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.74 `wmemset`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wmemset.html>

GnuLib module: `wmemset`

Portability problems fixed by GnuLib:

- This function is missing on some platforms: HP-UX 11.00, MSVC 9.

Portability problems not fixed by GnuLib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.71.75 `wprintf`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wprintf.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, HP-UX 11.00.
- This function exists, but does not support wide arguments on some platforms: Cygwin 1.5.x.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.
- This function does not support size specifiers as in C23 (`w8`, `w16`, `w32`, `w64`, `wf8`, `wf16`, `wf32`, `wf64`) on some platforms: glibc 2.37, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC, Android 9.0.
- This function doesn't support the ' flag at all on some platforms: mingw without `__USE_MINGW_ANSI_STDIO`, MSVC 14, Haiku.
- This function doesn't support the ' flag correctly for integers (it uses the decimal point instead of the thousands separator!) on some platforms: mingw with `__USE_MINGW_ANSI_STDIO`.
- This function doesn't support the ' flag together with a precision on an integer argument correctly on some platforms: glibc 2.36, FreeBSD 14.0, NetBSD 10.0, Solaris 11, Cygwin 3.6.0, Haiku.
- This function doesn't support the ' flag correctly in locales where the thousands separator is a multibyte character on some platforms: NetBSD 10.0, Solaris 11 OpenIndiana.
- This function does not support the 'b' directive, required by ISO C23, on some platforms: glibc 2.34, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, HP-UX 11, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android 9.0.

- This function does not support the ‘B’ directive on some platforms: glibc 2.34, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC, Android 9.0.
- `printf "%#.0x" or "%#.0X"` with a zero argument yields an incorrect result (non-empty) on some platforms: Mac OS X 10.6.
- The `%m` directive is not portable, use `%s` mapped to an argument of `strerror(errno)` (or a version of `strerror_r`) instead.
- In the C or POSIX locales, the `%c` and `%s` conversions may fail on some platforms: glibc 2.35.
- When formatting an integer with grouping flag, this function inserts thousands separators even in the "C" locale on some platforms: NetBSD 5.1.
- On some platforms, this function does not set `errno` or the stream error indicator on attempts to write to a read-only stream: Cygwin 1.7.9.

10.71.76 wscanf

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wscanf.html>

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-wscanf.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, HP-UX 11.00, Cygwin 1.5.x.
- This function does not support size specifiers as in C23 (`w8`, `w16`, `w32`, `w64`, `wf8`, `wf16`, `wf32`, `wf64`) on some platforms: glibc 2.38, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.2, AIX 7.2, HP-UX 11, Solaris 11.4, Cygwin 2.9.0, mingw, MSVC 14.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.72 Functions in <wctype.h>

10.72.1 iswalnum

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/iswalnum.html>

Gnulib module: `wctype-h`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8.
- This function cannot be called from plain inline or extern inline functions on some platforms: macOS 10.8.

Portability problems not fixed by Gnulib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters. However, the Gnulib function `c32isalnum`, provided by Gnulib module `c32isalnum`, operates on 32-bit wide characters and therefore does not have this limitation.

10.72.2 `iswalnum_l`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/iswalnum_l.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 6.0, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.3, Cygwin 1.7.x, mingw, MSVC 14, Android 4.4.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.72.3 `iswalpha`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/iswalpha.html>

Gnulib module: `wctype-h`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8.
- This function cannot be called from plain inline or extern inline functions on some platforms: macOS 10.8.

Portability problems not fixed by Gnulib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters. However, the Gnulib function `c32isalpha`, provided by Gnulib module `c32isalpha`, operates on 32-bit wide characters and therefore does not have this limitation.

10.72.4 `iswalpha_l`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/iswalpha_l.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 6.0, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.3, Cygwin 1.7.x, mingw, MSVC 14, Android 4.4.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.72.5 iswblank

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/iswblank.html>

Gnulib module: iswblank

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Solaris 9, mingw, MSVC 9, Android 4.4.
- This function is not declared (without `-D_GNU_SOURCE`) on some platforms: glibc 2.8.

Portability problems not fixed by Gnulib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters. However, the Gnulib function `c32isblank`, provided by Gnulib module `c32isblank`, operates on 32-bit wide characters and therefore does not have this limitation.

10.72.6 iswblank_l

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/iswblank_l.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 6.0, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.3, Cygwin 1.7.x, mingw, MSVC 14, Android 4.4.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.72.7 iswcntrl

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/iswcntrl.html>

Gnulib module: wctype-h

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8.
- This function cannot be called from plain inline or extern inline functions on some platforms: macOS 10.8.

Portability problems not fixed by Gnulib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters. However, the Gnulib function `c32iscntrl`, provided by Gnulib module `c32iscntrl`, operates on 32-bit wide characters and therefore does not have this limitation.
- This function returns 0 for U+2028 (LINE SEPARATOR) and U+2029 (PARAGRAPH SEPARATOR) on some platforms: macOS 14.

10.72.8 `iswcntrl_1`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/iswcntrl_1.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 6.0, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.3, Cygwin 1.7.x, mingw, MSVC 14, Android 4.4.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.72.9 `iswctype`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/iswctype.html>

Gnulib module: `iswctype`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8.
- This function is declared in `<wchar.h>`, not in `<wctype.h>`, on some platforms: HP-UX 11.00.
- This function cannot be called from plain inline or extern inline functions on some platforms: macOS 10.8.

Portability problems not fixed by Gnulib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters. However, the Gnulib function `c32_apply_type_test`, provided by Gnulib module `c32_apply_type_test`, operates on 32-bit wide characters and therefore does not have this limitation.

10.72.10 `iswctype_1`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/iswctype_1.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 6.0, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.3, Cygwin 1.7.x, mingw, MSVC 14, Android 4.4.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.72.11 iswdigit

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/iswdigit.html>

Gnulib module: iswdigit

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8.
- This function cannot be called from plain inline or extern inline functions on some platforms: macOS 10.8.
- This function is not ISO C 99 compliant on some platforms: FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Solaris 11.4, mingw, MSVC 14, Haiku, Android.

Portability problems not fixed by Gnulib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters. However, the Gnulib function `c32isdigit`, provided by Gnulib module `c32isdigit`, operates on 32-bit wide characters and therefore does not have this limitation.

10.72.12 iswdigit_l

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/iswdigit_l.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 6.0, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.3, Cygwin 1.7.x, mingw, MSVC 14, Android 4.4.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.72.13 iswgraph

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/iswgraph.html>

Gnulib module: wctype-h

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8.
- This function cannot be called from plain inline or extern inline functions on some platforms: macOS 10.8.

Portability problems not fixed by Gnulib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters. However, the Gnulib function `c32isgraph`, provided by Gnulib module `c32isgraph`, operates on 32-bit wide characters and therefore does not have this limitation.

10.72.14 `iswgraph_l`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/iswgraph_l.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 6.0, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.3, Cygwin 1.7.x, mingw, MSVC 14, Android 4.4.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.72.15 `iswlower`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/iswlower.html>

Gnulib module: `wctype-h`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8.
- This function cannot be called from plain inline or extern inline functions on some platforms: macOS 10.8.

Portability problems not fixed by Gnulib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters. However, the Gnulib function `c32islower`, provided by Gnulib module `c32islower`, operates on 32-bit wide characters and therefore does not have this limitation.

10.72.16 `iswlower_l`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/iswlower_l.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 6.0, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.3, Cygwin 1.7.x, mingw, MSVC 14, Android 4.4.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.72.17 `iswprint`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/iswprint.html>

Gnulib module: `wctype-h`

Portability problems fixed by GnuLib:

- This function is missing on some platforms: Minix 3.1.8.
- This function cannot be called from plain inline or extern inline functions on some platforms: macOS 10.8.
- This function returns true for the tab (`'\t'`) character on some platforms: mingw.

Portability problems not fixed by GnuLib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters. However, the GnuLib function `c32isprint`, provided by GnuLib module `c32isprint`, operates on 32-bit wide characters and therefore does not have this limitation.

10.72.18 `iswprint_l`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/iswprint_l.html

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on many platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 6.0, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.3, Cygwin 1.7.x, mingw, MSVC 14, Android 4.4.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.72.19 `iswpunct`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/iswpunct.html>

GnuLib module: `iswpunct`

Portability problems fixed by GnuLib:

- This function is missing on some platforms: Minix 3.1.8.
- This function cannot be called from plain inline or extern inline functions on some platforms: macOS 10.8.
- This function is inconsistent with the `ispunct` function, because it returns false for the characters `'$'`, `'+'`, `'<'`, `'='`, `'>'`, `'^'`, `'\`'`, `'|'`, `'~'` on some platforms: Android 11.

Portability problems not fixed by GnuLib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters. However, the GnuLib function `c32iswpunct`, provided by GnuLib module `c32iswpunct`, operates on 32-bit wide characters and therefore does not have this limitation.

10.72.20 `iswpunct_l`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/iswpunct_l.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 6.0, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.3, Cygwin 1.7.x, mingw, MSVC 14, Android 4.4.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.72.21 `iswspace`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/iswspace.html>

Gnulib module: `wctype-h`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8.
- This function cannot be called from plain inline or extern inline functions on some platforms: macOS 10.8.

Portability problems not fixed by Gnulib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters. However, the Gnulib function `c32isspace`, provided by Gnulib module `c32isspace`, operates on 32-bit wide characters and therefore does not have this limitation.

10.72.22 `iswspace_l`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/iswspace_l.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 6.0, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.3, Cygwin 1.7.x, mingw, MSVC 14, Android 4.4.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.72.23 `iswupper`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/iswupper.html>

Gnulib module: `wctype-h`

Portability problems fixed by GnuLib:

- This function is missing on some platforms: Minix 3.1.8.
- This function cannot be called from plain inline or extern inline functions on some platforms: macOS 10.8.

Portability problems not fixed by GnuLib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters. However, the GnuLib function `c32isupper`, provided by GnuLib module `c32isupper`, operates on 32-bit wide characters and therefore does not have this limitation.

10.72.24 `iswupper_l`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/iswupper_l.html

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on many platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 6.0, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.3, Cygwin 1.7.x, mingw, MSVC 14, Android 4.4.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.72.25 `iswxdigit`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/iswxdigit.html>

GnuLib module: `iswxdigit`

Portability problems fixed by GnuLib:

- This function is missing on some platforms: Minix 3.1.8.
- This function cannot be called from plain inline or extern inline functions on some platforms: macOS 10.8.
- This function is not ISO C 99 compliant on some platforms: FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Solaris 11.4, MSVC 14, Haiku, Android.

Portability problems not fixed by GnuLib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters. However, the GnuLib function `c32isxdigit`, provided by GnuLib module `c32isxdigit`, operates on 32-bit wide characters and therefore does not have this limitation.

10.72.26 `iswxdigit_l`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/iswxdigit_l.html

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on many platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 6.0, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.3, Cygwin 1.7.x, mingw, MSVC 14, Android 4.4.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.72.27 towctrans

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/towctrans.html>

GnuLib module: `towctrans`

Portability problems fixed by GnuLib:

- This function is missing on some platforms: Minix 3.1.8, HP-UX 11.00, mingw, MSVC 9, Android 7.1.

Portability problems not fixed by GnuLib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters. However, the GnuLib function `c32_apply_mapping`, provided by GnuLib module `c32_apply_mapping`, operates on 32-bit wide characters and therefore does not have this limitation.

10.72.28 towctrans_l

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/towctrans_l.html

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on many platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 6.0, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.3, Cygwin 1.7.x, mingw, MSVC 14, Android 7.1.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.72.29 tolower

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/tolower.html>

GnuLib module: `wctype-h`

Portability problems fixed by GnuLib:

- This function returns values of which the upper 16 bits are incorrect on some platforms: mingw.
- This function cannot be called from plain inline or extern inline functions on some platforms: macOS 10.8.

Portability problems not fixed by Gnulib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters. However, the Gnulib function `c32tolower`, provided by Gnulib module `c32tolower`, operates on 32-bit wide characters and therefore does not have this limitation.
- On native Windows, in an UTF-8 locale, this function does not even do the simple expected mappings, such as from 0x00C9 to 0x00E9.
- This function returns wrong values even for the ASCII characters in a zh_CN.GB18030 locale on some platforms: NetBSD 10.0.

10.72.30 `towlower_l`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/towlower_l.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 6.0, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.3, Cygwin 1.7.x, mingw, MSVC 14, Android 4.4.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.72.31 `towupper`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/towupper.html>

Gnulib module: `wctype-h`

Portability problems fixed by Gnulib:

- This function returns values of which the upper 16 bits are incorrect on some platforms: mingw.
- This function cannot be called from plain inline or extern inline functions on some platforms: macOS 10.8.

Portability problems not fixed by Gnulib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters. However, the Gnulib function `c32toupper`, provided by Gnulib module `c32toupper`, operates on 32-bit wide characters and therefore does not have this limitation.
- On native Windows, in an UTF-8 locale, this function does not even do the simple expected mappings, such as from 0x00E9 to 0x00C9.
- This function returns wrong values even for the ASCII characters in a zh_CN.GB18030 locale on some platforms: NetBSD 10.0.

10.72.32 `towupper_1`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/towupper_1.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 6.0, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.3, Cygwin 1.7.x, mingw, MSVC 14, Android 4.4.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.72.33 `wctrans`

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wctrans.html>

Gnulib module: `wctrans`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, HP-UX 11.00, mingw, MSVC 9, Android 7.1.
- This function does not support the `"tolower"` and `"toupper"` mappings on some platforms: NetBSD 10.0. (It returns non-null values for the arguments `"tolower"` and `"toupper"`, but with these values, the function `towctrans` always crashes.)

Portability problems not fixed by Gnulib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters. However, the Gnulib function `c32_get_mapping`, provided by Gnulib module `c32_get_mapping`, operates on 32-bit wide characters and therefore does not have this limitation.

10.72.34 `wctrans_1`

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/wctrans_1.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 6.0, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.3, Cygwin 1.7.x, mingw, MSVC 14, Android 7.1.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.72.35 wctype

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wctype.html>

Gnulib module: wctype

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, MSVC 9.
- This function is declared in `<wchar.h>`, not in `<wctype.h>`, on some platforms: HP-UX 11.00.
- This function does not support the "blank" character class on some platforms: mingw.
- The object returned by this function for the "blank" character class is inconsistent with the `iswblank` and `isblank` functions on some platforms: MSVC 14.
- The object returned by this function for the "punct" character class is inconsistent with the `ispunct` function on some platforms: Android 11.

Portability problems not fixed by Gnulib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters. However, the Gnulib function `c32_get_type_test`, provided by Gnulib module `c32_get_type_test`, operates on 32-bit wide characters and therefore does not have this limitation.

10.72.36 wctype_l

POSIX specification:

https://pubs.opengroup.org/onlinepubs/9799919799/functions/wctype_l.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 6.0, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.3, Cygwin 1.7.x, mingw, MSVC 14, Android 4.4.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

10.73 Functions in <wordexp.h>

10.73.1 wordexp

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wordexp.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: OpenBSD 7.5, Minix 3.1.8, Cygwin 1.5.x, mingw, MSVC 14, Android 9.0.

- On some platforms, this function does not set `errno` or the stream error indicator on attempts to write to a read-only stream: Cygwin 1.7.9.

10.73.2 wordfree

POSIX specification:

<https://pubs.opengroup.org/onlinepubs/9799919799/functions/wordfree.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: OpenBSD 7.5, Minix 3.1.8, Cygwin 1.5.x, mingw, MSVC 14, Android 9.0.

11 Past POSIX Function Substitutes

This chapter describes which functions and function-like macros specified by older versions of POSIX are substituted by Gnulib, which portability pitfalls are fixed by Gnulib, and which (known) portability problems are not worked around by Gnulib.

The notation “Gnulib module: —” means that Gnulib does not provide a module providing a substitute for the function. When the list “Portability problems not fixed by Gnulib” is empty, such a module is not needed: No portability problems are known. Otherwise, it indicates that such a module would be useful but is not available: No one so far found this function important enough to contribute a substitute for it. If you need this particular function, you may write to <bug-gnulib at gnu dot org>.

11.1 `_longjmp`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/_longjmp.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.
- This function is removed in POSIX.1-2024.

Note: Despite its being removed from POSIX, in 2024 on all systems which have `_setjmp`, it is the fastest way to save the registers but not the signal mask (up to 30 times faster than `setjmp` on some systems).

11.2 `_setjmp`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/_setjmp.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is removed in POSIX.1-2024.

Note: Despite its being removed from POSIX, in 2024 on all systems which have `_setjmp`, it is the fastest way to save the registers but not the signal mask (up to 30 times faster than `setjmp` on some systems).

11.3 `_tolower`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/_tolower.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, Minix 3.1.8, Android 4.4.
- This function is removed in POSIX.1-2024. Use the function `tolower` instead.

11.4 `_toupper`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/_toupper.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, Minix 3.1.8, Android 4.4.
- This function is removed in POSIX.1-2024. Use the function `toupper` instead.

11.5 `bcmp`

Removed in POSIX.1-2008. POSIX.1-2004 specification: <https://pubs.opengroup.org/onlinepubs/009695399/functions/bcmp.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14, Android 9.0.
- This function is removed in POSIX.1-2008. Better use `memcmp` instead.

11.6 `bcopy`

Removed in POSIX.1-2008. POSIX.1-2004 specification: <https://pubs.opengroup.org/onlinepubs/009695399/functions/bcopy.html>

Gnulib module: `bcopy`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14, Android 9.0.

Portability problems not fixed by Gnulib:

- This function is removed in POSIX.1-2008. Better use `memcpy` or `memmove` instead.
- This module does not arrange to declare this function.

11.7 `bsd_signal`

Removed in POSIX.1-2008. POSIX.1-2004 specification: https://pubs.opengroup.org/onlinepubs/009695399/functions/bsd_signal.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Mac OS X 10.3, FreeBSD 14.0, NetBSD 7.1, Cygwin 2.9, mingw, MSVC 14.

11.8 `bzero`

Removed in POSIX.1-2008. POSIX.1-2004 specification: <https://pubs.opengroup.org/onlinepubs/009695399/functions/bzero.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14, Android 9.0.
- This function is removed in POSIX.1-2008. Better use `memset` instead.

11.9 `ecvt`

Removed in POSIX.1-2008. POSIX.1-2004 specification: <https://pubs.opengroup.org/onlinepubs/009695399/functions/ecvt.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 14.0, NetBSD 10.0, Android 9.0.
- This function is not declared on some platforms: Cygwin 2.9.
- This function is removed in POSIX.1-2008. Better use `sprintf` instead.

11.10 `fattach`

Removed in POSIX.1-2024. POSIX.1-2017 specification: <https://pubs.opengroup.org/onlinepubs/9699919799/functions/fattach.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.11 fcvt

Removed in POSIX.1-2008. POSIX.1-2004 specification: <https://pubs.opengroup.org/onlinepubs/009695399/functions/fcvt.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 14.0, NetBSD 10.0, Android 9.0.
- This function is not declared on some platforms: Cygwin 2.9.
- This function is removed in POSIX.1-2008. Better use `sprintf` instead.

11.12 fdetach

Removed in POSIX.1-2024. POSIX.1-2017 specification:

<https://pubs.opengroup.org/onlinepubs/9699919799/functions/fdetach.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.13 ftime

Removed in POSIX.1-2008. POSIX.1-2004 specification: <https://pubs.opengroup.org/onlinepubs/009695399/functions/ftime.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Mac OS X 10.3, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Android 9.0.
- This function is not declared on some platforms: Android 13.
- On native Windows platforms (mingw, MSVC), this function works incorrectly when the environment variable TZ has been set by Cygwin.
- This function is removed in POSIX.1-2008. Better use `clock_gettime` instead, and use `ftime` only as a fallback for portability to Windows platforms.

11.14 ftw

Removed in POSIX.1-2024. POSIX.1-2017 specification:

<https://pubs.opengroup.org/onlinepubs/9699919799/functions/ftw.html>

Gnulib module: —

Portability problems fixed by Gnulib:

- On platforms where `off_t` is a 32-bit type, this function may not work correctly on files 2 GiB and larger. See [Section 14.2 \[Large File Support\]](#), page 842.

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 3.0, Minix 3.1.8, mingw, MSVC 14, Android 4.1.
- This function is removed in POSIX.1-2024. Use the GnuLib module `fts` instead.

11.15 `gcvt`

Removed in POSIX.1-2008. POSIX.1-2004 specification: <https://pubs.opengroup.org/onlinepubs/009695399/functions/gcvt.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: Mac OS X 10.3, FreeBSD 14.0, NetBSD 10.0, Android 9.0.
- This function is not declared on some platforms: Cygwin 2.9.
- This function is removed in POSIX.1-2008. Better use `sprintf` instead.

11.16 `getcontext`

Removed in POSIX.1-2008. POSIX.1-2004 specification: <https://pubs.opengroup.org/onlinepubs/009695399/functions/getcontext.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: OpenBSD 7.5, Cygwin 1.7.x, mingw, MSVC 14, Android 9.0.

Note: A third-party implementation is available at <https://github.com/kaniini/libucontext/>.

11.17 `gethostbyaddr`

Removed in POSIX.1-2008. POSIX.1-2004 specification: <https://pubs.opengroup.org/onlinepubs/009695399/functions/gethostbyaddr.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is declared in `<winsock2.h>` instead of `<netdb.h>` on some platforms: mingw, MSVC 14.

11.18 gethostbyname

Removed in POSIX.1-2008. POSIX.1-2004 specification: <https://pubs.opengroup.org/onlinepubs/009695399/functions/gethostbyname.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is declared in `<winsock2.h>` instead of `<netdb.h>` on some platforms: mingw, MSVC 14.

11.19 getitimer

Removed in POSIX.1-2024. POSIX.1-2017 specification: <https://pubs.opengroup.org/onlinepubs/9699919799/functions/getitimer.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.
- This function is removed in POSIX.1-2024. Use the function `timer_gettime` instead.

11.20 getmsg

Removed in POSIX.1-2024. POSIX.1-2017 specification: <https://pubs.opengroup.org/onlinepubs/9699919799/functions/getmsg.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.21 getpmsg

Removed in POSIX.1-2024. POSIX.1-2017 specification: <https://pubs.opengroup.org/onlinepubs/9699919799/functions/getpmsg.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.22 gets

Removed in POSIX.1-2024. POSIX.1-2017 specification:

<https://pubs.opengroup.org/onlinepubs/9699919799/functions/gets.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is removed in POSIX.1-2024. Use the function `fgets` instead.
- This function should never be used, because it can overflow any given buffer.
- This function is missing on some platforms: OpenBSD 7.5, Solaris 11.4.
- When reading from a non-blocking pipe whose buffer is empty, this function fails with `errno` being set to `EINVAL` instead of `EAGAIN` on some platforms: mingw, MSVC 14.
- On Windows platforms (excluding Cygwin), this function does not set `errno` upon failure.

11.23 gettimeofday

Removed in POSIX.1-2024. POSIX.1-2017 specification:

<https://pubs.opengroup.org/onlinepubs/9699919799/functions/gettimeofday.html>

Gnulib module: `gettimeofday`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: MSVC 14.
- This function is declared with a nonstandard function prototype (only one argument, or “...” after the first argument) on some platforms.
- On some platforms, the second argument has type `struct timezone*` rather than `void *`, making it an error to redeclare the function with the POSIX signature: glibc. However, rather than penalize these systems with a replacement function, gnulib defines `GETTIMEOFDAY_TIMEZONE` to the appropriate type for use in avoiding a compiler warning if assigning `gettimeofday` to a function pointer.
- This function has only a precision of 15.6 milliseconds on some platforms: mingw.

Portability problems not fixed by Gnulib:

- Behavior is non-portable if the second argument to `gettimeofday` is not `NULL`.
- This function is removed in POSIX.1-2024. Use the Gnulib module `gettime` or `timespec_get` instead. (POSIX recommends to use the function `clock_gettime`, but there is no corresponding Gnulib module for it yet.)

11.24 getwd

Removed in POSIX.1-2008. POSIX.1-2004 specification: <https://pubs.opengroup.org/onlinepubs/009695399/functions/getwd.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14, Android 9.0.
- The size of the buffer required for this function is not a compile-time constant. Also, the function truncates a result that would be larger than the minimum buffer size. For these reasons, this function is removed in POSIX.1-2008. Better use the `getcwd` function instead.

11.25 `h_errno`

Removed in POSIX.1-2008. POSIX.1-2004 specification: https://pubs.opengroup.org/onlinepubs/009695399/functions/h_errno.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.34, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.26 `index`

Removed in POSIX.1-2008. POSIX.1-2004 specification: <https://pubs.opengroup.org/onlinepubs/009695399/functions/index.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14, Android 9.0.
- This function is not declared on some platforms: Android 13.
- This function is removed in POSIX.1-2008. Better use `strchr` instead.

11.27 `ioctl`

Removed in POSIX.1-2024. POSIX.1-2017 specification: <https://pubs.opengroup.org/onlinepubs/9699919799/functions/ioctl.html>

LSB specification:

- https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-ioctl-2.html
- https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-ttyio-2.html
- https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-sockio-2.html

Gnulib module: `ioctl`

Portability problems fixed by Gnulib:

- On Windows platforms (excluding Cygwin), `ioctl` is called `ioctlsocket`, and error codes from this function are not placed in `errno`, and `WSAGetLastError` must be used instead.
- The second parameter is of type `unsigned long` rather than `int` on some platforms: glibc 2.26, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Haiku 2017.

Portability problems not fixed by Gnulib:

- Most `ioctl` requests are platform and hardware specific.

11.28 `isascii`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

<https://pubs.opengroup.org/onlinepubs/9699919799/functions/isascii.html>

Gnulib module: —

Portability problems not fixed by Gnulib:

- This function cannot be called from plain inline or extern inline functions on some platforms: macOS 10.8.
- This function is removed in POSIX.1-2024.

Note: This function's behaviour depends on the locale, but requires special handling for the multibyte characters that occur in strings in locales with `MB_CUR_MAX > 1` (this includes all the common UTF-8 locales). There are two alternative APIs:

`c_isascii`

This function operates in a locale independent way and returns true only for ASCII characters. It is provided by the Gnulib module '`c-ctype`'.

`mb_isascii`

This function operates in a locale dependent way, on multibyte characters. It is provided by the Gnulib module '`mbchar`'.

11.29 `isastream`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

<https://pubs.opengroup.org/onlinepubs/9699919799/functions/isastream.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.30 makecontext

Removed in POSIX.1-2008. POSIX.1-2004 specification: <https://pubs.opengroup.org/onlinepubs/009695399/functions/makecontext.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: OpenBSD 7.5, Cygwin 1.7.x, mingw, MSVC 14, Android 9.0.

Note: A third-party implementation is available at <https://github.com/kaniini/libucontext/>.

11.31 mktemp

Removed in POSIX.1-2008. POSIX.1-2004 specification: <https://pubs.opengroup.org/onlinepubs/009695399/functions/mktemp.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is not appropriate for creating temporary files. (It has security risks.) Therefore it is removed in POSIX.1-2008. Better use `mkstemp` instead.

11.32 posix_trace_attr_destroy

Removed in POSIX.1-2024. POSIX.1-2017 specification: https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_attr_destroy.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.33 posix_trace_attr_getclockres

Removed in POSIX.1-2024. POSIX.1-2017 specification: https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_attr_getclockres.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.34 `posix_trace_attr_getcreatetime`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_attr_getcreatetime.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.35 `posix_trace_attr_getgenversion`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_attr_getgenversion.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.36 `posix_trace_attr_getinherited`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_attr_getinherited.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.37 `posix_trace_attr_getlogfullpolicy`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_attr_getlogfullpolicy.html

Gnulib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.38 `posix_trace_attr_getlogsize`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_attr_getlogsize.html

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.39 `posix_trace_attr_getmaxdatasize`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_attr_getmaxdatasize.html

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.40 `posix_trace_attr_getmaxsystemeventsizesize`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_attr_getmaxsystemeventsizesize.html

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.41 `posix_trace_attr_getmaxusereventsize`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_attr_getmaxusereventsize.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.42 `posix_trace_attr_getname`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_attr_getname.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.43 `posix_trace_attr_getstreamfullpolicy`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_attr_getstreamfullpolicy.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.44 `posix_trace_attr_getstreamsize`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_attr_getstreamsize.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.45 `posix_trace_attr_init`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_attr_init.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.46 `posix_trace_attr_setinherited`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_attr_setinherited.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.47 `posix_trace_attr_setlogfullpolicy`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_attr_setlogfullpolicy.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.48 `posix_trace_attr_setlogsize`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_attr_setlogsize.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.49 `posix_trace_attr_setmaxdatasize`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_attr_setmaxdatasize.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.50 `posix_trace_attr_setname`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_attr_setname.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.51 `posix_trace_attr_setstreamfullpolicy`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_attr_setstreamfullpolicy.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.52 `posix_trace_attr_setstreamsize`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_attr_setstreamsize.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.53 `posix_trace_clear`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_clear.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.54 `posix_trace_close`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_close.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.55 `posix_trace_create`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_create.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.56 `posix_trace_create_withlog`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_create_withlog.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.57 `posix_trace_event`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_event.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.58 `posix_trace_eventid_equal`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_eventid_equal.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.59 `posix_trace_eventid_get_name`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_eventid_get_name.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.60 `posix_trace_eventid_open`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_eventid_open.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.61 `posix_trace_eventset_add`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_eventset_add.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.62 `posix_trace_eventset_del`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_eventset_del.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.63 `posix_trace_eventset_empty`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_eventset_empty.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.64 `posix_trace_eventset_fill`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_eventset_fill.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.65 `posix_trace_eventset_ismember`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_eventset_ismember.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.66 `posix_trace_eventtypelist_getnext_id`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_eventtypelist_getnext_id.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.67 `posix_trace_eventtypelist_rewind`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_eventtypelist_rewind.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.68 `posix_trace_flush`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_flush.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.69 `posix_trace_get_attr`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_get_attr.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.70 `posix_trace_get_filter`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_get_filter.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.71 `posix_trace_get_status`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_get_status.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.72 `posix_trace_getnext_event`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_getnext_event.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.73 `posix_trace_open`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_open.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.74 `posix_trace_rewind`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_rewind.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.75 `posix_trace_set_filter`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_set_filter.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.76 `posix_trace_shutdown`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_shutdown.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.77 `posix_trace_start`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_start.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.78 `posix_trace_stop`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_stop.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.79 `posix_trace_timedgetnext_event`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_timedgetnext_event.html

Gnulib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.80 `posix_trace_trid_eventid_open`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_trid_eventid_open.html

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.81 `posix_trace_trygetnext_event`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/posix_trace_trygetnext_event.html

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.82 `pthread_attr_getstackaddr`

Removed in POSIX.1-2008. POSIX.1-2004 specification:

https://pubs.opengroup.org/onlinepubs/009695399/functions/pthread_attr_getstackaddr.html

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: HP-UX 11, Cygwin 1.7.9, mingw, MSVC 14, Android 9.0.
- This function is not declared on some platforms: Android 13.

11.83 pthread_attr_setstackaddr

Removed in POSIX.1-2008. POSIX.1-2004 specification:

https://pubs.opengroup.org/onlinepubs/009695399/functions/pthread_attr_setstackaddr.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: HP-UX 11, Cygwin 1.7.9, mingw, MSVC 14, Android 9.0.
- This function is not declared on some platforms: Android 13.

11.84 pthread_getconcurrency

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/pthread_getconcurrency.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: NetBSD 10.0, Minix 3.1.8, mingw, MSVC 14, Android 9.0.
- This function is removed in POSIX.1-2024. You can remove calls to this function.

11.85 pthread_setconcurrency

Removed in POSIX.1-2024. POSIX.1-2017 specification:

https://pubs.opengroup.org/onlinepubs/9699919799/functions/pthread_setconcurrency.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: NetBSD 10.0, Minix 3.1.8, mingw, MSVC 14, Android 9.0.
- This function is removed in POSIX.1-2024. You can remove calls to this function.

11.86 putmsg

Removed in POSIX.1-2024. POSIX.1-2017 specification:

<https://pubs.opengroup.org/onlinepubs/9699919799/functions/putmsg.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.87 putpmsg

Removed in POSIX.1-2024. POSIX.1-2017 specification:

<https://pubs.opengroup.org/onlinepubs/9699919799/functions/putpmsg.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

11.88 rindex

Removed in POSIX.1-2008. POSIX.1-2004 specification: <https://pubs.opengroup.org/onlinepubs/009695399/functions/rindex.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14, Android 9.0.
- This function is removed in POSIX.1-2008. Better use `strrchr` instead.

11.89 scalb

Removed in POSIX.1-2008. POSIX.1-2004 specification: <https://pubs.opengroup.org/onlinepubs/009695399/functions/scalb.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

11.90 setcontext

Removed in POSIX.1-2008. POSIX.1-2004 specification: <https://pubs.opengroup.org/onlinepubs/009695399/functions/setcontext.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: OpenBSD 7.5, Cygwin 1.7.x, mingw, MSVC 14, Android 9.0.
- The effects of this call are system and compiler optimization dependent, since it restores the contents of register-allocated variables but not the contents of stack-allocated variables.

Note: A third-party implementation is available at <https://github.com/kaniini/libucontext/>.

11.91 `setitimer`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

<https://pubs.opengroup.org/onlinepubs/9699919799/functions/setitimer.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.
- This function is removed in POSIX.1-2024. Use the functions `timer_create` and `timer_settime` instead.

11.92 `setpgrp`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

<https://pubs.opengroup.org/onlinepubs/9699919799/functions/setpgrp.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.
- This function is removed in POSIX.1-2024. Use the function `setpgid` or `setsid` instead, as appropriate.

11.93 `sighold`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

<https://pubs.opengroup.org/onlinepubs/9699919799/functions/sighold.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, OpenBSD 7.5, Minix 3.1.8, mingw, MSVC 14, Android 7.1.
- This function is removed in POSIX.1-2024. Use the function `sigprocmask` instead.

11.94 `sigignore`

Removed in POSIX.1-2024. POSIX.1-2017 specification:

<https://pubs.opengroup.org/onlinepubs/9699919799/functions/sigignore.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 1.5.x, mingw, MSVC 14, Android 7.1.
- This function is removed in POSIX.1-2024. Use the function `sigaction` instead.

11.95 siginterrupt

Removed in POSIX.1-2024. POSIX.1-2017 specification:

<https://pubs.opengroup.org/onlinepubs/9699919799/functions/siginterrupt.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.
- This function is removed in POSIX.1-2024. Use the function `sigaction` instead.

Note: POSIX recommends using `sigaction` with `SA_RESTART` instead of `siginterrupt (sig, 0)`.

11.96 sigpause

Removed in POSIX.1-2024. POSIX.1-2017 specification:

<https://pubs.opengroup.org/onlinepubs/9699919799/functions/sigpause.html>

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-sigpause-3.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14, Android 7.1.
- This function is removed in POSIX.1-2024. Use the function `sigsuspend` instead.

11.97 sigrelse

Removed in POSIX.1-2024. POSIX.1-2017 specification:

<https://pubs.opengroup.org/onlinepubs/9699919799/functions/sigrelse.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, OpenBSD 7.5, Minix 3.1.8, mingw, MSVC 14, Android 7.1.
- This function is removed in POSIX.1-2024. Use the function `sigprocmask` instead.

11.98 sigset

Removed in POSIX.1-2024. POSIX.1-2017 specification:

<https://pubs.opengroup.org/onlinepubs/9699919799/functions/sigset.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 1.5.x, mingw, MSVC 14, Android 7.1.
- This function is removed in POSIX.1-2024. Use the function `sigaction` instead.

11.99 swapcontext

Removed in POSIX.1-2008. POSIX.1-2004 specification: <https://pubs.opengroup.org/onlinepubs/009695399/functions/swapcontext.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: OpenBSD 7.5, Cygwin 1.7.x, mingw, MSVC 14, Android 9.0.

Note: A third-party implementation is available at <https://github.com/kaniini/libucontext/>.

11.100 tempnam

Removed in POSIX.1-2024. POSIX.1-2017 specification:

<https://pubs.opengroup.org/onlinepubs/9699919799/functions/tempnam.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8.
- This function is removed in POSIX.1-2024. Use the function `mkstemp` instead.
- This function is not appropriate for creating temporary files. (It has security risks.) Better use `mkstemp` instead.

11.101 toascii

Removed in POSIX.1-2024. POSIX.1-2017 specification:

<https://pubs.opengroup.org/onlinepubs/9699919799/functions/toascii.html>

Gnulib module: `ctype`

Portability problems fixed by Gnulib:

- This function cannot be called from plain inline or extern inline functions on some platforms: macOS 10.8.

Portability problems not fixed by Gnulib:

11.102 ualarm

Removed in POSIX.1-2008. POSIX.1-2004 specification: <https://pubs.opengroup.org/onlinepubs/009695399/functions/ualarm.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14, Android 9.0.

11.103 ulimit

Removed in POSIX.1-2024. POSIX.1-2017 specification: <https://pubs.opengroup.org/onlinepubs/9699919799/functions/ulimit.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.
- This function is removed in POSIX.1-2024. Use the functions `getrlimit` and `setrlimit` instead.

11.104 usleep

Removed in POSIX.1-2008. POSIX.1-2004 specification: <https://pubs.opengroup.org/onlinepubs/009695399/functions/usleep.html>

Gnulib module: `usleep`

Portability problems fixed by Gnulib:

- On some systems, `usleep` rejects attempts to sleep longer than 1 second, as allowed by POSIX: NetBSD 10.0, mingw.

Portability problems not fixed by Gnulib:

- According to POSIX, the `usleep` function may interfere with the program's use of the `SIGALRM` signal. On Linux, it doesn't; on other platforms, it may.
- On some systems, `usleep` may sleep for less than the indicated duration: Cygwin 3.5.3.

11.105 utime

Removed in POSIX.1-2024. POSIX.1-2017 specification: <https://pubs.opengroup.org/onlinepubs/9699919799/functions/utime.html>

Gnulib module: `utime`

Portability problems fixed by Gnulib:

- The times that are set on the file are affected by the current time zone and by the DST flag of the current time zone on some platforms: mingw, MSVC 14 (when the environment variable `TZ` is set).

- On some platforms, the prototype for `utime` omits `const` for the second argument: mingw, MSVC 14.
- On some platforms, `utime("link-to-file/",buf)` succeeds instead of failing with `ENOTDIR`. macOS 14.

Portability problems not fixed by GnuLib:

- On some platforms, this function mis-handles a trailing slash: Solaris 9.
- This function cannot set full timestamp resolution. Use `utimensat(AT_FDCWD,file,times,0)`, or the gnuLib module `utimens`, instead.
- This function sets a slightly different time stamp than the requested one: Haiku.
- This function is removed in POSIX.1-2024. You can use GnuLib module `utimens` instead.

11.106 `vfork`

Removed in POSIX.1-2008. POSIX.1-2004 specification: <https://pubs.opengroup.org/onlinepubs/009695399/functions/vfork.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: mingw, MSVC 14.

11.107 `wcswcs`

Removed in POSIX.1-2008. POSIX.1-2004 specification: <https://pubs.opengroup.org/onlinepubs/009695399/functions/wcswcs.html>

GnuLib module: —

Portability problems fixed by GnuLib:

Portability problems not fixed by GnuLib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, Cygwin 2.9, mingw, MSVC 14, Android 9.0.
- This function is not declared on some platforms: Android 13.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.
- This function is removed in POSIX.1-2008. Better use `wcsstr` instead.

12 Glibc Header File Substitutes

This chapter describes which header files contained in GNU libc but not specified by ISO C or POSIX are substituted by Gnulib, which portability pitfalls are fixed by Gnulib, and which (known) portability problems are not worked around by Gnulib.

The notation “Gnulib module: —” means that Gnulib does not provide a module providing a substitute for the header file. When the list “Portability problems not fixed by Gnulib” is empty, such a module is not needed: No portability problems are known. Otherwise, it indicates that such a module would be useful but is not available: No one so far found this header file important enough to contribute a substitute for it. If you need this particular header file, you may write to <bug-gnulib at gnu dot org>.

12.1 a.out.h

Describes the structure of executables (and object files?) in the old `a.out` format.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on some platforms: macOS 14, Solaris 11.4, mingw, MSVC 14, Android 9.0.
- This header file is useless because most executables and object files are in ELF format on some platforms: glibc 2.3.6, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5.

12.2 aliases.h

Defines the type `struct aliasent` and declares the functions `setaliasent`, `endaliasent`, `getaliasent`, `getaliasent_r`, `getaliasbyname`, `getaliasbyname_r`.

Documentation:

- [man setaliasent](#).

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin, mingw, MSVC 14, Android 9.0.

12.3 alloca.h

Declares the `alloca` function of function-like macro.

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Variable-Size-Automatic.html,
- [man alloca](#).

Gnulib module: `alloca`

Portability problems fixed by Gnulib:

- This header file is missing on some platforms: FreeBSD 6.0, NetBSD 10.0, OpenBSD 7.5, mingw, MSVC 14.

Portability problems not fixed by Gnulib:

12.4 `ar.h`

Describes the structure of files produced by the ‘`ar`’ program. Defines the type `struct ar_hdr` and the macros `ARMAG`, `SARMAG`, `ARFMAG`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on some platforms: mingw, MSVC 14.

12.5 `argp.h`

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Argp.html.

Gnulib module: `argp`

Portability problems fixed by Gnulib:

- This header file is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin, mingw, MSVC 14, Android 9.0.

Portability problems not fixed by Gnulib:

12.6 `argz.h`

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Argz-Functions.html,
- `man argz`.

Gnulib module: `argz`

Portability problems fixed by Gnulib:

- This header file is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, mingw, MSVC 14, Android 9.0.
- The `argz` functions do not work on some platforms: Cygwin.

Portability problems not fixed by Gnulib:

12.7 `byteswap.h`

Defines the functions or function-like macros `bswap_16`, `bswap_32`, `bswap_64`.

Gnulib module: `byteswap`

Portability problems fixed by Gnulib:

- This header file is missing on some platforms: macOS 14, FreeBSD 13.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, mingw, MSVC 14.

Portability problems not fixed by Gnulib:

12.8 `crypt.h`

Defines the type `struct crypt_data` and declares the functions `crypt`, `crypt_r`, `setkey`, `setkey_r`, `encrypt`, `encrypt_r`.

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/crypt.html, https://www.gnu.org/software/libc/manual/html_node/DES-Encryption.html,
- `man crypt`, `man encrypt`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, mingw, MSVC 14, Android 9.0.
- The functions `crypt`, `setkey`, `encrypt` are missing on some platforms: HP-UX 11.
- The type `struct crypt_data` and the functions `crypt_r`, `setkey_r`, `encrypt_r` are missing on some platforms: Solaris 11.4, Cygwin.

12.9 `envz.h`

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Envz-Functions.html,
- `man envz`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, mingw, MSVC 14, Android 9.0.

12.10 `err.h`

Declares the functions `warn`, `vwarn`, `warnx`, `vwarnx`, `err`, `verr`, `errx`, `verrx`.

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Error-Messages.html,
- `man err`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on some platforms: AIX 5.1, HP-UX 11, Solaris 10, mingw, MSVC 14.

12.11 `error.h`

Declares the functions `error`, `error_at_line` and the variables `error_print_progname`, `error_message_count`, `error_one_per_line`.

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Error-Messages.html,
- `man error`.

Gnulib module: `error-h`

Portability problems fixed by Gnulib:

- This header file is missing on many platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.9, mingw, MSVC 14.

Portability problems not fixed by Gnulib:

12.12 `execinfo.h`

Declares the functions `backtrace`, `backtrace_symbols`, `backtrace_symbols_fd`.

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Backtraces.html,
- `man backtrace`.

Gnulib module: `execinfo`

Portability problems fixed by Gnulib:

- This header file is missing on some platforms: musl libc, FreeBSD 9.3, NetBSD 6.1, OpenBSD 6.9, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 10, Cygwin, mingw, MSVC 14, Android 9.0.

Portability problems not fixed by Gnulib:

- On platforms where the header file is missing, the Gnulib substitute implementation is just a stub, and does nothing.

12.13 `fpu_control.h`

Handling of the FPU control word. Defines the `fpu_control_t` type, declares the `__fpu_control` variable, and defines the `_FPU_GETCW`, `_FPU_SETCW` macros.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin, mingw, MSVC 14, Android 9.0.

12.14 `fstab.h`

Defines the type `struct fstab`, the macros `FSTAB_*`, `_PATH_FSTAB`, and declares the functions `setfsent`, `endfsent`, `getfsent`, `getfsspec`, `getfsfile`.

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/fstab.html,
- `man setfsent`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on some platforms: Minix 3.1.8, Solaris 11.4, Cygwin, mingw, MSVC 14, Android 9.0.
- The macro `_PATH_FSTAB` is missing on some platforms: AIX 5.1, HP-UX 11.

12.15 `fts.h`

Defines the types `FTS`, `FTSENT` and the macros `FTS_*`, and declares the functions `fts_open`, `fts_read`, `fts_children`, `fts_set`, `fts_close`.

Documentation:

- `man fts`.

Gnulib module: `fts`

Portability problems fixed by Gnulib:

- This header file is missing on some platforms: AIX 5.1, HP-UX 11, Solaris 11.3, mingw, MSVC 14.

Portability problems not fixed by Gnulib:

12.16 getopt.h

Defines the type `struct option` and declares the variables `optarg`, `optind`, `opterr`, `optopt` and the functions `getopt`, `getopt_long`, `getopt_long_only`.

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Getopt.html,
- `man getopt`.

Gnulib module: `getopt-gnu`

Portability problems fixed by Gnulib:

- This header file is missing on some platforms: AIX 7.2, HP-UX 11, MSVC 14.
- The function `getopt_long` is missing on some platforms: Solaris 9.
- The function `getopt_long_only` is missing on some platforms: FreeBSD 5.2.1, NetBSD 10.0, Solaris 9, mingw.
- The method to reset options is incompatible on some platforms: FreeBSD 6.0, NetBSD 10.0(?), OpenBSD 7.5, Cygwin 1.5.x, mingw.
- The function `getopt` does not handle a leading ‘+’ character in the options string on some platforms: Solaris 11 2010-11.

Portability problems not fixed by Gnulib:

12.17 gshadow.h

Defines the type `struct sgrp` and declares the functions `setsgent`, `endsgent`, `getsgent`, `getsgnam`, `sgetsgent`, `fgetsgent`, `putsgent`, `getsgent_r`, `getsgnam_r`, `sgetsgent_r`, `fgetsgent_r`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin, mingw, MSVC 14, Android 9.0.

12.18 ieee754.h

Defines the types `union ieee754_float`, `union ieee754_double`, `union ieee854_long_double`.

Gnulib module: `ieee754-h`

Portability problems fixed by Gnulib:

- This header file is missing on all non-glibc platforms: glibc/sparc64 in 32-bit mode, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin, mingw, MSVC 14, Android 9.0.

Portability problems not fixed by Gnulib:

- The substitute for this header file has not been tested for `long double` and does not work on some platforms.

- The substitute for this header file returns nonsense on (now-quite-rare) platforms that do not use IEEE floating point, such as Linux/m68k.

12.19 ifaddrs.h

Defines the type `struct ifaddrs` and declares the functions `getifaddrs`, `freeifaddrs`.

Documentation:

- <https://www.kernel.org/doc/man-pages/online/pages/man3/getifaddrs.3.html>.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on some platforms: AIX 5.1, HP-UX 11, Solaris 10, Cygwin 1.5.x, mingw, MSVC 14.

12.20 link.h

Defines the type `struct dl_phdr_info`, and declares the function `dl_iterate_phdr`.

Documentation: —

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on some platforms: macOS 14, AIX 7.1, HP-UX 11.31, Cygwin 2.9.0, mingw, MSVC 14.

12.21 malloc.h

Declares the function `memalign` and functions for customizing the `malloc` behavior.

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Unconstrained-Allocation.html.

Gnulib module: `malloc-h`

Portability problems fixed by Gnulib:

- This header file is missing on some platforms: macOS 14, OpenBSD 7.5.

Portability problems not fixed by Gnulib:

12.22 mcheck.h

Defines the type `enum mcheck_status` and declares the functions `mcheck`, `mcheck_pedantic`, `mcheck_check_all`, `mprobe`, `mtrace`, `muntrace`.

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Heap-Consistency-Checking.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin, mingw, MSVC 14, Android 9.0.

12.23 mntent.h

Defines the type `struct mntent` and the macros `MNTTAB`, `MOUNTED`, `MNTTYPE_*`, `MNTOPT_*`, and declares the functions `setmntent`, `getmntent`, `getmntent_r`, `addmntent`, `endmntent`, `hasmntopt`.

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/mntab.html,
- `man setmntent`.

Gnulib module: `mntent-h`

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Solaris 11.4, mingw, MSVC 14.
- The function `getmntent_r` is missing on all non-glibc platforms: AIX 5.1, HP-UX 11, Cygwin, Android 9.0.

Gnulib module `mountlist` provides a higher-level abstraction.

12.24 obstack.h

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Obstacks.html.

Gnulib module: `obstack`

Portability problems fixed by Gnulib:

- This header file is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin, mingw, MSVC 14, Android 9.0.

Portability problems not fixed by Gnulib:

12.25 `paths.h`

Defines the macros `_PATH_*`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on some platforms: Minix 3.1.8, HP-UX 11, Solaris 11 2010-11, mingw, MSVC 14.
- The set of `_PATH_*` macros is platform dependent.

12.26 `printf.h`

Defines the type `struct printf_info` and the macros and enum values `PA_*`, and declares the functions `printf_function`, `printf_arginfo_function`, `register_printf_function`, `parse_printf_format`, `printf_size`, `printf_size_info`.

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Parsing-a-Template-String.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on some platforms: Mac OS X 10.5, FreeBSD 6.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin, mingw, MSVC 14, Android 9.0.

12.27 `pty.h`

Declares the functions `openpty` and `forkpty`.

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Pseudo_002dTerminal-Pairs.html,
- `man openpty`.

Gnulib module: `pty-h`

Portability problems fixed by Gnulib:

- This header file is missing on some platforms that declare the `forkpty` and `openpty` functions in `util.h` or `libutil.h` instead: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5.
- This header file is missing on some platforms: Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4.

Portability problems not fixed by Gnulib:

- This header file is missing on some platforms: mingw, MSVC 14.

12.28 resolv.h

Defines the types `res_sendhookact`, `res_send_qhook`, `res_send_rhook`, `res_state`, `struct res_sym` and the macros `_PATH_RESCONF`, `RES_*`, and declares the functions `res_close`, `res_init`, `res_mkquery`, `res_query`, `res_querydomain`, `res_search`, `res_send`.

Documentation:

- [man res_init](#).

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on some platforms: Cygwin 1.5.x, mingw, MSVC 14.
- The functions are missing on some platforms: HP-UX 11.

12.29 shadow.h

Defines the type `struct spwd` and declares the functions `setspent`, `endspent`, `getspent`, `getspent_r`, `getspnam`, `getspnam_r`, `sgetspent`, `sgetspent_r`, `fgetspent`, `fgetspent_r`, `putspent`, `lckpword`, `ulckpword`.

Documentation:

- [man setspent](#).

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, Cygwin, mingw, MSVC 14, Android 9.0.
- The functions `getspent_r`, `getspnam_r`, `sgetspent_r`, `fgetspent`, `fgetspent_r`, `putspent` are missing on some platforms: HP-UX 11.
- The functions `sgetspent`, `sgetspent_r` are missing on some platforms: HP-UX 11, Solaris 11.4.

12.30 sys/file.h

Declares the function `flock`.

Gnulib module: `sys_file-h`

Portability problems fixed by Gnulib:

- This header file is missing on some platforms: MSVC 14.

Portability problems not fixed by Gnulib:

- This header file does not declare the `flock` function on some platforms: AIX 7.1, HP-UX 11.23, Solaris 11.4.

12.31 `sys/ioctl.h`

Declares the function `ioctl`.

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/IOCTLs.html,
- `man ioctl`.

Gnulib module: `sys_ioctl-h`

Portability problems fixed by Gnulib:

- This header file is missing on some platforms: mingw, MSVC 14.
- This header file does not declare the `ioctl` function on some platforms: AIX 5.1, Solaris 11.4, Haiku 2017.

Portability problems not fixed by Gnulib:

12.32 `sys/random.h`

Declares the function `getrandom` and the flags for it.

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Unpredictable-Bytes.html,
- `man getrandom`.

Gnulib module: `sys_random-h`

Portability problems fixed by Gnulib:

- This header file is missing on some platforms: glibc 2.24, NetBSD 9.3, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.11, Cygwin 2.6.x, mingw, MSVC 14.
- This header file is not self-contained on some platforms: uClibc, macOS 11.1.

Portability problems not fixed by Gnulib:

12.33 `sysexits.h`

Defines the `EX_*` macros, including `EX_OK`.

Gnulib module: `sysexits-h`

Portability problems fixed by Gnulib:

- This header file is missing on some platforms: mingw, MSVC 14.
- This header file defines no `EX_*` macros if `_BSD_SOURCE` is not defined on some platforms: Haiku.
- The macro `EX_CONFIG` is missing on some platforms: HP-UX 11.

Portability problems not fixed by Gnulib:

12.34 `ttyent.h`

Defines the type `struct ttyent` and declares the functions `setttyent`, `endttyent`, `getttyent`, `getttynam`.

Documentation:

- [man setttyent](#).

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This header file is missing on some platforms: HP-UX 11, Solaris 11.4, Cygwin, mingw, MSVC 14, Android 9.0.

12.35 `utmp.h`

Defines functions for login and logout (to a tty session) and for examining the history of logins and logouts.

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Manipulating-the-Database.html,
- [man utmp](#).

Gnulib module: `utmp-h`

Portability problems fixed by Gnulib:

- This header file is missing on some platforms: FreeBSD 14.0, mingw, MSVC 14.
- `<sys/types.h>` is a prerequisite of `<utmp.h>` on some platforms: FreeBSD 8.0, OpenBSD 7.2.

Portability problems not fixed by Gnulib:

- While some platforms have the `struct utmp` field `ut_user`, older platforms have the field `ut_name`.
- The `struct utmp` fields `ut_id`, `ut_pid`, `ut_type` do not exist on some platforms: macOS, old FreeBSD, NetBSD, OpenBSD, Minix.
- The `struct utmp` field `ut_host` does not exist on some platforms: Solaris.
- The `struct utmp` field `ut_exit` does not exist on some platforms: macOS, old FreeBSD, NetBSD, OpenBSD, Minix, Cygwin.
- The `struct utmp` field `ut_session` does not exist on some platforms: macOS, old FreeBSD, NetBSD, OpenBSD, Minix, AIX, HP-UX, Solaris, Cygwin.
- The `struct utmp` field `ut_addr` or `ut_addr_v6` does not exist on some platforms: macOS, old FreeBSD, NetBSD, OpenBSD, Minix, AIX, Solaris.
- On some platforms, this API does not support timestamps past the year 2038: glibc 2.38 on 32-bit platforms like x86 and ARM where `time_t` was historically 32 bits.
- On some platforms, this header misbehaves if the `year2038` or `year2038-recommended` modules are used and the program is configured without the `--disable-year2038` option. The `readutmp` module works around this problem: glibc 2.38 on 32-bit platforms

like x86 and ARM where `time_t` was historically 32 bits. See [Section 14.5 \[Avoiding the year 2038 problem\]](#), page 843.

13 Glibc Function Substitutes

This chapter describes which functions and function-like macros provided as extensions by at least GNU libc are also supported by Gnulib, which portability pitfalls are fixed by Gnulib, and which (known) portability problems are not worked around by Gnulib.

The notation “Gnulib module: —” means that Gnulib does not provide a module providing a substitute for the function. When the list “Portability problems not fixed by Gnulib” is empty, such a module is not needed: No portability problems are known. Otherwise, it indicates that such a module would be useful but is not available: No one so far found this function important enough to contribute a substitute for it. If you need this particular function, you may write to <bug-gnulib at gnu dot org>.

This list of functions is sorted according to the header that declares them.

13.1 Glibc Extensions to <aio.h>

13.1.1 aio_init

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Configuration-of-AIO.html,
- `man aio_init`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.2 Glibc <aliases.h>

13.2.1 endaliasent

Documentation:

`man endaliasent`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.2.2 getaliasbyname

Documentation:

`man getaliasbyname`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.2.3 getaliasbyname_r

Documentation:

`man getaliasbyname_r`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.2.4 getaliasent

Documentation:

`man getaliasent`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.2.5 getaliasent_r

Documentation:

`man getaliasent_r`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.2.6 setaliasent

Documentation:

`man setaliasent`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.3 Glibc <argp.h>

13.3.1 argp_err_exit_status

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Argp-Global-Variables.html.

Gnulib module: argp

Portability problems fixed by Gnulib:

- This variable is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

Portability problems not fixed by Gnulib:

13.3.2 argp_error

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Argp-Helper-Functions.html.

Gnulib module: argp

Portability problems fixed by Gnulib:

- This function is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

Portability problems not fixed by Gnulib:

13.3.3 argp_failure

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Argp-Helper-Functions.html.

Gnulib module: argp

Portability problems fixed by Gnulib:

- This function is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

Portability problems not fixed by Gnulib:

13.3.4 `argp_help`

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Argp-Help.html.

Gnulib module: `argp`

Portability problems fixed by Gnulib:

- This function is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

Portability problems not fixed by Gnulib:

13.3.5 `argp_parse`

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Argp.html.

Gnulib module: `argp`

Portability problems fixed by Gnulib:

- This function is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

Portability problems not fixed by Gnulib:

13.3.6 `argp_program_bug_address`

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Argp-Global-Variables.html.

Gnulib module: `argp`

Portability problems fixed by Gnulib:

- This variable is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

Portability problems not fixed by Gnulib:

13.3.7 `argp_program_version`

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Argp-Global-Variables.html.

Gnulib module: `argp`

Portability problems fixed by Gnulib:

- This variable is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

Portability problems not fixed by Gnulib:

13.3.8 `argp_program_version_hook`

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Argp-Global-Variables.html.

Gnulib module: `argp`

Portability problems fixed by Gnulib:

- This variable is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

Portability problems not fixed by Gnulib:

13.3.9 `argp_state_help`

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Argp-Helper-Functions.html.

Gnulib module: `argp`

Portability problems fixed by Gnulib:

- This function is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

Portability problems not fixed by Gnulib:

13.3.10 `argp_usage`

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Argp-Helper-Functions.html.

Gnulib module: `argp`

Portability problems fixed by Gnulib:

- This function is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

Portability problems not fixed by Gnulib:

13.4 Glibc `<argz.h>`

13.4.1 `argz_add`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-argz-add.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Argz-Functions.html,
- `man argz-add`.

Gnulib module: `argz`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, mingw, MSVC 14, Android 9.0.
- This function is broken on some platforms: Cygwin 1.5.24.

Portability problems not fixed by Gnulib:

13.4.2 `argz_add_sep`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-argz-add.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Argz-Functions.html,
- `man argz-add_sep`.

Gnulib module: `argz`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, mingw, MSVC 14, Android 9.0.
- This function is broken on some platforms: Cygwin 1.5.24.

Portability problems not fixed by Gnulib:

13.4.3 `argz_append`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-argz-add.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Argz-Functions.html,
- `man argz-append`.

Gnulib module: `argz`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, mingw, MSVC 14, Android 9.0.
- This function is broken on some platforms: Cygwin 1.5.24.

Portability problems not fixed by Gnulib:

13.4.4 `argz_count`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-argz-add.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Argz-Functions.html,
- `man argz_count`.

Gnulib module: `argz`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, mingw, MSVC 14, Android 9.0.
- This function is broken on some platforms: Cygwin 1.5.24.

Portability problems not fixed by Gnulib:

13.4.5 `argz_create`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-argz-add.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Argz-Functions.html,
- `man argz_create`.

Gnulib module: `argz`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, mingw, MSVC 14, Android 9.0.
- This function is broken on some platforms: Cygwin 1.5.24.

Portability problems not fixed by Gnulib:

13.4.6 `argz_create_sep`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-argz-add.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Argz-Functions.html,
- `man argz_create_sep`.

Gnulib module: `argz`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, mingw, MSVC 14, Android 9.0.

- This function is broken on some platforms: Cygwin 1.5.24.

Portability problems not fixed by Gnulib:

13.4.7 `argz_delete`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-argz-add.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Argz-Functions.html,
- `man argz_delete`.

Gnulib module: `argz`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, mingw, MSVC 14, Android 9.0.
- This function is broken on some platforms: Cygwin 1.5.24.

Portability problems not fixed by Gnulib:

13.4.8 `argz_extract`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-argz-add.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Argz-Functions.html,
- `man argz_extract`.

Gnulib module: `argz`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, mingw, MSVC 14, Android 9.0.
- This function is broken on some platforms: Cygwin 1.5.24.

Portability problems not fixed by Gnulib:

13.4.9 `argz_insert`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-argz-add.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Argz-Functions.html,
- `man argz_insert`.

Gnulib module: `argz`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, mingw, MSVC 14, Android 9.0.
- This function is broken on some platforms: Cygwin 1.5.24.

Portability problems not fixed by Gnulib:

13.4.10 `argz_next`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-argz-add.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Argz-Functions.html,
- `man argz_next`.

Gnulib module: `argz`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, mingw, MSVC 14, Android 9.0.
- This function is broken on some platforms: Cygwin 1.5.24.

Portability problems not fixed by Gnulib:

13.4.11 `argz_replace`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-argz-add.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Argz-Functions.html,
- `man argz_replace`.

Gnulib module: `argz`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, mingw, MSVC 14, Android 9.0.
- This function is broken on some platforms: Cygwin 1.5.24.

Portability problems not fixed by Gnulib:

13.4.12 argz_stringify

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-argz-add.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Argz-Functions.html,
- `man argz_stringify`.

Gnulib module: `argz`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, mingw, MSVC 14, Android 9.0.
- This function is broken on some platforms: Cygwin 1.5.24.

Portability problems not fixed by Gnulib:

13.5 Glibc Extensions to `<arpa/inet.h>`

13.5.1 inet_aton

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-inet-aton-3.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Host-Address-Functions.html,
- `man inet_aton`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

13.5.2 inet_lnaof

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Host-Address-Functions.html,
- `man inet_lnaof`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 4.4.

13.5.3 inet_makeaddr

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Host-Address-Functions.html,
- `man inet_makeaddr`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14, Android 4.4.

13.5.4 inet_net_ntop

Documentation:

`man inet_net_ntop`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, HP-UX 11.00, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.5.5 inet_net_pton

Documentation:

`man inet_net_pton`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, HP-UX 11.00, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.5.6 inet_netof

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Host-Address-Functions.html,
- `man inet_netof`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14, Android 4.4.

13.5.7 inet_network

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Host-Address-Functions.html,
- `man inet_network`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14, Android 4.4.

13.5.8 inet_nsap_addr

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: OpenBSD 7.5, Minix 3.1.8, HP-UX 11.00, Cygwin 2.9, mingw, MSVC 14.

13.5.9 inet_nsap_ntoa

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: OpenBSD 7.5, Minix 3.1.8, HP-UX 11.00, Cygwin 2.9, mingw, MSVC 14.

13.6 Glibc Extensions to <arpa/nameser.h>

13.6.1 ns_name_compress

Documentation:

https://web.deu.edu.tr/doc/oreilly/networking/dnsbind/ch14_02.htm

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.33, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, HP-UX 11.11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.6.2 ns_name_ntop

Documentation: —

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.33, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, HP-UX 11.11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.6.3 ns_name_pack

Documentation: —

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.33, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, HP-UX 11.11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.6.4 ns_name_pton

Documentation: —

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.33, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, HP-UX 11.11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.6.5 ns_name_skip

Documentation:

https://web.deu.edu.tr/doc/oreily/networking/dnsbind/ch14_02.htm

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.33, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, HP-UX 11.11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.6.6 ns_name_uncompress

Documentation:

https://web.deu.edu.tr/doc/oreilly/networking/dnsbind/ch14_02.htm

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.33, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, HP-UX 11.11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.6.7 ns_name_unpack

Documentation: —

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.33, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, HP-UX 11.11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

<https://man7.org/linux/man-pages/man3/libres.3.html> <https://www.kernel.org/doc/man-pages/online/pages/man3/>

13.7 Glibc <byteswap.h>

13.7.1 bswap_16

Documentation:

[man bswap_16](#)

Gnulib module: byteswap

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.7.2 bswap_32

Documentation:

[man bswap_32](#)

Gnulib module: byteswap

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.7.3 bswap_64

Documentation:

`man bswap_64`

Gnulib module: byteswap

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.8 Glibc Extensions to <complex.h>

13.8.1 clog10

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib---clog10.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Exponents-and-Logarithms.html,
- `man clog10`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 9, Android 9.0.

13.8.2 clog10f

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib---clog10f.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Exponents-and-Logarithms.html,
- `man clog10f`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 9, Android 9.0.

13.8.3 clog10l

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib---clog10l.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Exponents-and-Logarithms.html,
- `man clog10l`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 9, Android 9.0.

13.9 Glibc Extensions to <ctype.h>

13.9.1 isctype

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.10 Glibc Extensions to <dirent.h>

13.10.1 getdirentries

Documentation:

`man getdirentries`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: OpenBSD 7.5, Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.
- On platforms where `off_t` is a 32-bit type, this function may not work correctly on huge directories larger than 2 GB. The fix is to use the `AC_SYS_LARGEFILE` macro.

13.10.2 scandirat

Documentation:

`man scandirat`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: glibc 2.14, macOS 14, FreeBSD 13.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 1.7.10, mingw, MSVC 14, Android 6.0.

13.10.3 versionsort

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Scanning-Directory-Content.html,
- `man versionsort`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: macOS 14, FreeBSD 13.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android 9.0.

13.11 Glibc Extensions to <dlfcn.h>

13.11.1 dladdr1

Documentation:

`man dladdr1`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.11.2 dlinfo

Documentation:

`man dlinfo`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.11.3 dlmopen

Documentation:

[man dlmopen](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.11.4 dlvsym

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-dlvsym-1.html

Documentation:

[man dlvsym](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 6.0, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 6.0.

13.11.5 _dl_find_object

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Dynamic-Linker-Introspection.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.34, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

13.12 Glibc <envz.h>

13.12.1 envz_add

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-envz-add.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Envz-Functions.html,

- [man envz_add](#).

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, mingw, MSVC 14, Android 9.0.

13.12.2 envz_entry

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-envz-add.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Envz-Functions.html,
- [man envz_entry](#).

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, mingw, MSVC 14, Android 9.0.

13.12.3 envz_get

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-envz-add.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Envz-Functions.html,
- [man envz_get](#).

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, mingw, MSVC 14, Android 9.0.

13.12.4 envz_merge

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-envz-add.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Envz-Functions.html,

- `man envz_merge`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, mingw, MSVC 14, Android 9.0.

13.12.5 `envz_remove`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-envz-add.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Envz-Functions.html,
- `man envz_remove`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, mingw, MSVC 14, Android 9.0.

13.12.6 `envz_strip`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-envz-add.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Envz-Functions.html,
- `man envz_strip`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, mingw, MSVC 14, Android 9.0.

13.13 Glibc <err.h>

13.13.1 err

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-err-3.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Error-Messages.html,
- `man err`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: AIX 5.1, HP-UX 11, Solaris 10, mingw, MSVC 14.

13.13.2 errx

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-errx-3.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Error-Messages.html,
- `man errx`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: AIX 5.1, HP-UX 11, Solaris 10, mingw, MSVC 14.

13.13.3 verr

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Error-Messages.html,
- `man verr`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: AIX 5.1, HP-UX 11, Solaris 10, mingw, MSVC 14.

13.13.4 verrx

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-verrx-3.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Error-Messages.html,
- `man verrx`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: AIX 5.1, HP-UX 11, Solaris 10, mingw, MSVC 14.

13.13.5 vwarn

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Error-Messages.html,
- `man vwarn`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: AIX 5.1, HP-UX 11, Solaris 10, mingw, MSVC 14.

13.13.6 vwarnx

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Error-Messages.html,
- `man vwarnx`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: AIX 5.1, HP-UX 11, Solaris 10, mingw, MSVC 14.

13.13.7 warn

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-warn-3.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Error-Messages.html,
- `man warn`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: AIX 5.1, HP-UX 11, Solaris 10, mingw, MSVC 14.

13.13.8 warnx

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-warnx-3.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Error-Messages.html,
- `man warnx`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: AIX 5.1, HP-UX 11, Solaris 10, mingw, MSVC 14.

13.14 Glibc Extensions to <errno.h>

13.14.1 program_invocation_name

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Error-Messages.html,
- `man program_invocation_name`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This variable is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.7, mingw, MSVC 14, Android 9.0.

13.14.2 program_invocation_short_name

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Error-Messages.html,
- `man program_invocation_short_name`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This variable is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.7, mingw, MSVC 14, Android 9.0.

13.15 Glibc <error.h>

13.15.1 error

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-error-n.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Error-Messages.html,
- `man error`.

Gnulib module: error

Portability problems fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.9, mingw, MSVC 14, Android 5.1.
- This function outputs an extra space if `error_print_progname` is set, on some platforms: Android 11.

Portability problems not fixed by Gnulib:

13.15.2 error_at_line

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Error-Messages.html,
- `man error_at_line`.

Gnulib module: error

Portability problems fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.9, mingw, MSVC 14, Android 5.1.

Portability problems not fixed by Gnulib:

13.15.3 error_message_count

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Error-Messages.html,
- `man error_message_count`.

Gnulib module: error

Portability problems fixed by Gnulib:

- This variable is missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.9, mingw, MSVC 14, Android 5.1.

Portability problems not fixed by Gnulib:

13.15.4 error_one_per_line

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Error-Messages.html,
- `man error_one_per_line`.

Gnulib module: error

Portability problems fixed by Gnulib:

- This variable is missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.9, mingw, MSVC 14, Android 5.1.

Portability problems not fixed by Gnulib:

13.15.5 error_print_progname

Documentation:

`man error_print_progname`

Gnulib module: error

Portability problems fixed by Gnulib:

- This variable is missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.9, mingw, MSVC 14, Android 5.1.

Portability problems not fixed by Gnulib:

13.16 Glibc <execinfo.h>

13.16.1 backtrace

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-backtrace-1.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Backtraces.html,
- `man backtrace`.

Gnulib module: execinfo

Portability problems fixed by Gnulib:

- This function is missing on many platforms: musl libc, FreeBSD 9.3, NetBSD 6.1, OpenBSD 6.9, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 10, Cygwin 2.9, mingw, MSVC 14, Android API level 32.

Portability problems not fixed by Gnulib:

- The second argument is of type `size_t`, not `int`, on some platforms: FreeBSD, NetBSD, OpenBSD.
- The return value may be `(size_t)(-1)` instead of 0 on some platforms: FreeBSD 12.4/i386.

- The returned stack trace is empty if it would not entirely fit into the provided buffer on some platforms: FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5.
- The returned stack trace is always empty on some platforms: FreeBSD 11.0/i386, FreeBSD 12.2/sparc64.
- On platforms where the function is missing, the Gnulib substitute implementation is just a stub, and does nothing.

13.16.2 backtrace_symbols

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-backtrace-1.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Backtraces.html,
- `man backtrace_symbols`.

Gnulib module: `execinfo`

Portability problems fixed by Gnulib:

- This function is missing on many platforms: musl libc, FreeBSD 9.3, NetBSD 6.1, OpenBSD 6.9, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 10, Cygwin 2.9, mingw, MSVC 14, Android API level 32.

Portability problems not fixed by Gnulib:

- The second argument is of type `size_t`, not `int`, on some platforms: FreeBSD, NetBSD, OpenBSD.
- On platforms where the function is missing, the Gnulib substitute implementation is just a stub, and does nothing.

13.16.3 backtrace_symbols_fd

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-backtrace-1.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Backtraces.html,
- `man backtrace_symbols_fd`.

Gnulib module: `execinfo`

Portability problems fixed by Gnulib:

- This function is missing on many platforms: musl libc, FreeBSD 9.3, NetBSD 6.1, OpenBSD 6.9, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 10, Cygwin 2.9, mingw, MSVC 14, Android API level 32.

Portability problems not fixed by Gnulib:

- The second argument is of type `size_t`, not `int`, on some platforms: FreeBSD, NetBSD, OpenBSD.
- The strings printed by this function not the same as the strings returned by `backtrace_symbols` on some platforms: glibc 2.39.

- On platforms where the function is missing, the Gnulib substitute implementation is just a stub, and does nothing.

13.17 Glibc Extensions to <fcntl.h>

13.17.1 `fallocate`

Documentation:

`man fallocate`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc 2.9, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 3.4.x, mingw, MSVC 14, Android 4.4.
- On platforms where `off_t` is a 32-bit type, this function may not work correctly across the entire data range of files larger than 2 GB. The fix is to use the `AC_SYS_LARGEFILE` macro.

13.17.2 `name_to_handle_at`

Documentation:

`man name_to_handle_at`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on all non-glibc platforms: glibc 2.13, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.17.3 `readahead`

Documentation:

`man readahead`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 4.0.4.

13.17.4 `open_by_handle_at`

Documentation:

`man open_by_handle_at`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on all non-glibc platforms: glibc 2.13, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.17.5 sync_file_range

Documentation:

[man sync_file_range](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on many non-glibc platforms: glibc 2.5, GNU/Hurd, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 4.3.

13.18 Glibc Extensions to <fenv.h>

13.18.1 fedisableexcept

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-fedisableexcept.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Control-Functions.html,
- [man fedisableexcept](#).

Gnulib module: fenv-exceptions-trapping

Portability problems fixed by Gnulib:

- This function is missing on some platforms: macOS 14, musl libc, FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.7, mingw, MSVC 14, Android 4.4.

Portability problems not fixed by Gnulib:

13.18.2 feenableexcept

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-feenableexcept.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Control-Functions.html,
- [man feenableexcept](#).

Gnulib module: fenv-exceptions-trapping

Portability problems fixed by Gnulib:

- This function is missing on some platforms: macOS 14, musl libc, FreeBSD 5.2.1, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.7, mingw, MSVC 14, Android 4.4.
- This function does not detect failures on glibc 2.19/aarch64, FreeBSD 12.2/arm, FreeBSD 12.2/arm64, NetBSD 10.0/arm64.

Portability problems not fixed by Gnulib:

- The resulting signal is SIGILL instead of SIGFPE on some platforms: Linux/hppa, macOS 14/arm64.

13.18.3 fegetexcept

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-fegetexcept.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Control-Functions.html,
- `man fegetexcept`.

Gnulib module: fenv-exceptions-trapping

Portability problems fixed by Gnulib:

- This function is missing on many non-glibc platforms: Mac OS X 10.4, musl libc, FreeBSD 14.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.7, mingw, MSVC 14, Android 4.4.

Portability problems not fixed by Gnulib:

13.19 Glibc Extensions to <fmtmsg.h>

13.19.1 addseverity

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Adding-Severity-Classes.html,
- `man addseverity`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.20 Glibc <fstab.h>

13.20.1 endfsent

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/fstab.html,
- `man endfsent`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.20.2 getfsent

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/fstab.html,
- `man getfsent`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.20.3 getfsfile

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/fstab.html,
- `man getfsfile`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.20.4 getfsspec

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/fstab.html,
- `man getfsspec`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.20.5 setfsent

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/fstab.html,
- `man setfsent`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.21 Glibc <fts.h>

13.21.1 fts_children

Documentation:

`man fts_children`

Gnulib module: fts

Portability problems fixed by Gnulib:

- This function is missing on some platforms: AIX 5.1, HP-UX 11, Solaris 11.3, mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- On platforms where `off_t` is a 32-bit type, this function may not correctly report the size of files or block devices larger than 2 GB and may not work correctly on huge directories larger than 2 GB. Also, on platforms where `ino_t` is a 32-bit type, this function may report inode numbers incorrectly. The fix is to use the `AC_SYS_LARGEFILE` macro (only on Mac OS X systems).

13.21.2 fts_close

Documentation:

`man fts_close`

Gnulib module: fts

Portability problems fixed by Gnulib:

- This function is missing on some platforms: AIX 5.1, HP-UX 11, Solaris 11.3, mingw, MSVC 14.

Portability problems not fixed by Gnulib:

13.21.3 fts_open

Documentation:

`man fts_open`

Gnulib module: fts

Portability problems fixed by Gnulib:

- This function is missing on some platforms: AIX 5.1, HP-UX 11, Solaris 11.3, mingw, MSVC 14.

Portability problems not fixed by Gnulib:

13.21.4 `fts_read`

Documentation:

`man fts_read`

Gnulib module: `fts`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: AIX 5.1, HP-UX 11, Solaris 11.3, mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- On platforms where `off_t` is a 32-bit type, this function may not correctly report the size of files or block devices larger than 2 GB and may not work correctly on huge directories larger than 2 GB. Also, on platforms where `ino_t` is a 32-bit type, this function may report inode numbers incorrectly. The fix is to use the `AC_SYS_LARGEFILE` macro (only on Mac OS X systems).

13.21.5 `fts_set`

Documentation:

`man fts_set`

Gnulib module: `fts`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: AIX 5.1, HP-UX 11, Solaris 11.3, mingw, MSVC 14.

Portability problems not fixed by Gnulib:

13.22 Glibc `<getopt.h>`

13.22.1 `getopt_long`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-getopt-long-3.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Getopt-Long-Options.html,
- `man getopt_long`.

Gnulib module: `getopt-gnu`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: AIX 5.1, HP-UX 11, MSVC 14.
- The function `getopt_long` does not obey the combination of ‘+’ and ‘:’ flags in the options string on some platforms: glibc 2.11.
- The use of ‘W;’ in the optstring argument to does not always allow `-W foo` to behave synonymously with `--foo`: glibc 2.11.

- The function `getopt_long` does not support the '+' flag in the options string on some platforms: macOS 14, AIX 5.2, Solaris 10.
- The value of `optind` after a missing required argument is wrong on some platforms: macOS 14.
- The function `getopt_long` does not obey the '-' flag in the options string when `POSIXLY_CORRECT` is set on some platforms: Cygwin 1.7.0.
- Some implementations fail to reset state, including re-checking `POSIXLY_CORRECT`, when `optind` is set to '0': NetBSD, Cygwin 1.7.0.
- The function `getopt_long` does not support options with optional arguments on some platforms: macOS 14, OpenBSD 4.0, AIX 5.2, Solaris 11 2010-11, Cygwin 1.5.x.
- This function crashes if the option string includes W; but there are no long options, on some platforms: glibc 2.14.

Portability problems not fixed by Gnulib:

Gnulib provides also a module `options`, that fixes the following shortcomings of the `getopt_long` API.

These shortcomings are best illustrated with an example:

```
static struct option const long_options[] =
{
    { "width", required_argument, NULL, 'w' },
    { "help", no_argument, &show_help, 1 },
    { "version", no_argument, &show_version, 1 },
    { NULL, 0, NULL, 0 }
};

while ((optchar = getopt_long (argc, argv, "w:xhV", long_options, NULL))
    != -1)
    switch (optchar)
    {
        case '\0':          /* Long option with flag != NULL.  */
            break;
        case 'w':
            set_width (optarg);
            break;
        case 'x':
            do_x = true;
            break;
        case 'h':
            show_help = 1;   /* Action code duplication!  */
            break;
        case 'V':
            show_version = 1; /* Action code duplication!  */
            break;
        default:
            usage (EXIT_FAILURE);
    }
```

```
}
```

- The information whether an option takes a required vs. optional argument needs to be specified twice: in the `option[]` array for the long option and in the string argument for the short option. It is too easy to forget to add the ":" or "::" part in the string argument and thus get inconsistent behaviour between the long option and the short option.
- This information needs to be specified twice, but in different ways:

In the array	In the string
<code>no_argument</code>	<code>""</code>
<code>required_argument</code>	<code>":"</code>
<code>optional_argument</code>	<code>:::"</code>

- For an action that merely sets an `int`-typed variable to a value, you can specify this action in the `options[]` array, and thus omit the handling in the `switch` statement. But this works only for options that are long options without a corresponding short option. As soon as the option has a corresponding short option, you do need to handle it in the `switch` statement. Here again, there is the opportunity for inconsistent behaviour between the long option and the short option.
- The `val` field in a `struct option` has different meanings, depending on another field: If the `flag` field is non-NULL, `val` is a value to be stored in a variable. If the `flag` field is NULL, `val` is a key to be returned from `getopt_long` and subject to the `switch` statement.
- The handling of non-option arguments is specified by prepending a certain character ('+' or '-') to the string argument. This is not one of the usual ways to specify things in an API. The conventional way in an API is an argument of `enum` type, or a flags word.
- The handling of errors consists of two independent flags, and each of the flags has to be specified in a different way: one flag is specified by prepending a certain character (':') to the string argument; the other flag is specified through the global variable `opterr`.
- The `struct option` is a misnomer: It cannot encode short options. Therefore, it would have better been called `struct long_option`.
- The `getopt_long` function is expected to receive the same arguments in each call, in the `while` loop. The effects are undefined if you don't follow this (unwritten!) constraint.
- The fifth argument to `getopt_long`, `indexptr`, is redundant, because when the `flag` is non-NULL, the `switch` statement does not need to handle the option, and when the `flag` is NULL, `getopt_long` returns the value of `val`, as a way to identify which option was seen.

13.22.2 getopt_long_only

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-getopt-long-only-3.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Getopt-Long-Options.html,

- `man getopt_long_only`.

Gnulib module: `getopt-gnu`

Portability problems fixed by Gnulib:

- The function `getopt_long_only` does not obey the combination of ‘+’ and ‘:’ flags in the options string on some platforms: glibc 2.11.
- The use of ‘W;’ in the `optstring` argument to does not always allow `-W foo` to behave synonymously with `--foo`: glibc 2.11.
- The function `getopt_long_only` does not support the ‘+’ flag in the options string on some platforms: macOS 14, AIX 5.2, Solaris 10.
- The value of `optind` after a missing required argument is wrong on some platforms: macOS 14.
- The function `getopt_long_only` does not obey the ‘-’ flag in the options string when `POSIXLY_CORRECT` is set on some platforms: Cygwin 1.7.0.
- Some implementations fail to reset state, including re-checking `POSIXLY_CORRECT`, when `optind` is set to ‘0’: NetBSD, Cygwin 1.7.0.
- The function `getopt_long_only` does not support options with optional arguments on some platforms: macOS 14, OpenBSD 4.0, AIX 5.2, Solaris 11 2010-11, Cygwin 1.5.x.
- This function is missing on some platforms: FreeBSD 5.2.1, NetBSD 10.0, Minix 3.1.8, AIX 5.1, HP-UX 11, mingw, MSVC 14.
- This function crashes if the option string includes `W;` but there are no long options, on some platforms: glibc 2.14.

Portability problems not fixed by Gnulib:

- Some implementations return success instead of reporting an ambiguity if user’s option is a prefix of two long options with the same outcome: FreeBSD.
- The GNU Coding Standards discourage the use of `getopt_long_only` in new programs.

13.23 Glibc Extensions to `<glob.h>`

13.23.1 `glob_pattern_p`

Gnulib module: `glob`

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on most non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.5.x, mingw, MSVC 14, Android 9.0.

13.24 Glibc Extensions to `<gnu/libc-version.h>`

13.24.1 gnu_get_libc_release

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-gnu-get-libc-version-1.html

Documentation:

`man gnu_get_libc_release`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.24.2 gnu_get_libc_version

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-gnu-get-libc-version-1.html

Documentation:

`man gnu_get_libc_version`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.25 Glibc Extensions to <grp.h>

13.25.1 fgetgrent

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Scanning-All-Groups.html,
- `man fgetgrent`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.25.2 fgetgrent_r

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Scanning-All-Groups.html,
- `man fgetgrent_r`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.25.3 getgrent_r

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-getgrent-r-1.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Scanning-All-Groups.html,
- `man getgrent_r`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.25.4 getgrouplist

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-getgrouplist-3.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Setting-Groups.html,
- `man getgrouplist`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function takes `int` instead of `gid_t` parameters on some platforms: OS X 10.11.
- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.0, Cygwin 1.7.9, mingw, MSVC 14.

The Gnulib module `getugroups` provides a similar API.

13.25.5 `initgroups`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-initgroups-3.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Setting-Groups.html,
- `man initgroups`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is unsafe to call between `fork` and `exec` if the parent process is multi-threaded. Instead, use `getgroups` or `getgrouplist` (or use the gnulib module `mgetgroups`) before forking, and `setgroups` in the child.
- This function is missing on some platforms: mingw, MSVC 14.

13.25.6 `putgrent`

Documentation:

`man putgrent`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.25.7 `setgroups`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-setgroups-2.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Setting-Groups.html,
- `man setgroups`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: AIX 5.1, mingw, MSVC 14.
- On very old systems, this function operated on an array of `'int'`, even though that was a different size than an array of `'gid_t'`; you can use `autoconf's AC_TYPE_GETGROUPS` to set `GETGROUPS_T` to the appropriate size (since `getgroups` and `setgroups` share the same bug).

13.26 Glibc <gshadow.h>

13.26.1 endsgent

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.9, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.26.2 fgetsgent

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.9, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.26.3 fgetsgent_r

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.9, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.26.4 getsgent

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.9, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.26.5 getsgent_r

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.9, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.26.6 getsnam

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.9, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.26.7 getsnam_r

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.9, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.26.8 putsgent

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.9, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.26.9 setsgent

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.9, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.26.10 sgetsgent

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.9, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.26.11 sgetsgent_r

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.9, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.27 Glibc <ifaddrs.h>

13.27.1 getifaddrs

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-getifaddrs.html

Documentation:

`man getifaddrs`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: AIX 5.1, HP-UX 11, Solaris 10, Cygwin 1.5.x, mingw, MSVC 14, Android 6.0.

13.27.2 freeifaddrs

Documentation:

`man freeifaddrs`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: AIX 5.1, HP-UX 11, Solaris 10, Cygwin 1.5.x, mingw, MSVC 14, Android 6.0.

13.28 Glibc Extensions to <inttypes.h>

13.28.1 uimaxabs

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Absolute-Value.html,
- `man uimaxabs`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on most platforms: glibc 2.41, macOS 15, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.7, Minix, AIX 7.3, HP-UX 11.31, Solaris 11.4, Cygwin, mingw, MSVC 14.

13.29 Glibc <link.h>

13.29.1 dl_iterate_phdr

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-dl-iterate-phdr-1.html

Documentation:

`man dl_iterate_phdr`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 6.0, NetBSD 5.0, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 10, Cygwin 2.9, mingw, MSVC 14, Android 4.4.

13.30 Glibc <malloc.h>

13.30.1 mallinfo

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Statistics-of-Malloc.html,
- `man mallinfo`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Solaris 11 2010-11, mingw, MSVC 14.

13.30.2 mallinfo2

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Statistics-of-Malloc.html,
- `man mallinfo2`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.32, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14.

13.30.3 malloc_info

Documentation:

`man malloc_info`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: glibc 2.9, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 5.1.

13.30.4 malloc_stats

Documentation:

`man malloc_stats`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, mingw, MSVC 14, Android 9.0.

13.30.5 malloc_trim

Documentation:

`man malloc_trim`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, mingw, MSVC 14, Android 9.0.

13.30.6 malloc_usable_size

Documentation:

`man malloc_usable_size`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 6.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11.00, Solaris 11.4, mingw, MSVC 14, Android 4.1.

13.30.7 mallopt

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Malloc-Tunable-Parameters.html,
- `man mallopt`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Solaris 11 2010-11, mingw, MSVC 14, Android 7.1.

13.30.8 memalign

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Aligned-Memory-Blocks.html,
- `man memalign`.

Gnulib module: memalign

Portability problems fixed by Gnulib:

- This function is declared in `<stdlib.h>` instead of `<malloc.h>` on some platforms: Solaris 11.
- This function doesn't accept an alignment of 1 or 2 on some platforms: Solaris 11.

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 12.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11.00, mingw, MSVC 14.

The Gnulib module `aligned-malloc` provides functions for allocating and freeing blocks of suitably aligned memory.

13.30.9 pvalloc

Documentation:

`man pvalloc`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14.

13.31 Glibc Extensions to <math.h>

13.31.1 drem

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-drem.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Remainder-Functions.html,
- `man drem`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, HP-UX 11, Solaris 11.4, mingw, MSVC 14.

13.31.2 dremf

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-dremf.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Remainder-Functions.html,
- `man dremf`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, mingw, MSVC 14.

13.31.3 dreml

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-dreml.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Remainder-Functions.html,
- `man dreml`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android 9.0.

13.31.4 `finite`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-finite.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Floating-Point-Classes.html,
- `man finite`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, HP-UX 11, MSVC 14.

13.31.5 `finitef`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-finitef.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Floating-Point-Classes.html,
- `man finitef`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, mingw, MSVC 14.

13.31.6 `finitel`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-finitel.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Floating-Point-Classes.html,
- `man finitel`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android 9.0.

13.31.7 gamma

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-gamma.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Special-Functions.html,
- `man gamma`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

13.31.8 gammaf

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-gammaf.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Special-Functions.html,
- `man gammaf`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, Minix 3.1.8, AIX 5.1, HP-UX 11, mingw, MSVC 14.

13.31.9 gammal

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-gammal.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Special-Functions.html,

- `man gammal`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.31.10 `isinff`

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Floating-Point-Classes.html,
- `man isinff`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, mingw, MSVC 14, Android 4.4.
- This function is not declared on some platforms: Android 9.0.

13.31.11 `isinfl`

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Floating-Point-Classes.html,
- `man isinfl`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android 4.4.
- This function is not declared on some platforms: Android 9.0.

13.31.12 `isnanf`

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Floating-Point-Classes.html,
- `man isnanf`.

Gnulib module: isnanf

Portability problems fixed by Gnulib:

- This function is missing on some platforms: macOS 14, Minix 3.1.8, AIX 5.1, HP-UX 11, MSVC 14.

Portability problems not fixed by Gnulib:

13.31.13 isnanl

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Floating-Point-Classes.html,
- `man isnanl`.

Gnulib module: isnanl

Portability problems fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 1.7.x, MSVC 14, Android 4.4.
- This function is not declared on some platforms: Android 9.0.

Portability problems not fixed by Gnulib:

13.31.14 j0f

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-j0f.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Special-Functions.html,
- `man j0f`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, Minix 3.1.8, AIX 5.1, HP-UX 11, mingw, MSVC 14.

13.31.15 j0l

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-j0l.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Special-Functions.html,
- `man j0l`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.31.16 j1f

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-j1f.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Special-Functions.html,
- `man j1f`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, Minix 3.1.8, AIX 5.1, HP-UX 11, mingw, MSVC 14.

13.31.17 j1l

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-j1l.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Special-Functions.html,
- `man j1l`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.31.18 jnf

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-jnf.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Special-Functions.html,
- `man jnf`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, Minix 3.1.8, AIX 5.1, HP-UX 11, mingw, MSVC 14.

13.31.19 jnl

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-jnl.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Special-Functions.html,
- `man jnl`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.31.20 lgamma_r

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-lgamma-r.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Special-Functions.html,
- `man lgamma_r`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Mac OS X 10.5 x86, Minix 3.1.8, mingw, MSVC 14.

13.31.21 lgammaf_r

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-lgammaf-r.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Special-Functions.html,
- `man lgammaf_r`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Mac OS X 10.5, Minix 3.1.8, AIX 5.1, HP-UX 11, mingw, MSVC 14.

13.31.22 lgammal_r

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-lgammal-r.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Special-Functions.html,
- `man lgammal_r`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Mac OS X 10.5, FreeBSD 6.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Cygwin 1.7.x, mingw, MSVC 14, Android 5.1.

13.31.23 matherr

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-matherr-1.html

Documentation:

`man matherr`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 14.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, mingw, MSVC 14, Android 9.0.

13.31.24 pow10

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-pow10.html

Documentation:

`man pow10`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.5.x, mingw, MSVC 14, Android 9.0.

13.31.25 pow10f

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-pow10f.html

Documentation:

`man pow10f`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.5.x, mingw, MSVC 14, Android 9.0.

13.31.26 pow10l

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-pow10l.html

Documentation:

`man pow10l`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android 9.0.

13.31.27 scalbf

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-scalbf.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Normalization-Functions.html,
- `man scalbf`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, Minix 3.1.8, AIX 5.1, HP-UX 11, mingw, MSVC 14.

13.31.28 scalbl

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-scalbl.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Normalization-Functions.html,
- `man scalbl`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 1.7.x, mingw, MSVC 14, Android 9.0.

13.31.29 significand

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-significand.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Normalization-Functions.html,
- `man significand`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, HP-UX 11, mingw, MSVC 14.

13.31.30 `significandf`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-significandf.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Normalization-Functions.html,
- `man significandf`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, Minix 3.1.8, AIX 5.1, HP-UX 11, mingw, MSVC 14.

13.31.31 `significandl`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-significandl.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Normalization-Functions.html,
- `man significandl`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 2.9, mingw, MSVC 14.

13.31.32 `sincos`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-sincos.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Trig-Functions.html,
- `man sincos`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 11.0, NetBSD 9.3, OpenBSD 6.0, Minix 3.1.8, AIX 5.1, HP-UX 11, mingw, MSVC 14.

13.31.33 sincosf

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-sincosf.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Trig-Functions.html,
- `man sincosf`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 11.0, NetBSD 9.3, OpenBSD 6.0, Minix 3.1.8, AIX 5.1, HP-UX 11, mingw, MSVC 14.

13.31.34 sincosl

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-sincosl.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Trig-Functions.html,
- `man sincosl`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 11.0, NetBSD 9.3, OpenBSD 6.0, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 1.7.x, mingw, MSVC 14.

13.31.35 y0f

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-y0f.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Special-Functions.html,
- `man y0f`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, Minix 3.1.8, AIX 5.1, HP-UX 11, mingw, MSVC 14.

13.31.36 y0l

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-y0l.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Special-Functions.html,
- `man y0l`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.31.37 y1f

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-y1f.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Special-Functions.html,
- `man y1f`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, Minix 3.1.8, AIX 5.1, HP-UX 11, mingw, MSVC 14.

13.31.38 y1l

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-y1l.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Special-Functions.html,
- `man y1l`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.31.39 ynf

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-ynf.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Special-Functions.html,
- `man ynf`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, Minix 3.1.8, AIX 5.1, HP-UX 11, mingw, MSVC 14.

13.31.40 ynl

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-ynl.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Special-Functions.html,
- `man ynl`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.32 Glibc <mcheck.h>

13.32.1 mcheck

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Heap-Consistency-Checking.html,

- `man mcheck`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.32.2 `mcheck_check_all`

Documentation:

`man mcheck_check_all`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.32.3 `mcheck_pedantic`

Documentation:

`man mcheck_pedantic`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.32.4 `mprobe`

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Heap-Consistency-Checking.html,

- `man mprobe`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.32.5 mtrace

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Tracing-malloc.html,
- `man mtrace`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.32.6 muntrace

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Tracing-malloc.html,
- `man muntrace`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.33 Glibc <mntent.h>

13.33.1 addmntent

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/mtab.html,
- `man addmntent`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.33.2 endmntent

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/mtab.html,
- `man endmntent`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Solaris 11.4, mingw, MSVC 14, Android API level 20.

13.33.3 getmntent

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/mtab.html,
- `man getmntent`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, mingw, MSVC 14.

13.33.4 getmntent_r

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/mtab.html,
- `man getmntent_r`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android API level 20.

13.33.5 hasmntopt

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/mtab.html,
- `man hasmntopt`.

Gnulib module: hasmntopt

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Cygwin 3.5.3, Android API level 25.
- This function returns wrong results on some platforms: musl libc 1.2.5.

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, mingw, MSVC 14.

13.33.6 setmntent

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/mtab.html,
- `man setmntent`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Solaris 11.4, mingw, MSVC 14, Android API level 20.

13.34 Glibc Extensions to <netdb.h>

13.34.1 endnetgrent

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Lookup-Netgroup.html,
- `man endnetgrent`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.34.2 gethostbyaddr_r

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-gethostbyaddr-r-3.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Host-Names.html,
- `man gethostbyaddr_r`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 6.0, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 5.1.

13.34.3 gethostbyname2

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-gethostbyname2-3.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Host-Names.html,
- `man gethostbyname2`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, HP-UX 11, Solaris 11.4, Cygwin 1.5.x, mingw, MSVC 14.

13.34.4 gethostbyname2_r

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-gethostbyname2-r-3.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Host-Names.html,
- `man gethostbyname2_r`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 6.0, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 5.1.

13.34.5 gethostbyname_r

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-gethostbyname-r-3.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Host-Names.html,
- `man gethostbyname_r`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 6.0, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Cygwin 2.9, mingw, MSVC 14.

13.34.6 gethostent_r

Documentation:

`man gethostent_r`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 6.0, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.34.7 getnetbyaddr_r

Documentation:

`man getnetbyaddr_r`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 6.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.34.8 getnetbyname_r

Documentation:

`man getnetbyname_r`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 6.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.34.9 getnetent_r

Documentation:

`man getnetent_r`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 6.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.34.10 getnetgrent

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Lookup-Netgroup.html,
- `man getnetgrent`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.34.11 `getnetgrent_r`

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Lookup-Netgroup.html,
- `man getnetgrent_r`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 6.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.34.12 `getprotobyname_r`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-getprotobyname-r.html

Documentation:

`man getprotobyname_r`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 6.0, Minix 3.1.8, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.34.13 `getprotobynumber_r`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-getprotobynumber-r.html

Documentation:

`man getprotobynumber_r`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 6.0, Minix 3.1.8, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.34.14 getprotoent_r

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-getprotoent-r.html

Documentation:

`man getprotoent_r`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 6.0, Minix 3.1.8, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.34.15 getservbyname_r

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-getservbyname-r.html

Documentation:

`man getservbyname_r`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 6.0, Minix 3.1.8, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.34.16 getservbyport_r

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-getservbyport-r.html

Documentation:

`man getservbyport_r`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 6.0, Minix 3.1.8, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.34.17 getservent_r

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-getservent-r.html

Documentation:

`man getservent_r`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 6.0, Minix 3.1.8, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.34.18 `herror`

Documentation:

`man herror`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

13.34.19 `hstrerror`

Documentation:

`man hstrerror`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, HP-UX 11, mingw, MSVC 14.

13.34.20 `innetgr`

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Netgroup-Membership.html,
- `man innetgr`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.34.21 `rcmd`

Documentation:

`man rcmd`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14, Android 9.0.

13.34.22 `rcmd_af`

Documentation:

`man rcmd_af`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, HP-UX 11.11, Cygwin 1.5.x, mingw, MSVC 14, Android 9.0.

13.34.23 `rexec`

Documentation:

`man rexec`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, mingw, MSVC 14, Android 9.0.

13.34.24 `rexec_af`

Documentation:

`man rexec_af`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11.11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.34.25 `rresvport`

Documentation:

`man rresvport`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14, Android 9.0.

13.34.26 rresvport_af

Documentation:

[man rresvport_af](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, HP-UX 11.11, Cygwin 1.5.x, mingw, MSVC 14, Android 9.0.

13.34.27 ruserok

Documentation:

[man ruserok](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14, Android 9.0.

13.34.28 ruserok_af

Documentation:

[man ruserok_af](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.34.29 setnetgrent

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Lookup-Netgroup.html,
- [man setnetgrent](#).

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.35 Glibc <netinet/ether.h>

13.35.1 ether_aton

Documentation:

[man ether_aton](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 2.3.

13.35.2 ether_aton_r

Documentation:

[man ether_aton_r](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 6.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 2.3.

13.35.3 ether_hostton

Documentation:

[man ether_hostton](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.35.4 ether_line

Documentation:

[man ether_line](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.35.5 ether_ntoa

Documentation:

`man ether_ntoa`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 2.3.

13.35.6 ether_ntoa_r

Documentation:

`man ether_ntoa_r`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 6.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 2.3.

13.35.7 ether_ntohost

Documentation:

`man ether_ntohost`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.36 Glibc Extensions to <netinet/in.h>

13.36.1 bindresvport

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-bindresvport-3.html

Documentation:

`man bindresvport`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, HP-UX 11, Cygwin 1.5.x, mingw, MSVC 14.

13.36.2 getipv4sourcefilter

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: Mac OS X 10.5, FreeBSD 6.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11.23, Solaris 10, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.36.3 getsourcefilter

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: Mac OS X 10.5, FreeBSD 6.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11.23, Solaris 10, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.36.4 inet6_option_alloc

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.36.5 inet6_option_append

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.36.6 inet6_option_find

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.36.7 `inet6_option_init`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.36.8 `inet6_option_next`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.36.9 `inet6_option_space`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.36.10 `inet6_opt_append`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: glibc 2.4, Mac OS X 10.5, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.2, HP-UX 11.31, Solaris 9, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.36.11 `inet6_opt_find`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: glibc 2.4, Mac OS X 10.5, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.2, HP-UX 11.31, Solaris 9, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.36.12 `inet6_opt_finish`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: glibc 2.4, Mac OS X 10.5, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.2, HP-UX 11.31, Solaris 9, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.36.13 `inet6_opt_get_val`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: glibc 2.4, Mac OS X 10.5, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.2, HP-UX 11.31, Solaris 9, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.36.14 `inet6_opt_init`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: glibc 2.4, Mac OS X 10.5, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.2, HP-UX 11.31, Solaris 9, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.36.15 `inet6_opt_next`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: glibc 2.4, Mac OS X 10.5, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.2, HP-UX 11.31, Solaris 9, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.36.16 `inet6_opt_set_val`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: glibc 2.4, Mac OS X 10.5, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.2, HP-UX 11.31, Solaris 9, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.36.17 inet6_rth_add

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: glibc 2.4, Mac OS X 10.5, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.2, HP-UX 11.31, Solaris 9, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.36.18 inet6_rth_getaddr

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: glibc 2.4, Mac OS X 10.5, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.2, HP-UX 11.31, Solaris 9, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.36.19 inet6_rth_init

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: glibc 2.4, Mac OS X 10.5, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.2, HP-UX 11.31, Solaris 9, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.36.20 inet6_rth_reverse

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: glibc 2.4, Mac OS X 10.5, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.2, HP-UX 11.31, Solaris 9, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.36.21 inet6_rth_segments

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: glibc 2.4, Mac OS X 10.5, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.2, HP-UX 11.31, Solaris 9, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.36.22 inet6_rth_space

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: glibc 2.4, Mac OS X 10.5, NetBSD 3.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.2, HP-UX 11.31, Solaris 9, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.36.23 setip4sourcefilter

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: Mac OS X 10.5, FreeBSD 6.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11.23, Solaris 10, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.36.24 setsourcefilter

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: Mac OS X 10.5, FreeBSD 6.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11.23, Solaris 10, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.37 Glibc <obstack.h>

13.37.1 obstack_alloc_failed_handler

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Preparing-for-Obstacks.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This variable is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.37.2 obstack_exit_failure

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This variable is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.37.3 obstack_free

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Freeing-Obstack-Objects.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.37.4 obstack_printf

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Dynamic-Output.html.

Gnulib module: obstack-printf or obstack-printf-posix or obstack-printf-gnu

Portability problems fixed by either Gnulib module `obstack-printf` or `obstack-printf-posix` or `obstack-printf-gnu`:

- This function is missing on all non-glibc platforms: musl libc, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.3, HP-UX 11, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android 9.0.

Portability problems fixed by either Gnulib module `obstack-printf-posix` or `obstack-printf-gnu`:

- This function does not support size specifiers as in C99 (`hh`, `ll`, `j`, `t`, `z`) on some platforms: AIX 5.1, HP-UX 11.23, Solaris 9, Cygwin 1.5.24, old mingw, MSVC 9.
- This function does not support size specifiers as in C23 (`w8`, `w16`, `w32`, `w64`, `wf8`, `wf16`, `wf32`, `wf64`) on some platforms: glibc 2.37.
- printf of ‘long double’ numbers is unsupported on some platforms: mingw, MSVC 14.
- printf “%f”, “%e”, “%g” of Infinity and NaN yields an incorrect result on some platforms: AIX 7.3, Solaris 11.4, mingw, MSVC 14.
- This function doesn’t support the ‘`'`’ flag at all on some platforms: mingw without `__USE_MINGW_ANSI_STDIO`, MSVC 14.
- This function doesn’t support the ‘`'`’ flag correctly for integers (it uses the decimal point instead of the thousands separator!) on some platforms: mingw with `__USE_MINGW_ANSI_STDIO`.
- This function doesn’t support the ‘`'`’ flag together with a precision on an integer argument correctly on some platforms: glibc 2.36, FreeBSD 14.0, NetBSD 10.0, Solaris 11, Cygwin 3.6.0, Haiku.

- This function doesn't support the ' flag correctly in locales where the thousands separator is a multibyte character on some platforms: NetBSD 10.0, Solaris 11 OpenIndiana.
- This function does not support the 'a' and 'A' directives on some platforms: FreeBSD 5.2.1, OpenBSD 4.0, AIX 5.2, HP-UX 11, Solaris 11.4, Cygwin 1.5.x, mingw, MSVC 14.
- This function does not support the 'b' directive, required by ISO C23, on some platforms: glibc 2.34, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, HP-UX 11, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android 9.0.
- This function does not support the 'F' directive on some platforms: NetBSD 3.0, AIX 5.1, HP-UX 11.23, Solaris 9, Cygwin 1.5.x, mingw, MSVC 14.
- This function does not support the 'ls' directive on some platforms: OpenBSD 4.0, Cygwin 1.5.x, Haiku.
- This function does not support precisions in the 'ls' directive correctly on some platforms: Solaris 11.4.
- This function does not support format directives that access arguments in an arbitrary order, such as "%2\$s", on some platforms: NetBSD 3.0, mingw, MSVC 14.
- This function doesn't support the ' flag on some platforms: NetBSD 3.0, Cygwin 1.5.24, mingw, MSVC 14.
- This function does not round the argument of the 'a' directive correctly on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4.
- printf "%010f" of NaN and Infinity yields an incorrect result (padded with zeroes) on some platforms: Mac OS X 10.5, FreeBSD 6.0, NetBSD 5.0, AIX 7.3, Solaris 11.4, Cygwin 1.5.x, mingw, MSVC/clang.
- printf "%#.0x" or "%#.0X" with a zero argument yields an incorrect result (non-empty) on some platforms: Mac OS X 10.6.
- This function does not support precisions larger than 512 or 1024 in integer, floating-point and pointer output on some platforms: Solaris 10/x86, mingw, MSVC/clang.
- This function produces wrong output for the 'lc' directive with a NUL wide character argument on some platforms: musl libc 1.2.4.
- This function can crash in out-of-memory conditions on some platforms: FreeBSD 14.0, NetBSD 5.0.

Portability problems fixed by Gnulib module **obstack-printf-gnu**:

- This function does not support the 'B' directive on some platforms: glibc 2.34, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC, Android 9.0.

Portability problems fixed by either Gnulib module **obstack-printf-posix** or **obstack-printf-gnu** and additionally the Gnulib module **printf-with-n-directive**:

- This function does not support the 'n' directive on some platforms: MSVC 14.
- This function does not fully support the 'n' directive on some platforms: HP-UX 11, mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- The %m directive is not portable, use %s mapped to an argument of **strerror(errno)** (or a version of **strerror_r**) instead.

13.37.5 obstack_vprintf

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Variable-Arguments-Output.html.

Gnulib module: `obstack-printf` or `obstack-printf-posix` or `obstack-printf-gnu`

Portability problems fixed by either Gnulib module `obstack-printf` or `obstack-printf-posix` or `obstack-printf-gnu`:

- This function is missing on all non-glibc platforms: musl libc, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.3, HP-UX 11, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android 9.0.

Portability problems fixed by either Gnulib module `obstack-printf-posix` or `obstack-printf-gnu`:

- This function does not support size specifiers as in C99 (`hh`, `ll`, `j`, `t`, `z`) on some platforms: AIX 5.1, HP-UX 11.23, Solaris 9, Cygwin 1.5.24, old mingw, MSVC 9.
- This function does not support size specifiers as in C23 (`w8`, `w16`, `w32`, `w64`, `wf8`, `wf16`, `wf32`, `wf64`) on some platforms: glibc 2.37.
- `printf` of ‘long double’ numbers is unsupported on some platforms: mingw, MSVC 14.
- `printf "%f", "%e", "%g"` of Infinity and NaN yields an incorrect result on some platforms: AIX 7.3, Solaris 11.4, mingw, MSVC 14.
- This function doesn’t support the ‘`'`’ flag at all on some platforms: mingw without `__USE_MINGW_ANSI_STDIO`, MSVC 14.
- This function doesn’t support the ‘`'`’ flag correctly for integers (it uses the decimal point instead of the thousands separator!) on some platforms: mingw with `__USE_MINGW_ANSI_STDIO`.
- This function doesn’t support the ‘`'`’ flag together with a precision on an integer argument correctly on some platforms: glibc 2.36, FreeBSD 14.0, NetBSD 10.0, Solaris 11, Cygwin 3.6.0, Haiku.
- This function doesn’t support the ‘`'`’ flag correctly in locales where the thousands separator is a multibyte character on some platforms: NetBSD 10.0, Solaris 11 OpenIndiana.
- This function does not support the ‘`a`’ and ‘`A`’ directives on some platforms: FreeBSD 5.2.1, OpenBSD 4.0, AIX 5.2, HP-UX 11, Solaris 11.4, Cygwin 1.5.x, mingw, MSVC 14.
- This function does not support the ‘`b`’ directive, required by ISO C23, on some platforms: glibc 2.34, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, HP-UX 11, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android 9.0.
- This function does not support the ‘`F`’ directive on some platforms: NetBSD 3.0, AIX 5.1, HP-UX 11.23, Solaris 9, Cygwin 1.5.x, mingw, MSVC 14.
- This function does not support the ‘`ls`’ directive on some platforms: OpenBSD 4.0, Cygwin 1.5.x, Haiku.
- This function does not support precisions in the ‘`ls`’ directive correctly on some platforms: Solaris 11.4.
- This function does not support format directives that access arguments in an arbitrary order, such as `"%2$s"`, on some platforms: NetBSD 3.0, mingw, MSVC 14.

- This function doesn't support the ' flag on some platforms: NetBSD 3.0, Cygwin 1.5.24, mingw, MSVC 14.
- This function does not round the argument of the 'a' directive correctly on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4.
- printf "%010f" of NaN and Infinity yields an incorrect result (padded with zeroes) on some platforms: Mac OS X 10.5, FreeBSD 6.0, NetBSD 5.0, AIX 7.3, Solaris 11.4, Cygwin 1.5.x, mingw, MSVC/clang.
- printf "%#.0x" or "%#.0X" with a zero argument yields an incorrect result (non-empty) on some platforms: Mac OS X 10.6.
- This function does not support precisions larger than 512 or 1024 in integer, floating-point and pointer output on some platforms: Solaris 10/x86, mingw, MSVC/clang.
- This function produces wrong output for the 'lc' directive with a NUL wide character argument on some platforms: musl libc 1.2.4.
- This function can crash in out-of-memory conditions on some platforms: FreeBSD 14.0, NetBSD 5.0.

Portability problems fixed by Gnulib module `obstack-printf-gnu`:

- This function does not support the 'B' directive on some platforms: glibc 2.34, musl libc, macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, AIX 7.3, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC, Android 9.0.

Portability problems fixed by either Gnulib module `obstack-printf-posix` or `obstack-printf-gnu` and additionally the Gnulib module `printf-with-n-directive`:

- This function does not support the 'n' directive on some platforms: MSVC 14.
- This function does not fully support the 'n' directive on some platforms: HP-UX 11, mingw, MSVC 14.

Portability problems not fixed by Gnulib:

- The %m directive is not portable, use %s mapped to an argument of `strerror(errno)` (or a version of `strerror_r`) instead.

13.38 Glibc <printf.h>

13.38.1 parse_printf_format

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Parsing-a-Template-String.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.38.2 printf_size

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Predefined-Printf-Handlers.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.38.3 printf_size_info

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Predefined-Printf-Handlers.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.38.4 register_printf_function

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Registering-New-Conversions.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 6.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.38.5 register_printf_modifier

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.9, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.38.6 register_printf_specifier

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.9, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.38.7 register_printf_type

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.9, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.39 Glibc Extensions to <pthread.h>

13.39.1 pthread_attr_getaffinity_np

Documentation:

[man pthread_attr_getaffinity_np](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: glibc 2.3.2, macOS 14, FreeBSD 6.4, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.
- This function has a different signature on some platforms: glibc 2.3.3.

13.39.2 pthread_attr_setaffinity_np

Documentation:

[man pthread_attr_setaffinity_np](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: glibc 2.3.2, macOS 14, FreeBSD 6.4, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.
- This function has a different signature on some platforms: glibc 2.3.3.

13.39.3 pthread_attr_getsigmask_np

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Initial-Thread-Signal-Mask.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.31, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.39.4 pthread_attr_setsigmask_np

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Initial-Thread-Signal-Mask.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.31, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.39.5 pthread_clockjoin_np

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Default-Thread-Attributes.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.39.6 pthread_getaffinity_np

Documentation:

`man pthread_getaffinity_np`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: glibc 2.3.2, macOS 14, FreeBSD 6.4, NetBSD 3.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.0.x, mingw, MSVC 14, Android API level 35.

- This function has a different signature on some platforms: glibc 2.3.3.
- The third parameter has a different type on some platforms: FreeBSD 14.0, NetBSD 10.0.

13.39.7 pthread_getattr_default_np

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Default-Thread-Attributes.html,
- `man pthread_getattr_default_np`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.17, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.39.8 pthread_getattr_np

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-pthread-getattr-np-1.html

Documentation:

`man pthread_getattr_np`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.3, Cygwin 1.7.9, mingw, MSVC 14.

13.39.9 pthread_getname_np

Documentation:

`man pthread_getname_np`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: glibc 2.11, Mac OS X 10.5, FreeBSD 12.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.0, Cygwin 1.7.x, mingw, MSVC 14, Android 7.1. See also <https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3554.htm>.

13.39.10 pthread_gettid_np

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Process-Identification.html,
- `man pthread_gettid_np`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on most non-glibc platforms: glibc 2.41, macOS 15, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.7, Minix, AIX 7.3, HP-UX 11.31, Solaris 11.4, Cygwin, mingw, MSVC 14.

13.39.11 pthread_kill_other_threads_np

Documentation:

`man pthread_kill_other_threads_np`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.39.12 pthread_rwlockattr_getkind_np

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-pthread_rwlockattr_getkind_np-1.html

Documentation:

`man pthread_rwlockattr_getkind_np`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 5.1.

13.39.13 pthread_rwlockattr_setkind_np

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-pthread_rwlockattr_getkind_np-1.html

Documentation:

`man pthread_rwlockattr_setkind_np`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 5.1.

13.39.14 pthread_setaffinity_np

Documentation:

`man pthread_setaffinity_np`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: glibc 2.3.2, macOS 14, FreeBSD 6.4, NetBSD 3.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.0.x, mingw, MSVC 14, Android API level 35.
- This function has a different signature on some platforms: glibc 2.3.3.
- The third parameter has a different type on some platforms: FreeBSD 14.0, NetBSD 10.0.

13.39.15 pthread_setattr_default_np

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Default-Thread-Attributes.html,
- `man pthread_setattr-default_np`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.17, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.39.16 pthread_setname_np

Documentation:

`man pthread_setname_np`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: glibc 2.11, Mac OS X 10.5, FreeBSD 12.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.0, Cygwin 1.7.x, mingw, MSVC 14. See also <https://www.open-std.org/jtc1/sc22/wg14/docs/n3554.htm>.

- This function has a different signature on some platforms: macOS 14, NetBSD 10.0. On macOS the function takes only one argument, the name. On NetBSD the second argument is interpreted as a `printf` format string, with the third argument as parameter.

13.39.17 `pthread_sigqueue`

Documentation:

`man pthread_sigqueue`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: glibc 2.10, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.0, Cygwin 1.7.9, mingw, MSVC 14, Android 9.0.

13.39.18 `pthread_timedjoin_np`

Documentation:

`man pthread_timedjoin_np`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: glibc 2.3.2, macOS 14, FreeBSD 6.4, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.10.x, mingw, MSVC 14, Android 9.0.
- FreeBSD 6.4 has a function of this name in `libthr` but not in `libpthread`, and it also is missing a declaration.

13.39.19 `pthread_tryjoin_np`

Documentation:

`man pthread_tryjoin_np`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: glibc 2.3.2, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.10.x, mingw, MSVC 14, Android 9.0.

13.40 Glibc `<pty.h>`

13.40.1 forkpty

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Pseudo_002dTerminal-Pairs.html,
- `man forkpty`.

Gnulib module: forkpty

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.3, Android 5.1.
- On some systems (at least including Cygwin and Mac OS X) linking with `-lutil` is not required.
- On glibc, OpenBSD, NetBSD and FreeBSD linking with `-lutil` is required.
- The function is declared in `pty.h` on glibc and Cygwin. It is declared in `util.h` on Mac OS X, OpenBSD, and NetBSD. It is declared in `libutil.h` on FreeBSD. It is declared in `termios.h` on Solaris.
- Some platforms declare the function without marking the last two parameters `const`. FreeBSD 14.0, Solaris 11.4, Cygwin 1.7.1.

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

13.40.2 openpty

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Pseudo_002dTerminal-Pairs.html,
- `man openpty`.

Gnulib module: openpty

Portability problems fixed by Gnulib:

- This function is missing on some platforms: AIX 5.1, HP-UX 11, Solaris 11.3, Android 5.1.
- On some systems (at least including Cygwin and Mac OS X) linking with `-lutil` is not required.
- On glibc, OpenBSD, NetBSD and FreeBSD linking with `-lutil` is required.
- The function is declared in `pty.h` on glibc and Cygwin. It is declared in `util.h` on Mac OS X, OpenBSD, and NetBSD. It is declared in `libutil.h` on FreeBSD. It is declared in `termios.h` on Solaris.
- Some platforms declare the function without marking the last two parameters `const`. FreeBSD 14.0, Solaris 11.4, Cygwin 1.7.1.

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

13.41 Glibc Extensions to <pwd.h>

13.41.1 fgetpwent

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Scanning-All-Users.html,
- `man fgetpwent`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.41.2 fgetpwent_r

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Scanning-All-Users.html,
- `man fgetpwent_r`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.41.3 getpw

Documentation:

`man getpw`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.41.4 getpwent_r

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-getpwent-r-1.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Scanning-All-Users.html,
- `man getpwent_r`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.41.5 putpwent

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Writing-a-User-Entry.html,
- `man putpwent`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.42 Glibc Extensions to <regex.h>

13.42.1 re_comp

Documentation:

`man re_comp`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.42.2 re_compile_fastmap

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.42.3 `re_compile_pattern`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.42.4 `re_exec`

Documentation:

`man re_exec`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.42.5 `re_match`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.42.6 `re_match_2`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.42.7 `re_search`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.42.8 re_search_2

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.42.9 re_set_registers

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.42.10 re_set_syntax

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.42.11 re_syntax_options

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This variable is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.43 Glibc <regex.h>

13.43.1 advance

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 10, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.43.2 loc1

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This variable is missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 10, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.43.3 loc2

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This variable is missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 10, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.43.4 locs

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This variable is missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 10, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.43.5 step

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 10, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.44 Glibc <resolv.h>

13.44.1 dn_comp

Documentation:

[man dn_comp](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, OpenBSD 7.5, Minix 3.3.0, Cygwin 1.5.x, mingw, MSVC 14, Android 9.0.

13.44.2 dn_expand

Documentation:

[man dn_expand](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Cygwin 1.5.x, mingw, MSVC 14.

13.44.3 dn_skipname

Documentation:

[man dn_skipname](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, Cygwin 1.5.x, mingw, MSVC 14, Android 9.0.

13.44.4 res_dnok

Documentation:

[man res_dnok](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, HP-UX 11.11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.44.5 res_hnok

Documentation:

[man res_hnok](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, HP-UX 11.11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.44.6 res_init

Documentation:

[man res_init](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Cygwin 1.5.x, mingw, MSVC 14.

13.44.7 `res_mailok`

Documentation:

`man res_mailok`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, HP-UX 11.11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.44.8 `res_mkquery`

Documentation:

`man res_mkquery`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, Cygwin 1.5.x, mingw, MSVC 14.

13.44.9 `res_nmkquery`

Documentation:

`man res_nmkquery`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, HP-UX 11.31, Cygwin 1.5.x, mingw, MSVC 14, Android 9.0.

13.44.10 `res_nquery`

Documentation:

`man res_nquery`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, HP-UX 11.31, Cygwin 1.5.x, mingw, MSVC 14, Android 9.0.

13.44.11 `res_nquerydomain`

Documentation:

`man res_nquerydomain`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, HP-UX 11.31, Cygwin 1.5.x, mingw, MSVC 14, Android 9.0.

13.44.12 `res_nsearch`

Documentation:

`man res_nsearch`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, HP-UX 11.31, Cygwin 1.5.x, mingw, MSVC 14, Android 9.0.

13.44.13 `res_nsend`

Documentation:

`man res_nsend`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, HP-UX 11.31, Cygwin 1.5.x, mingw, MSVC 14, Android 9.0.

13.44.14 `res_ownok`

Documentation:

`man res_ownok`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, HP-UX 11.11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.44.15 `res_query`

Documentation:

`man res_query`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Cygwin 1.5.x, mingw, MSVC 14.

13.44.16 `res_querydomain`

Documentation:

`man res_querydomain`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, Minix 3.3.0, Cygwin 1.5.x, mingw, MSVC 14, Android 9.0.

13.44.17 `res_search`

Documentation:

`man res_search`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Cygwin 1.5.x, mingw, MSVC 14.

13.44.18 `res_send`

Documentation:

`man res_send`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, OpenBSD 7.5, Minix 3.3.0, Cygwin 1.5.x, mingw, MSVC 14, Android 9.0.

13.45 Glibc `<rpc/auth.h>`

13.45.1 `authdes_create`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.45.2 authdes_pk_create

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.45.3 authnone_create

Documentation:

[man authnone_create](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.45.4 authunix_create

Documentation:

[man authunix_create](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.45.5 authunix_create_default

Documentation:

[man authunix_create_default](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.45.6 getnetname

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.45.7 host2netname

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.45.8 key_decryptsession

Documentation:

[man key_decryptsession](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.45.9 key_decryptsession_pk

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.45.10 key_encryptsession

Documentation:

[man key_encryptsession](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.45.11 key_encryptsession_pk

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.45.12 `key_gendes`

Documentation:

`man key_gendes`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.45.13 `key_get_conv`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.45.14 `key_secretkey_is_set`

Documentation:

`man key_secretkey_is_set`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.45.15 `key_setsecret`

Documentation:

`man key_setsecret`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.45.16 `netname2host`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.45.17 netname2user

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.45.18 user2netname

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.45.19 xdr_des_block

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.45.20 xdr_opaque_auth

Documentation:

[`man xdr\_opaque\_auth`](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.46 Glibc <rpc/auth_des.h>

13.46.1 authdes_getucred

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.46.2 getpublickey

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.46.3 getsecretkey

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.46.4 rtime

Documentation:

[man rtime](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.47 Glibc <rpc/auth_unix.h>

13.47.1 xdr_authunix_parms

Documentation:

[man xdr_authunix_parms](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.48 Glibc <rpc/clnt.h>

13.48.1 callrpc

Documentation:

[man callrpc](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.48.2 `clnt_create`

Documentation:

`man clnt_create`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.48.3 `clnt_pcreateerror`

Documentation:

`man clnt_pcreateerror`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.48.4 `clnt_perrno`

Documentation:

`man clnt_perrno`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.48.5 `clnt_perror`

Documentation:

`man clnt_perror`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.48.6 `clnt_spcreateerror`

Documentation:

`man clnt_spcreateerror`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.48.7 `clnt_sperrno`

Documentation:

`man clnt_sperrno`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.48.8 `clnt_sperror`

Documentation:

`man clnt_sperror`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.48.9 `clntraw_create`

Documentation:

`man clntraw_create`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.48.10 `clnttcp_create`

Documentation:

`man clnttcp_create`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.48.11 `clntudp_bufcreate`

Documentation:

[man clntudp_bufcreate](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.48.12 `clntudp_create`

Documentation:

[man clntudp_create](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.48.13 `clntunix_create`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.48.14 `get_myaddress`

Documentation:

[man get_myaddress](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.48.15 getrpcport

Documentation:

`man getrpcport`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.48.16 rpc_createerr

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This variable is missing on some platforms: Minix 3.1.8, AIX 5.1, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.49 Glibc <rpc/key_prot.h>

13.49.1 xdr_cryptkeyarg

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.49.2 xdr_cryptkeyarg2

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.49.3 xdr_cryptkeyres

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.49.4 xdr_getcredres

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.49.5 xdr_key_netstarg

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.49.6 xdr_key_netstres

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.49.7 xdr_keybuf

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.49.8 xdr_keystatus

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.49.9 xdr_netnamestr

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.49.10 xdr_unixcred

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.50 Glibc <rpc/netdb.h>

13.50.1 endrpcent

Documentation:

[man endrpcent](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.50.2 getrpcbyname

Documentation:

[man getrpcbyname](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.50.3 getrpcbyname_r

Documentation:

[man getrpcbyname_r](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.50.4 getrpcbynumber

Documentation:

[man getrpcbynumber](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.50.5 getrpcbynumber_r

Documentation:

[man getrpcbynumber_r](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.50.6 getrpcent

Documentation:

[man getrpcent](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.50.7 getrpcent_r

Documentation:

[man getrpcent_r](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.50.8 setrpcent

Documentation:

[man setrpcent](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.51 Glibc <rpc/pmap_clnt.h>

13.51.1 clnt_broadcast

Documentation:

[man clnt_broadcast](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.51.2 pmap_getmaps

Documentation:

[man pmap_getmaps](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.51.3 pmap_getport

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-pmap-getport-3.html

Documentation:

[man pmap_getport](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.51.4 pmap_rmtcall

Documentation:

[man pmap_rmtcall](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.51.5 pmap_set

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-pmap-set-3.html

Documentation:

`man pmap_set`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.51.6 pmap_unset

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-pmap-unset-3.html

Documentation:

`man pmap_unset`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.52 Glibc <rpc/pmap_prot.h>

13.52.1 xdr_pmap

Documentation:

`man xdr_pmap`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.52.2 xdr_pmaplist

Documentation:

`man xdr_pmaplist`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.53 Glibc <rpc/pmap_rmt.h>

13.53.1 xdr_rmtcall_args

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.53.2 xdr_rmtcallres

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.54 Glibc <rpc/rpc_msg.h>

13.54.1 xdr_callhdr

Documentation:

[man xdr_callhdr](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.54.2 xdr_callmsg

Documentation:

[man xdr_callmsg](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.54.3 xdr_replymsg

Documentation:

`man xdr_replymsg`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.55 Glibc <rpc/svc.h>

13.55.1 svc_exit

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.55.2 svc_fdset

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This variable is missing on some platforms: Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.55.3 svc_getreq

Documentation:

`man svc_getreq`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.55.4 svc_getreq_common

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, Minix 3.1.8, AIX 5.1, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.55.5 `svc_getreq_poll`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.55.6 `svc_getreqset`

Documentation:

[man `svc\_getreqset`](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.55.7 `svc_max_pollfd`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This variable is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.55.8 `svc_pollfd`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This variable is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.55.9 `svc_register`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-svc-register-3.html

Documentation:

[man `svc\_register`](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.55.10 `svc_run`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-svc-run-3.html

Documentation:

`man svc_run`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.55.11 `svc_sendreply`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-svc-sendreply-3.html

Documentation:

`man svc_sendreply`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.55.12 `svc_unregister`

Documentation:

`man svc_unregister`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.55.13 `svcerr_auth`

Documentation:

`man svcerr_auth`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.55.14 `svcerr_decode`

Documentation:

`man svcerr_decode`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.55.15 `svcerr_noproc`

Documentation:

`man svcerr_noproc`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.55.16 `svcerr_noprog`

Documentation:

`man svcerr_noprog`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.55.17 `svcerr_progvers`

Documentation:

`man svcerr_progvers`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.55.18 `svcerr_systemerr`

Documentation:

`man svcerr_systemerr`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.55.19 svcerr_weakauth

Documentation:

`man svcerr_weakauth`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.55.20 svcraw_create

Documentation:

`man svcraw_create`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.55.21 svctcp_create

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-svctcp-create-3.html

Documentation:

`man svctcp_create`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.55.22 svcudp_bufcreate

Documentation:

`man svcudp_bufcreate`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.55.23 `svcudp_create`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-svcudp-create-3.html

Documentation:

`man svcudp_create`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.55.24 `svcunix_create`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.55.25 `xprt_register`

Documentation:

`man xprt_register`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.55.26 `xprt_unregister`

Documentation:

`man xprt_unregister`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.56 Glibc <rpc/xdr.h>

13.56.1 xdr_array

Documentation:

[man xdr_array](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.7.4, mingw, MSVC 14, Android 9.0.

13.56.2 xdr_bool

Documentation:

[man xdr_bool](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.7.4, mingw, MSVC 14, Android 9.0.

13.56.3 xdr_bytes

Documentation:

[man xdr_bytes](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.7.4, mingw, MSVC 14, Android 9.0.

13.56.4 xdr_char

Documentation:

[man xdr_char](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.7.4, mingw, MSVC 14, Android 9.0.

13.56.5 xdr_double

Documentation:

[man xdr_double](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.7.4, mingw, MSVC 14, Android 9.0.

13.56.6 xdr_enum

Documentation:

[man xdr_enum](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.7.4, mingw, MSVC 14, Android 9.0.

13.56.7 xdr_float

Documentation:

[man xdr_float](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.7.4, mingw, MSVC 14, Android 9.0.

13.56.8 xdr_free

Documentation:

[man xdr_free](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.7.4, mingw, MSVC 14, Android 9.0.

13.56.9 xdr_hyper

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: OpenBSD 7.5, Minix 3.1.8, Cygwin 1.7.4, mingw, MSVC 14, Android 9.0.

13.56.10 xdr_int

Documentation:

`man xdr_int`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.7.4, mingw, MSVC 14, Android 9.0.

13.56.11 xdr_int16_t

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 1.7.4, mingw, MSVC 14, Android 9.0.

13.56.12 xdr_int32_t

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 1.7.4, mingw, MSVC 14, Android 9.0.

13.56.13 xdr_int64_t

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 1.7.4, mingw, MSVC 14, Android 9.0.

13.56.14 xdr_int8_t

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 1.7.4, mingw, MSVC 14, Android 9.0.

13.56.15 xdr_long

Documentation:

[man xdr_long](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.7.4, mingw, MSVC 14, Android 9.0.

13.56.16 xdr_longlong_t

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: OpenBSD 7.5, Minix 3.1.8, Cygwin 1.7.4, mingw, MSVC 14, Android 9.0.

13.56.17 xdr_netobj

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.7.4, mingw, MSVC 14, Android 9.0.

13.56.18 xdr_opaque

Documentation:

[man xdr_opaque](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.7.4, mingw, MSVC 14, Android 9.0.

13.56.19 xdr_pointer

Documentation:

[man xdr_pointer](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.7.4, mingw, MSVC 14, Android 9.0.

13.56.20 xdr_quad_t

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.56.21 xdr_reference

Documentation:

[man xdr_reference](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.7.4, mingw, MSVC 14, Android 9.0.

13.56.22 xdr_short

Documentation:

[man xdr_short](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.7.4, mingw, MSVC 14, Android 9.0.

13.56.23 xdr_sizeof

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 1.7.4, mingw, MSVC 14, Android 9.0.

13.56.24 xdr_string

Documentation:

[man xdr_string](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.7.4, mingw, MSVC 14, Android 9.0.

13.56.25 xdr_u_char

Documentation:

`man xdr_u_char`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.7.4, mingw, MSVC 14, Android 9.0.

13.56.26 xdr_u_hyper

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: OpenBSD 7.5, Minix 3.1.8, Cygwin 1.7.4, mingw, MSVC 14, Android 9.0.

13.56.27 xdr_u_int

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-xdr-u-int-3.html

Documentation:

`man xdr_u_int`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.7.4, mingw, MSVC 14, Android 9.0.

13.56.28 xdr_u_long

Documentation:

`man xdr_u_long`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.7.4, mingw, MSVC 14, Android 9.0.

13.56.29 xdr_u_longlong_t

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: OpenBSD 7.5, Minix 3.1.8, Cygwin 1.7.4, mingw, MSVC 14, Android 9.0.

13.56.30 xdr_u_quad_t

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.56.31 xdr_u_short

Documentation:

`man xdr_u_short`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.7.4, mingw, MSVC 14, Android 9.0.

13.56.32 xdr_uint16_t

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 6.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 1.7.4, mingw, MSVC 14, Android 9.0.
- This function is not declared in the header on some platforms: Cygwin 1.7.5.

13.56.33 xdr_uint32_t

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 6.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.
- This function is not declared in the header on some platforms: Cygwin 1.7.5.

13.56.34 xdr_uint64_t

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 6.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.
- This function is not declared in the header on some platforms: Cygwin 1.7.5.

13.56.35 xdr_uint8_t

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.
- This function is not declared in the header on some platforms: Cygwin 1.7.5.

13.56.36 xdr_union

Documentation:

[man xdr_union](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.7.4, mingw, MSVC 14, Android 9.0.

13.56.37 xdr_vector

Documentation:

[man xdr_vector](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.7.4, mingw, MSVC 14, Android 9.0.

13.56.38 xdr_void

Documentation:

[man xdr_void](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.7.4, mingw, MSVC 14, Android 9.0.

13.56.39 xdr_wrapstring

Documentation:

[man xdr_wrapstring](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.7.4, mingw, MSVC 14, Android 9.0.

13.56.40 xdrmem_create

Documentation:

[man xdrmem_create](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.7.4, mingw, MSVC 14, Android 9.0.

13.56.41 xdrrec_create

Documentation:

[man xdrrec_create](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.7.4, mingw, MSVC 14, Android 9.0.

13.56.42 xdrrec_endofrecord

Documentation:

[man xdrrec_endofrecord](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.7.4, mingw, MSVC 14, Android 9.0.

13.56.43 xdrrec_eof

Documentation:

`man xdrrec_eof`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.7.4, mingw, MSVC 14, Android 9.0.

13.56.44 xdrrec_skiprecord

Documentation:

`man xdrrec_skiprecord`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.7.4, mingw, MSVC 14, Android 9.0.

13.56.45 xdrstdio_create

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-xdrstdio-create-3.html

Documentation:

`man xdrstdio_create`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.7.4, mingw, MSVC 14, Android 9.0.

13.57 Glibc <rpcsvc/nislib.h>

13.57.1 nis_add

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.2 nis_add_entry

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.3 nis_addmember

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.4 nis_checkpoint

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.5 nis_clone_object

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.6 nis_creategroup

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.7 nis_destroy_object

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.8 nis_destroygroup

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.9 nis_dir_cmp

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.10 nis_domain_of

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.11 nis_domain_of_r

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.12 nis_first_entry

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.13 nis_freenames

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.14 nis_freeresult

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.15 nis_freeservlist

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.16 nis_freetags

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.17 nis_getnames

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.18 nis_getservlist

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.19 nis_ismember

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.20 nis_leaf_of

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.21 nis_leaf_of_r

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.22 nis_lerror

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.23 nis_list

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.24 nis_local_directory

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.25 nis_local_group

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.26 nis_local_host

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.27 nis_local_principal

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.28 nis_lookup

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.29 nis_mkdir

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.30 nis_modify

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.31 nis_modify_entry

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.32 nis_name_of

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.33 nis_name_of_r

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.34 nis_next_entry

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.35 nis_perror

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.36 nis_ping

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.37 nis_print_directory

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.38 nis_print_entry

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.39 nis_print_group

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.40 nis_print_group_entry

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.41 nis_print_link

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.42 nis_print_object

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.43 nis_print_result

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.44 nis_print_rights

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.45 nis_print_table

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.46 nis_remove

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.47 nis_remove_entry

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.48 nis_removemember

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.49 nis_rmdir

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.50 nis_servstate

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.51 nis_sperrno

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.52 nis_sperror

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.53 `nis_sperror_r`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.54 `nis_stats`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.57.55 `nis_verifygroup`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.58 Glibc `<rpcsvc/nis_callback.h>`

13.58.1 `xdr_cback_data`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.58.2 `xdr_obj_p`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.59 Glibc <rpcsvc/yp.h>

13.59.1 xdr_domainname

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.59.2 xdr_keydat

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: NetBSD 5.0, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.59.3 xdr_valdat

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: NetBSD 5.0, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.59.4 xdr_ypbind_resptype

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: NetBSD 5.0, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.59.5 xdr_ypmap_parms

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, OpenBSD 3.8, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.59.6 xdr_ypmaplist

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.59.7 xdr_yppushresp_xfr

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, OpenBSD 3.8, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.59.8 xdr_ypreq_key

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.59.9 xdr_ypreq_nokey

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.59.10 xdr_ypreq_xfr

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, OpenBSD 3.8, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.59.11 xdr_ypresp_all

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: NetBSD 5.0, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.59.12 xdr_ypresp_key_val

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.59.13 xdr_ypresp_maplist

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.59.14 xdr_ypresp_master

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.59.15 xdr_ypresp_order

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.59.16 xdr_ypresp_val

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.59.17 xdr_ypresp_xfr

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.59.18 xdr_ypstat

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: NetBSD 5.0, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.59.19 xdr_ypxfrstat

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.60 Glibc <rpcsvc/ypclnt.h>

13.60.1 yp_all

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.60.2 yp_bind

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.60.3 yp_first

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.60.4 yp_get_default_domain

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.60.5 yp_master

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.60.6 yp_match

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.60.7 yp_next

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.60.8 yp_order

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.60.9 yp_unbind

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.60.10 ypbinderr_string

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.60.11 yperr_string

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.60.12 ypprot_err

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.61 Glibc Extensions to <sched.h>

13.61.1 clone

Documentation:

[man clone](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14.

13.61.2 getcpu

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/CPU-Affinity.html,
- [man getcpu](#).

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on all non-glibc platforms: glibc 2.28, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.61.3 sched_getaffinity

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-sched-getaffinity.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/CPU-Affinity.html,
- `man sched_getaffinity`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 13.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 3.0.x, mingw, MSVC 14, Android 3.0.

13.61.4 sched_getattr

Documentation:

`man sched_getattr`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on most non-glibc platforms: glibc 2.40, macOS 14, FreeBSD 13.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.10.x, mingw, MSVC 14, Android API level 36.

13.61.5 sched_getcpu

Documentation:

`man sched_getcpu`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and Cygwin and is therefore missing on most non-glibc platforms: glibc 2.5, macOS 14, FreeBSD 13.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.10.x, mingw, MSVC 14.

13.61.6 sched_setaffinity

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-sched-setaffinity.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/CPU-Affinity.html,
- `man sched_setaffinity`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 13.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 3.0.x, mingw, MSVC 14, Android 3.0.

13.61.7 sched_setattr

Documentation:

`man sched_setattr`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on most non-glibc platforms: glibc 2.40, macOS 14, FreeBSD 13.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.10.x, mingw, MSVC 14, Android API level 36.

13.61.8 setns

Documentation:

`man setns`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on many non-glibc platforms: glibc 2.13, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 4.4.

13.62 Glibc Extensions to <search.h>

13.62.1 hcreate_r

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-hcreate-r.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Hash-Search-Function.html,
- `man hcreate_r`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 6.0, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Solaris 11.4, mingw, MSVC 14, Android 8.1.

13.62.2 hdestroy_r

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-hdestroy-r.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Hash-Search-Function.html,
- `man hdestroy_r`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 6.0, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Solaris 11.4, mingw, MSVC 14, Android 8.1.

13.62.3 hsearch_r

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-hsearch-r.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Hash-Search-Function.html,
- `man hsearch_r`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 6.0, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Solaris 11.4, mingw, MSVC 14, Android 8.1.

13.62.4 tdestroy

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Tree-Search-Function.html,
- `man tdestroy`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.0, mingw, MSVC 14, Android 4.0.4.

13.62.5 twalk_r

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Tree-Search-Function.html,
- `man twalk_r`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.63 Glibc Extensions to <selinux/selinux.h>

13.63.1 fgetfilecon

Documentation:

`man fgetfilecon`

Gnulib module: selinux-h

Portability problems fixed by Gnulib:

- This function exists only on Linux and is therefore missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0. On those platforms, this module provides a stub that always sets `errno` to `ENOTSUP` and returns ‘-1’.
- On systems with SELinux support, this module provides a wrapper for the `fgetfilecon` function that insulates the caller from API-nonconforming behavior. Without this wrapper, `fgetfilecon` can return ‘0’ and set the `context` pointer to `NULL`, and in another scenario can return ‘10’ and set the `context` pointer to ‘unlabeled’. This wrapper returns ‘-1’ in each case and sets `errno` to `ENOTSUP` and `ENODATA` respectively. While the conditions that can provoke such behavior are rare,

the average caller does not handle them because the possibility of such behavior is not documented.

Portability problems not fixed by Gnulib:

13.63.2 `getfilecon`

Documentation:

`man getfilecon`

Gnulib module: `selinux-h`

Portability problems fixed by Gnulib:

- This function exists only on Linux and is therefore missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0. On those platforms, this module provides a stub that always sets `errno` to `ENOTSUP` and returns `‘-1’`.
- On systems with SELinux support, this module provides a wrapper for the `getfilecon` function that insulates the caller from API-nonconforming behavior. Without this wrapper, `getfilecon` can return `‘0’` and set the `context` pointer to `NULL`, and in another scenario can return `‘10’` and set the `context` pointer to `‘unlabeled’`. This wrapper returns `‘-1’` in each case and sets `errno` to `ENOTSUP` and `ENODATA` respectively. While the conditions that can provoke such behavior are rare, the average caller does not handle them because the possibility of such behavior is not documented.

Portability problems not fixed by Gnulib:

13.63.3 `lgetfilecon`

Documentation:

`man lgetfilecon`

Gnulib module: `selinux-h`

Portability problems fixed by Gnulib:

- This function exists only on Linux and is therefore missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0. On those platforms, this module provides a stub that always sets `errno` to `ENOTSUP` and returns `‘-1’`.
- On systems with SELinux support, this module provides a wrapper for the `lgetfilecon` function that insulates the caller from API-nonconforming behavior. Without this wrapper, `lgetfilecon` can return `‘0’` and set the `context` pointer to `NULL`, and in another scenario can return `‘10’` and set the `context` pointer to `‘unlabeled’`. This wrapper returns `‘-1’` in each case and sets `errno` to `ENOTSUP` and `ENODATA` respectively. While the conditions that can provoke such behavior are rare, the average caller does not handle them because the possibility of such behavior is not documented.

Portability problems not fixed by Gnulib:

13.64 Glibc <shadow.h>

13.64.1 endspent

Documentation:

[man endspent](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.64.2 fgetspent

Documentation:

[man fgetspent](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.64.3 fgetspent_r

Documentation:

[man fgetspent_r](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.64.4 getspent

Documentation:

[man getspent](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.64.5 getspent_r

Documentation:

[man getspent_r](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.64.6 getspnam

Documentation:

[man getspnam](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.64.7 getspnam_r

Documentation:

[man getspnam_r](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.64.8 lckpwordf

Documentation:

[man lckpwordf](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.64.9 putspent

Documentation:

`man putspent`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.64.10 setspent

Documentation:

`man setspent`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.64.11 sgetspent

Documentation:

`man sgetspent`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.64.12 sgetspent_r

Documentation:

`man sgetspent_r`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.64.13 `ulckpddf`

Documentation:

`man ulckpddf`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.65 Glibc Extensions to `<signal.h>`

13.65.1 `gsignal`

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Signaling-Yourself.html,
- `man gsignal`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.65.2 `sigandset`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-sigandset.html

Documentation:

`man sigandset`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 12.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.65.3 `sigblock`

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/BSD-Signal-Handling.html,
- `man sigblock`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14.

13.65.4 siggetmask

Documentation:

`man siggetmask`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.65.5 sigisemptyset

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-sigisemptyset.html

Documentation:

`man sigisemptyset`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 12.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.65.6 sigorset

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-sigorset.html

Documentation:

`man sigorset`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 12.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.65.7 sigreturn

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-sigreturn-2.html

Documentation:

`man sigreturn`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, NetBSD 10.0, OpenBSD 7.5, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.65.8 sigsetmask

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/BSD-Signal-Handling.html,
- `man sigsetmask`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14.

13.65.9 sigstack

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Signal-Stack.html,
- `man sigstack`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.65.10 sigvec

Documentation:

`man sigvec`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.65.11 `ssignal`

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Basic-Signal-Handling.html,
- `man ssignal`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.65.12 `sys_siglist`

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Signal-Messages.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This constant is missing on some platforms: Minix 3.1.8, HP-UX 11, Solaris 11.4, Cygwin 1.7.9, mingw, MSVC 14.

13.65.13 `sysv_signal`

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Basic-Signal-Handling.html,
- `man sysv_signal`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.65.14 `tgkill`

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Signaling-Another-Process.html,
- `man tgkill`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on most non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14.

13.66 Glibc Extensions to <spawn.h>

13.66.1 pidfd_spawn

Documentation:

<https://lwn.net/Articles/943022/>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.38, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

13.66.2 pidfd_spawnnp

Documentation:

<https://lwn.net/Articles/943022/>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.38, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

13.66.3 posix_spawn_file_actions_addchdir_np

`posix_spawn_file_actions_addchdir` is specified in POSIX.1-2024. Draft specification:

<https://www.austingroupbugs.net/view.php?id=1208>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc 2.28, macOS 10.14, FreeBSD 13.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.0, Cygwin 3.4.x, mingw, MSVC 14, Android API level 33.

Note: Gnulib has a module `posix_spawn_file_actions_addchdir` that provides equivalent functionality, just without the suffix `_np`.

13.66.4 `posix_spawn_file_actions_addclosefrom_np`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: glibc 2.33, macOS 14, FreeBSD 13.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.1, HP-UX 11.31, Solaris 10, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.66.5 `posix_spawn_file_actions_addfchdir_np`

`posix_spawn_file_actions_addfchdir` is specified in POSIX.1-2024. Draft specification:

<https://www.austingroupbugs.net/view.php?id=1208>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc 2.28, macOS 10.14, FreeBSD 13.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.4.x, mingw, MSVC 14, Android API level 33.

Note: Gnulib has a module `posix_spawn_file_actions_addfchdir` that provides equivalent functionality, just without the suffix `_np`.

13.66.6 `posix_spawn_file_actions_addtcsetpgrp_np`

Documentation: —

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.34, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android.

13.66.7 `posix_spawnattr_getcgroup_np`

Documentation:

<https://sourceware.org/pipermail/libc-alpha/2023-August/151106.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.38, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

13.66.8 posix_spawnattr_setcgroup_np

Documentation:

<https://sourceware.org/pipermail/libc-alpha/2023-August/151106.html>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.38, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

13.67 Glibc Extensions to <stdio.h>

13.67.1 cuserid

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Who-Logged-In.html,
- `man cuserid`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, mingw, MSVC 14, Android 9.0.

13.67.2 clearerr_unlocked

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-clearerr-unlocked-1.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Error-Recovery.html,
- `man clearerr_unlocked`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android 5.1.

13.67.3 fcloseall

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Closing-Streams.html,
- `man fcloseall`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 6.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 10, mingw, Android 9.0.
- This functions returns void instead of int on some platforms: FreeBSD 14.0.

13.67.4 feof_unlocked

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-feof-unlocked-1.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/EOF-and-Errors.html,
- `man feof_unlocked`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android 5.1.

13.67.5 ferror_unlocked

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-ferror-unlocked-1.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/EOF-and-Errors.html,
- `man ferror_unlocked`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android 5.1.

13.67.6 fflush_unlocked

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-fflush-unlocked-1.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Flushing-Buffers.html,
- `man fflush_unlocked`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 12.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android 8.1.

13.67.7 fgetc_unlocked

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-fgetc-unlocked-1.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Character-Input.html,
- `man fgetc_unlocked`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android 8.1.

13.67.8 fgets_unlocked

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-fgets-unlocked-1.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Line-Input.html,
- `man fgets_unlocked`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android 8.1.

13.67.9 fileno_unlocked

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-fileno-unlocked-1.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Descriptors-and-Streams.html,

- [man fileno_unlocked](#).

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android 6.0.

13.67.10 fopencookie

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Streams-and-Cookies.html,
- [man fopencookie](#).

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 10.4, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.3, HP-UX 11, Solaris 11.4, Cygwin 1.5.x, mingw, MSVC 14, Android 9.0.

13.67.11 fputc_unlocked

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-fputc-unlocked-1.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Simple-Output.html,
- [man fputc_unlocked](#).

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 12.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android 8.1.

13.67.12 fputs_unlocked

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-fputs-unlocked-1.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Simple-Output.html,
- [man fputs_unlocked](#).

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 12.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android 8.1.

13.67.13 fread_unlocked

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-fread-unlocked-1.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Block-Input_002fOutput.html,
- `man fread_unlocked`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 12.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android 8.1.

13.67.14 fwrite_unlocked

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-fwrite-unlocked-1.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Block-Input_002fOutput.html,
- `man fwrite_unlocked`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 12.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android 8.1.

13.67.15 getw

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Character-Input.html,
- `man getw`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Android 9.0.

13.67.16 putw

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Simple-Output.html,
- `man putw`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is not declared on some platforms: Android 9.0.

13.67.17 renameat2

Documentation:

`man renameat2`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc 2.27, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.8.x, mingw, MSVC 14, Android API level 29.
- This function succeeds when the `RENAME_NOREPLACE` flag is specified and the source and destination are the same file, on some platforms: Cygwin 3.4.6.
- `renameat2(AT_FDCWD, "file", AT_FDCWD, "new/" ...)` succeeds instead of failing and setting `errno` to `ENOTDIR`, on some platforms: GNU/Hurd with glibc 2.41.
- `renameat2(AT_FDCWD, "dir", AT_FDCWD, "new/" ...)` fails with `errno` set to `ENOENT`, on some platforms: GNU/Hurd with glibc 2.42.
- `renameat2(AT_FDCWD, "dir/", AT_FDCWD, "new" ...)` fails with `errno` set to `ENOTDIR`, on some platforms: GNU/Hurd with glibc 2.42.

The `renameatu` module addresses some of these portability issues.

13.67.18 setbuffer

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-setbuffer-3.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Controlling-Buffering.html,
- `man setbuffer`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, HP-UX 11, mingw, MSVC 14.

13.67.19 `setlinebuf`

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Controlling-Buffering.html,
- `man setlinebuf`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, HP-UX 11.23, mingw, MSVC 14.

13.67.20 `sys_errlist`

Documentation:

`man sys_errlist`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This variable is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, Android 9.0.

13.67.21 `sys_nerr`

Documentation:

`man sys_nerr`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This variable is missing on some platforms: Minix 3.1.8, Cygwin 2.9, mingw, Android 9.0.

13.67.22 `tmpnam_r`

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Temporary-Files.html,
- `man tmpnam_r`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.68 Glibc Extensions to <stdlib.h>

13.68.1 arc4random

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/High-Quality-Random.html,
- `man arc4random`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.35, AIX 7.3.1, HP-UX 11.31, Solaris 11.3, Cygwin 1.7.x, mingw, MSVC 14.

13.68.2 arc4random_buf

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/High-Quality-Random.html,
- `man arc4random`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.35, AIX 7.3.1, HP-UX 11.31, Solaris 11.3, Cygwin 1.7.x, mingw, MSVC 14.

13.68.3 arc4random_uniform

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/High-Quality-Random.html,
- `man arc4random`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.35, AIX 7.3.1, HP-UX 11.31, Solaris 11.3, Cygwin 1.7.x, mingw, MSVC 14.

13.68.4 canonicalize_file_name

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Symbolic-Links.html,
- `man canonicalize_file_name`.

Gnulib module: `canonicalize-lgpl`

Portability problems fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 10, Cygwin 1.5.x, mingw, MSVC 14, Android 9.0.
- This function fails to detect trailing slashes on non-directories on some platforms: glibc 2.3.5.

Portability problems not fixed by Gnulib:

13.68.5 cfree

Documentation:

`man cfree`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.68.6 clearenv

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Environment-Access.html,
- `man clearenv`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 13.2, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, Solaris 10, Cygwin 2.10.x, mingw, MSVC 14.

13.68.7 drand48_r

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-drand48-r-1.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/SVID-Random.html,
- `man drand48_r`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Solaris 11.3, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.68.8 `ecvt_r`

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/System-V-Number-Conversion.html,
- `man ecvt_r`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.68.9 `erand48_r`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-erand48-r-1.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/SVID-Random.html,
- `man erand48_r`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Solaris 11.3, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.68.10 `fcvt_r`

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/System-V-Number-Conversion.html,
- `man fcvt_r`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.68.11 getloadavg

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-getloadavg-3.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Processor-Resources.html,
- `man getloadavg`.

Gnulib module: getloadavg

Portability problems fixed by Gnulib:

- This function is missing on some platforms: AIX 5.1, HP-UX 11, Cygwin 1.7.x, mingw, MSVC 14, Android 9.0.
- This function is declared in `<sys/loadavg.h>`, not `<stdlib.h>`, on some platforms: Solaris 11.4.

Portability problems not fixed by Gnulib:

13.68.12 getpt

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Allocation.html,
- `man getpt`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.9, mingw, MSVC 14.

13.68.13 initstate_r

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-initstate-r-1.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/BSD-Random.html,
- `man initstate_r`.

Gnulib module: `random_r`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.
- This function has an incompatible declaration on some platforms: AIX 7.1.

Portability problems not fixed by Gnulib:

- This function crashes if the state buffer is unaligned on some platforms: glibc 2.40/sparc.
- This function has a slightly different (but compatible) declaration on some platforms: Haiku.

13.68.14 `jrand48_r`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-jrand48-r-1.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/SVID-Random.html,
- `man jrand48_r`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Solaris 11.3, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.68.15 `lcong48_r`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-lcong48-r-1.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/SVID-Random.html,
- `man lcong48_r`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Solaris 11.3, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.68.16 lrand48_r

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-lrand48-r-1.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/SVID-Random.html,
- `man lrand48_r`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Solaris 11.3, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.68.17 mkostemps

Documentation:

`man mkostemps`

Gnulib module: `mkostemps`

Portability problems fixed by Gnulib:

- This function is missing on many non-glibc platforms: glibc 2.10, Mac OS X 10.5, FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.5, mingw, MSVC 14, Android 5.1.
- This function is declared in `<unistd.h>` instead of `<stdlib.h>` on some platforms: macOS 14.
- On platforms where `off_t` is a 32-bit type, `mkostemps` may not work correctly to create files larger than 2 GB. (Cf. `AC_SYS_LARGEFILE`.)

Portability problems not fixed by Gnulib:

The gnulib module `clean-temp` can create temporary files that will not be left behind after signals such as `SIGINT`.

13.68.18 mkstemps

Documentation:

`man mkstemps`

Gnulib module: `mkstemps`

Portability problems fixed by Gnulib:

- This function is missing on many non-glibc platforms: glibc 2.10, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 10, Cygwin 1.5.x, mingw, MSVC 14.
- This function is declared in `<unistd.h>` instead of `<stdlib.h>` on some platforms: macOS 14.
- On platforms where `off_t` is a 32-bit type, `mkstemps` may not work correctly to create files larger than 2 GB. (Cf. `AC_SYS_LARGEFILE`.)

Portability problems not fixed by Gnulib:

The gnulib module `clean-temp` can create temporary files that will not be left behind after signals such as SIGINT.

13.68.19 `rand48_r`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-mrand48-r-1.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/SVID-Random.html,
- `man mrand48_r`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Solaris 11.3, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.68.20 `nrnd48_r`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-nrand48-r-1.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/SVID-Random.html,
- `man nrnd48_r`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Solaris 11.3, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.68.21 `on_exit`

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Cleanups-on-Exit.html,
- `man on_exit`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Solaris 11.4, mingw, MSVC 14, Android 9.0.

13.68.22 qecvt

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/System-V-Number-Conversion.html,
- `man qecvt`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.68.23 qecvt_r

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/System-V-Number-Conversion.html,
- `man qecvt_r`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.68.24 qfcvt

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/System-V-Number-Conversion.html,
- `man qfcvt`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.68.25 qfcvt_r

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/System-V-Number-Conversion.html,

- [man qfcvt_r](#).

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.68.26 qgcvt

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/System-V-Number-Conversion.html,
- [man qgcvt](#).

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.68.27 random_r

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-random-r-1.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/BSD-Random.html,
- [man random_r](#).

Gnulib module: random_r

Portability problems fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.
- This function has an incompatible declaration on some platforms: AIX 7.1.

Portability problems not fixed by Gnulib:

- This function has a slightly different (but compatible) declaration on some platforms: Haiku 2017.

13.68.28 rpmatch

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Yes_002dor_002dNo-Questions.html,
- `man rpmatch`.

Gnulib module: rpmatch

Portability problems fixed by Gnulib:

- This function is missing on some platforms: macOS 10.14, FreeBSD 5.2.1, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android 9.0.

Portability problems not fixed by Gnulib:

13.68.29 seed48_r

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-seed48-r-1.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/SVID-Random.html,
- `man seed48_r`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Solaris 11.3, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.68.30 setstate_r

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-setstate-r-1.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/BSD-Random.html,
- `man setstate_r`.

Gnulib module: random_r

Portability problems fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.
- This function has an incompatible declaration on some platforms: AIX 7.1.

Portability problems not fixed by Gnulib:

- This function has a slightly different (but compatible) declaration on some platforms: Haiku.

13.68.31 srand48_r

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-srand48-r-1.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/SVID-Random.html,
- `man srand48_r`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Solaris 11.3, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.68.32 srandom_r

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-srandom-r-1.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/BSD-Random.html,
- `man srandom_r`.

Gnulib module: `random_r`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

Portability problems not fixed by Gnulib:

13.68.33 strtod_l

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android 7.1.

13.68.34 strtod_l

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android 7.1.

13.68.35 strtol_l

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android 7.1.

13.68.36 strtold_l

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android 4.4.

13.68.37 strtoll_l

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android 4.4.

13.68.38 strtouq

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-strtoq-3.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Parsing-of-Integers.html,
- `man strtouq`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14.

- This function is not declared on some platforms: Android 9.0.

13.68.39 strtoul_l

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android 7.1.

13.68.40 strtoull_l

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android 4.4.

13.68.41 strtouq

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-strtouq-3.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Parsing-of-Integers.html,
- `man strtouq`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14.
- This function is not declared on some platforms: Android 9.0.

13.68.42 uabs

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Absolute-Value.html,
- `man uabs`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on most platforms: glibc 2.41, macOS 15, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.7, Minix, AIX 7.3, HP-UX 11.31, Solaris 11.4, Cygwin, mingw, MSVC 14.

13.68.43 labs

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Absolute-Value.html,
- `man labs`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on most platforms: glibc 2.41, macOS 15, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.7, Minix, AIX 7.3, HP-UX 11.31, Solaris 11.4, Cygwin, mingw, MSVC 14.

13.68.44 ullabs

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Absolute-Value.html,
- `man ullabs`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on most platforms: glibc 2.41, macOS 15, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.7, Minix, AIX 7.3, HP-UX 11.31, Solaris 11.4, Cygwin, mingw, MSVC 14.

13.68.45 valloc

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Aligned-Memory-Blocks.html,
- `man valloc`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14.

13.69 Glibc Extensions to <string.h>

13.69.1 explicit_bzero

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Erasing-Sensitive-Data.html,
- `man explicit_bzero`.

Gnulib module: `explicit_bzero`

The `explicit_bzero` function is an approximation to what is needed, and does not suffice in general to erase information. Although calling `explicit_bzero` should clear the memory in question, the information that was in memory may still be available elsewhere on the machine. Proper implementation of information erasure requires support from levels below C code.

C23 specifies the function `memset_explicit`, which should be preferred to `explicit_bzero` in new code. See [Section 10.47.8 \[memset_explicit\]](#), page 410.

Portability problems fixed by Gnulib:

- This function is missing on some platforms: glibc 2.24, FreeBSD 10, NetBSD 7.1, OpenBSD 5.4, macOS 10.12, Solaris 11.4, Android 9.0, and many other systems.

Portability problems not fixed by Gnulib:

- Although the module's implementation should clear the memory on platforms compatible with GCC and on platforms using traditional linkers, it may not clear the memory on non-GCC platforms that use whole-program optimization.

13.69.2 memfrob

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Obfuscating-Data.html,
- `man memfrob`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.69.3 memcpy

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Copying-Strings-and-Arrays.html,
- `man memcpy`.

Gnulib module: `memcpy`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 13.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, mingw, MSVC 14, Android 5.1.

Portability problems not fixed by Gnulib:

13.69.4 memrchr

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-memrchr.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Search-Functions.html,
- `man memrchr`.

Gnulib module: `memrchr`

Portability problems fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 6.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14.

Portability problems not fixed by Gnulib:

13.69.5 rawmemchr

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Search-Functions.html,
- `man rawmemchr`.

Gnulib module: `rawmemchr`

Portability problems fixed by Gnulib:

- This function is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android 9.0.

Portability problems not fixed by Gnulib:

13.69.6 sigabbrev_np

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Signal-Messages.html.

Gnulib module: `sigabbrev_np`

Portability problems fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.31, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

Portability problems not fixed by Gnulib:

Note: Gnulib has a module `sig2str` that contains an equivalent function and also one that does the opposite conversion, from an abbreviated signal name to a signal number.

13.69.7 sigdescr_np

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Signal-Messages.html.

Gnulib module: `sigdescr_np`

Portability problems fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.31, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

Portability problems not fixed by Gnulib:

- The description of `SIGCHLD` is wrong on some platforms: glibc.
- The description of `SIGEMT` is completely unintelligible on some platforms: glibc.

Note: This function is hardly useful, because it returns English strings, not internationalized strings. Better use the function `strsignal`, which returns internationalized strings.

13.69.8 strcasestr

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-strcasestr.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Search-Functions.html,
- `man strcasestr`.

Gnulib module: `strcasestr` or `strcasestr-simple`

Portability problems fixed by either Gnulib module `strcasestr-simple` or `strcasestr`:

- This function is missing on some platforms: AIX 7.2, HP-UX 11, Solaris 10, Cygwin 1.5.x, mingw, MSVC 14.
- This function can trigger `memchr` bugs on some platforms: glibc 2.10.
- This function can trigger false positives for long periodic needles on some platforms: glibc 2.12, Cygwin 1.7.7.
- This function may fail to find matches on some platforms: glibc 2.28.

Portability problems fixed by Gnulib module `strcasestr`:

- This function has quadratic instead of linear worst-case complexity on some platforms: glibc 2.8, musl libc 1.2.3, macOS 14, FreeBSD 13.1, NetBSD 10.0, OpenBSD 7.2, Solaris 11.4.

Portability problems not fixed by Gnulib:

13.69.9 strchrnul

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Search-Functions.html,

- [man strchnul](#).

Gnulib module: strchnul

Portability problems fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 15.3, FreeBSD 6.0, NetBSD 7.1, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 10, Cygwin 1.7.8, mingw, MSVC 14, Android 6.0.
- This function crashes when no occurrence is found on some platforms: Cygwin 1.7.9.

Portability problems not fixed by Gnulib:

13.69.10 strerrordesc_np

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Error-Messages.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.31, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.69.11 strerrorname_np

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Error-Messages.html.

Gnulib module: strerrorname_np

Portability problems fixed by Gnulib:

- This function is missing on many platforms: glibc 2.31, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android API level 34.
- This function returns wrong values on some platforms: glibc 2.37, Solaris 11 OmniOS.

Portability problems not fixed by Gnulib:

13.69.12 strfry

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Shuffling-Bytes.html,
- [man strfry](#).

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.69.13 strsep

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-strsep-3.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Finding-Tokens-in-a-String.html,
- `man strsep`.

Gnulib module: `strsep`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: HP-UX 11, Solaris 10, mingw, MSVC 14.

Portability problems not fixed by Gnulib:

13.69.14 strverscmp

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/String_002fArray-Comparison.html,
- `man strverscmp`.

Gnulib module: `strverscmp`

Portability problems fixed by Gnulib:

- This function is missing on many platforms: macOS 14, FreeBSD 13.1, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android 9.0.
- This function treats ASCII letters as smaller than a digit sequence on some platforms: musl libc 1.2.3, Cygwin 3.4.x.

Portability problems not fixed by Gnulib:

- On older glibc platforms, this function does not always properly treat digit strings with leading zeros as fractions (for example, `strverscmp` ("B0075022800016.gbp.corp.com", "B007502357019.GBP.CORP.COM") does not yield a negative number as it should): glibc 2.9

13.70 Glibc <sys/auxv.h>

13.70.1 getauxval

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Auxiliary-Vector.html,
- `man getauxval`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: glibc 2.15, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14.

13.71 Glibc <sys/capability.h>

13.71.1 capget

Documentation:

`man capget`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14.

13.71.2 capset

Documentation:

`man capset`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14.

13.72 Glibc <sys/epoll.h>

13.72.1 epoll_create

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-epoll-create-1.html

Documentation:

`man epoll_create`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and illumos and is therefore missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14.

13.72.2 `epoll_create1`

Documentation:

[man `epoll\_create1`](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and illumos and is therefore missing on many non-glibc platforms: glibc 2.8, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 4.3.

13.72.3 `epoll_ctl`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-epoll-ctl-1.html

Documentation:

[man `epoll\_ctl`](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and illumos and is therefore missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14.

13.72.4 `epoll_pwait`

Documentation:

[man `epoll\_pwait`](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and illumos and is therefore missing on many non-glibc platforms: glibc 2.5, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 4.3.

13.72.5 `epoll_pwait2`

Documentation:

[man `epoll\_pwait2`](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on many non-glibc platforms: glibc 2.34, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android API level 34.

13.72.6 `epoll_wait`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-epoll-wait-1.html

Documentation:

`man epoll_wait`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and illumos and is therefore missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14.

13.73 Glibc `<sys/eventfd.h>`

13.73.1 `eventfd`

Documentation:

`man eventfd`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux, FreeBSD, NetBSD, and illumos and is therefore missing on many non-glibc platforms: glibc 2.6, macOS 14, FreeBSD 12.0, NetBSD 9.3, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14.

13.73.2 `eventfd_read`

Documentation:

`man eventfd_read`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux, FreeBSD, NetBSD, and illumos and is therefore missing on many non-glibc platforms: glibc 2.6, macOS 14, FreeBSD 12.0, NetBSD 9.3, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14.

13.73.3 `eventfd_write`

Documentation:

`man eventfd_write`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux, FreeBSD, NetBSD, and illumos and is therefore missing on many non-glibc platforms: glibc 2.6, macOS 14, FreeBSD 12.0, NetBSD 9.3, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14.

13.74 Glibc <sys/fanotify.h>

13.74.1 fanotify_init

Documentation:

`man fanotify_init`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on all non-glibc platforms: glibc 2.12, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.74.2 fanotify_mark

Documentation:

`man fanotify_mark`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on all non-glibc platforms: glibc 2.12, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.75 Glibc <sys/file.h>

13.75.1 flock

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-flock-2.html

Documentation:

`man flock`

Gnulib module: flock

Portability problems fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14
- This function is missing on some platforms: AIX 7.1, HP-UX 11.23, Solaris 11.4. But the replacement function does not really work; it exhibits test failures.

Portability problems not fixed by Gnulib:

13.76 Glibc <sys/fsuid.h>

13.76.1 setfsgid

Documentation:

`man setfsgid`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 4.4.

13.76.2 setfsuid

Documentation:

`man setfsuid`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 4.4.

13.77 Glibc <sys/gmon.h>

13.77.1 monstartup

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.78 Glibc <sys/inotify.h>

13.78.1 inotify_add_watch

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-inotify-add-watch.html

Documentation:

`man inotify_add_watch`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on most non-glibc platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14.

13.78.2 inotify_init

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-inotify-init.html

Documentation:

`man inotify_init`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on most non-glibc platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14.

13.78.3 inotify_init1

Documentation:

`man inotify_init1`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on most non-glibc platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 4.3.

13.78.4 inotify_rm_watch

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-inotify-rm-watch.html

Documentation:

`man inotify_rm_watch`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on most non-glibc platforms: glibc 2.3.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14.

13.79 Glibc <sys/io.h>, <sys/perm.h>

13.79.1 ioperm

Documentation:

[man ioperm](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.79.2 iopl

Documentation:

[man iopl](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.80 Glibc <sys/kdaemon.h>

13.80.1 bdflush

Documentation:

[man bdflush](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.81 Glibc <sys/klog.h>

13.81.1 klogctl

Documentation:

[man klogctl](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14.

13.82 Glibc Extensions to <sys/mman.h>

13.82.1 madvise

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Memory_002dmapped-I_002f0.html,
- `man madvise`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, Cygwin 1.7.7, mingw, MSVC 14.

13.82.2 memfd_create

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Memory_002dmapped-I_002f0.html,
- `man memfd_create`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and FreeBSD and is therefore missing on many non-glibc platforms: glibc 2.26, macOS 14, FreeBSD 12.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android API level 29.

13.82.3 mincore

Documentation:

`man mincore`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 2.9, mingw, MSVC 14.

13.82.4 mlock2

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Page-Lock-Functions.html,
- `man mlock2`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on many platforms: glibc 2.26, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android API level 29.

13.82.5 mremap

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-mremap.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Memory_002dmapped-I_002f0.html,
- `man mremap`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: macOS 14, FreeBSD 14.0, NetBSD 3.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14.

13.82.6 pkey_alloc

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Memory-Protection.html,
- `man pkey_alloc`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on all non-glibc platforms: glibc 2.26, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.82.7 pkey_free

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Memory-Protection.html,
- `man pkey_free`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on all non-glibc platforms: glibc 2.26, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.82.8 pkey_get

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Memory-Protection.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on all non-glibc platforms: glibc 2.26, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.82.9 pkey_mprotect

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Memory-Protection.html,
- `man pkey_mprotect`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on all non-glibc platforms: glibc 2.26, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.82.10 pkey_set

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Memory-Protection.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on all non-glibc platforms: glibc 2.26, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.82.11 process_madvise

Documentation:

[man process_madvise](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on many non-glibc platforms: glibc 2.35, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

13.82.12 process_mrelease

Documentation:

<https://lwn.net/Articles/864184/>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on many non-glibc platforms: glibc 2.35, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

13.82.13 remap_file_pages

Documentation:

[man remap_file_pages](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.83 Glibc <sys/mount.h>

13.83.1 fsconfig

Documentation:

<https://lore.kernel.org/all/159827191245.306468.4903071494263813779.stgit@warthog.procyon.org.uk/>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on many non-glibc platforms: glibc 2.35, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

13.83.2 fsmount

Documentation:

<https://lore.kernel.org/all/159827190508.306468.12755090833140558156.stgit@warthog.procyon.org.uk/>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on many non-glibc platforms: glibc 2.35, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

13.83.3 fsopen

Documentation:

<https://lore.kernel.org/all/159827190508.306468.12755090833140558156.stgit@warthog.procyon.org.uk/>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on many non-glibc platforms: glibc 2.35, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

13.83.4 fspick

Documentation:

<https://lore.kernel.org/all/159827189767.306468.1803062787718957199.stgit@warthog.procyon.org.uk/>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on many non-glibc platforms: glibc 2.35, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

13.83.5 mount

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Mount_002dUnmount_002dRemount.html,
- `man mount`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

13.83.6 mount_setattr

Documentation:

`man mount_setattr`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on many non-glibc platforms: glibc 2.35, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

13.83.7 move_mount

Documentation:

<https://lore.kernel.org/all/159827189025.306468.4916341547843731338.stgit@warthog.procyon.org.uk/>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on many non-glibc platforms: glibc 2.35, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

13.83.8 open_tree

Documentation:

<https://lore.kernel.org/all/159827188271.306468.16962617119460123110.stgit@warthog.procyon.org.uk/>

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on many non-glibc platforms: glibc 2.35, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

13.83.9 umount

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Mount_002dUnmount_002dRemount.html,
- `man umount`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, AIX 5.1, mingw, MSVC 14.

13.83.10 umount2

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Mount_002dUnmount_002dRemount.html,
- `man umount2`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11.23, Cygwin 2.9, mingw, MSVC 14.

13.84 Glibc <sys/personality.h>

13.84.1 personality

Documentation:

`man personality`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 4.0.2.

13.85 Glibc <sys/pidfd.h>

13.85.1 pidfd_getfd

Documentation:

`man pidfd_getfd`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on many non-glibc platforms: glibc 2.35, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

13.85.2 pidfd_getpid

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Querying-a-Process.html,
- <https://lwn.net/Articles/943022/>.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.38, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

13.85.3 pidfd_open

Documentation:

`man pidfd_open`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on many non-glibc platforms: glibc 2.35, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

13.85.4 pidfd_send_signal

Documentation:

`man pidfd_send_signal`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on many non-glibc platforms: glibc 2.35, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.5.3, mingw, MSVC 14, Android.

13.86 Glibc <sys/prctl.h>

13.86.1 prctl

Documentation:

[man prctl](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14.

13.87 Glibc <sys/profil.h>

13.87.1 sprofil

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11.11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.88 Glibc <sys/ptrace.h>

13.88.1 ptrace

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-ptrace-1.html

Documentation:

[man ptrace](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: AIX 5.1, Cygwin 2.9, mingw, MSVC 14.

13.89 Glibc <sys/quota.h>

13.89.1 quotactl

Documentation:

[man quotactl](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android 7.1.

13.90 Glibc <sys/random.h>

13.90.1 getrandom

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Unpredictable-Bytes.html,
- `man getrandom`.

Gnulib module: `getrandom`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: glibc 2.24, macOS 14, FreeBSD 11.0, NetBSD 9.3, OpenBSD 7.5, Minix 3.3, AIX 7.1, HP-UX 11.31, Solaris 11.0, Cygwin 1.7.x, mingw, MSVC 14, Android 8.1.
- This function has a different return type on some platforms: Solaris 11.4.

Portability problems not fixed by Gnulib:

- The `GRND_INSECURE` flag is missing on some platforms: glibc 2.34, macOS 10.15, GNU/kFreeBSD, FreeBSD 12.0, OpenBSD 7.5, Minix 3.3, Haiku.
- The `GRND_RANDOM` flag has different effects on different platforms. Some platforms ignore the flag, or yield data that can fail to be random in some cases.

Although this function is intended to produce random data, the data’s security properties may not be appropriate for your application. For example, identical “random” data streams might be produced by rebooted virtual machines. If this is of concern you may need to use additional techniques such as hedging.¹

Related modules include `getentropy`, which has a simpler but more-limited API, and `crypto/gc-random`, which is likely a better match for code already using the other `crypto` APIs.

13.91 Glibc <sys/reboot.h>

13.91.1 reboot

Documentation:

`man reboot`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: AIX 5.1, Cygwin 2.9, mingw, MSVC 14.

¹ Ristenpart T, Yilek S. When good randomness goes bad: virtual machine vulnerabilities and hedging deployed cryptography (<https://pages.cs.wisc.edu/~rist/papers/sslhedge.pdf>). NDSS 2010.

13.92 Glibc Extensions to <sys/resource.h>

13.92.1 prlimit

Documentation:

[man prlimit](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on many non-glibc platforms: glibc 2.12, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 6.0.

13.93 Glibc Extensions to <sys/sem.h>

13.93.1 semtimedop

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Semaphores.html,
- [man semtimedop](#).

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11.11, Cygwin 2.9, mingw, MSVC 14, Android 7.1.

13.94 Glibc <sys/sendfile.h>

13.94.1 sendfile

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-sendfile.html

Documentation:

[man sendfile](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, Solaris 11 2010-11, Cygwin 2.9, mingw, MSVC 14, Android 4.4.
- On platforms where `off_t` is a 32-bit type, this function may not work correctly on files larger than 2 GB. The fix is to use the `AC_SYS_LARGEFILE` macro.

13.95 Glibc <sys/signalfd.h>

13.95.1 signalfd

Documentation:

`man signalfd`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux, Cygwin, and illumos and is therefore missing on many non-glibc platforms: glibc 2.6, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.11.x, mingw, MSVC 14.

13.96 Glibc <sys/single_threaded.h>

13.96.1 __libc_single_threaded

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Single_002dThreaded.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This variable is missing on all non-glibc platforms: glibc 2.31, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.97 Glibc Extensions to <sys/socket.h>

13.97.1 isfdtype

Documentation:

`man isfdtype`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.97.2 recvmsg

Documentation:

`man recvmsg`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: glibc 2.11, macOS 14, FreeBSD 6.4, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.3, Cygwin 2.9, mingw, MSVC 14, Android 4.4.

13.97.3 sendmmsg

Documentation:

`man sendmmsg`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: glibc 2.13, macOS 14, FreeBSD 6.4, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.3, Cygwin 2.9, mingw, MSVC 14, Android 4.4.

13.98 Glibc Extensions to <sys/stat.h>

13.98.1 getumask

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Setting-Permissions.html,
- `man getumask`.

Gnulib module: getumask

Portability problems fixed by Gnulib:

- This function exists only on Hurd and is therefore missing on all non-glibc platforms: glibc/Linux, glibc/kFreeBSD, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3, AIX 7.2, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

Portability problems not fixed by Gnulib:

13.98.2 lchmod

Gnulib module: lchmod

Portability problems fixed by Gnulib:

- This function is missing on some platforms: OpenBSD 7.5, Minix 3.1.8, AIX 5.1, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android API level 35.
- This function is not declared on some platforms: HP-UX 11.31.
- This function always fails with `errno` set to `ENOSYS`, even when the file is not a symbolic link: GNU/Linux with glibc 2.31.
- This function does not fail when the file name argument ends in a slash and (without the slash) names a non-directory, on some platforms: AIX 7.2.

Portability problems not fixed by Gnulib:

- Some platforms do not allow changing the access bits on symbolic links.
- This function can fail with `errno` set to `EMFILE` or `ENFILE`, and it fails with `errno` set to `EOPNOTSUPP` if the `/proc` file system is not mounted: GNU/Linux with glibc 2.34.

13.98.3 statx

Documentation:

`man statx`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc 2.27, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android API level 29.
- There is an incompatible function of the same name on some platforms: AIX 5.2 or newer.

13.99 Glibc <sys/statfs.h>

13.99.1 fstatfs

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-fstatfs-2.html

Documentation:

`man fstatfs`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: AIX 5.1, mingw, MSVC 14.
- On platforms where `f_blocks` in `'struct statfs'` is a 32-bit value, this function may not work correctly on files systems larger than 4 TiB. The fix is to use the `AC_SYS_LARGEFILE` macro. This affects Mac OS X.

13.99.2 statfs

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-statfs-2.html

Documentation:

`man statfs`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, mingw, MSVC 14.

- On platforms where `f_blocks` in ‘`struct statfs`’ is a 32-bit value, this function may not work correctly on file systems larger than 4 TiB. The fix is to use the `AC_SYS_LARGEFILE` macro. This affects Mac OS X.

13.100 Glibc <sys/swap.h>

13.100.1 swapoff

Documentation:

`man swapoff`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 4.3.

13.100.2 swapon

Documentation:

`man swapon`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: OpenBSD 7.5, Minix 3.1.8, AIX 5.1, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 4.3.

13.101 Glibc <sys/sysctl.h>

13.101.1 sysctl

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/System-Parameters.html,
- `man sysctl`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.34, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.102 Glibc <sys/sysinfo.h>

13.102.1 get_avphys_pages

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Query-Memory-Parameters.html,
- `man get_avphys_pages`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.1, mingw, MSVC 14, Android 5.1.

13.102.2 get_nprocs

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Processor-Resources.html,
- `man get_nprocs`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 10, Cygwin 1.7.1, mingw, MSVC 14, Android 5.1.

Gnulib provides the module `nproc` that performs a similar function but is portable to more systems.

13.102.3 get_nprocs_conf

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Processor-Resources.html,
- `man get_nprocs_conf`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 10, Cygwin 1.7.1, mingw, MSVC 14, Android 5.1.

13.102.4 get_phys_pages

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Query-Memory-Parameters.html,
- `man get_phys_pages`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.1, mingw, MSVC 14, Android 5.1.

13.102.5 sysinfo

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-sysinfo-1.html

Documentation:

`man sysinfo`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, Cygwin 1.7.9, mingw, MSVC 14.
- This function is not declared and thus not part of the Android API for Android API levels < 9.

13.103 Glibc <sys/syslog.h>

13.103.1 vsyslog

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-vsyslog-3.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/syslog_003b-vsyslog.html,
- `man vsyslog`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, HP-UX 11, mingw, MSVC 14.

13.104 Glibc <sys/sysmacros.h>

13.104.1 gnu_dev_major

Documentation:

`man gnu_dev_major`

Gnulib module: —

Portability problems fixed by Gnulib:

- The `AC_HEADER_MAJOR` macro in Autoconf 2.69 and earlier fails to set `MAJOR_IN_SYSMACROS` when it detects namespace pollution in `sys/types.h`; which in turn provokes deprecation warnings in glibc 2.25.

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.104.2 gnu_dev_makedev

Documentation:

`man gnu_dev_makedev`

Gnulib module: —

Portability problems fixed by Gnulib:

- The `AC_HEADER_MAJOR` macro in Autoconf 2.69 and earlier fails to set `MAJOR_IN_SYSMACROS` when it detects namespace pollution in `sys/types.h`; which in turn provokes deprecation warnings in glibc 2.25.

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.104.3 gnu_dev_minor

Documentation:

`man gnu_dev_minor`

Gnulib module: —

Portability problems fixed by Gnulib:

- The `AC_HEADER_MAJOR` macro in Autoconf 2.69 and earlier fails to set `MAJOR_IN_SYSMACROS` when it detects namespace pollution in `sys/types.h`; which in turn provokes deprecation warnings in glibc 2.25.

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.105 Glibc Extensions to <sys/time.h>

13.105.1 adjtime

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-adjtime-2.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/High_002dResolution-Calendar.html,
- `man adjtime`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.105.2 futimes

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-futimes.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/File-Times.html,
- `man futimes`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, mingw, MSVC 14, Android 7.1.
- This function cannot set full timestamp resolution. Use `futimens(fd,times)` instead.

13.105.3 futimesat

Documentation:

`man futimesat`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.3.6, macOS 14, FreeBSD 6.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 1.5.x, mingw, MSVC 14, Android 7.1.
- On some platforms, this function mis-handles a trailing slash: Solaris 9.

- This function cannot set full timestamp resolution. Use

```
file ? utimensat(fd, file, times, 0)
      : futimens(fd, times)
```

or the glibc module `fdutimensat`, instead.

13.105.4 lutimes

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/File-Times.html,
- `man lutimes`.

Glibc module: —

Portability problems fixed by Glibc:

Portability problems not fixed by Glibc:

- This function is missing on some platforms: OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, mingw, MSVC 14, Android 7.1.
- This function cannot set full timestamp resolution. Use `utimensat(AT_FDCWD, file, times, AT_SYMLINK_NOFOLLOW)`, or the glibc module `utimens`, instead.
- The mere act of using `lstat` modifies the access time of symlinks on some platforms, so `lutimes` can only effectively change modification time: Cygwin.

13.105.5 settimeofday

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/High_002dResolution-Calendar.html,
- `man settimeofday`.

Glibc module: —

Portability problems fixed by Glibc:

Portability problems not fixed by Glibc:

- This function is missing on some platforms: mingw, MSVC 14.

13.106 Glibc <sys/timerfd.h>

13.106.1 timerfd_create

Documentation:

`man timerfd_create`

Glibc module: —

Portability problems fixed by Glibc:

Portability problems not fixed by Glibc:

- This function exists only on Linux, FreeBSD, NetBSD, Cygwin, and illumos and is therefore missing on many non-glibc platforms: glibc 2.7, macOS 14, FreeBSD 13.2, NetBSD 9.3, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.11.x, mingw, MSVC 14, Android 4.3.

13.106.2 timerfd_gettime

Documentation:

`man timerfd_gettime`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux, FreeBSD, NetBSD, Cygwin, and illumos and is therefore missing on many non-glibc platforms: glibc 2.7, macOS 14, FreeBSD 13.2, NetBSD 9.3, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.11.x, mingw, MSVC 14, Android 4.3.

13.106.3 timerfd_settime

Documentation:

`man timerfd_settime`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux, FreeBSD, NetBSD, Cygwin, and illumos and is therefore missing on many non-glibc platforms: glibc 2.7, macOS 14, FreeBSD 13.2, NetBSD 9.3, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.11.x, mingw, MSVC 14, Android 4.3.

13.107 Glibc <sys/timex.h>

13.107.1 adjtimex

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/High_002dResolution-Calendar.html,
- `man adjtimex`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 6.0.

13.107.2 ntp_adjtime

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/High-Accuracy-Clock.html,
- `man ntp_adjtime`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Mac OS X 10.5, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.107.3 ntp_gettime

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/High-Accuracy-Clock.html,
- `man ntp_gettime`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Mac OS X 10.5, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.107.4 ntp_gettimex

Documentation:

`man ntp_gettimex`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.11, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.108 Glibc Extensions to <sys/uio.h>

13.108.1 preadv

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Scatter_002dGather.html,
- `man preadv`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.9, macOS 10.15, FreeBSD 5.2.1, Minix 3.1.8, AIX 5.2, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 6.0.
- On platforms where `off_t` is a 32-bit type, this function may not work correctly on files larger than 2 GB. The fix is to use the `AC_SYS_LARGEFILE` macro.

13.108.2 preadv2

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Scatter_002dGather.html,
- `man preadv2`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on many platforms: glibc 2.25, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android API level 32.

13.108.3 process_vm_readv

Documentation:

`man process_vm_readv`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on many non-glibc platforms: glibc 2.14, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 5.1.

13.108.4 process_vm_writev

Documentation:

`man process_vm_writev`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on many non-glibc platforms: glibc 2.14, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 5.1.

13.108.5 pwritev

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Scatter_002dGather.html,
- `man pwritev`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.9, macOS 10.15, FreeBSD 5.2.1, Minix 3.1.8, AIX 5.2, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 6.0.
- On platforms where `off_t` is a 32-bit type, this function may not work correctly on files larger than 2 GB. The fix is to use the `AC_SYS_LARGEFILE` macro.

13.108.6 `pwritev2`

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Scatter_002dGather.html,
- `man pwritev2`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on many platforms: glibc 2.25, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android API level 32.

13.109 Glibc `<sys/ustat.h>`

13.109.1 `ustat`

Documentation:

`man ustat`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.110 Glibc `<sys/vlimit.h>`

13.110.1 `vlimit`

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Limits-on-Resources.html,
- `man vlimit`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.111 Glibc Extensions to <sys/wait.h>

13.111.1 wait3

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/BSD-Wait-Functions.html,
- `man wait3`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14, Android 9.0.
- This function is not declared on some platforms: Android 13.

13.111.2 wait4

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-wait4-2.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Process-Completion.html,
- `man wait4`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, HP-UX 11.11, mingw, MSVC 14, Android 4.2.

13.112 Glibc <sys/xattr.h>

13.112.1 fgetxattr

Documentation:

`man fgetxattr`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: FreeBSD 14.0, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.5.x, mingw, MSVC 14, Android 4.0.4.
- This function has extra `offset` and `options` parameters on some platforms: macOS 14.

13.112.2 flistxattr

Documentation:

`man flistxattr`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: FreeBSD 14.0, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.5.x, mingw, MSVC 14, Android 4.0.4.
- This function has an extra `options` parameter on some platforms: macOS 14.
- This function fails with `EACCES` on unreadable files accessed via NFS: Linux kernel 6.12.
- This function does not report synthetic `security.*` extended attributes on `tmpfs` and `kernfs`-based file systems: Linux kernel 6.14.
- This function can fail with `ERANGE` even if the buffer is large enough: Linux kernel 6.15.

13.112.3 fremovexattr

Documentation:

`man fremovexattr`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: FreeBSD 14.0, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.5.x, mingw, MSVC 14, Android 4.0.4.
- This function has an extra `options` parameter on some platforms: macOS 14.

13.112.4 fsetxattr

Documentation:

`man fsetxattr`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: FreeBSD 14.0, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.5.x, mingw, MSVC 14, Android 4.0.4.

- This function has extra `offset` and `options` parameters and no `flags` parameter on some platforms: macOS 14.

13.112.5 `getxattr`

Documentation:

`man getxattr`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: FreeBSD 14.0, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.5.x, mingw, MSVC 14, Android 4.0.4.
- This function has extra `offset` and `options` parameters on some platforms: macOS 14.

13.112.6 `lgetxattr`

Documentation:

`man lgetxattr`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.5.x, mingw, MSVC 14, Android 4.0.4.

13.112.7 `listxattr`

Documentation:

`man listxattr`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: FreeBSD 14.0, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.5.x, mingw, MSVC 14, Android 4.0.4.
- This function has an extra `options` parameter on some platforms: macOS 14.
- This function fails with `EACCES` on unreadable files accessed via NFS: Linux kernel 6.12.
- This function does not report synthetic `security.*` extended attributes on `tmpfs` and `kernfs`-based file systems: Linux kernel 6.14.
- This function can fail with `ERANGE` even if the buffer is large enough: Linux kernel 6.15.

13.112.8 `llistxattr`

Documentation:

`man llistxattr`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.5.x, mingw, MSVC 14, Android 4.0.4.
- This function fails with `EACCES` on unreadable files accessed via NFS: Linux kernel 6.12.
- This function does not report synthetic `security.*` extended attributes on `tmpfs` and `kernfs`-based file systems: Linux kernel 6.14.
- This function can fail with `ERANGE` even if the buffer is large enough: Linux kernel 6.15.

13.112.9 `lremovexattr`

Documentation:

`man lremovexattr`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.5.x, mingw, MSVC 14, Android 4.0.4.

13.112.10 `lsetxattr`

Documentation:

`man lsetxattr`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.5.x, mingw, MSVC 14, Android 4.0.4.

13.112.11 `removexattr`

Documentation:

`man removexattr`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: FreeBSD 14.0, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.5.x, mingw, MSVC 14, Android 4.0.4.

- This function has an extra `options` parameter on some platforms: macOS 14.

13.112.12 `setxattr`

Documentation:

`man setxattr`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: FreeBSD 14.0, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.5.x, mingw, MSVC 14, Android 4.0.4.
- This function has extra `offset` and `options` parameters and no `flags` parameter on some platforms: macOS 14.

13.113 Glibc Extensions to `<termios.h>`

13.113.1 `cfmakeraw`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-cfmakeraw-3.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Noncanonical-Input.html,
- `man cfmakeraw`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, HP-UX 11, Solaris 11.4, Cygwin 1.5.x, mingw, MSVC 14, Android 4.4.

13.113.2 `cfsetspeed`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-cfsetspeed-3.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Line-Speed.html,
- `man cfsetspeed`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android 4.4.

13.114 Glibc Extensions to <time.h>

13.114.1 clock_adjtime

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on many non-glibc platforms: glibc 2.13, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 6.0.

13.114.2 dysize

Documentation:

[man dysize](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.114.3 getdate_r

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/General-Time-String-Parsing.html,
- [man getdate_r](#).

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.114.4 stime

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-stime-2.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Simple-Calendar-Time.html,
- [man stime](#).

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Cygwin 1.7.x, mingw, MSVC 14, Android 9.0.

13.114.5 `strptime_1`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android 8.1.

13.114.6 `timelocal`

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Broken_002ddown-Time.html,
- `man timelocal`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: AIX 5.1, HP-UX 11, Solaris 11.3, mingw, MSVC 14, Android 3.0.

13.115 Glibc `<ttyent.h>`

13.115.1 `endttyent`

Documentation:

`man endttyent`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.115.2 `getttyent`

Documentation:

`man getttyent`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.115.3 getttynam

Documentation:

`man getttynam`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.115.4 setttyent

Documentation:

`man setttyent`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.116 Glibc Extensions to <unistd.h>

13.116.1 acct

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-acct-3.html

Documentation:

`man acct`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, Cygwin 2.9, mingw, MSVC 14.

13.116.2 brk

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Resizing-the-Data-Segment.html,

- `man brk`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 12.2/arm64, AIX 5.1, Cygwin 2.9, mingw, MSVC 14.

13.116.3 chroot

Documentation:

`man chroot`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: AIX 5.1, mingw, MSVC 14.

13.116.4 closefrom

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Opening-and-Closing-Files.html

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.33, macOS 14, FreeBSD 7.2, Minix 3.3.0, AIX 7.1, HP-UX 11.31, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

Note (quoted from `man close`): “The [POSIX] standard developers rejected a proposal to add `closefrom()` to the [POSIX] standard. Because the standard permits implementations to use inherited file descriptors as a means of providing a conforming environment for the child process, it is not possible to standardize an interface that closes arbitrary file descriptors above a certain value while still guaranteeing a conforming environment.”

13.116.5 close_range

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Opening-and-Closing-Files.html,
- `man close_range`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: glibc 2.33, macOS 14, FreeBSD 12.4, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.3.1, HP-UX 11.31, Solaris 11.4, Cygwin 3.4.x, mingw, MSVC 14, Android API level 33.

13.116.6 `copy_file_range`

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Copying-File-Data.html,
- `man copy_file_range`.

Gnulib module: `copy-file-range`

Portability problems fixed by Gnulib:

- This function exists only on Linux and FreeBSD and is therefore missing on many platforms: glibc 2.26, macOS 14, FreeBSD 12.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android API level 33. But the replacement function is only a stub: It always fails with error `ENOSYS`.
- This function has many problems on Linux kernel versions before 5.3. See <https://lwn.net/Articles/789527/> (```Fixes for major copy_file_range() issues```). On these kernel versions, the replacement function always fails with error `ENOSYS`.
- This function returns an incorrect value when the number of bytes copied exceeds `INT_MAX` on systems using glibc version 2.41 or 2.42. See <https://sourceware.org/PR33245> (glibc bug 33245).
- This function is provided on GNU/Hurd but it is only a stub: It always fails with error `ENOSYS`.

Portability problems not fixed by Gnulib:

- If built with include files for Linux kernel 5.3 or later, the Gnulib substitute may misbehave when run on Linux kernel 5.2 or earlier. A similar problem exists if built with include files for glibc 2.43 or later, and run on glibc 2.41 or 2.42.

13.116.7 `daemon`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-daemon-3.html

Documentation:

`man daemon`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, HP-UX 11, mingw, MSVC 14.

13.116.8 `eaccess`

Documentation:

`man eaccess`

This function is an alias of `euidaccess`. See [Section 13.116.10 \[euidaccess\]](#), page 822.

13.116.9 endusershell

Documentation:

`man endusershell`

Gnulib module: `getusershell`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14, Android 9.0.
- This function is missing a declaration on some platforms: Solaris 9.

Portability problems not fixed by Gnulib:

13.116.10 euidaccess

Documentation:

`man euidaccess`

Gnulib module: `euidaccess`

Portability problems fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.3, Cygwin 1.5.x, mingw, MSVC 14, Android 9.0.

Portability problems not fixed by Gnulib:

Other problems of this function:

- There is an inherent race between calling this function and performing some action based on the results; you should think twice before trusting this function, especially in a set-uid or set-gid program.
- This function does not have an option for not following symbolic links (like `stat` versus `lstat`). If you need this option, use the Gnulib module `faccessat` with the `AT_EACCESS` flag.
- On Solaris, for the root user, any file is `X_OK` even if the file does not have the `x` permission bit set.

13.116.11 execveat

Documentation: `man execveat`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: glibc 2.33/Linux, glibc 2.34/Hurd, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.3.0, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.116.12 `execvpe`

Documentation:

`man execvpe`

Gnulib module: `execvpe`

Portability problems fixed by Gnulib:

- This function is missing on many non-glibc platforms: glibc 2.10, macOS 14, FreeBSD 14.0, NetBSD 7.1, OpenBSD 3.8, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.3, Cygwin 1.5.x, mingw, Android 4.4.
- This function is not declared on some platforms: AIX 7.1.
- On Windows platforms (excluding Cygwin), this function does not pass command-line arguments correctly if they contain space, tab, backslash, or double-quote characters.
- On Windows platforms (excluding Cygwin), this function spawns an asynchronous child process and then exits the current process immediately. As a consequence, the parent of the current process 1. may incorrectly proceed as if its child had exited, and 2. will never see the child's exit status.
- On Windows platforms (excluding Cygwin), the return type of this function is `intptr_t`, not `int`.

Portability problems not fixed by Gnulib:

13.116.13 `get_current_dir_name`

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Working-Directory.html,
- `man get_current_dir_name`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Solaris 11.4, Cygwin 1.7.9, mingw, MSVC 14, Android 9.0.

13.116.14 `getdomainname`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-getdomainname.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Host-Identification.html,
- `man getdomainname`.

Gnulib module: `getdomainname`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Solaris 11.3, mingw, MSVC 14, Android 7.1.
- This function is declared in `netdb.h`, not in `unistd.h`, on some platforms: AIX 7.1.
- The second argument is of type `int`, not `size_t`, on some platforms: macOS 14, FreeBSD 14.0, AIX 7.1.

Portability problems not fixed by Gnulib:

13.116.15 getdtablesize

SUSv2 specification: <https://pubs.opengroup.org/onlinepubs/7908799/xsh/getdtablesize.html>

LSB specification:
https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-getdtablesize.html

Documentation:
[man getdtablesize](#)

Gnulib module: `getdtablesize`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Android LP64, mingw, MSVC 14, Android 9.0.
- This function is not declared on some platforms: Android 13 (LP32?).
- This function does not represent the true `RLIMIT_NOFILE` soft limit on some platforms: Android 13 (LP32?), Cygwin 1.7.25.

Portability problems not fixed by Gnulib:

- On OpenVMS, this function returns the maximum number of open file descriptors in a process. The possible values of file descriptors are not constrained by this function.

13.116.16 getpagesize

LSB specification:
https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-getpagesize.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Query-Memory-Parameters.html,
- [man getpagesize](#).

Gnulib module: `getpagesize`

Portability problems fixed by Gnulib:

- This function is missing on some platforms: MSVC 14, Android 4.4.
- This function is not declared on some platforms: HP-UX 11 with compiler option `-D_XOPEN_SOURCE=600`.
- This function is broken on some platforms: mingw.

Portability problems not fixed by Gnulib:

- The function has never been in POSIX and was removed from the Single UNIX Specification in SUSv3 (2001). Although portable programs are supposed to use `sysconf (_SC_PAGESIZE)` instead, in practice `getpagesize` can be a little faster.
- On typical platforms this function's API limits page sizes to at most 1 GiB, even though `sysconf (_SC_PAGESIZE)` does not have this limitation. However, this has not been a practical problem yet, and so many programs use `getpagesize` that it's unlikely that new platforms would introduce larger page sizes any time soon.

13.116.17 `getpass`

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/getpass.html,
- `man getpass`.

Gnulib module: `getpass` or `getpass-gnu`

Portability problems fixed by either Gnulib module `getpass` or `getpass-gnu`:

- This function is missing on some platforms: mingw, MSVC 14, Android 9.0.
- This function cannot be called from plain inline or extern inline functions on some platforms: Android 13.

Portability problems fixed by Gnulib module `getpass-gnu`:

- The returned password is truncated to `PASS_MAX` characters on some platforms: uClibc (256), musl (128), macOS 14 (128), FreeBSD 14.0 (128), NetBSD 3.0 (128), OpenBSD 4.0 (128), AIX 5.1 (32), HP-UX 11 (8), Solaris 11 2010-11 (8, even less than `PASS_MAX`), Cygwin (128). The gnulib implementation returns the password untruncated.

Portability problems not fixed by Gnulib:

13.116.18 `gettid`

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Process-Identification.html,
- `man gettid`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function exists only on Linux and is therefore missing on most non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14.

13.116.19 getusershell

Documentation:

`man getusershell`

Gnulib module: getusershell

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14, Android 9.0.
- This function is missing a declaration on some platforms: Solaris 9.
- This function mistakenly returns comments and empty lines on some platforms: musl libc 1.2.4

Portability problems not fixed by Gnulib:

13.116.20 group_member

Documentation:

`man group_member`

Gnulib module: group-member

Portability problems fixed by Gnulib:

- This function is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

Portability problems not fixed by Gnulib:

13.116.21 profil

Documentation:

`man profil`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.116.22 revoke

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, mingw, MSVC 14, Android 9.0.

13.116.23 sbrk

Documentation:

[man sbrk](#)

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 12.2/arm64, AIX 5.1, mingw, MSVC 14.

13.116.24 setlogin

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.116.25 setdomainname

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Host-Identification.html,
- [man setdomainname](#).

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, Cygwin 2.9, mingw, MSVC 14, Android 7.1.

13.116.26 sethostid

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Host-Identification.html,
- [man sethostid](#).

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.116.27 sethostname

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-sethostname-2.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Host-Identification.html,
- `man sethostname`.

Gnulib module: sethostname

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, Cygwin 1.7.x, mingw, MSVC 14, Android 5.1. Note that the Gnulib replacement may fail with `ENOSYS` on some platforms.
- This function is not declared on some platforms: AIX 7.1, Solaris 10.

Portability problems not fixed by Gnulib:

- The first parameter is `char *` instead of `const char *` on some platforms: Solaris 11 2010-11.
- The second parameter is `int` instead of `size_t` on some platforms: macOS 14, FreeBSD 14.0, MidnightBSD 3.0, Solaris 11 2010-11, Solaris 11 OpenIndiana, Solaris 11 OmniOS.
- This function does not reject a too long host name on some platforms: Cygwin 3.5.3, Haiku.

13.116.28 setusershell

Documentation:

`man setusershell`

Gnulib module: getusershell

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, mingw, MSVC 14, Android 9.0.
- This function is missing a declaration on some platforms: Solaris 9.

Portability problems not fixed by Gnulib:

13.116.29 syncfs

Documentation:

`man syncfs`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: glibc 2.13, macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 8.1.

13.116.30 `syscall`

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/System-Calls.html,
- `man syscall`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, Cygwin 2.9, mingw, MSVC 14.

13.116.31 `ttyslot`

Documentation:

`man ttyslot`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14, Android 9.0.

13.116.32 `vhangup`

Documentation:

`man vhangup`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, mingw, MSVC 14, Android 9.0.

13.117 Glibc `<utmp.h>`

13.117.1 `endutent`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-endutent-3.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Manipulating-the-Database.html,
- `man endutent`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, OpenBSD 7.5, Minix 3.1.8, mingw, MSVC 14.

- This function is not declared on some platforms: Android before ca. 2015.

13.117.2 `getutent`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-getutent-3.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Manipulating-the-Database.html,
- `man getutent`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, OpenBSD 7.5, Minix 3.1.8, mingw, MSVC 14.

13.117.3 `getutent_r`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-getutent-r-3.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Manipulating-the-Database.html,
- `man getutent_r`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.117.4 `getutid`

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Manipulating-the-Database.html,
- `man getutid`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, mingw, MSVC 14, Android 9.0.

13.117.5 `getutid_r`

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Manipulating-the-Database.html,
- `man getutid_r`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.117.6 `getutline`

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Manipulating-the-Database.html,
- `man getutline`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, mingw, MSVC 14, Android 9.0.

13.117.7 `getutline_r`

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Manipulating-the-Database.html,
- `man getutline_r`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.

13.117.8 `pututline`

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Manipulating-the-Database.html,
- `man pututline`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, mingw, MSVC 14.

13.117.9 setutent

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-setutent-3.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Manipulating-the-Database.html,
- `man setutent`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, OpenBSD 7.5, Minix 3.1.8, mingw, MSVC 14.

13.117.10 updwtmp

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Manipulating-the-Database.html,
- `man updwtmp`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, HP-UX 11, mingw, MSVC 14, Android 9.0.

13.117.11 utmpname

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-utmpname-3.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Manipulating-the-Database.html,
- `man utmpname`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, OpenBSD 7.5, Minix 3.1.8, mingw, MSVC 14.

13.117.12 login

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Logging-In-and-Out.html,
- `man login`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: FreeBSD 14.0, NetBSD 5.0, OpenBSD 3.8, Minix 3.1.8, AIX 7.1, HP-UX 11.31, Solaris 11.4, mingw, MSVC 14, Android 9.0.

13.117.13 login_tty

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Logging-In-and-Out.html,
- `man login_tty`.

Gnulib module: login_tty

Portability problems fixed by Gnulib:

- This function is missing on some platforms: Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.3, Android 5.1.
- This function is declared in `<utmp.h>` on glibc, Cygwin, Android, in `<util.h>` on macOS 14, NetBSD 5.0, OpenBSD 3.8, in `<libutil.h>` on FreeBSD 14.0, Haiku, and in `<termios.h>` on Solaris 11.4.
- This function requires linking with `-lutil` on some platforms: glibc 2.3.6, FreeBSD 14.0, NetBSD 5.0, OpenBSD 3.8. It is available without link options on other platforms: macOS 14, Cygwin.

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: mingw, MSVC 14.

13.118 Glibc Extensions to `<utmpx.h>`

13.118.1 getutmp

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/XPG-Functions.html,
- `man getutmp`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 14.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.
- On some platforms, this function does not support timestamps past the year 2038: glibc 2.38 on 32-bit platforms like x86 and ARM where `time_t` was historically 32 bits.
- On some platforms, this function misbehaves if the `year2038` or `year2038-recommended` modules are used and the program is configured without the `--disable-year2038` option. The `readutmp` module works around this problem: glibc 2.38 on 32-bit platforms like x86 and ARM where `time_t` was historically 32 bits. See [Section 14.5 \[Avoiding the year 2038 problem\]](#), page 843.

13.118.2 getutmpx

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/XPG-Functions.html,
- `man getutmpx`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 14.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 2.9, mingw, MSVC 14, Android 9.0.
- On some platforms, this function does not support timestamps past the year 2038: glibc 2.38 on 32-bit platforms like x86 and ARM where `time_t` was historically 32 bits.
- On some platforms, this function misbehaves if the `year2038` or `year2038-recommended` modules are used and the program is configured without the `--disable-year2038` option. The `readutmp` module works around this problem: glibc 2.38 on 32-bit platforms like x86 and ARM where `time_t` was historically 32 bits. See [Section 14.5 \[Avoiding the year 2038 problem\]](#), page 843.

13.118.3 updwtmpx

Documentation:

`man updwtmpx`

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Cygwin 1.5.x, mingw, MSVC 14, Android 9.0.

13.118.4 utmpxname

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/XPG-Functions.html,
- `man utmpxname`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: FreeBSD 14.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, mingw, MSVC 14, Android 9.0.

13.119 Glibc Extensions to `<wchar.h>`

13.119.1 fgetwc_unlocked

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-fgetwc-unlocked-1.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Character-Input.html,
- `man fgetwc_unlocked`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android 9.0.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

13.119.2 fgetws_unlocked

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-fgetws-unlocked-1.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Line-Input.html,
- `man fgetws_unlocked`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android 9.0.

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

13.119.3 `fputwc_unlocked`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-fputwc-unlocked-1.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Simple-Output.html,
- `man fputwc_unlocked`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android 9.0.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

13.119.4 `fputws_unlocked`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-fputws-unlocked-1.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Simple-Output.html,
- `man fputws_unlocked`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android 9.0.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

13.119.5 `getwc_unlocked`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-getwc-unlocked-1.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Character-Input.html,

- [man getwc_unlocked](#).

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android 9.0.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

13.119.6 getwchar_unlocked

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-getwchar-unlocked-1.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Character-Input.html,
- [man getwchar_unlocked](#).

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on all non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android 9.0.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

13.119.7 putwc_unlocked

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-putwc-unlocked-1.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Simple-Output.html,
- [man putwc_unlocked](#).

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on some platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android 9.0.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

13.119.8 putwchar_unlocked

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-putwchar-unlocked-1.html

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Simple-Output.html,
- `man putwchar_unlocked`.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android 9.0.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

13.119.9 wcschrnul

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Search-Functions.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

13.119.10 wcsftime_l

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android 8.1.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

13.119.11 wcstod_l

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 6.0, NetBSD 7.1, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android 8.1.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

13.119.12 wcstof_l

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 6.0, NetBSD 7.1, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android 8.1.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

13.119.13 wcstol_l

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android 8.1.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

13.119.14 wcstold_l

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 6.0, NetBSD 7.1, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android 4.4.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

13.119.15 wcstoll_l

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android 4.4.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

13.119.16 wcstoq

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-wcstoq.html

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Parsing-of-Integers.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

13.119.17 wcstoul_l

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android 8.1.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

13.119.18 wcstoull_l

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many platforms: FreeBSD 6.0, NetBSD 5.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 1.7.x, mingw, MSVC 14, Android 4.4.

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

13.119.19 `wcstouq`

LSB specification:

https://refspecs.linuxbase.org/LSB_5.0.0/LSB-Core-generic/LSB-Core-generic/baselib-wcstouq.html

Documentation:

https://www.gnu.org/software/libc/manual/html_node/Parsing-of-Integers.html.

Gnulib module: —

Portability problems fixed by Gnulib:

Portability problems not fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 14.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9, mingw, MSVC 14, Android 9.0.
- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

13.119.20 `wmempcpy`

Documentation:

- https://www.gnu.org/software/libc/manual/html_node/Copying-Strings-and-Arrays.html,
- `man wmempcpy`.

Gnulib module: `wmempcpy`

Portability problems fixed by Gnulib:

- This function is missing on many non-glibc platforms: macOS 14, FreeBSD 13.0, NetBSD 10.0, OpenBSD 7.5, Minix 3.1.8, AIX 5.1, HP-UX 11, Solaris 11.4, Cygwin 2.9.x, mingw, MSVC 14, Android 5.1.

Portability problems not fixed by Gnulib:

- On Windows and 32-bit AIX platforms, `wchar_t` is a 16-bit type and therefore cannot accommodate all Unicode characters.

14 Native Windows Support

There are three ways to create binaries that run on Microsoft Windows:

- Native binaries, built using the MinGW tool chain.
- Native binaries, built using the MSVC (MS Visual C/C++) tool chain.
- Binaries for the Cygwin environment.

This chapter deals with the MinGW and MSVC platforms, commonly called “native Windows” platforms. Cygwin, on the other hand, is close enough to POSIX that it can be treated like any other Unix-like platform.

14.1 Libtool and Windows

If you want it to be possible to compile your program for a native Windows platform and you use Libtool, you need to use the `win32-dll` option of `LT_INIT`. In other words, put:

```
LT_INIT([win32-dll])
```

in your `configure.ac`. This sets the correct names for the `OBJDUMP`, `DLLTOOL`, and `AS` tools for the build.

If you are building a library, you will also need to pass `-no-undefined` to make sure Libtool produces a DLL for your library. From a `Makefile.am`:

```
libgassl_la_LDFLAGS += -no-undefined
```

14.2 Large File Support

This module provides support for files 2 GiB and larger, or with device or inode numbers exceeding 32 bits. To this effect, it attempts to ensure that types like `off_t` and `ino_t` are 64-bit, at least on the following platforms: glibc, Mac OS X, FreeBSD, NetBSD, OpenBSD, AIX, HP-UX, Solaris, Cygwin, mingw, MSVC.

If the types cannot be made 64-bit, `configure` issues a warning and proceeds.

This module gives `configure` an option `--disable-largefile` that suppresses support for large files. This may be useful if the package links to other libraries whose user-facing ABIs still require `off_t` or most other file-related types to be 32-bit on your platform.

This module also adds to `configure` an option `--enable-year2038`, needed on some platforms to access files with timestamps past the year 2038. See [Section 14.5 \[Avoiding the year 2038 problem\]](#), page 843.

14.3 Inode numbers on Windows

The module `'windows-stat-inodes'` ensures that, on native Windows platforms, `struct stat` contains `st_dev`, `st_ino` fields that are able to distinguish different inodes.

Note: Such values can only be provided for most files on the file system. For a few files (such as inaccessible files), `st_dev` and `st_ino` are set to 0. Therefore, you should test whether `st_dev != 0 && st_ino != 0`, before going to make inferences based on the file identity based on `st_dev` and `st_ino`.

14.4 Precise file timestamps on Windows

The module ‘`windows-stat-timespec`’ ensures that, on native Windows platforms, `struct stat` contains `st_atim`, `st_mtim`, `st_ctim` fields of type `struct timespec`, providing 100 ns resolution for the timestamps of files.

Note: On some types of file systems, the timestamp resolution is limited by the file system. For example, on FAT file systems, `st_mtim` only has a resolution of 2 seconds. For more details, see <https://docs.microsoft.com/en-us/windows/desktop/SysInfo/file-times>.

14.5 Avoiding the year 2038 problem

The *year 2038 problem* denotes unpredictable behaviour that will likely occur in the year 2038, for programs that use a 32-bit signed integer ‘`time_t`’ type that cannot represent timestamps on or after 2038-01-19 03:14:08 UTC. See [Year 2038 problem \(https://en.wikipedia.org/wiki/Year_2038_problem\)](https://en.wikipedia.org/wiki/Year_2038_problem) for details.

The Gnulib module ‘`year2038`’ fixes this problem on some platforms, by making `time_t` wide enough to represent timestamps after 2038. This has no effect on most current platforms, which have timestamps that are already wide enough. However, ‘`year2038`’ by default arranges for builds on legacy 32-bit Linux kernels running glibc 2.34 and later to compile with ‘`_TIME_BITS=64`’ to get wider timestamps. On older platforms that do not support timestamps after the year 2038, ‘`year2038`’ causes `configure` to issue a warning but still proceed. On platforms that appear to support post-2038 timestamps but where something prevents this from working, `configure` fails.

The default behavior of ‘`year2038`’ can be overridden by using the `configure` option `--disable-year2038`, which suppresses support for post-2038 timestamps. This may be useful if the package links to other libraries whose user-facing ABIs still require `time_t` to be 32-bit on your platform.

The Gnulib module ‘`year2038-recommended`’ is like ‘`year2038`’, except it by default rejects platforms where `time_t` cannot represent timestamps after 2038. If this module is used and a 32-bit platform cannot support 64-bit `time_t`, one can still fix the year-2038 problem by using a 64-bit instead of a 32-bit build, as noted in the architecture list below. If all else fails one can configure with `--disable-year2038`; however, the resulting programs will mishandle timestamps after 2038.

The Gnulib module ‘`year2038-recommended`’ is designed for packages intended for use on 32-bit platforms after the year 2038. If your package is commonly built on 32-bit platforms that will not be used after the year 2038, you can use the ‘`year2038`’ module instead, to save builders the trouble of configuring with `--disable-year2038`.

If the Gnulib module ‘`largefile`’ is used but neither ‘`year2038`’ nor ‘`year2038-recommended`’ is used, `configure` will have an option `--enable-year2038` that causes `configure` to behave as if ‘`year2038`’ was used. This is for packages that have long used ‘`largefile`’ but have not gotten around to upgrading their Gnulib module list to include ‘`year2038`’ or ‘`year2038-recommended`’. See [Section 14.2 \[Large File Support\]](#), page 842.

With the ‘`year2038-recommended`’ module, `configure` by default should work on the following 32-bit platforms (or 32-bit ABIs in bi-arch systems):

- Linux kernel 5.1 (2019) and later with glibc 2.34 (2021) and later on x86, arm, mips (o32 or n32 ABI), powerpc32, sparc32, s390, hppa, m68k, sh, csky, microblaze, nios2,
- Linux kernel 5.1 (2019) and later with musl libc 1.2 (2020) and later on x86,
- Linux on arc, loong32, or1k, riscv32 and x86_64-x32,
- NetBSD 6.0 (2012) and later on x86 and sparc,
- OpenBSD 5.5 (2014) and later on x86,
- FreeBSD/arm,
- Minix 3.3 (2014).

Whereas with ‘`year2038-recommended`’, `configure` should by default fail on earlier versions of the abovementioned platforms if a version is listed, and it should also by default fail on all versions of the following older 32-bit platforms or ABIs:

- Android on ARMv7 or x86,
- Mac OS X 10.6 (2009) and earlier on x86 and powerpc,
- GNU/Hurd/x86,
- GNU/kFreeBSD/x86,
- FreeBSD/x86 (this port demoted to Tier 2 in FreeBSD 13 [2021] and planned to never have 64-bit `time_t`),
- MidnightBSD/x86,
- AIX/powerpc (to fix, configure with ‘`CC='gcc -maix64' AR='ar -X64'`’),
- Solaris 11.4 (2018) and earlier on x86 and sparc (to fix, configure with ‘`CC='gcc -m64'`’),
- Cygwin 3.3.6 (2022) and earlier on x86,
- Haiku/x86.

If you use the ‘`year2038`’ or ‘`year2038-recommended`’ modules, and configure to support timestamps after the year 2038, your code should not include ‘`<utmp.h>`’ or ‘`<utmpx.h>`’ directly, because these include files do not work with 64-bit timestamps if the platform’s `time_t` was traditionally 32 bits. Your code can instead use the ‘`readutmp`’ module, which works around this problem.

14.6 Windows sockets

One of the portability problems for native Windows are sockets and networking functions.

14.6.1 Getaddrinfo and WINVER

This was written for the `getaddrinfo` module, but may be applicable to other functions too.

The `getaddrinfo` function exists in `ws2tcpip.h` and `-lws2_32` on Windows XP. The function declaration is present if `WINVER >= 0x0501`. Windows 2000 does not have `getaddrinfo` in its `WS2_32.DLL`.

Thus, if you want to assume Windows XP or later, you can add `AC_DEFINE([WINVER], [0x0501])` to avoid compiling the (partial) `getaddrinfo` implementation.

If you want to support Windows 2000, don’t do anything. The replacement function will open `WS2_32.DLL` during run-time to see if there is a `getaddrinfo` function available, and use it when available.

14.7 Native Windows Support without MSVC Support

If your package does not desire to have MSVC support, that is, if MinGW shall be the only native Windows platform that you wish to get support for from Gnulib, and you wish to minimize the number of files imported from Gnulib accordingly, you can do so by passing the options ‘`--avoid=msvc-inval --avoid=msvc-nothrow`’ to `gnulib-tool`.

14.8 Visual Studio Compatibility

The `lib-msvc-compat` module detects whether the linker supports `--output-def` when building a library. That parameter is used to generate a DEF file for a shared library (DLL). DEF files are useful for developers that use Visual Studio to develop programs that links to your library. See the GNU LD manual for more information.

There are other ways to create a DEF file, but we believe they are all sub-optimal to using `--output-def` during the build process. The variants we have considered include:

- Use DUMPBIN /EXPORTS. This is explained in <https://docs.microsoft.com/en-us/cpp/build/reference/dash-exports>. The tool does not generate DEF files directly, so its output needs to be post processed manually:

```
$ { echo EXPORTS; \
  dumpbin /EXPORTS libfoo-0.dll | tail -n+20 | awk '{ print $4 }'; \
} > libfoo-0.def
$ lib /def:libfoo-0.def
```

- Use IMPDEF. There is a tool called IMPDEF that can generate DEF files. However, it is not part of a standard Visual Studio installation. Further, it is documented as being an unreliable process.
- Use DLLTOOL. The `dlltool` is part of the MinGW suite, and thus not part of a standard Visual Studio installation. The documentation for the IMPDEF tool claims that DLLTOOL is the wrong tool for this job. Finally, DLLTOOL does not generate DEF files directly, so it requires post-processing of the output.

If you are using `libtool` to build your shared library, here is how to use this module. Import `lib-msvc-compat` to your project, and then add the following lines to the `Makefile.am` that builds the library:

```
if HAVE_LD_OUTPUT_DEF
libfoo_la_LDFLAGS += -Wl,--output-def,libfoo-$(DLL_VERSION).def
libfoo-$(DLL_VERSION).def: libfoo.la
defexecdir = $(libdir)
defexec_DATA = libfoo-$(DLL_VERSION).def
DISTCLEANFILES += $(defexec_DATA)
endif
```

The `DLL_VERSION` variable needs to be defined. It should be the shared library version number used in the DLL filename. For Windows targets you compute this value from the values you pass to `Libtool's -version-info`. Assuming you have variables `LT_CURRENT` and `LT_AGE` defined for the `CURRENT` and `AGE` libtool version integers, you compute `DLL_VERSION` as follows:

```
DLL_VERSION=`expr ${LT_CURRENT} - ${LT_AGE}`
AC_SUBST(DLL_VERSION)
```

15 Multithreading

Multithreading is a programming paradigm. In a multithreaded program, multiple threads execute concurrently (or quasi concurrently) at different places in the program.

There are three motivations for using multithreading in a program:

- Exploiting CPU hardware with multiple execution units. Nowadays, many CPUs have 2 to 8 execution cores in a single chip. Additionally, often multiple CPU chips are combined in a single package. Thus, some CPU packages support 64 or 96 simultaneous threads of execution.
- Simplifying program architecture. When a program has to read from different file descriptors, network sockets, or event channels at the same time, the classical single-threaded architecture is to have a main loop which uses `select` or `poll` on all the descriptors and then dispatches according to from which descriptor input arrived. In a multi-threaded program, you allocate one thread for each descriptor, and these threads can be programmed and managed independently.
- Offloading work from signal handlers. A signal handler is not allowed to call `malloc`; therefore you are very limited in what you can do in a signal handler. But a signal handler can notify a thread, and the thread can then do the appropriate processing, as complex as it needs to be.

A multithreading API offers

- Primitives for creating threads, for waiting until threads are terminated, and for reaping their results.
- Primitives through which different threads can operate on the same data or use some data structures for communicating between the threads. These are called “mutexes” or “locks”.
- Primitives for executing a certain (initialization) code at most once.
- Primitives for notifying one or more other threads. These are called wait queues or “condition variables”.
- Primitives for allowing different threads to have different values for a variable. Such a variable is said to reside in “thread-local storage” or “thread-specific storage”.
- Primitives for relinquishing control for some time and letting other threads go.

Note: Programs that achieve multithreading through OpenMP (cf. the glibc module ‘`openmp`’) don’t create and manage their threads themselves. Nevertheless, they need to use mutexes/locks in many cases.

15.1 The three multithreading APIs

Three multithreading APIs are available to Glibc users:

- POSIX multithreading,
- ISO C multithreading,
- Glibc multithreading.

They are supported on all platforms that have multithreading in one form or the other. Currently, these are all platforms supported by Glibc, except for Minix.

The main differences are:

- The exit code of a thread is a pointer in the POSIX and Gnulib APIs, but only an `int` in the ISO C API.
- The POSIX API has additional facilities for detaching threads, setting the priority of a thread, assigning a thread to a certain set of processors, and much more.
- In the POSIX and ISO C APIs, most functions have a return code, and you are supposed to check the return code; even locking and unlocking a lock can fail. In the Gnulib API, many functions don't have a return code; if they cannot complete, the program aborts. This sounds harsh, but such aborts have not been reported in 12 years.
- In the ISO C API, the initialization of a statically allocated lock is clumsy: You have to initialize it through a once-only function.

15.2 Choosing the right multithreading API

Here are guidelines for determining which multithreading API is best for your code.

In programs that use advanced POSIX APIs, such as detached threads (`pthread_detach`), signal blocking (`pthread_sigmask`), priorities (`pthread_setschedparam`), processor affinity (`pthread_setaffinity_np`), it is best to use the POSIX API. This is because you cannot convert an ISO C `thr_t` or a Gnulib `gl_thread_t` to a POSIX `pthread_t`.

In code that is shared with glibc, it is best to use the POSIX API as well.

In libraries, it is best to use the Gnulib API. This is because it gives the person who builds the library an option `'--enable-threads={isoc,posix,windows}'`, that determines on which native multithreading API of the platform to rely. In other words, with this choice, you can minimize the amount of glue code that your library needs to contain.

In the other cases, the POSIX API and the Gnulib API are equally well suited.

The ISO C API is never the best choice, as of this writing (2020).

15.3 The POSIX multithreading API

The POSIX multithreading API is documented in POSIX <https://pubs.opengroup.org/onlinepubs/9799919799/>.

To make use of POSIX multithreading, even on platforms that don't support it natively (most prominently, native Windows), use the following Gnulib modules:

Purpose

For thread creation and management:
 For simple and recursive locks:
 For read-write locks:
 For once-only execution:
 For “condition variables” (wait queues):
 For thread-local storage:
 For relinquishing control:
 For spin locks:

Module

`pthread-thread`
`pthread-mutex`
`pthread-rwlock`
`pthread-once`
`pthread-cond`
`pthread-tss`
`sched_yield`
`pthread-spin`

There is also a convenience module named `pthread` which depends on all of these (except `sched_yield`); so you don't need to enumerate these modules one by one.

15.4 The ISO C multithreading API

The ISO C multithreading API is documented in ISO C 11 <https://www.open-std.org/jtc1/sc22/wg14/www/docs/n1570.pdf>.

To make use of ISO C multithreading, even on platforms that don't support it or have severe bugs, use the following Gnulib modules:

Purpose	Module
For thread creation and management:	<code>thrd</code>
For simple locks, recursive locks, and read-write locks:	<code>mtx</code>
For once-only execution:	<code>mtx</code>
For “condition variables” (wait queues):	<code>cnd</code>
For thread-local storage:	<code>tss</code>

There is also a convenience module named `threads` which depends on all of these; so you don't need to enumerate these modules one by one.

15.5 The Gnulib multithreading API

The Gnulib multithreading API is documented in the respective include files:

- `<glthread/thread.h>`
- `<glthread/lock.h>`
- `<glthread/once.h>`
- `<glthread/cond.h>`
- `<glthread/tls.h>`
- `<glthread/yield.h>`
- `<glthread/spin.h>`

To make use of Gnulib multithreading, use the following Gnulib modules:

Purpose	Module
For thread creation and management:	<code>thread</code>
For simple locks, recursive locks, and read-write locks:	<code>lock</code>
For once-only execution:	<code>once</code>
For “condition variables” (wait queues):	<code>cond</code>
For thread-local storage:	<code>tls</code>
For relinquishing control:	<code>yield</code>
For spin locks:	<code>spin</code>

The Gnulib multithreading supports a configure option ‘`--enable-threads={isoc,posix,windows}`’, that chooses the underlying thread implementation. Currently (2020):

- `--enable-threads=posix` is supported and is the best choice on all platforms except for native Windows. It may also work, to a limited extent, on mingw with the `winpthreads` library, but is not recommended there.
- `--enable-threads=windows` is supported and is the best choice on native Windows platforms (mingw and MSVC).
- `--enable-threads=isoc` is supported on all platforms that have the ISO C multithreading API. However, `--enable-threads=posix` is always a better choice.

15.6 Optimizations of multithreaded code

Despite all the optimizations of multithreading primitives that have been implemented over the years – from `atomic operations in hardware` (<https://en.wikipedia.org/wiki/Compare-and-swap>), over `futexes` (<https://en.wikipedia.org/wiki/Futex>) and `restartable sequences` (<https://www.efficios.com/blog/2019/02/08/linux-restartable-sequences/>) in the Linux kernel, to lock elision [1] (<https://lwn.net/Articles/534758/>) [2] (https://www.gnu.org/software/libc/manual/html_node/Elision-Tunables.html)) – single-threaded programs can still profit performance-wise from the assertion that they are single-threaded.

Gnulib defines four facilities that help optimizing for the single-threaded case.

- The Gnulib multithreading API, when used on `glibc ≤ 2.32` and `*BSD` systems, uses weak symbols to detect whether the program is linked with `libpthread`. If not, the program has no way to create additional threads and must therefore be single-threaded. This optimization applies to all the Gnulib multithreading API (locks, thread-local storage, and more).
- The `thread-optim` module, on `glibc ≥ 2.32` systems, allows your code to skip locking between threads (regardless which of the three multithreading APIs you use). You need extra code for this: include the `"thread-optim.h"` header file, and use the macro `gl_multithreaded` like this:

```
bool mt = gl_multithreaded ();
if (mt) gl_lock_lock (some_lock);
...
if (mt) gl_lock_unlock (some_lock);
```

- You may use the `unlocked-io` module if you want the `FILE` stream functions `getc`, `putc`, etc. to use unlocked I/O if available, throughout the package. Unlocked I/O can improve performance, sometimes dramatically. But unlocked I/O is safe only in single-threaded programs, as well as in multithreaded programs for which you can guarantee that every `FILE` stream, including `stdin`, `stdout`, `stderr`, is used only in a single thread.

You need extra code for this optimization to be effective: include the `"unlocked-io.h"` header file. Some Gnulib modules that do operations on `FILE` streams have these preparations already included.

- You may define the C macro `GNULIB_REGEX_SINGLE_THREAD`, if all the programs in your package invoke the functions of the `regex` module only from a single thread.
- You may define the C macro `GNULIB_MBRTOWC_SINGLE_THREAD`, if all the programs in your package invoke the functions `mbrtowc`, `mbrtoc32`, and the functions of the `regex` module only from a single thread. (The `regex` module uses `mbrtowc` under the hood.)
- You may define the C macro `GNULIB_WCHAR_SINGLE_LOCALE`, if all the programs in your package set the locale early and
 - don't change the locale after it has been initialized, and
 - don't call locale sensitive functions (`mbrtowc`, `wcwidth`, etc.) before the locale has been initialized.

This macro optimizes the functions `mbrtowc`, `mbrtoc32`, and `wcwidth`. You can get this macro defined by including the Gnulib module `wchar-single`.

- You may define the C macro `GNULIB_EXCLUDE_SINGLE_THREAD`, if all the programs in your package invoke the functions of the `exclude` module only from a single thread.

16 Strings and Characters

This chapter describes the APIs for strings and characters, provided by Gnulib.

16.1 Strings

Several possible representations exist for the representation of strings in memory of a running C program.

16.1.1 The C string representation

The classical representation of a string in C is a sequence of characters, where each character takes up one or more bytes, followed by a terminating NUL byte. This representation is used for strings that are passed by the operating system (in the `argv` argument of `main`, for example) and for strings that are passed to the operating system (in system calls such as `open`). The C type to hold such strings is `'char *'` or, in places where the string shall not be modified, `'const char *'`. There are many C library functions, standardized by ISO C and POSIX, that assume this representation of strings.

A *character encoding*, or *encoding* for short, describes how the elements of a character set are represented as a sequence of bytes. For example, in the ASCII encoding, the UNDERSCORE character is represented by a single byte, with value 0x5F. As another example, the COPYRIGHT SIGN character is represented:

- in the ISO-8859-1 encoding, by the single byte 0xA9,
- in the UTF-8 encoding, by the two bytes 0xC2 0xA9,
- in the GB18030 encoding, by the four bytes 0x81 0x30 0x84 0x38.

Note: The `'char'` type may be signed or unsigned, depending on the platform. When we talk about the "byte 0xA9" we actually mean the `char` object whose value is `(char) 0xA9`; we omit the cast to `char` in this documentation, for brevity.

In POSIX, the character encoding is determined by the locale. The locale is some environmental attribute that the user can choose.

Depending on the encoding, in general, every character is represented by one or more bytes (up to 4 bytes in practice – but use `MB_LEN_MAX` instead of the number 4 in the code). When every character is represented by only 1 byte, we speak of an “unibyte locale”, otherwise of a “multibyte locale”.

It is important to realize that the majority of Unix installations nowadays use UTF-8 as locale encoding; therefore, the majority of users are using multibyte locales.

Three important facts to remember are:

A `'char'` is a byte, not a character.

As a consequence:

- The `<ctype.h>` API, that was designed only with unibyte encodings in mind, is useless nowadays for general text processing; it does not work in multibyte locales.
- The `strlen` function does not return the number of characters in a string. Nor does it return the number of screen columns occupied by a string after it is output. It merely returns the number of *bytes* occupied by a string.

- Truncating a string, for example, with `strncpy`, can have the effect of truncating it in the middle of a multibyte character. Such a string will, when output, have a garbled character at its end, often represented by a hollow box.

Multibyte does not imply UTF-8 encoding.

While UTF-8 is the most common multibyte encoding, GB18030 is also a supported locale encoding on GNU systems (mostly because it is a Chinese government standard, last revised in 2022).

Searching for a character in a string is not the same as searching for a byte in the string.

Take the above example of COPYRIGHT SIGN in the GB18030 encoding: A byte search will find the bytes '0' and '8' in this string. But a search for the *character* "0" or "8" in the string "©" must, of course, report “not found”.

As a consequence:

- `strchr` and `strrchr` do not work with multibyte strings if the locale encoding is GB18030 and the character to be searched is a digit.
- `strstr` does not work with multibyte strings if the locale encoding is different from UTF-8.
- `strcspn`, `strpbrk`, `strspn` cannot work correctly in multibyte locales: they assume the second argument is a list of single-byte characters. Even in this simple case, they do not work with multibyte strings if the locale encoding is GB18030 and one of the characters to be searched is a digit.
- `strsep` and `strtok_r` do not work with multibyte strings unless all of the delimiter characters are ASCII characters < 0x30.
- The `strcasemp`, `strncasemp`, and `strcasestr` functions do not work with multibyte strings.

Workarounds can be found in Gnulib, in the form of `mbs*` API functions:

- Gnulib has functions `mbslen` and `mbswidth` that can be used instead of `strlen` when the number of characters or the number of screen columns of a string is requested.
- Gnulib has functions `mbschr` and `mbsrchr` that are like `strchr` and `strrchr`, but work in multibyte locales.
- Gnulib has a function `mbsstr` that is like `strstr`, but works in multibyte locales.
- Gnulib has functions `mbscspn`, `mbspbrk`, `mbsspn` that are like `strcspn`, `strpbrk`, `strspn`, but work in multibyte locales.
- Gnulib has functions `mbssep` and `mbstok_r` that are like `strsep` and `strtok_r` but work in multibyte locales.
- Gnulib has functions `mbscasemp`, `mbsncasemp`, `mbspcasemp`, and `mbscasestr` that are like `strcasemp`, `strncasemp`, and `strcasestr`, but work in multibyte locales. Still, the function `ulc_casemp` is preferable to these functions.

A C string can contain encoding errors.

Not every NUL-terminated byte sequence represents a valid multibyte string. Byte sequences can contain encoding errors, that is, bytes or byte sequences that are invalid and do not represent characters.

String functions like `mbstowcs` and `strcoll` whose behavior depends on encoding have unspecified behavior on strings containing encoding errors, unless the behavior is specifically documented. If an application needs a particular behavior on these strings it can iterate through them itself, as described in the next subsection.

16.1.2 Iterating through strings

For complex string processing, string functions may not be enough, and you need to iterate through a string while processing each (possibly multibyte) character or encoding error in turn. Gnuilib has several modules for iterating forward through a string in this way. Backward iteration, that is, from the string's end to start, is not provided, as it is too hairy in general.

- The `mbiter` module iterates through a string whose length is already known. The string can contain NULs and encoding errors.
- The `mbiterf` module is like `mbiter` except it is more complex and typically faster.
- The `mbuiter` module iterates through a C string whose length is not a-priori known. The string can contain encoding errors and is terminated by the first NUL.
- The `mbuiterf` module is like `mbuiter` except it is more complex and typically faster.
- The `mcel` module is simpler than `mbiter` and `mbuiter` and can be faster than even `mbiterf` and `mbuiterf`. It can iterate through either strings whose length is known, or C strings, or strings terminated by other ASCII characters $< 0x30$.
- The `mcel-prefer` module is like `mcel` except that it causes some other modules to be based on `mcel` instead of on the `mbiter` family.

The choice of modules depends on the application's needs. The `mbiter` module family is more suitable for applications that treat some sequences of two or more bytes as a single encoding error, and for applications that need to support obsolescent encodings on non-GNU platforms, such as CP864, EBCDIC, Johab, and Shift JIS. In this module family, `mbuiter` and `mbuiterf` are more suitable than `mbiter` and `mbiterf` when arguments are C strings, lengths are not already known, and it is highly likely that only the first few multibyte characters need to be inspected.

The `mcel` module is simpler and can be faster than the `mbiter` family, and is more suitable for applications that do not need the `mbiter` family's special features.

The `mcel-prefer` module is like `mcel` except that it also causes some other modules, such as `mbstowcs`, to use `mcel` rather than the `mbiter` family. This can be simpler and faster. However, it does not support the obsolescent encodings, and it may behave differently on data containing encoding errors where behavior is unspecified or undefined, because in `mcel` each encoding error is a single byte whereas in the `mbiter` family a single encoding error can contain two or more bytes.

If a package uses `mcel-prefer`, it may also want to give `gnulib-tool` one or more of the options `--avoid=mbiter`, `--avoid=mbiterf`, `--avoid=mbuiter` and `--avoid=mbuiterf`, to avoid packaging modules that are not needed.

16.1.3 Strings with NUL characters

The GNU Coding Standards, section https://www.gnu.org/prep/standards/html_node/Semantics.html, specifies:

Utilities reading files should not drop NUL characters, or any other nonprinting characters.

When it is a requirement to store NUL characters in strings, a variant of the C strings is needed. Gnulib offers a “string descriptor” type for this purpose. See [Section 17.11 \[Handling strings with NUL characters\]](#), page 881.

All remarks regarding encodings and multibyte characters in the previous section apply to string descriptors as well.

16.1.4 Character and String Functions in C Locale

The functions in this section are similar to the generic string functions from the standard C library, except that

- They behave as if the locale was set to the "C" locale, even when the locale is different, and/or
- They are specially optimized for the case where all characters are plain ASCII characters.

The functions are provided by the following modules.

16.1.4.1 c-ctype

The `c-ctype` module contains functions operating on single-byte characters, like the functions in `<ctype.h>`, that operate as if the locale encoding was ASCII. (The "C" locale on many systems has the locale encoding "ASCII".)

The functions are:

```
extern bool c_isascii (int c);

extern bool c_isalnum (int c);
extern bool c_isalpha (int c);
extern bool c_isblank (int c);
extern bool c_iscntrl (int c);
extern bool c_isdigit (int c);
extern bool c_islower (int c);
extern bool c_isgraph (int c);
extern bool c_isprint (int c);
extern bool c_ispunct (int c);
extern bool c_isspace (int c);
extern bool c_isupper (int c);
extern bool c_isxdigit (int c);

extern int c_tolower (int c);
extern int c_toupper (int c);
```

These functions assign properties only to ASCII characters.

The `c` argument can be a `char` or `unsigned char` value, whereas the corresponding functions in `<ctype.h>` take an argument that is actually an `unsigned char` value.

The `c_is*` functions return ‘bool’, where the corresponding functions in `<ctype.h>` return ‘int’ for historical reasons.

Note: The `<ctype.h>` functions support only unibyte locales.

16.1.4.2 c-strcaseeq

The `c-strcaseeq` module contains an optimized case-insensitive string comparison function operating on single-byte character strings, that operate as if the locale encoding was ASCII. (The "C" locale on many systems has the locale encoding "ASCII".)

The functions is actually implemented as a macro:

```
extern int STRCASEEQ (const char *s1, const char *s2,
                     int s20, int s21, int s22, int s23, int s24, int s25,
                     int s26, int s27, int s28);
```

`s2` should be a short literal ASCII string, and `s20`, `s21`, ... the individual characters of `s2`.

For case conversion here, only ASCII characters are considered to be upper case or lower case.

16.1.4.3 c-strcasecmp

The `c-strcasecmp` module contains a case-insensitive string comparison function operating on single-byte character strings, like the functions in `<strings.h>`, that operate as if the locale encoding was ASCII. (The "C" locale on many systems has the locale encoding "ASCII".)

The function is:

```
extern int c_strcasecmp (const char *s1, const char *s2);
```

For case conversion here, only ASCII characters are considered to be upper case or lower case.

Note: The function `strcasecmp` from `<strings.h>` supports only unibyte locales; for multibyte locales, you need the function `mbstrcasecmp`.

16.1.4.4 c-strncasecmp

The `c-strncasecmp` module contains a case-insensitive string comparison function operating on single-byte character strings, like the functions in `<strings.h>`, that operate as if the locale encoding was ASCII. (The "C" locale on many systems has the locale encoding "ASCII".)

The function is:

```
extern int c_strncasecmp (const char *s1, const char *s2, size_t n);
```

For case conversion here, only ASCII characters are considered to be upper case or lower case.

Note: The function `strncasecmp` from `<strings.h>` supports only unibyte locales; for multibyte locales, you need the function `mbstrncasecmp` or `mbcasecmp`.

16.1.4.5 c-strcasestr

The `c-strcasestr` module contains a case-insensitive string search function operating on single-byte character strings, that operate as if the locale encoding was ASCII. (The "C" locale on many systems has the locale encoding "ASCII".)

The function is:

```
extern char *c_strcasestr (const char *haystack, const char *needle);
```

For case conversion here, only ASCII characters are considered to be upper case or lower case.

Note: The function `strcasestr` from `<string.h>` supports only unibyte locales; for multibyte locales, you need the function `mbstrcasestr`.

16.1.4.6 c-strstr

The `c-strstr` module contains a substring search function operating on single-byte character strings, that operate as if the locale encoding was ASCII. (The "C" locale on many systems has the locale encoding "ASCII".)

The function is:

```
extern char *c_strstr (const char *haystack, const char *needle);
```

Note: The function `strstr` from `<string.h>` supports only unibyte locales; for multibyte locales, you need the function `mbstrstr`.

16.1.4.7 c-strtod

The `c-strtod` module contains a string to number ('double') conversion function operating on single-byte character strings, that operates as if the locale encoding was ASCII. (The "C" locale on many systems has the locale encoding "ASCII".)

The function is:

```
extern double c_strtod (const char *string, char **endp);
```

In particular, only a period '.' is accepted as decimal point, even when the current locale's notion of decimal point is a comma ',', and no characters outside the basic character set are accepted.

On platforms without `strtod_l`, this function is not safe for use in multi-threaded applications since it calls `setlocale`.

16.1.4.8 c-strtold

The `c-strtold` module contains a string to number ('long double') conversion function operating on single-byte character strings, that operates as if the locale encoding was ASCII. (The "C" locale on many systems has the locale encoding "ASCII".)

The function is:

```
extern long double c_strtold (const char *string, char **endp);
```

In particular, only a period '.' is accepted as decimal point, even when the current locale's notion of decimal point is a comma ',',.

16.1.5 Comparison of string APIs

This table summarizes the API functions available for strings, in POSIX and in GnuLib.

unibyte strings only	assume locale	C	multibyte strings	multibyte strings with NULs	wide character strings	32-bit wide character strings
strlen	strlen		mbslen	sd_length	wcslen	u32_ strlen
strnlen	strnlen		mbsnlen	—	wcsnlen	u32_ strnlen, u32_ mbsnlen
strcmp	strcmp		strcmp	sd_cmp	wscmp	u32_ strcmp
strncmp	strncmp		strncmp	—	wcsncmp	u32_ strncmp
strcasemp	strcasemp		mbscasemp	—	wscasemp	u32_ casemp
strncasemp	strncasemp		mbsncasemp, mbspcasemp	—	wcsncasemp	u32_ casemp
strcoll	strcmp		strcoll	—	wscoll	u32_ strcoll
strxfrm	—		strxfrm	—	wcsxfrm	—
strchr	strchr		mbschr	sd_index	wcschr	u32_ strchr
strrchr	strrchr		mbsrchr	sd_last_ index	wcsrchr	u32_ strrchr
strstr	strstr		mbsstr	sd_ contains	wcsstr	u32_ strstr
strcasestr	strcasestr		mbscasestr	—	—	—
str_ startswith	str_ startswith		mbs_ startswith	sd_ startswith	—	u32_ startswith

<code>str_endswith</code>	<code>str_endsWith</code>	<code>mbs_endsWith</code>	<code>sd_endsWith</code>	—	<code>u32_endsWith</code>
<code>strspn</code>	<code>strspn</code>	<code>mbsspn</code>	—	<code>wcsspn</code>	<code>u32_strspn</code>
<code>strcspn</code>	<code>strcspn</code>	<code>mbscspn</code>	—	<code>wcscspn</code>	<code>u32_strcspn</code>
<code>strpbrk</code>	<code>strpbrk</code>	<code>mbspbrk</code>	—	<code>wcspbrk</code>	<code>u32_strpbrk</code>
<code>strtok_r</code>	<code>strtok_r</code>	<code>mbstok_r</code>	—	<code>wcstok</code>	<code>u32_strtok</code>
<code>strsep</code>	<code>strsep</code>	<code>mbssep</code>	—	—	—
<code>strcpy</code>	<code>strcpy</code>	<code>strcpy</code>	<code>sd_copy</code>	<code>wscopy</code>	<code>u32_strcpy</code>
<code>stpcpy</code>	<code>stpcpy</code>	<code>stpcpy</code>	—	<code>wcpcpy</code>	<code>u32_stpcpy</code>
<code>strncpy</code>	<code>strncpy</code>	<code>strncpy</code>	—	<code>wcsncpy</code>	<code>u32_strncpy</code>
<code>stpncpy</code>	<code>stpncpy</code>	<code>stpncpy</code>	—	<code>wcpncpy</code>	<code>u32_stpncpy</code>
<code>strcat</code>	<code>strcat</code>	<code>strcat</code>	<code>sd_concat</code>	<code>wscat</code>	<code>u32_strcat</code>
<code>strncat</code>	<code>strncat</code>	<code>strncat</code>	—	<code>wcsncat</code>	<code>u32_strncat</code>
<code>free</code>	<code>free</code>	<code>free</code>	<code>sd_free</code>	<code>free</code>	<code>free</code>
<code>strdup</code>	<code>strdup</code>	<code>strdup</code>	<code>sd_copy</code>	<code>wcsdup</code>	<code>u32_strdup</code>
<code>strndup</code>	<code>strndup</code>	<code>strndup</code>	—	—	—

<code>mbswidth</code>	<code>mbswidth</code>	<code>mbswidth</code>	—	<code>wcswidth</code>	<code>c32swidth</code> , <code>u32_</code> <code>strwidth</code>
<code>strtol</code>	<code>strtol</code>	<code>strtol</code>	—	—	—
<code>strtoul</code>	<code>strtoul</code>	<code>strtoul</code>	—	—	—
<code>strtoll</code>	<code>strtoll</code>	<code>strtoll</code>	—	—	—
<code>strtoull</code>	<code>strtoull</code>	<code>strtoull</code>	—	—	—
<code>strtoimax</code>	<code>strtoimax</code>	<code>strtoimax</code>	—	<code>wcstoimax</code>	—
<code>strtoumax</code>	<code>strtoumax</code>	<code>strtoumax</code>	—	<code>wcstoumax</code>	—
<code>strtof</code>	—	<code>strtof</code>	—	—	—
<code>strtod</code>	<code>c_strtod</code>	<code>strtod</code>	—	—	—
<code>strtold</code>	<code>c_strtold</code>	<code>strtold</code>	—	—	—
<code>strfromf</code>	—	<code>strfromf</code>	—	—	—
<code>strfromd</code>	—	<code>strfromd</code>	—	—	—
<code>strfroml</code>	—	<code>strfroml</code>	—	—	—
—	—	—	—	<code>mbstowcs</code>	<code>mbstoc32s</code>
—	—	—	—	<code>mbsrtowcs</code>	<code>mbsrtoc32s</code>
—	—	—	—	<code>mbsnrtowcs</code>	<code>mbsnrtoc32s</code>
—	—	—	—	<code>wcstombs</code>	<code>c32stombs</code>
—	—	—	—	<code>wcsrtombs</code>	<code>c32srtombs</code>
—	—	—	—	<code>wcsnrtombs</code>	<code>c32snrtombs</code>

16.2 Characters

A *character* is the elementary unit that strings are made of.

What is a character? “A character is an element of a character set” is sort of a circular definition, but it highlights the fact that it is not merely a number. Although many characters are visually represented by a single glyph, there are characters that, for example, have a different glyph when used at the end of a word than when used inside a word. A character is also not the minimal rendered text processing unit; that is a grapheme cluster and in general consists of one or more characters. If you want to know more about the concept of character and various concepts associated with characters, refer to the Unicode standard.

For the representation in memory of a character, various types have been in use, and some of them were failures: `char` and `wchar_t` were invented for this purpose, but are not the right types. `char32_t` is the right type (successor of `wchar_t`); and `mbchar_t` (defined by Gnulib) is an alternative for specific kinds of processing.

16.2.1 The `char` type

The `char` type is in the C language since the beginning in the 1970ies, but – due to its limitation of 256 possible values – is no longer the adequate type for storing a character.

Technically, it is still adequate in unibyte locales. But since most locales nowadays are multibyte locales, it makes no sense to write a program that runs only in unibyte locales.

ISO C and POSIX standardized an API for characters of type `char`, in `<ctype.h>`. This API is nowadays useless and obsolete, when it comes to general text processing.

The important lessons to remember are:

A ‘char’ is just the elementary storage unit for a string, not a character.

Never use `<ctype.h>`!

16.2.2 The `wchar_t` type

The ISO C and POSIX standard creators made an attempt to overcome the dead end regarding the `char` type. They introduced

- a type ‘`wchar_t`’, designed to encapsulate a character,
- a “wide string” type ‘`wchar_t *`’, with some API functions declared in `<wchar.h>`, and
- functions declared in `<wctype.h>` that were meant to supplant the ones in `<ctype.h>`.

Unfortunately, this API and its implementation has numerous problems:

- On Windows platforms and on AIX in 32-bit mode, `wchar_t` is a 16-bit type. This means that it can never accommodate an entire Unicode character. Either the `wchar_t *` strings are limited to characters in UCS-2 (the “Basic Multilingual Plane” of Unicode), or – if `wchar_t *` strings are encoded in UTF-16 – a `wchar_t` represents only half of a character in the worst case, making the `<wctype.h>` functions pointless.
- On Solaris and FreeBSD, the `wchar_t` encoding is locale dependent and undocumented. This means, if you want to know any property of a `wchar_t` character, other than the properties defined by `<wctype.h>` – such as whether it’s a dash, currency symbol, paragraph separator, or similar –, you have to convert it to `char *` encoding first, by use of the function `wctomb`.

- When you read a stream of wide characters, through the functions `fgetwc` and `fgetws`, and when the input stream/file is not in the expected encoding, you have no way to determine the invalid byte sequence and do some corrective action. If you use these functions, your program becomes “garbage in - more garbage out” or “garbage in - abort”.

As a consequence, it is better to use multibyte strings. Such multibyte strings can bypass limitations of the `wchar_t` type, if you use functions defined in Gnulib and GNU libunistring for text processing. They can also faithfully transport malformed characters that were present in the input, without requiring the program to produce garbage or abort.

16.2.3 The `char32_t` type

The ISO C and POSIX standard creators then introduced the `char32_t` type. In ISO C 11, it was conceptually a “32-bit wide character” type. In ISO C 23, its semantics has been further specified: A `char32_t` value is a Unicode code point.

Thus, the `char32_t` type is not affected the problems that plague the `wchar_t` type.

The `char32_t` type and its API are defined in the `<uchar.h>` header file.

ISO C and POSIX specify only the basic functions for the `char32_t` type, namely conversion of a single character (`mbrtoc32` and `c32rtomb`). For convenience, Gnulib adds API for classification and case conversion of characters.

GNU libunistring can also be used on `char32_t` values. Since `char32_t` is the same as `uint32_t`, all `u32_*` functions of GNU libunistring are applicable to arrays of `char32_t` values.

On glibc systems, use of the 32-bit wide strings (`char32_t[]`) is exactly as efficient as the use of the older wide strings (`wchar_t[]`). This is possible because on glibc, `wchar_t` values already always were 32-bit and Unicode code points. `mbrtoc32` is just an alias of `mbrtowc`. The Gnulib `*c32*` functions are optimized so that on glibc systems they immediately redirect to the corresponding `*wc*` functions.

Gnulib implements the ISO C 23 semantics of `char32_t` when you import the ‘`uchar-h-c23`’ module. Without this module, it implements only the ISO C 11 semantics; the effect is that on some platforms (macOS, FreeBSD, NetBSD, Solaris) a `char32_t` value is the same as a `wchar_t` value, not a Unicode code point. Thus, when you want to pass `char32_t` values to GNU libunistring or to some Unicode centric Gnulib functions, you need the ‘`uchar-h-c23`’ module in order to do so without portability problems.

16.2.4 The `mbchar_t` type

Gnulib defines an alternate way to encode a multibyte character: `mbchar_t`. Its main feature is the ability to process a string or stream with some malformed characters without reporting an error.

The type `mbchar_t`, defined in “`mbchar.h`”, holds a character in both the multibyte and the 32-bit wide character representation. In case of a malformed character only the multibyte representation is used.

16.2.4.1 Reading multibyte strings

If you want to process (possibly multibyte) characters while reading them from a `FILE *` stream, without reading them into a string first, the `mbfile` module is made for this purpose.

16.2.5 Comparison of character APIs

This table summarizes the API functions available for characters, in POSIX and in Glibc.

unibyte character	assume C locale	wide character	32-bit character	wide mbchar_t character
<code>== '\0'</code>	<code>== '\0'</code>	<code>== L'\0'</code>	<code>== 0</code>	<code>mb_isnul</code>
<code>==</code>	<code>==</code>	<code>==</code>	<code>==</code>	<code>mb_equal</code>
<code>isalnum</code>	<code>c_isalnum</code>	<code>iswalnum</code>	<code>c32isalnum</code>	<code>mb_isalnum</code>
<code>isalpha</code>	<code>c_isalpha</code>	<code>iswalpha</code>	<code>c32isalpha</code>	<code>mb_isalpha</code>
<code>isblank</code>	<code>c_isblank</code>	<code>iswblank</code>	<code>c32isblank</code>	<code>mb_isblank</code>
<code>iscntrl</code>	<code>c_iscntrl</code>	<code>iswcntrl</code>	<code>c32iscntrl</code>	<code>mb_iscntrl</code>
<code>isdigit</code>	<code>c_isdigit</code>	<code>iswdigit</code>	<code>c32isdigit</code>	<code>mb_isdigit</code>
<code>isgraph</code>	<code>c_isgraph</code>	<code>iswgraph</code>	<code>c32isgraph</code>	<code>mb_isgraph</code>
<code>islower</code>	<code>c_islower</code>	<code>iswlower</code>	<code>c32islower</code>	<code>mb_islower</code>
<code>isprint</code>	<code>c_isprint</code>	<code>iswprint</code>	<code>c32isprint</code>	<code>mb_isprint</code>
<code>ispunct</code>	<code>c_ispunct</code>	<code>iswpunct</code>	<code>c32ispunct</code>	<code>mb_ispunct</code>
<code>isspace</code>	<code>c_isspace</code>	<code>iswspace</code>	<code>c32isspace</code>	<code>mb_isspace</code>
<code>isupper</code>	<code>c_isupper</code>	<code>iswupper</code>	<code>c32isupper</code>	<code>mb_isupper</code>
<code>isxdigit</code>	<code>c_isxdigit</code>	<code>iswxdigit</code>	<code>c32isxdigit</code>	<code>mb_isxdigit</code>
—	—	<code>wctype</code>	<code>c32_get_type_test</code>	—
—	—	<code>iswctype</code>	<code>c32_apply_type_test</code>	—
<code>tolower</code>	<code>c_tolower</code>	<code>towlower</code>	<code>c32tolower</code>	—

<code>toupper</code>	<code>c_toupper</code>	<code>toupper</code>	<code>c32toupper</code>	—
—	—	<code>wctrans</code>	<code>c32_get_ mapping</code>	—
—	—	<code>towctrans</code>	<code>c32_apply_ mapping</code>	—
—	—	<code>wcwidth</code>	<code>c32width</code>	<code>mb_width</code>

17 Particular Modules

17.1 `alloca`

The `alloca` module provides for a function `alloca` which allocates memory on the stack, where the system allows it. A memory block allocated with `alloca` exists only until the function that calls `alloca` returns or exits abruptly.

There are a few systems where this is not possible: HP-UX systems, and some other platforms when the C++ compiler is used. On these platforms the `alloca` module provides a `malloc` based emulation. This emulation will not free a memory block immediately when the calling function returns, but rather will wait until the next `alloca` call from a function with the same or a shorter stack length. Thus, in some cases, a few memory blocks will be kept although they are not needed any more.

The user can `#include <alloca.h>` and use `alloca` on all platforms. Note that the `#include <alloca.h>` must be the first one after the autoconf-generated `config.h`, for AIX 3 compatibility. Thanks to IBM for this nice restriction!

Note that GCC 3.1 and 3.2 can *inline* functions that call `alloca`. When this happens, the memory blocks allocated with `alloca` will not be freed until *the end of the calling function*. If this calling function runs a loop calling the function that uses `alloca`, the program easily gets a stack overflow and crashes. To protect against this compiler behaviour, you can mark the function that uses `alloca` with the following attribute:

```
#ifdef __GNUC__
__attribute__((__noinline__))
#endif
```

An alternative to this module is the ‘`alloca-opt`’ module.

17.2 `alloca-opt`

The `alloca-opt` module provides for a function `alloca` which allocates memory on the stack, where the system allows it. A memory block allocated with `alloca` exists only until the function that calls `alloca` returns or exits abruptly.

There are a few systems where this is not possible: HP-UX systems, and some other platforms when the C++ compiler is used. On these platforms the `alloca-opt` module provides no replacement, just a preprocessor macro `HAVE_ALLOCA`.

The user can `#include <alloca.h>` on all platforms, and use `alloca` on those platforms where the preprocessor macro `HAVE_ALLOCA` evaluates to true. If `HAVE_ALLOCA` is false, the code should use a heap-based memory allocation based on `malloc` or (in C++) `new`. Note that the `#include <alloca.h>` must be the first one after the autoconf-generated `config.h`, for AIX 3 compatibility. Thanks to IBM for this nice restriction!

Note that GCC 3.1 and 3.2 can *inline* functions that call `alloca`. When this happens, the memory blocks allocated with `alloca` will not be freed until *the end of the calling function*. If this calling function runs a loop calling the function that uses `alloca`, the program easily gets a stack overflow and crashes. To protect against this compiler behaviour, you can mark the function that uses `alloca` with the following attribute:

```
#ifdef __GNUC__
__attribute__((__noinline__))
#endif
```

17.3 Safe Allocation Macros

The standard C library `malloc`/`realloc`/`calloc`/`free` APIs are prone to a number of common coding errors. The `safe-alloc` module provides macros that make it easier to avoid many of them. It still uses the standard C allocation functions behind the scenes.

This module is obsolete, as it does not seem to have caught on in practice and some of its features could not be ported to unusual platforms.

Some of the memory allocation mistakes that are commonly made are

- passing the incorrect number of bytes to `malloc`, especially when allocating an array,
- unchecked integer overflow when calculating array sizes,
- fail to check the return value of `malloc` and `realloc` for errors,
- forget to fully initialize memory just allocated with `malloc`,
- duplicate calls to `free` by forgetting to set the pointer variable to `NULL`,
- leaking memory in calls to `realloc` when that call fails.

The `safe-alloc` module addresses these problems in the following way:

- It defines macros that wrap around the standard C allocation functions. That makes it possible to use the compiler's knowledge of the size of objects for allocation; it also allows setting pointers passed in as arguments when appropriate.
- It uses return values only for a success/failure error condition flag, and annotates them with GCC's `__warn_unused_result__` attribute.
- When allocating a fresh array, it uses `calloc` instead of `malloc` so that the array's contents are zeroed. However, memory added to an already-existing array is uninitialized.

`int ALLOC(ptr)` [Macro]

Allocate `sizeof *ptr` bytes of memory and store the address of allocated memory in `ptr`. Fill the newly allocated memory with zeros.

Returns `-1` on failure, `0` on success.

`int ALLOC_N(ptr, count)` [Macro]

Allocate an array of `count` elements, each `sizeof *ptr` bytes long, and store the address of allocated memory in `ptr`. Fill the newly allocated memory with zeros.

Returns `-1` on failure, `0` on success.

`int ALLOC_N_UNINITIALIZED(ptr, count)` [Macro]

Allocate an array of `count` elements, each `sizeof *ptr` bytes long, and store the address of allocated memory in `ptr`. The allocated memory is not initialized.

Returns `-1` on failure, `0` on success.

`void FREE(ptr)` [Macro]

Free the memory stored in `ptr` and set `ptr` to `NULL`.

17.4 Attributes

The module ‘`attribute`’ provides a header file `attribute.h` that defines macros related to C and C++ attributes and the GCC `__attribute__` keyword.

Here is an example of its use:

```
#include <attribute.h>

NODISCARD
extern char *crypt (char const *, char const *)
    ATTRIBUTE_NOTHROW ATTRIBUTE_LEAF ATTRIBUTE_NONNULL ((1, 2));
```

`NODISCARD` expands to `[[nodiscard]]` if the compiler supports this C23 syntax, otherwise to `__attribute__ ((__warn_unused_result__))` if the compiler is a recent-enough GCC or GCC-like compiler, otherwise to nothing. `ATTRIBUTE_NOTHROW` expands to `__attribute__ ((__nothrow__))` if the compiler is a recent-enough GCC or GCC-like compiler, and to nothing otherwise. Similarly for `ATTRIBUTE_LEAF`. `ATTRIBUTE_NONNULL ((1, 2))` expands to `__attribute__ ((__nonnull__ (1, 2)))` if the compiler is recent-enough GCC, and to nothing otherwise.

Most of these attribute names begin with `ATTRIBUTE_`. A few do not, because they are part of C23 and their names are not likely to clash with other macro names. These macros are `DEPRECATED`, `FALLTHROUGH`, `MAYBE_UNUSED`, and `NODISCARD`, which can be defined to `[[deprecated]]` etc. on C23 platforms. Also, these exceptional macros should be placed at the start of function declarations, whereas the `ATTRIBUTE_*` macros can be placed at the end.

The module also defines the macro `UNNAMED`, designed for function parameters that are never used, whereas the `MAYBE_UNUSED` attribute is better suited for parameters that might or might not be unused depending on preprocessor macro settings. For example:

```
int
f (int a, MAYBE_UNUSED int b, int UNNAMED (c))
{
    int r = a;
#ifdef B_FEATURE
    r |= b;
#endif
    return r;
}
```

17.5 Compile-time Assertions

The module ‘`verify`’ provides a header file `verify.h` that defines macros related to compile-time verification.

Two of these macros are `verify (V)` and `verify_expr (V, EXPR)`. Both accept an integer constant expression argument `V` and verify that it is nonzero. If not, a compile-time error results.

These two macros implement compile-time tests, as opposed to the standard `assert` macro which supports only runtime tests. Since the tests occur at compile-time, they are more reliable, and they require no runtime overhead.

`verify (V);` is a declaration; it can occur outside of functions. In contrast, `verify_expr (V, EXPR)` is an expression that returns the value of `EXPR`; it can be used in macros that expand to expressions. If `EXPR` is an integer constant expression, then `verify_expr (V, EXPR)` is also an integer constant expression. Although `EXPR` and `verify_expr (V, EXPR)` are guaranteed to have the same side effects and value and type (after integer promotion), they need not have the same type if `EXPR`'s type is an integer that is narrower than `int` or `unsigned int`.

`V` should be an integer constant expression in the sense of the C standard. Its leaf operands should be integer, enumeration, or character constants; or `sizeof` expressions that return constants; or floating constants that are the immediate operands of casts. Outside a `sizeof` subexpression, `V` should not contain any assignments, function calls, comma operators, casts to non-integer types, or subexpressions whose values are outside the representable ranges for their types. If `V` is not an integer constant expression, then a compiler might reject a usage like `'verify (V);'` even when `V` is nonzero.

Although the standard `assert` macro is a runtime test, C23 and C++17 specify a builtin `static_assert (V)`, which differs from `verify` in two major ways. First, it can also be used within a `struct` or `union` specifier, in place of an ordinary member declaration. Second, it allows the programmer to specify, as an optional second argument, a compile-time diagnostic as a string literal. If your program is not intended to be portable to compilers that lack C23 or C++17 `static_assert`, the only advantage of `verify` is that its name is a bit shorter.

The `verify.h` header defines one more macro, `assume (E)`, which expands to an expression of type `void` that causes the compiler to assume that `E` yields a nonzero value. `E` should be a scalar expression, and should not have side effects; it may or may not be evaluated. The behavior is undefined if `E` would yield zero. The main use of `assume` is optimization, as the compiler may be able to generate better code if it assumes `E`. For best results, `E` should be simple enough that a compiler can determine that it has no side effects: if `E` calls an external function or accesses volatile storage the compiler may not be able to optimize `E` away and `assume (E)` may therefore slow down the program.

Here are some example uses of these macros.

```
#include <verify.h>

#include <limits.h>
#include <time.h>

/* Verify that time_t is an integer type. */
verify ((time_t) 1.5 == 1);

/* Verify that time_t is no smaller than int. */
verify (sizeof (int) <= sizeof (time_t));

/* Verify that time_t is signed. */
verify ((time_t) -1 < 0);

/* Verify that time_t uses two's complement representation. */
verify (~ (time_t) -1 == 0);
```

```

/* Return the maximum value of the integer type T,
   verifying that T is an unsigned integer type.
   The cast to (T) is outside the call to verify_expr
   so that the result is of type T
   even when T is narrower than unsigned int.  */
#define MAX_UNSIGNED_VAL(t) \
    ((t) verify_expr (0 < (t) -1, -1))

/* Return T divided by CHAR_MAX + 1, where behavior is
   undefined if T < 0.  In the common case where CHAR_MAX
   is 127 the compiler can therefore implement the division
   by shifting T right 7 bits, an optimization that would
   not be valid if T were negative.  */
time_t
time_index (time_t t)
{
    assume (0 <= t);
    return t / (CHAR_MAX + 1);
}

```

17.6 Non-returning Functions

A "non-returning" function is a function which cannot return normally. Instead of returning, it can loop forever, or it can transfer control via `abort`, `execvp`, `exit`, `longjmp`, `throw` (in C++), or similar mechanisms. Non-returning functions are declared with a `void` return type.

It helps the compiler's ability to emit sensible warnings, following data-flow analysis, to declare which functions are non-returning. It can also help generate more-efficient code, as there is no need to save a return address when calling a non-returning function.

Gnulib has multiple ways to support such a declaration:

- The `_Noreturn` keyword. No modules are needed, as Gnulib arranges for `<config.h>` to define `_Noreturn` to an appropriate replacement on platforms lacking it. Unfortunately, although this approach works for all current C versions, the `_Noreturn` keyword is obsolescent in C23.
- The `'noreturn'` module. It provides a way to put this declaration at function declarations, at function definitions, and in function pointer types. The identifiers to use are:
 - `_GL_NORETURN_FUNC` for use in function declarations and function definitions.
 - `_GL_NORETURN_FUNCPTR` for use on function pointers.

The include file is `<noreturn.h>`.

Which of the approaches to use? If the non-returning functions you have to declare are unlikely to be accessed through function pointers, you should use `_Noreturn`; otherwise the module `noreturn` provides for better data-flow analysis and thus for better warnings.

There is also an obsolete `stdnoreturn-h` module, but its use is no longer recommended.

17.7 Integer Properties

The `intprops` module consists of an include file `<intprops.h>` that defines several macros useful for testing properties of integer types.

Integer overflow is a common source of problems in programs written in C and other languages. In some cases, such as signed integer arithmetic in C programs, the resulting behavior is undefined, and practical platforms do not always behave as if integers wrap around reliably. In other cases, such as unsigned integer arithmetic in C, the resulting behavior is well-defined, but programs may still misbehave badly after overflow occurs.

Many techniques have been proposed to attack these problems. These include precondition testing, wraparound behavior where signed integer arithmetic is guaranteed to be modular, saturation semantics where overflow reliably yields an extreme value, undefined behavior sanitizers where overflow is guaranteed to trap, and various static analysis techniques.

Gnulib supports wraparound arithmetic and precondition testing, as these are relatively easy to support portably and efficiently. There are two families of precondition tests: the first, for integer types, is easier to use, while the second, for integer ranges, has a simple and straightforward portable implementation.

Like other Gnulib modules, the implementation of the `intprops` module assumes that integers use a two's complement representation, but it does not assume that signed integer arithmetic wraps around. See [Section 1.5.4 \[Other portability assumptions\]](#), page 6.

17.7.1 Arithmetic Type Properties

`TYPE_IS_INTEGER (t)` is an arithmetic constant expression that yields 1 if the arithmetic type `t` is an integer type, 0 otherwise. `bool` counts as an integer type.

`TYPE_SIGNED (t)` is an arithmetic constant expression that yields 1 if the real type `t` is a signed integer type or a floating type, 0 otherwise. If `t` is an integer type, `TYPE_SIGNED (t)` is an integer constant expression.

`EXPR_SIGNED (e)` yields 1 if the real expression `e` has a signed integer type or a floating type, 0 otherwise. If `e` is an integer constant expression or an arithmetic constant expression, `EXPR_SIGNED (e)` is likewise. The expression `e` is not evaluated, and `EXPR_SIGNED (e)` is typically optimized to a constant.

Example usage:

```
#include <intprops.h>
#include <sys/types.h>

enum
{
    clock_t_is_integer = TYPE_IS_INTEGER (clock_t),
    uid_t_is_signed = TYPE_SIGNED (uid_t)
};

int
CLOCKS_PER_SEC_is_signed (void)
{
```

```

    return EXPR_SIGNED (CLOCKS_PER_SEC);
}

```

17.7.2 Arithmetic Type Conversion

Here are some ways in C to convert an arithmetic expression *e* to a possibly different arithmetic type *t*.

- An explicit conversion like `((t) e)` is powerful and can therefore be dangerous, because the conversion can succeed even if neither *e* nor *t* happens to have an arithmetic type. For example, if *e* is a pointer, `((long int) e)` will do a conversion even if that was not the intent.
- An implicit conversion like `t v = e;` is less powerful, as it does not convert from pointers. However, it can lose information and sign, as in for example `int v = -0.9;` which sets *v* to zero.
- A no-op arithmetic expression like `+e` is even less powerful, as it preserves value (including sign) because it does only integer promotions. That is, it converts to `int` if that can represent all values of *e*'s underlying type, otherwise to `unsigned int` if that can represent all values, and otherwise it does no conversion.

`INT_PROMOTE (e)` is an expression with the same value as the arithmetic expression *e* but with *e*'s type after any integer promotion. It behaves like `+e`.

In the following example, using `INT_PROMOTE` pacifies GCC's `-Wswitch-enum` option, and may help human readers see what is going on even if they are not expert in C's integer promotion rules and might be confused by the simpler `switch (+v)`.

```

enum { A = 1, B, C, D, E } v = ...;
switch (INT_PROMOTE (v))
{
    case A: case C:
        return true;
    default:
        /* Handle all other cases,
           even cases like v == 0.  */
        return false;
}

```

17.7.3 Integer Bounds

`INT_BUFSIZE_BOUND (t)` is an integer constant expression that is a bound on the size of the string representing an integer type or expression *t* in decimal notation, including the terminating null character and any leading `-` character. For example, if `INT_BUFSIZE_BOUND (int)` is 12, any value of type `int` can be represented in 12 bytes or less, including the terminating null. The bound is not necessarily tight.

Example usage:

```

#include <intprops.h>
#include <stdio.h>
int
int_strlen (int i)
{

```

```

    char buf[INT_BUFSIZE_BOUND (int)];
    return sprintf (buf, "%d", i);
}

```

`INT_STRLEN_BOUND (t)` is an integer constant expression that is a bound on the length of the string representing an integer type or expression *t* in decimal notation, including any leading `-` character. This is one less than `INT_BUFSIZE_BOUND (t)`.

`TYPE_MINIMUM (t)` and `TYPE_MAXIMUM (t)` are integer constant expressions equal to the minimum and maximum values of the integer type *t*. These expressions are of the type *t*.

Example usage:

```

#include <sys/types.h>
#include <intprops.h>
bool
in_off_t_range (long long int a)
{
    return TYPE_MINIMUM (off_t) <= a && a <= TYPE_MAXIMUM (off_t);
}

```

17.7.4 Checking Integer Overflow

Signed integer arithmetic has undefined behavior on overflow in C. Although almost all modern computers use two's complement signed arithmetic that is well-defined to wrap around, C compilers routinely optimize assuming that signed integer overflow cannot occur, which means that a C program cannot easily get at the underlying machine arithmetic. For example:

```

if ((a + b < b) == (a < 0))
    a += b;
else
    printf ("overflow\n");

```

might not work as expected if `a` and `b` are signed, because a compiler can assume that signed overflow cannot occur and treat the entire `if` expression as if it were true. And even if `a` is unsigned, the expression might not work as expected if `b` is negative or is wider than `a`.

The following macros work around this problem by yielding an overflow indication while computing the sum, difference, or product of two integers. For example, if `i` is of type `int`, `INT_ADD_OK (INT_MAX - 1, 1, &i)` sets `i` to `INT_MAX` and yields 1, whereas `INT_ADD_OK (INT_MAX, 1, &i)` yields 0.

Example usage:

```

#include <intprops.h>
#include <stdio.h>

/* Compute A * B, reporting whether overflow occurred. */
void
print_product (long int a, long int b)
{
    long int r;
    if (INT_MULTIPLY_OK (a, b, &r))
        printf ("result is %ld\n", r);
}

```

```

    else
        printf ("overflow\n");
}

```

These macros work for both signed and unsigned integers, so they can be used with integer types like `time_t` that may or may not be signed, depending on the platform.

These macros have the following restrictions:

- Their first two arguments must be integer expressions.
- Their last argument must be a non-null pointer to an integer.
- They may evaluate their arguments zero or multiple times, so the arguments should not have side effects.
- They are not necessarily constant expressions, even if all their arguments are constant expressions.

INT_ADD_OK (*a*, *b*, *r*)

Compute the sum of *a* and *b*. If it fits into **r*, store it there and yield 1. Otherwise yield 0, possibly modifying **r* to an unspecified value. See above for restrictions.

INT_SUBTRACT_OK (*a*, *b*, *r*)

Compute the difference between *a* and *b*. If it fits into **r*, store it there and yield 1. Otherwise yield 0, possibly modifying **r* to an unspecified value. See above for restrictions.

INT_MULTIPLY_OK (*a*, *b*, *r*)

Compute the product of *a* and *b*. If it fits into **r*, store it there and yield 1. Otherwise yield 0, possibly modifying **r* to an unspecified value. See above for restrictions.

Other macros are available if you need wrapped-around results when overflow occurs (see [Section 17.7.5 \[Wraparound Arithmetic\]](#), page 872), or if you need to check for overflow in operations other than addition, subtraction, and multiplication (see [Section 17.7.6 \[Integer Type Overflow\]](#), page 874).

17.7.5 Wraparound Arithmetic with Integers

Signed integer arithmetic has undefined behavior on overflow in C. Although almost all modern computers use two's complement signed arithmetic that is well-defined to wrap around, C compilers routinely optimize assuming that signed integer overflow cannot occur, which means that a C program cannot easily get at the underlying machine arithmetic. For example, on a typical machine with 32-bit two's complement `int` the expression `INT_MAX + 1` does not necessarily yield `INT_MIN`, because the compiler may do calculations with a 64-bit register, or may generate code that traps on signed integer overflow.

The following macros work around this problem by storing the wraparound value, i.e., the low-order bits of the correct answer, and by yielding an overflow indication. For example, if *i* is of type `int`, `INT_ADD_WRAPV (INT_MAX, 1, &i)` sets *i* to `INT_MIN` and yields 1 on a two's complement machine. See [Section 17.7.6 \[Integer Type Overflow\]](#), page 874.

Example usage:

```
#include <intprops.h>
```

```

#include <stdio.h>

/* Print the low order bits of A * B,
   reporting whether overflow occurred.  */
void
print_product (long int a, long int b)
{
    long int r;
    int overflow = INT_MULTIPLY_WRAPV (a, b, &r);
    printf ("result is %ld (%s)\n", r,
           (overflow
            ? "after overflow"
            : "no overflow"));
}

```

These macros work for both signed and unsigned integers, so they can be used with integer types like `time_t` that may or may not be signed, depending on the platform.

These macros have the following restrictions:

- Their first two arguments must be integer expressions.
- Their last argument must be a non-null pointer to an integer.
- They may evaluate their arguments zero or multiple times, so the arguments should not have side effects.
- They are not necessarily constant expressions, even if all their arguments are constant expressions.

INT_ADD_WRAPV (a, b, r)

Store the low-order bits of the sum of *a* and *b* into **r*. Yield 1 if overflow occurred, 0 if the low-order bits are the mathematically-correct sum. See above for restrictions.

INT_SUBTRACT_WRAPV (a, b, r)

Store the low-order bits of the difference between *a* and *b* into **r*. Yield 1 if overflow occurred, 0 if the low-order bits are the mathematically-correct difference. See above for restrictions.

INT_MULTIPLY_WRAPV (a, b, r)

Store the low-order bits of the product of *a* and *b* into **r*. Yield 1 if overflow occurred, 0 if the low-order bits are the mathematically-correct product. See above for restrictions.

If your code includes `<intprops.h>` only for these `_WRAPV` macros, you may prefer to use GnuLib's `stdckdint.h` module instead, as it supports similar macros that were standardized in C23 and are therefore independent of GnuLib if your code can assume C23 or later. See [Section 9.52 \[stdckdint.h\], page 79](#).

Other macros are available if you do not need wrapped-around results when overflow occurs (see [Section 17.7.4 \[Checking Integer Overflow\], page 871](#)), or if you need to check for overflow in operations other than addition, subtraction, and multiplication (see [Section 17.7.6 \[Integer Type Overflow\], page 874](#)).

17.7.6 Integer Type Overflow

Although unsigned integer arithmetic wraps around modulo a power of two, signed integer arithmetic has undefined behavior on overflow in C. Almost all modern computers use two's complement signed arithmetic that is well-defined to wrap around, but C compilers routinely optimize based on the assumption that signed integer overflow cannot occur, which means that a C program cannot easily get at the underlying machine behavior. For example, the signed integer expression $(a + b < b) \neq (a < 0)$ is not a reliable test for whether $a + b$ overflows, because a compiler can assume that signed overflow cannot occur and treat the entire expression as if it were false.

These macros yield 1 if the corresponding C operators overflow, 0 otherwise. They work correctly on all known practical hosts, and do not rely on undefined behavior due to signed arithmetic overflow. They are integer constant expressions if their arguments are. They are typically easier to use than the integer range overflow macros (see [Section 17.7.7 \[Integer Range Overflow\]](#), page 875), and they support more operations and evaluation contexts than the integer overflow checking macros (see [Section 17.7.4 \[Checking Integer Overflow\]](#), page 871) or the wraparound macros (see [Section 17.7.5 \[Wraparound Arithmetic\]](#), page 872).

These macros can be tricky to use with arguments narrower than `int`. For example, in the common case with 16-bit `short int` and 32-bit `int`, if `a` and `b` are of type `short int` then `INT_MULTIPLY_OVERFLOW(a, b)` always yields 0, as $a * b$ cannot overflow due to C's rule that `a` and `b` are widened to `int` before multiplying. For this reason, often it is better to use the integer overflow checking macros (see [Section 17.7.4 \[Checking Integer Overflow\]](#), page 871) or the wraparound macros (see [Section 17.7.5 \[Wraparound Arithmetic\]](#), page 872) when checking for overflow in addition, subtraction, or multiplication.

Example usage:

```
#include <intprops.h>
#include <limits.h>
#include <stdio.h>

/* Print A * B if in range, an overflow
   indicator otherwise. */
void
print_product (long int a, long int b)
{
    if (INT_MULTIPLY_OVERFLOW (a, b))
        printf ("multiply would overflow");
    else
        printf ("product is %ld", a * b);
}

/* Does the product of two ints always fit
   in a long int? */
enum {
    INT_PRODUCTS_FIT_IN_LONG
    = ! (INT_MULTIPLY_OVERFLOW
        ((long int) INT_MIN, INT_MIN))
}
```

```
};
```

These macros have the following restrictions:

- Their arguments must be integer expressions.
- They may evaluate their arguments zero or multiple times, so the arguments should not have side effects.

These macros are tuned for their last argument being a constant.

`INT_ADD_OVERFLOW (a, b)`

Yield 1 if $a + b$ would overflow, 0 otherwise. See above for restrictions.

`INT_SUBTRACT_OVERFLOW (a, b)`

Yield 1 if $a - b$ would overflow, 0 otherwise. See above for restrictions.

`INT_NEGATE_OVERFLOW (a)`

Yields 1 if $-a$ would overflow, 0 otherwise. See above for restrictions.

`INT_MULTIPLY_OVERFLOW (a, b)`

Yield 1 if $a * b$ would overflow, 0 otherwise. See above for restrictions.

`INT_DIVIDE_OVERFLOW (a, b)`

Yield 1 if a / b would overflow, 0 otherwise. See above for restrictions. Division overflow can happen on two's complement hosts when dividing the most negative integer by -1 . This macro does not check for division by zero.

`INT_REMAINDER_OVERFLOW (a, b)`

Yield 1 if $a \% b$ would overflow, 0 otherwise. See above for restrictions. Remainder overflow can happen on two's complement hosts when dividing the most negative integer by -1 ; although the mathematical result is always 0, in practice some implementations trap, so this counts as an overflow. This macro does not check for division by zero.

`INT_LEFT_SHIFT_OVERFLOW (a, b)`

Yield 1 if $a \ll b$ would overflow, 0 otherwise. See above for restrictions. The C standard says that behavior is undefined for shifts unless $0 \leq b < w$ where w is a 's word width, and that when a is negative then $a \ll b$ has undefined behavior, but this macro does not check these other restrictions.

17.7.7 Integer Range Overflow

These macros yield 1 if the corresponding C operators might not yield numerically correct answers due to arithmetic overflow, and 0 if the operators do not overflow. They do not rely on undefined or implementation-defined behavior. They are integer constant expressions if their arguments are. Their implementations are simple and straightforward, but they are typically harder to use than the integer type overflow macros. See [Section 17.7.6 \[Integer Type Overflow\]](#), page 874.

Although the implementation of these macros is similar to that suggested in the SEI CERT C Secure Coding Standard, in its two sections “[INT30-C. Ensure that unsigned integer operations do not wrap](https://www.securecoding.cert.org/confluence/display/c/INT30-C.+Ensure+that+unsigned+integer+operations+do+not+wrap) (<https://www.securecoding.cert.org/confluence/display/c/INT30-C.+Ensure+that+unsigned+integer+operations+do+not+wrap>)” and “[INT32-C. Ensure that operations on signed integers do not result in overflow](https://www.securecoding.cert.org/confluence/display/c/INT32-C) (<https://www.securecoding.cert.org/confluence/display/c/INT32-C>).

`+Ensure+that+operations+on+signed+integers+do+not+result+in+overflow)`",
Gnulib's implementation was derived independently of CERT's suggestions.

Example usage:

```
#include <intprops.h>
#include <limits.h>
#include <stdio.h>

void
print_product (long int a, long int b)
{
    if (INT_MULTIPLY_RANGE_OVERFLOW (a, b, LONG_MIN, LONG_MAX))
        printf ("multiply would overflow");
    else
        printf ("product is %ld", a * b);
}

/* Does the product of two ints always fit
   in a long int? */
enum {
    INT_PRODUCTS_FIT_IN_LONG
    = ! (INT_MULTIPLY_RANGE_OVERFLOW
        ((long int) INT_MIN, (long int) INT_MIN,
         LONG_MIN, LONG_MAX))
};
```

These macros have the following restrictions:

- Their arguments must be integer expressions.
- They may evaluate their arguments zero or multiple times, so the arguments should not have side effects.
- The arithmetic arguments (including the *min* and *max* arguments) must be of the same integer type after the usual arithmetic conversions, and the type must have minimum value *min* and maximum *max*. Unsigned values should use a zero *min* of the proper type, for example, `(unsigned int) 0`.

These macros are tuned for constant *min* and *max*. For commutative operations such as `a + b`, they are also tuned for constant *b*.

`INT_ADD_RANGE_OVERFLOW (a, b, min, max)`
Yield 1 if `a + b` would overflow in `[min,max]` integer arithmetic, 0 otherwise.
See above for restrictions.

`INT_SUBTRACT_RANGE_OVERFLOW (a, b, min, max)`
Yield 1 if `a - b` would overflow in `[min,max]` integer arithmetic, 0 otherwise.
See above for restrictions.

`INT_NEGATE_RANGE_OVERFLOW (a, min, max)`
Yield 1 if `-a` would overflow in `[min,max]` integer arithmetic, 0 otherwise. See above for restrictions.

INT_MULTIPLY_RANGE_OVERFLOW (*a*, *b*, *min*, *max*)

Yield 1 if *a* * *b* would overflow in [*min*,*max*] integer arithmetic, 0 otherwise. See above for restrictions.

INT_DIVIDE_RANGE_OVERFLOW (*a*, *b*, *min*, *max*)

Yield 1 if *a* / *b* would overflow in [*min*,*max*] integer arithmetic, 0 otherwise. See above for restrictions. Division overflow can happen on two's complement hosts when dividing the most negative integer by -1 . This macro does not check for division by zero.

INT_REMAINDER_RANGE_OVERFLOW (*a*, *b*, *min*, *max*)

Yield 1 if *a* % *b* would overflow in [*min*,*max*] integer arithmetic, 0 otherwise. See above for restrictions. Remainder overflow can happen on two's complement hosts when dividing the most negative integer by -1 ; although the mathematical result is always 0, in practice some implementations trap, so this counts as an overflow. This macro does not check for division by zero.

INT_LEFT_SHIFT_RANGE_OVERFLOW (*a*, *b*, *min*, *max*)

Yield 1 if *a* << *b* would overflow in [*min*,*max*] integer arithmetic, 0 otherwise. See above for restrictions. Here, *min* and *max* are for *a* only, and *b* need not be of the same type as the other arguments. The C standard says that behavior is undefined for shifts unless $0 \leq b < w$ where *w* is *a*'s word width, and that when *a* is negative then *a* << *b* has undefined behavior, but this macro does not check these other restrictions.

17.8 Static inline functions

In order to mark functions as **static inline** in *.c* files, the only prerequisite you need is an **AC_REQUIRE([AC_C_INLINE])**. No GnuLib module is needed.

When marking functions as **static inline** in *.h* files, on the other hand, it is useful to avoid the following problem: When gcc or clang is not optimizing, it generates the same object code for every **static inline** function that is actually used in the current compilation unit, thus bloating the size of the resulting binary. (Code of **static inline** functions that are unused in the current compilation unit gets dropped; that is not an issue.) To avoid this problem, use the GnuLib module 'inline' and make the **static inline** function definition conditional on the value of the C macro **HAVE_INLINE**.

17.9 Extern inline functions

The **extern-inline** module supports the use of C99-style **extern inline** functions so that the code still runs on compilers that do not support this feature correctly.

C code ordinarily should not use **inline**. Typically it is better to let the compiler figure out whether to inline, as compilers are pretty good about optimization nowadays. In this sense, **inline** is like **register**, another keyword that is typically no longer needed.

Functions defined (not merely declared) in headers are an exception, as avoiding **inline** would commonly cause problems for these functions. Suppose *aaa.h* defines the function *aaa_fun*, and *aaa.c*, *bbb.c* and *ccc.c* all include *aaa.h*. If code is intended to be portable to non-C99 compilers, *aaa_fun* cannot be declared with the C99 **inline** keyword. This problem cannot be worked around by making *aaa_fun* an ordinary function, as it would

be defined three times with external linkage and the definitions would clash. Although `aaa_fun` could be a static function, with separate compilation if `aaa_fun` is not inlined its code will appear in the executable three times.

To avoid this code bloat, `aaa.h` can do this:

```
/* aaa.h */
/* #include any other headers here */
#ifndef _GL_INLINE_HEADER_BEGIN
    #error "Please include config.h first."
#endif
_GL_INLINE_HEADER_BEGIN
#ifndef AAA_INLINE
# define AAA_INLINE _GL_INLINE
#endif
...
AAA_INLINE int
aaa_fun (int i)
{
    return i + 1;
}
...
_GL_INLINE_HEADER_END
```

and `aaa.c` can do this:

```
/* aaa.c */
#define AAA_INLINE _GL_EXTERN_INLINE
#include <config.h>
#include <aaa.h>
```

whereas `bbb.c` and `ccc.c` can include `aaa.h` in the usual way. C99 compilers expand `AAA_INLINE` to C99-style `inline` usage, where `aaa_fun` is declared `extern inline` in `aaa.c` and plain `inline` in other modules. Non-C99 compilers that are compatible with GCC use GCC-specific syntax to accomplish the same ends. Other non-C99 compilers use `static inline` so they suffer from code bloat, but they are not mainline platforms and will die out eventually.

In this coding idiom, the compilation unit should define `AAA_INLINE` before including the `aaa.h` header that conditionally defines it. In the unusual and not-recommended case where `config.h` itself includes `aaa.h`, the compilation unit should define `AAA_INLINE` before including `config.h`, not merely before including `aaa.h`. Also, you need one `AAA_INLINE`-like macro per compilation unit, not one per header file. In other words, if the header file `aaa.h` defines functions defined in `aaa-foo.c` and `aaa-bar.c`, you need different macros `AAA_FOO_INLINE` and `AAA_BAR_INLINE`. Use `AAA_FOO_INLINE` for the functions defined in `aaa-foo.c`, and use `AAA_BAR_INLINE` for the functions defined in `aaa-bar.c`.

Functions defined in this way should not refer to identifiers with static linkage, as C99 through C23 prohibit this. Although a future C standard will likely relax this restriction, it is still an issue on many practical platforms.

`_GL_INLINE` is a portable alternative to C99 plain `inline`.

`_GL_EXTERN_INLINE` is a portable alternative to C99 `extern inline`.

Invoke `_GL_INLINE_HEADER_BEGIN` before all uses of `_GL_INLINE` in an include file. This suppresses some bogus warnings in GCC versions before 5.1. If an include file includes other files, it is better to invoke this macro after including the other files.

Invoke `_GL_INLINE_HEADER_END` after all uses of `_GL_INLINE` in an include file.

17.10 Handling closed standard file descriptors

Usually, when a program gets invoked, its file descriptors 0 (for standard input), 1 (for standard output), and 2 (for standard error) are open. But there are situations when some of these file descriptors are closed. These situations can arise when

- The invoking process invokes `close()` on the file descriptor before `exec`, or
- The invoking process invokes `posix_spawn_file_actions_addclose()` for the file descriptor before `posix_spawn` or `posix_spawnp`, or
- The invoking process is a Bourne shell, and the shell script uses the POSIX syntax for closing the file descriptor: `<&-` for closing standard input, `>&-` for closing standard output, or `2>&-` for closing standard error.

When a closed file descriptor is accessed through a system call, such as `fcntl()`, `fstat()`, `read()`, or `write()`, the system call fails with error `EBADF` ("Bad file descriptor").

When a new file descriptor is allocated, the operating system chooses the smallest non-negative integer that does not yet correspond to an open file descriptor. So, when a given fd (0, 1, or 2) is closed, opening a new file descriptor may assign the new file descriptor to this fd. This can have unintended effects, because now standard input/output/error of your process is referring to a file that was not meant to be used in that role.

This situation is a security risk because the behaviour of the program in this situation was surely never tested, therefore anything can happen then – from overwriting precious files of the user to endless loops.

To deal with this situation, you first need to determine whether your program is affected by the problem.

- Does your program invoke functions that allocate new file descriptors? These are the system calls
 - `open()`, `openat()`, `creat()`
 - `dup()`
 - `fopen()`, `freopen()`
 - `pipe()`, `pipe2()`, `popen()`
 - `opendir()`
 - `tmpfile()`, `mkstemp()`, `mkstemp()`, `mkostemp()`, `mkostemp()`

Note that you also have to consider the libraries that your program uses.

- If your program may open two or more file descriptors or FILE streams for reading at the same time, and some of them may reference standard input, your program *is affected*.
- If your program may open two or more file descriptors or FILE streams for writing at the same time, and some of them may reference standard output or standard error, your program *is affected*.

- If your program does not open new file descriptors or FILE streams, it is *not affected*.
- If your program opens only one new file descriptor or FILE stream at a time, it is *not affected*. This is often the case for programs that are structured in simple phases: first a phase where input is read from a file into memory, then a phase of processing in memory, finally a phase where the result is written to a file.
- If your program opens only two new file descriptors or FILE streams at a time, out of which one is for reading and the one is for writing, it is *not affected*. This is because if the first file descriptor is allocated and the second file descriptor is picked as 0, 1, or 2, and both happen to be the same, writing to the one opened in `O_RDONLY` mode will produce an error `EBADF`, as desired.

If your program is affected, what is the mitigation?

Some operating systems install open file descriptors in place of the closed ones, either in the `exec` system call or during program startup. When such a file descriptor is accessed through a system call, it behaves like an open file descriptor opened for the “wrong” direction: the system calls `fcntl()` and `fstat()` succeed, whereas `read()` from fd 0 and `write()` to fd 1 or 2 fail with error `EBADF` ("Bad file descriptor"). The important point here is that when your program allocates a new file descriptor, it will have a value greater than 2.

This mitigation is enabled on HP-UX, for all programs, and on glibc, FreeBSD, NetBSD, OpenBSD, but only for `setuid` or `setgid` programs. Since it is operating system dependent, it is not a complete mitigation.

For a complete mitigation, Gnulib provides two alternative sets of modules:

- The `xstdopen` module.
- The `*-safer` modules: `fcntl-safer`, `openat-safer`, `unistd-safer`, `fopen-safer`, `freopen-safer`, `pipe2-safer`, `popen-safer`, `dirent-safer`, `tmpfile-safer`, `stdlib-safer`.

The approach with the `xstdopen` module is simple, but it adds three system calls to program startup. Whereas the approach with the `*-safer` modules is more complicated and error-prone, and does not fix the problem if system library functions call one of the affected functions, but adds no overhead (no additional system calls) in the normal case.

To use the approach with the `xstdopen` module:

1. Import the module `xstdopen` from Gnulib.
2. In the compilation unit that contains the `main` function, include "`xstdopen.h`".
3. In the `main` function, near the beginning, namely right after the `__i18n` related initializations (`setlocale`, `bindtextdomain`, `textdomain` invocations, if any) and the `closeout` initialization (if any), insert the invocation:

```
/* Ensure that stdin, stdout, stderr are open. */
xstdopen ();
```

To use the approach with the `*-safer` modules:

1. Import the relevant modules from Gnulib.
2. In the compilation units that contain these function calls, include the replacement header file.

Do so according to this table:

Function	Module	Header file
<code>open()</code>	<code>fcntl-safer</code>	<code>"fcntl--.h"</code>
<code>openat()</code>	<code>openat-safer</code>	<code>"fcntl--.h"</code>
<code>creat()</code>	<code>fcntl-safer</code>	<code>"fcntl--.h"</code>
<code>dup()</code>	<code>unistd-safer</code>	<code>"unistd--.h"</code>
<code>fopen()</code>	<code>fopen-safer</code>	<code>"stdio--.h"</code>
<code>freopen()</code>	<code>freopen-safer</code>	<code>"stdio--.h"</code>
<code>pipe()</code>	<code>unistd-safer</code>	<code>"unistd--.h"</code>
<code>pipe2()</code>	<code>pipe2-safer</code>	<code>"unistd--.h"</code>
<code>popen()</code>	<code>popen-safer</code>	<code>"stdio--.h"</code>
<code>opendir()</code>	<code>dirent-safer</code>	<code>"dirent--.h"</code>
<code>tmpfile()</code>	<code>tmpfile-safer</code>	<code>"stdio--.h"</code>
<code>mkstemp()</code>	<code>stdlib-safer</code>	<code>"stdlib--.h"</code>
<code>mkstemps()</code>	<code>stdlib-safer</code>	<code>"stdlib--.h"</code>
<code>mkostemp()</code>	<code>stdlib-safer</code>	<code>"stdlib--.h"</code>
<code>mkostemps()</code>	<code>stdlib-safer</code>	<code>"stdlib--.h"</code>

17.11 Handling strings with NUL characters

Strings in C are usually represented by a character sequence with a terminating NUL character. A `'char *'`, pointer to the first byte of this character sequence, is what gets passed around as function argument or return value.

The major restriction of this string representation is that it cannot handle strings that contain NUL characters: such strings will appear shorter than they were meant to be. In most application areas, this is not a problem, and the `char *` type is well usable.

A second problem of this string representation is that taking a substring is not cheap: it either requires a memory allocation or a destructive modification of the string. The former has a runtime cost; the latter complicates the logic of the program. This matters for application areas that analyze text, such as parsers.

In areas where strings with embedded NUL characters need to be handled or where taking substrings is a recurrent operation, the common approach is to use a `char *ptr` pointer variable together with a `size_t nbytes` variable (or an `idx_t nbytes` variable, if you want to avoid problems due to integer overflow). This works fine in code that constructs or manipulates strings with embedded NUL characters. But when it comes to *storing* them, for example in an array or as key or value of a hash table, one needs a type that combines these two fields.

The Gnulib modules `string-desc`, `xstring-desc`, and `string-desc-quotearg` provide such a type. We call it a “string descriptor” and name it `string_desc_t`.

The type `string_desc_t` is a struct that contains a pointer to the first byte and the number of bytes of the memory region that make up the string. An additional terminating NUL byte, that may be present in memory, is not included in this byte count. This type implements the same concept as `std::string_view` in C++, or the `String` type in Java.

`string_desc_t` is a string descriptor to a string that cannot be written to. There is also a type `rw_string_desc_t`, that is a descriptor for a writable string. `rw_string_desc_t`

compares to `string_desc_t`, like the pointer type `'char *'` compares to the pointer type `'const char *'`.

A `string_desc_t` or `rw_string_desc_t` can be passed to a function as an argument, or can be the return value of a function. This is type-safe: If, by mistake, a programmer passes a `string_desc_t` to a function that expects a `char *` argument, or vice versa, or assigns a `string_desc_t` value to a variable of type `char *`, or vice versa, the compiler will report an error.

Unfortunately, `string_desc_t` and `rw_string_desc_t` being different types, there is no implicit conversion from `rw_string_desc_t` to `string_desc_t`. In places where such a conversion is desired, the (inline) function `sd_readonly` needs to be called.

Functions related to string descriptors are provided:

- Side-effect-free operations in `"string-desc.h"`,
- Memory-allocating operations in `"string-desc.h"`,
- Memory-allocating operations with out-of-memory checking in `"xstring-desc.h"`,
- Operations with side effects in `"string-desc.h"`.

For outputting a string descriptor, the `*printf` family of functions cannot be used directly. A format string directive such as `"%.*s"` would not work:

- it would stop the output at the first encountered NUL character,
- it would require to cast the number of bytes to `int`, and thus would not work for strings longer than `INT_MAX` bytes.

Therefore Gnulib offers

- a function `sd_fwrite` that outputs a string descriptor to a `FILE` stream,
- a function `sd_write` that outputs a string descriptor to a file descriptor,
- and for those applications where the NUL characters should become visible as `'\0'`, a family of `quotearg` based functions, that allow to specify the escaping rules in detail.

The functionality is thus split across three modules as follows:

- The module `string-desc`, under LGPL, defines the type and elementary functions.
- The module `xstring-desc`, under GPL, defines the memory-allocating functions with out-of-memory checking.
- The module `string-desc-quotearg`, under GPL, defines the `quotearg` based functions.

17.12 Container data types

Gnulib provides several generic container data types. They can be used to organize collections of application-defined objects.

17.12.1 Ordinary container data types

gl_list_node_value	$O(1)$	$O(1)$	$O(1)$	$O(1)$	$O(1)$	$O(1)$	$O(1)$	$O(1)$
gl_list_node_set_value	$O(1)$	$O(1)$	$O(1)$	$O(1)$	$O(1)$	$O(1)$	$O((\log n)\mathfrak{O}(1))$	
gl_list_next_node	$O(1)$	$O(1)$	$O(1)$	$O(\log n)$	$O(1)$	$O(1)$	$O(\log n)$	$O(\log n)$
gl_list_previous_node	$O(1)$	$O(1)$	$O(1)$	$O(\log n)$	$O(1)$	$O(1)$	$O(\log n)$	$O(\log n)$
gl_list_first_node	$O(1)$	$O(1)$	$O(1)$	$O(\log n)$	$O(1)$	$O(1)$	$O(\log n)$	$O(\log n)$
gl_list_last_node	$O(1)$	$O(1)$	$O(1)$	$O(\log n)$	$O(1)$	$O(1)$	$O(\log n)$	$O(\log n)$
gl_list_get_at	$O(1)$	$O(1)$	$O(n)$	$O(\log n)$	$O(n)$	$O(n)$	$O(\log n)$	$O(\log n)$
gl_list_get_first	$O(1)$	$O(1)$	$O(1)$	$O(\log n)$	$O(1)$	$O(1)$	$O(\log n)$	$O(\log n)$
gl_list_get_last	$O(1)$	$O(1)$	$O(1)$	$O(\log n)$	$O(1)$	$O(1)$	$O(\log n)$	$O(\log n)$
gl_list_set_at	$O(1)$	$O(1)$	$O(n)$	$O(\log n)$	$O(n)$	$O(n)$	$O((\log n)\mathfrak{O}(\log n))$	
gl_list_set_first	$O(1)$	$O(1)$	$O(1)$	$O(\log n)$	$O(n)$	$O(1)$	$O((\log n)\mathfrak{O}(\log n))$	
gl_list_set_last	$O(1)$	$O(1)$	$O(1)$	$O(\log n)$	$O(n)$	$O(1)$	$O((\log n)\mathfrak{O}(\log n))$	
gl_list_search	$O(n)$	$O(n)$	$O(n)$	$O(n)$	$O(n)$	$O(1)$	$O(\log n)$	$O(1)$
gl_list_search_from	$O(n)$	$O(n)$	$O(n)$	$O(n)$	$O(n)$	$O(1)$	$O((\log n)\mathfrak{O}(\log n))$	
gl_list_search_from_to	$O(n)$	$O(n)$	$O(n)$	$O(n)$	$O(n)$	$O(1)$	$O((\log n)\mathfrak{O}(\log n))$	
gl_list_indexof	$O(n)$	$O(n)$	$O(n)$	$O(n)$	$O(n)$	$O(n)$	$O(\log n)$	$O(\log n)$
gl_list_indexof_from	$O(n)$	$O(n)$	$O(n)$	$O(n)$	$O(n)$	$O(n)$	$O((\log n)\mathfrak{O}(\log n))$	
gl_list_indexof_from_to	$O(n)$	$O(n)$	$O(n)$	$O(n)$	$O(n)$	$O(n)$	$O((\log n)\mathfrak{O}(\log n))$	
gl_list_add_first	$O(n)$	$O(1)$	$O(1)$	$O(\log n)$	$O(1)$	$O(1)$	$O((\log n)\mathfrak{O}(\log n))$	
gl_list_add_last	$O(1)$	$O(1)$	$O(1)$	$O(\log n)$	$O(1)$	$O(1)$	$O((\log n)\mathfrak{O}(\log n))$	
gl_list_add_before	$O(n)$	$O(n)$	$O(1)$	$O(\log n)$	$O(1)$	$O(1)$	$O((\log n)\mathfrak{O}(\log n))$	
gl_list_add_after	$O(n)$	$O(n)$	$O(1)$	$O(\log n)$	$O(1)$	$O(1)$	$O((\log n)\mathfrak{O}(\log n))$	
gl_list_add_at	$O(n)$	$O(n)$	$O(n)$	$O(\log n)$	$O(n)$	$O(n)$	$O((\log n)\mathfrak{O}(\log n))$	
gl_list_remove_node	$O(n)$	$O(n)$	$O(1)$	$O(\log n)$	$O(n)$	$O(1)$	$O((\log n)\mathfrak{O}(\log n))$	
gl_list_remove_at	$O(n)$	$O(n)$	$O(n)$	$O(\log n)$	$O(n)$	$O(n)$	$O((\log n)\mathfrak{O}(\log n))$	

<code>gl_list_remove_first</code>	$O(n)$	$O(1)$	$O(1)$	$O(\log n)$	$O(n)$	$O(1)$	$O((\log n)\mathcal{O}(\log n))$
<code>gl_list_remove_last</code>	$O(1)$	$O(1)$	$O(1)$	$O(\log n)$	$O(n)$	$O(1)$	$O((\log n)\mathcal{O}(\log n))$
<code>gl_list_remove</code>	$O(n)$	$O(n)$	$O(n)$	$O(n)$	$O(n)$	$O(1)$	$O((\log n)\mathcal{O}(\log n))$
<code>gl_list_iterator</code>	$O(1)$	$O(1)$	$O(1)$	$O(\log n)$	$O(1)$	$O(1)$	$O(\log n) O(\log n)$
<code>gl_list_iterator_from_to</code>	$O(1)$	$O(1)$	$O(n)$	$O(\log n)$	$O(n)$	$O(n)$	$O(\log n) O(\log n)$
<code>gl_list_iterator_next</code>	$O(1)$	$O(1)$	$O(1)$	$O(\log n)$	$O(1)$	$O(1)$	$O(\log n) O(\log n)$
<code>gl_sortedlist_search</code>	$O(\log n)$	$O(\log n)$	$O(n)$	$O(\log n)$	$O(n)$	$O(n)$	$O(\log n) O(\log n)$
<code>gl_sortedlist_search_from</code>	$O(\log n)$	$O(\log n)$	$O(n)$	$O(\log n)$	$O(n)$	$O(n)$	$O(\log n) O(\log n)$
<code>gl_sortedlist_indexof</code>	$O(\log n)$	$O(\log n)$	$O(n)$	$O(\log n)$	$O(n)$	$O(n)$	$O(\log n) O(\log n)$
<code>gl_sortedlist_indexof_from</code>	$O(\log n)$	$O(\log n)$	$O(n)$	$O(\log n)$	$O(n)$	$O(n)$	$O(\log n) O(\log n)$
<code>gl_sortedlist_add</code>	$O(n)$	$O(n)$	$O(n)$	$O(\log n)$	$O(n)$	$O(n)$	$O((\log n)\mathcal{O}(\log n))$
<code>gl_sortedlist_remove</code>	$O(n)$	$O(n)$	$O(n)$	$O(\log n)$	$O(n)$	$O(n)$	$O((\log n)\mathcal{O}(\log n))$

The GnuLib modules for sequential lists are:

Implementation	Modules
Abstract	<code>list</code> , <code>xlist</code>
ARRAY	<code>array-list</code>
CARRAY	<code>carray-list</code>
LINKED	<code>linked-list</code>
TREE	<code>avltree-list</code> , <code>rbtree-list</code>
LINKEDHASH	<code>linkedhash-list</code>
TREEHASH	<code>avltreehash-list</code> , <code>rbtreehash-list</code>
Portion of a list	<code>sublist</code> , <code>xsublist</code>

17.12.1.2 Sets

The implementations and the guaranteed average performance for the operations for the “set” data type are:

Operation	ARRAY	LINKEDHASH, HASH
<code>gl_set_size</code>	$O(1)$	$O(1)$
<code>gl_set_add</code>	$O(n)$	$O(1)$
<code>gl_set_remove</code>	$O(n)$	$O(1)$
<code>gl_set_search</code>	$O(n)$	$O(1)$
<code>gl_set_iterator</code>	$O(1)$	$O(1)$
<code>gl_set_iterator_next</code>	$O(1)$	$O(1)$

The Gnulib modules for sets are:

Implementation	Modules
Abstract	set, xset
ARRAY	array-set
LINKEDHASH	linkedhash-set
HASH	hash-set

17.12.1.3 Ordered sets

The implementations and the guaranteed average performance for the operations for the “ordered set” data type are:

Operation	ARRAY	TREE
gl_oset_size	$O(1)$	$O(1)$
gl_oset_add	$O(n)$	$O(\log n)$
gl_oset_remove	$O(n)$	$O(\log n)$
gl_oset_search	$O(\log n)$	$O(\log n)$
gl_oset_search_atleast	$O(\log n)$	$O(\log n)$
gl_oset_iterator	$O(1)$	$O(\log n)$
gl_oset_iterator_next	$O(1)$	$O(\log n)$

The Gnulib modules for ordered sets are:

Implementation	Modules
Abstract	oset, xoset
ARRAY	array-oset
TREE	avltree-oset, rbtree-oset

17.12.1.4 Maps

The implementations and the guaranteed average performance for the operations for the “map” data type are:

Operation	ARRAY	LINKEDHASH, HASH
gl_map_size	$O(1)$	$O(1)$
gl_map_get	$O(n)$	$O(1)$
gl_map_put	$O(n)$	$O(1)$
gl_map_remove	$O(n)$	$O(1)$
gl_map_search	$O(n)$	$O(1)$
gl_map_iterator	$O(1)$	$O(1)$
gl_map_iterator_next	$O(1)$	$O(1)$

The Gnulib modules for maps are:

Implementation	Modules
Abstract	map, xmap
ARRAY	array-map
LINKEDHASH	linkedhash-map
HASH	hash-map

17.12.1.5 Ordered maps

The implementations and the guaranteed average performance for the operations for the “ordered map” data type are:

Operation	ARRAY	TREE
<code>gl_omap_size</code>	$O(1)$	$O(1)$
<code>gl_omap_get</code>	$O(\log n)$	$O(\log n)$
<code>gl_omap_put</code>	$O(n)$	$O(\log n)$
<code>gl_omap_remove</code>	$O(n)$	$O(\log n)$
<code>gl_omap_search</code>	$O(\log n)$	$O(\log n)$
<code>gl_omap_search_atleast</code>	$O(\log n)$	$O(\log n)$
<code>gl_omap_iterator</code>	$O(1)$	$O(\log n)$
<code>gl_omap_iterator_next</code>	$O(1)$	$O(\log n)$

The Gnulib modules for ordered maps are:

Implementation	Modules
Abstract	<code>omap</code> , <code>xomap</code>
ARRAY	<code>array-omap</code>
TREE	<code>avltree-omap</code> , <code>rbtree-omap</code>

17.12.1.6 C++ classes for container data types

For C++, Gnulib provides a C++ template class for each of these container data types.

Data type	C++ class	Module	Include file
Sequential list	<code>gl_List</code>	<code>list-c++</code>	<code>"gl_list.hh"</code>
Set	<code>gl_Set</code>	<code>set-c++</code>	<code>"gl_set.hh"</code>
Ordered set	<code>gl_OSet</code>	<code>oset-c++</code>	<code>"gl_oset.hh"</code>
Map	<code>gl_Map</code>	<code>map-c++</code>	<code>"gl_map.hh"</code>
Ordered map	<code>gl_OMap</code>	<code>omap-c++</code>	<code>"gl_omap.hh"</code>

17.12.2 Specialized container data types

The `hamt` module implements the hash array mapped trie (HAMT) data structure. This is a data structure that contains (key, value) pairs. Lookup of a (key, value) pair given the key is on average an $O(1)$ operation, assuming a good hash function for the keys is employed.

The HAMT data structure is useful when you want modifications (additions of pairs, removal, value changes) to be visible only to some part of your program, whereas other parts of the program continue to use the unmodified HAMT. The HAMT makes this possible in a space-efficient manner: the modified and the unmodified HAMT share most of their allocated memory. It is also time-efficient: Every such modification is $O(1)$ on average, again assuming a good hash function for the keys.

A HAMT can be used whenever an ordinary hash table would be used. It does however, provide non-destructive updating operations without the need to copy the whole container. On the other hand, a hash table is simpler so that its performance may be better when non-destructive update operations are not needed.

For example, a HAMT can be used to model the dynamic environment in a LISP interpreter. Updating a value in the dynamic environment of one continuation frame would not modify values in earlier frames.

To use the module, include `hamt.h` in your code. The public interface is documented in that header file. You have to provide a hash function and an equivalence relation, which

defines key equality. The module includes a test file `test-hamt.c`, which demonstrates how the API can be used.

In the current implementation, each inner node of the HAMT can store up to $32 = 2^5$ entries and subtries. Whenever a collision between the initial bits of the hash values of two entries would happen, the next 5 bits of the hash values are examined and the two entries pushed down one level in the trie.

HAMTs have the same average access times as hash tables but grow and shrink dynamically, so they use memory more economically and do not have to be periodically resized.

They were described and analyzed in *Phil Bagwell (2000). Ideal Hash Trees (Report). Infoscience Department, École Polytechnique Fédérale de Lausanne.*

The persistence aspect of the HAMT data structure, which means that each updating operation (like inserting, replacing, or removing an entry) returns a new HAMT while leaving the original one intact, is achieved through structure sharing, which is even safe in the presence of multiple threads when the used C compiler supports atomics.

17.13 Modernized printf

The `*zprintf` family of functions is a modernized form of the `*printf` family of functions.

The problem

The `*printf` functions have a return type ‘`int`’ and therefore can only produce results that are up to (2 GiB - 1 byte) long.

The problem with this is not so much that it is an arbitrary limitation (that persists even in processes that have, say, 50 GiB of RAM available). The bigger problem is that in reliable programs, it requires handling of an error code `EOverflow` that indicates a result whose size would be 2 GiB or larger.

How does a reliable program do error handling of `*printf` function calls? For output to strings and file descriptors (such as `sprintf` and `dprintf`), there is no other way than to check each such call.

For output to `FILE` streams (such as `fprintf`), beginners are tempted to ignore the return value of each call and instead check `ferror(stream)` at the end. The problem with this approach is that at the moment the error is detected, incorrect output has already been sent onto the stream. So, in this case as well, the reliable approach is to check each such call.

The *simple format strings* that most programs use in 99% of the places, namely with no wide string or wide character arguments, nor with widths passed as `int` argument, can only fail with two possible error codes:

- `ENOMEM`, when the result would be too large to allocate in the process’ memory.
- `EOverflow`, when the result is 2 GiB or larger but still allocatable.

Many GNU programs use “checking” wrappers (functions `xvasprintf`, etc.) that check for `ENOMEM` and call `xalloc_die`, thus aborting the program in that case. The problem is that `EOverflow` is not handled, even with such wrappers.

Should `EOverflow` be handled like `ENOMEM`, by aborting the program? No, as mentioned above, that would be an arbitrary limitation, which the [GNU Coding Standards](#) urge us to avoid.

The solution

The `*zprintf` functions are like the `*printf` functions, except that the return type is

- `ptrdiff_t` instead of `int`, for output to strings,
- `off64_t` (which is always equivalent to `int64_t`) instead of `int`, for output to file streams and file descriptors.

Thus, for these functions, `EOverflow` cannot occur (except for format strings which take widths as argument, which we have excluded above), and the “checking” wrappers (functions `xvasprintf`, etc.) are thus sufficient for ensuring an error-free result.

Note: In 64-bit processes, `ptrdiff_t` is 64 bits wide, i.e. equivalent to `int64_t`. In 32-bit processes, `ptrdiff_t` is only 32 bits wide, but since in these environments, memory regions of 2 GiB or larger cannot be allocated anyway (`malloc` would fail with `ENOMEM`), this type is sufficient.

The following GnuLib functions and modules exist:

Original function	Modernized function	Modules
<code>sprintf</code>	<code>szprintf</code>	<code>szprintf</code> , <code>szprintf-posix</code> , <code>szprintf-gnu</code>
<code>vsprintf</code>	<code>vszprintf</code>	<code>vszprintf</code> , <code>vszprintf-posix</code> , <code>vszprintf-gnu</code>
<code>snprintf</code>	<code>snzprintf</code>	<code>snzprintf</code> , <code>snzprintf-posix</code> , <code>snzprintf-gnu</code>
<code>vsnprintf</code>	<code>vsnzprintf</code>	<code>vsnzprintf</code> , <code>vsnzprintf-posix</code> , <code>vsnzprintf-gnu</code>
<code>asprintf</code>	<code>aszprintf</code>	<code>vaszprintf</code> , <code>vaszprintf-posix</code> , <code>vaszprintf-gnu</code>
<code>vasprintf</code>	<code>vaszprintf</code>	<code>vaszprintf</code> , <code>vaszprintf-posix</code> , <code>vaszprintf-gnu</code>
<code>c_snprintf</code>	<code>c_snzprintf</code>	<code>c-snzprintf</code> , <code>c-snzprintf-gnu</code>
<code>c_vsnprintf</code>	<code>c_vsnzprintf</code>	<code>c-vsnzprintf</code> , <code>c-vsnzprintf-gnu</code>
<code>c_asprintf</code>	<code>c_aszprintf</code>	<code>c-vaszprintf</code> , <code>c-vaszprintf-gnu</code>
<code>c_vasprintf</code>	<code>c_vaszprintf</code>	<code>c-vaszprintf</code> , <code>c-vaszprintf-gnu</code>
<code>fprintf</code>	<code>fzprintf</code>	<code>fzprintf</code> , <code>fzprintf-posix</code> , <code>fzprintf-gnu</code>
<code>vfprintf</code>	<code>vfzprintf</code>	<code>vfzprintf</code> , <code>vfzprintf-posix</code> , <code>vfzprintf-gnu</code>
<code>printf</code>	<code>zprintf</code>	<code>zprintf</code> , <code>zprintf-posix</code> , <code>zprintf-gnu</code>
<code>vprintf</code>	<code>vzprintf</code>	<code>vzprintf</code> , <code>vzprintf-posix</code> , <code>vzprintf-gnu</code>
<code>dprintf</code>	<code>dzprintf</code>	<code>dzprintf</code> , <code>dzprintf-posix</code> , <code>dzprintf-gnu</code>

<code>vdprintf</code>	<code>vdzprintf</code>	<code>vdzprintf,</code> <code>vdzprintf-gnu</code>	<code>vdzprintf-posix,</code>
<code>obstack_printf</code>	<code>obstack_zprintf</code>	<code>obstack-zprintf,</code> <code>posix, obstack-zprintf-gnu</code>	<code>obstack-zprintf-</code>
<code>obstack_vprintf</code>	<code>obstack_vzprintf</code>	<code>obstack-zprintf,</code> <code>posix, obstack-zprintf-gnu</code>	<code>obstack-zprintf-</code>

The following functions use the `*zprintf` functions under the hood and thus don't need a `*zprintf` variant:

Function	Modules
<code>xasprintf</code>	<code>xvasprintf,</code> <code>xvasprintf-gnu</code> <code>xvasprintf-posix,</code>
<code>xvasprintf</code>	<code>xvasprintf,</code> <code>xvasprintf-gnu</code>
<code>c_xasprintf</code>	<code>c-xvasprintf</code>
<code>c_xvasprintf</code>	<code>c-xvasprintf</code>
<code>xprintf</code>	<code>xprintf, xprintf-posix, xprintf-gnu</code>
<code>xvprintf</code>	<code>xprintf, xprintf-posix, xprintf-gnu</code>
<code>xfprintf</code>	<code>xprintf, xprintf-posix, xprintf-gnu</code>
<code>xvfprintf</code>	<code>xprintf, xprintf-posix, xprintf-gnu</code>

Note: Even with the `*zprintf` functions, you need to be prepared to handle specific error codes when you use non-simple format strings:

- `EILSEQ` when the format string takes wide strings or wide characters as arguments,
- `EOVERFLOW` when the format string takes a width as argument and you cannot ensure that its value is in the range `0...INT_MAX`.

17.14 Advanced functions on FILE streams

Gnulib provides a few functions that operate on `FILE *` streams and that go beyond what ISO C and POSIX specify.

Each such function is provided in a Gnulib module of the same name.

When Gnulib implemented them in 2007, some people considered them “unportable”, because they required digging in undocumented internals of the `FILE` structure. But Gnulib made them portable, by providing implementations for all platforms, from `glibc` to native Windows, and from Minix to QNX. Nowadays, new platforms already implement them in their `libc`; this has been seen for the Android `libc` and `musl libc`.

The stream argument of these functions must not be wide-character oriented. But this is not an actual restriction, since no programs use wide-character orientation on `FILE` streams anyway: no one uses the `fwide` function.

17.14.1 Determining the state of a FILE stream

Gnulib modules: `freadable`, `fwritable`, `freading`, `fwriting`

The `freadable` function returns `true` if the given stream supports reading, `false` if it supports only writing, i.e. if it was opened write-only or append-only.

The `fwritable` function returns `true` if the given stream supports writing, `false` if it supports only reading.

The `freading` function returns `true` if the given stream is opened read-only, or if the last operation on the stream was a read operation.

The `fwriting` function returns `true` if the given stream is opened write-only or append-only, or if the last operation on the stream was a write operation.

`freading` and `fwriting` will never both be true on the same stream at the same time. If the stream supports both reads and writes, then

- both `freading` and `fwriting` might be false when the stream is first opened, after read encounters EOF, or after `fflush`,
- `freading` might be false or true and `fwriting` might be false after repositioning (such as `fseek`, `fsetpos`, or `rewind`),

depending on the underlying implementation.

17.14.2 Setting an error in a FILE stream

A `FILE *` stream has an error indicator, that can be tested through the `ferror` function and cleared through the `clearerr` function. The error indicator is set when an I/O operation on the stream fails. But ISO C and POSIX don't provide a way to set the error indicator!

The `fseterr` function sets the error indicator in the given stream.

17.14.3 Dropping the contents of the buffers of a FILE stream

The `fpurge` function drops the contents of the input and output buffers of the given `FILE *` stream:

- It discards the contents of the output buffer, if the stream is currently writing. That is the opposite of what the `fflush` function does.
- It discards the contents of the input buffer, if the stream is currently reading. That is the same as what the `fflush` function does.

17.14.4 Accessing the write buffer of a FILE stream

The following two functions provide information about the output buffer of a `FILE *` stream.

The `fpending` function returns the number of pending (a.k.a. buffered, unflushed) bytes in the given stream.

The `fbufmode` function returns the buffering mode of the given stream:

Result	means
<code>_IONBF</code>	unbuffered
<code>_IOLBF</code>	line buffered
<code>_IOFBF</code>	fully buffered

Note that on some platforms, it is impossible to distinguish `_IOFBF` and `_IOLBF`.

17.14.5 Accessing the read buffer of a FILE stream

The following functions allow code to scan the contents of the input buffer of a `FILE *` stream, without first copying it into a separate memory area via `fread`.

The function `freadahead` returns the number of bytes waiting in the input buffer of the given stream. This includes both the bytes that have been read from the underlying input source and the bytes that have been pushed back through `ungetc`.

A function call `freadptr (stream, &size)` returns a pointer to the input buffer of the given `stream`, or NULL. If the returned pointer is non-NULL, `size` is set to the (positive) size of the input buffer. If the returned pointer is NULL, you should use `getc (stream)`, `fgetc (stream)`, or `fread (... , stream)` to access the input from `stream`.

A function call `freadseek (stream, offset)` is the same as `fseek (stream, offset, SEEK_CUR)`, except that the latter does not work on non-seekable input streams (such as pipes).

17.15 libunistring

GNU libunistring provides functions for manipulating Unicode strings and for manipulating C strings according to the Unicode standard.

It consists of the following parts:

<code><unistr.h></code>	elementary string functions
<code><uniconv.h></code>	conversion from/to legacy encodings
<code><unistdio.h></code>	formatted output to strings
<code><uniname.h></code>	character names
<code><unictype.h></code>	character classification and properties
<code><uniwidth.h></code>	string width when using nonproportional fonts
<code><unigbrk.h></code>	grapheme cluster breaks
<code><uniwbrk.h></code>	word breaks
<code><unilbrk.h></code>	line breaking algorithm
<code><uninorm.h></code>	normalization (composition and decomposition)
<code><unicase.h></code>	case folding

Nearly all of the source code of GNU libunistring resides in Gnulib.

17.15.1 How to use libunistring

There are three ways to use GNU libunistring:

- You can declare GNU libunistring as a prerequisite of your package and then, at build time, link with the installed libunistring. To achieve this, use the Gnulib module `libunistring`.
- You can include the source code of specific libunistring modules in your package, through the usual Gnulib mechanism. Then, there is no prerequisite.
- A combination of both: You can declare GNU libunistring as an optional prerequisite of your package and include the source code of specific libunistring modules in your package. At configure time, when the user has specified the `configure` option `--with-included-libunistring`, the included modules will be used and the package will not link against an installed libunistring. Otherwise, the package will link against an installed libunistring if found, and only those modules will be built that are not contained or that are too old in the found libunistring. To achieve this, request from Gnulib the specific modules and the Gnulib module `libunistring-optional`.

17.15.2 The libunistring modules

Every function or variable that is part of the GNU libunistring API is available through a specific Gnulib module. For the documentation, refer to the [GNU libunistring documentation](#).

17.15.2.1 Declarations in <unitypes.h>

The <unitypes.h> header file is provided by the module `unitypes-h`.

17.15.2.2 Declarations in <unistr.h>

Function	Module
<code>u8_check</code>	<code>unistr/u8-check</code>
<code>u16_check</code>	<code>unistr/u16-check</code>
<code>u32_check</code>	<code>unistr/u32-check</code>
<code>u8_to_u16</code>	<code>unistr/u8-to-u16</code>
<code>u8_to_u32</code>	<code>unistr/u8-to-u32</code>
<code>u16_to_u8</code>	<code>unistr/u16-to-u8</code>
<code>u16_to_u32</code>	<code>unistr/u16-to-u32</code>
<code>u32_to_u8</code>	<code>unistr/u32-to-u8</code>
<code>u32_to_u16</code>	<code>unistr/u32-to-u16</code>
<code>u8_mblen</code>	<code>unistr/u8-mblen</code>
<code>u16_mblen</code>	<code>unistr/u16-mblen</code>
<code>u32_mblen</code>	<code>unistr/u32-mblen</code>
<code>u8_mbtouc</code>	<code>unistr/u8-mbtouc</code>
<code>u16_mbtouc</code>	<code>unistr/u16-mbtouc</code>
<code>u32_mbtouc</code>	<code>unistr/u32-mbtouc</code>
<code>u8_mbtouc_unsafe</code>	<code>unistr/u8-mbtouc-unsafe</code>
<code>u16_mbtouc_unsafe</code>	<code>unistr/u16-mbtouc-unsafe</code>
<code>u32_mbtouc_unsafe</code>	<code>unistr/u32-mbtouc-unsafe</code>
<code>u8_mbtoucr</code>	<code>unistr/u8-mbtoucr</code>
<code>u16_mbtoucr</code>	<code>unistr/u16-mbtoucr</code>
<code>u32_mbtoucr</code>	<code>unistr/u32-mbtoucr</code>
<code>u8_uctomb</code>	<code>unistr/u8-uctomb</code>
<code>u16_uctomb</code>	<code>unistr/u16-uctomb</code>
<code>u32_uctomb</code>	<code>unistr/u32-uctomb</code>
<code>u8_cpy</code>	<code>unistr/u8-cpy</code>
<code>u16_cpy</code>	<code>unistr/u16-cpy</code>
<code>u32_cpy</code>	<code>unistr/u32-cpy</code>
<code>u8_pcpy</code>	<code>unistr/u8-pcpy</code>
<code>u16_pcpy</code>	<code>unistr/u16-pcpy</code>
<code>u32_pcpy</code>	<code>unistr/u32-pcpy</code>
<code>u8_move</code>	<code>unistr/u8-move</code>
<code>u16_move</code>	<code>unistr/u16-move</code>
<code>u32_move</code>	<code>unistr/u32-move</code>
<code>u8_set</code>	<code>unistr/u8-set</code>
<code>u16_set</code>	<code>unistr/u16-set</code>
<code>u32_set</code>	<code>unistr/u32-set</code>

u8_cmp	unistr/u8-cmp
u16_cmp	unistr/u16-cmp
u32_cmp	unistr/u32-cmp
u8_cmp2	unistr/u8-cmp2
u16_cmp2	unistr/u16-cmp2
u32_cmp2	unistr/u32-cmp2
u8_chr	unistr/u8-chr
u16_chr	unistr/u16-chr
u32_chr	unistr/u32-chr
u8_mbsnlen	unistr/u8-mbsnlen
u16_mbsnlen	unistr/u16-mbsnlen
u32_mbsnlen	unistr/u32-mbsnlen
u8_cpy_alloc	unistr/u8-cpy-alloc
u16_cpy_alloc	unistr/u16-cpy-alloc
u32_cpy_alloc	unistr/u32-cpy-alloc
u8_strmblen	unistr/u8-strmblen
u16_strmblen	unistr/u16-strmblen
u32_strmblen	unistr/u32-strmblen
u8_strmbtouc	unistr/u8-strmbtouc
u16_strmbtouc	unistr/u16-strmbtouc
u32_strmbtouc	unistr/u32-strmbtouc
u8_next	unistr/u8-next
u16_next	unistr/u16-next
u32_next	unistr/u32-next
u8_prev	unistr/u8-prev
u16_prev	unistr/u16-prev
u32_prev	unistr/u32-prev
u8_strlen	unistr/u8-strlen
u16_strlen	unistr/u16-strlen
u32_strlen	unistr/u32-strlen
u8_strnlen	unistr/u8-strnlen
u16_strnlen	unistr/u16-strnlen
u32_strnlen	unistr/u32-strnlen
u8_strcpy	unistr/u8-strcpy
u16_strcpy	unistr/u16-strcpy
u32_strcpy	unistr/u32-strcpy
u8_stpcpy	unistr/u8-stpcpy
u16_stpcpy	unistr/u16-stpcpy
u32_stpcpy	unistr/u32-stpcpy
u8_strncpy	unistr/u8-strncpy
u16_strncpy	unistr/u16-strncpy
u32_strncpy	unistr/u32-strncpy
u8_stpncpy	unistr/u8-stpncpy
u16_stpncpy	unistr/u16-stpncpy
u32_stpncpy	unistr/u32-stpncpy
u8_strcat	unistr/u8-strcat
u16_strcat	unistr/u16-strcat

u32_strcat	unistr/u32-strcat
u8_strncat	unistr/u8-strncat
u16_strncat	unistr/u16-strncat
u32_strncat	unistr/u32-strncat
u8_strcmp	unistr/u8-strcmp
u16_strcmp	unistr/u16-strcmp
u32_strcmp	unistr/u32-strcmp
u8_strcoll	unistr/u8-strcoll
u16_strcoll	unistr/u16-strcoll
u32_strcoll	unistr/u32-strcoll
u8_strncmp	unistr/u8-strncmp
u16_strncmp	unistr/u16-strncmp
u32_strncmp	unistr/u32-strncmp
u8_strdup	unistr/u8-strdup
u16_strdup	unistr/u16-strdup
u32_strdup	unistr/u32-strdup
u8_strchr	unistr/u8-strchr
u16_strchr	unistr/u16-strchr
u32_strchr	unistr/u32-strchr
u8_strrchr	unistr/u8-strrchr
u16_strrchr	unistr/u16-strrchr
u32_strrchr	unistr/u32-strrchr
u8_strerror	unistr/u8-strcspn
u16_strerror	unistr/u16-strcspn
u32_strerror	unistr/u32-strcspn
u8_strerror	unistr/u8-strspn
u16_strerror	unistr/u16-strspn
u32_strerror	unistr/u32-strspn
u8_strpbrk	unistr/u8-strpbrk
u16_strpbrk	unistr/u16-strpbrk
u32_strpbrk	unistr/u32-strpbrk
u8_strstr	unistr/u8-strstr
u16_strstr	unistr/u16-strstr
u32_strstr	unistr/u32-strstr
u8_startswith	unistr/u8-startswith
u16_startswith	unistr/u16-startswith
u32_startswith	unistr/u32-startswith
u8_endswith	unistr/u8-endswith
u16_endswith	unistr/u16-endswith
u32_endswith	unistr/u32-endswith
u8_strtok	unistr/u8-strtok
u16_strtok	unistr/u16-strtok
u32_strtok	unistr/u32-strtok

17.15.2.3 Declarations in <uniconv.h>

Function	Module
locale_charset	localcharset
u8_conv_from_encoding	uniconv/u8-conv-from-enc
u16_conv_from_encoding	uniconv/u16-conv-from-enc
u32_conv_from_encoding	uniconv/u32-conv-from-enc
u8_conv_to_encoding	uniconv/u8-conv-to-enc
u16_conv_to_encoding	uniconv/u16-conv-to-enc
u32_conv_to_encoding	uniconv/u32-conv-to-enc
u8_strconv_from_encoding	uniconv/u8-strconv-from-enc
u16_strconv_from_encoding	uniconv/u16-strconv-from-enc
u32_strconv_from_encoding	uniconv/u32-strconv-from-enc
u8_strconv_to_encoding	uniconv/u8-strconv-to-enc
u16_strconv_to_encoding	uniconv/u16-strconv-to-enc
u32_strconv_to_encoding	uniconv/u32-strconv-to-enc
u8_strconv_from_locale	uniconv/u8-strconv-from-locale
u16_strconv_from_locale	uniconv/u16-strconv-from-locale
u32_strconv_from_locale	uniconv/u32-strconv-from-locale
u8_strconv_to_locale	uniconv/u8-strconv-to-locale
u16_strconv_to_locale	uniconv/u16-strconv-to-locale
u32_strconv_to_locale	uniconv/u32-strconv-to-locale

17.15.2.4 Declarations in <unistdio.h>

Function	Module
ulc_sprintf	unistdio/ulc-sprintf
ulc_snprintf	unistdio/ulc-snprintf
ulc_asprintf	unistdio/ulc-asprintf
ulc_asnprintf	unistdio/ulc-asnprintf
ulc_vsprintf	unistdio/ulc-vsprintf
ulc_vsnprintf	unistdio/ulc-vsnprintf
ulc_vasprintf	unistdio/ulc-vasprintf
ulc_vasnprintf	unistdio/ulc-vasnprintf
u8_sprintf	unistdio/u8-sprintf
u8_snprintf	unistdio/u8-snprintf
u8_asprintf	unistdio/u8-asprintf
u8_asnprintf	unistdio/u8-asnprintf
u8_vsprintf	unistdio/u8-vsprintf
u8_vsnprintf	unistdio/u8-vsnprintf
u8_vasprintf	unistdio/u8-vasprintf
u8_vasnprintf	unistdio/u8-vasnprintf
u8_u8_sprintf	unistdio/u8-u8-sprintf
u8_u8_snprintf	unistdio/u8-u8-snprintf
u8_u8_asprintf	unistdio/u8-u8-asprintf
u8_u8_asnprintf	unistdio/u8-u8-asnprintf
u8_u8_vsprintf	unistdio/u8-u8-vsprintf
u8_u8_vsnprintf	unistdio/u8-u8-vsnprintf

u8_u8_vasprintf	unistdio/u8-u8-vasprintf
u8_u8_vasnprintf	unistdio/u8-u8-vasnprintf
u16_sprintf	unistdio/u16-sprintf
u16_snprintf	unistdio/u16-snprintf
u16_asprintf	unistdio/u16-asprintf
u16_asnprintf	unistdio/u16-asnprintf
u16_vsprintf	unistdio/u16-vsprintf
u16_vsnprintf	unistdio/u16-vsnprintf
u16_vasprintf	unistdio/u16-vasprintf
u16_vasnprintf	unistdio/u16-vasnprintf
u16_u16_sprintf	unistdio/u16-u16-sprintf
u16_u16_snprintf	unistdio/u16-u16-snprintf
u16_u16_asprintf	unistdio/u16-u16-asprintf
u16_u16_asnprintf	unistdio/u16-u16-asnprintf
u16_u16_vsprintf	unistdio/u16-u16-vsprintf
u16_u16_vsnprintf	unistdio/u16-u16-vsnprintf
u16_u16_vasprintf	unistdio/u16-u16-vasprintf
u16_u16_vasnprintf	unistdio/u16-u16-vasnprintf
u32_sprintf	unistdio/u32-sprintf
u32_snprintf	unistdio/u32-snprintf
u32_asprintf	unistdio/u32-asprintf
u32_asnprintf	unistdio/u32-asnprintf
u32_vsprintf	unistdio/u32-vsprintf
u32_vsnprintf	unistdio/u32-vsnprintf
u32_vasprintf	unistdio/u32-vasprintf
u32_vasnprintf	unistdio/u32-vasnprintf
u32_u32_sprintf	unistdio/u32-u32-sprintf
u32_u32_snprintf	unistdio/u32-u32-snprintf
u32_u32_asprintf	unistdio/u32-u32-asprintf
u32_u32_asnprintf	unistdio/u32-u32-asnprintf
u32_u32_vsprintf	unistdio/u32-u32-vsprintf
u32_u32_vsnprintf	unistdio/u32-u32-vsnprintf
u32_u32_vasprintf	unistdio/u32-u32-vasprintf
u32_u32_vasnprintf	unistdio/u32-u32-vasnprintf
ulc_fprintf	unistdio/ulc-fprintf
ulc_vfprintf	unistdio/ulc-vfprintf

17.15.2.5 Declarations in <unistd.h>

Function	Module
unicode_character_name	unistdio/unistdio
unicode_name_character	unistdio/unistdio

17.15.2.6 Declarations in <unistd.h>

Function or constant	Module
UC_CATEGORY_L	unistdio/category-L
UC_CATEGORY_LC	unistdio/category-LC

UC_CATEGORY_Lu	unictype/category-Lu
UC_CATEGORY_Ll	unictype/category-Ll
UC_CATEGORY_Lt	unictype/category-Lt
UC_CATEGORY_Lm	unictype/category-Lm
UC_CATEGORY_Lo	unictype/category-Lo
UC_CATEGORY_M	unictype/category-M
UC_CATEGORY_Mn	unictype/category-Mn
UC_CATEGORY_Mc	unictype/category-Mc
UC_CATEGORY_Me	unictype/category-Me
UC_CATEGORY_N	unictype/category-N
UC_CATEGORY_Nd	unictype/category-Nd
UC_CATEGORY_Nl	unictype/category-Nl
UC_CATEGORY_No	unictype/category-No
UC_CATEGORY_P	unictype/category-P
UC_CATEGORY_Pc	unictype/category-Pc
UC_CATEGORY_Pd	unictype/category-Pd
UC_CATEGORY_Ps	unictype/category-Ps
UC_CATEGORY_Pe	unictype/category-Pe
UC_CATEGORY_Pi	unictype/category-Pi
UC_CATEGORY_Pf	unictype/category-Pf
UC_CATEGORY_Po	unictype/category-Po
UC_CATEGORY_S	unictype/category-S
UC_CATEGORY_Sm	unictype/category-Sm
UC_CATEGORY_Sc	unictype/category-Sc
UC_CATEGORY_Sk	unictype/category-Sk
UC_CATEGORY_So	unictype/category-So
UC_CATEGORY_Z	unictype/category-Z
UC_CATEGORY_Zs	unictype/category-Zs
UC_CATEGORY_Zl	unictype/category-Zl
UC_CATEGORY_Zp	unictype/category-Zp
UC_CATEGORY_C	unictype/category-C
UC_CATEGORY_Cc	unictype/category-Cc
UC_CATEGORY_Cf	unictype/category-Cf
UC_CATEGORY-Cs	unictype/category-Cs
UC_CATEGORY-Co	unictype/category-Co
UC_CATEGORY_Cn	unictype/category-Cn
uc_general_category_or	unictype/category-or
uc_general_category_and	unictype/category-and
uc_general_category_and_not	unictype/category-and-not
uc_general_category_name	unictype/category-name
uc_general_category_long_name	unictype/category-longname
uc_general_category_byname	unictype/category-byname
uc_general_category	unictype/category-of
uc_is_general_category	unictype/category-test
uc_is_general_category_withtable	unictype/category-test-withtable
uc_combining_class_name	unictype/combining-class-name
uc_combining_class_long_name	unictype/combining-class-longname

uc_combining_class_byname	unictype/combining-class-byname
uc_combining_class	unictype/combining-class
uc_bidi_class_name	unictype/bidiclass-name
uc_bidi_category_name	unictype/bidiclass-name
uc_bidi_class_long_name	unictype/bidiclass-longname
uc_bidi_class_byname	unictype/bidiclass-byname
uc_bidi_category_byname	unictype/bidiclass-byname
uc_bidi_class	unictype/bidiclass-of
uc_bidi_category	unictype/bidiclass-of
uc_is_bidi_class	unictype/bidiclass-test
uc_is_bidi_category	unictype/bidiclass-test
uc_decimal_value	unictype/decimal-digit
uc_digit_value	unictype/digit
uc_numeric_value	unictype/numeric
uc_mirror_char	unictype/mirror
uc_joining_type_name	unictype/joiningtype-name
uc_joining_type_long_name	unictype/joiningtype-longname
uc_joining_type_byname	unictype/joiningtype-byname
uc_joining_type	unictype/joiningtype-of
uc_joining_group_name	unictype/joininggroup-name
uc_joining_group_byname	unictype/joininggroup-byname
uc_joining_group	unictype/joininggroup-of
UC_PROPERTY_WHITE_SPACE	unictype/property-white-space
UC_PROPERTY_ALPHABETIC	unictype/property-alphabetic
UC_PROPERTY_OTHER_ALPHABETIC	unictype/property-other-alphabetic
UC_PROPERTY_NOT_A_CHARACTER	unictype/property-not-a-character
UC_PROPERTY_DEFAULT_IGNOREABLE_CODE_POINT	unictype/property-default-ignorable-code-point
UC_PROPERTY_OTHER_DEFAULT_IGNOREABLE_CODE_POINT	unictype/property-other-default-ignorable-code-point
UC_PROPERTY_DEPRECATED	unictype/property-deprecated
UC_PROPERTY_LOGICAL_ORDER_EXCEPTION	unictype/property-logical-order-exception
UC_PROPERTY_VARIATION_SELECTOR	unictype/property-variation-selector
UC_PROPERTY_PRIVATE_USE	unictype/property-private-use
UC_PROPERTY_UNASSIGNED_CODE_VALUE	unictype/property-unassigned-code-value
UC_PROPERTY_UPPERCASE	unictype/property-uppercase
UC_PROPERTY_OTHER_UPPERCASE	unictype/property-other-uppercase
UC_PROPERTY_LOWERCASE	unictype/property-lowercase
UC_PROPERTY_OTHER_LOWERCASE	unictype/property-other-lowercase
UC_PROPERTY_TITLECASE	unictype/property-titlecase
UC_PROPERTY_CASED	unictype/property-cased
UC_PROPERTY_CASE_IGNOREABLE	unictype/property-case-ignorable

UC_PROPERTY_CHANGES_WHEN_LOWERCASED	unictype/property-changes-when-lowercased
UC_PROPERTY_CHANGES_WHEN_UPPERCASED	unictype/property-changes-when-uppercased
UC_PROPERTY_CHANGES_WHEN_TITLECASED	unictype/property-changes-when-titlecased
UC_PROPERTY_CHANGES_WHEN_CASEFOLDED	unictype/property-changes-when-casefolded
UC_PROPERTY_CHANGES_WHEN_CASEMAPPED	unictype/property-changes-when-casemapped
UC_PROPERTY_SOFT_DOTTED	unictype/property-soft-dotted
UC_PROPERTY_ID_START	unictype/property-id-start
UC_PROPERTY_OTHER_ID_START	unictype/property-other-id-start
UC_PROPERTY_ID_CONTINUE	unictype/property-id-continue
UC_PROPERTY_OTHER_ID_CONTINUE	unictype/property-other-id-continue
UC_PROPERTY_XID_START	unictype/property-xid-start
UC_PROPERTY_XID_CONTINUE	unictype/property-xid-continue
UC_PROPERTY_ID_COMPAT_MATH_START	unictype/property-id-compatible-math-start
UC_PROPERTY_ID_COMPAT_MATH_CONTINUE	unictype/property-id-compatible-math-continue
UC_PROPERTY_PATTERN_WHITE_SPACE	unictype/property-pattern-white-space
UC_PROPERTY_PATTERN_SYNTAX	unictype/property-pattern-syntax
UC_PROPERTY_JOIN_CONTROL	unictype/property-join-control
UC_PROPERTY_GRAPHEME_BASE	unictype/property-grapheme-base
UC_PROPERTY_GRAPHEME_EXTEND	unictype/property-grapheme-extend
UC_PROPERTY_OTHER_GRAPHEME_EXTEND	unictype/property-other-grapheme-extend
UC_PROPERTY_GRAPHEME_LINK	unictype/property-grapheme-link
UC_PROPERTY_MODIFIER_COMBINING_MARK	unictype/property-modifier-combining-mark
UC_PROPERTY_BIDI_CONTROL	unictype/property-bidi-control
UC_PROPERTY_BIDI_LEFT_TO_RIGHT	unictype/property-bidi-left-to-right
UC_PROPERTY_BIDI_HEBREW_RIGHT_TO_LEFT	unictype/property-bidi-hebrew-right-to-left
UC_PROPERTY_BIDI_ARABIC_RIGHT_TO_LEFT	unictype/property-bidi-arabic-right-to-left
UC_PROPERTY_BIDI_EUROPEAN_DIGIT	unictype/property-bidi-european-digit
UC_PROPERTY_BIDI_EUR_NUM_SEPARATOR	unictype/property-bidi-eur-num-separator
UC_PROPERTY_BIDI_EUR_NUM_TERMINATOR	unictype/property-bidi-eur-num-terminator

UC_PROPERTY_BIDI_ARABIC_DIGIT	unictype/property-bidi-arabic-digit
UC_PROPERTY_BIDI_COMMON_SEPARATOR	unictype/property-bidi-common-separator
UC_PROPERTY_BIDI_BLOCK_SEPARATOR	unictype/property-bidi-block-separator
UC_PROPERTY_BIDI_SEGMENT_SEPARATOR	unictype/property-bidi-segment-separator
UC_PROPERTY_BIDI_WHITESPACE	unictype/property-bidi-whitespace
UC_PROPERTY_BIDI_NON_SPACING_MARK	unictype/property-bidi-non-spacing-mark
UC_PROPERTY_BIDI_BOUNDARY_NEUTRAL	unictype/property-bidi-boundary-neutral
UC_PROPERTY_BIDI_PDF	unictype/property-bidi-pdf
UC_PROPERTY_BIDI_EMBEDDING_OR_OVERRIDE	unictype/property-bidi-embedding-or-override
UC_PROPERTY_BIDI_OTHER_NEUTRAL	unictype/property-bidi-other-neutral
UC_PROPERTY_HEX_DIGIT	unictype/property-hex-digit
UC_PROPERTY_ASCII_HEX_DIGIT	unictype/property-ascii-hex-digit
UC_PROPERTY_IDEOGRAPHIC	unictype/property-ideographic
UC_PROPERTY_UNIFIED_IDEOGRAPH	unictype/property-unified-ideograph
UC_PROPERTY_RADICAL	unictype/property-radical
UC_PROPERTY_IDS_UNARY_OPERATOR	unictype/property-ids-unary-operator
UC_PROPERTY_IDS_BINARY_OPERATOR	unictype/property-ids-binary-operator
UC_PROPERTY_IDS_TRINARY_OPERATOR	unictype/property-ids-trinary-operator
UC_PROPERTY_EMOJI	unictype/property-emoji
UC_PROPERTY_EMOJI_PRESENTATION	unictype/property-emoji-presentation
UC_PROPERTY_EMOJI_MODIFIER	unictype/property-emoji-modifier
UC_PROPERTY_EMOJI_MODIFIER_BASE	unictype/property-emoji-modifier-base
UC_PROPERTY_EMOJI_COMPONENT	unictype/property-emoji-component
UC_PROPERTY_EXTENDED_PICTOGRAPHIC	unictype/property-extended-pictographic
UC_PROPERTY_ZERO_WIDTH	unictype/property-zero-width
UC_PROPERTY_SPACE	unictype/property-space
UC_PROPERTY_NON_BREAK	unictype/property-non-break
UC_PROPERTY_ISO_CONTROL	unictype/property-iso-control
UC_PROPERTY_FORMAT_CONTROL	unictype/property-format-control
UC_PROPERTY_PREPENDED_CONCATENATION_MARK	unictype/property-prepended-concatenation-mark
UC_PROPERTY_DASH	unictype/property-dash

UC_PROPERTY_HYPHEN	unictype/property-hyphen
UC_PROPERTY_PUNCTUATION	unictype/property-punctuation
UC_PROPERTY_LINE_SEPARATOR	unictype/property-line-separator
UC_PROPERTY_PARAGRAPH_SEPARATOR	unictype/property-paragraph-separator
UC_PROPERTY_QUOTATION_MARK	unictype/property-quotation-mark
UC_PROPERTY_SENTENCE_TERMINAL	unictype/property-sentence-terminal
UC_PROPERTY_TERMINAL_PUNCTUATION	unictype/property-terminal-punctuation
UC_PROPERTY_CURRENCY_SYMBOL	unictype/property-currency-symbol
UC_PROPERTY_MATH	unictype/property-math
UC_PROPERTY_OTHER_MATH	unictype/property-other-math
UC_PROPERTY_PAISED_PUNCTUATION	unictype/property-paired-punctuation
UC_PROPERTY_LEFT_OF_PAIR	unictype/property-left-of-pair
UC_PROPERTY_COMBINING	unictype/property-combining
UC_PROPERTY_COMPOSITE	unictype/property-composite
UC_PROPERTY_DECIMAL_DIGIT	unictype/property-decimal-digit
UC_PROPERTY_NUMERIC	unictype/property-numeric
UC_PROPERTY_DIACRITIC	unictype/property-diacritic
UC_PROPERTY_EXTENDER	unictype/property-extender
UC_PROPERTY_IGNOREABLE_CONTROL	unictype/property-ignorable-control
UC_PROPERTY_REGIONAL_INDICATOR	unictype/property-regional-indicator
uc_property_byname	unictype/property-byname
uc_property_is_valid	unictype/base
uc_is_property	unictype/property-test
uc_is_property_white_space	unictype/property-white-space
uc_is_property_alphabetic	unictype/property-alphabetic
uc_is_property_other_alphabetic	unictype/property-other-alphabetic
uc_is_property_not_a_character	unictype/property-not-a-character
uc_is_property_default_ignorable_code_point	unictype/property-default-ignorable-code-point
uc_is_property_other_default_ignorable_code_point	unictype/property-other-default-ignorable-code-point
uc_is_property_deprecated	unictype/property-deprecated
uc_is_property_logical_order_exception	unictype/property-logical-order-exception
uc_is_property_variation_selector	unictype/property-variation-selector
uc_is_property_private_use	unictype/property-private-use
uc_is_property_unassigned_code_value	unictype/property-unassigned-code-value
uc_is_property_uppercase	unictype/property-uppercase
uc_is_property_other_uppercase	unictype/property-other-uppercase
uc_is_property_lowercase	unictype/property-lowercase

uc_is_property_other_lowercase	unictype/property-other-lowercase
uc_is_property_titlecase	unictype/property-titlecase
uc_is_property_cased	unictype/property-cased
uc_is_property_case_ignorable	unictype/property-case-ignorable
uc_is_property_changes_when_lowercased	unictype/property-changes-when-lowercased
uc_is_property_changes_when_uppercased	unictype/property-changes-when-uppercased
uc_is_property_changes_when_titlecased	unictype/property-changes-when-titlecased
uc_is_property_changes_when_casefolded	unictype/property-changes-when-casefolded
uc_is_property_changes_when_casemapped	unictype/property-changes-when-casemapped
uc_is_property_soft_dotted	unictype/property-soft-dotted
uc_is_property_id_start	unictype/property-id-start
uc_is_property_other_id_start	unictype/property-other-id-start
uc_is_property_id_continue	unictype/property-id-continue
uc_is_property_other_id_continue	unictype/property-other-id-continue
uc_is_property_xid_start	unictype/property-xid-start
uc_is_property_xid_continue	unictype/property-xid-continue
uc_is_property_id_compat_math_start	unictype/property-id-compatible-math-start
uc_is_property_id_compat_math_continue	unictype/property-id-compatible-math-continue
uc_is_property_pattern_white_space	unictype/property-pattern-white-space
uc_is_property_pattern_syntax	unictype/property-pattern-syntax
uc_is_property_join_control	unictype/property-join-control
uc_is_property_grapheme_base	unictype/property-grapheme-base
uc_is_property_grapheme_extend	unictype/property-grapheme-extend
uc_is_property_other_grapheme_extend	unictype/property-other-grapheme-extend
uc_is_property_grapheme_link	unictype/property-grapheme-link
uc_is_property_modifier_combining_mark	unictype/property-modifier-combining-mark
uc_is_property_bidi_control	unictype/property-bidi-control
uc_is_property_bidi_left_to_right	unictype/property-bidi-left-to-right
uc_is_property_bidi_hebrew_right_to_left	unictype/property-bidi-hebrew-right-to-left
uc_is_property_bidi_arabic_right_to_left	unictype/property-bidi-arabic-right-to-left
uc_is_property_bidi_european_digit	unictype/property-bidi-european-digit

<code>uc_is_property_bidi_eur_num_separator</code>	<code>unictype/property-bidi-eur-num-separator</code>
<code>uc_is_property_bidi_eur_num_terminator</code>	<code>unictype/property-bidi-eur-num-terminator</code>
<code>uc_is_property_bidi_arabic_digit</code>	<code>unictype/property-bidi-arabic-digit</code>
<code>uc_is_property_bidi_common_separator</code>	<code>unictype/property-bidi-common-separator</code>
<code>uc_is_property_bidi_block_separator</code>	<code>unictype/property-bidi-block-separator</code>
<code>uc_is_property_bidi_segment_separator</code>	<code>unictype/property-bidi-segment-separator</code>
<code>uc_is_property_bidi_whitespace</code>	<code>unictype/property-bidi-whitespace</code>
<code>uc_is_property_bidi_non_spacing_mark</code>	<code>unictype/property-bidi-non-spacing-mark</code>
<code>uc_is_property_bidi_boundary_neutral</code>	<code>unictype/property-bidi-boundary-neutral</code>
<code>uc_is_property_bidi_pdf</code>	<code>unictype/property-bidi-pdf</code>
<code>uc_is_property_bidi_embedding_or_override</code>	<code>unictype/property-bidi-embedding-or-override</code>
<code>uc_is_property_bidi_other_neutral</code>	<code>unictype/property-bidi-other-neutral</code>
<code>uc_is_property_hex_digit</code>	<code>unictype/property-hex-digit</code>
<code>uc_is_property_ascii_hex_digit</code>	<code>unictype/property-ascii-hex-digit</code>
<code>uc_is_property_ideographic</code>	<code>unictype/property-ideographic</code>
<code>uc_is_property_unified_ideograph</code>	<code>unictype/property-unified-ideograph</code>
<code>uc_is_property_radical</code>	<code>unictype/property-radical</code>
<code>uc_is_property_ids_unary_operator</code>	<code>unictype/property-ids-unary-operator</code>
<code>uc_is_property_ids_binary_operator</code>	<code>unictype/property-ids-binary-operator</code>
<code>uc_is_property_ids_trinary_operator</code>	<code>unictype/property-ids-trinary-operator</code>
<code>uc_is_property_emoji</code>	<code>unictype/property-emoji</code>
<code>uc_is_property_emoji_presentation</code>	<code>unictype/property-emoji-presentation</code>
<code>uc_is_property_emoji_modifier</code>	<code>unictype/property-emoji-modifier</code>
<code>uc_is_property_emoji_modifier_base</code>	<code>unictype/property-emoji-modifier-base</code>
<code>uc_is_property_emoji_component</code>	<code>unictype/property-emoji-component</code>
<code>uc_is_property_extended_pictographic</code>	<code>unictype/property-extended-pictographic</code>
<code>uc_is_property_zero_width</code>	<code>unictype/property-zero-width</code>
<code>uc_is_property_space</code>	<code>unictype/property-space</code>
<code>uc_is_property_non_break</code>	<code>unictype/property-non-break</code>
<code>uc_is_property_iso_control</code>	<code>unictype/property-iso-control</code>

uc_is_property_format_control	unictype/property-format-control
uc_is_property_prepend_	unictype/property-prepend-
concatenation_mark	concatenation-mark
uc_is_property_dash	unictype/property-dash
uc_is_property_hyphen	unictype/property-hyphen
uc_is_property_punctuation	unictype/property-punctuation
uc_is_property_line_separator	unictype/property-line-separator
uc_is_property_paragraph_separator	unictype/property-paragraph-
	separator
uc_is_property_quotation_mark	unictype/property-quotation-mark
uc_is_property_sentence_terminal	unictype/property-sentence-terminal
uc_is_property_terminal_punctuation	unictype/property-terminal-
	punctuation
uc_is_property_currency_symbol	unictype/property-currency-symbol
uc_is_property_math	unictype/property-math
uc_is_property_other_math	unictype/property-other-math
uc_is_property_paired_punctuation	unictype/property-paired-
	punctuation
uc_is_property_left_of_pair	unictype/property-left-of-pair
uc_is_property_combining	unictype/property-combining
uc_is_property_composite	unictype/property-composite
uc_is_property_decimal_digit	unictype/property-decimal-digit
uc_is_property_numeric	unictype/property-numeric
uc_is_property_diacritic	unictype/property-diacritic
uc_is_property_extender	unictype/property-extender
uc_is_property_ignorable_control	unictype/property-ignorable-control
uc_is_property_regional_indicator	unictype/property-regional-
	indicator
uc_indic_conjunct_break_name	unictype/incb-name
uc_indic_conjunct_break_byname	unictype/incb-byname
uc_indic_conjunct_break	unictype/incb-of
uc_script	unictype/scripts
uc_script_byname	unictype/scripts
uc_is_script	unictype/scripts
uc_all_scripts	unictype/scripts
uc_block	unictype/block-of
uc_is_block	unictype/block-test
uc_all_blocks	unictype/block-list
uc_is_c_whitespace	unictype/syntax-c-whitespace
uc_is_java_whitespace	unictype/syntax-java-whitespace
uc_c_ident_category	unictype/syntax-c-ident
uc_java_ident_category	unictype/syntax-java-ident
uc_is_alnum	unictype/ctype-alnum
uc_is_alpha	unictype/ctype-alpha
uc_is_cntrl	unictype/ctype-cntrl
uc_is_digit	unictype/ctype-digit

<code>uc_is_graph</code>	<code>unictype/ctype-graph</code>
<code>uc_is_lower</code>	<code>unictype/ctype-lower</code>
<code>uc_is_print</code>	<code>unictype/ctype-print</code>
<code>uc_is_punct</code>	<code>unictype/ctype-punct</code>
<code>uc_is_space</code>	<code>unictype/ctype-space</code>
<code>uc_is_upper</code>	<code>unictype/ctype-upper</code>
<code>uc_is_xdigit</code>	<code>unictype/ctype-xdigit</code>
<code>uc_is_blank</code>	<code>unictype/ctype-blank</code>

17.15.2.7 Declarations in <uniwidth.h>

Function	Module
<code>uc_width</code>	<code>uniwidth/width</code>
<code>u8_width</code>	<code>uniwidth/u8-width</code>
<code>u16_width</code>	<code>uniwidth/u16-width</code>
<code>u32_width</code>	<code>uniwidth/u32-width</code>
<code>u8_strwidth</code>	<code>uniwidth/u8-strwidth</code>
<code>u16_strwidth</code>	<code>uniwidth/u16-strwidth</code>
<code>u32_strwidth</code>	<code>uniwidth/u32-strwidth</code>

17.15.2.8 Declarations in <unigbrk.h>

Function	Module
<code>u8_grapheme_next</code>	<code>unigbrk/u8-grapheme-next</code>
<code>u16_grapheme_next</code>	<code>unigbrk/u16-grapheme-next</code>
<code>u32_grapheme_next</code>	<code>unigbrk/u32-grapheme-next</code>
<code>u8_grapheme_prev</code>	<code>unigbrk/u8-grapheme-prev</code>
<code>u16_grapheme_prev</code>	<code>unigbrk/u16-grapheme-prev</code>
<code>u32_grapheme_prev</code>	<code>unigbrk/u32-grapheme-prev</code>
<code>u8_grapheme_breaks</code>	<code>unigbrk/u8-grapheme-breaks</code>
<code>u16_grapheme_breaks</code>	<code>unigbrk/u16-grapheme-breaks</code>
<code>u32_grapheme_breaks</code>	<code>unigbrk/u32-grapheme-breaks</code>
<code>ulc_grapheme_breaks</code>	<code>unigbrk/ulc-grapheme-breaks</code>
<code>uc_grapheme_breaks</code>	<code>unigbrk/uc-grapheme-breaks</code>
<code>uc_graphemeclusterbreak_property</code>	<code>unigbrk/uc-gbrk-prop</code>
<code>uc_is_grapheme_break</code>	<code>unigbrk/uc-is-grapheme-break</code>

17.15.2.9 Declarations in <uniwbrk.h>

Function	Module
<code>u8_wordbreaks</code>	<code>uniwbrk/u8-wordbreaks</code>
<code>u16_wordbreaks</code>	<code>uniwbrk/u16-wordbreaks</code>
<code>u32_wordbreaks</code>	<code>uniwbrk/u32-wordbreaks</code>
<code>ulc_wordbreaks</code>	<code>uniwbrk/ulc-wordbreaks</code>
<code>uc_wordbreak_property</code>	<code>uniwbrk/wordbreak-property</code>

17.15.2.10 Declarations in <unilbrk.h>

Function	Module
<code>u8_possible_linebreaks</code>	<code>unilbrk/u8-possible-linebreaks</code>

u16_possible_linebreaks	unilbrk/u16-possible-linebreaks
u32_possible_linebreaks	unilbrk/u32-possible-linebreaks
ulc_possible_linebreaks	unilbrk/ulc-possible-linebreaks
u8_width_linebreaks	unilbrk/u8-width-linebreaks
u16_width_linebreaks	unilbrk/u16-width-linebreaks
u32_width_linebreaks	unilbrk/u32-width-linebreaks
ulc_width_linebreaks	unilbrk/ulc-width-linebreaks

17.15.2.11 Declarations in <uninorm.h>

Function or macro	Module
uc_decomposition	uninorm/decomposition
uc_canonical_decomposition	uninorm/canonical-decomposition
uc_composition	uninorm/composition
UNINORM_NFD	uninorm/nfd
UNINORM_NFC	uninorm/nfc
UNINORM_NFKD	uninorm/nfkd
UNINORM_NFKC	uninorm/nfkc
uninorm_is_compat_decomposing	uninorm/base
uninorm_is_composing	uninorm/base
uninorm_decomposing_form	uninorm/decomposing-form
u8_normalize	uninorm/u8-normalize
u16_normalize	uninorm/u16-normalize
u32_normalize	uninorm/u32-normalize
u8_normcmp	uninorm/u8-normcmp
u16_normcmp	uninorm/u16-normcmp
u32_normcmp	uninorm/u32-normcmp
u8_normxfrm	uninorm/u8-normxfrm
u16_normxfrm	uninorm/u16-normxfrm
u32_normxfrm	uninorm/u32-normxfrm
u8_normcoll	uninorm/u8-normcoll
u16_normcoll	uninorm/u16-normcoll
u32_normcoll	uninorm/u32-normcoll
uninorm_filter_create	uninorm/filter
uninorm_filter_write	uninorm/filter
uninorm_filter_flush	uninorm/filter
uninorm_filter_free	uninorm/filter

17.15.2.12 Declarations in <unicase.h>

Function or constant	Module
uc_toupper	unicase/toupper
uc_tolower	unicase/tolower
uc_totitle	unicase/totitle
uc_locale_language	unicase/locale-language
u8_toupper	unicase/u8-toupper
u16_toupper	unicase/u16-toupper
u32_toupper	unicase/u32-toupper

u8_tolower	unicase/u8-tolower
u16_tolower	unicase/u16-tolower
u32_tolower	unicase/u32-tolower
u8_totitle	unicase/u8-totitle
u16_totitle	unicase/u16-totitle
u32_totitle	unicase/u32-totitle
unicase_empty_prefix_context	unicase/empty-prefix-context
u8_casing_prefix_context	unicase/u8-prefix-context
u16_casing_prefix_context	unicase/u16-prefix-context
u32_casing_prefix_context	unicase/u32-prefix-context
u8_casing_prefixes_context	unicase/u8-prefix-context
u16_casing_prefixes_context	unicase/u16-prefix-context
u32_casing_prefixes_context	unicase/u32-prefix-context
unicase_empty_suffix_context	unicase/empty-suffix-context
u8_casing_suffix_context	unicase/u8-suffix-context
u16_casing_suffix_context	unicase/u16-suffix-context
u32_casing_suffix_context	unicase/u32-suffix-context
u8_casing_suffixes_context	unicase/u8-suffix-context
u16_casing_suffixes_context	unicase/u16-suffix-context
u32_casing_suffixes_context	unicase/u32-suffix-context
u8_ct_toupper	unicase/u8-ct-toupper
u16_ct_toupper	unicase/u16-ct-toupper
u32_ct_toupper	unicase/u32-ct-toupper
u8_ct_tolower	unicase/u8-ct-tolower
u16_ct_tolower	unicase/u16-ct-tolower
u32_ct_tolower	unicase/u32-ct-tolower
u8_ct_totitle	unicase/u8-ct-totitle
u16_ct_totitle	unicase/u16-ct-totitle
u32_ct_totitle	unicase/u32-ct-totitle
u8_casefold	unicase/u8-casefold
u16_casefold	unicase/u16-casefold
u32_casefold	unicase/u32-casefold
u8_ct_casefold	unicase/u8-ct-casefold
u16_ct_casefold	unicase/u16-ct-casefold
u32_ct_casefold	unicase/u32-ct-casefold
u8_casecmp	unicase/u8-casecmp
u16_casecmp	unicase/u16-casecmp
u32_casecmp	unicase/u32-casecmp
ulc_casecmp	unicase/ulc-casecmp
u8_casexfrm	unicase/u8-casexfrm
u16_casexfrm	unicase/u16-casexfrm
u32_casexfrm	unicase/u32-casexfrm
ulc_casexfrm	unicase/ulc-casexfrm
u8_casecoll	unicase/u8-casecoll
u16_casecoll	unicase/u16-casecoll
u32_casecoll	unicase/u32-casecoll
ulc_casecoll	unicase/ulc-casecoll

<code>u8_is_uppercase</code>	<code>unicase/u8-is-uppercase</code>
<code>u16_is_uppercase</code>	<code>unicase/u16-is-uppercase</code>
<code>u32_is_uppercase</code>	<code>unicase/u32-is-uppercase</code>
<code>u8_is_lowercase</code>	<code>unicase/u8-is-lowercase</code>
<code>u16_is_lowercase</code>	<code>unicase/u16-is-lowercase</code>
<code>u32_is_lowercase</code>	<code>unicase/u32-is-lowercase</code>
<code>u8_is_titlecase</code>	<code>unicase/u8-is-titlecase</code>
<code>u16_is_titlecase</code>	<code>unicase/u16-is-titlecase</code>
<code>u32_is_titlecase</code>	<code>unicase/u32-is-titlecase</code>
<code>u8_is_casefolded</code>	<code>unicase/u8-is-casefolded</code>
<code>u16_is_casefolded</code>	<code>unicase/u16-is-casefolded</code>
<code>u32_is_casefolded</code>	<code>unicase/u32-is-casefolded</code>
<code>u8_is_cased</code>	<code>unicase/u8-is-cased</code>
<code>u16_is_cased</code>	<code>unicase/u16-is-cased</code>
<code>u32_is_cased</code>	<code>unicase/u32-is-cased</code>

17.16 Stack traces

Printing a stack trace was traditionally seen as a feature of the debugging environment and thus only implemented in the debuggers (gdb etc.). However, they are also useful in production code, for use in two circumstances:

- When a problem occurs on a user’s machine, when the user is merely a user, not a programmer.
- In unit tests that run in continuous-integration environments. In such environments, the virtual machine is discarded immediately after the tests have run. It is not possible to run a debugger in such environments.

And in fact, printing a stack trace is part of the basic runtime system in programming languages such as Java (`printStackTrace` method (<https://docs.oracle.com/javase/8/docs/api/java/lang/Throwable.html#printStackTrace-->)), Python (`print_exception` method (<https://docs.python.org/3/library/traceback.html>)), Go (`PrintStack` function (<https://pkg.go.dev/runtime/debug#PrintStack>)), and ISO C++ 23 (`std::stacktrace` class (<https://en.cppreference.com/w/cpp/utility/basic/stacktrace>)).

Gnulib provides a module ‘stack-trace’ with this feature: `print_stack_trace()` prints a stack trace of the current thread to standard error.

For it to work best, three requirements need to be met:

- The `libbacktrace` library (<https://github.com/ianlancetaylor/libbacktrace>) or GCC’s sanitizer library `libasan` needs to be installed.
- The program needs to be compiled with debugging information (option `-g`).
- On macOS, where debugging information is stored in a separate directory rather than in the compiled binary (see <https://stackoverflow.com/questions/10044697/#12827463>), the `dsymutil` program needs to be used when linking, and the debugging information needs to be copied when the program is installed. Cf. <https://github.com/ianlancetaylor/libbacktrace/issues/122#issuecomment-2122589147>.

When these requirements are not met, the function `print_stack_trace ()` either prints a stack trace without source file names and line numbers, or prints nothing at all.

Gnulib also provides a module ‘`abort-debug`’, that overrides the `abort` function so that it prints the stack trace of the current thread, before actually aborting. Thus, `abort ()` remains the idiom of choice for signaling a fatal situation that requires developer attention: it is useful both in debugging environments and production code.

Note: While the original `abort` function is safe to call in signal handlers, the overridden `abort` function is not. In signal handlers, you will need to call the original `abort` function, by doing `#undef abort` first.

17.17 Printing `--version` and bug-reporting information

Gnulib provides a few modules for printing `--version` and bug-reporting information according to the GNU Coding Standards; see https://www.gnu.org/prep/standards/html_node/_002d_002dversion.html#g_t_002d_002dversion

The ‘`version-etc`’ module defines functions used to emit `--version` and bug-reporting input in a standard way. The module expects the string `version_etc_copyright` to be defined to display the copyright holder along with the package release year. The `version_etc` function can then be used like so:

```
const char version_etc_copyright[] = "Copyright %s %d copyright-holder";

int
main (void)
{
    version_etc (stdout, "my-program", "my-package", "1.0",
                "author1", "author2", (const char *) NULL);
    return 0;
}
```

The above example generates the following output:

```
my-program (my-package) 1.0
Copyright (C) 2025 copyright-holder
License GPLv3+: GNU GPL version 3 or later <https://gnu.org/licenses/gpl.html>.
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
```

```
Written by author1 and author2.
```

The ‘`version-etc`’ module also defines the function `emit_bug_reporting_address` which emits the bug-reporting location based on arguments passed to `AC_INIT`. For example, using the following in `configure.ac`:

```
AC_INIT([gnulib], [0.0], [bug-gnulib@gnu.org], [gnulib],
        [https://www.gnu.org/software/gnulib/])
```

And the following program:

```
#include <config.h>

#include "version-etc.h"

int
main (void)
{
```

```

    emit_bug_reporting_address ();
    return 0;
}

```

Will output:

```

Report bugs to: bug-gnulib@gnu.org
gnulib home page: <https://www.gnu.org/software/gnulib/>
General help using GNU software: <https://www.gnu.org/gethelp/>

```

The ‘version-etc-fsf’ module defines and links to the variable `version_etc_copyright` containing copyright information for Free Software Foundation. This module should be used by programs which are FSF-copyrighted and use the ‘version-etc’ module.

The ‘argp-version-etc’ defines the `argp_version_setup` function takes the program name and authors as arguments. This function should be called before `argp_parse` since it modifies argp global variables; see https://www.gnu.org/software/libc/manual/html_node/Argp-Global-Variables.html.

17.18 Recognizing Option Arguments

The module ‘argmatch’ provides a simple textual user interface to a finite choice. It is for example well suited to recognize arguments of options or values of environment variables that accept a fixed set of valid choices.

These choices may be denoted by synonyms, such as ‘none’ and ‘off’ below.

```

$ my_cp --backup=none foo bar
$ my_cp --backup=non foo bar
$ my_cp --backup=no foo bar
$ my_cp --backup=n foo bar
my_cp: ambiguous argument 'n' for 'backup type'
Valid arguments are:
- 'no', 'none', 'off'
- 'numbered', 't', 'newstyle'
- 'existing', 'nil', 'numbered-existing'
- 'simple', 'never', 'single'
Try 'my_cp --help' for more information.
$ my_cp --backup=num foo bar
$ my_cp --backup=true foo bar
my_cp: invalid argument 'true' for 'backup type'
Valid arguments are:
- 'no', 'none', 'off'
- 'numbered', 't', 'newstyle'
- 'existing', 'nil', 'numbered-existing'
- 'simple', 'never', 'single'
Try 'my_cp --help' for more information.

```

To set up `argmatch`, first call ‘`ARGMATCH_DEFINE_GROUP (name, type)`’ with the name of the `argmatch` group name, and the value type. For instance:

```

enum backup_type
{
    no_backups,
    simple_backups,
    numbered_existing_backups,
    numbered_backups
};

```

```
ARGMATCH_DEFINE_GROUP (backup, enum backup_type);
```

This defines a few types and functions named `argmatch_name_*`. Introduce the array that defines the mapping from user-input to actual value, with a terminator:

```
static const argmatch_backup_arg argmatch_backup_args[] =
{
    { "no",                no_backups },
    { "none",              no_backups },
    { "off",               no_backups },
    { "simple",             simple_backups },
    { "never",             simple_backups },
    { "single",            simple_backups },
    { "existing",           numbered_existing_backups },
    { "nil",               numbered_existing_backups },
    { "numbered-existing", numbered_existing_backups },
    { "numbered",          numbered_backups },
    { "t",                 numbered_backups },
    { "newstyle",          numbered_backups },
    { NULL, no_backups }
};
```

Then introduce the array that defines the values, also with a terminator. Document only once per group of synonyms:

```
static const argmatch_backup_doc argmatch_backup_docs[] =
{
    { "no",      N_("never make backups (even if --backup is given)") },
    { "numbered", N_("make numbered backups") },
    { "existing", N_("numbered if numbered backups exist, simple otherwise") },
    { "simple",   N_("always make simple backups") },
    { NULL, NULL }
};
```

Finally, define the `argmatch` group:

```
const argmatch_backup_group_type argmatch_backup_group =
{
    argmatch_backup_args,
    argmatch_backup_docs,
    N_("\n
The backup suffix is '~', unless set with --suffix or SIMPLE_BACKUP_SUFFIX.\n\
The version control method may be selected via the --backup option or through\n\
the VERSION_CONTROL environment variable. Here are the values:\n"),
    NULL
};
```

To use the `argmatch` group:

```
ptrdiff_t i = argmatch_backup_choice ("--backup", "none");
// argmatch_backup_group.args[i].arg is "none", so its value
// is argmatch_backup_group.args[i].val.
// Return -1 on invalid argument, and -2 on ambiguity.
```

```

enum backup_type val = *argmatch_backup_value ("--backup", "none");
// Returns a pointer to the value, and exit on errors.
// So argmatch_backup_group.args[i].val == val.

const char *arg = argmatch_backup_argument (&no_backups);
// arg is "no".

// Print the documentation on stdout.
argmatch_backup_usage (stdout);
// Gives:
//
// The backup suffix is '~', unless set with --suffix or SIMPLE_BACKUP_SUFFIX.
// The version control method may be selected via the --backup option or through
// the VERSION_CONTROL environment variable. Here are the values:
//
// no, none, off  never make backups (even if --backup is given)
// numbered, t, newstyle
//                make numbered backups
// existing, nil, numbered-existing
//                numbered if numbered backups exist, simple otherwise
// simple, never, single
//                always make simple backups

```

17.19 Quoting

Gnulib provides ‘quote’ and ‘quotearg’ modules to help with quoting text, such as file names, in messages to the user. Here’s an example of using ‘quote’:

```

#include <quote.h>

...
error (0, errno, _("cannot change owner of %s"), quote (fname));

```

This differs from

```

error (0, errno, _("cannot change owner of '%s'"), fname);

```

in that `quote` escapes unusual characters in `fname`, e.g., ‘`'`’ and control characters like ‘`\n`’.

However, a caveat: `quote` reuses the storage that it returns. Hence if you need more than one thing quoted at the same time, you need to use `quote_n`.

Also, the `quote` module is not suited for multithreaded applications. In that case, you have to use `quotearg_alloc`, defined in the ‘quotearg’ module, which is decidedly less convenient.

17.20 progname and getprogname

Gnulib has two modules for retrieving the name of the currently executing program: `progname` and `getprogname`.

The `progname` module defines a variable `program_name`. It contains the name of the currently executing program, on all platforms. But it cannot be used implicitly: It requires that every `main` function be modified to invoke `set_program_name (argv[0])` as one of its first actions.

The `getprogname` module defines a function `getprogname()`. It returns the name of the currently executing program, on most platforms. The advantage of this module is that it

can be used without prior initializations. But it has limitations: In some rare situations, it cannot determine the name; then it returns "?" instead. And on some platforms, it returns a truncated program name.

The `error` function uses the `getprogname` module.

17.21 gcd: greatest common divisor

The `gcd` function returns the greatest common divisor of two numbers $a > 0$ and $b > 0$. It is the caller's responsibility to ensure that the arguments are non-zero.

If you need a `gcd` function for an integer type larger than 'unsigned long', you can include the `gcd.c` implementation file with parametrization. The parameters are:

- `WORD_T` Define this to the unsigned integer type that you need this function for.
- `GCD` Define this to the name of the function to be created.

The created function has the prototype

```
WORD_T GCD (WORD_T a, WORD_T b);
```

If you need the least common multiple of two numbers, it can be computed like this: $\text{lcm}(a,b) = (a / \text{gcd}(a,b)) * b$ or $\text{lcm}(a,b) = a * (b / \text{gcd}(a,b))$. Avoid the formula $\text{lcm}(a,b) = (a * b) / \text{gcd}(a,b)$ because—although mathematically correct—it can yield a wrong result, due to integer overflow.

In some applications it is useful to have a function taking the gcd of two signed numbers. In this case, the gcd function result is usually normalized to be non-negative (so that two gcd results can be compared in magnitude or compared against 1, etc.). Note that in this case the prototype of the function has to be

```
unsigned long gcd (long a, long b);
```

and not

```
long gcd (long a, long b);
```

because `gcd(LONG_MIN, LONG_MIN) = -LONG_MIN = LONG_MAX + 1` does not fit into a signed 'long'.

17.22 Profiling of program phases

The module 'timevar' provides a simple self-profiling facility, based on timers.

```
Execution times (seconds)
read          : 0.09 (19%) usr 0.08 (80%) sys 0.09 (18%) wall
read: scan    : 0.04 ( 9%) usr 0.08 (80%) sys 0.12 (26%) wall
read: parse   : 0.05 (10%) usr 0.00 ( 0%) sys 0.05 (10%) wall
work          : 0.33 (70%) usr 0.00 ( 0%) sys 0.35 (71%) wall
work: phase 1 : 0.30 (64%) usr 0.00 ( 0%) sys 0.30 (64%) wall
work: phase 2 : 0.13 (28%) usr 0.00 ( 0%) sys 0.14 (29%) wall
output        : 0.04 ( 9%) usr 0.02 (20%) sys 0.04 ( 8%) wall
total time    : 0.47          0.10          0.49
```

To set up `timevar`, copy the stub file `gnulib/lib/timevar.def` next to where `timevar.h` and `timevar.c` were imported in your project, and define your timers there. For instance:

```
/* The total execution time. Mandatory. */
DEFTIMEVAR (tv_total, "total time")
```

```

/* Examples. */
DEFTIMEVAR (tv_read,      "read")
DEFTIMEVAR (tv_work,      "work")
DEFTIMEVAR (tv_work_1,    "work: phase 1")
DEFTIMEVAR (tv_work_2,    "work: phase 2")
DEFTIMEVAR (tv_output,    "output")

```

Do not remove `tv_total`, it is mandatory. You may change its associated string.

Use `timevar_push/timevar_pop` to start/stop timers, as in the following example.

```

#include <config.h>
#include "timevar.h"

#include <stdio.h>
#include "read.h"
#include "work.h"
#include "output.h"

int
main (void)
{
    timevar_enabled = true;
    timevar_init ();
    timevar_start (tv_total);

    timevar_push (tv_read);
    reader ();
    timevar_pop (tv_read);

    timevar_push (tv_work);
    work ();
    timevar_pop (tv_work);

    timevar_push (tv_output);
    output ();
    timevar_pop (tv_output);

    timevar_stop (tv_total);
    timevar_print (stderr);
}

```

with, for instance, in `work.c`

```

#include <config.h>
#include "work.h"

void
work (void)
{
    timevar_push (tv_work_phase1);
    work1 ();
    timevar_pop (tv_work_phase1);

    timevar_push (tv_work_phase2);
    work2 ();
    timevar_pop (tv_work_phase2);
}

```

17.23 Library version handling

The module ‘`check-version`’ can be useful when your gnulib application is a system library. You will typically wrap the call to the `check_version` function through a library API, your library header file may contain:

```
#define STRINGPREP_VERSION "0.5.18"
...
extern const char *stringprep_check_version (const char *req_version);
```

To avoid ELF symbol collisions with other libraries that use the ‘`check-version`’ module, add to `config.h` through a `AC_DEFINE` something like:

```
AC_DEFINE(check_version, stringprep_check_version,
[Rename check_version.])
```

The `stringprep_check_version` function will thus be implemented by the `check_version` module.

There are two uses of the interface. The first is a way to provide for applications to find out the version number of the library it uses. The application may contain diagnostic code such as:

```
printf ("Stringprep version: header %s library %s",
        STRINGPREP_VERSION,
        stringprep_check_version (NULL));
```

Separating the library and header file version can be useful when searching for version mismatch related problems.

The second uses is as a rudimentary test of proper library version, by making sure the application get a library version that is the same, or newer, than the header file used when building the application. This doesn’t catch all problems, libraries may change backwards incompatibly in later versions, but enable applications to require a certain minimum version before it may proceed.

Typical uses look like:

```
/* Check version of libgcrypt. */
if (!gcry_check_version (GCRYPT_VERSION))
    die ("version mismatch\n");
```

17.24 Supporting Relocation

It has been a pain for many users of GNU packages for a long time that packages are not relocatable. It means a user cannot copy a program, installed by another user on the same machine, to his home directory, and have it work correctly (including `libn`). So many users need to go through `configure; make; make install` with all its dependencies, options, and hurdles.

Red Hat, Debian, and other binary distributions solve the “ease of installation” problem, but they hardwire path names, usually to `/usr` or `/usr/local`. This means that users need root privileges to install a binary package, and prevents installing two different versions of the same binary package.

A relocatable program can be moved or copied to a different location on the file system. It is possible to make symlinks to the installed and moved programs, and invoke them

through the symlink. It is possible to do the same thing with a hard link *only* if the hard link file is in the same directory as the real program.

The **relocatable-prog** module aims to ease the process of making a GNU program relocatable. It helps overcome two obstacles. First, it aids with relocating the hard-coded references to absolute file names that GNU programs often contain. These references must be fixed up at runtime if a program is to be successfully relocated. The **relocatable-prog** module provides a function **relocate** that does this job.

Second, the loader must be able to find shared libraries linked to relocatable executables or referenced by other shared libraries linked to relocatable executables. The **relocatable-prog** module helps out here in a platform-specific way:

- On most operating systems, it adds a linker option (**-rpath**) that causes the dynamic linker to search for libraries in a directory relative to the location of the invoked executable. This works on GNU/Linux and modern versions of GNU/Hurd, GNU/kFreeBSD, macOS, FreeBSD, NetBSD, OpenBSD, Solaris, Haiku.
- On other Unix systems, it installs a trampoline executable. The trampoline sets the environment variable that controls shared library searching (usually **LD_LIBRARY_PATH**) and then invokes the real executable. This applies to operating systems such as AIX, HP-UX, or Minix.
- On Windows, the executable's own directory is searched for libraries, so installing shared libraries into the executable's directory is sufficient.

You can make your program relocatable by following these steps:

1. Import the **relocatable-prog** module. For libraries, use the **relocatable-lib** or **relocatable-lib-lgpl** module, if the libraries are independent. For installing multiple libraries, at least one of which depends on another one, use the **relocatable-prog** module. If you need more than one module, or you need to use them with different settings, you will need multiple copies of gnuilib (see [Section 3.10 \[Multiple instances\]](#), page 32).
2. In every program, add to **main** as the first statement (even before setting the locale or doing anything related to **libintl**):

```
set_program_name (argv[0]);
```

The prototype for this function is in **progrname.h**.

3. If you want your code to be portable to platforms that do not support automatic initialization, call **set_relocation_prefix**.
4. Everywhere where you use a constant pathname from installation-time, wrap it in **relocate** so it gets translated to the run-time situation. Example:

```
bindtextdomain (PACKAGE, LOCALEDIR);
```

becomes:

```
bindtextdomain (PACKAGE, relocate (LOCALEDIR));
```

The prototype for this function is in **relocatable.h**.

There is also a variant of this function, named **relocate2**, that makes it easy to reclaim the memory allocated by the call.

5. The `set_program_name` function can also configure some additional libraries to relocate files that they access, by defining corresponding C preprocessor symbols to 1. The libraries for which this is supported and the corresponding preprocessor symbols are:

```
libcharset  DEPENDS_ON_LIBCHARSET
libiconv    DEPENDS_ON_LIBICONV
libintl     DEPENDS_ON_LIBINTL
```

Defining the symbol for a library makes every program in the package depend on that library, whether the program really uses the library or not, so this feature should be used with some caution.

6. If your package installs shell scripts, also import the `relocatable-script` module. Then, near the beginning of each shell script that your package installs, add the following:

```
@relocatable_sh@

prefix="@prefix@"
exec_prefix="@exec_prefix@" # usually needs $prefix.
datarootdir="@datarootdir@" # usually needs $prefix.

if test "@RELOCATABLE@" = yes; then
    bindir="@bindir@"
    orig_installdir="$bindir" # see Makefile.am's *_SCRIPTS variables
    func_find_curr_installdir # determine curr_installdir
    func_find_prefixes
    relocate () {
        echo "$1/" \
        | sed -e "s%~${orig_installprefix}%/${curr_installprefix}%/" \
        | sed -e 's,/,$,,'
    }
else
    relocate () {
        echo "$1"
    }
fi

# Get some relocated directory names.
sysconfdir=`relocate "@sysconfdir@"` # usually needs $prefix.
some_datadir=`relocate "@datadir@/something"` # usually needs $datarootdir.
bindir=`relocate "@bindir@"` # usually needs $exec_prefix, hence $prefix.
```

You must adapt the definition of `orig_installdir`, depending on where the script gets installed. Also, at the end, instead of `sysconfdir` and `some_datadir`, transform those variables that you need.

7. If your package installs Perl scripts, also import the `relocatable-perl` module. Then, near the beginning of each Perl script that your package installs, add the following:

```
@relocatable_pl@
if ("@RELOCATABLE@" eq "yes") {
    my $exec_prefix = "@exec_prefix@";
    my $orig_installdir = "@bindir@"; # see Makefile.am's *_SCRIPTS variables
    my ($orig_installprefix, $curr_installprefix) =
        find_prefixes($orig_installdir, find_curr_installdir());

    # the subroutine is defined whether or not the enclosing block is executed
```

```

sub relocate {
    my ($dir) = @_;
    if ("@RELOCATABLE@" eq "yes") {
        $dir =~ s/~/ $orig_installprefix/%$curr_installprefix/%;
        $dir =~ s,/$/,;
    }
    return $dir;
}

# Get some relocated directory names.
# (The gnuilib module 'configmake' can help with this.)
$sysconfdir = relocate("@sysconfdir@");
$some_datadir = relocate("@datadir@/something");

```

You must adapt the definition of `$orig_installdir`, depending on where the script gets installed. Also, at the end, instead of `sysconfdir` and `some_datadir`, transform those variables that you need.

8. In your `Makefile.am`, for every program `foo` that gets installed in, say, `$(bindir)`, you add:

```

foo_CPPFLAGS = -DINSTALLDIR=$(bindir_c_make)
if RELOCATABLE_VIA_LD
foo_LDFLAGS = `$(RELOCATABLE_LDFLAGS) $(bindir)`
endif

```

Similarly, if a program `foo` gets installed in `$(pkglibdir)`, you add:

```

foo_CPPFLAGS = -DINSTALLDIR=$(pkglibdir_c_make)
if RELOCATABLE_VIA_LD
foo_LDFLAGS = `$(RELOCATABLE_LDFLAGS) $(pkglibdir)`
endif

```

The `Makefile` variables `bindir_c_make` or `pkglibdir_c_make` get defined by the Autoconf macros `gl_BUILD_TO_HOST_BINDIR` or `gl_BUILD_TO_HOST_PKGLIBDIR`, respectively. These Autoconf macros are defined in the file `m4/build-to-host.m4`. You need to

- Import this file `m4/build-to-host.m4` into your package, for example by using of a command like `'gnulib-tool --copy m4/build-to-host.m4'`.
 - Invoke the corresponding macro(s) from your package's `configure.ac`.
9. When building `gnulib` to use with a relocatable library, you need to define the preprocessor symbol `IN_LIBRARY`. You may also want to build with `ENABLE_COSTLY_RELOCATABLE`, in which case you will also need to define `INSTALLDIR`. The following fragment can be added to an override `Makefile.am` used to build `gnulib` (see [Section 3.8 \[Modified build rules\]](#), page 30).

```

AM_CPPFLAGS += -DIN_LIBRARY -DENABLE_COSTLY_RELOCATABLE

if SHLIBS_IN_BINDIR
AM_CPPFLAGS += -DINSTALLDIR=$(bindir_c_make)
else
AM_CPPFLAGS += -DINSTALLDIR=$(libdir_c_make)
endif

```

SHLIBS_IN_BINDIR is defined in `configure.ac` as follows:

```
AM_CONDITIONAL([SHLIBS_IN_BINDIR],
               [case "$host_os" in mingw* | cygwin*) true;; *) false;; esac])
```

10. In your `Makefile.am`, for every library `libfoo` that gets installed in, say, `$(libdir)`, you add:

```
if RELOCATABLE_VIA_LD
libfoo_la_LDFLAGS = `$(RELOCATABLE_LDFLAGS) $(libdir)`
endif
```

11. Add a couple of variable assignments to your `Makefile.am`.

If your package (or any package you rely on, e.g. `gettext-runtime`) will be relocated together with a set of installed shared libraries, then set `RELOCATABLE_LIBRARY_PATH` to a colon-separated list of those libraries' directories, e.g.

```
RELOCATABLE_LIBRARY_PATH = $(libdir)
```

If your `config.h` is not in `$(top_builddir)`, then set `RELOCATABLE_CONFIG_H_DIR` to its directory, e.g.

```
RELOCATABLE_CONFIG_H_DIR = $(top_builddir)/src
```

17.25 func

The `func` module makes sure that you can use the predefined identifier `__func__` as defined by C99 in your code.

A small example is:

```
#include <config.h>
#include <stdio.h> /* for printf */

int main (void)
{
    printf ("%s: hello world\n", __func__);
}
```

Note that `sizeof` cannot be applied to `__func__`: On SunPRO C compiler, `sizeof __func__` evaluates to 0.

17.26 stat-size

The `stat-size` module provides a small number of macros intended for interpreting the file size information in an instance of `struct stat`.

On POSIX systems, the `st_blocks` member of `struct stat` contains the number of disk blocks occupied by a file. The `ST_NBLOCKS` and `STP_NBLOCKS` macros estimate this quantity on systems which don't actually have `st_blocks`. Each of these blocks contains `ST_NBLOCKSIZE` bytes.

The value of `ST_NBLOCKSIZE` is often quite small, small enough that performing I/O in chunks that size would be inefficient. The `ST_BLKSIZE` and `STP_BLKSIZE` macros give the I/O block size recommended for I/O to this file. This is not guaranteed to give optimum performance, but it should be reasonably efficient.

18 Regular expressions

18.1 Overview

A *regular expression* (or *regexp*, or *pattern*) is a text string that describes some (mathematical) set of strings. A regexp *r* *matches* a string *s* if *s* is in the set of strings described by *r*.

Using the Regex library, you can:

- see if a string matches a specified pattern as a whole, and
- search within a string for a substring matching a specified pattern.

Some regular expressions match only one string, i.e., the set they describe has only one member. For example, the regular expression ‘foo’ matches the string ‘foo’ and no others. Other regular expressions match more than one string, i.e., the set they describe has more than one member. For example, the regular expression ‘f*’ matches the set of strings made up of any number (including zero) of ‘f’s. As you can see, some characters in regular expressions match themselves (such as ‘f’) and some don’t (such as ‘*’); the ones that don’t match themselves instead let you specify patterns that describe many different strings.

To either match or search for a regular expression with the Regex library functions, you must first compile it with a Regex pattern compiling function. A *compiled pattern* is a regular expression converted to the internal format used by the library functions. Once you’ve compiled a pattern, you can use it for matching or searching any number of times.

The Regex library is used by including `regex.h`. Regex provides three groups of functions with which you can operate on regular expressions. One group—the GNU group—is more powerful but not completely compatible with the other two, namely the POSIX and Berkeley Unix groups; its interface was designed specifically for GNU.

We wrote this chapter with programmers in mind, not users of programs—such as Emacs—that use Regex. We describe the Regex library in its entirety, not how to write regular expressions that a particular program understands.

18.2 Regular Expression Syntax

Characters are things you can type. *Operators* are things in a regular expression that match one or more characters. You compose regular expressions from operators, which in turn you specify using one or more characters.

Most characters represent what we call the match-self operator, i.e., they match themselves; we call these characters *ordinary*. Other characters represent either all or parts of fancier operators; e.g., ‘.’ represents what we call the match-any-character operator (which, no surprise, matches (almost) any character); we call these characters *special*. Two different things determine what characters represent what operators:

1. the regular expression syntax your program has told the Regex library to recognize, and
2. the context of the character in the regular expression.

In the following sections, we describe these things in more detail.

18.2.1 Syntax Bits

In any particular syntax for regular expressions, some characters are always special, others are sometimes special, and others are never special. The particular syntax that `Regex` recognizes for a given regular expression depends on the current syntax (as set by `re_set_syntax`) when the pattern buffer of that regular expression was compiled.

You get a pattern buffer by compiling a regular expression. See [Section 18.6.1.1 \[GNU Pattern Buffers\]](#), page 936, for more information on pattern buffers. See [Section 18.6.1.2 \[GNU Regular Expression Compiling\]](#), page 936, and [Section 18.6.2.1 \[BSD Regular Expression Compiling\]](#), page 943, for more information on compiling.

`Regex` considers the current syntax to be a collection of bits; we refer to these bits as *syntax bits*. In most cases, they affect what characters represent what operators. We describe the meanings of the operators to which we refer in [Section 18.3 \[Common Operators\]](#), page 927, and [Section 18.4 \[GNU Operators\]](#), page 935.

For reference, here is the complete list of syntax bits, in alphabetical order:

`RE_BACKSLASH_ESCAPE_IN_LISTS`

If this bit is set, then ‘\’ inside a list (see [Section 18.3.6 \[List Operators\]](#), page 930) quotes (makes ordinary, if it’s special) the following character; if this bit isn’t set, then ‘\’ is an ordinary character inside lists. (See [Section 18.2.4 \[The Backslash Character\]](#), page 926, for what ‘\’ does outside of lists.)

`RE_BK_PLUS_QM`

If this bit is set, then ‘\+’ represents the match-one-or-more operator and ‘\?’ represents the match-zero-or-more operator; if this bit isn’t set, then ‘+’ represents the match-one-or-more operator and ‘?’ represents the match-zero-or-one operator. This bit is irrelevant if `RE_LIMITED_OPS` is set.

`RE_CHAR_CLASSES`

If this bit is set, then you can use character classes in lists; if this bit isn’t set, then you can’t.

`RE_CONTEXT_INDEP_ANCHORS`

If this bit is set, then ‘^’ and ‘\$’ are special anywhere outside a list; if this bit isn’t set, then these characters are special only in certain contexts. See [Section 18.3.9.1 \[Match-beginning-of-line Operator\]](#), page 934, and [Section 18.3.9.2 \[Match-end-of-line Operator\]](#), page 934.

`RE_CONTEXT_INDEP_OPS`

If this bit is set, then certain characters are special anywhere outside a list; if this bit isn’t set, then those characters are special only in some contexts and are ordinary elsewhere. Specifically, if this bit isn’t set then ‘*’, and (if the syntax bit `RE_LIMITED_OPS` isn’t set) ‘+’ and ‘?’ (or ‘\+’ and ‘\?’), depending on the syntax bit `RE_BK_PLUS_QM`) represent repetition operators only if they’re not first in a regular expression or just after an open-group or alternation operator. The same holds for ‘{’ (or ‘\{’, depending on the syntax bit `RE_NO_BK_BRACES`) if it is the beginning of a valid interval and the syntax bit `RE_INTERVALS` is set.

RE_CONTEXT_INVALID_DUP

If this bit is set, then an open-interval operator cannot occur at the start of a regular expression, or immediately after an alternation, open-group or close-interval operator.

RE_CONTEXT_INVALID_OPS

If this bit is set, then repetition and alternation operators can't be in certain positions within a regular expression. Specifically, the regular expression is invalid if it has:

- a repetition operator first in the regular expression or just after a match-beginning-of-line, open-group, or alternation operator; or
- an alternation operator first or last in the regular expression, just before a match-end-of-line operator, or just after an alternation or open-group operator.

If this bit isn't set, then you can put the characters representing the repetition and alternation characters anywhere in a regular expression. Whether or not they will in fact be operators in certain positions depends on other syntax bits.

RE_DEBUG If this bit is set, and the regex library was compiled with `-DDEBUG`, then internal debugging is turned on; if unset, then it is turned off.

RE_DOT_NEWLINE

If this bit is set, then the match-any-character operator matches a newline; if this bit isn't set, then it doesn't.

RE_DOT_NOT_NULL

If this bit is set, then the match-any-character operator doesn't match a null character; if this bit isn't set, then it does.

RE_HAT_LISTS_NOT_NEWLINE

If this bit is set, nonmatching lists `['^...']` do not match newline; if not set, they do.

RE_ICASE If this bit is set, then ignore case when matching; otherwise, case is significant.

RE_INTERVALS

If this bit is set, then Regex recognizes interval operators; if this bit isn't set, then it doesn't.

RE_INVALID_INTERVAL_ORD

If this bit is set, a syntactically invalid interval is treated as a string of ordinary characters. For example, the extended regular expression `'a{1}'` is treated as `'a\{1'`.

RE_LIMITED_OPS

If this bit is set, then Regex doesn't recognize the match-one-or-more, match-zero-or-one or alternation operators; if this bit isn't set, then it does.

RE_NEWLINE_ALT

If this bit is set, then newline represents the alternation operator; if this bit isn't set, then newline is ordinary.

RE_NO_BK_BRACES

If this bit is set, then ‘{’ represents the open-interval operator and ‘}’ represents the close-interval operator; if this bit isn’t set, then ‘\{’ represents the open-interval operator and ‘\}’ represents the close-interval operator. This bit is relevant only if **RE_INTERVALS** is set.

RE_NO_BK_PARENS

If this bit is set, then ‘(’ represents the open-group operator and ‘)’ represents the close-group operator; if this bit isn’t set, then ‘\(’ represents the open-group operator and ‘\)’ represents the close-group operator.

RE_NO_BK_REFS

If this bit is set, then Regex doesn’t recognize ‘\digit’ as the back-reference operator; if this bit isn’t set, then it does.

RE_NO_BK_VBAR

If this bit is set, then ‘|’ represents the alternation operator; if this bit isn’t set, then ‘\|’ represents the alternation operator. This bit is irrelevant if **RE_LIMITED_OPS** is set.

RE_NO_EMPTY_RANGES

If this bit is set, then a regular expression with a range whose ending point collates lower than its starting point is invalid; if this bit isn’t set, then Regex considers such a range to be empty.

RE_NO_GNU_OPS

If this bit is set, GNU regex operators are not recognized; otherwise, they are.

RE_NO_POSIX_BACKTRACKING

If this bit is set, succeed as soon as we match the whole pattern, without further backtracking. This means that a match may not be the leftmost longest; see [Section 18.5 \[What Gets Matched?\]](#), [page 936](#), for what this means.

RE_NO_SUB

If this bit is set, then **no_sub** will be set to one during **re_compile_pattern**. This causes matching and searching routines not to record substring match information.

RE_UNMATCHED_RIGHT_PAREN_ORD

If this bit is set and the regular expression has no matching open-group operator, then Regex considers what would otherwise be a close-group operator (based on how **RE_NO_BK_PARENS** is set) to match ‘)’.

18.2.2 Predefined Syntaxes

If you’re programming with Regex, you can set a pattern buffer’s (see [Section 18.6.1.1 \[GNU Pattern Buffers\]](#), [page 936](#)) syntax either to an arbitrary combination of syntax bits (see [Section 18.2.1 \[Syntax Bits\]](#), [page 922](#)) or else to the configurations defined by Regex. These configurations define the syntaxes used by certain programs—GNU Emacs, POSIX Awk, traditional Awk, Grep, Egrep—in addition to syntaxes for POSIX basic and extended regular expressions.

The predefined syntaxes—taken directly from **regex.h**—are:

```
# define RE_SYNTAX_EMACS \
```



```

(RE_CHAR_CLASSES | RE_INTERVALS)

#define RE_SYNTAX_AWK \
    (RE_BACKSLASH_ESCAPE_IN_LISTS | RE_DOT_NOT_NULL \
    | RE_NO_BK_PARENS | RE_NO_BK_REFS \
    | RE_NO_BK_VBAR | RE_NO_EMPTY_RANGES \
    | RE_UNMATCHED_RIGHT_PAREN_ORD)

#define RE_SYNTAX_POSIX_AWK \
    (RE_SYNTAX_POSIX_EXTENDED | RE_BACKSLASH_ESCAPE_IN_LISTS)

#define RE_SYNTAX_GREP \
    (RE_BK_PLUS_QM | RE_CHAR_CLASSES \
    | RE_HAT_LISTS_NOT_NEWLINE | RE_INTERVALS \
    | RE_NEWLINE_ALT)

#define RE_SYNTAX_EGREP \
    (RE_CHAR_CLASSES | RE_CONTEXT_INDEP_ANCHORS \
    | RE_CONTEXT_INDEP_OPS | RE_HAT_LISTS_NOT_NEWLINE \
    | RE_NEWLINE_ALT | RE_NO_BK_PARENS \
    | RE_NO_BK_VBAR)

#define RE_SYNTAX_POSIX_EGREP \
    (RE_SYNTAX_EGREP | RE_INTERVALS | RE_NO_BK_BRACES)

/* P1003.2/D11.2, section 4.20.7.1, lines 5078ff. */
#define RE_SYNTAX_ED RE_SYNTAX_POSIX_BASIC

#define RE_SYNTAX_SED RE_SYNTAX_POSIX_BASIC

/* Syntax bits common to both basic and extended POSIX regex syntax. */
#define _RE_SYNTAX_POSIX_COMMON \
    (RE_CHAR_CLASSES | RE_DOT_NEWLINE | RE_DOT_NOT_NULL \
    | RE_INTERVALS | RE_NO_EMPTY_RANGES)

#define RE_SYNTAX_POSIX_BASIC \
    (_RE_SYNTAX_POSIX_COMMON | RE_BK_PLUS_QM)

/* Differs from ..._POSIX_BASIC only in that RE_BK_PLUS_QM becomes
   RE_LIMITED_OPS, i.e., \? \+ \| are not recognized. Actually, this
   isn't minimal, since other operators, such as \`, aren't disabled. */
#define RE_SYNTAX_POSIX_MINIMAL_BASIC \
    (_RE_SYNTAX_POSIX_COMMON | RE_LIMITED_OPS)

#define RE_SYNTAX_POSIX_EXTENDED \
    (_RE_SYNTAX_POSIX_COMMON | RE_CONTEXT_INDEP_ANCHORS \
    | RE_CONTEXT_INDEP_OPS | RE_NO_BK_BRACES \
    | RE_NO_BK_PARENS | RE_NO_BK_VBAR \
    | RE_UNMATCHED_RIGHT_PAREN_ORD)

/* Differs from ..._POSIX_EXTENDED in that RE_CONTEXT_INVALID_OPS
   replaces RE_CONTEXT_INDEP_OPS and RE_NO_BK_REFS is added. */
#define RE_SYNTAX_POSIX_MINIMAL_EXTENDED \
    (_RE_SYNTAX_POSIX_COMMON | RE_CONTEXT_INDEP_ANCHORS \
    | RE_CONTEXT_INVALID_OPS | RE_NO_BK_BRACES \
    | RE_NO_BK_PARENS | RE_NO_BK_REFS \
    | RE_NO_BK_VBAR | RE_UNMATCHED_RIGHT_PAREN_ORD)

```

18.2.3 Collating Elements vs. Characters

POSIX generalizes the notion of a character to that of a collating element. It defines a *collating element* to be “a sequence of one or more bytes defined in the current collating sequence as a unit of collation.”

This generalizes the notion of a character in two ways. First, a single character can map into two or more collating elements. For example, the German “ß” collates as the collating element ‘s’ followed by another collating element ‘s’. Second, two or more characters can map into one collating element. For example, the Czech ‘ch’ collates after ‘h’ and before ‘i’.

Since POSIX’s “collating element” preserves the essential idea of a “character,” we use the latter, more familiar, term in this document.

18.2.4 The Backslash Character

The ‘\’ character has one of four different meanings, depending on the context in which you use it and what syntax bits are set (see [Section 18.2.1 \[Syntax Bits\]](#), page 922). It can: 1) stand for itself, 2) quote the next character, 3) introduce an operator, or 4) do nothing.

1. It stands for itself inside a list (see [Section 18.3.6 \[List Operators\]](#), page 930) if the syntax bit `RE_BACKSLASH_ESCAPE_IN_LISTS` is not set. For example, ‘[\\]’ would match ‘\’.
2. It quotes (makes ordinary, if it’s special) the next character when you use it either:
 - outside a list,¹ or
 - inside a list and the syntax bit `RE_BACKSLASH_ESCAPE_IN_LISTS` is set.
3. It introduces an operator when followed by certain ordinary characters—sometimes only when certain syntax bits are set. See the cases `RE_BK_PLUS_QM`, `RE_NO_BK_BRACES`, `RE_NO_BK_VAR`, `RE_NO_BK_PARENS`, `RE_NO_BK_REF` in [Section 18.2.1 \[Syntax Bits\]](#), page 922. Also:
 - ‘\b’ represents the match-word-boundary operator (see [Section 18.4.1.1 \[Match-word-boundary Operator\]](#), page 935).
 - ‘\B’ represents the match-within-word operator (see [Section 18.4.1.2 \[Match-within-word Operator\]](#), page 935).
 - ‘\<’ represents the match-beginning-of-word operator (see [Section 18.4.1.3 \[Match-beginning-of-word Operator\]](#), page 935).
 - ‘\>’ represents the match-end-of-word operator (see [Section 18.4.1.4 \[Match-end-of-word Operator\]](#), page 935).
 - ‘\w’ represents the match-word-constituent operator (see [Section 18.4.1.5 \[Match-word-constituent Operator\]](#), page 935).

¹ Sometimes you don’t have to explicitly quote special characters to make them ordinary. For instance, most characters lose any special meaning inside a list (see [Section 18.3.6 \[List Operators\]](#), page 930). In addition, if the syntax bits `RE_CONTEXT_INVALID_OPS` and `RE_CONTEXT_INDEP_OPS` aren’t set, then (for historical reasons) the matcher considers special characters ordinary if they are in contexts where the operations they represent make no sense; for example, then the match-zero-or-more operator (represented by ‘*’) matches itself in the regular expression ‘*foo’ because there is no preceding expression on which it can operate. It is poor practice, however, to depend on this behavior; if you want a special character to be ordinary outside a list, it’s better to always quote it, regardless.

- ‘\W’ represents the match-non-word-constituent operator (see [Section 18.4.1.6 \[Match-non-word-constituent Operator\]](#), page 935).
- ‘\sclass’ is equivalent to `[[:space:]]` (see [Section 18.4.2.1 \[Match-space Operator\]](#), page 935).
- ‘\Sclass’ is equivalent to `[^[:space:]]` (see [Section 18.4.2.2 \[Match-non-space Operator\]](#), page 935).
- ‘\’ represents the match-beginning-of-string operator and ‘\’ represents the match-end-of-string operator (see [Section 18.4.3 \[Whole-string Operators\]](#), page 935).

4. In all other cases, Regex ignores ‘\’. For example, ‘\n’ matches ‘n’.

18.3 Common Operators

You compose regular expressions from operators. In the following sections, we describe the regular expression operators specified by POSIX; GNU also uses these. Most operators have more than one representation as characters. See [Section 18.2 \[Regular Expression Syntax\]](#), page 921, for what characters represent what operators under what circumstances.

For most operators that can be represented in two ways, one representation is a single character and the other is that character preceded by ‘\’. For example, either ‘(’ or ‘\(’ represents the open-group operator. Which one does depends on the setting of a syntax bit, in this case `RE_NO_BK_PARENS`. Why is this so? Historical reasons dictate some of the varying representations, while POSIX dictates others.

Finally, almost all characters lose any special meaning inside a list (see [Section 18.3.6 \[List Operators\]](#), page 930).

18.3.1 The Match-self Operator (*ordinary character*)

This operator matches the character itself. All ordinary characters (see [Section 18.2 \[Regular Expression Syntax\]](#), page 921) represent this operator. For example, ‘f’ is always an ordinary character, so the regular expression ‘f’ matches only the string ‘f’. In particular, it does *not* match the string ‘ff’.

18.3.2 The Match-any-character Operator (.)

This operator matches any single printing or nonprinting character except it won’t match a:

newline if the syntax bit `RE_DOT_NEWLINE` isn’t set.

null if the syntax bit `RE_DOT_NOT_NULL` is set.

The ‘.’ (period) character represents this operator. For example, ‘a.b’ matches any three-character string beginning with ‘a’ and ending with ‘b’.

18.3.3 The Concatenation Operator

This operator concatenates two regular expressions *a* and *b*. No character represents this operator; you simply put *b* after *a*. The result is a regular expression that will match a string if *a* matches its first part and *b* matches the rest. For example, ‘xy’ (two match-self operators) matches ‘xy’.

18.3.4 Repetition Operators

Repetition operators repeat the preceding regular expression a specified number of times.

18.3.4.1 The Match-zero-or-more Operator (*)

This operator repeats the smallest possible preceding regular expression as many times as necessary (including zero) to match the pattern. ‘*’ represents this operator. For example, ‘o*’ matches any string made up of zero or more ‘o’s. Since this operator operates on the smallest preceding regular expression, ‘fo*’ has a repeating ‘o’, not a repeating ‘fo’. So, ‘fo*’ matches ‘f’, ‘fo’, ‘foo’, and so on.

Since the match-zero-or-more operator is a suffix operator, it may be useless as such when no regular expression precedes it. This is the case when it:

- is first in a regular expression, or
- follows a match-beginning-of-line, open-group, or alternation operator.

Three different things can happen in these cases:

1. If the syntax bit `RE_CONTEXT_INVALID_OPS` is set, then the regular expression is invalid.
2. If `RE_CONTEXT_INVALID_OPS` isn’t set, but `RE_CONTEXT_INDEP_OPS` is, then ‘*’ represents the match-zero-or-more operator (which then operates on the empty string).
3. Otherwise, ‘*’ is ordinary.

The matcher processes a match-zero-or-more operator by first matching as many repetitions of the smallest preceding regular expression as it can. Then it continues to match the rest of the pattern.

If it can’t match the rest of the pattern, it backtracks (as many times as necessary), each time discarding one of the matches until it can either match the entire pattern or be certain that it cannot get a match. For example, when matching ‘ca*ar’ against ‘caaar’, the matcher first matches all three ‘a’s of the string with the ‘a*’ of the regular expression. However, it cannot then match the final ‘ar’ of the regular expression against the final ‘r’ of the string. So it backtracks, discarding the match of the last ‘a’ in the string. It can then match the remaining ‘ar’.

18.3.4.2 The Match-one-or-more Operator (+ or \+)

If the syntax bit `RE_LIMITED_OPS` is set, then Regex doesn’t recognize this operator. Otherwise, if the syntax bit `RE_BK_PLUS_QM` isn’t set, then ‘+’ represents this operator; if it is, then ‘\+’ does.

This operator is similar to the match-zero-or-more operator except that it repeats the preceding regular expression at least once; see [Section 18.3.4.1 \[Match-zero-or-more Operator\]](#), [page 928](#), for what it operates on, how some syntax bits affect it, and how Regex backtracks to match it.

For example, supposing that ‘+’ represents the match-one-or-more operator; then ‘ca+r’ matches, e.g., ‘car’ and ‘caaar’, but not ‘cr’.

18.3.4.3 The Match-zero-or-one Operator (? or \?)

If the syntax bit `RE_LIMITED_OPS` is set, then Regex doesn’t recognize this operator. Otherwise, if the syntax bit `RE_BK_PLUS_QM` isn’t set, then ‘?’ represents this operator; if it is, then ‘\?’ does.

This operator is similar to the match-zero-or-more operator except that it repeats the preceding regular expression once or not at all; see [Section 18.3.4.1 \[Match-zero-or-more Operator\]](#), page 928, to see what it operates on, how some syntax bits affect it, and how Regex backtracks to match it.

For example, supposing that ‘?’ represents the match-zero-or-one operator; then ‘ca?r’ matches both ‘car’ and ‘cr’, but nothing else.

18.3.4.4 Interval Operators (`{ ... }` or `\{ ... \}`)

If the syntax bit `RE_INTERVALS` is set, then Regex recognizes *interval expressions*. They repeat the smallest possible preceding regular expression a specified number of times.

If the syntax bit `RE_NO_BK_BRACES` is set, ‘{’ represents the *open-interval operator* and ‘}’ represents the *close-interval operator*; otherwise, ‘\{’ and ‘\}’ do.

Specifically, supposing that ‘{’ and ‘}’ represent the open-interval and close-interval operators; then:

- `{count}` matches exactly *count* occurrences of the preceding regular expression.
- `{min,}` matches *min* or more occurrences of the preceding regular expression.
- `{min, max}` matches at least *min* but no more than *max* occurrences of the preceding regular expression.

The interval expression (but not necessarily the regular expression that contains it) is invalid if:

- *min* is greater than *max*, or
- any of *count*, *min*, or *max* are outside the range zero to `RE_DUP_MAX` (which symbol `regex.h` defines).

If the interval expression is invalid and the syntax bit `RE_NO_BK_BRACES` is set, then Regex considers all the characters in the would-be interval to be ordinary. If that bit isn’t set, then the regular expression is invalid.

If the interval expression is valid but there is no preceding regular expression on which to operate, then if the syntax bit `RE_CONTEXT_INVALID_OPS` is set, the regular expression is invalid. If that bit isn’t set, then Regex considers all the characters—other than backslashes, which it ignores—in the would-be interval to be ordinary.

18.3.5 The Alternation Operator (`|` or `\|`)

If the syntax bit `RE_LIMITED_OPS` is set, then Regex doesn’t recognize this operator. Otherwise, if the syntax bit `RE_NO_BK_VBAR` is set, then ‘|’ represents this operator; otherwise, ‘\|’ does.

Alternatives match one of a choice of regular expressions: if you put the character(s) representing the alternation operator between any two regular expressions *a* and *b*, the result matches the union of the strings that *a* and *b* match. For example, supposing that ‘|’ is the alternation operator, then ‘foo|bar|quux’ would match any of ‘foo’, ‘bar’ or ‘quux’.

The alternation operator operates on the *largest* possible surrounding regular expressions. (Put another way, it has the lowest precedence of any regular expression operator.) Thus, the only way you can delimit its arguments is to use grouping. For example, if ‘(’ and

‘)’ are the open and close-group operators, then ‘fo(o|b)ar’ would match either ‘fooar’ or ‘fobar’. (‘foo|bar’ would match ‘foo’ or ‘bar’.)

The matcher usually tries all combinations of alternatives so as to match the longest possible string. For example, when matching ‘(fooq|foo)*(qbarquux|bar)’ against ‘fooqbarquux’, it cannot take, say, the first (“depth-first”) combination it could match, since then it would be content to match just ‘fooqbar’.

Note that since the default behavior is to return the leftmost longest match, when more than one of a series of alternatives matches the actual match will be the longest matching alternative, not necessarily the first in the list.

18.3.6 List Operators ([...] and [^ ...])

Lists, also called *bracket expressions*, are a set of one or more items. An *item* is a character, a collating symbol, an equivalence class expression, a character class expression, or a range expression. The syntax bits affect which kinds of items you can put in a list. We explain the last four items in subsections below. Empty lists are invalid.

A *matching list* matches a single character represented by one of the list items. You form a matching list by enclosing one or more items within an *open-matching-list operator* (represented by ‘[’) and a *close-list operator* (represented by ‘]’).

For example, ‘[ab]’ matches either ‘a’ or ‘b’. ‘[ad]*’ matches the empty string and any string composed of just ‘a’s and ‘d’s in any order. Regex considers invalid a regular expression with a ‘[’ but no matching ‘]’.

Nonmatching lists are similar to matching lists except that they match a single character *not* represented by one of the list items. You use an *open-nonmatching-list operator* (represented by ‘[^’²) instead of an open-matching-list operator to start a nonmatching list.

For example, ‘[^ab]’ matches any character except ‘a’ or ‘b’.

If the syntax bit RE_HAT_LISTS_NOT_NEWLINE is set, then nonmatching lists do not match a newline.

Most characters lose any special meaning inside a list. The special characters inside a list follow.

- ‘]’ ends the list if it’s not the first list item. So, if you want to make the ‘]’ character a list item, you must put it first.
- ‘\’ quotes the next character if the syntax bit RE_BACKSLASH_ESCAPE_IN_LISTS is set.
- ‘[.’ represents the open-collating-symbol operator (see [Section 18.3.6.1 \[Collating Symbol Operators\]](#), page 931).
- ‘.]’ represents the close-collating-symbol operator.
- ‘[=’ represents the open-equivalence-class operator (see [Section 18.3.6.2 \[Equivalence Class Operators\]](#), page 931).
- ‘=]’ represents the close-equivalence-class operator.

² Regex therefore doesn’t consider the ‘^’ to be the first character in the list. If you put a ‘^’ character first in (what you think is) a matching list, you’ll turn it into a nonmatching list.

- ‘[:’ represents the open-character-class operator (see [Section 18.3.6.3 \[Character Class Operators\]](#), page 931) if the syntax bit `RE_CHAR_CLASSES` is set and what follows is a valid character class expression.
- ‘:]’ represents the close-character-class operator if the syntax bit `RE_CHAR_CLASSES` is set and what precedes it is an open-character-class operator followed by a valid character class name.
- ‘-’ represents the range operator (see [Section 18.3.6.4 \[Range Operator\]](#), page 932) if it’s not first or last in a list or the ending point of a range.

All other characters are ordinary. For example, ‘[.*]’ matches ‘.’ and ‘*’.

18.3.6.1 Collating Symbol Operators ([. . . .])

Collating symbols can be represented inside lists. You form a *collating symbol* by putting a collating element between an *open-collating-symbol operator* and a *close-collating-symbol operator*. ‘[.]’ represents the open-collating-symbol operator and ‘.]’ represents the close-collating-symbol operator. For example, if ‘11’ is a collating element, then ‘[[.11.]’ would match ‘11’.

18.3.6.2 Equivalence Class Operators ([= . . . =])

Regex recognizes equivalence class expressions inside lists. A *equivalence class expression* is a set of collating elements which all belong to the same equivalence class. You form an equivalence class expression by putting a collating element between an *open-equivalence-class operator* and a *close-equivalence-class operator*. ‘[=’ represents the open-equivalence-class operator and ‘=]’ represents the close-equivalence-class operator. For example, if ‘a’ and ‘A’ were an equivalence class, then both ‘[[=a=]]’ and ‘[[=A=]]’ would match both ‘a’ and ‘A’. If the collating element in an equivalence class expression isn’t part of an equivalence class, then the matcher considers the equivalence class expression to be a collating symbol.

18.3.6.3 Character Class Operators ([: . . . :])

If the syntax bit `RE_CHAR_CLASSES` is set, then Regex recognizes character class expressions inside lists. A *character class expression* matches one character from a given class. You form a character class expression by putting a character class name between an *open-character-class operator* (represented by ‘[:]’ and a *close-character-class operator* (represented by ‘:]’). The character class names and their meanings are:

<code>alnum</code>	letters and digits
<code>alpha</code>	letters
<code>blank</code>	system-dependent; for GNU, a space or tab
<code>cntrl</code>	control characters (in the ASCII encoding, code 0177 and codes less than 040)
<code>digit</code>	digits
<code>graph</code>	same as <code>print</code> except omits space
<code>lower</code>	lowercase letters
<code>print</code>	printable characters (in the ASCII encoding, space tilde—codes 040 through 0176)

<code>punct</code>	neither control nor alphanumeric characters
<code>space</code>	space, carriage return, newline, vertical tab, and form feed
<code>upper</code>	uppercase letters
<code>xdigit</code>	hexadecimal digits: 0–9, a–f, A–F

These correspond to the definitions in the C library’s `<ctype.h>` facility. For example, `[:alpha:]` corresponds to the standard facility `isalpha`. Regex recognizes character class expressions only inside of lists; so `[[[:alpha:]]]` matches any letter, but `[:alpha:]` outside of a bracket expression and not followed by a repetition operator matches just itself.

18.3.6.4 The Range Operator (-)

Regex recognizes *range expressions* inside a list. They represent those characters that fall between two elements in the current collating sequence. You form a range expression by putting a *range operator* between two of any of the following: characters, collating elements, collating symbols, and equivalence class expressions. The starting point of the range and the ending point of the range don’t have to be the same kind of item, e.g., the starting point could be a collating element and the ending point could be an equivalence class expression. If a range’s ending point is an equivalence class, then all the collating elements in that class will be in the range.³ ‘-’ represents the range operator. For example, ‘a–f’ within a list represents all the characters from ‘a’ through ‘f’ inclusively.

If the syntax bit `RE_NO_EMPTY_RANGES` is set, then if the range’s ending point collates less than its starting point, the range (and the regular expression containing it) is invalid. For example, the regular expression `[z–a]` would be invalid. If this bit isn’t set, then Regex considers such a range to be empty.

Since ‘-’ represents the range operator, if you want to make a ‘-’ character itself a list item, you must do one of the following:

- Put the ‘-’ either first or last in the list.
- Include a range whose starting point collates strictly lower than ‘-’ and whose ending point collates equal or higher. Unless a range is the first item in a list, a ‘-’ can’t be its starting point, but *can* be its ending point. That is because Regex considers ‘-’ to be the range operator unless it is preceded by another ‘-’. For example, in the ASCII encoding, ‘)’, ‘*’, ‘+’, ‘,’ , ‘-’, ‘.’, and ‘/’ are contiguous characters in the collating sequence. You might think that `[)-+--/]` has two ranges: ‘)-+’ and ‘--/’. Rather, it has the ranges ‘)-+’ and ‘+--’, plus the character ‘/’, so it matches, e.g., ‘,’ , not ‘.’.
- Put a range whose starting point is ‘-’ first in the list.

For example, `[-a–z]` matches a lowercase letter or a hyphen (in English, in ASCII).

18.3.7 Grouping Operators ((...) or \ (... \))

A *group*, also known as a *subexpression*, consists of an *open-group operator*, any number of other operators, and a *close-group operator*. Regex treats this sequence as a unit, just as mathematics and programming languages treat a parenthesized expression as a unit.

³ You can’t use a character class for the starting or ending point of a range, since a character class is not a single character.

Therefore, using *groups*, you can:

- delimit the argument(s) to an alternation operator (see [Section 18.3.5 \[Alternation Operator\]](#), page 929) or a repetition operator (see [Section 18.3.4 \[Repetition Operators\]](#), page 928).
- keep track of the indices of the substring that matched a given group. See [Section 18.6.1.8 \[Using Registers\]](#), page 941, for a precise explanation. This lets you:
 - use the back-reference operator (see [Section 18.3.8 \[Back-reference Operator\]](#), page 933).
 - use registers (see [Section 18.6.1.8 \[Using Registers\]](#), page 941).

If the syntax bit `RE_NO_BK_PARENS` is set, then `'(` represents the open-group operator and `)'` represents the close-group operator; otherwise, `'\('` and `'\)'` do.

If the syntax bit `RE_UNMATCHED_RIGHT_PAREN_ORD` is set and a close-group operator has no matching open-group operator, then Regex considers it to match `)'`.

18.3.8 The Back-reference Operator (`\digit`)

If the syntax bit `RE_NO_BK_REF` isn't set, then Regex recognizes back-references. A back-reference matches a specified preceding group. The back-reference operator is represented by `'\digit'` anywhere after the end of a regular expression's *digit*-th group (see [Section 18.3.7 \[Grouping Operators\]](#), page 932).

digit must be between `'1'` and `'9'`. The matcher assigns numbers 1 through 9 to the first nine groups it encounters. By using one of `'\1'` through `'\9'` after the corresponding group's close-group operator, you can match a substring identical to the one that the group does.

Back-references match according to the following (in all examples below, `'(` represents the open-group, `)'` the close-group, `'{'` the open-interval and `'}'` the close-interval operator):

- If the group matches a substring, the back-reference matches an identical substring. For example, `'(a)\1'` matches `'aa'` and `'(bana)na\1bo\1'` matches `'bananabanabobana'`. Likewise, `'(.*)\1'` matches any (newline-free if the syntax bit `RE_DOT_NEWLINE` isn't set) string that is composed of two identical halves; the `'(.*)'` matches the first half and the `'\1'` matches the second half.
- If the group matches more than once (as it might if followed by, e.g., a repetition operator), then the back-reference matches the substring the group *last* matched. For example, `'((a*)b)*\1\2'` matches `'aabababa'`; first group 1 (the outer one) matches `'aab'` and group 2 (the inner one) matches `'aa'`. Then group 1 matches `'ab'` and group 2 matches `'a'`. So, `'\1'` matches `'ab'` and `'\2'` matches `'a'`.
- If the group doesn't participate in a match, i.e., it is part of an alternative not taken or a repetition operator allows zero repetitions of it, then the back-reference makes the whole match fail. For example, `'(one()|two())-and-(three\2|four\3)'` matches `'one-and-three'` and `'two-and-four'`, but not `'one-and-four'` or `'two-and-three'`. For example, if the pattern matches `'one-and-'`, then its group 2 matches the empty string and its group 3 doesn't participate in the match. So, if it then matches `'four'`, then when it tries to back-reference group 3—which it will attempt to do because `'\3'` follows the `'four'`—the match will fail because group 3 didn't participate in the match.

You can use a back-reference as an argument to a repetition operator. For example, `(a(b))\2*` matches `a` followed by two or more `b`'s. Similarly, `(a(b))\2{3}` matches `abbbb`.

If there is no preceding *digit*-th subexpression, the regular expression is invalid.

Back-references can greatly slow down matching, as they can generate exponentially many matching possibilities that can consume both time and memory to explore. Also, the POSIX specification for back-references is at times unclear. Furthermore, many regular expression implementations have back-reference bugs that can cause programs to return incorrect answers or even crash, and fixing these bugs has often been low-priority: for example, as of 2020 the [GNU C library bug database \(https://sourceware.org/bugzilla/\)](https://sourceware.org/bugzilla/) contained back-reference bugs [52](#), [10844](#), [11053](#), [24269](#) and [25322](#), with little sign of forthcoming fixes. Luckily, back-references are rarely useful and it should be little trouble to avoid them in practical applications.

18.3.9 Anchoring Operators

These operators can constrain a pattern to match only at the beginning or end of the entire string or at the beginning or end of a line.

18.3.9.1 The Match-beginning-of-line Operator (^)

This operator can match the empty string either at the beginning of the string or after a newline character. Thus, it is said to *anchor* the pattern to the beginning of a line.

In the cases following, `^` represents this operator. (Otherwise, `^` is ordinary.)

- It (the `^`) is first in the pattern, as in `^foo`.
- The syntax bit `RE_CONTEXT_INDEP_ANCHORS` is set, and it is outside a bracket expression.
- It follows an open-group or alternation operator, as in `a\(^b\)` and `a\|^b`. See [Section 18.3.7 \[Grouping Operators\]](#), page 932, and [Section 18.3.5 \[Alternation Operator\]](#), page 929.

These rules imply that some valid patterns containing `^` cannot be matched; for example, `foo^bar` if `RE_CONTEXT_INDEP_ANCHORS` is set.

If the `not_bol` field is set in the pattern buffer (see [Section 18.6.1.1 \[GNU Pattern Buffers\]](#), page 936), then `^` fails to match at the beginning of the string. This lets you match against pieces of a line, as you would need to if, say, searching for repeated instances of a given pattern in a line; it would work correctly for patterns both with and without match-beginning-of-line operators.

18.3.9.2 The Match-end-of-line Operator (\$)

This operator can match the empty string either at the end of the string or before a newline character in the string. Thus, it is said to *anchor* the pattern to the end of a line.

It is always represented by `$`. For example, `foo$` usually matches, e.g., `foo` and, e.g., the first three characters of `foo\nbar`.

Its interaction with the syntax bits and pattern buffer fields is exactly the dual of `^`'s; see the previous section. (That is, `“^”` becomes `“$”`, `“beginning”` becomes `“end”`, `“next”` becomes `“previous”`, `“after”` becomes `“before”`, and `“not_bol”` becomes `“not_eol”`.)

18.4 GNU Operators

The following are operators that GNU defines (and POSIX doesn't) that you can use unless the syntax bit `RE_NO_GNU_OPS` is set.

18.4.1 Word Operators

The operators in this section require Regex to recognize parts of words. Characters that are part of words, which are called *word-constituent*, are letters, digits, and the underscore ('_'); more precisely, any character in the POSIX class `alnum` in the current locale, or underscore.

18.4.1.1 The Match-word-boundary Operator (`\b`)

This operator (represented by '`\b`') matches the empty string at either the beginning or the end of a word. For example, '`\brat\b`' matches the separate word '`rat`'.

18.4.1.2 The Match-within-word Operator (`\B`)

This operator (represented by '`\B`') matches the empty string within a word. For example, '`c\Brat\Be`' matches '`crate`', but '`dirty \Brat`' doesn't match '`dirty rat`'.

18.4.1.3 The Match-beginning-of-word Operator (`\<`)

This operator (represented by '`\<`') matches the empty string at the beginning of a word.

18.4.1.4 The Match-end-of-word Operator (`\>`)

This operator (represented by '`\>`') matches the empty string at the end of a word.

18.4.1.5 The Match-word-constituent Operator (`\w`)

This operator (represented by '`\w`') matches any word-constituent character.

18.4.1.6 The Match-non-word-constituent Operator (`\W`)

This operator (represented by '`\W`') matches any character that is not word-constituent.

18.4.2 Space Operators

18.4.2.1 The Match-space Operator (`\s`)

This operator (represented by '`\s`') matches any space character (that is, in the POSIX class `[:space:]`).

18.4.2.2 The Match-non-space Operator (`\S`)

This operator (represented by '`\S`') matches any character that is not a space (that is, in the POSIX class `[:space:]`).

18.4.3 Whole-string Operators

Following are operators which work on the whole string.

18.4.3.1 The Match-beginning-of-string Operator (`\^`)

This operator (represented by '`\^`') matches the empty string at the beginning of the string.

18.4.3.2 The Match-end-of-string Operator (`\'`)

This operator (represented by `\'`) matches the empty string at the end of the string.

18.5 What Gets Matched?

Regex usually matches strings according to the “leftmost longest” rule; that is, it chooses the longest of the leftmost matches. This does not mean that for a regular expression containing subexpressions that it simply chooses the longest match for each subexpression, left to right; the overall match must also be the longest possible one.

For example, `(ac*)(c*d[ac]*)\1` matches `acdacaaa`, not `acdac`, as it would if it were to choose the longest match for the first subexpression.

18.6 Programming with Regex

Here we describe how you use the Regex data structures and functions in C programs. Regex has three interfaces: one designed for GNU, one compatible with POSIX (as specified by POSIX, draft 1003.2/D11.2), and one compatible with Berkeley Unix. The POSIX interface is not documented here; see the documentation of GNU libc, or the POSIX man pages. The Berkeley Unix interface is documented here for convenience, since its documentation is not otherwise readily available on GNU systems.

18.6.1 GNU Regex Functions

If you’re writing code that doesn’t need to be compatible with either POSIX or Berkeley Unix, you can use these functions. They provide more options than the other interfaces.

18.6.1.1 GNU Pattern Buffers

To compile, match, or search for a given regular expression, you must supply a pattern buffer. A *pattern buffer* holds one compiled regular expression.⁴

You can have several different pattern buffers simultaneously, each holding a compiled pattern for a different regular expression.

`regex.h` defines the pattern buffer `struct` with the following public fields:

```
unsigned char *buffer;
unsigned long allocated;
char *fastmap;
char *translate;
size_t re_nsub;
unsigned no_sub : 1;
unsigned not_bol : 1;
unsigned not_eol : 1;
```

18.6.1.2 GNU Regular Expression Compiling

In GNU, you can both match and search for a given regular expression. To do either, you must first compile it in a pattern buffer (see [Section 18.6.1.1 \[GNU Pattern Buffers\]](#), [page 936](#)).

⁴ Regular expressions are also referred to as “patterns,” hence the name “pattern buffer.”

Regular expressions match according to the syntax with which they were compiled; with GNU, you indicate what syntax you want by setting the variable `re_syntax_options` (declared in `regex.h`) before calling the compiling function, `re_compile_pattern` (see below). See [Section 18.2.1 \[Syntax Bits\]](#), page 922, and [Section 18.2.2 \[Predefined Syntaxes\]](#), page 924.

You can change the value of `re_syntax_options` at any time. Usually, however, you set its value once and then never change it.

`re_compile_pattern` takes a pattern buffer as an argument. You must initialize the following fields:

`translate` initialization
`translate`

Initialize this to point to a translate table if you want one, or to zero if you don't. We explain translate tables in [Section 18.6.1.7 \[GNU Translate Tables\]](#), page 940.

`fastmap` Initialize this to nonzero if you want a fastmap, or to zero if you don't.

`buffer`
`allocated`

If you want `re_compile_pattern` to allocate memory for the compiled pattern, set both of these to zero. If you have an existing block of memory (allocated with `malloc`) you want Regex to use, set `buffer` to its address and `allocated` to its size (in bytes).

`re_compile_pattern` uses `realloc` to extend the space for the compiled pattern as necessary.

To compile a pattern buffer, use:

```
char *
re_compile_pattern (const char *regex, const int regex_size,
                   struct re_pattern_buffer *pattern_buffer)
```

`regex` is the regular expression's address, `regex_size` is its length, and `pattern_buffer` is the pattern buffer's address.

If `re_compile_pattern` successfully compiles the regular expression, it returns zero and sets `*pattern_buffer` to the compiled pattern. It sets the pattern buffer's fields as follows:

`buffer` to the compiled pattern.
`syntax` to the current value of `re_syntax_options`.
`re_nsub` to the number of subexpressions in `regex`.

If `re_compile_pattern` can't compile `regex`, it returns an error string corresponding to a POSIX error code.

18.6.1.3 GNU Matching

Matching the GNU way means trying to match as much of a string as possible starting at a position within it you specify. Once you've compiled a pattern into a pattern buffer (see [Section 18.6.1.2 \[GNU Regular Expression Compiling\]](#), page 936), you can ask the matcher to match that pattern against a string using:

```
int
```

```

re_match (struct re_pattern_buffer *pattern_buffer,
          const char *string, const int size,
          const int start, struct re_registers *regs)

```

pattern_buffer is the address of a pattern buffer containing a compiled pattern. *string* is the string you want to match; it can contain newline and null characters. *size* is the length of that string. *start* is the string index at which you want to begin matching; the first character of *string* is at index zero. See [Section 18.6.1.8 \[Using Registers\]](#), page 941, for an explanation of *regs*; you can safely pass zero.

re_match matches the regular expression in *pattern_buffer* against the string *string* according to the syntax of *pattern_buffer*. (See [Section 18.6.1.2 \[GNU Regular Expression Compiling\]](#), page 936, for how to set it.) The function returns -1 if the compiled pattern does not match any part of *string* and -2 if an internal error happens; otherwise, it returns how many (possibly zero) characters of *string* the pattern matched.

An example: suppose *pattern_buffer* points to a pattern buffer containing the compiled pattern for `'a*'`, and *string* points to `'aaaaab'` (whereupon *size* should be 6). Then if *start* is 2, **re_match** returns 3, i.e., `'a*'` would have matched the last three `'a'`s in *string*. If *start* is 0, **re_match** returns 5, i.e., `'a*'` would have matched all the `'a'`s in *string*. If *start* is either 5 or 6, it returns zero.

If *start* is not between zero and *size*, then **re_match** returns -1 .

18.6.1.4 GNU Searching

Searching means trying to match starting at successive positions within a string. The function **re_search** does this.

Before calling **re_search**, you must compile your regular expression. See [Section 18.6.1.2 \[GNU Regular Expression Compiling\]](#), page 936.

Here is the function declaration:

```

int
re_search (struct re_pattern_buffer *pattern_buffer,
          const char *string, const int size,
          const int start, const int range,
          struct re_registers *regs)

```

whose arguments are the same as those to **re_match** (see [Section 18.6.1.3 \[GNU Matching\]](#), page 937) except that the two arguments *start* and *range* replace **re_match**'s argument *start*.

If *range* is positive, then **re_search** attempts a match starting first at index *start*, then at *start* + 1 if that fails, and so on, up to *start* + *range*; if *range* is negative, then it attempts a match starting first at index *start*, then at *start* - 1 if that fails, and so on.

If *start* is not between zero and *size*, then **re_search** returns -1 . When *range* is positive, **re_search** adjusts *range* so that *start* + *range* - 1 is between zero and *size*, if necessary; that way it won't search outside of *string*. Similarly, when *range* is negative, **re_search** adjusts *range* so that *start* + *range* + 1 is between zero and *size*, if necessary.

If the **fastmap** field of *pattern_buffer* is zero, **re_search** matches starting at consecutive positions; otherwise, it uses **fastmap** to make the search more efficient. See [Section 18.6.1.6 \[Searching with Fastmaps\]](#), page 939.

If no match is found, `re_search` returns `-1`. If a match is found, it returns the index where the match began. If an internal error happens, it returns `-2`.

18.6.1.5 Matching and Searching with Split Data

Using the functions `re_match_2` and `re_search_2`, you can match or search in data that is divided into two strings.

The function:

```
int
re_match_2 (struct re_pattern_buffer *buffer,
            const char *string1, const int size1,
            const char *string2, const int size2,
            const int start,
            struct re_registers *regs,
            const int stop)
```

is similar to `re_match` (see [Section 18.6.1.3 \[GNU Matching\], page 937](#)) except that you pass *two* data strings and sizes, and an index *stop* beyond which you don't want the matcher to try matching. As with `re_match`, if it succeeds, `re_match_2` returns how many characters of *string* it matched. Regard *string1* and *string2* as concatenated when you set the arguments *start* and *stop* and use the contents of *regs*; `re_match_2` never returns a value larger than *size1* + *size2*.

The function:

```
int
re_search_2 (struct re_pattern_buffer *buffer,
            const char *string1, const int size1,
            const char *string2, const int size2,
            const int start, const int range,
            struct re_registers *regs,
            const int stop)
```

is similarly related to `re_search`.

18.6.1.6 Searching with Fastmaps

If you're searching through a long string, you should use a fastmap. Without one, the searcher tries to match at consecutive positions in the string. Generally, most of the characters in the string could not start a match. It takes much longer to try matching at a given position in the string than it does to check in a table whether or not the character at that position could start a match. A *fastmap* is such a table.

More specifically, a fastmap is an array indexed by the characters in your character set. Under the ASCII encoding, therefore, a fastmap has 256 elements. If you want the searcher to use a fastmap with a given pattern buffer, you must allocate the array and assign the array's address to the pattern buffer's `fastmap` field. You either can compile the fastmap yourself or have `re_search` do it for you; when `fastmap` is nonzero, it automatically compiles a fastmap the first time you search using a particular compiled pattern.

By setting the buffer's `fastmap` field before calling `re_compile_pattern`, you can reuse a buffer data structure across multiple searches with different patterns, and allocate the

fastmap only once. Nonetheless, the fastmap must be recompiled each time the buffer has a new pattern compiled into it.

To compile a fastmap yourself, use:

```
int
re_compile_fastmap (struct re_pattern_buffer *pattern_buffer)
```

pattern_buffer is the address of a pattern buffer. If the character *c* could start a match for the pattern, `re_compile_fastmap` makes *pattern_buffer*→fastmap[*c*] nonzero. It returns 0 if it can compile a fastmap and -2 if there is an internal error. For example, if '|' is the alternation operator and *pattern_buffer* holds the compiled pattern for 'a|b', then `re_compile_fastmap` sets fastmap['a'] and fastmap['b'] (and no others).

`re_search` uses a fastmap as it moves along in the string: it checks the string's characters until it finds one that's in the fastmap. Then it tries matching at that character. If the match fails, it repeats the process. So, by using a fastmap, `re_search` doesn't waste time trying to match at positions in the string that couldn't start a match.

If you don't want `re_search` to use a fastmap, store zero in the **fastmap** field of the pattern buffer before calling `re_search`.

Once you've initialized a pattern buffer's **fastmap** field, you need never do so again—even if you compile a new pattern in it—provided the way the field is set still reflects whether or not you want a fastmap. `re_search` will still either do nothing if **fastmap** is null or, if it isn't, compile a new fastmap for the new pattern.

18.6.1.7 GNU Translate Tables

If you set the **translate** field of a pattern buffer to a translate table, then the GNU Regex functions to which you've passed that pattern buffer use it to apply a simple transformation to all the regular expression and string characters at which they look.

A *translate table* is an array indexed by the characters in your character set. Under the ASCII encoding, therefore, a translate table has 256 elements. The array's elements are also characters in your character set. When the Regex functions see a character *c*, they use **translate**[*c*] in its place, with one exception: the character after a '\' is not translated. (This ensures that, the operators, e.g., '\B' and '\b', are always distinguishable.)

For example, a table that maps all lowercase letters to the corresponding uppercase ones would cause the matcher to ignore differences in case.⁵ Such a table would map all characters except lowercase letters to themselves, and lowercase letters to the corresponding uppercase ones. Under the ASCII encoding, here's how you could initialize such a table (we'll call it *case_fold*):

```
for (i = 0; i < 256; i++)
    case_fold[i] = i;
for (i = 'a'; i <= 'z'; i++)
    case_fold[i] = i - ('a' - 'A');
```

You tell Regex to use a translate table on a given pattern buffer by assigning that table's address to the **translate** field of that buffer. If you don't want Regex to do any translation, put zero into this field. You'll get weird results if you change the table's contents anytime

⁵ A table that maps all uppercase letters to the corresponding lowercase ones would work just as well for this purpose.

between compiling the pattern buffer, compiling its fastmap, and matching or searching with the pattern buffer.

18.6.1.8 Using Registers

A group in a regular expression can match a (possibly empty) substring of the string that regular expression as a whole matched. The matcher remembers the beginning and end of the substring matched by each group.

To find out what they matched, pass a nonzero *regs* argument to a GNU matching or searching function (see [Section 18.6.1.3 \[GNU Matching\]](#), page 937, and [Section 18.6.1.4 \[GNU Searching\]](#), page 938), i.e., the address of a structure of this type, as defined in `regex.h`:

```
struct re_registers
{
    unsigned num_regs;
    regoff_t *start;
    regoff_t *end;
};
```

Except for (possibly) the *num_regs*'th element (see below), the *i*th element of the **start** and **end** arrays records information about the *i*th group in the pattern. (They're declared as C pointers, but this is only because not all C compilers accept zero-length arrays; conceptually, it is simplest to think of them as arrays.)

The **start** and **end** arrays are allocated in one of two ways. The simplest and perhaps most useful is to let the matcher (re)allocate enough space to record information for all the groups in the regular expression. If **re_set_registers** is not called before searching or matching, then the matcher allocates two arrays each of $1 + re_nsub$ elements (*re_nsub* is another field in the pattern buffer; see [Section 18.6.1.1 \[GNU Pattern Buffers\]](#), page 936). The extra element is set to -1 . Then on subsequent calls with the same pattern buffer and *regs* arguments, the matcher reallocates more space if necessary.

The function:

```
void
re_set_registers (struct re_pattern_buffer *buffer,
                 struct re_registers *regs,
                 size_t num_regs,
                 regoff_t *starts, regoff_t *ends)
```

sets *regs* to hold *num_regs* registers, storing them in *starts* and *ends*. Subsequent matches using *buffer* and *regs* will use this memory for recording register information. *starts* and *ends* must be allocated with `malloc`, and must each be at least $num_regs * sizeof(regoff_t)$ bytes long.

If *num_regs* is zero, then subsequent matches should allocate their own register data.

Unless this function is called, the first search or match using *buffer* will allocate its own register data, without freeing the old data.

The following examples illustrate the information recorded in the **re_registers** structure. (In all of them, '(' represents the open-group and ')' the close-group operator. The first character in the string *string* is at index 0.)

- If the regular expression has an i -th group that matches a substring of *string*, then the function sets `regs->start[i]` to the index in *string* where the substring matched by the i -th group begins, and `regs->end[i]` to the index just beyond that substring's end. The function sets `regs->start[0]` and `regs->end[0]` to analogous information about the entire pattern.

For example, when you match `'((a)(b))'` against `'ab'`, you get:

- 0 in `regs->start[0]` and 2 in `regs->end[0]`
- 0 in `regs->start[1]` and 2 in `regs->end[1]`
- 0 in `regs->start[2]` and 1 in `regs->end[2]`
- 1 in `regs->start[3]` and 2 in `regs->end[3]`
- If a group matches more than once (as it might if followed by, e.g., a repetition operator), then the function reports the information about what the group *last* matched.

For example, when you match the pattern `'(a)*'` against the string `'aa'`, you get:

- 0 in `regs->start[0]` and 2 in `regs->end[0]`
- 1 in `regs->start[1]` and 2 in `regs->end[1]`
- If the i -th group does not participate in a successful match, e.g., it is an alternative not taken or a repetition operator allows zero repetitions of it, then the function sets `regs->start[i]` and `regs->end[i]` to `-1`.

For example, when you match the pattern `'(a)*b'` against the string `'b'`, you get:

- 0 in `regs->start[0]` and 1 in `regs->end[0]`
- `-1` in `regs->start[1]` and `-1` in `regs->end[1]`
- If the i -th group matches a zero-length string, then the function sets `regs->start[i]` and `regs->end[i]` to the index just beyond that zero-length string.

For example, when you match the pattern `'(a*)b'` against the string `'b'`, you get:

- 0 in `regs->start[0]` and 1 in `regs->end[0]`
- 0 in `regs->start[1]` and 0 in `regs->end[1]`
- If an i -th group contains a j -th group in turn not contained within any other group within group i and the function reports a match of the i -th group, then it records in `regs->start[j]` and `regs->end[j]` the last match (if it matched) of the j -th group.

For example, when you match the pattern `'((a*)b)*'` against the string `'abb'`, group 2 last matches the empty string, so you get what it previously matched:

- 0 in `regs->start[0]` and 3 in `regs->end[0]`
- 2 in `regs->start[1]` and 3 in `regs->end[1]`
- 2 in `regs->start[2]` and 2 in `regs->end[2]`

When you match the pattern `'((a)*b)*'` against the string `'abb'`, group 2 doesn't participate in the last match, so you get:

- 0 in `regs->start[0]` and 3 in `regs->end[0]`
- 2 in `regs->start[1]` and 3 in `regs->end[1]`
- 0 in `regs->start[2]` and 1 in `regs->end[2]`

- If an i -th group contains a j -th group in turn not contained within any other group within group i and the function sets `regs->start[i]` and `regs->end[i]` to -1 , then it also sets `regs->start[j]` and `regs->end[j]` to -1 .

For example, when you match the pattern ‘`((a)*b)*c`’ against the string ‘`c`’, you get:

- 0 in `regs->start[0]` and 1 in `regs->end[0]`
- -1 in `regs->start[1]` and -1 in `regs->end[1]`
- -1 in `regs->start[2]` and -1 in `regs->end[2]`

18.6.1.9 Freeing GNU Pattern Buffers

To free any allocated fields of a pattern buffer, use the POSIX function `regfree`:

```
void
regfree (regex_t *preg)
```

`preg` is the pattern buffer whose allocated fields you want freed; this works because since the type `regex_t`—the type for POSIX pattern buffers—is equivalent to the type `re_pattern_buffer`.

`regfree` also sets `preg`’s `allocated` field to zero. After a buffer has been freed, it must have a regular expression compiled in it before passing it to a matching or searching function.

18.6.2 BSD Regex Functions

If you’re writing code that has to be Berkeley Unix compatible, you’ll need to use these functions whose interfaces are the same as those in Berkeley Unix.

18.6.2.1 BSD Regular Expression Compiling

With Berkeley Unix, you can only search for a given regular expression; you can’t match one. To search for it, you must first compile it. Before you compile it, you must indicate the regular expression syntax you want it compiled according to by setting the variable `re_syntax_options` (declared in `regex.h`) to some syntax (see [Section 18.2 \[Regular Expression Syntax\]](#), page 921).

To compile a regular expression use:

```
char *
re_comp (char *regex)
```

`regex` is the address of a null-terminated regular expression. `re_comp` uses an internal pattern buffer, so you can use only the most recently compiled pattern buffer. This means that if you want to use a given regular expression that you’ve already compiled—but it isn’t the latest one you’ve compiled—you’ll have to recompile it. If you call `re_comp` with the null string (*not* the empty string) as the argument, it doesn’t change the contents of the pattern buffer.

If `re_comp` successfully compiles the regular expression, it returns zero. If it can’t compile the regular expression, it returns an error string. `re_comp`’s error messages are identical to those of `re_compile_pattern` (see [Section 18.6.1.2 \[GNU Regular Expression Compiling\]](#), page 936).

18.6.2.2 BSD Searching

Searching the Berkeley Unix way means searching in a string starting at its first character and trying successive positions within it to find a match. Once you've compiled a pattern using `re_comp` (see [Section 18.6.2.1 \[BSD Regular Expression Compiling\]](#), page 943), you can ask `Regex` to search for that pattern in a string using:

```
int
re_exec (char *string)
```

string is the address of the null-terminated string in which you want to search.

`re_exec` returns either 1 for success or 0 for failure. It automatically uses a GNU fastmap (see [Section 18.6.1.6 \[Searching with Fastmaps\]](#), page 939).

18.7 Regular expression syntaxes

Gnlib supports many different types of regular expressions; although the underlying features are the same or identical, the syntax used varies. The descriptions given here for the different types are generated automatically.

18.7.1 'awk' regular expression syntax

The character `'.'` matches any single character except the null character.

- `'+'` indicates that the regular expression should match one or more occurrences of the previous atom or regexp.
- `'?'` indicates that the regular expression should match zero or one occurrence of the previous atom or regexp.
- `'\+'` matches a `'+'`
- `'\?'` matches a `'?'`.

Bracket expressions are used to match ranges of characters. Bracket expressions where the range is backward, for example `'[z-a]'`, are invalid. Within square brackets, `'\'` can be used to quote the following character. Character classes are supported; for example `'[:digit:]'` will match a single decimal digit.

GNU extensions are not supported and so `'\w'`, `'\W'`, `'\<'`, `'\>'`, `'\b'`, `'\B'`, `'\^'`, and `'\''` match `'w'`, `'W'`, `'<'`, `'>'`, `'b'`, `'B'`, `'^'`, and `''` respectively.

Grouping is performed with parentheses `'()'`. An unmatched `'()'` matches just itself. A backslash followed by a digit matches that digit.

The alternation operator is `'|'`.

The characters `'^'` and `'$'` always represent the beginning and end of a string respectively, except within square brackets. Within brackets, `'^'` can be used to invert the membership of the character class being specified.

`'*'`, `'+'` and `'?'` are special at any point in a regular expression except:

1. At the beginning of a regular expression
2. After an open-group, signified by `'('`
3. After the alternation operator `'|'`

The longest possible match is returned; this applies to the regular expression as a whole and (subject to this constraint) to subexpressions within groups.

18.7.2 ‘ed’ regular expression syntax

The character ‘.’ matches any single character except the null character.

- ‘\+’ indicates that the regular expression should match one or more occurrences of the previous atom or regexp.
- ‘\?’ indicates that the regular expression should match zero or one occurrence of the previous atom or regexp.
- ‘+ and ?’ match themselves.

Bracket expressions are used to match ranges of characters. Bracket expressions where the range is backward, for example ‘[z-a]’, are invalid. Within square brackets, ‘\’ is taken literally. Character classes are supported; for example ‘[:digit:]’ will match a single decimal digit.

GNU extensions are supported:

1. ‘\w’ matches a character within a word
2. ‘\W’ matches a character which is not within a word
3. ‘\<’ matches the beginning of a word
4. ‘\>’ matches the end of a word
5. ‘\b’ matches a word boundary
6. ‘\B’ matches characters which are not a word boundary
7. ‘\`’ matches the beginning of the whole input
8. ‘\’ matches the end of the whole input

Grouping is performed with backslashes followed by parentheses ‘\(', ‘\)’. A backslash followed by a digit acts as a back-reference and matches the same thing as the previous grouped expression indicated by that number. For example ‘\2’ matches the second group expression. The order of group expressions is determined by the position of their opening parenthesis ‘\('.

The alternation operator is ‘\|’.

The character ‘^’ only represents the beginning of a string when it appears:

1. At the beginning of a regular expression
2. After an open-group, signified by ‘\('
3. After the alternation operator ‘\|’

The character ‘\$’ only represents the end of a string when it appears:

1. At the end of a regular expression
2. Before a close-group, signified by ‘\)’
3. Before the alternation operator ‘\|’

‘*’, ‘\+’ and ‘\?’ are special at any point in a regular expression except:

1. At the beginning of a regular expression
2. After an open-group, signified by ‘\('
3. After the alternation operator ‘\|’

Intervals are specified by ‘\{’ and ‘\}’. Invalid intervals such as ‘a\{1z’ are not accepted.

The longest possible match is returned; this applies to the regular expression as a whole and (subject to this constraint) to subexpressions within groups.

18.7.3 ‘egrep’ regular expression syntax

The character ‘.’ matches any single character.

- ‘+’ indicates that the regular expression should match one or more occurrences of the previous atom or regexp.
- ‘?’ indicates that the regular expression should match zero or one occurrence of the previous atom or regexp.
- ‘\+’ matches a ‘+’
- ‘\?’ matches a ‘?’.

Bracket expressions are used to match ranges of characters. Bracket expressions where the range is backward, for example ‘[z-a]’, are invalid. Within square brackets, ‘\’ is taken literally. Character classes are supported; for example ‘[:digit:]’ will match a single decimal digit.

GNU extensions are supported:

1. ‘\w’ matches a character within a word
2. ‘\W’ matches a character which is not within a word
3. ‘\<’ matches the beginning of a word
4. ‘\>’ matches the end of a word
5. ‘\b’ matches a word boundary
6. ‘\B’ matches characters which are not a word boundary
7. ‘\`’ matches the beginning of the whole input
8. ‘\’ matches the end of the whole input

Grouping is performed with parentheses ‘()’. An unmatched ‘)’ matches just itself. A backslash followed by a digit acts as a back-reference and matches the same thing as the previous grouped expression indicated by that number. For example ‘\2’ matches the second group expression. The order of group expressions is determined by the position of their opening parenthesis ‘(’.

The alternation operator is ‘|’.

The characters ‘^’ and ‘\$’ always represent the beginning and end of a string respectively, except within square brackets. Within brackets, ‘^’ can be used to invert the membership of the character class being specified.

The characters ‘*’, ‘+’ and ‘?’ are special anywhere in a regular expression.

Intervals are specified by ‘{’ and ‘}’. Invalid intervals are treated as literals, for example ‘a{1’ is treated as ‘a\{1’

The longest possible match is returned; this applies to the regular expression as a whole and (subject to this constraint) to subexpressions within groups.

18.7.4 ‘emacs’ regular expression syntax

The character ‘.’ matches any single character except newline.

- ‘+’ indicates that the regular expression should match one or more occurrences of the previous atom or regexp.

- ‘?’ indicates that the regular expression should match zero or one occurrence of the previous atom or regexp.
- ‘\+’ matches a ‘+’
- ‘\?’ matches a ‘?’.

Bracket expressions are used to match ranges of characters. Bracket expressions where the range is backward, for example ‘[z-a]’, are ignored. Within square brackets, ‘\’ is taken literally. Character classes are supported; for example ‘[:digit:]’ will match a single decimal digit.

GNU extensions are supported:

1. ‘\w’ matches a character within a word
2. ‘\W’ matches a character which is not within a word
3. ‘\<’ matches the beginning of a word
4. ‘\>’ matches the end of a word
5. ‘\b’ matches a word boundary
6. ‘\B’ matches characters which are not a word boundary
7. ‘\`’ matches the beginning of the whole input
8. ‘\’ matches the end of the whole input

Grouping is performed with backslashes followed by parentheses ‘\(', ‘\)’. A backslash followed by a digit acts as a back-reference and matches the same thing as the previous grouped expression indicated by that number. For example ‘\2’ matches the second group expression. The order of group expressions is determined by the position of their opening parenthesis ‘\('.

The alternation operator is ‘\|’.

The character ‘^’ only represents the beginning of a string when it appears:

1. At the beginning of a regular expression
2. After an open-group, signified by ‘\('
3. After the alternation operator ‘\|’

The character ‘\$’ only represents the end of a string when it appears:

1. At the end of a regular expression
2. Before a close-group, signified by ‘\)’
3. Before the alternation operator ‘\|’

‘*’, ‘+’ and ‘?’ are special at any point in a regular expression except:

1. At the beginning of a regular expression
2. After an open-group, signified by ‘\('
3. After the alternation operator ‘\|’

Intervals are specified by ‘\{’ and ‘\}’. Invalid intervals such as ‘a\{1z’ are not accepted.

The longest possible match is returned; this applies to the regular expression as a whole and (subject to this constraint) to subexpressions within groups.

18.7.5 ‘gnu-awk’ regular expression syntax

The character ‘.’ matches any single character.

- ‘+’ indicates that the regular expression should match one or more occurrences of the previous atom or regexp.
- ‘?’ indicates that the regular expression should match zero or one occurrence of the previous atom or regexp.
- ‘\+’ matches a ‘+’
- ‘\?’ matches a ‘?’.

Bracket expressions are used to match ranges of characters. Bracket expressions where the range is backward, for example ‘[z-a]’, are invalid. Within square brackets, ‘\’ can be used to quote the following character. Character classes are supported; for example ‘[[digit:]]’ will match a single decimal digit.

GNU extensions are supported:

1. ‘\w’ matches a character within a word
2. ‘\W’ matches a character which is not within a word
3. ‘\<’ matches the beginning of a word
4. ‘\>’ matches the end of a word
5. ‘\b’ matches a word boundary
6. ‘\B’ matches characters which are not a word boundary
7. ‘\`’ matches the beginning of the whole input
8. ‘\’ matches the end of the whole input

Grouping is performed with parentheses ‘()’. An unmatched ‘)’ matches just itself. A backslash followed by a digit acts as a back-reference and matches the same thing as the previous grouped expression indicated by that number. For example ‘\2’ matches the second group expression. The order of group expressions is determined by the position of their opening parenthesis ‘(’.

The alternation operator is ‘|’.

The characters ‘^’ and ‘\$’ always represent the beginning and end of a string respectively, except within square brackets. Within brackets, ‘^’ can be used to invert the membership of the character class being specified.

‘*’, ‘+’ and ‘?’ are special at any point in a regular expression except:

1. At the beginning of a regular expression
2. After an open-group, signified by ‘(’
3. After the alternation operator ‘|’

Intervals are specified by ‘{’ and ‘}’. Invalid intervals are treated as literals, for example ‘a{1’ is treated as ‘a\{1’

The longest possible match is returned; this applies to the regular expression as a whole and (subject to this constraint) to subexpressions within groups.

18.7.6 ‘grep’ regular expression syntax

The character ‘.’ matches any single character.

- ‘\+’ indicates that the regular expression should match one or more occurrences of the previous atom or regexp.
- ‘\?’ indicates that the regular expression should match zero or one occurrence of the previous atom or regexp.
- ‘+ and ?’ match themselves.

Bracket expressions are used to match ranges of characters. Bracket expressions where the range is backward, for example ‘[z-a]’, are invalid. Within square brackets, ‘\’ is taken literally. Character classes are supported; for example ‘[:digit:]’ will match a single decimal digit.

GNU extensions are supported:

1. ‘\w’ matches a character within a word
2. ‘\W’ matches a character which is not within a word
3. ‘\<’ matches the beginning of a word
4. ‘\>’ matches the end of a word
5. ‘\b’ matches a word boundary
6. ‘\B’ matches characters which are not a word boundary
7. ‘\^’ matches the beginning of the whole input
8. ‘\’ matches the end of the whole input

Grouping is performed with backslashes followed by parentheses ‘\(', ‘\)’. A backslash followed by a digit acts as a back-reference and matches the same thing as the previous grouped expression indicated by that number. For example ‘\2’ matches the second group expression. The order of group expressions is determined by the position of their opening parenthesis ‘\('.

The alternation operator is ‘\|’.

The character ‘^’ only represents the beginning of a string when it appears:

1. At the beginning of a regular expression
2. After an open-group, signified by ‘\('
3. After a newline
4. After the alternation operator ‘\|’

The character ‘\$’ only represents the end of a string when it appears:

1. At the end of a regular expression
2. Before a close-group, signified by ‘\)’
3. Before a newline
4. Before the alternation operator ‘\|’

‘*’, ‘\+’ and ‘\?’ are special at any point in a regular expression except:

1. At the beginning of a regular expression
2. After an open-group, signified by ‘\('

3. After a newline
4. After the alternation operator ‘|’

Intervals are specified by ‘{’ and ‘}’. Invalid intervals such as ‘a{1z’ are not accepted.

The longest possible match is returned; this applies to the regular expression as a whole and (subject to this constraint) to subexpressions within groups.

18.7.7 ‘posix-awk’ regular expression syntax

The character ‘.’ matches any single character except the null character.

- | | |
|------|--|
| ‘+’ | indicates that the regular expression should match one or more occurrences of the previous atom or regexp. |
| ‘?’ | indicates that the regular expression should match zero or one occurrence of the previous atom or regexp. |
| ‘\+’ | matches a ‘+’ |
| ‘\?’ | matches a ‘?’. |

Bracket expressions are used to match ranges of characters. Bracket expressions where the range is backward, for example ‘[z-a]’, are invalid. Within square brackets, ‘\’ can be used to quote the following character. Character classes are supported; for example ‘[[digit:]]’ will match a single decimal digit.

GNU extensions are not supported and so ‘\w’, ‘\W’, ‘\<’, ‘\>’, ‘\b’, ‘\B’, ‘\`’, and ‘\’ match ‘w’, ‘W’, ‘<’, ‘>’, ‘b’, ‘B’, ‘`’, and ‘\’ respectively.

Grouping is performed with parentheses ‘()’. An unmatched ‘)’ matches just itself. A backslash followed by a digit acts as a back-reference and matches the same thing as the previous grouped expression indicated by that number. For example ‘\2’ matches the second group expression. The order of group expressions is determined by the position of their opening parenthesis ‘(’.

The alternation operator is ‘|’.

The characters ‘^’ and ‘\$’ always represent the beginning and end of a string respectively, except within square brackets. Within brackets, ‘^’ can be used to invert the membership of the character class being specified.

‘*’, ‘+’ and ‘?’ are special at any point in a regular expression except the following places, where they are not allowed:

1. At the beginning of a regular expression
2. After an open-group, signified by ‘(’
3. After the alternation operator ‘|’

Intervals are specified by ‘{’ and ‘}’. Invalid intervals are treated as literals, for example ‘a{1’ is treated as ‘a\{1’

The longest possible match is returned; this applies to the regular expression as a whole and (subject to this constraint) to subexpressions within groups.

18.7.8 ‘posix-basic’ regular expression syntax

This is a synonym for ed.

18.7.9 ‘posix-egrep’ regular expression syntax

This is a synonym for egrep.

18.7.10 ‘posix-extended’ regular expression syntax

The character ‘.’ matches any single character except the null character.

- ‘+’ indicates that the regular expression should match one or more occurrences of the previous atom or regexp.
- ‘?’ indicates that the regular expression should match zero or one occurrence of the previous atom or regexp.
- ‘\+’ matches a ‘+’
- ‘\?’ matches a ‘?’.

Bracket expressions are used to match ranges of characters. Bracket expressions where the range is backward, for example ‘[z-a]’, are invalid. Within square brackets, ‘\’ is taken literally. Character classes are supported; for example ‘[[digit:]]’ will match a single decimal digit.

GNU extensions are supported:

1. ‘\w’ matches a character within a word
2. ‘\W’ matches a character which is not within a word
3. ‘\<’ matches the beginning of a word
4. ‘\>’ matches the end of a word
5. ‘\b’ matches a word boundary
6. ‘\B’ matches characters which are not a word boundary
7. ‘\`’ matches the beginning of the whole input
8. ‘\’ matches the end of the whole input

Grouping is performed with parentheses ‘()’. An unmatched ‘)’ matches just itself. A backslash followed by a digit acts as a back-reference and matches the same thing as the previous grouped expression indicated by that number. For example ‘\2’ matches the second group expression. The order of group expressions is determined by the position of their opening parenthesis ‘(’.

The alternation operator is ‘|’.

The characters ‘^’ and ‘\$’ always represent the beginning and end of a string respectively, except within square brackets. Within brackets, ‘^’ can be used to invert the membership of the character class being specified.

‘*’, ‘+’ and ‘?’ are special at any point in a regular expression except the following places, where they are not allowed:

1. At the beginning of a regular expression
2. After an open-group, signified by ‘(’
3. After the alternation operator ‘|’

Intervals are specified by ‘{’ and ‘}’. Invalid intervals such as ‘a{1z’ are not accepted.

The longest possible match is returned; this applies to the regular expression as a whole and (subject to this constraint) to subexpressions within groups.

18.7.11 ‘posix-minimal-basic’ regular expression syntax

The character ‘.’ matches any single character except the null character.

Bracket expressions are used to match ranges of characters. Bracket expressions where the range is backward, for example ‘[z-a]’, are invalid. Within square brackets, ‘\’ is taken literally. Character classes are supported; for example ‘[:digit:]’ will match a single decimal digit.

GNU extensions are supported:

1. ‘\w’ matches a character within a word
2. ‘\W’ matches a character which is not within a word
3. ‘\<’ matches the beginning of a word
4. ‘\>’ matches the end of a word
5. ‘\b’ matches a word boundary
6. ‘\B’ matches characters which are not a word boundary
7. ‘\^’ matches the beginning of the whole input
8. ‘\’ matches the end of the whole input

Grouping is performed with backslashes followed by parentheses ‘\(', ‘\)’. A backslash followed by a digit acts as a back-reference and matches the same thing as the previous grouped expression indicated by that number. For example ‘\2’ matches the second group expression. The order of group expressions is determined by the position of their opening parenthesis ‘\('.

The character ‘^’ only represents the beginning of a string when it appears:

1. At the beginning of a regular expression
2. After an open-group, signified by ‘\('

The character ‘\$’ only represents the end of a string when it appears:

1. At the end of a regular expression
2. Before a close-group, signified by ‘\)’

Intervals are specified by ‘\{’ and ‘\}’. Invalid intervals such as ‘a\{1z’ are not accepted.

The longest possible match is returned; this applies to the regular expression as a whole and (subject to this constraint) to subexpressions within groups.

18.7.12 ‘sed’ regular expression syntax

This is a synonym for ed.

19 Build Infrastructure Modules

Gnulib has a couple of modules that don't provide code, but rather extend the GNU Build System. That is, they are convenience facilities for use with GNU Automake (in particular).

19.1 Searching for Libraries

The following macros check for the presence or location of certain C, C++, or Fortran library archive files.

Simple Library Tests

The macros `AC_CHECK_LIB`, `AC_SEARCH_LIBS` from GNU Autoconf check for the presence of certain C, C++, or Fortran library archive files. The libraries are looked up in the default linker path—a system dependent list of directories, that usually contains the `/usr/lib` directory—and those directories given by `-L` options in the `LDFLAGS` variable.

Locating Libraries

The following macros, defined in the Gnulib module `have-lib`, search for the location of certain C, C++, or Fortran library archive files and make the found location available to the compilation process and to further Autoconf tests.

`AC_LIB_LINKFLAGS(name, [dependencies])` [Macro]

Searches for `lib<name>` and the libraries corresponding to explicit and implicit dependencies. Sets and `AC_SUBST`s the `LIB<NAME>` and `LTLIB<NAME>` variables (with `<NAME>` in upper case) and augments the `CPPFLAGS` variable by `-I` options.

This macro should be used when `lib<name>` is expected to be found.

`AC_LIB_HAVE_LINKFLAGS(name, [dependencies], [includes], [testcode], [missing-message])` [Macro]

Searches for `lib<name>` and the libraries corresponding to explicit and implicit dependencies, together with the specified include files and the ability to compile and link the specified *testcode*. The *missing-message* defaults to `no` and may contain additional hints for the user. If found, it sets and `AC_SUBST`s `HAVE_LIB<NAME>=yes` and the `LIB<NAME>` and `LTLIB<NAME>` variables (with `<NAME>` in upper case) and augments the `CPPFLAGS` variable by `-I` options, and `#defines` `HAVE_LIB<NAME>` to 1. Otherwise, it sets and `AC_SUBST`s `HAVE_LIB<NAME>=no` and `LIB<NAME>` and `LTLIB<NAME>` to empty.

These macros assume that when a library is installed in *some_directory*/lib, its include files are installed in *some_directory*/include.

The complexities that `AC_LIB_LINKFLAGS` and `AC_LIB_HAVE_LINKFLAGS` deal with are the following:

- The library is not necessarily already in the search path (`CPPFLAGS` for the include file search path, `LDFLAGS` for the library search path). The macro provides a `--with-lib<name>` option. The user of the `'configure'` script can use this option to indicate the location of the library and its include files. If not provided, the `--prefix` directory is searched as well.

- The library is not necessarily already in the run time library search path. To avoid the need for setting an environment variable like `LD_LIBRARY_PATH`, the macro adds the appropriate run time search path options to the `LIB<NAME>` variable. This works on most systems. It can also be inhibited: The user of ‘`configure`’ can use the `--disable-rpath` option to force an installation that doesn’t contain hardcoded library search paths but instead may require the use of an environment variable like `LD_LIBRARY_PATH`.

The macros also set a variable `LTLIB<NAME>`, that should be used when linking with `libtool`. Both `LTLIB<NAME>` and `LIB<NAME>` contain essentially the same option, but where `LIB<NAME>` contains platform dependent flags like ‘`-Wl,-rpath`’, `LTLIB<NAME>` contains platform independent flags like ‘`-R`’.

If you, by mistake, use `LIB<NAME>` instead of `LTLIB<NAME>` when linking with `libtool`, you will observe that the binaries created in the build dir will prefer the shared libraries in the installation directories over the shared libraries in the build dir; this can lead to all sorts of build failures, test failures, or crashes!

If you, on the other hand, by mistake, use `LTLIB<NAME>` instead of `LIB<NAME>` when linking *without* `libtool`, you will observe build failures, because the ‘`-R`’ options contained in `LTLIB<NAME>` are not valid options to compilers such as `GCC`.

Example of using `AC_LIB_LINKFLAGS`

Suppose you want to use `libz`, the compression library.

1. In `configure.ac` you add the line

```
AC_CONFIG_AUX_DIR([build-aux])
AC_LIB_LINKFLAGS([z])
```

Note that since the `AC_LIB_LINKFLAGS` invocation modifies the `CPPFLAGS`, it should precede all tests that check for header files, declarations, structures or types.

2. To the package’s `build-aux` directory you add the file `config.rpath`, also part of the Gnulib `havelib` module. (`gnulib-tool` will usually do this for you automatically.)
3. In `Makefile.in` you add `@LIBZ@` to the link command line of your program. Or, if you are using Automake, you add `$(LIBZ)` to the `LDADD` variable that corresponds to your program.

Dependencies

The dependencies list is a space separated list of library names that `libname` is known to depend upon. Example: If `libfooy` depends on `libfoox`, and `libfooz` depends on `libfoox` and `libfooy`, you can write:

```
AC_LIB_LINKFLAGS([foox])
AC_LIB_LINKFLAGS([fooy], [foox])
AC_LIB_LINKFLAGS([fooz], [foox fooy])
```

Explicit dependencies are necessary if you cannot assume that a `.la` file, created by `libtool`, is installed. If you can assume that `libfooy.la` is installed by `libtool` (and has not been omitted by the package distributor!), you can omit the explicit dependency and just write

```
AC_LIB_LINKFLAGS([fooy])
```

This way, you don’t need to know in advance which libraries the needed library depends upon.

Static vs. shared

The macros find the libraries regardless whether they are installed as shared or static libraries.

CPPFLAGS vs. LDFLAGS

The macros determine the directories that should be added to the compiler preprocessor's search path and to the linker's search path. For the compiler preprocessor, `-I` options with the necessary directories are added to the `CPPFLAGS` variable, for use by the whole package. For the linker, appropriate options are added to the `LIB<NAME>` and `LTLIB<NAME>` variables, for use during linking by those programs and libraries that need the dependency on `lib<name>`. You need to use the value of `LIB<NAME>` or `LTLIB<NAME>` in the Makefiles. `LTLIB<NAME>` is for use with `libtool`, whereas `LIB<NAME>` is for when `libtool` is not involved in linking.

The macros do not check whether the include files and the library found match. If you want to verify this at configure time, one technique is to have a version number in the include files and a version number in the library, like this:

```
#define LIBNAME_VERSION 10203
extern int libname_version; /* initialized to LIBNAME_VERSION */
```

and use a test like

```
AC_TRY_RUN([int main () { return libname_version != LIBNAME_VERSION; }])
```

Bi-arch systems

A bi-arch system is one where

- the processor has a 32-bit execution mode and a 64-bit execution mode (for example, x86_64, ia64, sparc64, powerpc64), and
- 32-bit mode libraries and executables and 64-bit mode libraries are both installed, and
- 32-bit mode libraries and object files cannot be mixed with 64-bit mode ones.

On several types of such systems, for historical reasons, the 32-bit libraries are installed in `prefix/lib`, whereas the 64-bit libraries are installed in

- `prefix/lib64` on many glibc systems,
- `prefix/lib/64` on Solaris systems.

On such systems, in 64-bit mode, `configure` will search for the libraries in `prefix/lib64` or `prefix/lib/64`, respectively, not in `prefix/lib`. A user can adhere to these system-wide conventions by using the `--libdir` option when installing packages. When a user has already installed packages in 64-bit mode using the GNU default `--libdir=prefix/lib`, he can make this directory adhere to the system-wide convention by placing a symbolic link:

On glibc systems:

```
ln -s lib prefix/lib64
```

On Solaris systems:

```
ln -s . prefix/lib/64
```

19.2 Controlling the Exported Symbols of Shared Libraries

The `lib-symbol-visibility` module allows precise control of the symbols exported by a shared library. This is useful because

- It prevents abuse of undocumented APIs of your library. Symbols that are not exported from the library cannot be used. This eliminates the problem that when the maintainer of the library changes internals of the library, maintainers of other projects cry “breakage”. Instead, these maintainers are forced to negotiate the desired API from the maintainer of the library.
- It reduces the risk of symbol collision between your library and other libraries. For example, the symbol `'readline'` is defined in several libraries, most of which don't have the same semantics and the same calling convention as the GNU readline library.
- It reduces the startup time of programs linked to the library. This is because the dynamic loader has less symbols to process.
- It allows the compiler to generate better code. Within a shared library, a call to a function that is a global symbol costs a “call” instruction to a code location in the so-called PLT (procedure linkage table) which contains a “jump” instruction to the actual function's code. (This is needed so that the function can be overridden, for example by a function with the same name in the executable or in a shared library interposed with `LD_PRELOAD`.) Whereas a call to a function for which the compiler can assume that it is in the same shared library is just a direct “call” instructions. Similarly for variables: A reference to a global variable fetches a pointer in the so-called GOT (global offset table); this is a pointer to the variable's memory. So the code to access it is two memory load instructions. Whereas for a variable which is known to reside in the same shared library, it is just a direct memory access: one memory load instruction.

There are traditionally three ways to specify the exported symbols of a shared library.

- The programmer specifies the list of symbols to be exported when the shared library is created. Usually a command-line option is passed to the linker, with the name of a file containing the symbols.

The upside of this approach is flexibility: it allows the same code to be used in different libraries with different export lists. The downsides are: 1. it's a lot of maintenance overhead when the symbol list is platform dependent, 2. it doesn't work well with C++, due to name mangling.

- The programmer specifies a “hidden” attribute for every variable and function that shall not be exported.

The drawbacks of this approach are: Symbols are still exported from the library by default. It's a lot of maintenance work to mark every non- exported variable and function. But usually the exported API is quite small, compared to the internal API of the library. And it's the wrong paradigm: It doesn't force thinking when introducing new exported API.

- The programmer specifies a “hidden” attribute for all files that make up the shared library, and an “exported” attribute for those symbols in these files that shall be exported.

This is perfect: It burdens the maintainer only for exported API, not for library-internal API. And it keeps the annotations in the source code.

GNU libtool’s `-export-symbols` option implements the first approach. The script `declared.sh` from Gnulib can help to produce the list of symbols.

This gnulib module implements the third approach. For this it relies on GNU GCC 4.0 or newer, namely on its `-fvisibility=hidden` command-line option and the “visibility” attribute. (The “visibility” attribute was already supported in GCC 3.4, but without the command line option, introduced in GCC 4.0, the third approach could not be used.)

More explanations on this subject can be found in <https://gcc.gnu.org/wiki/Visibility>, which contains more details on the GCC features and additional advice for C++ libraries, and in Ulrich Drepper’s paper <https://www.akkadia.org/drepper/dsohowto.pdf>, which also explains other tricks for reducing the startup time impact of shared libraries.

The gnulib autoconf macro `gl_VISIBILITY` tests for GCC 4.0 or newer. It defines a Makefile variable `@CFLAG_VISIBILITY@` containing `-fvisibility=hidden` or nothing. It also defines as a C macro and as a substituted variable: `@HAVE_VISIBILITY@`. Its value is 1 when symbol visibility control is supported, and 0 otherwise.

As of 2022, symbol visibility control is supported on

- ELF platforms (glibc, Linux, *BSD, Solaris) with GCC or clang,
- macOS,
- AIX with gcc or xlclang.

It is not supported on

- Other compilers on ELF platforms or AIX,
- Windows.

To use this module in a library, say `libfoo`, you will do these steps:

1. Add `@CFLAG_VISIBILITY@` or (in a `Makefile.am`) `$(CFLAG_VISIBILITY)` to the `CFLAGS` for the compilation of the sources that make up the library.
2. Add a C macro definition, say `-DBUILDING_LIBFOO`, to the `CPPFLAGS` for the compilation of the sources that make up the library.
3. Define a macro specific to your library like this.

```
#if HAVE_VISIBILITY && BUILDING_LIBFOO
# define LIBFOO_SHLIB_EXPORTED __attribute__((__visibility__("default")))
#else
# define LIBFOO_SHLIB_EXPORTED
#endif
```

This macro should be enabled in all public header files of your library.

4. Annotate all variable, function and class declarations in all public header files of your library with `'LIBFOO_SHLIB_EXPORTED'`. This annotation can occur at different locations: between the `'extern'` and the type or return type, or just before the entity being declared, or after the entire declarator. My preference is to put it right after `'extern'`, so that the declarations in the header files remain halfway readable.

Note that the precise control of the exported symbols will not work with other compilers than GCC ≥ 4.0 , and will not work on systems where the assembler or linker lack the support of “hidden” visibility. Therefore, it’s good if, in order to reduce the risk of collisions with symbols in other libraries, you continue to use a prefix specific to your library for all non-static variables and functions and for all C++ classes in your library.

Note about other compilers: MSVC support can be added easily, by extending the definition of the macro mentioned above, to something like this:

```
#if HAVE_VISIBILITY && BUILDING_LIBFOO
# define LIBFOO_SHLIB_EXPORTED __attribute__((__visibility__("default")))
#elif (defined _WIN32 && !defined __CYGWIN__) && @BUILDING_SHARED@ && BUILDING_LIBFOO
# if defined DLL_EXPORT
# define LIBFOO_SHLIB_EXPORTED __declspec(dllexport)
# else
# define LIBFOO_SHLIB_EXPORTED
# endif
#elif (defined _WIN32 && !defined __CYGWIN__) && @BUILDING_SHARED@
# define LIBFOO_SHLIB_EXPORTED __declspec(dllimport)
#else
# define LIBFOO_SHLIB_EXPORTED
#endif
```

Here `BUILDING_SHARED` is an Autoconf variable that you have to define. It ought to evaluate to 1 in a build configured with ‘`--enable-shared`’, or to 0 in a build configured with ‘`--disable-shared`’. You may use the following ‘`configure.ac`’ snippet:

```
if test "$enable_shared" = yes; then
    BUILDING_SHARED=1
else
    BUILDING_SHARED=0
fi
AC_SUBST([BUILDING_SHARED])
```

And `DLL_EXPORT` is defined by Libtool, on Windows platforms, when compiling for a shared library (called DLL under Windows). It is not defined when Libtool compiles an object file meant to be linked statically into some executable.

19.3 LD Version Scripts

The `lib-symbol-versions` module can be used to add shared library versioning support. Currently, only GNU LD and the Solaris linker supports this.

Version scripts provides information that can be used by GNU/Linux distribution packaging tools. For example, Debian has a tool `dpkg-shlibdeps` that can determine the minimal required version of each dependency (by looking at the symbol list) and stuff the information into the Debian specific packaging files.

For more information and other uses of version scripts, see Ulrich Drepper’s paper <https://www.akkadia.org/drepper/dsohowto.pdf>

You use the module by importing it to your library, and then add the following lines to the `Makefile.am` that builds the library:

```
if HAVE_LD_VERSION_SCRIPT
libfoo_la_LDFLAGS += -Wl,--version-script=$(srcdir)/libfoo.map
endif
```

The version script file format is documented in the GNU LD manual, but a small example would be:

```
LIBFOO_1.0 {
    global:
        libfoo_init; libfoo_doit; libfoo_done;

    local:
        *;
```

```
};
```

If you target platforms that do not support linker scripts (i.e., all platforms that doesn't use GNU LD) you may want to consider a more portable but less powerful alternative: `libtool -export-symbols`. It will hide internal symbols from your library, but will not add ELF versioning symbols. Your usage would then be something like:

```
if HAVE_LD_VERSION_SCRIPT
libfoo_la_LDFLAGS += -Wl,--version-script=$(srcdir)/libfoo.map
else
libfoo_la_LDFLAGS += -export-symbols $(srcdir)/libfoo.sym
endif
```

See the Libtool manual for the file syntax, but a small example would be:

```
libfoo_init
libfoo_doit
libfoo_done
```

To avoid the need for a `*.sym` file if your symbols are easily expressed using a regular expression, you may use `-export-symbols-regex`:

```
if HAVE_LD_VERSION_SCRIPT
libfoo_la_LDFLAGS += -Wl,--version-script=$(srcdir)/libfoo.map
else
libfoo_la_LDFLAGS += -export-symbols-regex '^libfoo_.*'
endif
```

For more discussions about symbol visibility, rather than shared library versioning, see the `lib-symbol-visibility` module (see [Section 19.2 \[Exported Symbols of Shared Libraries\]](#), page 956).

19.4 configmake

The `configmake` module builds a C include file named `configmake.h` containing the usual installation directory values; for example, those specified by `--prefix` or `--libdir` to configure. Each variable is given a `#define` with an all-uppercase macro name, such as `PREFIX` and `LIBDIR`. (Automake cannot create this file directly because the user might override directory values at `make` time.)

Specifically, the module retrieves values of the variables through `configure` followed by `make`, not directly through `configure`, so that a user who sets some of these variables consistently on the `make` command line gets correct results.

One advantage of this approach, compared to the classical approach of adding `-DLIBDIR=\"$(libdir)\"` etc. to `AM_CPPFLAGS`, is that it protects against the use of undefined variables. That is, if, say, `$(libdir)` is not set in the Makefile, `LIBDIR` is not defined by this module, and code using `LIBDIR` gives a compilation error.

Another advantage is that `make` output is shorter.

For the complete list of variables which are `#defined` this way, see the file `gnulib/modules/configmake`, or inspect your resulting `gnulib` Makefile.

19.5 warnings

The `warnings` module allows regularly building a package with more GCC warnings than the default warnings emitted by GCC. It is often used indirectly through the `manywarnings` module (see [Section 19.6 \[manywarnings\]](#), page 960).

It provides the following functionality:

- You can select some warning options, such as ‘-Wall’, to be enabled whenever building with a GCC version that supports these options. The user can choose to override these warning options by providing the opposite options in the `CFLAGS` variable at configuration time.
- You can make these warnings apply to selected directories only. In projects where subprojects are maintained by different people, or where parts of the source code are imported from external sources (for example from gnuilib), it is useful to apply different warning options to different directories.
- It lets you use ‘-Werror’ at ‘make distcheck’ time, to verify that on the maintainer’s system, no warnings remain. (Note that use of ‘-Werror’ in `CFLAGS` does not work in general, because it may break autoconfiguration.)
- Similarly, it lets you use ‘-Werror’ when the builder runs `configure` with an option such as `--enable-gcc-warnings`.

To use this module, you need the following:

1. In `configure.ac`, use for example


```
gl_WARN_ADD([-Wall], [WARN_CFLAGS])
gl_WARN_ADD([-Wpointer-arith], [WARN_CFLAGS])
```
2. In the directories which shall use `WARN_CFLAGS`, use it in the definition of `AM_CFLAGS`, like this:

```
AM_CFLAGS = $(WARN_CFLAGS)
```

Note that the `AM_CFLAGS` is used in combination with `CFLAGS` and before `CFLAGS` in build rules emitted by Automake. This allows the user to provide `CFLAGS` that override the `WARN_CFLAGS`.

‘`gl_WARN_ADD([-Werror])`’ is intended for developers, and should be avoided in contexts where it would affect ordinary installation builds. The warnings emitted by GCC depend, to some extent, on the contents of the system header files, on the size and signedness of built-in types, etc. Use of ‘-Werror’ would cause frustration to all users on platforms that the maintainer has not tested before the release. It is better if ‘-Werror’ is off by default, and is enabled only by developers. For example, ‘-Werror’ could affect ‘make distcheck’ or ‘configure --enable-gcc-warnings’ as mentioned above.

19.6 manywarnings

The `manywarnings` module enables many GCC warnings for your package. Here is an example use:

```
AC_ARG_ENABLE([gcc-warnings],
  [AS_HELP_STRING([--enable-gcc-warnings[=TYPE]],
    [control generation of GCC warnings. The TYPE 'no' disables
    warnings; 'yes' (default) generates cheap warnings;
    'expensive' in addition generates expensive warnings.])])

AS_IF([test "$enable_gcc_warnings" != no],
  [
    # Set up the list of unwanted warning options.
    nw=
    if test "$enable_gcc_warnings" != expensive; then
```

```

    nw="$nw -fanalyzer"
fi
nw="$nw -Wbad-function-cast" # Casting a function's result is not more
                             # dangerous than casting any other value.
nw="$nw -Winline"           # It's OK to not inline.
nw="$nw -Wsign-compare"     # Too many false alarms.
nw="$nw -Wstrict-overflow"  # It's OK to optimize strictly.
nw="$nw -Wsystem-headers"   # Don't warn in system headers.

# Setup the list of meaningful warning options for the C compiler.
# The list comes from manywarnings.m4. Warning options that are not
# generally meaningful have already been filtered out (cf.
# build-aux/gcc-warning.spec).
gl_MANYWARN_ALL_GCC([possible_warning_options])

# Compute the list of warning options that are desired.
gl_MANYWARN_COMPLEMENT([desired_warning_options],
                       [$possible_warning_options], [$nw])
# Compute the list of remaining undesired warning options.
# Namely those, that were not in manywarnings.m4 because they were
# already listed in build-aux/gcc-warning.spec; this includes those
# that are implied by -Wall.
gl_MANYWARN_COMPLEMENT([remaining_undesired_warning_options],
                       [$nw], [$possible_warning_options])

# Add the desired warning options to WARN_CFLAGS.
for w in $desired_warning_options; do
    gl_WARN_ADD([$w])
done

# Add the opposites of the remaining undesired warning options to
# WARN_CFLAGS.
for w in `echo "$remaining_undesired_warning_options" | sed -e 's/-W/-Wno-/g`; do
    gl_WARN_ADD([$w])
done
})

```

This module sets up many GCC warning options.

When you use it for the first time, it is common practice to do it as follows:

- Start with the newest major release of GCC. This will save you time, because some warning options produce many false alarms with older versions of GCC (such as `-Wstrict-overflow` or `-Wunsafe-loop-optimizations`).
- Consider the platforms commonly used when enabling GCC warnings. This includes not only target architectures and operating systems, but also optimization options, which can greatly affect the warnings generated. Makefiles generated by `configure` default to `-O2` optimization. If you also commonly build with `-O0` or other optimization options, you can compile again with those options. Using more optimizations catches more bugs, because the compiler does a better static analysis of the program when optimizing more. Also, some warning options that diagnose suboptimal code generation, such as `-Winline`, are not effective when not optimizing. On the other hand, if it's frequent to build the package with warnings but without optimizations, for debugging purposes, then you don't want to see undesired warnings in these phases of development either.
- Compile the package with an empty `nw` value, that is, with all possible warnings enabled.

- Then you will go through the list of warnings. Since there are likely many warnings, the first time, it's a good idea to sort them by warning option first:

```
$ grep warning: make-output.log \
| sed -e 's/^\(.*\) \[(-W.*)\]$/\2 \1/' | sort -k1
```

- You will likely deactivate warnings that occur often and don't point to mistakes in the code, by adding them to the 'nw' variable, then reconfiguring and recompiling. When warnings point to real mistakes and bugs in the code, you will of course not disable them but fix your code to silence the warning instead.

Many GCC warning options usually don't point to mistakes in the code; these warnings enforce a certain programming style. It is a project management decision whether you want your code to follow any of these styles. Note that some of these programming styles are conflicting. You cannot have them all; you have to choose among them.

When a warning option pinpoints real bugs occasionally, but it also whines about a few code locations which are fine, we recommend to leave the warning option enabled. Whether you then live with the remaining few warnings, or choose to disable them one-by-one through `#pragma GCC diagnostic ignored "option"` (see [Section “Diagnostic Pragmas” in *Using the GNU Compiler Collection*, <https://gcc.gnu.org/onlinedocs/gcc/Diagnostic-Pragmas.html>](https://gcc.gnu.org/onlinedocs/gcc/Diagnostic-Pragmas.html)), is again a project management decision.

When a new major version of GCC is released, the Gnulib maintainers add the newly available warning options into the `gl_MANYWARN_ALL_GCC` macro. You will then enjoy the benefits of the new warnings, simply by updating to the newest Gnulib. If some of the new warnings are undesired, you can add them to the 'nw' variable, as described above.

Comments on particular warning flags:

'-fanalyzer'

The `manywarnings` module by default uses GCC's `-fanalyzer` option, as this issues some useful warnings. (It can also help GCC generate better code.) However, `-fanalyzer` can greatly slow down compilation, and in programs with large modules it can be so slow as to be unusable, so it is common for `configure` to disable it unless `configure` is given an option like `--enable-gcc-warnings=expensive`.

'-fstrict-aliasing'

Although the `manywarnings` module does not enable GCC's `-fstrict-aliasing` option, it is enabled by default if you compile with `-O2` or higher optimization, and can help GCC generate better warnings.

'-Wanalyzer-malloc-leak'

The `-fanalyzer` option generates many false alarms about `malloc` leaks, which `manywarnings` suppresses by also using `-Wno-analyzer-malloc-leak`.

'-fstrict-flex-arrays'

The `manywarnings` module by default uses GCC's `-fstrict-flex-arrays` option if available, so that GCC can warn about nonportable usage of flexible array members. In a few cases this can help GCC generate better code, so it is not strictly a warning option.

`'-Wsign-compare'`

GCC and Clang generate too many false alarms with `-Wsign-compare`, and we don't recommend that warning. You can disable it by using `gl_WARN_ADD([-Wno-sign-compare])` as illustrated above. Programs using Gnulib generally don't enable that warning when compiling Gnulib code. If you happen to find a real bug with that warning we'd like to know it.

19.7 Running self-tests under valgrind

For projects written in C or similar languages, running the self-tests under Valgrind can reveal hard to find memory issues. Gnulib supports two ways to make use of Valgrind: one that enables use of Valgrind at configure time, when `configure` found it to be present; and one at the discretion of the developer.

19.7.1 Using valgrind without developer intervention

The `valgrind-tests` module searches for Valgrind at configure time and declares the `LOG_VALGRIND` automake variable for use with automake's `LOG_COMPILER`.

After importing the `valgrind-tests` module to your project, you use it by adding the following to the `Makefile.am` that runs the self-tests:

```
LOG_COMPILER = $(LOG_VALGRIND)
```

This will run all self-checks under valgrind.

Replace `LOG_COMPILER` with `TESTS_ENVIRONMENT` if you are using the old serial test harness. The parallel test harness has been the default in automake since version 1.11.3, but if you are using an older automake, or put `'serial-tests'` in `'AM_INIT_AUTOMAKE'/'AUTOMAKE_OPTIONS'` you would still be using the serial test harness.

If you desire a project-wide decision that valgrind is not enabled by default, but still allow users to enable it with `--enable-valgrind-tests` you may put the following in `configure.ac` before `gl_INIT`.

```
gl_VALGRIND_TESTS_DEFAULT_NO
```

19.7.2 Valgrind options

The `VALGRIND` variable holds the name of the valgrind binary and some options passed to valgrind. You may provide additional options that are passed to valgrind using the `'VALGRINDFLAGS'` variable, for example:

```
./configure VALGRINDFLAGS="--suppressions=~/local.supp"
```

Alternatively during build phase:

```
make check VALGRINDFLAGS="--suppressions=~/local.supp"
```

This is useful if you have a valgrind suppression files that are needed to avoid triggering errors for known errors, typically in system libraries.

The `VALGRIND` variable include options that are useful when valgrind is run non-interactively through the test harness. The default parameters are `-q` to silence the output, `--error-exitcode=1` to cause valgrind errors to be treated as fatal errors, and `--leak-check=full` to check for memory leaks.

These options can be controlled through the `DEFAULT_VALGRINDFLAGS` variable. For example, when configuring the package:

```
./configure DEFAULT_VALGRINDFLAGS="--quiet"
```

Alternatively, during the build phase:

```
make check DEFAULT_VALGRINDFLAGS="--quiet"
```

That would have the effect of removing `--error-exitcode=1` and `--leak-check=full` from the default options, thus causing any valgrind errors to be silently ignored, instead of causing fatal test failures.

As a developer you may use the variables in `configure.ac` before calling `gl_INIT`, like this if your program has deeply-nested call chains:

```
gl_EARLY
...
VALGRINDFLAGS="$VALGRINDFLAGS --num-callers=42"
...
gl_INIT
```

Note that any user-supplied `VALGRINDFLAGS` value is preserved, which is usually what you want.

Finally, as a developer you may want to provide additional per-directory options to valgrind and the `AM_VALGRINDFLAGS` variable can be used for this. For example:

```
AM_VALGRINDFLAGS = --suppressions=$(srcdir)/local-valgrind.supp
LOG_COMPILER = $(LOG_VALGRIND)
```

19.7.3 Using valgrind at the developer's discretion

In this approach, you define a `Makefile.am` variable ‘`VALGRIND`’ (or, more abstractly, ‘`CHECKER`’), that is usually set to empty. When you have configured and built the package and you decide that you want to run the tests with valgrind, you do so by modifying the definition of ‘`VALGRIND`’ in the Makefile.

19.7.4 How to use Valgrind with shell scripts

It is not desirable to apply valgrind to shell scripts or other non-binaries, because

- It is wasteful, and you usually don't want to look for memory leaks in bash.
- On a bi-arch system, you may get an error message such as "valgrind: wrong executable class (eg. 32-bit instead of 64-bit)".

There are two ways to avoid this:

- Using the Automake `parallel-tests` feature, you can use the following instead:

```
TEST_EXTENSIONS = .pl .sh
LOG_COMPILER = $(LOG_VALGRIND)
```

Then valgrind will only be used for the non-`.sh` and non-`.pl` tests.

For old automake (before 1.11.3), you will need `AUTOMAKE_OPTIONS = parallel-tests` to enable the parallel test harness.

- You can make use of the `build-aux/run-test` script from Gnulib. Add these lines to your `Makefile.am`:

```
LOG_COMPILER += $(SHELL) $(top_srcdir)/build-aux/run-test '$(LOG_VALGRIND)'
```

Replace `LOG_COMPILER` with `TESTS_ENVIRONMENT` if you use the old serial test harness.

However, with this measure in place, binaries invoked through scripts will not be invoked under valgrind. This can be solved by defining environment variables in the `TESTS_ENVIRONMENT` variable that are then used by the shell scripts. For example, add the following:

```
TESTS_ENVIRONMENT = VALGRIND='$(LOG_VALGRIND)'
```

And then modify the shell scripts to invoke the binary prefixed with `$VALGRIND`.

19.8 Package version management

The goals of a package maintainer, when dealing with the version number of a package, are:

- Programs should identify themselves with that version number when invoked as *program --version*.
- Every package tarball should contain the version number, in up to four ways:
 - The tarball name should be *package-version.tar.gz*.
 - The tarball should unpack into a directory named *package-version*.
 - Programs built from that tarball should identify themselves when invoked as *program --version*.
 - Documentation (especially in PDF format) and man pages may contain the version number, as an important bit of meta information.
- The package maintainer should be able to easily set the version number before making a release, and doing so should be quick.
- When a contributor is building from the git repository (as opposed to a tarball), the version number should, by default, reflect the git commit, in order to avoid confusion w.r.t. official releases.

Here's how the GNU Build System and Gnulib achieve these goals.

19.8.1 Setting the package version

When the maintainer has decided about the version number for the next build (and likely also the next tarball), they implement that decision by storing this version number in a file named *.tarball-version* at the top level of the package. This is a simple one-line text file.

The maintainer can, alternatively, decide to let the git version be the version number. To implement this choice, they remove the file *.tarball-version* at the top level of the package.

The actual version number comes from an invocation of the program *git-version-gen*, part of the Gnulib module *git-version-gen*. It looks at the file *.tarball-version*, if that exists, and at the current checkout, otherwise.

The file *.tarball-version* should not be put under version control. Therefore you may want to list it in the package's top-level *.gitignore* file.

But you need a *Makefile.am* rule that will make sure that *.tarball-version* will exist in distribution tarballs:

```
dist-hook: dist-tarball-version
.PHONY: dist-tarball-version
dist-tarball-version:
echo '$(VERSION)' > $(distdir)/.tarball-version
```

Typically, the maintainer will set the version number in a clean directory (i.e. after “make distclean”). In order to adapt to alternative workflows, the file *GNUmakefile*, part of the Gnulib module *gnumakefile*, contains a rule that will print a warning or possibly call *autoreconf* if it finds that the version has changed but has not yet been propagated. Reminder: Not every package supports *autoreconf*.

19.8.2 Propagating the package version

At the end of a `configure` run, `config.status` is run, that creates various files with embedded pieces of information. It thus propagates the values of various variables to various files in the build tree (most notably, `Makefiles` and `config.h`).

One such propagated value is the value of the Automake-defined variable `$(VERSION)`. (There is also the Autoconf-defined variable `$(PACKAGE_VERSION)`, but nothing except Automake ought to use it.)

For `$(VERSION)` to have a sensible value, a few lines are needed in `configure.ac`.

The recommended code pattern is

```
AC_INIT([package], [package])
AC_CONFIG_SRCDIR([unique-file-in-source-dir])
AC_CONFIG_AUX_DIR([build-aux])
VERSION_NUMBER=`cd $srcdir \
                && build-aux/git-version-gen .tarball-version`
gl_INIT_PACKAGE_VERSION([$VERSION_NUMBER])
AM_INIT_AUTOMAKE([options])
```

With this code pattern, the contents of the file `.tarball-version` and the git status are considered when `configure` is run.

Two older code patterns are deprecated, because they read the contents of the file `.tarball-version` and the git status when `autoconf` is run, not when `configure` is run. These older code patterns thus require a longer turnaround cycle when the maintainer has changed the version number.

The first such old code pattern is to set the version number directly in `configure.ac`:

```
AC_INIT([package, version])
AC_CONFIG_SRCDIR([unique-file-in-source-dir])
AC_CONFIG_AUX_DIR([build-aux])
AM_INIT_AUTOMAKE([options])
```

The second such old code pattern is to invoke `git-version-gen` at `autoconf` time:

```
AC_INIT([package],
        m4_esyscmd([build-aux/git-version-gen .tarball-version]))
AC_CONFIG_SRCDIR([unique-file-in-source-dir])
AC_CONFIG_AUX_DIR([build-aux])
AM_INIT_AUTOMAKE([options])
```

The macro `gl_INIT_PACKAGE_VERSION` is defined in the Gnulib module `package-version`.

19.8.3 Using the package version during the build

The unoptimized way

Once propagated through `config.status`, the value of `$(VERSION)` can appear anywhere in built files. For built files that are distributed (that is, not erased by “make distclean”), it is important to add a dependency in the `Makefile.am`, so that the file gets rebuilt:

```
built-file: $(top_builddir)/config.status
```

For files that are erased by “make distclean”, this is not really necessary, because

- In the main workflow, where the developer changes the version only between “make distclean” and “./configure”, the built files have been remade.
- Most `.o` files depend on `config.h`, which is rebuilt when `configure` runs.

Thus, only in packages that don’t use Gnulib would a dependency such as

```
hello-hello.$(OBJEXT): $(top_builddir)/config.status
```

be needed.

The optimized way

Notice that `config.status` usually changes much more often than the version number. Therefore, for built files which depend *only* on `$(VERSION)` and not on other variables, the following optimized technique can be used.

The technique consists of keeping a file named `$(top_srcdir)/.version`, which is a timestamp file. Its modification time represents the last time the value of `$(VERSION)` was changed. Its contents is an undocumented implementation detail.

With such a file, the dependency in `Makefile.am` becomes:

```
built-file: $(top_srcdir)/.version
```

In order to prepare for using `$(top_srcdir)/.version`, three modifications in the package are needed:

- In the top-level `configure.ac` file, add an invocation of the macro `gl_CONFIG_VERSION_STAMP`, near the end (before `AC_OUTPUT`). This macro is defined in file `m4/version-stamp.m4`, part of the Gnulib module `version-stamp`.
- In the top-level `Makefile.am`, add:

```
EXTRA_DIST += $(top_srcdir)/.version
BUILT_SOURCES += $(top_srcdir)/.version
```

so that the file `.version` will be present in tarballs and so that “make maintainer-clean” will erase it.

- Add `.version` to the package’s top-level `.gitignore` file.

19.9 VCS To ChangeLog

Gnulib provides the ‘`vcs-to-changelog`’ module to generate an output similar to the GNU ChangeLog format from metadata of source control software such as git. Here’s an example of using ‘`vcs-to-changelog`’:

```
build-aux/vcs-to-changelog.py <from_ref> <to_ref>
```

where `<from_ref>` and `<to_ref>` refer to the range of commits to generate the output.

VCS To ChangeLog currently recognises changes in C source code and can traverse commits in git. Additional source frontends and source control backends may be added to the module. ‘`vcs-to-changelog`’ takes the following optional arguments:

- `-d`: Run the parser debugger, used for debugging ‘`vcs-to-changelog`’
- `-q filename`: Load *filename* as the quirks file for the project.

The `quirks` file is a python module that must minimally implement a `get_project_quirks` function that returns an object of type `ProjectQuirks` or its subclass. The subclass may override the following members of `ProjectQuirks`:

- `repo`: Specify the project repo source control. The default value is `git`.
- `IGNORE_LIST`: A list of files to ignore in the changesets, either because they are not needed (such as the `ChangeLog`) or because they are not parseable. For example, the GNU C Library has a header file that is only assembly code, which breaks the C parser.
- `MACRO_QUIRKS`: A list of dictionary entries with indexes as `orig` and `sub` where `orig` is a Python regular expression pattern to match and `sub` is the substitution. These substitutions are used to work around C/C++ macros that are known to break parsing of C programs.
- `C_MACROS`: This is a list of C preprocessor macro definitions that are extensively used and are known to break parsing due to some characteristic, mainly the lack of a semi-colon at the end.

19.10 gitlog-to-changelog

Gnulib has a module `gitlog-to-changelog` to parse `git log` output and generate `ChangeLog` files; see https://www.gnu.org/prep/standards/html_node/Change-Logs.html.

You can use it by extending the `dist-hook` rule in the top-level `Makefile.am` like this:

```
dist-hook: gen-ChangeLog
.PHONY: gen-ChangeLog
gen-ChangeLog:
    $(AM_V_GEN)if test -e $(srcdir)/.git; then \
        LC_ALL=en_US.UTF-8 TZ=UTC0 \
        $(top_srcdir)/build-aux/gitlog-to-changelog \
        --srcdir=$(srcdir) > $(distdir)/ChangeLog \
        || exit 1; \
    fi
```

See `gitlog-to-changelog --help` for complete documentation.

The `LC_ALL=en_US.UTF-8 TZ=UTC0` line in this recipe assumes that you want to generate reproducible `ChangeLog` files that do not depend on the developer's locale and time zone. Omit this line if you prefer `ChangeLog` files that depend on these developer settings.

If you wish to output the `ChangeLog` with dates respecting the time zone each individual commit was made in you can use the `--commit-timezone` option. For example:

```
dist-hook: gen-ChangeLog
.PHONY: gen-ChangeLog
gen-ChangeLog:
    $(AM_V_GEN)if test -e $(srcdir)/.git; then \
        $(top_srcdir)/build-aux/gitlog-to-changelog \
        --srcdir=$(srcdir) --commit-timezone \
        > $(distdir)/ChangeLog \
        || exit 1; \
    fi
```

The use of `--commit-timezone` means that `ChangeLog` dates correctly represent when a committer pushed a change according to their time zone. However, as a consequence `ChangeLog` dates will no longer be monotonically increasing, if the developers are spread across different time zones. For example, the following three commits were made in a short period of time across two different time zones. This behavior may be undesired.

```
2024-06-19 Bruno Haible <bruno@clisp.org>

...

2024-06-18 Collin Funk <collin.funk1@gmail.com>

...

2024-06-19 Bruno Haible <bruno@clisp.org>

...
```

If you wish to limit the `ChangeLog` entries (perhaps for size issues) to contain only entries since a particular git tag, you can use a `gen-ChangeLog` rule like the following:

```
gen_start_ver = 8.31
gen-ChangeLog:
    $(AM_V_GEN)if test -e $(srcdir)/.git; then \
        log_fix='$(srcdir)/build-aux/git-log-fix'; \
        test -e "$$log_fix" \
        && amend_git_log=--amend=$$log_fix \
        || amend_git_log=; \
        { LC_ALL=en_US.UTF-8 TZ=UTC0 \
            $(top_srcdir)/build-aux/gitlog-to-changelog \
            --srcdir=$(srcdir) \
            "$$amend_git_log" -- 'v$(gen_start_ver)~..' && \
            printf '\n\nSee the source repo for older entries.\n'; \
        } > $(distdir)/ChangeLog \
        || exit 1; \
    fi
```

20 Build Infrastructure Files

Gnulib contains also a small number of files that are not part of modules. They are meant to be imported into packages by means of ‘`gnulib-tool --copy-file`’, not ‘`gnulib-tool --import`’. For example, the commands to import the files `config.guess` and `config.sub` are

```
for file in config.guess config.sub; do
  $GNULIB_TOOL --copy-file build-aux/$file \
    && chmod a+x build-aux/$file \
    || exit $?
done
```

Packages that don’t use Gnulib can get hold of these files through direct download from Gnulib’s git repository. The commands to do this look as follows:

```
for file in config.guess config.sub; do
  echo "$0: getting $file..."
  wget -q --timeout=5 -O build-aux/$file.tmp "https://git.savannah.gnu.org/gitweb/?p=gnulib.git;a=blob_plain;f=$file"
  && mv build-aux/$file.tmp build-aux/$file \
  && chmod a+x build-aux/$file
  retval=$?
  rm -f build-aux/$file.tmp
  test $retval -eq 0 || exit $retval
done
```

20.1 Recognizing platforms

`build-aux/config.guess`

`build-aux/config.sub`

These files are helper scripts, invoked by the ‘`configure`’ script. `config.guess` recognizes the platform on which the script is running, and produces a triplet of the form *cpu-type-vendor-operating_system*. `config.sub` receives a possibly abbreviated triplet and produces a canonical triplet for a platform. For more information, see https://www.gnu.org/prep/standards/html_node/Configuration.html.

It is important that you always include the newest versions of these two files in your tarball, because people who work on emerging platforms otherwise have a hard time building your package.

20.2 Utilities for Makefiles

These programs can be used in Makefiles. Some of them are also described in https://www.gnu.org/software/automake/manual/html_node/Auxiliary-Programs.html.

`build-aux/ar-lib`

`build-aux/compile`

These two scripts are necessary for supporting portability to native Windows with the MSVC compiler. `compile` is a wrapper script that invokes the compiler and provides a command-line interface compatible with Unix compilers. Similarly, `ar-lib` is a wrapper script that provides a command-line interface compatible with Unix `ar`.

build-aux/depcomp

This is a helper script, used by Makefile rules generated by GNU Automake. It generates Makefile dependencies while compiling a file.

build-aux/install-sh

This is a helper script, used by Makefile rules generated by GNU Automake. It installs files during the **make install** phase. In the Makefile, don't use this file directly; always use `$(INSTALL_PROGRAM)` or `$(INSTALL_DATA)` instead.

build-aux/mdate-sh

This script determines the modification time of a file and pretty-prints it. The typical use is to add a "Last modified" line to the documentation.

build-aux/mkinstalldirs

This is a helper script, used by Makefile rules generated by GNU Automake. It creates directories during the **make install** phase. It is roughly equivalent to `'mkdir -p'` (except that the latter is not portable). In the Makefile, don't use this file directly; always use `$(MKDIR_P)` instead.

build-aux/mktempd

This script creates a temporary directory. It is roughly equivalent to `'mktemp -d'` (except that the latter is not portable).

build-aux/move-if-change

This script moves a freshly generated file to a destination file, with a special optimization for the case that both files are identical. In this case the freshly generated file is deleted, and the time stamp of the destination file is *not* changed. This is useful when updating a file that rarely actually changes and which many Makefile targets depend upon.

20.3 Programs for developing in Git checkouts

These programs can help when developing in a Git checkout. The maintainer of the package copies these programs into the version control of the package, so that co-developers can use these tools right away.

top/gitsub.sh

This program manages the subdirectories of a Git checkout that come from other packages, including Gnulib.

top/bootstrap**top/autopull.sh****top/autogen.sh****top/bootstrap-funclib.sh**

This is a set of three programs and a function library, that manage the source directory of a package, preparing for the state where `'./configure'` can be used.

autopull.sh is a program for fetching dependencies that may require network accesses. It manages the Git submodules, including Gnulib – assuming that Gnulib is a Git submodule. It also can fetch the PO files for internationalized packages.

`autogen.sh` is a program that is meant to be run after `autopull.sh`. It generates all autogeneratable files that are omitted from version control. Usually this means that it invokes `gnulib-tool` and `automake`, that generate files from other files.

`bootstrap` is a wrapper around both: `./bootstrap --pull` is equivalent to `./autopull.sh`, `./bootstrap --gen` is equivalent to `./autogen.sh`. Plain `./bootstrap` is equivalent to `./autopull.sh` immediately followed by `./autogen.sh`; however, because plain `./bootstrap` mixes version control management and generation of files in non-obvious ways, it has a number of usability issues for the advanced developer.

`bootstrap-funclib.sh` is a function library for these three programs. It is not meant to be used directly.

All three programs make use of a configuration file, called `bootstrap.conf`.

`build-aux/bootstrap`

This acts like `top/bootstrap`, except it does not need the companion files `autogen.sh`, `autopull.sh`, and `bootstrap-funclib.sh` so it avoids some clutter in your project's top level directory. With this approach, you update via `./bootstrap --pull` and `./bootstrap --gen` instead of via `./autopull.sh` and `./autogen.sh`. Otherwise this approach acts similarly, and uses the same `bootstrap.conf` file.

`build-aux/bootstrap.conf`

This is the template configuration file. After copying it into the top-level directory of your package, you need to customize it.

`build-aux/po/Makefile.in.in`

`build-aux/po/remove-potcdate.sed`

These are auxiliary files used by `bootstrap`. You don't have to copy them yourself; `bootstrap` will do that.

20.4 Utilities for building documentation

These are auxiliary files for building documentation.

`build-aux/texinfo.tex`

This file is needed for the conversion of Texinfo-format documentation to PDF, PostScript, or DVI formats. It implements the GNU Texinfo commands on top of plain TeX.

`build-aux/x-to-1.in`

This file, once processed, gives a program `x-to-1`, that produces a manual page for a program, by combining a skeleton with the program's `--help` output.

20.5 Utilities for building libraries

`build-aux/declared.sh`

This program extracts the declared global symbols of a C header file. It is useful when you want to control the set of symbols exported by a library. See [Section 19.2 \[Exported Symbols of Shared Libraries\]](#), page 956.

20.6 Utilities for running test suites

`build-aux/run-test`

This file is a test driver that supports running a test under `valgrind`.

`build-aux/test-driver.diff`

This is a patch, against Automake's test driver, that supports running a test suite on Android.

21 Release Management Files

Gnulib also contain a few scripts that are useful for the release management of a package. They can be used directly off the Gnulib checkout; they don't need to be copied first.

21.1 Tools for releasing packages with shared libraries

`build-aux/libtool-next-version`

This program is a wizard that helps a maintainer update the libtool version of a shared library, without making mistakes in this process. For background documentation, see https://www.gnu.org/software/libtool/manual/html_node/Updating-version-info.html.

21.2 Tools for uploading release tarballs

`build-aux/gnupload`

This program is a user-friendly way to upload a release tarball to one of the GNU servers (<ftp.gnu.org> or <alpha.gnu.org>). It implements the interface described in https://www.gnu.org/prep/maintain/html_node/Automated-FTP-Uploads.html.

`build-aux/ncftpput-ftp`

This is a helper program that mimics the `ncftpput` program used by `gnupload`. If you want to use `gnupload` but don't have `ncftp` installed, copy this file into your `$PATH`, renaming it to `ncftpput`.

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```

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If your document contains nontrivial examples of program code, we recommend releasing these examples in parallel under your choice of free software license, such as the GNU General Public License, to permit their use in free software.

Appendix B Gnulib history

In the beginning, Richard Stallman maintained the regular expression engine of GNU and made it available to other GNU packages, such as `sed`, `awk`, `grep`, so that they could use the same engine. Recall that there was no GNU C library at the time.

A couple of years later, Jim Meyering, who was maintaining the shell utilities, the file utilities, and the text processing utilities, collected the common parts of these three packages in the same repository. Paul Eggert joined in, with code coming from a few GNU packages that he maintained, and so did Bruno Haible, with reusable code from the GNU `gettext` package.

As they cared for portability, many changes in the C code were accompanied by changes in the build infrastructure. Copying these changes into packages — it was all done manually — became cumbersome. So they wrote a program, called ‘`gnulib-tool`’, that does this job of copying the requested shared code into a particular package. This was in 2002.

Providing a substitute / override for a system function was relatively easy. Providing a substitute / override for a system header file was significantly harder, but was done successively: for `stdint.h` in 2004, for `stdarg.h`, `sys/socket.h`, `sys/stat.h` in 2006, for `sys/time.h`, `wchar.h` in 2007, and the development of the corresponding idioms took until 2010.

Unicode string modules (that make up GNU `libunistring`) were added in 2007–2009.

Modules for numeric functions (`<math.h>`) were added in 2010–2011.

Modules for container data structures were added between 2006 and 2018.

Versatile bit-set modules were added in 2018.

POSIX threads on non-POSIX platforms, as well as ISO C threads on all platforms, were added in 2019.

The `posix_spawn` facility was brought to completion on native Windows in 2022, providing the world’s first `posix_spawn` implementation for this platform.

Support for Android was added in 2023 and immediately used by GNU Emacs for Android.

Functions for working with Unicode characters in multibyte representation, based on `mbrtoc32`, were added in 2023.

Modules for manipulating the floating-point environment (`fenv.h`) were added in 2023.

The ‘`gnulib-tool`’ rewrite in Python, that was started by Dmitry Selyutin in 2012 but lay unfinished for many years, was completed by Collin Funk and Bruno Haible in 2024.

Module Index

—
_Exit..... 381

A

abort-debug..... 910
accept..... 431
accept4..... 432
access..... 477
acos..... 169
acosf..... 169
acosl..... 170
agpl-3.0..... 53
alignalloc..... 383
alignasof..... 57, 76
aligned-malloc..... 383, 393, 640
aligned_alloc..... 382
alloca..... 585, 864
alloca-opt..... 864
alphasort..... 134
areadlink..... 501, 502
areadlink-with-size..... 88
argmatch..... 911
argp..... 585, 599, 600, 601
argp-version-etc..... 911
argz..... 585, 602, 603, 604, 605, 606
arpa_inet-h..... 59
array-list..... 885
array-map..... 886
array-omap..... 887
array-oset..... 886
array-set..... 886
asin..... 171
asinf..... 171
asinl..... 172
assert..... 59, 103
assert-h..... 58, 59
atan..... 173
atan2..... 173
atan2f..... 174
atanf..... 175
atanl..... 176
atexit..... 383
atoll..... 385
attribute..... 866
avltree-list..... 885
avltree-omap..... 887
avltree-oset..... 886
avltreehash-list..... 885

B

basename-lgpl..... 160
bcopy..... 554
bind..... 432
bool..... 5, 57, 79
bsearch..... 385
btoc32..... 512
btowc..... 511
byteswap..... 586, 610, 611

C

c-ctype..... 120, 121, 122, 123, 124, 125, 126, 127,
128, 129, 130, 131, 132, 133, 561, 854, 862
c-snzprintf..... 889
c-snzprintf-gnu..... 889
c-strcasecmp..... 855
c-strcaseeq..... 855
c-strcasestr..... 856
c-strncasecmp..... 855
c-strstr..... 856
c-strtod..... 856, 857
c-strtold..... 856, 857
c-vaszprintf..... 889
c-vaszprintf-gnu..... 889
c-vsnzprintf..... 889
c-vsnzprintf-gnu..... 889
c-xvasprintf..... 890
c32_apply_mapping..... 548, 862
c32_apply_type_test..... 542, 862
c32_get_mapping..... 550, 862
c32_get_type_test..... 551, 862
c32isalnum..... 120, 540, 862
c32isalpha..... 121, 540, 862
c32isblank..... 122, 541, 862
c32iscntrl..... 123, 541, 862
c32isdigit..... 124, 543, 862
c32isgraph..... 125, 543, 862
c32islower..... 126, 544, 862
c32isprint..... 127, 545, 862
c32ispunct..... 128, 545, 862
c32isspace..... 129, 546, 862
c32isupper..... 130, 547, 862
c32isxdigit..... 131, 547, 862
c32rtomb..... 475, 525
c32snrtombs..... 531, 857
c32srtombs..... 532, 857
c32stombs..... 407, 857
c32swidth..... 535, 857
c32tob..... 536
c32tolower..... 132, 549, 862
c32toupper..... 133, 549, 862
c32width..... 536, 862
call_once..... 82, 453

calloc-gnu..... 385
 calloc-posix..... 385
 canonicalize-lgpl..... 399, 763
 carray-list..... 885
 cbrt..... 178
 cbrt-ieee..... 178
 cbrtf..... 178
 cbrtf-ieee..... 178
 cbrtl..... 178
 cbrtl-ieee..... 178
 ceil..... 178
 ceil-ieee..... 178
 ceilf..... 179
 ceilf-ieee..... 179
 ceill..... 179
 ceill-ieee..... 179
 chdir..... 478
 check-version..... 916
 chmod..... 437
 chown..... 479
 clean-temp..... 392, 767, 768
 close..... 479
 closedir..... 134
 cnd..... 453, 454, 455, 848
 cond..... 848
 configmake..... 959
 connect..... 432
 copy-file-range..... 821
 copysign..... 180
 copysignf..... 180
 copysignl..... 180
 cos..... 181
 cosf..... 181
 cosh..... 181
 coshf..... 181
 cosl..... 182
 creat..... 143
 crypto/gc-random..... 489, 797
 ctime..... 465
 ctype.. 120, 121, 123, 124, 125, 126, 127, 128, 129,
 130, 131, 132, 133, 581
 ctype-h..... 60

D

dirent-h..... 61
 dirent-safer..... 879
 dirfd..... 134
 dirname..... 69, 160
 dprintf..... 339
 dprintf-gnu..... 339
 dprintf-posix..... 339
 dup..... 480
 dup2..... 480
 dup3..... 481
 duplocale..... 165
 dzprintf..... 889
 dzprintf-gnu..... 889

dzprintf-posix..... 889

E

endian..... 62, 139, 140, 141, 142
 environ..... 482
 erf..... 185
 erfc..... 185
 errno-h..... 62
 error..... 620, 621
 error-h..... 587
 euidaccess..... 822
 execinfo..... 587, 621, 622
 execl..... 482
 execl_e..... 483
 execlp..... 483
 execv..... 483
 execve..... 484
 execvp..... 484
 execvpe..... 823
 exp..... 186
 exp-ieee..... 186
 exp2..... 188
 exp2-ieee..... 188
 exp2f..... 188
 exp2f-ieee..... 188
 exp2l..... 189
 exp2l-ieee..... 189
 expf..... 190
 expf-ieee..... 190
 expl..... 190
 expl-ieee..... 190
 explicit_bzero..... 776
 expm1..... 190
 expm1-ieee..... 190
 expm1f..... 191
 expm1f-ieee..... 191
 expm1l..... 191
 expm1l-ieee..... 191
 extensions..... 136, 155, 298, 299, 315, 462, 466
 extern-inline..... 877

F

fabs..... 191
 fabs-ieee..... 191
 fabsf..... 191
 fabsf-ieee..... 191
 fabsl..... 192
 fabsl-ieee..... 192
 faccessat..... 485
 fbufmode..... 891
 fchdir..... 144, 485
 fchmodat..... 438
 fchownat..... 486
 fclose..... 340
 fcntl..... 143
 fcntl-h..... 63

fcntl-safer..... 879
 fdatsync..... 486
 fdopen..... 341
 fdopendir..... 135
 fdutimensat..... 446, 807
 fenv-environment..... 147, 148, 149, 151
 fenv-exceptions-state-c23..... 151
 fenv-exceptions-state-c99..... 147, 149
 fenv-exceptions-tracking-c23..... 149
 fenv-exceptions-tracking-c99..... 147, 148, 150
 fenv-exceptions-trapping..... 624, 625
 fenv-h..... 65
 fenv-rounding..... 148, 150
 fflush..... 341
 ffs..... 419
 ffsll..... 420
 ffsll..... 420
 flexmember..... 5
 float-h..... 65
 flock..... 784
 floor..... 194
 floor-ieee..... 194
 floorf..... 194
 floorf-ieee..... 194
 floorl..... 195
 floorl-ieee..... 195
 fma..... 195
 fma-ieee..... 195
 fmaf..... 195
 fmaf-ieee..... 195
 fmal..... 196
 fmal-ieee..... 196
 fmod..... 206
 fmod-ieee..... 206
 fmodf..... 206
 fmodf-ieee..... 206
 fmodl..... 206
 fmodl-ieee..... 206
 fnmatch..... 152
 fnmatch-gnu..... 66, 152
 fnmatch-h..... 66
 fopen..... 344
 fopen-gnu..... 344
 fopen-safer..... 879
 forkpty..... 684
 fpending..... 891
 fprintf-gnu..... 344
 fprintf-posix..... 344
 fpurge..... 891
 freadable..... 890
 freadahead..... 891
 freading..... 890
 freadptr..... 892
 freadseek..... 892
 free-posix..... 386
 freelocale..... 166
 freopen..... 348
 freopen-safer..... 348, 879

frexp..... 207
 frexp-ieee..... 207
 frexpf..... 208
 frexpf-ieee..... 208
 frexpl..... 208
 frexpl-ieee..... 208
 fseek..... 349
 fseeko..... 349
 fseterr..... 891
 fstat..... 439
 fstatat..... 439
 fsusage..... 446
 fsync..... 487
 ftell..... 350
 ftello..... 350, 367, 368
 ftruncate..... 488
 fts..... 557, 588, 627, 628
 func..... 5, 920
 futimens..... 440
 fwritable..... 890
 fwriting..... 890
 fzprintf..... 889
 fzprintf-gnu..... 889
 fzprintf-posix..... 889

G

gcd..... 914
 getaddrinfo..... 264, 265, 844
 getcwd..... 488
 getcwd-lgpl..... 488
 getdelim..... 353
 getdomainname..... 823
 getdtablesize..... 824
 getentropy..... 489, 797
 getgroups..... 490
 gethostname..... 69, 491
 getline..... 354
 getloadavg..... 765
 getlocalename_l-simple..... 166
 getlogin..... 491
 getlogin_r..... 491
 getopt-gnu..... 492, 589, 628, 631
 getopt-posix..... 492
 getpagesize..... 824
 getpass..... 825
 getpass-gnu..... 825
 getpayload..... 211
 getpayloadf..... 211
 getpayloadl..... 212
 getpeername..... 433
 getprogname..... 913
 getrandom..... 489, 797
 getrusage..... 427
 getsockname..... 433
 getsockopt..... 434
 getsubopt..... 387
 gettext..... 34, 69

gettext-h 34
 gettime 464, 559
 gettime-res 464, 473
 gettimeofday 559
 getugroups 633
 getumask 800
 getusershell 822, 826, 828
 git-version-gen 965
 gitlog-to-changelog 968
 glob 153, 154, 631
 glob-h 67
 gnumakefile 965
 gnupload 974
 gpl-2.0 53
 gpl-3.0 53
 grantpt 387
 group-member 826

H

hamt 887
 hash-map 886
 hash-set 886
 hasmntopt 657
 havelib 953
 htonl 100, 102
 hypot 212
 hypot-ieee 212
 hypotf 212
 hypotf-ieee 212
 hypotl 213
 hypotl-ieee 213

I

iconv 17, 67, 156
 iconv_open 156
 iconv_open-utf 156
 ieee754-h 589
 ilogb 213
 ilogbf 213
 ilogbl 214
 imaxabs 157
 imaxdiv 157
 inet_ntop 101
 inet_pton 102
 inline 877
 intprops 869
 inttypes-h 68
 ioctl 560
 isalnum_l 121
 isalpha_l 122
 isatty 495
 isblank 122, 862
 isblank_l 123
 iscntrl_l 124
 isdigit_l 125
 isfinite 214

isgraph_l 126
 isinf 215
 islower_l 127
 isnan 216
 isnanf 645
 isnanl 645
 isprint_l 128
 ispunct_l 129
 isspace_l 130
 isupper_l 131
 iswblank 541, 862
 iswctype 542, 862
 iswdigit 543, 862
 iswpunct 545, 862
 iswxdigit 547, 862
 isxdigit_l 132

J

j0 217
 j1 217
 jn 217

L

langinfo-h 68
 largefile 90, 843
 lchmod 800
 lchown 495
 ldexp 217
 ldexp-ieee 217
 ldexpf 217
 ldexpf-ieee 217
 ldexpl 218
 ldexpl-ieee 218
 lgamma 218
 lgpl-2.1 53
 lgpl-3.0 53
 lib-msvc-compat 845
 lib-symbol-versions 958
 lib-symbol-visibility 956
 libunistring 892
 libunistring-optional 892
 limits-h 69
 link 496
 linkat 496
 linked-list 885
 linkedhash-list 885
 linkedhash-map 886
 linkedhash-set 886
 list 882, 885
 list-c++ 887
 listen 434
 localcharset 896
 locale-h 70
 localeconv 167
 localtime 468
 lock 848

log.....	221
log-ieee.....	221
log10.....	221
log10-ieee.....	221
log10f.....	221
log10f-ieee.....	221
log10l.....	222
log10l-ieee.....	222
log1p.....	223
log1p-ieee.....	223
log1pf.....	223
log1pf-ieee.....	223
log1pl.....	223
log1pl-ieee.....	223
log2.....	224
log2-ieee.....	224
log2f.....	224
log2f-ieee.....	224
log2l.....	224
log2l-ieee.....	224
logb.....	225
logb-ieee.....	225
logbf.....	226
logbf-ieee.....	226
logbl.....	226
logbl-ieee.....	226
logf.....	226
logf-ieee.....	226
login_tty.....	833
logl.....	227
logl-ieee.....	227
logp1.....	227
logp1f.....	227
logp1l.....	227
lseek.....	497
lstat.....	441

M

maintainer-makefile	17
malloc-gnu	390
malloc-h	590
malloc-posix	390
manywarnings	959, 960
map	882, 886
map-c++	887
math-h	71
mbchar	120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 561, 862
mbfile	862
mbiter	853
mbiterf	853
mbrlen	515
mbrtoc16	475
mbrtoc32	476, 516
mbrtoc32-regular	476
mbrtowc	516
mbs_endswith	857

```

mbs_startswith..... 857
mbscasecmp..... 421, 855, 857
mbscasestr..... 856, 857
mbschr..... 411, 857
mbscspn..... 412, 857
mbsinit..... 516
mbslen..... 857
mbsncasecmp..... 421, 855, 857
mbsnlen..... 857
mbsnrtoc32s..... 517, 857
mbsnrtowcs..... 517, 857
mbspbrk..... 417, 857
mbspcasecmp..... 421, 855, 857
mbsrchr..... 417, 857
mbsrtoc32s..... 517, 857
mbsrtowcs..... 517, 857
mbssep..... 857
mbsspn..... 418, 857
mbsstr..... 418, 856, 857
mbstoc32s..... 390, 857
mbstok_r..... 419, 857
mbstowcs..... 390, 857
mbswidth..... 857
mbtowc..... 390
mbuiter..... 853
mbuiterf..... 853
mcel..... 853
mcel-prefer..... 853
memalign..... 640
memccpy..... 408
memchr..... 408
memcmp..... 408
memcpy..... 409
memmem..... 409
memmem-simple..... 409
memmove..... 409
mempcpy..... 776
memrchr..... 777
memset..... 410
memset_explicit..... 410
mgetgroups..... 490, 634
mkdir..... 441
mkdirat..... 442
mkdtemp..... 391
mkfifo..... 442
mkfifoat..... 442, 443
mknod..... 443
mkostemp..... 391
mkostemps..... 767
mkstemp..... 392
mkstemps..... 767
mktime..... 469
mntent-h..... 591
modf..... 229
modf-ieee..... 229
modff..... 230
modff-ieee..... 230
modfl..... 230

```


modfl-ieee..... 230
 monetary-h..... 71
 mountlist..... 591
 mtx..... 456, 457, 848

N

nanosleep..... 469
 net_if-h..... 72
 netdb-h..... 72
 netinet_in-h..... 73
 newlocale..... 167
 nextafter..... 232
 nl_langinfo..... 159
 nonblocking. 63, 144, 342, 343, 344, 346, 347, 349,
 351, 352, 353, 355, 358, 359, 362, 372, 374, 375, 377,
 499, 500, 509
 noreturn..... 868
 nproc..... 803
 nstrftime..... 470
 nullptr..... 57

O

obstack..... 591
 obstack-printf..... 673, 675
 obstack-printf-gnu..... 673, 675
 obstack-printf-posix..... 673, 675
 obstack-zprintf..... 889
 obstack-zprintf-gnu..... 889
 obstack-zprintf-posix..... 889
 omap..... 882, 887
 omap-c++..... 887
 once..... 848
 open..... 144
 openat..... 145
 openat-safer..... 879
 opendir..... 135
 openmp..... 846
 openpty..... 393, 684
 options..... 629
 oset..... 882, 886
 oset-c++..... 887

P

package-version..... 966
 pagealign_alloc..... 383, 393
 parse-datetime..... 467
 pathmax..... 70
 pclose..... 354
 perror..... 355
 pipe-posix..... 499
 pipe2..... 499
 pipe2-safer..... 879
 poll..... 270
 poll-h..... 73
 popen..... 355

popen-safer..... 879
 posix_memalign..... 393
 posix_openpt..... 393
 posix_spawn..... 316
 posix_spawn_file_actions_addchdir... 316, 322,
 753
 posix_spawn_file_actions_addclose..... 316
 posix_spawn_file_actions_adddup2..... 317
 posix_spawn_file_actions_addfchdir.. 316, 322,
 754
 posix_spawn_file_actions_addopen..... 317
 posix_spawn_file_actions_destroy..... 318
 posix_spawn_file_actions_init..... 318
 posix_spawnattr_destroy..... 318
 posix_spawnattr_getflags..... 318
 posix_spawnattr_getpgroup..... 319
 posix_spawnattr_getschedparam..... 319
 posix_spawnattr_getschedpolicy..... 319
 posix_spawnattr_getsigdefault..... 320
 posix_spawnattr_getsigmask..... 320
 posix_spawnattr_init..... 320
 posix_spawnattr_setflags..... 320
 posix_spawnattr_setpgroup..... 321
 posix_spawnattr_setschedparam..... 321
 posix_spawnattr_setschedpolicy..... 321
 posix_spawnattr_setsigdefault..... 321
 posix_spawnattr_setsigmask..... 322
 posix_spawnnp..... 322
 posixcheck..... 30
 pow..... 235
 powf..... 235
 pread..... 499
 printf-gnu..... 355
 printf-posix..... 355
 printf-with-n-directive..... 340, 346, 357, 365,
 366, 372, 373, 376, 379, 380, 674, 676
 progname..... 913
 pselect..... 429
 pthread..... 847
 pthread-cond..... 278, 279, 280, 281, 847
 pthread-h..... 74
 pthread-mutex... 284, 285, 286, 287, 288, 289, 847
 pthread-once..... 290, 847
 pthread-rwlock..... 291, 292, 293, 294, 847
 pthread-spin..... 296, 297, 847
 pthread-thread. 271, 273, 274, 281, 282, 283, 294,
 847
 pthread-tss..... 283, 284, 296, 847
 pthread_mutex_timedlock..... 286
 pthread_sigmask..... 310
 ptsname..... 393
 ptsname_r..... 394
 pty-h..... 592
 putenv-gnu..... 394
 pwrite..... 500

Q

qsort..... 395
 qsort_r..... 395
 quote..... 913
 quotearg..... 913

R

raise..... 311
 rand..... 396
 random..... 388, 396, 400, 401
 random_r..... 396, 766, 770, 771, 772
 rawmemchr..... 777
 rbtree-list..... 885
 rbtree-omap..... 887
 rbtree-oset..... 886
 rbtreehash-list..... 885
 read..... 500
 readdir..... 136
 readline..... 17
 readlink..... 501
 readlinkat..... 501
 readutmp..... 96, 595, 834
 realloc-posix..... 397
 reallocarray..... 398
 recv..... 434
 recvfrom..... 435
 regex..... 74, 299, 300
 relocatable-lib..... 917
 relocatable-lib-lgpl..... 917
 relocatable-perl..... 918
 relocatable-prog..... 917
 relocatable-script..... 918
 remainder..... 237
 remainder-ieee..... 237
 remainderf..... 237
 remainderf-ieee..... 237
 remainderl..... 237
 remainderl-ieee..... 237
 remove..... 360
 rename..... 360
 renameat..... 361
 renameatu..... 760
 rewinddir..... 137
 rint..... 238
 rint-ieee..... 238
 rintf..... 239
 rintf-ieee..... 239
 rintl..... 239
 rintl-ieee..... 239
 rmdir..... 502
 round..... 240
 round-ieee..... 240
 roundf..... 241
 roundf-ieee..... 241
 roundl..... 241
 roundl-ieee..... 241
 rpmatch..... 771

S

safe-alloc..... 865
 safe-read..... 500
 safe-write..... 510
 same-inode..... 88
 scandir..... 137
 sched-h..... 75
 sched_yield..... 302, 847
 search-h..... 75
 secure_getenv..... 399
 select..... 429
 selinux-h..... 744, 745
 send..... 435
 sendto..... 436
 servent..... 267
 set..... 882, 886
 set-c++..... 887
 setenv..... 400
 sethostname..... 828
 setlocale..... 167
 setlocale-null..... 167
 setpayload..... 244
 setpayloadf..... 244
 setpayloadl..... 244
 setpayloadsig..... 244
 setpayloadsigf..... 245
 setpayloadsigl..... 245
 setsockopt..... 436
 sf-istream..... 343
 shutdown..... 436
 sig2str..... 311, 315, 777
 sigabbrev_np..... 777
 sigaction..... 311
 sigdescr_np..... 778
 signal-h..... 76
 signbit..... 245
 sigpipe.... 344, 346, 347, 351, 355, 358, 359, 372, 375, 509
 sigprocmask..... 312, 313, 314
 sin..... 246
 sinf..... 246
 sinh..... 246
 sinhf..... 246
 sinl..... 247
 sleep..... 505
 snprintf..... 363
 snprintf-gnu..... 363
 snprintf-posix..... 363
 snzprintf..... 889
 snzprintf-gnu..... 889
 snzprintf-posix..... 889
 socket..... 437
 spawn-h..... 76
 spin..... 848
 sprintf-gnu..... 365
 sprintf-posix..... 365
 sqrt..... 248
 sqrt-ieee..... 248

sqrtf.....	248	strings-h.....	83
sqrtf-ieee.....	248	strncasecmp.....	421, 857
sqrtl.....	249	strncasecmp_l.....	421
sqrtl-ieee.....	249	strncat.....	415, 857
stack-trace.....	909	strncmp.....	416
stat.....	444	strncpy.....	416
stat-size.....	920	strndup.....	416, 857
stat-time.....	87	strnlen.....	417, 857
stdalign-h.....	77	strpbrk.....	417, 857
stdarg-h.....	78, 323	strptime.....	470
stdbit.....	78	strsep.....	780, 857
stdbool-h-c99.....	79	strsignal.....	417
stdc_bit_ceil.....	333	strstr.....	418, 857
stdc_bit_floor.....	334	strstr-simple.....	418
stdc_bit_width.....	334	strtod.....	402, 857
stdc_count_ones.....	334	strtod-obsolete.....	402
stdc_count_zeros.....	334	strtof.....	403, 857
stdc_first_leading_one.....	335	strtoimax.....	158, 857
stdc_first_leading_zero.....	335	strtok_r.....	419, 857
stdc_first_trailing_one.....	335	strtol.....	404, 857
stdc_first_trailing_zero.....	335	strtold.....	404, 857
stdc_has_single_bit.....	336	strtoll.....	405, 857
stdc_leading_ones.....	336	strtoul.....	406, 857
stdc_leading_zeros.....	336	strtoull.....	406, 857
stdc_trailing_ones.....	336	strtoumax.....	158, 857
stdc_trailing_zeros.....	337	strverscmp.....	780
stdckdint-h.....	79, 873	sublist.....	885
stddef-h.....	80	symlink.....	505
stdint-h.....	5, 80	symlinkat.....	506
stdio-h.....	81, 342, 343, 344, 346, 347, 349, 351, 352, 353, 355, 358, 359, 362, 372, 375, 500	sys_file-h.....	593
stdlib-h.....	82, 386	sys_ioctl-h.....	594
stdlib-safer.....	879	sys_msg-h.....	84
stdnoreturn-h.....	83, 868	sys_random-h.....	594
stpcpy.....	410, 857	sys_resource-h.....	85
stpncpy.....	411, 857	sys_select-h.....	85
str_endswith.....	857	sys_sem-h.....	85
str_startswith.....	857	sys_shm-h.....	86
strcasecmp.....	420, 857	sys_socket-h.....	86
strcasecmp_l.....	421	sys_stat-h.....	87, 441
strcasestr.....	778, 857	sys_time-h.....	89
strcasestr-simple.....	778	sys_times-h.....	89
strchrnul.....	779	sys_types-h.....	89
strcspn.....	412, 857	sys_uio-h.....	90
strdup.....	413, 857	sys_un-h.....	90
strdup-posix.....	413	sys_utsname-h.....	91
strerror.....	413	sys_wait-h.....	91
strerror_l.....	413	sys_exit-h.....	594
strerror_r-posix.....	414	system-posix.....	82, 406
strerrorname_np.....	779	szprintf.....	889
strfmon_l.....	257	szprintf-gnu.....	889
strftime-fixes.....	469	szprintf-posix.....	889
striconv.....	156		
striconveh.....	156, 157		
string-buffer.....	354		
string-desc.....	857, 881		
string-desc-quotearg.....	881		
string-h.....	83, 411, 412, 416		

T

tan..... 249
 tanf..... 249
 tanh..... 249
 tanhf..... 250
 tanl..... 250
 tcgetsid..... 452
 termios-h..... 92
 thrd..... 458, 459, 460, 848
 thread..... 848
 thread-optim..... 849
 threads..... 848
 threads-h..... 92
 time..... 470
 time-h..... 93
 time_r..... 467, 468
 time_rz..... 468
 timegm..... 471
 timer-time..... 471, 472
 times..... 447
 timespec_get..... 464, 472, 559
 timespec_getres..... 473
 timevar..... 914
 tls..... 848
 tmpfile..... 368
 tmpfile-safer..... 879
 tolower_l..... 132
 totalorder..... 252
 totalorderf..... 252
 totalorderl..... 253
 totalordermag..... 253
 totalordermagf..... 253
 totalordermagl..... 254
 toupper_l..... 133
 towctrans..... 548, 862
 trunc..... 254
 trunc-ieee..... 254
 truncate..... 507
 truncf..... 254
 truncf-ieee..... 254
 trunc1..... 255
 trunc1-ieee..... 255
 tsearch..... 304, 305
 tss..... 461, 462, 848
 ttyname_r..... 508
 tzname..... 474
 tzset..... 474

U

uchar-h..... 94
 uchar-h-c23..... 94, 477, 861
 uname..... 448
 unicast/empty-prefix-context..... 907
 unicast/empty-suffix-context..... 907
 unicast/locale-language..... 907
 unicast/tolower..... 132, 907
 unicast/totitle..... 907

unicast/toupper..... 133, 907
 unicast/u16-casecmp..... 907
 unicast/u16-casecoll..... 907
 unicast/u16-casefold..... 907
 unicast/u16-casexfrm..... 907
 unicast/u16-ct-casefold..... 907
 unicast/u16-ct-tolower..... 907
 unicast/u16-ct-totitle..... 907
 unicast/u16-ct-toupper..... 907
 unicast/u16-is-cased..... 907
 unicast/u16-is-casefolded..... 907
 unicast/u16-is-lowercase..... 907
 unicast/u16-is-titlecase..... 907
 unicast/u16-is-uppercase..... 907
 unicast/u16-prefix-context..... 907
 unicast/u16-suffix-context..... 907
 unicast/u16-tolower..... 907
 unicast/u16-totitle..... 907
 unicast/u16-toupper..... 907
 unicast/u32-casecmp..... 857, 907
 unicast/u32-casecoll..... 907
 unicast/u32-casefold..... 907
 unicast/u32-casexfrm..... 907
 unicast/u32-ct-casefold..... 907
 unicast/u32-ct-tolower..... 907
 unicast/u32-ct-totitle..... 907
 unicast/u32-ct-toupper..... 907
 unicast/u32-is-cased..... 907
 unicast/u32-is-casefolded..... 907
 unicast/u32-is-lowercase..... 907
 unicast/u32-is-titlecase..... 907
 unicast/u32-is-uppercase..... 907
 unicast/u32-prefix-context..... 907
 unicast/u32-suffix-context..... 907
 unicast/u32-tolower..... 907
 unicast/u32-totitle..... 907
 unicast/u32-toupper..... 907
 unicast/u8-casecmp..... 907
 unicast/u8-casecoll..... 907
 unicast/u8-casefold..... 907
 unicast/u8-casexfrm..... 907
 unicast/u8-ct-casefold..... 907
 unicast/u8-ct-tolower..... 907
 unicast/u8-ct-totitle..... 907
 unicast/u8-ct-toupper..... 907
 unicast/u8-is-cased..... 907
 unicast/u8-is-casefolded..... 907
 unicast/u8-is-lowercase..... 907
 unicast/u8-is-titlecase..... 907
 unicast/u8-is-uppercase..... 907
 unicast/u8-prefix-context..... 907
 unicast/u8-suffix-context..... 907
 unicast/u8-tolower..... 907
 unicast/u8-totitle..... 907
 unicast/u8-toupper..... 907
 unicast/ulc-casecmp..... 907
 unicast/ulc-casecoll..... 907
 unicast/ulc-casexfrm..... 907

unicnv/u16-conv-from-enc.....	896	unictype/category-Pe.....	897
unicnv/u16-conv-to-enc.....	896	unictype/category-Pf.....	897
unicnv/u16-strconv-from-enc.....	896	unictype/category-Pi.....	897
unicnv/u16-strconv-from-locale.....	896	unictype/category-Po.....	897
unicnv/u16-strconv-to-enc.....	896	unictype/category-Ps.....	897
unicnv/u16-strconv-to-locale.....	896	unictype/category-S.....	897
unicnv/u32-conv-from-enc.....	896	unictype/category-Sc.....	897
unicnv/u32-conv-to-enc.....	896	unictype/category-Sk.....	897
unicnv/u32-strconv-from-enc.....	896	unictype/category-Sm.....	897
unicnv/u32-strconv-from-locale.....	896	unictype/category-So.....	897
unicnv/u32-strconv-to-enc.....	896	unictype/category-test.....	897
unicnv/u32-strconv-to-locale.....	896	unictype/category-test-withtable.....	897
unicnv/u8-conv-from-enc.....	896	unictype/category-Z.....	897
unicnv/u8-conv-to-enc.....	896	unictype/category-Zl.....	897
unicnv/u8-strconv-from-enc.....	896	unictype/category-Zp.....	897
unicnv/u8-strconv-from-locale.....	896	unictype/category-Zs.....	897
unicnv/u8-strconv-to-enc.....	896	unictype/combining-class.....	897
unicnv/u8-strconv-to-locale.....	896	unictype/combining-class-byname.....	897
unictype/base.....	897	unictype/combining-class-longname.....	897
unictype/bidiclass-byname.....	897	unictype/combining-class-name.....	897
unictype/bidiclass-longname.....	897	unictype/ctype-alnum.....	120, 897
unictype/bidiclass-name.....	897	unictype/ctype-alpha.....	121, 897
unictype/bidiclass-of.....	897	unictype/ctype-blank.....	122, 897
unictype/bidiclass-test.....	897	unictype/ctype-cntrl.....	123, 897
unictype/block-list.....	897	unictype/ctype-digit.....	124, 897
unictype/block-of.....	897	unictype/ctype-graph.....	125, 897
unictype/block-test.....	897	unictype/ctype-lower.....	126, 897
unictype/category-and.....	897	unictype/ctype-print.....	127, 897
unictype/category-and-not.....	897	unictype/ctype-punct.....	128, 897
unictype/category-byname.....	897	unictype/ctype-space.....	129, 897
unictype/category-C.....	897	unictype/ctype-upper.....	130, 897
unictype/category-Cc.....	897	unictype/ctype-xdigit.....	131, 897
unictype/category-Cf.....	897	unictype/decimal-digit.....	897
unictype/category-Cn.....	897	unictype/digit.....	897
unictype/category-Co.....	897	unictype/incb-byname.....	897
unictype/category-Cs.....	897	unictype/incb-name.....	897
unictype/category-L.....	897	unictype/incb-of.....	897
unictype/category-LC.....	897	unictype/joininggroup-byname.....	897
unictype/category-Ll.....	897	unictype/joininggroup-name.....	897
unictype/category-Lm.....	897	unictype/joininggroup-of.....	897
unictype/category-Lo.....	897	unictype/joiningtype-byname.....	897
unictype/category-longname.....	897	unictype/joiningtype-longname.....	897
unictype/category-Lt.....	897	unictype/joiningtype-name.....	897
unictype/category-Lu.....	897	unictype/joiningtype-of.....	897
unictype/category-M.....	897	unictype/mirror.....	897
unictype/category-Mc.....	897	unictype/numeric.....	897
unictype/category-Me.....	897	unictype/property-alphabetic.....	897
unictype/category-Mn.....	897	unictype/property-ascii-hex-digit.....	897
unictype/category-name.....	897	unictype/property-bidi-arabic-digit.....	897
unictype/category-N.....	897	unictype/property-bidi-arabic-right-to- left.....	897
unictype/category-Nd.....	897	unictype/property-bidi-block-separator...	897
unictype/category-Nl.....	897	unictype/property-bidi-boundary-neutral...	897
unictype/category-No.....	897	unictype/property-bidi-common-separator...	897
unictype/category-of.....	897	unictype/property-bidi-control.....	897
unictype/category-or.....	897	unictype/property-bidi-embedding-or- override.....	897
unictype/category-P.....	897	unictype/property-bidi-eur-num-separator...	897
unictype/category-Pc.....	897		
unictype/category-Pd.....	897		

unictype/property-bidi-eur-num-terminator	897	unictype/property-left-of-pair	897
unictype/property-bidi-european-digit	897	unictype/property-line-separator	897
unictype/property-bidi-hebrew-right-to-left	897	unictype/property-logical-order-exception	897
unictype/property-bidi-left-to-right	897	unictype/property-lowercase	897
unictype/property-bidi-non-spacing-mark	897	unictype/property-math	897
unictype/property-bidi-other-neutral	897	unictype/property-modifier-combining-mark	897
unictype/property-bidi-pdf	897	unictype/property-non-break	897
unictype/property-bidi-segment-separator	897	unictype/property-not-a-character	897
unictype/property-bidi-whitespace	897	unictype/property-numeric	897
unictype/property-byname	897	unictype/property-other-alphabetic	897
unictype/property-case-ignorable	897	unictype/property-other-default-ignorable-code-point	897
unictype/property-cased	897	unictype/property-other-grapheme-extend	897
unictype/property-changes-when-casfolded	897	unictype/property-other-id-continue	897
unictype/property-changes-when-casemapped	897	unictype/property-other-id-start	897
unictype/property-changes-when-lowercased	897	unictype/property-other-lowercase	897
unictype/property-changes-when-titlecased	897	unictype/property-other-math	897
unictype/property-changes-when-uppercased	897	unictype/property-other-uppercase	897
unictype/property-combining	897	unictype/property-paired-punctuation	897
unictype/property-composite	897	unictype/property-paragraph-separator	897
unictype/property-currency-symbol	897	unictype/property-pattern-syntax	897
unictype/property-dash	897	unictype/property-pattern-white-space	897
unictype/property-decimal-digit	897	unictype/property-prepended-concatenation-mark	897
unictype/property-default-ignorable-code-point	897	unictype/property-private-use	897
unictype/property-deprecated	897	unictype/property-punctuation	897
unictype/property-diacritic	897	unictype/property-quotation-mark	897
unictype/property-emoji	897	unictype/property-radical	897
unictype/property-emoji-component	897	unictype/property-regional-indicator	897
unictype/property-emoji-modifier	897	unictype/property-sentence-terminal	897
unictype/property-emoji-modifier-base	897	unictype/property-soft-dotted	897
unictype/property-emoji-presentation	897	unictype/property-space	897
unictype/property-extended-pictographic	897	unictype/property-terminal-punctuation	897
unictype/property-extender	897	unictype/property-test	897
unictype/property-format-control	897	unictype/property-titlecase	897
unictype/property-grapheme-base	897	unictype/property-unassigned-code-value	897
unictype/property-grapheme-extend	897	unictype/property-unified-ideograph	897
unictype/property-grapheme-link	897	unictype/property-uppercase	897
unictype/property-hex-digit	897	unictype/property-variation-selector	897
unictype/property-hyphen	897	unictype/property-white-space	897
unictype/property-id-compat-math-continue	897	unictype/property-xid-continue	897
unictype/property-id-compat-math-start	897	unictype/property-xid-start	897
unictype/property-id-continue	897	unictype/property-zero-width	897
unictype/property-id-start	897	unictype/scripts	897
unictype/property-ideographic	897	unictype/syntax-c-ident	897
unictype/property-ids-binary-operator	897	unictype/syntax-c-whitespace	897
unictype/property-ids-ternary-operator	897	unictype/syntax-java-ident	897
unictype/property-ids-unary-operator	897	unictype/syntax-java-whitespace	897
unictype/property-ignorable-control	897	unigbrk/u16-grapheme-breaks	906
unictype/property-iso-control	897	unigbrk/u16-grapheme-next	906
unictype/property-join-control	897	unigbrk/u16-grapheme-prev	906
		unigbrk/u32-grapheme-breaks	906
		unigbrk/u32-grapheme-next	906
		unigbrk/u32-grapheme-prev	906
		unigbrk/u8-grapheme-breaks	906
		unigbrk/u8-grapheme-next	906

unigbrk/u8-grapheme-prev.....	906	unistdio/u32-u32-asnprintf.....	896
unigbrk/uc-gbrk-prop.....	906	unistdio/u32-u32-asprintf.....	896
unigbrk/uc-grapheme-breaks.....	906	unistdio/u32-u32-snprintf.....	896
unigbrk/uc-is-grapheme-break.....	906	unistdio/u32-u32-sprintf.....	896
unigbrk/ulc-grapheme-breaks.....	906	unistdio/u32-u32-vasnprintf.....	896
unilbrk/u16-possible-linebreaks.....	906	unistdio/u32-u32-vasprintf.....	896
unilbrk/u16-width-linebreaks.....	906	unistdio/u32-u32-vsnprintf.....	896
unilbrk/u32-possible-linebreaks.....	906	unistdio/u32-u32-vsprintf.....	896
unilbrk/u32-width-linebreaks.....	906	unistdio/u32-vasnprintf.....	896
unilbrk/u8-possible-linebreaks.....	906	unistdio/u32-vasprintf.....	896
unilbrk/u8-width-linebreaks.....	906	unistdio/u32-vsnprintf.....	896
unilbrk/ulc-possible-linebreaks.....	906	unistdio/u32-vsprintf.....	896
unilbrk/ulc-width-linebreaks.....	906	unistdio/u8-asnprintf.....	896
uniname/uniname.....	897	unistdio/u8-asprintf.....	896
uninorm/base.....	907	unistdio/u8-snprintf.....	896
uninorm/canonical-decomposition.....	907	unistdio/u8-sprintf.....	896
uninorm/composition.....	907	unistdio/u8-u8-asnprintf.....	896
uninorm/decomposing-form.....	907	unistdio/u8-u8-asprintf.....	896
uninorm/decomposition.....	907	unistdio/u8-u8-snprintf.....	896
uninorm/filter.....	907	unistdio/u8-u8-sprintf.....	896
uninorm/nfc.....	907	unistdio/u8-u8-vasnprintf.....	896
uninorm/nfd.....	907	unistdio/u8-u8-vasprintf.....	896
uninorm/nfkc.....	907	unistdio/u8-u8-vsnprintf.....	896
uninorm/nfkd.....	907	unistdio/u8-u8-vsprintf.....	896
uninorm/u16-normalize.....	907	unistdio/u8-vasnprintf.....	896
uninorm/u16-normcmp.....	907	unistdio/u8-vasprintf.....	896
uninorm/u16-normcoll.....	907	unistdio/u8-vsnprintf.....	896
uninorm/u16-normxfrm.....	907	unistdio/u8-vsprintf.....	896
uninorm/u32-normalize.....	907	unistdio/ulc-asnprintf.....	896
uninorm/u32-normcmp.....	907	unistdio/ulc-asprintf.....	896
uninorm/u32-normcoll.....	907	unistdio/ulc-fprintf.....	896
uninorm/u32-normxfrm.....	907	unistdio/ulc-snprintf.....	896
uninorm/u8-normalize.....	907	unistdio/ulc-sprintf.....	896
uninorm/u8-normcmp.....	907	unistdio/ulc-vasnprintf.....	896
uninorm/u8-normcoll.....	907	unistdio/ulc-vasprintf.....	896
uninorm/u8-normxfrm.....	907	unistdio/ulc-vfprintf.....	896
unistd-h.....	95, 477	unistdio/ulc-vsnprintf.....	896
unistd-safer.....	879	unistdio/ulc-vsprintf.....	896
unistdio/u16-asnprintf.....	896	unistr/u16-check.....	893
unistdio/u16-asprintf.....	896	unistr/u16-chr.....	893
unistdio/u16-snprintf.....	896	unistr/u16-cmp.....	893
unistdio/u16-sprintf.....	896	unistr/u16-cmp2.....	893
unistdio/u16-u16-asnprintf.....	896	unistr/u16-cpy.....	893
unistdio/u16-u16-asprintf.....	896	unistr/u16-cpy-alloc.....	893
unistdio/u16-u16-snprintf.....	896	unistr/u16-endswith.....	893
unistdio/u16-u16-sprintf.....	896	unistr/u16-mblen.....	893
unistdio/u16-u16-vasnprintf.....	896	unistr/u16-mbsnlen.....	893
unistdio/u16-u16-vasprintf.....	896	unistr/u16-mbtouc.....	893
unistdio/u16-u16-vsnprintf.....	896	unistr/u16-mbtouc-unsafe.....	893
unistdio/u16-u16-vsprintf.....	896	unistr/u16-mbtoucr.....	893
unistdio/u16-vasnprintf.....	896	unistr/u16-move.....	893
unistdio/u16-vasprintf.....	896	unistr/u16-next.....	893
unistdio/u16-vsnprintf.....	896	unistr/u16-pcpy.....	893
unistdio/u16-vsprintf.....	896	unistr/u16-prev.....	893
unistdio/u32-asnprintf.....	896	unistr/u16-set.....	893
unistdio/u32-asprintf.....	896	unistr/u16-startswith.....	893
unistdio/u32-snprintf.....	896	unistr/u16-stpcpy.....	893
unistdio/u32-sprintf.....	896	unistr/u16-stpncpy.....	893

unistr/u16-strcat.....	893	unistr/u32-strspn.....	857, 893
unistr/u16-strchr.....	893	unistr/u32-strstr.....	857, 893
unistr/u16-strcmp.....	893	unistr/u32-strtok.....	857, 893
unistr/u16-strcoll.....	893	unistr/u32-to-u16.....	893
unistr/u16-strcpy.....	893	unistr/u32-to-u8.....	893
unistr/u16-strcspn.....	893	unistr/u32-uctomb.....	893
unistr/u16-strdup.....	893	unistr/u8-check.....	893
unistr/u16-strlen.....	893	unistr/u8-chr.....	893
unistr/u16-strmblen.....	893	unistr/u8-cmp.....	893
unistr/u16-strmbtouc.....	893	unistr/u8-cmp2.....	893
unistr/u16-strncat.....	893	unistr/u8-cpy.....	893
unistr/u16-strncmp.....	893	unistr/u8-cpy-alloc.....	893
unistr/u16-strncpy.....	893	unistr/u8-endswith.....	893
unistr/u16-strnlen.....	893	unistr/u8-mblen.....	893
unistr/u16-strpbrk.....	893	unistr/u8-mbsnlen.....	893
unistr/u16-strrchr.....	893	unistr/u8-mbtouc.....	893
unistr/u16-strspn.....	893	unistr/u8-mbtouc-unsafe.....	893
unistr/u16-strstr.....	893	unistr/u8-mbtoucr.....	893
unistr/u16-strtok.....	893	unistr/u8-move.....	893
unistr/u16-to-u32.....	893	unistr/u8-next.....	893
unistr/u16-to-u8.....	893	unistr/u8-pcpy.....	893
unistr/u16-uctomb.....	893	unistr/u8-prev.....	893
unistr/u32-check.....	893	unistr/u8-set.....	893
unistr/u32-chr.....	893	unistr/u8-startswith.....	893
unistr/u32-cmp.....	893	unistr/u8-stpcpy.....	893
unistr/u32-cmp2.....	893	unistr/u8-stpnpcpy.....	893
unistr/u32-cpy.....	893	unistr/u8-strcat.....	893
unistr/u32-cpy-alloc.....	893	unistr/u8-strchr.....	893
unistr/u32-endswith.....	857, 893	unistr/u8-strcmp.....	893
unistr/u32-mblen.....	893	unistr/u8-strcoll.....	893
unistr/u32-mbsnlen.....	857, 893	unistr/u8-strcpy.....	893
unistr/u32-mbtouc.....	893	unistr/u8-strcspn.....	893
unistr/u32-mbtouc-unsafe.....	893	unistr/u8-strdup.....	893
unistr/u32-mbtoucr.....	893	unistr/u8-strlen.....	893
unistr/u32-move.....	893	unistr/u8-strmblen.....	893
unistr/u32-next.....	893	unistr/u8-strmbtouc.....	893
unistr/u32-pcpy.....	893	unistr/u8-strncat.....	893
unistr/u32-prev.....	893	unistr/u8-strncmp.....	893
unistr/u32-set.....	893	unistr/u8-strncpy.....	893
unistr/u32-startswith.....	857, 893	unistr/u8-strnlen.....	893
unistr/u32-stpcpy.....	857, 893	unistr/u8-strpbrk.....	893
unistr/u32-stpnpcpy.....	857, 893	unistr/u8-strrchr.....	893
unistr/u32-strcat.....	857, 893	unistr/u8-strspn.....	893
unistr/u32-strchr.....	857, 893	unistr/u8-strstr.....	893
unistr/u32-strcmp.....	857, 893	unistr/u8-strtok.....	893
unistr/u32-strcoll.....	857, 893	unistr/u8-to-u16.....	893
unistr/u32-strcpy.....	857, 893	unistr/u8-to-u32.....	893
unistr/u32-strcspn.....	857, 893	unistr/u8-uctomb.....	893
unistr/u32-strdup.....	857, 893	unitypes-h.....	893
unistr/u32-strlen.....	857, 893	uniwbrk/u16-wordbreaks.....	906
unistr/u32-strmblen.....	893	uniwbrk/u32-wordbreaks.....	906
unistr/u32-strmbtouc.....	893	uniwbrk/u8-wordbreaks.....	906
unistr/u32-strncat.....	857, 893	uniwbrk/ulc-wordbreaks.....	906
unistr/u32-strncmp.....	857, 893	uniwbrk/wordbreak-property.....	906
unistr/u32-strncpy.....	857, 893	uniwidth/u16-strwidth.....	906
unistr/u32-strnlen.....	857, 893	uniwidth/u16-width.....	906
unistr/u32-strpbrk.....	857, 893	uniwidth/u32-strwidth.....	857, 906
unistr/u32-strrchr.....	857, 893	uniwidth/u32-width.....	906

uniwidth/u8-strwidth 906
 uniwidth/u8-width 906
 uniwidth/width 906
 unlink 508
 unlinkat 509
 unlinkdir 508, 509
 unlocked-io 849
 unlockpt 407
 unsetenv 407
 usleep 582
 utime 582
 utime-h 96
 utimens 441, 447, 583, 807
 utimensat 445
 utmp-h 595

V

valgrind-tests 963
 vararrays 4
 vasprintf 337, 369
 vasprintf-gnu 337, 369
 vasprintf-posix 337, 369
 vaszprintf 889
 vaszprintf-gnu 889
 vaszprintf-posix 889
 vcs-to-changelog 967
 vdprintf 371
 vdprintf-gnu 371
 vdprintf-posix 371
 vdzprintf 889
 vdzprintf-gnu 889
 vdzprintf-posix 889
 verify 59, 866
 version-etc 910
 version-etc-fsf 911
 vfprintf-gnu 372
 vfprintf-posix 372
 vfscanf 374
 vfzprintf 889
 vfzprintf-gnu 889
 vfzprintf-posix 889
 vla 5
 vprintf-gnu 375
 vprintf-posix 375
 vscanf 377
 vsnprintf 377
 vsnprintf-gnu 377
 vsnprintf-posix 377
 vsnzprintf 889
 vsnzprintf-gnu 889
 vsnzprintf-posix 889
 vsprintf-gnu 379
 vsprintf-posix 379
 vszprintf 889
 vszprintf-gnu 889
 vszprintf-posix 889
 vzprintf 889

vzprintf-gnu 889
 vzprintf-posix 889

W

waitpid 449
 warnings 959
 wchar-h 97
 wchar-single 849
 wcpcpy 525, 857
 wcpncpy 525, 857
 wcrtoomb 525
 wcscasecmp 526, 857
 wcscat 526, 857
 wcschr 526, 857
 wcscmp 527, 857
 wcscoll 527, 857
 wcscpy 527, 857
 wcscspn 528, 857
 wcsdup 528, 857
 wcsftime 528
 wcslen 529, 857
 wcsncasecmp 529, 857
 wcsncat 530, 857
 wcsncmp 530, 857
 wcsncpy 530, 857
 wcsnlen 531, 857
 wcsnrtombs 531, 857
 wcsrchr 532, 857
 wcsrtombs 532, 857
 wcssp 532, 857
 wcsstr 533, 857
 wcsstr-simple 533
 wcstok 533, 857
 wcswidth 535, 857
 wcsxfrm 535, 857
 wctob 536
 wctomb 408
 wctrans 550, 862
 wctype. 120, 121, 122, 123, 124, 125, 126, 127, 128,
 129, 130, 131, 132, 133, 551, 862
 wctype-h. 97, 539, 540, 541, 543, 544, 546, 548, 549
 wcwidth 536, 862
 windows-stat-inodes 87, 842
 windows-stat-timespec 843
 wmemchr 537
 wmemcmp 537
 wmemcpy 537
 wmemmove 537
 wmemcpy 841
 wmemset 538
 write 509

X

xalloc-die.....	49
xlist.....	885
xmap.....	886
xomap.....	887
xoset.....	886
xprintf.....	890
xprintf-gnu.....	890
xprintf-posix.....	890
xset.....	886
xstdopen.....	879
xstrerror.....	415
xstring-desc.....	881
xsublist.....	885
xvasprintf.....	890
xvasprintf-gnu.....	890

xvasprintf-posix.....	890
-----------------------	-----

Y

y0.....	256
y1.....	257
year2038.....	93, 96, 595, 834, 843
year2038-recommended.....	96, 595, 834, 843
yield.....	848
yn.....	257

Z

zprintf.....	889
zprintf-gnu.....	889
zprintf-posix.....	889

Undocumented Module Index

These are the modules that are not explicitly documented in this manual. Please look at the respective module description. (Ideally, this list should be as small as possible.)

A

acl.....	999
acl-permissions.....	999
alignof.....	999
allocator.....	999
amemxfrm.....	999
announce-gen.....	999
ansi-c++-opt.....	999
areadlinkat.....	999
areadlinkat-with-size.....	999
argv-iter.....	999
array-mergesort.....	999
assure.....	999
astrxfrm.....	999
asynsafe-spin.....	999
at-init.....	999
autobuild.....	999

B

backup-rename.....	999
backupfile.....	999
base32.....	999
base64.....	999
bcp47.....	999
binary-io.....	999
bison.....	999
bison-i18n.....	999
bitrotate.....	999
bitsetv.....	999
boot-time.....	999
builtin-expect.....	999

C

c-dtoastr.....	999
c-file-type.....	999
c-ldtoastr.....	999
c-nstrftime.....	999
c-snprintf.....	999
c-stack.....	999
c-strcase.....	999
c-strtof.....	999
c-vasnprintf.....	999
c-vasprintf.....	999
c-vsnprintf.....	999
c99.....	999
canon-host.....	999
canonicalize.....	999
careadlinkat.....	999
chdir-long.....	999

chmodat.....	999
chownat.....	999
classpath.....	999
clean-temp-simple.....	999
clock-time.....	999
cloexec.....	999
close-stream.....	999
closein.....	999
closeout.....	999
concat-filename.....	999
config-h.....	999
copy-file.....	999
count-leading-zeros.....	999
count-one-bits.....	999
count-trailing-zeros.....	999
cpu-supports.....	999
crc.....	999
crc-x86_64.....	999
crypto/af_alg.....	999
crypto/arcfour.....	999
crypto/arctwo.....	999
crypto/des.....	999
crypto/gc.....	999
crypto/gc-arcfour.....	999
crypto/gc-arctwo.....	999
crypto/gc-camellia.....	999
crypto/gc-des.....	999
crypto/gc-hmac-md5.....	999
crypto/gc-hmac-sha1.....	999
crypto/gc-hmac-sha256.....	999
crypto/gc-hmac-sha512.....	999
crypto/gc-md2.....	999
crypto/gc-md4.....	999
crypto/gc-md5.....	999
crypto/gc-pbkdf2.....	999
crypto/gc-rijndael.....	999
crypto/gc-sha1.....	999
crypto/gc-sha256.....	999
crypto/gc-sha512.....	999
crypto/gc-sm3.....	999
crypto/hmac-md5.....	999
crypto/hmac-sha1.....	999
crypto/hmac-sha256.....	999
crypto/hmac-sha512.....	999
crypto/md2.....	999
crypto/md2-buffer.....	999
crypto/md4.....	999
crypto/md4-buffer.....	999
crypto/md5.....	999
crypto/md5-buffer.....	999
crypto/rijndael.....	999
crypto/sha1.....	999

crypto/sha1-buffer..... 999
 crypto/sha256..... 999
 crypto/sha256-buffer..... 999
 crypto/sha3..... 999
 crypto/sha3-buffer..... 999
 crypto/sha512..... 999
 crypto/sha512-buffer..... 999
 crypto/sm3..... 999
 crypto/sm3-buffer..... 999
 csharpcomp..... 999
 csharpcomp-script..... 999
 csharpexec..... 999
 csharpexec-script..... 999
 cycle-check..... 999
 cygpath..... 999

D

d-ino..... 999
 d-type..... 999
 dcomp-script..... 999
 dev-ino..... 999
 dfa..... 999
 di-set..... 999
 diffseq..... 999
 dirname-lgpl..... 999
 do-release-commit-and-tag..... 999
 double-slash-root..... 999
 dtoastr..... 999
 dtotimespec..... 999

E

eloop-threshold..... 999
 errno-iter..... 999
 exclude..... 999
 execute..... 999
 exitfail..... 999

F

fatal-signal..... 999
 fd-safer-flag..... 999
 fdl..... 999
 fdl-1.3..... 999
 file-has-acl..... 999
 file-remote..... 999
 file-set..... 999
 file-type..... 999
 fileblocks..... 999
 filemode..... 999
 filename..... 999
 filenameecat..... 999
 filenameecat-lgpl..... 999
 filevercmp..... 999
 findprog..... 999
 findprog-in..... 999
 findprog-lgpl..... 999

fpe-trapping..... 999
 fpieee..... 999
 fprintftime..... 999
 fpucw..... 999
 frexp-nolibm..... 999
 frexpl-nolibm..... 999
 fstrcmp..... 999
 ftoastr..... 999
 full-read..... 999
 full-write..... 999
 fwriteerror..... 999

G

gendocs..... 999
 get-rusage-as..... 999
 get-rusage-data..... 999
 get_ppid_of..... 999
 get_progname_of..... 999
 gethrxtime..... 999
 getlocalename_l-unsafe..... 999
 getlocalename_l-unsafe-limited..... 999
 getndelim2..... 999
 getnline..... 999
 gnu-make..... 999
 gnu-web-doc-update..... 999
 gnulib-i18n..... 999
 gocomp-script..... 999
 gperf..... 999

H

hard-locale..... 999
 hash..... 999
 hashcode-file-inode..... 999
 hashcode-mem..... 999
 hashcode-named-file..... 999
 hashcode-string1..... 999
 hashcode-string2..... 999
 hashkey-string..... 999
 host-cpu-c-abi..... 999
 host-os..... 999
 hostent..... 999
 human..... 999

I

i-ring 999
 ialloc 999
 iconv-h 999
 idcache 999
 idpriv-drop 999
 idpriv-droptemp 999
 idx 999
 ignore-value 999
 immutable 999
 include_next 999
 ino-map 999
 integer_length 999
 integer_length_1 999
 integer_length_ll 999
 inttostr 999
 inttypes-h-incomplete 999
 isapipe 999
 isdir 999
 isfinite-no-c++ 999
 isinf-no-c++ 999
 isnan-no-c++ 999
 isnand 999
 isnand-nolibm 999
 isnanf-nolibm 999
 isnanl-nolibm 999
 issymlink 999
 issymlinkat 999

J

javacomp 999
 javacomp-script 999
 javaexec 999
 javaexec-script 999
 javaversion 999
 jit/cache 999

K

kwset 999

L

ldd 999
 ldtoastr 999
 lib-ignore 999
 libgmp 999
 libgmp-mpq 999
 libgmp-mpz 999
 libsigsegv 999
 libtextstyle 999
 libtextstyle-optional 999
 linebuffer 999
 link-follow 999
 localename 999
 localename-envron 999
 long-options 999

longlong 999

M

malloca 999
 math1 999
 mbmemcasecmp 999
 mbmemcasecoll 999
 mbszero 999
 mem-hash-map 999
 memcasecmp 999
 memchr2 999
 memcmp2 999
 memcoll 999
 memxor 999
 minmax 999
 mkancesdirs 999
 mkdir-p 999
 mktempd 999
 modechange 999
 modula2comp-script 999
 mpsort 999
 msvc-inval 999
 msvc-nothrow 999
 multiarch 999

N

nan 999
 next-prime 999
 nlcanon 999
 no-c++ 999
 nocrash 999
 nstrftime-limited 999

O

openat-h 999
 openat2 999
 opendirat 999
 openmp-init 999

P

parse-datetime2 999
 parse-duration 999
 passfd 999
 perl 999
 physmem 999
 pipe-filter-gi 999
 pipe-filter-ii 999
 pmccabe2html 999
 posix-shell 999
 posixtm 999
 posixver 999
 printf-frexp 999
 printf-frexpl 999
 printf-safe 999
 priv-set 999
 propername 999
 propername-lite 999
 pt_chown 999

Q

qacl 999
 qcopy-acl 999
 qset-acl 999
 quotearg-simple 999

R

read-file 999
 readme-release 999
 readtokens 999
 readtokens0 999
 regex-quote 999
 regexprops-generic 999
 relocatable-prog-wrapper 999
 rmdir-errno 999
 root-uid 999

S

same 999
 save-cwd 999
 savedir 999
 savewd 999
 selinux-at 999
 setlocale-fixes 999
 settime 999
 sfl-istream 999
 sh-filename 999
 sh-quote 999
 signbit-no-c++ 999
 signed-nan 999
 signed-snan 999
 sigpipe-die 999
 sigsegv 999
 simple-atomic 999
 size_max 999

snan 999
 sockets 999
 socklen 999
 sparcv8+ 999
 spawn-pipe 999
 ssfmalloc 999
 ssize_t 999
 stack 999
 stat-macros 999
 statat 999
 std-gnu11 999
 std-gnu23 999
 stdbit-h 999
 stdcountof-h 999
 stdio-windows 999
 stdopen 999
 strcase 999
 streq 999
 striconveha 999
 string-buffer-reversed 999
 stringeq 999
 strlen1 999
 supersede 999
 system-quote 999

T

tempname 999
 term-style-control 999
 terminfo 999
 terminfo-h 999
 test-framework-sh 999
 test-xfail 999
 timespec 999
 timespec-add 999
 timespec-sub 999
 tmpdir 999
 trim 999

U

u64 999
 ucs4-utf16 999
 ucs4-utf8 999
 unicodeio 999
 unictype/bidiclass-all 999
 unictype/block-all 999
 unictype/category-all 999
 unictype/combining-class-all 999
 unictype/incb-all 999
 unictype/joininggroup-all 999
 unictype/joiningtype-all 999
 unictype/property-all 999
 unictype/scripts-all 999
 unlink-busy 999
 update-copyright 999
 useless-if-before-free 999
 userspec 999

utf16-ucs4..... 999
 utf16-ucs4-unsafe..... 999
 utf8-ucs4..... 999
 utf8-ucs4-unsafe..... 999
 utimecmp..... 999

V

va-args..... 999
 vasnprintf..... 999
 vasnprintf-gnu..... 999
 vasnprintf-posix..... 999
 vasnwprintf..... 999
 vasnwprintf-gnu..... 999
 vasnwprintf-posix..... 999
 vc-list-files..... 999
 vc-mtime..... 999
 verror..... 999
 version-stamp..... 999
 vma-iter..... 999
 vma-prot..... 999

W

wait-process..... 999
 wgetcwd-lgpl..... 999
 windows-cygpath..... 999
 winsz-ioctl..... 999
 winsz-termios..... 999
 write-any-file..... 999

X

xalignalloc..... 999
 xalloc..... 999
 xalloc-oversized..... 999
 xbinary-io..... 999
 xconcat-filename..... 999
 xfreopen..... 999
 xgetcwd..... 999
 xgetcwd-lgpl..... 999
 xgetdomainname..... 999
 xgetgroups..... 999
 xgethostname..... 999
 xhash..... 999
 xmalloca..... 999
 xmemcoll..... 999
 xmemdup0..... 999
 xnanosleep..... 999
 xreadlink..... 999
 xreadlinkat..... 999
 xsetenv..... 999
 xsize..... 999
 xstriconv..... 999
 xstriconveh..... 999
 xstring-buffer..... 999
 xstring-buffer-reversed..... 999
 xstrndup..... 999
 xstrtod..... 999
 xstrtoimax..... 999
 xstrtol..... 999
 xstrtol-error..... 999
 xstrtold..... 999
 xstrtoll..... 999
 xstrtoumax..... 999

Y

yesno..... 999

General Index

\$

\$ 934

(

(..... 932

)

) 932

*

* 928

+

+ 928

_

_ 930

.

. 927

.tarball-version 965

=

[=] in regex 931

?

? 928

[

[..... 930

[=] in regex 931

[^ 930

[colon] in regex 931

]

] 930

^

^ 930, 934

-

--attribute_ 866

--libc_single_threaded 799

--VA_ARGS_ 5

_dl_find_object 614

_exit 477

_Exit 381

_Fork 477

_GL_EXTERN_INLINE 878

_GL_INLINE 878

_GL_INLINE_HEADER_BEGIN 878

_GL_INLINE_HEADER_END 879

_longjmp 553

_Noreturn 868

_setjmp 553

_tolower 554

_toupper 554

\

\ 926, 930

\' 936

\(..... 932

\) 932

\< 935

\> 935

\' 935

\{ 929

\} 929

\| 929

\b 935

\B 935

\s 935

\S 935

\w 935

\W 935

{

{ 929

}

} 929

|

| 929

A

a64l	381	argp_state_help	601
abort	382	argp_usage	601
Aborting execution	49	argz_add	601
abs	382	argz_add_sep	602
AC_LIB_HAVE_LINKFLAGS(<i>name</i> , [<i>dependencies</i>], [<i>includes</i>], [<i>testcode</i>], [<i>missing-message</i>])	953	argz_append	602
AC_LIB_LINKFLAGS(<i>name</i> , [<i>dependencies</i>])	953	argz_count	603
accept	431	argz_create	603
accept4	432	argz_create_sep	603
access	477	argz_delete	604
acct	819	argz_extract	604
acos	169	argz_insert	604
acosf	169	argz_next	605
acosh	169	argz_replace	605
acoshf	170	argz_stringify	606
acoshl	170	arithmetic type conversion	870
acosl	170	asctime	462
acospi	170	asctime_r	462
acospif	171	asin	171
acospil	171	asinf	171
addmntent	656	asinh	172
addseverity	625	asinhf	172
adjtime	806	asinhhl	172
adjtimex	808	asinl	172
advance	688	asinpi	173
aio_cancel	98	asinpif	173
aio_error	98	asinpil	173
aio_fsync	98	asprintf	337
aio_init	597	assert	103
aio_read	99	assertion	866
aio_return	99	at_quick_exit	383
aio_suspend	99	atan	173
aio_write	100	atan2	173
alarm	478	atan2f	174
aligned_alloc	382	atan2l	174
ALLOC_N	865	atan2pi	174
ALLOC_N_UNINITIALIZED	865	atan2pif	174
alloca	864	atan2pil	175
allocated initialization	937	atanf	175
ALLOC	865	atanh	175
alphasort	134	atanhf	175
alternation operator	929	atanhl	176
alternation operator and ‘~’	934	atanl	176
anchoring	934	atanpi	176
anchors	934	atanpif	176
arc4random	762	atanpil	177
arc4random_buf	762	atexit	383
arc4random_uniform	762	atof	384
argp_err_exit_status	599	atoi	384
argp_error	599	atol	384
argp_failure	599	atoll	385
argp_help	600	atomic_compare_exchange_strong	324
argp_parse	600	atomic_compare_exchange_strong_explicit	324
argp_program_bug_address	600	atomic_compare_exchange_weak	324
argp_program_version	600	atomic_compare_exchange_weak_explicit	325
argp_program_version_hook	601	atomic_exchange	325
		atomic_exchange_explicit	325
		atomic_fetch_add	326
		atomic_fetch_add_explicit	326

atomic_fetch_and..... 326
 atomic_fetch_and_explicit..... 327
 atomic_fetch_or..... 327
 atomic_fetch_or_explicit..... 327
 atomic_fetch_sub..... 328
 atomic_fetch_sub_explicit..... 328
 atomic_fetch_xor..... 328
 atomic_fetch_xor_explicit..... 329
 atomic_flag_clear..... 329
 atomic_flag_clear_explicit..... 329
 atomic_flag_test_and_set..... 330
 atomic_flag_test_and_set_explicit..... 330
 atomic_init..... 330
 atomic_is_lock_free..... 331
 atomic_load..... 331
 atomic_load_explicit..... 331
 atomic_signal_fence..... 332
 atomic_store..... 332
 atomic_store_explicit..... 332
 atomic_thread_fence..... 333
 Attributes..... 866
 authdes_create..... 693
 authdes_getucred..... 697
 authdes_pk_create..... 694
 authnone_create..... 694
 authunix_create..... 694
 authunix_create_default..... 694
 autopoint, caveat..... 34
 Awk..... 924

B

back-references..... 933
 backtrace..... 621
 backtrace_symbols..... 622
 backtrace_symbols_fd..... 622
 backtracking..... 928, 930
 basename..... 160
 bcmp..... 554
 bcopy..... 554
 bdflush..... 787
 be16toh..... 139
 be32toh..... 140
 be64toh..... 140
 beginning-of-line operator..... 934
 bind..... 432
 bind_textdomain_codeset..... 161
 bindresvport..... 667
 bindtextdomain..... 161
 block size..... 920
 bracket expression..... 930
 brk..... 819
 bsd_signal..... 555
 bsearch..... 385
 bswap_16..... 610
 bswap_32..... 610
 bswap_64..... 611
 btowc..... 511

buffer field, set by re_compile_pattern..... 937
 buffer initialization..... 937
 bzero..... 555

C

c16rtomb..... 475
 c32rtomb..... 475
 c8rtomb..... 474
 C++ header files..... 40
 C++ tests modules..... 49
 cabs..... 104
 cabsf..... 104
 cabsl..... 104
 cacos..... 104
 cacosf..... 105
 cacosh..... 105
 cacoshf..... 105
 cacoshl..... 105
 cacosl..... 106
 call_once..... 453
 calloc..... 385
 callrpc..... 698
 canonicalize..... 177
 canonicalize_file_name..... 763
 canonicalizef..... 177
 canonicalizel..... 177
 capget..... 781
 capset..... 781
 carg..... 106
 cargf..... 106
 cargl..... 106
 casin..... 107
 casinl..... 107
 casinh..... 107
 casinhf..... 107
 casinhl..... 108
 casinl..... 108
 catan..... 108
 catanf..... 108
 catanh..... 109
 catanhf..... 109
 catanhl..... 109
 catanl..... 109
 catclose..... 269
 catgets..... 270
 catopen..... 270
 cbrt..... 178
 cbrtf..... 178
 cbrtl..... 178
 ccos..... 110
 ccosf..... 110
 ccosh..... 110
 ccoshf..... 110
 ccoshl..... 110
 ccosl..... 111
 ceil..... 178
 ceilf..... 179

ceil1.....	179	cnd_broadcast.....	453
cexp.....	111	cnd_destroy.....	454
cexpf.....	111	cnd_init.....	454
cexpl.....	111	cnd_signal.....	454
cfgetispeed.....	450	cnd_timedwait.....	455
cfgetospeed.....	450	cnd_wait.....	455
cfmakeraw.....	816	‘colon]’ in regex.....	931
cfree.....	763	comments describing functions.....	42
cfsetispeed.....	450	compoundn.....	179
cfsetospeed.....	450	compoundnf.....	180
cfsetspeed.....	816	compoundnl.....	180
changelog.....	968	conditional dependencies.....	38
character classes.....	931	configmake.h, module for updating.....	959
chdir.....	478	confstr.....	479
chmod.....	437	conj.....	113
chown.....	479	conjf.....	113
chroot.....	820	conjl.....	114
cimag.....	112	connect.....	432
cimagf.....	112	copy_file_range.....	821
cimagl.....	112	copysign.....	180
clearenv.....	763	copysignf.....	180
clearerr.....	338	copysignl.....	180
clearerr_unlocked.....	755	cos.....	181
clnt_broadcast.....	706	cosf.....	181
clnt_create.....	699	cosh.....	181
clnt_pcreateerror.....	699	coshf.....	181
clnt_perrno.....	699	coshl.....	182
clnt_perror.....	699	cosl.....	182
clnt_spccreateerror.....	700	cospi.....	182
clnt_sperrno.....	700	cospif.....	182
clnt_sperror.....	700	cospil.....	183
clntraw_create.....	700	cpow.....	114
clnttcp_create.....	700	cpowf.....	114
clntudp_bufcreate.....	701	cpowl.....	114
clntudp_create.....	701	cproj.....	115
clntunix_create.....	701	cprojf.....	115
clock.....	463	cprojl.....	115
clock_adjtime.....	817	creal.....	115
clock_getcpuclockid.....	463	crealf.....	116
clock_getres.....	463	creall.....	116
clock_gettime.....	464	creat.....	143
clock_nanosleep.....	464	crypt.....	480
clock_settime.....	464	csin.....	116
clog.....	112	csinf.....	116
clog10.....	611	csinh.....	117
clog10f.....	611	csinhf.....	117
clog10l.....	612	csinhl.....	117
clogf.....	113	csinl.....	117
clogl.....	113	csqrt.....	118
clone.....	740	csqrtf.....	118
close.....	479	csqrtl.....	118
close_range.....	820	ctan.....	118
closedir.....	134	ctanf.....	119
closefrom.....	820	ctanh.....	119
closelog.....	449	ctanhf.....	119
CMPLX.....	103	ctanh1.....	119
CMPLXF.....	103	ctanl.....	120
CMPLXL.....	103	ctermid.....	339

ctime..... 465
 ctime_r..... 466
 cuserid..... 755

D

daddl..... 183
 daemon..... 821
 daylight..... 466
 dbm_clearerr..... 260
 dbm_close..... 260
 dbm_delete..... 261
 dbm_error..... 261
 dbm_fetch..... 261
 dbm_firstkey..... 261
 dbm_nextkey..... 262
 dbm_open..... 262
 dbm_store..... 262
 dcgettext..... 161
 dcgettext_l..... 162
 dcgettext..... 162
 dcgettext_l..... 162
 ddivl..... 183
 DEF files..... 845
 describing functions, locating..... 42
 dfmal..... 183
 dgettext..... 163
 dgettext_l..... 163
 difftime..... 466
 dirfd..... 134
 dirname..... 160
 div..... 385
 dl_iterate_phdr..... 638
 dladdr..... 138
 dladdr1..... 613
 dlclose..... 138
 dlerror..... 138
 dlinfo..... 613
 dlmopen..... 614
 dlopen..... 139
 dlsym..... 139
 dlvsym..... 614
 dmull..... 184
 dn_comp..... 689
 dn_expand..... 690
 dn_skipname..... 690
 dngettext..... 163
 dngettext_l..... 164
 double inclusion of header files..... 40
 dprintf..... 339
 drand48..... 386
 drand48_r..... 763
 drem..... 641
 dremf..... 641
 drem1..... 641
 dsqrt1..... 184
 dsub1..... 184
 dup..... 480

dup2..... 480
 dup3..... 481
 duplocale..... 165
 dysize..... 817

E

eaccess..... 821
 ecvt..... 555
 ecvt_r..... 764
 Egrep..... 924
 Emacs..... 924
 encrypt..... 481
 end in struct re_registers..... 941
 end-of-line operator..... 934
 endaliasent..... 597
 endfsent..... 626
 endgrent..... 154
 endhostent..... 263
 endmntent..... 656
 endnetent..... 264
 endnetgrent..... 658
 endprotoent..... 264
 endpwent..... 298
 endrpcent..... 704
 endservent..... 264
 endsent..... 635
 endspent..... 746
 endttyent..... 818
 endusershell..... 822
 endutent..... 829
 endutxent..... 510
 environ..... 482
 envz_add..... 614
 envz_entry..... 615
 envz_get..... 615
 envz_merge..... 615
 envz_remove..... 616
 envz_strip..... 616
 epoll_create..... 781
 epoll_create1..... 782
 epoll_ctl..... 782
 epoll_pwait..... 782
 epoll_pwait2..... 782
 epoll_wait..... 783
 equivalence class expression in regex..... 931
 erand48..... 386
 erand48_r..... 764
 erf..... 185
 erfc..... 185
 erfcf..... 185
 erfc1..... 185
 erff..... 185
 erfl..... 186
 err..... 617
 errno..... 143
 error..... 620, 914
 error_at_line..... 620

```

error_message_count ..... 620
error_one_per_line ..... 621
error_print_progname ..... 621
errx ..... 617
ether_aton ..... 666
ether_aton_r ..... 666
ether_hostton ..... 666
ether_line ..... 666
ether_ntoa ..... 667
ether_ntoa_r ..... 667
ether_ntohost ..... 667
euidaccess ..... 822
eventfd ..... 783
eventfd_read ..... 783
eventfd_write ..... 783
execl ..... 482
execle ..... 483
execlp ..... 483
execv ..... 483
execve ..... 484
execveat ..... 822
execvp ..... 484
execvpe ..... 823
exit ..... 386
exp ..... 186
exp10 ..... 186
exp10f ..... 187
exp10l ..... 187
exp10m1 ..... 187
exp10m1f ..... 188
exp10m1l ..... 188
exp2 ..... 188
exp2f ..... 188
exp2l ..... 189
exp2m1 ..... 189
exp2m1f ..... 189
exp2m1l ..... 189
expf ..... 190
expl ..... 190
explicit_bzero ..... 775
expm1 ..... 190
expm1f ..... 191
expm1l ..... 191
EXPR_SIGNED ..... 869
extern inline ..... 877
Extra tests modules ..... 49

```

F

fabs.....	191
fabsf.....	191
fabsl.....	192
faccessat.....	485
fadd.....	192
faddl.....	192
fallocate.....	623
fanotify_init.....	784
fanotify_mark.....	784

fastmap initialization	937
fastmaps	939
fattach	555
fchdir	485
fchmod	438
fchmodat	438
fchown	485
fchownat	486
fclose	340
fcloseall	755
fcntl	143
fcvt	556
fcvt_r	764
FD_CLR	428
FD_ISSET	428
FD_SET	428
FD_ZERO	429
fdatasync	486
fdetach	556
fdim	192
fdimf	193
fdiml	193
fdiv	193
fdivl	193
fdopen	341
fdopendir	135
feclearexcept	147
fedisableexcept	624
feenableexcept	624
fegetenv	147
fegetexcept	625
fegetexceptflag	147
fegetmode	148
fegetround	148
fehldexcept	148
feof	341
feof_unlocked	756
feraiseexcept	148
ferror	341
ferror_unlocked	756
fesetenv	149
fesetexcept	149
fesetexceptflag	149
fesetmode	150
fesetround	150
fetestexcept	150
fetestexceptflag	151
feupdateenv	151
fexecve	487
fflush	341
fflush_unlocked	756
ffma	194
ffmal	194
ffs	419
ffsl	420
ffsll	420
fgetc	342
fgetc_unlocked	757

fgetfilecon.....	744	fminimum_magf.....	202
fgetgrent.....	632	fminimum_magl.....	202
fgetgrent_r.....	633	fminimum_num.....	203
fgetpos.....	342	fminimum_numf.....	203
fgetpwent.....	685	fminimum_numl.....	204
fgetpwent_r.....	685	fminimumf.....	204
fgets.....	343	fminimuml.....	204
fgets_unlocked.....	757	fminl.....	205
fgetsgent.....	635	fminmag.....	205
fgetsgent_r.....	635	fminmagf.....	205
fgetspent.....	746	fminmagl.....	205
fgetspent_r.....	746	fmod.....	206
fgetwc.....	512	fmodf.....	206
fgetwc_unlocked.....	835	fmodl.....	206
fgetws.....	512	fmtmsg.....	151
fgetws_unlocked.....	835	fmul.....	207
fgetxattr.....	812	fmull.....	207
fileno.....	343	fnmatch.....	152
fileno_unlocked.....	757	fopen.....	344
Finding modules.....	20	fopencookie.....	758
finite.....	642	fork.....	487
finitef.....	642	forkpty.....	684
finitel.....	642	fpathconf.....	487
flistxattr.....	813	fpclassify.....	207
flock.....	784	fprintf.....	344
flockfile.....	343	fputc.....	346
floor.....	194	fputc_unlocked.....	758
floorf.....	194	fputs.....	347
floorl.....	195	fputs_unlocked.....	758
fma.....	195	fputwc.....	512
fmaf.....	195	fputwc_unlocked.....	836
fmal.....	196	fputws.....	513
fmax.....	196	fputws_unlocked.....	836
fmaxf.....	196	fread.....	347
fmaximum.....	196	fread_unlocked.....	759
fmaximum_mag.....	197	free.....	386
fmaximum_mag_num.....	197	free_aligned_sized.....	386
fmaximum_mag_numf.....	198	free_sized.....	387
fmaximum_mag_numl.....	198	freeaddrinfo.....	264
fmaximum_magf.....	197	freeifaddrs.....	637
fmaximum_magl.....	197	freelocale.....	166
fmaximum_num.....	198	FREE.....	865
fmaximum_numf.....	199	fremovexattr.....	813
fmaximum_numl.....	199	freopen.....	348
fmaximumf.....	199	frexp.....	207
fmaximuml.....	199	frexpf.....	208
fmaxl.....	200	frexpl.....	208
fmaxmag.....	200	fromfp.....	208
fmaxmagf.....	200	fromfpf.....	209
fmaxmagl.....	200	fromfpl.....	209
fmemopen.....	343	fromfpx.....	209
fmin.....	201	fromfpxf.....	209
fminf.....	201	fromfpxl.....	210
fminimum.....	201	fscanf.....	348
fminimum_mag.....	201	fsconfig.....	792
fminimum_mag_num.....	202	fseek.....	349
fminimum_mag_numf.....	203	fseeko.....	349
fminimum_mag_numl.....	203	fsetpos.....	350

fssetxattr	813
fsmount	792
fsopen	792
fspick	792
fsqrt	210
fsqrtl	210
fstat	439
fstatat	439
fstatfs	801
fstatvfs	446
fsub	210
fsubl	211
fsync	487
ftell	350
ftello	350
ftime	556
ftok	422
ftruncate	488
ftrylockfile	351
fts_children	627
fts_close	627
fts_open	627
fts_read	628
fts_set	628
ftw	556
funlockfile	351
futimens	440
futimes	806
futimesat	806
fwide	513
fwprintf	513
fwrite	351
fwrite_unlocked	759
fwscanf	514

G

gai_strerror	265
gamma.....	643
gammaf	643
gammal	643
gcd.....	914
gcvrt.....	557
get_aphys_pages.....	803
get_current_dir_name	823
get_myaddress	701
get_nprocs	803
get_nprocs_conf	803
get_phys_pages	804
getaddrinfo	265
getaliasbyname	598
getaliasbyname_r.....	598
getaliasent.....	598
getaliasent_r	598
getauxval	780
getc.....	352
getc_unlocked	352
getchar	353

getchar_unlocked.....	353
getcontext.....	557
getcpu.....	740
getcwd.....	488
getdate.....	467
getdate_err.....	467
getdate_r.....	817
getdelim.....	353
getdirententries.....	612
getdomainname.....	823
getdtablesize.....	824
getegid.....	489
getentropy.....	489
getenv.....	387
geteuid.....	489
getfilecon.....	745
getfsent.....	626
getfsfile.....	626
getfsspec.....	626
getgid.....	490
getgrent.....	154
getgrent_r.....	633
getgrgid.....	154
getgrgid_r.....	155
getgrnam.....	155
getgrnam_r.....	155
getgrouplist.....	633
getgroups.....	490
gethostbyaddr.....	557
gethostbyaddr_r.....	658
gethostbyname.....	558
gethostbyname_r.....	659
gethostbyname2.....	659
gethostbyname2_r.....	659
gethostent.....	265
gethostent_r.....	660
gethostid.....	490
gethostname.....	491
getifaddrs.....	637
getip4sourcefilter.....	668
getitimer.....	558
getline.....	354
getloadavg.....	765
getlocalename_l.....	166
getlogin.....	491
getlogin_r.....	491
getmntent.....	657
getmntent_r.....	657
getmsg.....	558
getnameinfo.....	265
getnetbyaddr.....	266
getnetbyaddr_r.....	660
getnetbyname.....	266
getnetbyname_r.....	660
getnetent.....	266
getnetent_r.....	660
getnetgrent.....	660
getnetgrent_r.....	661

getnetname	694	getsockopt	434
getopt	492	getsourcefilter	668
getopt_long	628	getspent	746
getopt_long_only	630	getspent_r	747
getpagesize	824	getspnam	747
getpass	825	getspnam_r	747
getpayload	211	getsubopt	387
getpayloadf	211	gettext	164
getpayloadl	212	gettext_l	164
getpeername	433	gettextize, caveat	34
getpgid	493	gettid	825
getpgrp	493	gettimeofday	559
getpid	493	getttyent	818
getpmsg	558	getttynam	819
getppid	494	getuid	495
getpriority	427	getumask	800
getprogname	913	getusershell	826
getprotobyname	266	getutent	830
getprotobyname_r	661	getutent_r	830
getprotobynumber	267	getutid	830
getprotobynumber_r	661	getutid_r	831
getprotoent	267	getutline	831
getprotoent_r	662	getutline_r	831
getpt	765	getutmp	834
getpublickey	698	getutmpx	834
getpw	685	getutxent	510
getpwent	298	getutxid	510
getpwent_r	685	getutxline	511
getpwnam	298	getw	759
getpwnam_r	298	getwc	515
getpwuid	298	getwc_unlocked	836
getpwuid_r	299	getwchar	515
getrandom	797	getwchar_unlocked	837
getresgid	494	getwd	559
getresuid	494	getxattr	814
getrlimit	427	gitlog	968
getrpcbyname	704	glob	153
getrpcbyname_r	704	glob_pattern_p	631
getrpcbynumber	704	globfree	154
getrpcbynumber_r	705	gmtime	467
getrpccent	705	gmtime_r	467
getrpccent_r	705	gnu_dev_major	805
getrpcport	702	gnu_dev_makedev	805
getrusage	427	gnu_dev_minor	805
gets	559	gnu_get_libc_release	632
getsecretkey	698	gnu_get_libc_version	632
getservbyname	267	gnulib-tool	20
getservbyname_r	662	grantpt	387
getservbyport	267	Grep	924
getservbyport_r	662	group_member	826
getservent	268	grouping	932
getservent_r	662	gsignal	749
getsgent	635		
getsgent_r	635		
getsgnam	636		
getsgnam_r	636		
getsid	494		
getsockname	433		

H

<code>h_errno</code>	560
<code>hasmntopt</code>	657
<code>hcreate</code>	302
<code>hcreate_r</code>	742
<code>hdestroy</code>	303
<code>hdestroy_r</code>	743
header file include protection	40
Header files and C++	40
<code>herror</code>	663
<code>host2netname</code>	695
<code>hsearch</code>	303
<code>hsearch_r</code>	743
<code>hstrerror</code>	663
<code>htobe16</code>	140
<code>htobe32</code>	140
<code>htobe64</code>	141
<code>htole16</code>	141
<code>htole32</code>	141
<code>htole64</code>	142
<code>htonl</code>	100
<code>htons</code>	100
<code>hypot</code>	212
<code>hypotf</code>	212
<code>hypotl</code>	213

I

<code>iconv</code>	156
<code>iconv_close</code>	156
<code>iconv_open</code>	156
<code>if_freenameindex</code>	262
<code>if_indextoname</code>	263
<code>if_nameindex</code>	263
<code>if_nametoindex</code>	263
<code>ilogb</code>	213
<code>ilogbf</code>	213
<code>ilogbl</code>	214
<code>imaxabs</code>	157
<code>imaxdiv</code>	157
<code>in6addr_any</code>	269
<code>in6addr_loopback</code>	269
<code>index</code>	560
<code>inet_addr</code>	101
<code>inet_aton</code>	606
<code>inet_lnaof</code>	606
<code>inet_makeaddr</code>	607
<code>inet_net_ntop</code>	607
<code>inet_net_pton</code>	607
<code>inet_netof</code>	607
<code>inet_network</code>	608
<code>inet_nsap_addr</code>	608
<code>inet_nsap_ntoa</code>	608
<code>inet_ntoa</code>	101
<code>inet_ntop</code>	101
<code>inet_pton</code>	102
<code>inet6_opt_append</code>	669
<code>inet6_opt_find</code>	669

<code>inet6_opt_finish</code>	670
<code>inet6_opt_get_val</code>	670
<code>inet6_opt_init</code>	670
<code>inet6_opt_next</code>	670
<code>inet6_opt_set_val</code>	670
<code>inet6_option_alloc</code>	668
<code>inet6_option_append</code>	668
<code>inet6_option_find</code>	668
<code>inet6_option_init</code>	669
<code>inet6_option_next</code>	669
<code>inet6_option_space</code>	669
<code>inet6_rth_add</code>	671
<code>inet6_rth_getaddr</code>	671
<code>inet6_rth_init</code>	671
<code>inet6_rth_reverse</code>	671
<code>inet6_rth_segments</code>	671
<code>inet6_rth_space</code>	672
<code>initgroups</code>	634
initial import	20
<code>initstate</code>	388
<code>initstate_r</code>	765
<code>inline</code>	877
<code>innetgr</code>	663
<code>inotify_add_watch</code>	785
<code>inotify_init</code>	786
<code>inotify_init1</code>	786
<code>inotify_rm_watch</code>	786
<code>insque</code>	303
<code>int</code>	865
<code>INT_ADD_OK</code>	872
<code>INT_ADD_OVERFLOW</code>	875
<code>INT_ADD_RANGE_OVERFLOW</code>	876
<code>INT_ADD_WRAPV</code>	873
<code>INT_BUFSIZE_BOUND</code>	870
<code>INT_DIVIDE_OVERFLOW</code>	875
<code>INT_DIVIDE_RANGE_OVERFLOW</code>	877
<code>INT_LEFT_SHIFT_OVERFLOW</code>	875
<code>INT_LEFT_SHIFT_RANGE_OVERFLOW</code>	877
<code>INT_MULTIPLY_OK</code>	872
<code>INT_MULTIPLY_OVERFLOW</code>	875
<code>INT_MULTIPLY_RANGE_OVERFLOW</code>	877
<code>INT_MULTIPLY_WRAPV</code>	873
<code>INT_NEGATE_OVERFLOW</code>	875
<code>INT_NEGATE_RANGE_OVERFLOW</code>	876
<code>INT_PROMOTE</code>	870
<code>INT_REMAINDER_OVERFLOW</code>	875
<code>INT_REMAINDER_RANGE_OVERFLOW</code>	877
<code>INT_STRLEN_BOUND</code>	871
<code>INT_SUBTRACT_OK</code>	872
<code>INT_SUBTRACT_OVERFLOW</code>	875
<code>INT_SUBTRACT_RANGE_OVERFLOW</code>	876
<code>INT_SUBTRACT_WRAPV</code>	873
integer arithmetic portability	14
integer bounds	870
integer overflow	869
integer overflow checking	871
integer properties	869
integer range overflow	875

integer type conversion..... 870
 integer type overflow..... 874
 interval expression..... 929
 invoking `gnulib-tool`..... 20
`ioctl`..... 560
`ioperm`..... 787
`iopl`..... 787
`isalnum`..... 120
`isalnum_l`..... 121
`isalpha`..... 121
`isalpha_l`..... 122
`isascii`..... 561
`isastream`..... 561
`isatty`..... 495
`isblank`..... 122
`isblank_l`..... 123
`iscntrl`..... 123
`iscntrl_l`..... 124
`isctype`..... 612
`isdigit`..... 124
`isdigit_l`..... 125
`isfdtype`..... 799
`isfinite`..... 214
`isgraph`..... 125
`isgraph_l`..... 126
`isgreater`..... 214
`isgreaterequal`..... 215
`isinf`..... 215
`isinff`..... 644
`isinfl`..... 644
`isless`..... 215
`islessequal`..... 215
`islessgreater`..... 216
`islower`..... 126
`islower_l`..... 127
`isnan`..... 216
`isnanf`..... 644
`isnanl`..... 645
`isnormal`..... 216
`isprint`..... 127
`isprint_l`..... 128
`ispunct`..... 128
`ispunct_l`..... 129
`isspace`..... 129
`isspace_l`..... 130
`isunordered`..... 216
`isupper`..... 130
`isupper_l`..... 131
`iswalnum`..... 539
`iswalnum_l`..... 540
`iswalpha`..... 540
`iswalpha_l`..... 540
`iswblank`..... 541
`iswblank_l`..... 541
`iswcntrl`..... 541
`iswcntrl_l`..... 542
`iswctype`..... 542
`iswctype_l`..... 542

`iswdigit`..... 543
`iswdigit_l`..... 543
`iswgraph`..... 543
`iswgraph_l`..... 544
`iswlower`..... 544
`iswlower_l`..... 544
`iswprint`..... 544
`iswprint_l`..... 545
`iswpunct`..... 545
`iswpunct_l`..... 546
`iswspace`..... 546
`iswspace_l`..... 546
`iswupper`..... 546
`iswupper_l`..... 547
`iswxdigit`..... 547
`iswxdigit_l`..... 547
`isxdigit`..... 131
`isxdigit_l`..... 132

J

`j0`..... 217
`j0f`..... 645
`j0l`..... 645
`j1`..... 217
`j1f`..... 646
`j1l`..... 646
`jn`..... 217
`jnf`..... 646
`jnl`..... 647
`jrand48`..... 388
`jrand48_r`..... 766

K

`key_decryptsession`..... 695
`key_decryptsession_pk`..... 695
`key_encryptsession`..... 695
`key_encryptsession_pk`..... 695
`key_gendes`..... 696
`key_get_conv`..... 696
`key_secretkey_is_set`..... 696
`key_setsecret`..... 696
`kill`..... 309
`kill_dependency`..... 333
`killpg`..... 309
`klogctl`..... 787

L

l64a.....	388
labs.....	388
lchmod.....	800
lchown.....	495
lckpddf.....	747
lcong48.....	389
lcong48_r.....	766
LD DEF files.....	845
ldexp.....	217
ldexpf.....	217
ldexpl.....	218
ldiv.....	389
le16toh.....	142
le32toh.....	142
le64toh.....	142
lfind.....	303
lgamma.....	218
lgamma_r.....	647
lgammaf.....	218
lgammaf_r.....	648
lgammal.....	218
lgammal_r.....	648
lgetfilecon.....	745
lgetxattr.....	814
link.....	496
linkat.....	496
lio_listio.....	100
listen.....	434
listxattr.....	814
llabs.....	389
lldiv.....	389
llistxattr.....	815
llogb.....	219
llogbf.....	219
llogbl.....	219
llrint.....	219
llrintf.....	220
llrintl.....	220
llround.....	220
llroundf.....	220
llroundl.....	220
loc1.....	689
loc2.....	689
localeconv.....	167
localtime.....	468
localtime_r.....	468
lockf.....	496
locs.....	689
log.....	221
log10.....	221
log10f.....	221
log10l.....	222
log10p1.....	222
log10p1f.....	222
log10p1l.....	222
log1p.....	223
log1pf.....	223

log1p1.....	223
log2.....	224
log2f.....	224
log2l.....	224
log2p1.....	225
log2p1f.....	225
log2p1l.....	225
logb.....	225
logbf.....	226
logbl.....	226
logf.....	226
login.....	833
login_tty.....	833
logl.....	227
logp1.....	227
logp1f.....	227
logp1l.....	227
long-running tests modules.....	49
longjmp.....	308
lrand48.....	389
lrand48_r.....	767
lremovexattr.....	815
lrint.....	228
lrintf.....	228
lrintl.....	228
lround.....	228
lroundf.....	229
lroundl.....	229
lsearch.....	304
lseek.....	497
lsetxattr.....	815
lstat.....	441
lutimes.....	807

M

madvise.....	788
makecontext.....	562
mallinfo.....	638
mallinfo2.....	638
malloc.....	390, 937
malloc_info.....	639
malloc_stats.....	639
malloc_trim.....	639
malloc_usable_size.....	639
mallopt.....	640
matching list.....	930
matching newline.....	930
matching with GNU functions.....	937
math_errhandling.....	229
matherr.....	648
mblen.....	390
mbrlen.....	515
mbrtoc16.....	475
mbrtoc32.....	476
mbrtoc8.....	475
mbrtowc.....	516
mbsinit.....	516

mbsnrtowcs	517	mq_unlink	260
mbsrtowcs	517	mrnd48	392
mbstowcs	390	mrnd48_r	768
mbtowc	390	mremap	789
mcheck	654	msgctl	426
mcheck_check_all	655	msgget	426
mcheck_pedantic	655	msgrcv	426
memalign	640	msgsnd	426
memalignment	391	msync	423
memccpy	408	mtrace	656
memchr	408	mtx_destroy	456
memcmp	408	mtx_init	456
memcpy	409	mtx_lock	456
memfd_create	788	mtx_timedlock	457
memfrob	776	mtx_trylock	457
memmem	409	mtx_unlock	457
memmove	409	multibyte locale	851
Memory allocation failure	49	munlock	423
mempcpy	776	munlockall	424
memrchr	777	munmap	424
memset	410	muntrace	656
memset_explicit	410		
mincore	788	N	
mkdir	441	name_to_handle_at	623
mkdirat	442	nan	230
mkdtemp	391	nanf	231
mkfifo	442	nanl	231
mkfifoat	442	nanosleep	469
mknod	443	nearbyint	231
mknodat	443	nearbyintf	231
mkostemp	391	nearbyintl	231
mkostemps	767	netname2host	696
mkstemp	392	netname2user	697
mkstemps	767	newlocale	167
mktemp	562	nextafter	232
mktime	469	nextafterf	232
mlock	422	nextafterl	232
mlock2	789	nextdown	232
mlockall	422	nextdownf	233
mmap	422	nextdownl	233
modf	229	nexttoward	233
modff	230	nexttowardf	233
modfl	230	nexttowardl	234
monstartup	785	nextup	234
mount	793	nextupf	234
mount_setattr	793	nextupl	234
move_mount	793	nftw	153
mprobe	655	ngettext	164
mprotect	423	ngettext_l	165
mq_close	258	nice	497
mq_getattr	258	nis_add	724
mq_notify	258	nis_add_entry	725
mq_open	258	nis_addmember	725
mq_receive	259	nis_checkpoint	725
mq_send	259	nis_clone_object	725
mq_setattr	259	nis_creategroup	725
mq_timedreceive	259	nis_destroy_object	725
mq_timedsend	260		

nis_destroygroup 726
 nis_dir_cmp 726
 nis_domain_of 726
 nis_domain_of_r 726
 nis_first_entry 726
 nis_freenames 726
 nis_freeresult 727
 nis_freeservlist 727
 nis_freetags 727
 nis_getnames 727
 nis_getservlist 727
 nis_ismember 727
 nis_leaf_of 728
 nis_leaf_of_r 728
 nis_lerror 728
 nis_list 728
 nis_local_directory 728
 nis_local_group 728
 nis_local_host 729
 nis_local_principal 729
 nis_lookup 729
 nis_mkdir 729
 nis_modify 729
 nis_modify_entry 729
 nis_name_of 730
 nis_name_of_r 730
 nis_next_entry 730
 nis_perror 730
 nis_ping 730
 nis_print_directory 731
 nis_print_entry 731
 nis_print_group 731
 nis_print_group_entry 731
 nis_print_link 731
 nis_print_object 732
 nis_print_result 732
 nis_print_rights 732
 nis_print_table 732
 nis_remove 732
 nis_remove_entry 733
 nis_removemember 733
 nis_rmdir 733
 nis_servstate 733
 nis_sperrno 733
 nis_sperror 733
 nis_sperror_r 734
 nis_stats 734
 nis_verifygroup 734
 nl_langinfo 159
 nl_langinfo_l 160
 nonmatching list 930
 not_bol field in pattern buffer 934
 nrand48 392
 nrand48_r 768
 ns_name_compress 608
 ns_name_ntop 609
 ns_name_pack 609
 ns_name_pton 609

ns_name_skip 609
 ns_name_uncompress 610
 ns_name_unpack 610
 ntohl 102
 ntohs 102
 ntp_adjtime 808
 ntp_gettime 809
 ntp_gettimex 809
 null pointer 57
 null pointer type 80
 num_regs in struct re_registers 941

O

Obsolete modules 49
 obstack_alloc_failed_handler 672
 obstack_exit_failure 672
 obstack_free 673
 obstack_printf 673
 obstack_vprintf 675
 on_exit 768
 open 144
 open-group operator and ‘^’ 934
 open_by_handle_at 623
 open_memstream 354
 open_tree 793
 open_wmemstream 518
 openat 145
 opendir 135
 openlog 449
 openpty 684
 optarg 497
 opterr 498
 optind 498
 optopt 498
 or operator 929
 Out of Memory handling 49
 overflow, integer 869
 overflow, integer range 875
 overflow, integer type 874

P

parenthesizing 932
 parse_printf_format 676
 pathconf 498
 pattern buffer initialization 937
 pattern buffer, definition of 936
 pause 498
 pclose 354
 perror 355
 personality 794
 pidfd_getfd 795
 pidfd_getpid 795
 pidfd_open 795
 pidfd_send_signal 795
 pidfd_spawn 753
 pidfd_spawnnp 753

pipe.....	499	posix_trace_attr_getlogfullpolicy.....	563
pipe2.....	499	posix_trace_attr_getlogsize.....	564
pkey_alloc.....	789	posix_trace_attr_getmaxdatasize.....	564
pkey_free.....	790	posix_trace_attr_getmaxsystemeventsizesize... 564	
pkey_get.....	790	posix_trace_attr_getmaxusereventsizesize.... 565	
pkey_mprotect.....	790	posix_trace_attr_getname.....	565
pkey_set.....	790	posix_trace_attr_getstreamfullpolicy..... 565	
pmap_getmaps.....	706	posix_trace_attr_getstreamsize.....	565
pmap_getport.....	706	posix_trace_attr_init.....	566
pmap_rmtcall.....	706	posix_trace_attr_setinherited.....	566
pmap_set.....	707	posix_trace_attr_setlogfullpolicy.....	566
pmap_unset.....	707	posix_trace_attr_setlogsize.....	567
poll.....	270	posix_trace_attr_setmaxdatasize.....	567
popen.....	355	posix_trace_attr_setname.....	567
portability, integer arithmetic.....	14	posix_trace_attr_setstreamfullpolicy..... 567	
posix_close.....	499	posix_trace_attr_setstreamsize.....	568
posix_devctl.....	134	posix_trace_clear.....	568
posix_fadvise.....	146	posix_trace_close.....	568
posix_fallocate.....	146	posix_trace_create.....	569
posix_getdents.....	135	posix_trace_create_withlog.....	569
posix_madvise.....	424	posix_trace_event.....	569
posix_mem_offset.....	424	posix_trace_eventid_equal.....	569
posix_memalign.....	393	posix_trace_eventid_get_name.....	570
posix_openpt.....	393	posix_trace_eventid_open.....	570
posix_spawn.....	316	posix_trace_eventset_add.....	570
posix_spawn_file_actions_addchdir.....	316	posix_trace_eventset_del.....	571
posix_spawn_file_actions_addchdir_np.....	753	posix_trace_eventset_empty.....	571
posix_spawn_file_actions_addclose.....	316	posix_trace_eventset_fill.....	571
posix_spawn_file_actions_addclosefrom_np.....	754	posix_trace_eventset_ismember.....	571
posix_spawn_file_actions_adddup2.....	317	posix_trace_eventtypelist_getnext_id..... 572	
posix_spawn_file_actions_addfdchdir.....	317	posix_trace_eventtypelist_rewind.....	572
posix_spawn_file_actions_addfdchdir_np....	754	posix_trace_flush.....	572
posix_spawn_file_actions_addopen.....	317	posix_trace_get_attr.....	573
posix_spawn_file_actions_addtcsetpgrp_np.....	754	posix_trace_get_filter.....	573
posix_spawn_file_actions_destroy.....	318	posix_trace_get_status.....	573
posix_spawn_file_actions_init.....	318	posix_trace_getnext_event.....	573
posix_spawnattr_destroy.....	318	posix_trace_open.....	574
posix_spawnattr_getcgroup_np.....	754	posix_trace_rewind.....	574
posix_spawnattr_getflags.....	318	posix_trace_set_filter.....	574
posix_spawnattr_getpgroup.....	319	posix_trace_shutdown.....	575
posix_spawnattr_getschedparam.....	319	posix_trace_start.....	575
posix_spawnattr_getschedpolicy.....	319	posix_trace_stop.....	575
posix_spawnattr_getsigdefault.....	320	posix_trace_timedgetnext_event.....	575
posix_spawnattr_getsigmask.....	320	posix_trace_trid_eventid_open.....	576
posix_spawnattr_init.....	320	posix_trace_trygetnext_event.....	576
posix_spawnattr_setcgroup_np.....	755	posix_typed_mem_get_info.....	425
posix_spawnattr_setflags.....	320	posix_typed_mem_open.....	425
posix_spawnattr_setpgroup.....	321	POSIX Awk.....	924
posix_spawnattr_setschedparam.....	321	pow.....	235
posix_spawnattr_setschedpolicy.....	321	pow10.....	649
posix_spawnattr_setsigdefault.....	321	pow10f.....	649
posix_spawnattr_setsigmask.....	322	pow10l.....	649
posix_spawnnp.....	322	powf.....	235
posix_trace_attr_destroy.....	562	powl.....	235
posix_trace_attr_getclockres.....	562	pown.....	235
posix_trace_attr_getcreatetime.....	563	pownf.....	236
posix_trace_attr_getgenversion.....	563	pownl.....	236
posix_trace_attr_getinherited.....	563	powr.....	236

powrf.....	236	pthread_clockjoin_np.....	679
powrl.....	237	pthread_cond_broadcast.....	278
ppoll.....	270	pthread_cond_clockwait.....	278
prctl.....	796	pthread_cond_destroy.....	279
pread.....	499	pthread_cond_init.....	279
preadv.....	809	pthread_cond_signal.....	279
preadv2.....	810	pthread_cond_timedwait.....	279
printf.....	355	pthread_cond_wait.....	280
printf_size.....	677	pthread_condattr_destroy.....	280
printf_size_info.....	677	pthread_condattr_getclock.....	280
privileged tests modules.....	49	pthread_condattr_getpshared.....	280
prlimit.....	798	pthread_condattr_init.....	281
process_madvise.....	791	pthread_condattr_setclock.....	281
process_mrelease.....	791	pthread_condattr_setpshared.....	281
process_vm_readv.....	810	pthread_create.....	281
process_vm_writev.....	810	pthread_detach.....	282
profil.....	826	pthread_equal.....	282
program_invocation_name.....	619	pthread_exit.....	282
program_invocation_short_name.....	619	pthread_getaffinity_np.....	679
program_name.....	913	pthread_getattr_default_np.....	680
pselect.....	429	pthread_getattr_np.....	680
psiginfo.....	309	pthread_getconcurrency.....	577
psignal.....	310	pthread_getcpuclockid.....	282
pthread_atfork.....	271	pthread_getname_np.....	680
pthread_attr_destroy.....	271	pthread_getschedparam.....	283
pthread_attr_getaffinity_np.....	678	pthread_getspecific.....	283
pthread_attr_getdetachstate.....	271	pthread_gettid_np.....	681
pthread_attr_getguardsize.....	272	pthread_join.....	283
pthread_attr_getinheritsched.....	272	pthread_key_create.....	283
pthread_attr_getschedparam.....	272	pthread_key_delete.....	284
pthread_attr_getschedpolicy.....	272	pthread_kill.....	310
pthread_attr_getscope.....	273	pthread_kill_other_threads_np.....	681
pthread_attr_getsigmask_np.....	679	pthread_mutex_clocklock.....	284
pthread_attr_getstack.....	273	pthread_mutex_consistent.....	284
pthread_attr_getstackaddr.....	576	pthread_mutex_destroy.....	284
pthread_attr_getstacksize.....	273	pthread_mutex_getprioceiling.....	285
pthread_attr_init.....	273	pthread_mutex_init.....	285
pthread_attr_setaffinity_np.....	678	pthread_mutex_lock.....	285
pthread_attr_setdetachstate.....	274	pthread_mutex_setprioceiling.....	285
pthread_attr_setguardsize.....	274	pthread_mutex_timedlock.....	286
pthread_attr_setinheritsched.....	274	pthread_mutex_trylock.....	286
pthread_attr_setschedparam.....	274	pthread_mutex_unlock.....	286
pthread_attr_setschedpolicy.....	275	pthread_mutexattr_destroy.....	287
pthread_attr_setscope.....	275	pthread_mutexattr_getprioceiling.....	287
pthread_attr_setsigmask_np.....	679	pthread_mutexattr_getprotocol.....	287
pthread_attr_setstack.....	275	pthread_mutexattr_getpshared.....	287
pthread_attr_setstackaddr.....	577	pthread_mutexattr_getrobust.....	288
pthread_attr_setstacksize.....	275	pthread_mutexattr_gettype.....	288
pthread_barrier_destroy.....	276	pthread_mutexattr_init.....	288
pthread_barrier_init.....	276	pthread_mutexattr_setprioceiling.....	288
pthread_barrier_wait.....	276	pthread_mutexattr_setprotocol.....	289
pthread_barrierattr_destroy.....	276	pthread_mutexattr_setpshared.....	289
pthread_barrierattr_getpshared.....	277	pthread_mutexattr_setrobust.....	289
pthread_barrierattr_init.....	277	pthread_mutexattr_settype.....	289
pthread_barrierattr_setpshared.....	277	pthread_once.....	290
pthread_cancel.....	277	pthread_rwlock_clockrdlock.....	290
pthread_cleanup_pop.....	278	pthread_rwlock_clockwrlock.....	290
pthread_cleanup_push.....	278	pthread_rwlock_destroy.....	291

pthread_rwlock_init 291
 pthread_rwlock_rdlock 291
 pthread_rwlock_timedrdlock 291
 pthread_rwlock_timedwrlock 292
 pthread_rwlock_tryrdlock 292
 pthread_rwlock_trywrlock 292
 pthread_rwlock_unlock 293
 pthread_rwlock_wrlock 293
 pthread_rwlockattr_destroy 293
 pthread_rwlockattr_getkind_np 681
 pthread_rwlockattr_getpshared 294
 pthread_rwlockattr_init 294
 pthread_rwlockattr_setkind_np 681
 pthread_rwlockattr_setpshared 294
 pthread_self 294
 pthread_setaffinity_np 682
 pthread_setattr_default_np 682
 pthread_setcancelstate 295
 pthread_setcanceltype 295
 pthread_setconcurrency 577
 pthread_setname_np 682
 pthread_setschedparam 295
 pthread_setschedprio 295
 pthread_setspecific 296
 pthread_sigmask 310
 pthread_sigqueue 683
 pthread_spin_destroy 296
 pthread_spin_init 296
 pthread_spin_lock 296
 pthread_spin_trylock 297
 pthread_spin_unlock 297
 pthread_testcancel 297
 pthread_timedjoin_np 683
 pthread_tryjoin_np 683
 ptrace 796
 ptsname 393
 ptsname_r 394
 putc 358
 putc_unlocked 358
 putchar 358
 putchar_unlocked 359
 putenv 394
 putgrent 634
 putmsg 577
 putpmsg 578
 putpwent 686
 puts 359
 putsgent 636
 putspent 748
 pututline 831
 pututxline 511
 putw 760
 putwc 518
 putwc_unlocked 837
 putwchar 518
 putwchar_unlocked 838
 pvalloc 640
 pwrite 500

pwritev 810
 pwritev2 811

Q

qecvt 769
 qecvt_r 769
 qfcvt 769
 qfcvt_r 769
 qgcvt 770
 qsort 395
 qsort_r 395
 quick_exit 395
 quotactl 796
 quote 913
 quote_n 913
 quotearg 913
 quotearg_alloc 913
 Quoting 913

R

raise 311
 rand 396
 rand_r 396
 random 396
 random_r 770
 range argument to `re_search` 938
 rawmemchr 777
 rcmd 663
 rcmd_af 664
 re_comp 686, 943
 re_compile_fastmap 686, 940
 re_compile_pattern 687, 937
 re_exec 687, 944
 re_match 687, 937
 re_match_2 687, 939
 re_nsub field, set by `re_compile_pattern` 937
 re_pattern_buffer definition 936
 re_registers 941
 re_search 687, 938
 re_search_2 688, 939
 re_set_registers 688, 941
 re_set_syntax 688
 re_syntax_options 688
 re_syntax_options initialization 936
 RE_BACKSLASH_ESCAPE_IN_LIST 922
 RE_BK_PLUS_QM 922
 RE_CHAR_CLASSES 922
 RE_CONTEXT_INDEP_ANCHORS 922
 RE_CONTEXT_INDEP_ANCHORS (and ‘^’) 934
 RE_CONTEXT_INDEP_OPS 922
 RE_CONTEXT_INVALID_DUP 922
 RE_CONTEXT_INVALID_OPS 923
 RE_DEBUG 923
 RE_DOT_NEWLINE 923
 RE_DOT_NOT_NULL 923
 RE_HAT_LISTS_NOT_NEWLINE 923

RE_ICASE	923
RE_INTERVALS	923
RE_INVALID_INTERVAL_ORD	923
RE_LIMITED_OPS	923
RE_NEWLINE_ALT	923
RE_NO_BK_BRACES	923
RE_NO_BK_PARENS	924
RE_NO_BK_REFS	924
RE_NO_BK_VBAR	924
RE_NO_EMPTY_RANGES	924
RE_NO_GNU_OPS	924
RE_NO_POSIX_BACKTRACKING	924
RE_NO_SUB	924
RE_UNMATCHED_RIGHT_PAREN_ORD	924
read	500
readahead	623
readdir	136
readdir_r	136
readlink	501
readlinkat	501
readv	447
realloc	397
reallocarray	398
realpath	399
reboot	797
recv	434
recvfrom	435
recvmsg	799
recvmsg	435
regcomp	299
regerror	300
regex.h	921
regexexec	300
regex anchoring	934
regfree	300, 943
register_printf_function	677
register_printf_modifier	677
register_printf_specifier	678
register_printf_type	678
regular expressions, syntax of	921
remainder	237
remainderf	237
remainderl	237
remap_file_pages	791
remove	360
removexattr	815
remque	304
remquo	238
remquoof	238
remquoof	238
rename	360
renameat	361
renameat2	760
res_dnok	690
res_hnok	690
res_init	690
res_mailok	691
res_mkquery	691

```

res_nmkquery ..... 691
res_nquery ..... 691
res_nquerydomain ..... 692
res_nsearch ..... 692
res_nsend ..... 692
res_ownok ..... 692
res_query ..... 693
res_querydomain ..... 693
res_search ..... 693
res_send ..... 693
revoke ..... 826
rewind ..... 362
rewinddir ..... 137
rexec ..... 664
rexec_af ..... 664
rindex ..... 578
rint ..... 238
rintf ..... 239
rintl ..... 239
rmdir ..... 502
rootn ..... 239
rootnf ..... 239
rootnl ..... 240
round ..... 240
roundeven ..... 240
roundevenf ..... 240
roundevenl ..... 241
roundf ..... 241
roundl ..... 241
rpc_createerr ..... 702
rpmatch ..... 771
rresvport ..... 664
rresvport_af ..... 665
rsqrt ..... 242
rsqrtf ..... 242
rsqrtl ..... 242
rtime ..... 698
ruserok ..... 665
ruserok_af ..... 665

```

S

sbrk.....	827
scalb.....	578
scalbf.....	650
scalbl.....	650
scalbln.....	242
scalblnf.....	243
scalblnl.....	243
scalbn.....	243
scalbnf.....	243
scalbnl.....	243
scandir.....	137
scandirat.....	613
scanf.....	362
sched_get_priority_max.....	300
sched_get_priority_min.....	301
sched_getaffinity.....	741

<code>sched_getattr</code>	741	<code>setlocale</code>	167
<code>sched_getcpu</code>	741	<code>setlogin</code>	827
<code>sched_getparam</code>	301	<code>setlogmask</code>	449
<code>sched_getscheduler</code>	301	<code>setmntent</code>	658
<code>sched_rr_get_interval</code>	301	<code>setnetent</code>	268
<code>sched_setaffinity</code>	742	<code>setnetgrent</code>	665
<code>sched_setattr</code>	742	<code>setns</code>	742
<code>sched_setparam</code>	302	<code>setpayload</code>	244
<code>sched_setscheduler</code>	302	<code>setpayloadf</code>	244
<code>sched_yield</code>	302	<code>setpayloadl</code>	244
searching with GNU functions	938	<code>setpayloadsig</code>	244
<code>secure_getenv</code>	399	<code>setpayloadsigf</code>	245
<code>seed48</code>	399	<code>setpayloadsigl</code>	245
<code>seed48_r</code>	771	<code>setpgid</code>	503
<code>seekdir</code>	137	<code>setpgrp</code>	579
<code>select</code>	429	<code>setpriority</code>	427
<code>sem_clockwait</code>	305	<code>setprotoent</code>	268
<code>sem_close</code>	305	<code>setpwent</code>	299
<code>sem_destroy</code>	306	<code>setregid</code>	503
<code>sem_getvalue</code>	306	<code>setresgid</code>	504
<code>sem_init</code>	306	<code>setresuid</code>	504
<code>sem_open</code>	306	<code>setreuid</code>	504
<code>sem_post</code>	307	<code>setrlimit</code>	428
<code>sem_timedwait</code>	307	<code>setrpcent</code>	705
<code>sem_trywait</code>	307	<code>setservernt</code>	268
<code>sem_unlink</code>	307	<code>setsgent</code>	636
<code>sem_wait</code>	307	<code>setsid</code>	504
<code>semctl</code>	430	<code>setsockopt</code>	436
<code>semget</code>	430	<code>setsourcefilter</code>	672
<code>semop</code>	430	<code>setspent</code>	748
<code>semtimedop</code>	798	<code>setstate</code>	400
<code>send</code>	435	<code>setstate_r</code>	771
<code>sendfile</code>	798	<code>settimeofday</code>	807
<code>sendmmsg</code>	800	<code>settyent</code>	819
<code>sendmsg</code>	436	<code>setuid</code>	505
<code>sendto</code>	436	<code>setusershell</code>	828
<code>setaliasent</code>	599	<code>setutent</code>	832
<code>setbuf</code>	363	<code>setutxent</code>	511
<code>setbuffer</code>	760	<code>setvbuf</code>	363
<code>setcontext</code>	578	<code>setxattr</code>	816
<code>setdomainname</code>	827	<code>sgetsgent</code>	636
<code>setegid</code>	503	<code>sgetsgent_r</code>	637
<code>setenv</code>	400	<code>sgetspent</code>	748
<code>seteuid</code>	503	<code>sgetspent_r</code>	748
<code>setfsent</code>	627	<code>shm_open</code>	425
<code>setfsgid</code>	785	<code>shm_unlink</code>	425
<code>setfsuid</code>	785	<code>shmat</code>	430
<code>setgid</code>	503	<code>shmctl</code>	431
<code>setgrent</code>	155	<code>shmdt</code>	431
<code>setgroups</code>	634	<code>shmget</code>	431
<code>sethostent</code>	268	<code>shutdown</code>	436
<code>sethostid</code>	827	<code>sig2str</code>	311
<code>sethostname</code>	828	<code>sigabbrev_np</code>	777
<code>setipv4sourcefilter</code>	672	<code>sigaction</code>	311
<code>setitimer</code>	579	<code>sigaddset</code>	312
<code>setjmp</code>	308	<code>sigaltstack</code>	312
<code>setkey</code>	400	<code>sigandset</code>	749
<code>setlinebuf</code>	761	<code>sigblock</code>	749

sigdelset	312	srand48	401
sigdescr_np	778	srand48_r	772
sigemptyset	312	srandom	401
sigfillset	313	srandom_r	772
siggetmask	750	sscanf	367
sighold	579	ssignal	752
sigignore	579	ST_BLKSIZE	920
siginterrupt	580	ST_NBLOCKS	920
sigisemptyset	750	ST_NBLOCKSIZE	920
sigismember	313	start argument to re_search	938
siglongjmp	308	start in struct re_registers	941
signal	313	stat	444
signalfd	799	statfs	801
signbit	245	static inline	877
signgam	246	statvfs	446
significand	650	statx	801
significandf	651	stdc_bit_ceil	333
significandl	651	stdc_bit_floor	334
sigorset	750	stdc_bit_width	334
sigpause	580	stdc_count_ones	334
sigpending	314	stdc_count_zeros	334
sigprocmask	314	stdc_first_leading_one	335
sigqueue	314	stdc_first_leading_zero	335
sigrelse	580	stdc_first_trailing_one	335
sigreturn	751	stdc_first_trailing_zero	335
sigset	581	stdc_has_single_bit	336
sigsetjmp	309	stdc_leading_ones	336
sigsetmask	751	stdc_leading_zeros	336
sigstack	751	stdc_trailing_ones	336
sigsuspend	314	stdc_trailing_zeros	337
sigtimedwait	315	stderr	367
sigvec	751	stdin	368
sigwait	315	stdopen	879
sigwaitinfo	315	stdout	368
sin	246	step	689
sincos	651	stime	817
sincosf	652	STP_BLKSIZE	920
sincosl	652	STP_NBLOCKS	920
sinf	246	stpcpy	410
sinh	246	stpncpy	411
sinhf	246	str2sig	315
sinhl	247	strcasecmp	420
sinl	247	strcasecmp_l	421
sinpi	247	strcasestr	778
sinpif	248	strcat	411
sinpil	248	strchr	411
sleep	505	strchrnul	778
snprintf	363	strcmp	411
socketatmark	437	strcoll	412
socket	437	strcoll_l	412
socketpair	437	strcpy	412
specification	42	strcspn	412
sprintf	365	strdup	413
sprofil	796	strerror	413
sqrt	248	strerror_l	413
sqrtf	248	strerror_r	414
sqrtl	249	strerrordesc_np	779
srand	401	strerrorname_np	779

tcgetattr	451
tcgetpgrp	507
tcgetsid	452
tcgetwinsize	452
tcsendbreak	452
tcsetattr	452
tcsetpgrp	507
tcsetwinsize	453
tdelete	304
tdestroy	744
telldir	138
tempnam	581
tests modules, C++	49
tests modules, long-running	49
tests modules, privileged	49
tests modules, unportable	49
textdomain	165
tfind	304
tgamma	251
tgammaf	251
tgammaL	252
tgkill	752
thrd_create	458
thrd_current	458
thrd_detach	458
thrd_equal	459
thrd_exit	459
thrd_join	459
thrd_sleep	460
thrd_yield	460
time	470
timegm	471
timelocal	818
timer_create	471
timer_delete	471
timer_getoverrun	472
timer_gettime	472
timer_settime	472
timerfd_create	807
timerfd_gettime	808
timerfd_settime	808
times	447
timespec_get	472
timespec_getres	473
timezone	473
tmpfile	368
tmpnam	369
tmpnam_r	761
toascii	581
tolower	132
tolower_l	132
totalorder	252
totalorderf	252
totalorderl	253
totalordermag	253
totalordermagf	253
totalordermagl	254
toupper	133

toupper_1	133
towctrans	548
towctrans_1	548
tolower	548
tolower_1	549
towupper	549
towupper_1	550
translate initialization	937
trunc	254
truncate	507
truncf	254
truncl	255
tsearch	305
tss_create	461
tss_delete	461
tss_get	461
tss_set	462
ttyname	507
ttyname_r	508
ttyslot	829
twalk	305
twalk_r	744
type conversion, arithmetic	870
TYPE_IS_INTEGER	869
TYPE_MAXIMUM	871
TYPE_MINIMUM	871
TYPE_SIGNED	869
tzname	473
tzset	474

U

uabs.....	774
ualarm.....	582
ufromfp.....	255
ufromfpf.....	255
ufromfpl.....	255
ufromfpx.....	256
ufromfpxf.....	256
ufromfpxl.....	256
uimaxabs.....	637
ulabs.....	775
ulckpddf.....	749
ulimit.....	582
ullabs.....	775
umask.....	445
umount.....	794
umount2.....	794
uname.....	448
ungetc.....	369
ungetwc.....	520
unibyte locale.....	851
unlink.....	508
unlinkat.....	509
unlockpt.....	407
unportable tests modules.....	49
unsetenv.....	407
updwtmp.....	832

updwtmpx	834
uselocale	168
user2netname	697
usleep	582
ustat	811
utime	582
utimensat	445
utimes	446
utmpname	832
utmpxname	835

V

va_arg	323
va_copy	323
va_end	323
va_start	323
valgrind	963
valloc	775
vasprintf	369
VCS To ChangeLog	967
vdprintf	371
verify	866
verify_expr	866
verr	617
verrx	618
versionsort	613
vfork	583
vfprintf	372
vfscanf	374
vfwprintf	520
vfwscanf	521
vhangup	829
vlimit	811
void	865
vprintf	375
vscanf	377
vsnprintf	377
vsprintf	379
vsscanf	381
vswprintf	522
vswscanf	523
vsyslog	804
vwarn	618
vwarnx	618
vwprintf	523
vwscanf	524

W

wait	448
wait3	812
wait4	812
waitid	448
waitpid	449
warn	618
warnx	619
wcpcpy	525
wcpncpy	525
wcrtomb	525
wcscasecmp	526
wcscasecmp_l	526
wcscat	526
wcschr	526
wcschrnul	838
wcscmp	527
wcscoll	527
wcscoll_l	527
wcscpy	527
wcscspn	528
wcsdup	528
wcsftime	528
wcsftime_l	838
wcslcat	529
wcslcpy	529
wcslen	529
wcsncasecmp	529
wcsncasecmp_l	530
wcsncat	530
wcsncmp	530
wcsncpy	530
wcsnlen	531
wcsnrtombs	531
wcspbrk	532
wcsrchr	532
wcsrtombs	532
wcsspn	532
wcsstr	533
wcstod	533
wcstod_l	839
wcstof	533
wcstof_l	839
wcstoimax	158
wcstok	533
wcstol	534
wcstol_l	839
wcstold	534
wcstold_l	839
wcstoll	534
wcstoll_l	840
wcstombs	407
wcstoq	840
wcstoul	535
wcstoul_l	840
wcstoull	535
wcstoull_l	840
wcstoumax	159

wcstouq	841
wcswcs	583
wcswidth	535
wcsxfrm	535
wcsxfrm_l	536
wctob	536
wctomb	408
wctrans	550
wctrans_l	550
wctype	551
wctype_l	551
wcwidth	536
wmemchr	537
wmemcmp	537
wmemcpy	537
wmemmove	537
wmemcpy	841
wmemset	538
word boundaries, matching	935
wordexp	551
wordfree	552
wprintf	538
wraparound integer arithmetic	872
write	509
writew	448
wscanf	539

X

xalloc_die	49
xdr_array	715
xdr_authunix_parms	698
xdr_bool	715
xdr_bytes	715
xdr_callhdr	708
xdr_callmsg	708
xdr_cback_data	734
xdr_char	715
xdr_cryptkeyarg	702
xdr_cryptkeyarg2	702
xdr_cryptkeyres	702
xdr_des_block	697
xdr_domainname	735
xdr_double	716
xdr_enum	716
xdr_float	716
xdr_free	716
xdr_getcredres	703
xdr_hyper	716
xdr_int	717
xdr_int16_t	717
xdr_int32_t	717
xdr_int64_t	717
xdr_int8_t	717
xdr_key_netstarg	703
xdr_key_netstres	703
xdr_keybuf	703
xdr_keydat	735

xdr_keystatus	703
xdr_long	718
xdr_longlong_t	718
xdr_netnamestr	703
xdr_netobj	718
xdr_obj_p	734
xdr_opaque	718
xdr_opaque_auth	697
xdr_pmap	707
xdr_pmaplist	707
xdr_pointer	718
xdr_quad_t	719
xdr_reference	719
xdr_replymsg	709
xdr_rmtcall_args	708
xdr_rmtcallres	708
xdr_short	719
xdr_sizeof	719
xdr_string	719
xdr_u_char	720
xdr_u_hyper	720
xdr_u_int	720
xdr_u_long	720
xdr_u_longlong_t	721
xdr_u_quad_t	721
xdr_u_short	721
xdr_uint16_t	721
xdr_uint32_t	721
xdr_uint64_t	722
xdr_uint8_t	722
xdr_union	722
xdr_unixcred	704
xdr_valdat	735
xdr_vector	722
xdr_void	722
xdr_wrapstring	723
xdr_yplib_resptype	735
xdr_yplib_parms	735
xdr_yplibmaplist	736
xdr_yplibpushresp_xfr	736
xdr_yplibreq_key	736
xdr_yplibreq_nokey	736
xdr_yplibreq_xfr	736
xdr_yplibresp_all	736
xdr_yplibresp_key_val	737
xdr_yplibresp_maplist	737
xdr_yplibresp_master	737
xdr_yplibresp_order	737
xdr_yplibresp_val	737
xdr_yplibresp_xfr	737
xdr_yplibstat	738
xdr_yplibxfirststat	738
xdrmem_create	723
xdrrec_create	723
xdrrec_endofrecord	723
xdrrec_eof	724
xdrrec_skiprecord	724
xdrstdio_create	724

xprt_register 714
xprt_unregister 714

Y

y0 256
y0f 652
y0l 653
y1 257
y1f 653
y1l 653
yn 257
ynf 654

ynl 654
yp_all 738
yp_bind 738
yp_first 738
yp_get_default_domain 739
yp_master 739
yp_match 739
yp_next 739
yp_order 739
yp_unbind 739
ypbinderr_string 740
yperr_string 740
ypprot_err 740